

MHT-CET Biology Answer Key & Solutions

All 2,120 records | Repaired release v2.1.0

Questions	2,120
Standard XI / XII	436 / 1,684
Difficulty labels	Easy, Medium, Hard, Challenging
Answer-key balance	A = B = C = D = 530
Figures	44 assets linked to 82 questions
Validation	PASS

Each entry gives the keyed option, answer text, a concise solution and rationales for all three distractors. The answers retain record IDs so they can be reconciled directly with the JSON, workbook and generated papers.

Standard 11 | Chapter 01

Biomolecules

BIO-11-CH01-0001 | Easy | Bioelements

Correct answer: B. Carbon, hydrogen and oxygen

Solution: Only carbon, hydrogen and oxygen are placed in the prime element group. They are the elements from which the bulk of the organic molecules of the cell is built.

Distractor review: A: Nitrogen belongs to the secondary elements; the student stretches the prime group to include it because these four are usually quoted together as the main elements of living matter. C: These three are secondary elements, and the familiar fertiliser trio is mistaken for the group that builds the bulk of organic matter. D: Iron, zinc and copper are trace elements needed in very small amounts, so they cannot form the prime group.

BIO-11-CH01-0002 | Medium | Bioelements

Correct answer: A. Nitrogen is a secondary element and chlorine is a trace element

Solution: Nitrogen is listed with the secondary elements and chlorine with the trace elements, so this is the only pairing that matches the grouping of inorganic matter in the cell.

Distractor review: B: The prime group is restricted to carbon, hydrogen and oxygen, and chlorine is required only in trace amounts, so both placements fail. C: Chlorine is placed correctly, but nitrogen is promoted to the prime group, which contains only carbon, hydrogen and oxygen. D: The two groups are interchanged: nitrogen is a secondary element while chlorine is one of the trace elements.

BIO-11-CH01-0003 | Hard | Organic and inorganic matter of the cell

Correct answer: D. I and II only

Solution: Polynucleotides appear in the macromolecule list and glycerol in the micromolecule list, so both of those placements hold. Alkaloids belong to the third group, the secondary metabolites, so that placement fails.

Distractor review: A: Alkaloids are listed with the secondary metabolites, alongside phenolics and terpenoids, and not with the macromolecules. B: Polynucleotides are indeed macromolecules, so rejecting the first placement is wrong, and alkaloids are secondary metabolites. C: The placement of alkaloids is the faulty one; they belong to the secondary metabolites, which form a group of their own.

BIO-11-CH01-0004 | Medium | Macromolecules and micromolecules

Correct answer: D. Polypeptides and amino acids

Solution: Polypeptides are in the macromolecule list and amino acids, their building units, are in the micromolecule list, so this pair matches the order asked for.

Distractor review: A: The order is reversed: simple sugars are micromolecules while polysaccharides are macromolecules. B: Both fatty acids and glycerol appear in the micromolecule list, so the first member is not a macromolecule. C: Both members are macromolecules, so the second one cannot be the micromolecule asked for.

BIO-11-CH01-0005 | Medium | Organic and inorganic matter of the cell

Correct answer: B. Organic matter, as secondary metabolites

Solution: Essential oils and lectins are both named in the secondary metabolite group, which is the third subdivision of the organic matter of the cell.

Distractor review: A: The micromolecules are the simple sugars, nucleotides, fatty acids, glycerol and amino acids; essential oils and lectins form a separate group of their own. C: The macromolecules are the polysaccharides, polynucleotides, lipids and polypeptides, and neither essential oils nor lectins appears in that list. D: Trace elements are single elements such as zinc and boron, whereas essential oils and lectins are carbon compounds and therefore organic.

BIO-11-CH01-0006 | Hard | Secondary metabolites

Correct answer: A. I and III only (secondary-metabolite statements)

Solution: Secondary metabolites are small organic molecules made by several bacteria, fungi and plants, and they are not required for the growth, development or reproduction of the producer, which is what makes them secondary.

Distractor review: B: A secondary metabolite is by definition not essential for growth, development and reproduction, so the second statement contradicts the definition. C: The claim of essentiality is the false one; describing these molecules as small and organic is part of their definition. D: Accepting all three ignores that these molecules are singled out precisely because the organism can grow, develop and reproduce without them.

BIO-11-CH01-0007 | Easy | General formula of carbohydrates**Correct answer: B. 2:1, the ratio found in water**

Solution: Every repeating CH₂O unit contains two hydrogen atoms and one oxygen atom, giving a 2:1 ratio throughout the molecule, which is exactly the ratio of hydrogen to oxygen in water.

Distractor review: A: The ratio has been inverted; each CH₂O unit carries two hydrogen atoms for one oxygen atom. C: Water is H₂O and already has two hydrogen atoms per oxygen atom, so a 1:1 ratio matches neither water nor the general formula. D: The ratio is right but carbon dioxide contains no hydrogen at all, so it cannot show any hydrogen to oxygen ratio.

BIO-11-CH01-0008 | Medium | General formula of carbohydrates**Correct answer: C. C₅H₁₀O₅**

Solution: In C₅H₁₀O₅ the carbon and oxygen atoms are equal at five each and the hydrogen atoms are exactly twice that number, so the formula can be written as (CH₂O)₅.

Distractor review: A: Twelve hydrogen atoms against five oxygen atoms is not a 2:1 ratio; twelve hydrogen atoms would need six oxygen atoms. B: Here hydrogen is fewer than twice oxygen, so the repeating unit CH₂O does not fit. D: Carbon and oxygen must be equal in number for the unit CH₂O to repeat a whole number of times, and here they are not.

BIO-11-CH01-0009 | Medium | General formula of carbohydrates**Correct answer: C. Pentose, 5 oxygen atoms**

Solution: Ten hydrogen atoms means $2n = 10$, so $n = 5$. A sugar with five carbon atoms is a pentose, and it contains five oxygen atoms, as in C₅H₁₀O₅.

Distractor review: A: The series is worked out correctly but the oxygen count is taken equal to the hydrogen count instead of half of it. B: Ten hydrogen atoms give n equal to 5, not 6; a hexose would carry twelve hydrogen atoms. D: Halving the hydrogen count gives 5, so taking n as 4 understates the molecule, which would then have only eight hydrogen atoms.

BIO-11-CH01-0010 | Hard | Monosaccharides**Correct answer: D. Sedoheptulose, 14 hydrogen atoms**

Solution: Seven carbon atoms place the sugar in the heptose group, whose named example is sedoheptulose. With $n = 7$ the molecule carries $2 \times 7 = 14$ hydrogen atoms, giving C₇H₁₄O₇.

Distractor review: A: The sugar is named correctly, but the hydrogen count is taken equal to the carbon count, whereas each CH₂O unit carries two hydrogen atoms. B: Twelve hydrogen atoms belong to a six carbon sugar; the hexose value has been carried over to a seven carbon sugar. C: Erythrose is the four carbon tetrose, so it cannot be the seven carbon member of the series even though the arithmetic is right.

BIO-11-CH01-0011 | Easy | Monosaccharides**Correct answer: D. Triose**

Solution: Glyceraldehyde has three carbon atoms and is the example given for the triose group, the smallest member of the monosaccharide series.

Distractor review: A: The tetrose named in the classification is erythrose, a four carbon sugar. B: The pentoses named are ribose and deoxyribose, each with five carbon atoms. C: The heptose named is sedoheptulose, with seven carbon atoms, the largest of the series given.

BIO-11-CH01-0012 | Medium | Monosaccharides**Correct answer: B. Erythrose, 4 carbon atoms**

Solution: Erythrose is the named tetrose, so a molecule of it contains four carbon atoms, which is the only correct pairing among those offered.

Distractor review: A: Ribose is a pentose with five carbon atoms and is being confused with the hexose group. C: Galactose is a hexose with six carbon atoms, listed together with glucose and fructose. D: Sedoheptulose is the seven carbon heptose, not a six carbon sugar.

BIO-11-CH01-0013 | Challenging | Disaccharides**Correct answer: D. Z is sucrose, and its hydrogen to oxygen ratio is 2:1**

Solution: Dividing 22 hydrogen atoms by 11 oxygen atoms gives the 2:1 ratio expected of a carbohydrate. A disaccharide that yields two hexoses, one of them fructose, must be sucrose, which is glucose plus fructose.

Distractor review: A: The ratio is worked out correctly, but maltose yields two glucose units on hydrolysis and never yields fructose. B: The sugar is identified correctly, but 22 hydrogen to 11 oxygen reduces exactly to 2:1; a ratio greater than 2:1 is a feature of lipids, not of carbohydrates. C: Lactose yields glucose and galactose rather than fructose, and 22 hydrogen atoms to 11 oxygen atoms is 2:1 and not 1:1.

BIO-11-CH01-0014 | Easy | Disaccharides**Correct answer: C. Galactose and maltose**

Solution: Galactose is one of the named hexose monosaccharides and maltose is a disaccharide of two glucose units, so this pair follows the order asked for.

Distractor review: A: The order is reversed: maltose is the disaccharide and galactose the monosaccharide. B: Both members are disaccharides, so the first one cannot be the monosaccharide asked for. D: Both members are monosaccharides, a pentose and a hexose, so neither is a disaccharide.

BIO-11-CH01-0015 | Challenging | Disaccharides**Correct answer: A. X maltose, Y lactose, Z sucrose**

Solution: Only maltose gives two molecules of one and the same sugar, glucose, so it is X. Lactose is the disaccharide of glucose and galactose, so it is Y. Sucrose is the form in which carbohydrate travels in plants, so it is Z.

Distractor review: B: Sucrose yields two different monosaccharides, glucose and fructose, and maltose is not the transported form in plants. C: Sucrose yields glucose and fructose rather than glucose and galactose, and lactose is not moved through the plant body. D: The first two are interchanged: lactose gives two different sugars while maltose gives two glucose units.

BIO-11-CH01-0016 | Easy | Glycosidic bond**Correct answer: B. Glycosidic bond**

Solution: The bond formed on condensation between a sugar unit and the group next to it is the glycosidic bond, and it is found in disaccharides, in polysaccharides and in nucleosides alike.

Distractor review: A: A peptide bond joins the carboxyl group of one amino acid residue to the amino group of another and belongs to polypeptides. C: Ester linkages join fatty acids to alcohols in simple lipids and do not link sugar units to one another. D: Hydrogen bonds act between whole molecules, as in cellulose, but they do not join the sugar units within a chain.

BIO-11-CH01-0017 | Medium | Glycosidic bond**Correct answer: B. 249**

Solution: Each new unit added to an unbranched chain brings one new glycosidic bond, and the first unit brings none, so 250 units are held together by $250 - 1 = 249$ glycosidic bonds.

Distractor review: A: One bond is counted for every unit, but a chain of n units joined end to end has one bond fewer than the number of units. C: The number of units is halved, as though each bond used up two units and no unit were shared between successive bonds. D: An extra bond is added beyond the end of the chain, although the two terminal units are bonded on one side only.

BIO-11-CH01-0018 | Medium | Reducing sugars**Correct answer: D. Cu^{2+} is converted to Cu^+**

Solution: The free aldehyde or ketone group of the monosaccharide reduces the copper of Benedict's reagent, taking it from the cupric state Cu^{2+} to the cuprous state Cu^+ .

Distractor review: A: This is an oxidation of copper, whereas the sugar acts as the reducing agent and so the copper of the reagent must be reduced. B: The copper of the reagent is brought only from the cupric to the cuprous state, not all the way to the free metal. C: The reagent supplies copper in the cupric state, so the change must begin at Cu^{2+} and not at Cu^+ .

BIO-11-CH01-0019 | Hard | Blood glucose**Correct answer: B. 4.5 g**

Solution: 5 litres is 5000 ml, which contains $5000 / 100 = 50$ portions of 100 ml. Each portion holds about 90 mg of glucose, so the total is $50 \times 90 = 4500$ mg, that is 4.5 g.

Distractor review: A: This is the amount in one litre of blood; the total volume of 5 litres has not been used. C: The answer is ten times too large, as though 5 litres contained five hundred portions of 100 ml instead of fifty. D: This assumes 5 litres to be 10000 ml, doubling the number of 100 ml portions in the given volume.

BIO-11-CH01-0020 | Easy | Starch**Correct answer: D. Amylose and amylopectin**

Solution: Starch is not a single substance but is made up of the two components amylose and amylopectin.

Distractor review: A: Glycogen is a polysaccharide in its own right and is not a component of the starch molecule. B: Cellulose is a separate structural polysaccharide and forms no part of starch. C: Both named substances are complete polysaccharides themselves, not the components into which starch is resolved.

BIO-11-CH01-0021 | Medium | Starch**Correct answer: A. Both are components of starch**

Solution: Amylose and amylopectin are the two components of one and the same polysaccharide, starch, which is classed as a homopolysaccharide.

Distractor review: B: Both components belong to starch; glycogen is a separate polysaccharide and is not resolved into named components. C: Cellulose is not resolved into components either, so amylose cannot be assigned to it. D: They belong to starch, which is a homopolysaccharide; the heteropolysaccharides named are hyaluronic acid, heparin and chondroitin sulphate.

BIO-11-CH01-0022 | Hard | Cellulose**Correct answer: C. Glycosidic bonds within a molecule and hydrogen bonds between molecules**

Solution: Within one cellulose molecule the sugar units are held by glycosidic bonds, the linkage common to all polysaccharides. Separate cellulose molecules are then held side by side by hydrogen bonds, which gives cellulose its strength as a structural material.

Distractor review: A: The two are interchanged; the sugar units of any polysaccharide chain are joined by glycosidic bonds. B: Peptide bonds join amino acid residues in polypeptides and take no part in holding cellulose molecules together. D: Ester linkages occur in simple lipids between fatty acids and alcohols, not between the sugar units of a chain.

BIO-11-CH01-0023 | Hard | Polysaccharides**Correct answer: C. Cellulose**

Solution: Cellulose is the homopolysaccharide that does not act as a food reserve. It is a structural material, its molecules being held to one another by hydrogen bonds.

Distractor review: A: Glycogen is the storage polysaccharide of animals, so it follows the general rule rather than being the exception to it. B: Starch, made up of amylose and amylopectin, is a reserve of carbohydrate and not a structural material. D: Amylopectin is one of the two components of starch and shares its role as a reserve rather than a structural one.

BIO-11-CH01-0024 | Hard | Polysaccharides**Correct answer: B. Hyaluronic acid**

Solution: Hyaluronic acid is listed with heparin and chondroitin sulphate among the heteropolysaccharides, which are built from more than one kind of monosaccharide unit, unlike starch, cellulose and glycogen.

Distractor review: A: Glycogen is listed as a homopolysaccharide together with starch and cellulose. C: Starch is a homopolysaccharide; the fact that it has two components does not make it a heteropolysaccharide. D: Cellulose is a homopolysaccharide, set apart by its structural role and not by its composition.

BIO-11-CH01-0025 | Easy | Chitin**Correct answer: C. Chitin**

Solution: The exoskeleton of an insect such as a beetle is built chiefly of chitin, the polysaccharide named for that covering.

Distractor review: A: Cellulose is the structural polysaccharide of plants and does not build the covering of an insect. B: Glycogen is the storage polysaccharide of animals and forms no part of a body covering. D: Heparin is a heteropolysaccharide and is not a covering or supporting material.

BIO-11-CH01-0026 | Medium | Glycogen**Correct answer: A. Glycogen, a homopolysaccharide**

Solution: The carbohydrate reserve of animals is glycogen, and it belongs to the homopolysaccharides, the group built from a single kind of monosaccharide unit.

Distractor review: B: The polysaccharide is named correctly, but glycogen is grouped with starch and cellulose as a homopolysaccharide. C: Chitin occurs in the insect exoskeleton as a covering material and is not the reserve carbohydrate of animals. D: Heparin is indeed a heteropolysaccharide, but it is not a storage form of carbohydrate.

BIO-11-CH01-0027 | Easy | Elemental composition of lipids**Correct answer: A. X is a carbohydrate and Y is a lipid**

Solution: The chapter separates the two classes on the hydrogen to oxygen ratio. A value of exactly 2:1 identifies a carbohydrate and a value greater than 2:1 identifies a lipid, so X is the carbohydrate and Y is the lipid.

Distractor review: B: This reverses the criterion; the ratio of exactly 2:1, the same as in water, belongs to carbohydrates and the raised ratio belongs to lipids. C: A carbohydrate cannot show a hydrogen to oxygen ratio greater than 2:1, since the general formula $(CH_2O)_n$ fixes that ratio at 2:1. D: A ratio of exactly 2:1 is the carbohydrate value, so X cannot be assigned to the lipids on this evidence.

BIO-11-CH01-0028 | Easy | Fatty acids**Correct answer: C. Palmitic acid and stearic acid**

Solution: Palmitic acid and stearic acid are the two saturated fatty acids listed, and they are described as occurring in all animal and plant fats.

Distractor review: A: Both members of this pair are unsaturated; the student swaps the two categories of fatty acid entirely. B: Palmitic acid is saturated but oleic acid is unsaturated, so the pair is mixed rather than wholly saturated. D: Stearic acid is saturated but linoleic acid is unsaturated, so this pair also crosses the two categories.

BIO-11-CH01-0029 | Medium | Fatty acids**Correct answer: B. Oleic acid occurs in nearly all fats and linoleic acid occurs in many seed oils**

Solution: Among the unsaturated fatty acids, oleic acid is described as present in nearly all fats while linoleic acid is described as present in many seed oils.

Distractor review: A: The two descriptions are interchanged; the chapter attaches the wide occurrence in nearly all fats to oleic acid, not to linoleic acid. C: The occurrence in many seed oils is the description given for linoleic acid, so attaching it to oleic acid misplaces the pairing. D: Stearic acid is a saturated fatty acid described as occurring in all animal and plant fats, not as the seed oil example.

BIO-11-CH01-0030 | Easy | Simple lipids**Correct answer: C. Esters of fatty acids with various alcohols**

Solution: The defining feature of a simple lipid is that it is an ester formed between fatty acids and an alcohol, with no additional non-lipid group.

Distractor review: A: The presence of a phosphate group defines a phospholipid, which belongs to the compound lipids and not to the simple lipids. B: A sugar residue is the extra component of a glycolipid, again a compound lipid rather than a simple lipid. D: Sterols are placed under the derived lipids, a class kept separate from the simple lipids in the chapter's scheme.

BIO-11-CH01-0031 | Medium | Simple lipids**Correct answer: A. A fat is an ester of fatty acids with glycerol, while a wax is an ester of a long-chain fatty acid with a long-chain alcohol**

Solution: Both are esters of fatty acids, but the fat uses glycerol as the alcohol and is therefore a triglyceride, while the wax uses a long-chain alcohol together with a long-chain fatty acid.

Distractor review: B: The alcohol components are interchanged; glycerol is the alcohol of fats and the long-chain alcohol is that of waxes. C: A phosphate group would make the molecule a phospholipid, that is a compound lipid, so it cannot be part of the definition of a fat. D: A sugar residue belongs to glycolipids and a sterol nucleus to derived lipids, so neither feature defines a simple lipid.

BIO-11-CH01-0032 | Easy | Simple lipids**Correct answer: D. Triglycerides**

Solution: Because the alcohol in a fat is glycerol, the chapter refers to fats as triglycerides.

Distractor review: A: Phospholipids are compound lipids characterised by a phosphate group, not the alternative name for the fats. B: Cerebroside is the alternative name for a glycolipid, which is a compound lipid and not a fat. C: Phytosterols are plant sterols placed among the derived lipids, a class quite distinct from the fats.

BIO-11-CH01-0033 | Medium | Compound lipids**Correct answer: C. Phospholipid with a phosphate group and glycolipid with a sugar**

Solution: A phospholipid is recognised by its phosphate group, lecithin being the example given, while a glycolipid is recognised by its sugar component.

Distractor review: A: The two additional components are interchanged; the prefix of each name states which group it carries. B: A sterol nucleus belongs to the derived lipids; the glycolipid is defined by the sugar it contains. D: Glycerol alone with fatty acids gives a simple fat, so it cannot be the feature that marks out a phospholipid.

BIO-11-CH01-0034 | Medium | Derived lipids**Correct answer: B. Cholesterol, progesterone and vitamin D**

Solution: Cholesterol, progesterone and vitamin D are all listed under the sterols, and the sterols form the derived lipid class.

Distractor review: A: Lecithin is a phospholipid and therefore a compound lipid, so the set is not confined to derived lipids. C: Cerebroside is the other name for a glycolipid, which belongs to the compound lipids and breaks the set. D: Vanaspati ghee is given as an example of a fat, which is a simple lipid, so this set mixes two classes.

BIO-11-CH01-0035 | Hard | Compound lipids**Correct answer: C. Cerebrosides are glycolipids and carry a sugar, not phosphate**

Solution: Glycolipids are also called cerebrosides, and the extra group they carry is a sugar. Phosphate is the additional group of phospholipids such as lecithin, a different sub-class of compound lipid.

Distractor review: A: This keeps the error; cerebroside is the alternative name for a glycolipid, and the phospholipid example given is lecithin. B: Derived lipids are the sterols; a cerebroside is a compound lipid and does carry an additional group. D: A simple lipid has no additional group at all, whereas the sugar of the cerebroside is exactly such an additional group.

BIO-11-CH01-0036 | Medium | Classification of lipids**Correct answer: B. I is a simple lipid, II is a compound lipid and III is a derived lipid**

Solution: A triglyceride of oleic acid with glycerol is a fat and therefore a simple lipid; lecithin carries a phosphate group and is a compound lipid; a phytosterol is a plant sterol and belongs to the derived lipids.

Distractor review: A: The first two assignments are interchanged; the presence of phosphate in lecithin, not the presence of glycerol, marks a compound lipid. C: A triglyceride is the standard simple lipid and a phytosterol is a sterol, so these two assignments are reversed. D: Lecithin carries phosphate and so is a compound lipid, while a phytosterol belongs with the sterols of the derived class.

BIO-11-CH01-0037 | Medium | Membrane lipids**Correct answer: D. Phospholipids**

Solution: Phospholipids are the membrane lipid, and it is the phospholipid that is shown forming the lipid bilayer when placed in an aqueous medium.

Distractor review: A: Triglycerides are storage fats; the chapter names the phospholipid, not the fat, as the lipid of the membrane. B: A wax is a simple lipid of a long-chain fatty acid with a long-chain alcohol and is not given as a membrane-forming lipid. C: Phytosterols are plant sterols belonging to the derived lipids and are not described as forming the bilayer.

BIO-11-CH01-0038 | Hard | Derived lipids**Correct answer: B. Diosgenin from Dioscorea**

Solution: Diosgenin is the sterol the chapter attributes to the yam plant Dioscorea. Being a sterol, it belongs to the same derived lipid group as progesterone and testosterone, which makes it the appropriate starting material.

Distractor review: A: Cholesterol is listed among the sterols but is not the substance attributed to the yam plant Dioscorea. C: Lecithin is a phospholipid, not a sterol at all, so it cannot serve as a steroid starting material. D: Diosgenin is correctly a sterol, but flax is associated in the chapter with omega-3 fatty acid and not with diosgenin.

BIO-11-CH01-0039 | Hard | Elemental composition of lipids**Correct answer: B. X has 36 hydrogen atoms and Y has more than 36**

Solution: For the carbohydrate the ratio is fixed at 2:1, so 18 oxygen atoms are accompanied by $2 \times 18 = 36$ hydrogen atoms. For the lipid the ratio exceeds 2:1 with the same 18 oxygen atoms, so its hydrogen atoms must number more than 36.

Distractor review: A: The hydrogen count of X is right, but a ratio greater than 2:1 in the lipid must raise the hydrogen number above 36, not lower it. C: The ratio has been inverted; 2:1 means twice as many hydrogen atoms as oxygen atoms, giving 36 and not 9. D: If Y also had exactly 36 hydrogen atoms its ratio would be 2:1, which is the carbohydrate value and not the lipid value.

BIO-11-CH01-0040 | Hard | Essential fatty acids**Correct answer: A. Oleic acid**

Solution: Only linoleic, linolenic and arachidonic acids are named as the essential fatty acids. Oleic acid is unsaturated and occurs in nearly all fats, but it is not on that list, so it is the exception that disproves the assumption.

Distractor review: B: Linoleic acid is one of the three fatty acids listed as essential, so it supports the assumption instead of refuting it. C: Linolenic acid also appears in the list of essential fatty acids and therefore cannot be the exception. D: Arachidonic acid is the third member of the essential fatty acid list, so it is consistent with the student's assumption.

BIO-11-CH01-0041 | Medium | Essential fatty acids**Correct answer: A. Flax seeds**

Solution: Flax seeds are the seed source associated with omega-3 fatty acid, so they are the appropriate dietary suggestion.

Distractor review: B: Mustard seeds are not the seed the chapter links with omega-3 fatty acid; the student picks any oilseed from general familiarity. C: Sesame is a common oilseed but is not the source associated with omega-3 fatty acid in the chapter. D: Groundnut is an oil-yielding seed but the omega-3 association is made with a different seed.

BIO-11-CH01-0042 | Challenging | General properties of lipids

Correct answer: A. They are insoluble in water, their hydrogen to oxygen ratio is greater than 2:1, and their molecular weight is smaller than that of the macromolecules

Solution: All three properties must hold at once. Lipids are water-insoluble, their hydrogen to oxygen ratio exceeds the 2:1 of carbohydrates, and their molecular weight is smaller than that of the macromolecules.

Distractor review: B: A ratio of exactly 2:1 is the carbohydrate value; the lipid ratio is raised above it, so only two of the three parts hold. C: Lipids are described as water-insoluble and of smaller molecular weight than the macromolecules, so two parts of this statement fail. D: A ratio below 2:1 and a molecular weight above that of the macromolecules both contradict the description of lipids.

BIO-11-CH01-0043 | Easy | Introduction to proteins

Correct answer: C. Berzelius, in 1830

Solution: Berzelius suggested the term protein in 1830.

Distractor review: A: Buchner is associated with the demonstration of fermentation by yeast extract and with the term enzyme, not with the term protein. B: The scientist is right but the year is not; 1869 is associated with the isolation of nucleic acids from pus cells. D: Miescher is linked with nucleic acids rather than proteins, so both parts of this pairing are misplaced.

BIO-11-CH01-0044 | Easy | Peptide bond

Correct answer: A. The carboxyl group of one residue and the amino group of another residue

Solution: The peptide bond is formed between the carboxyl group of one amino acid residue and the amino group of the next residue in the chain.

Distractor review: B: Joining two amino groups would not release the linkage described; the bond requires one carboxyl and one amino group. C: Two carboxyl groups cannot form the peptide linkage, which is specifically between a carboxyl group and an amino group. D: Side chains are not involved in the backbone linkage; the peptide bond is formed by the amino group, not by a side chain.

BIO-11-CH01-0045 | Easy | Introduction to proteins

Correct answer: A. 100 to 3000 amino acid units, of 20 naturally occurring types

Solution: Proteins are stated to contain 100 to 3000 amino acid units, built from 20 naturally occurring types of amino acid.

Distractor review: B: The two figures have been exchanged; 20 is the number of amino acid types and 100 is the lower end of the chain length. C: The chain length is right but the number of naturally occurring amino acid types is overstated. D: The number of types is right but the lower limit of the chain length is ten times too high.

BIO-11-CH01-0046 | Easy | Classification of proteins

Correct answer: C. Legumelin

Solution: Legumelin is the albumin of pulses named among the simple proteins.

Distractor review: A: Globulin is indeed a simple protein but it is listed separately from the albumins and is not the protein of pulses. B: Mucin is a mucoprotein and therefore a conjugated protein, not a simple protein of the albumin group. D: Peptones belong to the derived proteins, a class produced by breakdown rather than an albumin of pulses.

BIO-11-CH01-0047 | Medium | Classification of proteins

Correct answer: D. Egg albumin is a simple protein and mucin is a conjugated protein

Solution: Egg albumin is one of the albumins listed among the simple proteins, while mucin is a mucoprotein and therefore belongs with the conjugated proteins.

Distractor review: A: The classes are interchanged; egg albumin has no non-protein component while mucin is a mucoprotein. B: Both placements are wrong: serum albumin is an albumin among the simple proteins and haemoglobin carries the non-protein haem group. C: Legumelin is correctly placed, but globulin is itself listed among the simple proteins, so the statement as a whole fails.

BIO-11-CH01-0048 | Medium | Protein structure

Correct answer: C. The secondary structure is held by hydrogen bonds, while the tertiary structure involves disulphide bonds

Solution: Hydrogen bonds maintain the alpha helix, beta helix and pleated sheet of the secondary structure, whereas the looping, twisting and folding back of the chain in the tertiary structure is due to disulphide bonds.

Distractor review: A: The two bond types are interchanged; the helices and pleated sheets are held by hydrogen bonds, not by disulphide bonds. B: Peptide bonds link the residues of the chain itself and are not what distinguishes the secondary level from the tertiary level. D: If only peptide bonds were involved there would be no basis for distinguishing the two structural levels at all.

BIO-11-CH01-0049 | Medium | Protein structure**Correct answer: D. Silk fibre**

Solution: Silk fibre is the example given for peptide chains linked side by side into pleated sheets by intermolecular hydrogen bonds.

Distractor review: A: Egg albumin is quoted as an example of a simple protein of the albumin group, not as the example of a pleated sheet. B: Haemoglobin is quoted for its conjugated nature of globin with haem, not for a pleated sheet arrangement. C: Heparin appears among the mucoproteins; it is not the example given for pleated sheet formation.

BIO-11-CH01-0050 | Medium | Amphoteric nature of proteins**Correct answer: D. pH 7.4**

Solution: The amphoteric behaviour of proteins is described at pH 7.4.

Distractor review: A: pH 2 is quoted elsewhere in the chapter as the optimum for pepsin, not as the pH of amphoteric protein behaviour. B: pH 9.5 is quoted as the optimum for trypsin; it is an enzyme optimum and not the pH used for amphoteric behaviour. C: Exact neutrality is assumed by the student, but the value stated for amphoteric behaviour is slightly on the alkaline side of 7.

BIO-11-CH01-0051 | Hard | Amino acids in proteins**Correct answer: B. Lysine and arginine are the basic amino acids, and cysteine forms the disulphide bond**

Solution: The two basic amino acids named are lysine and arginine, and the disulphide bond is formed by cysteine. The two roles belong to different residues and must not be interchanged.

Distractor review: A: Cysteine is the residue of the disulphide bond and arginine is a basic amino acid, so both roles have been swapped. C: Lysine is one of the two basic amino acids named and cannot be the residue that forms the disulphide bond. D: The basic pair is right, but arginine is not the residue that provides the disulphide bond.

BIO-11-CH01-0052 | Hard | Protein structure**Correct answer: B. Disulphide bond, provided by cysteine**

Solution: The tertiary structure arises when the chain is looped, twisted and folded back because of disulphide bonds, and cysteine is the amino acid that forms them. Hydrogen bonding is confined to the secondary level.

Distractor review: A: The amino acid is right but the bond is not; extending hydrogen bonding to the tertiary level is exactly the error in question. C: Peptide bonds link residues along the chain at every level and lysine is a basic amino acid, so neither part fits the tertiary folding. D: The bond is right but arginine is one of the basic amino acids; the disulphide bond is contributed by a different residue.

BIO-11-CH01-0053 | Medium | Classification of proteins**Correct answer: D. Conjugated proteins**

Solution: Because haemoglobin is globin combined with the non-protein haem group, it meets the definition of a conjugated protein, and its haem-bearing nature underlies its indispensable role in respiration.

Distractor review: A: A simple protein has no non-protein component, whereas haemoglobin carries the haem group in addition to globin. B: Derived proteins such as metaproteins and peptones arise from the breakdown of other proteins, which is not how haemoglobin is described. C: Metaproteins are one kind of derived protein; haemoglobin is not a breakdown product but a protein with a bound non-protein group.

BIO-11-CH01-0054 | Hard | Classification of proteins**Correct answer: A. I is a simple protein, II is a conjugated protein and III is a derived protein**

Solution: Serum albumin is one of the albumins among the simple proteins, a lipoprotein is listed among the conjugated proteins, and peptones are listed among the derived proteins.

Distractor review: B: The first two are interchanged; serum albumin is an albumin of the simple proteins and the lipid component makes a lipoprotein conjugated. C: Serum albumin is not a breakdown product and a peptone is precisely such a product, so these two assignments are reversed. D: A lipoprotein carries a non-protein lipid component and is therefore conjugated, while a peptone belongs with the derived proteins.

BIO-11-CH01-0055 | Hard | Proteins and their functions**Correct answer: A. Fibrinogen with the clotting of blood**

Solution: Fibrinogen is the protein associated with the clotting of blood. Collagen is structural, myoglobin is the oxygen-related muscle protein, and IgG acts in defence, so only the fibrinogen pairing holds.

Distractor review: B: Collagen is a structural protein of connective tissue; the clotting role belongs to a different protein of the list. C: IgG acts in the body's defence; oxygen storage in muscle is the role of a different named protein. D: Myoglobin is the muscle protein concerned with oxygen; defence against antigens is the role of the immunoglobulin.

BIO-11-CH01-0056 | Medium | Peptide bond**Correct answer: C. 2999**

Solution: Each peptide bond links one residue to the next, so a chain of n residues has $n - 1$ bonds. With $n = 3000$, the number of peptide bonds is 2999.

Distractor review: A: This assumes one peptide bond per amino acid unit; the chain has one bond fewer than the number of units because the two ends are free. B: This adds a bond beyond the two ends of the chain, which have no further residue to bond to. D: This halves the number of units, as though each peptide bond consumed two units without sharing them with neighbours.

BIO-11-CH01-0057 | Hard | Protein structure**Correct answer: B. 396**

Solution: The smallest number of amino acid units stated for a protein is 100, so each chain has $100 - 1 = 99$ peptide bonds. The four chains of the quaternary structure are separate polypeptides, so the total is $4 \times 99 = 396$.

Distractor review: A: This multiplies the residues by the number of chains without subtracting one bond for each separate chain. C: This treats the four chains as one continuous chain of 400 residues, but separate chains are not joined by peptide bonds. D: This gives the bonds of a single chain only and ignores the other three chains of the quaternary structure.

BIO-11-CH01-0058 | Challenging | Classification of proteins**Correct answer: A. I, II and III only**

Solution: Statements I, II and III agree with the chapter. Statement IV fails because egg albumin is one of the albumins listed among the simple proteins, not among the conjugated proteins.

Distractor review: B: Egg albumin is an albumin of the simple proteins, so including statement IV admits a wrong classification. C: This drops the correct description of haemoglobin and retains the wrong classification of egg albumin. D: This drops the correct placing of peptones among the derived proteins and keeps the incorrect statement about egg albumin.

BIO-11-CH01-0059 | Hard | History of nucleic acid research**Correct answer: D. II, IV, III, I**

Solution: Miescher isolated nucleic acids from pus cells in 1869, Fuelgen worked in 1924, the two types of nucleic acid were identified in 1938 and Chargaff gave the base ratio generalisation in 1950, so the order is Miescher, Fuelgen, identification, Chargaff.

Distractor review: A: This places the identification of the two nucleic acids in 1924 and the work of Fuelgen in 1938; the actual years are 1938 and 1924 respectively. B: This makes Fuelgen earlier than Miescher, whereas Miescher's isolation of 1869 is the earliest of the four landmarks. C: This places Chargaff's base ratio work before the identification of the two nucleic acids, but Chargaff's generalisation is of 1950 and the identification is of 1938.

BIO-11-CH01-0060 | Easy | Nitrogenous bases**Correct answer: B. Purines are double-ring bases and include adenine and guanine, while pyrimidines are single-ring bases and include cytosine, thymine and uracil**

Solution: Adenine and guanine carry two fused rings and are the purines, while cytosine, thymine and uracil carry a single ring and are the pyrimidines.

Distractor review: A: The membership of the two classes is right but the ring structures are interchanged; the double-ring bases are the purines. C: The ring structures are right but the bases are interchanged; cytosine and thymine are pyrimidines, not purines. D: Cytosine is wrongly grouped with adenine and guanine is wrongly grouped with the pyrimidines; only adenine and guanine are purines.

BIO-11-CH01-0061 | Medium | Nucleotide structure**Correct answer: D. Base at C-1, phosphate at C-5, chain linkage involving C-3**

Solution: In a nucleotide the base is attached at C-1 of the pentose sugar and the phosphate at C-5, while C-3 is involved in joining one nucleotide to the next so that the backbone runs as alternate sugar and phosphate groups.

Distractor review: A: The base and the phosphate positions are interchanged; the phosphate occupies C-5 and the base C-1. B: The phosphate is placed at C-3 and the chain linkage at C-5; in fact the phosphate occupies C-5 and C-3 takes part in the linkage. C: The base is placed at C-3, whereas C-3 is the carbon involved in the chain and C-1 carries the base.

BIO-11-CH01-0062 | Medium | Double helix model

Correct answer: C. Watson and Crick proposed the model using the results obtained by R. Franklin and M. Wilkins, and performed no experiments of their own

Solution: The double-helical structure was put forward by Watson and Crick, who worked from the results obtained by R. Franklin and M. Wilkins and did not perform experiments of their own.

Distractor review: A: The roles are reversed; Franklin and Wilkins supplied the experimental results and Watson and Crick built the model. B: Watson and Crick carried out no experiments themselves, so the model cannot rest on their own experimental work. D: Chargaff's contribution was the generalisation about base ratios, not the helical model.

BIO-11-CH01-0063 | Hard | Structure of the DNA double helix

Correct answer: A. I, III and IV only

Solution: The helix is 20 angstrom wide, its two chains run antiparallel with the 3' end of one beside the 5' end of the other, and a major and a minor groove are present; the spacing of neighbouring nucleotide pairs is 3.4 angstrom, not 34 angstrom, which is the length of one complete turn.

Distractor review: B: Statement II is wrong: 34 angstrom is the length of one complete turn, while neighbouring nucleotide pairs are 3.4 angstrom apart. C: This accepts 34 angstrom as the spacing between neighbouring nucleotide pairs and rejects the correct diameter of 20 angstrom. D: This rejects the antiparallel arrangement and the two grooves, both of which are features of the helix, and accepts the wrong spacing value.

BIO-11-CH01-0064 | Easy | Base pairing

Correct answer: D. Two and three

Solution: An adenine-thymine pair is held together by two hydrogen bonds while a guanine-cytosine pair is held by three.

Distractor review: A: The values are interchanged; three hydrogen bonds belong to the guanine-cytosine pair, not to the adenine-thymine pair. B: This treats every base pair alike, ignoring that the guanine-cytosine pair is held by an extra hydrogen bond. C: This treats every base pair alike at three bonds, ignoring that the adenine-thymine pair is held by only two.

BIO-11-CH01-0065 | Hard | Chargaff's observations

Correct answer: C. The ratio is the same in the two tissues of the one animal and may differ in the animal of the other species

Solution: All cells of one species carry DNA of the same base ratio, so the liver and muscle samples of one animal give the same value, while an animal of another species may give a different value.

Distractor review: A: This reverses the observation; the ratio is a property of the species and does not change from tissue to tissue. B: The ratio is constant only within a species; Chargaff found that it varies between species. D: This denies the constancy within a species, which is the central part of Chargaff's generalisation.

BIO-11-CH01-0066 | Medium | Chargaff arithmetic

Correct answer: C. 28 per cent

Solution: Cytosine 22 per cent means guanine is also 22 per cent, so guanine and cytosine together are 44 per cent. The remaining 56 per cent is shared equally by adenine and thymine, giving adenine 28 per cent.

Distractor review: A: This equates adenine with cytosine; adenine equals thymine, and cytosine equals guanine, so the two values need not be the same. B: This is obtained by subtracting only the cytosine from 100 and halving the remainder, forgetting that guanine is also present at 22 per cent. D: This is the combined percentage of adenine and thymine; it has not been divided between the two equal bases.

BIO-11-CH01-0067 | Hard | Chargaff arithmetic

Correct answer: A. 20 per cent

Solution: If $(A+T)$ is 1.5 times $(G+C)$, then the two sums together equal 2.5 times $(G+C)$, which must be 100 per cent. Hence $(G+C)$ is 40 per cent, and since guanine equals cytosine, cytosine is 20 per cent.

Distractor review: B: This is the combined percentage of guanine and cytosine; it has not been divided between the two equal bases. C: This is half of the adenine and thymine share, which belongs to adenine and to thymine, not to cytosine. D: This is obtained by dividing 100 by the ratio 1.5 and then halving twice, which does not follow from the relation between the two base sums.

BIO-11-CH01-0068 | Medium | Hydrogen bonding in DNA**Correct answer: A. 260**

Solution: The 40 adenine-thymine pairs contribute 40 multiplied by 2, that is 80 bonds, and the 60 guanine-cytosine pairs contribute 60 multiplied by 3, that is 180 bonds, giving 260 hydrogen bonds in all.

Distractor review: B: This is obtained by giving three hydrogen bonds to the adenine-thymine pairs and two to the guanine-cytosine pairs, that is by interchanging the two values. C: This counts two hydrogen bonds for all 100 pairs and ignores the extra bond of every guanine-cytosine pair. D: This counts three hydrogen bonds for all 100 pairs and ignores that an adenine-thymine pair has only two.

BIO-11-CH01-0069 | Easy | Dimensions of DNA**Correct answer: D. 1700 angstrom**

Solution: Each successive nucleotide pair adds 3.4 angstrom to the length, so 500 pairs give 500 multiplied by 3.4, that is 1700 angstrom.

Distractor review: A: This uses 1000, the total number of nucleotides in both chains, instead of the 500 nucleotide pairs stacked along the length. B: This is a tenfold arithmetical slip, as though the spacing were 0.34 angstrom. C: This is a tenfold arithmetical slip in the other direction, as though the spacing were 34 angstrom.

BIO-11-CH01-0070 | Medium | Dimensions of DNA**Correct answer: D. 30**

Solution: Dividing the length of 1020 angstrom by the 34 angstrom occupied by one complete turn gives 30 complete turns.

Distractor review: A: This divides the length by 3.4 angstrom, the distance between neighbouring nucleotide pairs, instead of by the length of one complete turn. B: This divides the length by 20 angstrom, which is the diameter of the helix and not a distance measured along its length. C: This simply repeats the length of one complete turn as though it were the number of turns.

BIO-11-CH01-0071 | Challenging | Hydrogen bonding in DNA**Correct answer: C. 540**

Solution: A length of 680 angstrom at 3.4 angstrom per nucleotide pair gives 200 nucleotide pairs, that is 400 bases in all. With 60 adenine bases there are 60 thymine bases, so 60 pairs are adenine-thymine and the remaining 140 pairs are guanine-cytosine. The bonds number 60 multiplied by 2 plus 140 multiplied by 3, that is 120 plus 420, giving 540.

Distractor review: A: This interchanges the two bond values, giving three bonds to each adenine-thymine pair and two to each guanine-cytosine pair. B: This treats all 200 nucleotide pairs as guanine-cytosine pairs and ignores the 60 adenine-thymine pairs. D: This gives two hydrogen bonds to each of the 200 nucleotide pairs and ignores the extra bond of every guanine-cytosine pair.

BIO-11-CH01-0072 | Hard | Exceptions to base equality**Correct answer: A. I, II and III (phi-x-174 claims)**

Solution: Bacteriophage phi-x-174 carries DNA, and that DNA is single stranded, so its bases have no partner chain to pair with and the purine and pyrimidine amounts need not be equal. All three claims therefore hold.

Distractor review: B: The first two claims hold, but rejecting the third misses the consequence: with no partner chain there is no pairing to force the two kinds of base into equal amounts. C: The single chain and the unequal amounts are both right, but denying that the genetic material is DNA is wrong; this bacteriophage carries DNA, and it is that DNA which has one chain. D: Accepting unequal amounts while denying the single chain removes the very reason for the exception, since a two-chain molecule would force the amounts to be equal.

BIO-11-CH01-0073 | Easy | DNA compared with RNA**Correct answer: C. DNA is usually double stranded and contains thymine, whereas RNA is usually single stranded and contains uracil in place of thymine**

Solution: DNA has two chains and carries thymine among its pyrimidines, while RNA has a single chain and carries uracil where DNA has thymine.

Distractor review: A: Both features are interchanged; the double-stranded molecule containing thymine is DNA. B: The number of chains is right but the bases are interchanged; uracil belongs to RNA and thymine to DNA. D: The bases are right but the number of chains is interchanged; RNA is the molecule that is usually single stranded.

BIO-11-CH01-0074 | Hard | Base equality and RNA

Correct answer: D. The hereditary material of this virus is single-stranded RNA, whose bases are not paired with those of a second chain

Solution: Tobacco mosaic virus has RNA as its hereditary material, and because this RNA is single stranded its bases are not paired with those of a partner chain, so purine and pyrimidine amounts are not equal.

Distractor review: A: Tobacco mosaic virus has RNA, not DNA, as its hereditary material, so the pairing of adenine with thymine does not arise. B: The RNA of this virus is single stranded, and adenine is a purine that does not pair with the purine guanine. C: It is thymine that is replaced by uracil in RNA, not the other way round, and this substitution is not the reason the amounts differ.

BIO-11-CH01-0075 | Medium | Forms of RNA in viruses

Correct answer: B. Reovirus and wound tumour virus

Solution: Double-stranded RNA is found in Reovirus and in wound tumour virus, which are the exceptions to the generally single-stranded nature of RNA.

Distractor review: A: The RNA of tobacco mosaic virus, although it serves as the hereditary material, is single stranded. C: Tobacco mosaic virus has single-stranded RNA and bacteriophage phi-x-174 has single-stranded DNA, so neither carries double-stranded RNA. D: Bacteriophage phi-x-174 is the example of single-stranded DNA and does not carry RNA at all.

BIO-11-CH01-0076 | Medium | Messenger RNA

Correct answer: C. It is a linear polynucleotide forming about 3 per cent of the cellular RNA and carrying initiation and termination codons

Solution: Messenger RNA carries the information for the amino acid sequence as a linear polynucleotide, makes up about 3 per cent of the cellular RNA and possesses initiation and termination codons.

Distractor review: A: The figure of 80 to 90 per cent of cellular RNA belongs to ribosomal RNA; messenger RNA accounts for about 3 per cent. B: The clover-leaf shape belongs to transfer RNA; messenger RNA is a linear polynucleotide. D: The anticodon loop is a part of transfer RNA, not of messenger RNA, which carries initiation and termination codons.

BIO-11-CH01-0077 | Hard | Ribosomal RNA

Correct answer: C. I, III and IV only (rRNA statements)

Solution: Ribosomal RNA makes up 50 to 60 per cent of the ribosome, is synthesised in the nucleus and shows secondary coiling, as studied by Kurland in 1960; its share of the cellular RNA is 80 to 90 per cent and not about 3 per cent.

Distractor review: A: Statement II is wrong: about 3 per cent of cellular RNA is the share of messenger RNA, while ribosomal RNA forms 80 to 90 per cent. B: This accepts the messenger RNA figure of 3 per cent for ribosomal RNA and rejects its correct 50 to 60 per cent share of the ribosome. D: This rejects the nuclear site of synthesis and the work of Kurland, both of which are correct, and accepts the wrong share of cellular RNA.

BIO-11-CH01-0078 | Challenging | Transfer RNA

Correct answer: B. I, II and III only

Solution: Transfer RNA is 70 to 80 nucleotides long, was given its clover-leaf representation by Holley in 1965 and has the DHU arm, amino acid binding site, anticodon loop and ribosome recognition site as its four parts; the amino acid is attached at the 3' acceptor stem and not at the 5' end.

Distractor review: A: Statement IV is wrong and statement III has been rejected; the amino acid attaches at the 3' acceptor stem, and the four named parts are correct. C: This accepts attachment at the 5' acceptor stem and rejects the size of 70 to 80 nucleotides, which is correct. D: This accepts attachment at the 5' acceptor stem and rejects the clover-leaf description of Holley in 1965, which is correct.

BIO-11-CH01-0079 | Easy | Concept of metabolism

Correct answer: D. The sum of all the chemical reactions taking place in an organism

Solution: Metabolism covers every chemical reaction occurring in an organism, both the reactions that break molecules down and those that build them up.

Distractor review: A: This describes only the catabolic part; metabolism includes the building-up reactions as well. B: This describes only the anabolic or biosynthetic part; metabolism includes the breaking-down reactions as well. C: This describes the metabolic pool of molecules rather than the reactions themselves.

BIO-11-CH01-0080 | Medium | Catabolic and anabolic pathways

Correct answer: A. A catabolic pathway breaks complex molecules down into simpler ones, whereas an anabolic pathway builds complex molecules from simpler ones

Solution: Breaking complex molecules down into simpler ones is catabolic, while assembling complex molecules from simpler ones is anabolic or biosynthetic.

Distractor review: B: The two directions are interchanged; the biosynthetic or building-up pathway is the anabolic one. C: Both halves are wrong: building up is anabolic, and anabolic pathways do not merely rearrange molecules of unchanged size. D: The first half is right, but an anabolic pathway is a biosynthetic pathway that assembles larger molecules, not one that rearranges molecules of unchanged size.

BIO-11-CH01-0081 | Medium | Metabolic pool

Correct answer: B. the metabolic pool

Solution: The shared collection of intermediate molecules that either kind of pathway can draw upon is the metabolic pool.

Distractor review: A: A catabolic pathway is a sequence of breaking-down reactions, not the collection of molecules on which such reactions draw. C: An anabolic pathway is a sequence of building-up reactions, not the collection of molecules available for them. D: A secondary metabolite is a single kind of small organic molecule not needed for growth, development and reproduction, not a shared collection of intermediates.

BIO-11-CH01-0082 | Hard | Catabolic and anabolic pathways

Correct answer: D. Hydrolysis of sucrose into glucose and fructose - catabolic

Solution: The splitting of sucrose into glucose and fructose converts one larger molecule into two smaller ones, which makes it a catabolic event, while the formation of glycosidic and peptide bonds and any assembly of a polysaccharide are anabolic.

Distractor review: A: Joining two monosaccharides builds a larger molecule and is therefore anabolic, not catabolic. B: Linking amino acid residues into a polypeptide is a building-up event and belongs to an anabolic pathway. C: Splitting a polysaccharide into simple sugars is a breaking-down event and belongs to a catabolic pathway.

BIO-11-CH01-0083 | Easy | Definition of secondary metabolites

Correct answer: B. Small organic molecules produced by several bacteria, fungi and plants that are not essential for their growth, development and reproduction

Solution: Secondary metabolites are small organic molecules made by several bacteria, fungi and plants, and the defining point is that the producer does not need them for its growth, development and reproduction.

Distractor review: A: Being essential for growth, development and reproduction is exactly what secondary metabolites are not; that description fits the ordinary biomolecules of the cell. C: Secondary metabolites are organic molecules; calling them inorganic misses the basis of the definition. D: They are small molecules and are produced by bacteria and fungi as well as by plants, not by green plants alone.

BIO-11-CH01-0084 | Easy | Examples of secondary metabolites

Correct answer: C. Alkaloids, flavonoids and terpenoids

Solution: Alkaloids, flavonoids and terpenoids all belong to the secondary metabolites, whereas polypeptides, polysaccharides and nucleotides are ordinary macromolecules or micromolecules of the cell.

Distractor review: A: Polypeptides are macromolecules of the cell built from amino acids and are not counted among the secondary metabolites. B: Nucleotides are micromolecules of the cell from which polynucleotides are built, and are not secondary metabolites. D: Polysaccharides are macromolecules of the cell and are not secondary metabolites, whatever their storage or structural role.

BIO-11-CH01-0085 | Easy | Sources of secondary metabolites

Correct answer: C. The plant *Papaver somniferum*

Solution: Morphine is the secondary metabolite of *Papaver somniferum* and is used both to relieve pain and to suppress cough.

Distractor review: A: Dioscorea is named in connection with diosgenin, a sterol of the derived lipids, and not with morphine. B: Cabbage is named in connection with glucosinolates, which give it its flavour and protect it from pests, not with morphine. D: Flax seeds are named in connection with omega-3 fatty acid and not with any pain-relieving secondary metabolite.

BIO-11-CH01-0086 | Medium | Groups of secondary metabolites

Correct answer: C. Terpenes are synthesised from mevalonic acid, whereas phenolics are synthesised from simple sugars and contain a benzene ring

Solution: The terpenes arise from mevalonic acid, while the phenolics arise from simple sugars and are marked by the presence of a benzene ring.

Distractor review: A: The two precursors are interchanged; mevalonic acid is the starting material of the terpenes. B: The precursors are interchanged, and the benzene ring is attached to the wrong precursor as well. D: Sulphur and nitrogen belong to the third group, the nitrogen-containing compounds, and not to terpenes and phenolics.

BIO-11-CH01-0087 | Medium | Classification of secondary metabolites

Correct answer: D. I, II and III only

Solution: Chemical structure, composition, solubility and pathway of synthesis are the bases of classification, the three groups are terpenes, phenolics and nitrogen-containing compounds, and the nitrogen-containing group may carry sulphur as well; the description of synthesis from simple sugars with a benzene ring belongs to the phenolics, not the terpenes.

Distractor review: A: Statement IV describes the phenolics; the terpenes come from mevalonic acid, and statement III about sulphur has been wrongly rejected. B: This accepts the wrong account of the terpenes and rejects the bases of classification, which are correctly stated. C: This rejects both the three recognised groups and the presence of sulphur, and accepts the wrong description of the terpenes.

BIO-11-CH01-0088 | Medium | Pathways of synthesis

Correct answer: B. the phenylpropanoid pathway

Solution: Tannins, the compounds that make wine and chocolate astringent, are formed through the phenylpropanoid pathway.

Distractor review: A: Mevalonic acid is the starting material of the terpenes and does not lead to tannins. C: Glucosinolates are secondary metabolites of cabbage; no pathway of that name is given for the formation of tannins. D: A benzene ring is a structural feature of the phenolics, not the name of a pathway of synthesis.

BIO-11-CH01-0089 | Medium | Uses of secondary metabolites

Correct answer: C. Antibiotics - used as food preservatives

Solution: Antibiotics are the secondary metabolites put to use as food preservatives, while astringency in wine and chocolate belongs to tannins and the flavour and pest protection of cabbage to glucosinolates.

Distractor review: A: Tannins are named for the astringency they give to wine and chocolate; it is the antibiotics that are used to preserve food. B: Glucosinolates give cabbage its flavour and protect it from pests; astringency in wine and chocolate is due to tannins. D: Morphine is the pain reliever and cough suppressant of *Papaver somniferum*; the flavour and pest protection of cabbage are due to glucosinolates.

BIO-11-CH01-0090 | Challenging | Groups and examples together

Correct answer: D. Cannabinol is a terpene synthesised from mevalonic acid, and glucosinolates give cabbage its flavour and protect it from pests

Solution: Cannabinol belongs to the terpenes, which are synthesised from mevalonic acid, while glucosinolates are the compounds that give cabbage its characteristic flavour and guard it against pests.

Distractor review: A: Cannabinol is a terpene and not a phenolic, and astringency in wine and chocolate is due to tannins rather than to glucosinolates. B: The group of cannabinol is right but its precursor is mevalonic acid, and it is the antibiotics that serve as food preservatives. C: The precursor is right but cannabinol belongs to the terpenes, and astringency in chocolate is due to tannins.

BIO-11-CH01-0091 | Easy | Discovery and naming of enzymes

Correct answer: A. In yeast

Solution: Buchner found that yeast extract free of living cells still fermented sugar. The word coined from this work is built on the Greek *en*, meaning in, and *zyma*, meaning yeast, so the literal sense is in yeast.

Distractor review: B: The Greek *zyma* means yeast, not sugar; the student names the term after the substance fermented rather than after the source of the extract Buchner worked with. C: The Greek *en* does mean in, but the second word is *zyma*, yeast. Reading the term as within the cell contradicts the very result Buchner obtained, which was fermentation without living cells. D: Living ferment describes the older view that fermentation required intact living cells; the word enzyme was introduced precisely because that view had been set aside.

BIO-11-CH01-0092 | Medium | Properties of enzymes**Correct answer: B. 100 g**

Solution: 1 mg multiplied by 100000 gives 100000 mg of sucrose. Since 1000 mg make 1 g, this is 100 g of sucrose, which shows how a very small quantity of enzyme acts on a very large quantity of substrate.

Distractor review: A: Obtained by using a factor of 10000 instead of 100000, or by dropping one power of ten while converting milligram to gram. C: Obtained by multiplying by 1000 rather than 100000, a common slip when the milligram to gram conversion factor is used as the catalytic factor. D: Obtained by carrying one extra power of ten, treating the factor as one million times the enzyme weight.

BIO-11-CH01-0093 | Medium | Properties of enzymes**Correct answer: B. Reversibility of action and specificity of action**

Solution: Catalysing both the forward and the backward conversion of the same pair of compounds is reversibility of action, and acting on that pair alone and on no other substrate is specificity of action.

Distractor review: A: Nothing in the description varies the pH of the medium, and the catalytic property alone does not account for the enzyme refusing all other substrates. C: Specificity is correctly picked up, but no temperature is altered in the description, so sensitivity to temperature is not demonstrated. D: Reversibility is correctly picked up, but the proteinaceous nature of the enzyme is a fact of composition and is not shown by the reactions described.

BIO-11-CH01-0094 | Hard | Properties of enzymes**Correct answer: A. The enzyme loses its catalytic ability, because the active site no longer exists**

Solution: The active site is created by the specific three-dimensional folding of the enzyme. Once that folding is lost the active site no longer exists, so the enzyme cannot bind its substrate or catalyse the reaction, even though the chain of amino acid residues is unbroken.

Distractor review: B: Treats catalytic ability as a property of the unbroken chain itself; the chapter ties activity to the three-dimensional conformation that carries the active site, not merely to intact peptide bonds. C: Assumes catalysis and specificity reside in different parts of the molecule, whereas both depend on the same active site formed by the folded conformation. D: Cofactors are inorganic ions loosely attached to the protein part of a conjugated enzyme, so an unfolded protein cannot become one.

BIO-11-CH01-0095 | Easy | Composition of enzymes**Correct answer: A. Coenzymes**

Solution: The chapter names NAD and FMN as the coenzymes, the organic non-protein components attached to the protein part of a conjugated enzyme, and reserves the term cofactor for inorganic ions.

Distractor review: B: A cofactor is an inorganic ion such as magnesium, copper, zinc, iron or manganese, whereas NAD and FMN are organic molecules. C: A prosthetic group is the component firmly bound to the protein component by chemical bonds, and the chapter lists it separately from the coenzymes, of which NAD and FMN are the named examples. D: An activator is an inorganic substance that increases enzyme activity; it is discussed among the substances affecting activity, not as a component of a conjugated enzyme.

BIO-11-CH01-0096 | Medium | Composition of enzymes**Correct answer: D. A coenzyme and a cofactor**

Solution: FMN is an organic molecule tightly attached to the protein part, which makes it a coenzyme, and zinc is an inorganic ion loosely attached, which makes it a cofactor. NAD is the other coenzyme example given, and magnesium, copper, zinc and iron are given as cofactor ions.

Distractor review: A: Reverses both criteria, calling the organic tightly attached FMN a cofactor and the loosely attached inorganic zinc ion a coenzyme. B: It calls the organic FMN a cofactor, a term this chapter reserves for inorganic ions, and it calls the bound zinc ion an activator, which is a substance affecting activity rather than a component carried on the protein part. C: FMN is correctly named, but an activator is an inorganic substance added to increase activity rather than a component held on the protein part of the enzyme.

BIO-11-CH01-0097 | Medium | Composition of enzymes**Correct answer: D. Catalase and iron**

Solution: Catalase is the enzyme named with iron as its cofactor. The other specific pairing given in the chapter is manganese with peptidases.

Distractor review: A: Zinc is listed among the cofactor ions, but the ion associated with catalase in the chapter is iron. B: Copper is listed among the cofactor ions, but the ion associated with peptidases in the chapter is manganese. C: Swaps the two named pairings, giving peptidases the ion that belongs to catalase.

BIO-11-CH01-0098 | Hard | Composition of enzymes**Correct answer: A. A prosthetic group**

Solution: Firm binding to the protein component through chemical bonds is the defining description of a prosthetic group. Tight attachment of an organic coenzyme is not described in terms of chemical bonds that must be broken, and a cofactor is only loosely attached.

Distractor review: B: Coenzymes are described as tightly attached, and the word tightly tempts the student here, but attachment by chemical bonds that must be broken to separate the component is the description reserved for the prosthetic group. C: Cofactors are inorganic ions loosely attached to the protein part, which is the opposite of firm attachment through chemical bonds. D: An activator is an inorganic substance that raises enzyme activity when supplied to the reaction, not a bound component of the enzyme molecule.

BIO-11-CH01-0099 | Easy | Functional classification of enzymes**Correct answer: C. Oxidoreductases**

Solution: Alcohol dehydrogenase is the example given in the chapter for the oxidoreductases.

Distractor review: A: Transferases are represented by glucokinase and glutamate pyruvate transaminase; alcohol dehydrogenase is not among them. B: Hydrolases are represented by amylases, proteases, lipases, sucrase and nuclease, none of which is a dehydrogenase. D: Xylose isomerase is the example given for isomerases; the name dehydrogenase does not indicate a rearrangement of a molecule.

BIO-11-CH01-0100 | Medium | Functional classification of enzymes**Correct answer: B. Fumarase and lyases**

Solution: Fumarase is listed in the chapter among the lyases, along with aldolases, decarboxylases and dehydratases.

Distractor review: A: Xylose isomerase is the example given for the isomerases; only aldolases, decarboxylases, dehydratases and fumarase are listed as lyases. C: Amylases are listed among the hydrolases, together with proteases, lipases, sucrase and nuclease. D: Glucokinase is listed as a transferase, not as a hydrolase.

BIO-11-CH01-0101 | Medium | Functional classification of enzymes**Correct answer: D. Ligases**

Solution: The ligases, also called synthetases, are the class described as creating covalent bonds with energy obtained from the hydrolysis of a compound. The chapter gives no named example for this class.

Distractor review: A: The similar name misleads the student; the lyases are represented in the chapter by aldolases, decarboxylases, dehydratases and fumarase, and are not described as bond-creating enzymes driven by hydrolysis of a compound. B: Transferases such as glucokinase and glutamate pyruvate transaminase are grouped by the transfer of a group, not by the creation of covalent bonds using energy from hydrolysis. C: Hydrolases such as amylases, proteases and sucrase carry out hydrolysis itself; the class in question uses the energy released by hydrolysis of a compound to make a bond.

BIO-11-CH01-0102 | Challenging | Functional classification of enzymes**Correct answer: A. Alcohol dehydrogenase and xylose isomerase**

Solution: Alcohol dehydrogenase is the example given for the oxidoreductases while xylose isomerase is the example given for the isomerases, so this is the only pair drawn from two different classes. Aldolase with fumarase are both lyases, amylase with nuclease are both hydrolases, and glucokinase with glutamate pyruvate transaminase are both transferases.

Distractor review: B: Both aldolases and fumarase are listed among the lyases, so this pair lies within a single class. C: Both amylases and nuclease are listed among the hydrolases, so this pair lies within a single class. D: Both are listed as transferases, so this pair lies within a single class.

BIO-11-CH01-0103 | Easy | Mechanism of enzyme action**Correct answer: A. The enzyme and the substrate**

Solution: In Fischer's analogy the enzyme is the lock and the substrate is the key, with the active site as the key hole, so that only a key of the correct size fits.

Distractor review: B: Reverses the analogy; the model treats the enzyme as the fixed lock and the substrate as the key that enters it. C: The key hole, not the whole lock, represents the active site, and the product plays no part in the analogy as the chapter states it. D: Neither element of the analogy stands for the product; the model describes how the enzyme-substrate complex is formed.

BIO-11-CH01-0104 | Medium | Mechanism of enzyme action

Correct answer: B. The approach of the substrate brings about a change in the conformation of the enzyme

Solution: In the induced fit or flexible model the enzyme is not rigid; the approaching substrate induces a change in the conformation of the enzyme so that the active site comes to fit it. Both models still describe the formation of the enzyme-substrate complex.

Distractor review: A: Places the flexibility in the substrate; the flexible model of Koshland places the conformational change in the enzyme. C: This is the position of the lock and key model, in which the key hole is of the correct size from the outset, so it cannot be the point of difference. D: Both models are accounts of how the enzyme-substrate complex is formed, so neither dispenses with it.

BIO-11-CH01-0105 | Hard | Mechanism of enzyme action

Correct answer: C. Incorrect, because a change of shape in the enzyme on substrate approach belongs to Koshland's model

Solution: The lock and key model holds the active site to be already of the correct size and shape, so no change in the enzyme is required or described. A change in the conformation of the enzyme brought about by the approaching substrate is the feature of the induced fit or flexible model, so the student's statement misplaces it.

Distractor review: A: Extends the analogy beyond what it states; in the lock and key model the key hole is already of the right size and does not widen. B: Reaches the right verdict for a wrong reason; both models are accounts of how the enzyme-substrate complex is formed. D: Collapses the two models into one; the conformational change induced by the substrate is what separates the flexible model from the lock and key model.

BIO-11-CH01-0106 | Easy | Michaelis-Menten constant

Correct answer: C. The substrate concentration, in moles per litre, that gives half of the maximum velocity

Solution: K_m is a substrate concentration expressed in moles per litre, and it is the particular concentration at which the reaction runs at half of its maximum velocity.

Distractor review: A: Complete saturation corresponds to the maximum velocity; K_m corresponds to 50 per cent saturation of the enzyme with substrate. B: K_m is a substrate concentration, not an enzyme concentration, and it is fixed at half maximal and not at maximal velocity. D: Confuses the constant with the maximum velocity itself; K_m is expressed as a concentration in moles per litre, not as a rate.

BIO-11-CH01-0107 | Hard | Michaelis-Menten constant

Correct answer: D. P has the greater affinity, because a lower substrate concentration gives half maximal velocity

Solution: The K_m of P, 2×10^{-5} moles per litre, is lower than the K_m of Q, 4×10^{-3} moles per litre. P therefore reaches half of its maximum velocity at a much smaller supply of substrate, which is what high affinity for the substrate means.

Distractor review: A: Reads a high K_m as high affinity; the chapter states that a low K_m means high affinity and a high K_m means low affinity. B: Treats K_m as a property of the substrate; two enzymes acting on the same substrate can and do have different K_m values. C: Misreads the powers of ten: 4×10^{-3} is the higher concentration, so Q needs more substrate, not less, to reach half maximal velocity.

BIO-11-CH01-0108 | Medium | Michaelis-Menten constant

Correct answer: B. 50 per cent saturated with substrate and working at half of its maximum velocity

Solution: K_m is by definition the substrate concentration producing half of the maximum velocity, and it corresponds to 50 per cent saturation of the enzyme with substrate. Since the assay concentration equals K_m , both conditions hold together.

Distractor review: A: Maximum velocity is reached only when the enzyme is saturated; at half saturation the velocity is half of the maximum. C: Complete saturation would require a substrate concentration well above K_m , and it corresponds to the maximum velocity, not to half of it. D: Halves the saturation a second time; the chapter links K_m directly with 50 per cent saturation of the enzyme with substrate.

BIO-11-CH01-0109 | Hard | Factors affecting enzyme activity

Correct answer: D. All the active sites of the fixed quantity of enzyme present are occupied by substrate

Solution: As substrate concentration rises, more and more active sites are occupied, so velocity climbs steeply at first. Once the fixed quantity of enzyme is saturated, no free active site is left, and further substrate leaves the velocity unchanged, which is why the curve becomes almost parallel to the x axis.

Distractor review: A: Loss of conformation follows unsuitable temperature or pH, both of which are held constant here; an excess of substrate does not unfold the enzyme. B: The initial velocity is measured in each case, and the flat part of the curve lies at the highest substrate concentrations, where substrate is most plentiful. C: A reverse reaction would show as a loss of product rather than as a ceiling on the initial velocity; the flat part reflects saturation of the enzyme.

BIO-11-CH01-0110 | Hard | Factors affecting enzyme activity

Correct answer: A. The optimum is 37 degree C, and above 40 degree C the enzyme denatures rapidly, most enzymes being destroyed at 60 to 70 degree C

Solution: Activity rises with temperature up to the optimum, which for reactions at normal mammalian body temperature is 37 degree C, the optimum range for most enzymes being 20 to 35 degree C. Beyond about 40 degree C the enzyme denatures rapidly, and by 60 to 70 degree C most enzymes are destroyed, which is why activity is negligible at 65 degree C.

Distractor review: B: Substrate is in large excess, and 65 degree C lies within the 60 to 70 degree C band in which most enzymes are destroyed, so it cannot be an optimum. C: Km is a substrate concentration characteristic of the enzyme, not a temperature threshold, so it cannot mark the point at which activity falls. D: The pH is held at the optimum throughout the experiment, so the loss of activity must be traced to the enzyme protein itself.

BIO-11-CH01-0111 | Medium | Factors affecting enzyme activity

Correct answer: A. Pepsin and trypsin

Solution: Pepsin has its optimum at pH 2 and trypsin at pH 9.5, so the enzyme peaking in strongly acidic conditions is pepsin and the one peaking in alkaline conditions is trypsin.

Distractor review: B: Interchanges the two optima; the chapter gives pH 2 for pepsin and pH 9.5 for trypsin. C: The first assignment is right, but the optimum of pH 9.5 is given for trypsin, and no pH optimum is given for amylase. D: Both assignments fail: trypsin has its optimum at pH 9.5, not at pH 2, and no pH optimum is given for sucrase.

BIO-11-CH01-0112 | Challenging | Factors affecting enzyme activity

Correct answer: B. Two

Solution: Raising the temperature to 65 degree C places the enzyme in the 60 to 70 degree C band in which most enzymes are destroyed, so activity falls, and an inhibitor by definition reduces enzyme activity, so activity falls there too. Adding more of the same enzyme does not depress activity, and an inorganic activator increases it. Hence two assays show lower activity than the control.

Distractor review: A: Counts the inhibitor alone and overlooks that 65 degree C lies within the 60 to 70 degree C band in which most enzymes are destroyed. C: Counts the addition of extra enzyme or of an activator as depressing activity, whereas an activator is an inorganic substance that increases enzyme activity. D: Assumes that every departure from the control conditions must lower activity, ignoring that an activator raises it.

BIO-11-CH01-0113 | Easy | Substrate saturation

Correct answer: A. Nearly all active sites of the available enzyme molecules are occupied

Solution: At high substrate concentration the fixed amount of enzyme becomes saturated, so nearly all active sites are occupied and adding more substrate produces little further rise in velocity.

Distractor review: B: The substrate has converted the enzyme into a lipid is a plausible distractor, but it conflicts with the required interpreting the plateau of an enzyme saturation curve and therefore cannot be the single correct response. C: The reaction has become independent of enzyme concentration at every substrate level is a plausible distractor, but it conflicts with the required interpreting the plateau of an enzyme saturation curve and therefore cannot be the single correct response. D: The enzyme has necessarily been denatured by high temperature is a plausible distractor, but it conflicts with the required interpreting the plateau of an enzyme saturation curve and therefore cannot be the single correct response.

BIO-11-CH01-0114 | Medium | Temperature and enzyme activity

Correct answer: B. High temperature may irreversibly alter the enzyme's three-dimensional active site

Solution: Heating well above the optimum can denature the protein and alter its active site. Cooling does not necessarily restore that three-dimensional structure.

Distractor review: A: Cooling removes every substrate molecule from the solution is a plausible distractor, but it conflicts with the required inferring persistent activity loss after heating above the optimum and therefore cannot be the single correct response. C: The enzyme changes permanently into a carbohydrate at 65 degree C is a plausible distractor, but it conflicts with the required inferring persistent activity loss after heating above the optimum and therefore cannot be the single correct response. D: Cooling forces all enzyme molecules to bind inhibitors is a plausible distractor, but it conflicts with the required inferring persistent activity loss after heating above the optimum and therefore cannot be the single correct response.

BIO-11-CH01-0115 | Hard | DNA base pairing

Correct answer: C. 240

Solution: There are 40 G-C pairs and 60 A-T pairs. The total is $40 \times 3 + 60 \times 2 = 120 + 120 = 240$ hydrogen bonds.

Distractor review: A: 180 is a plausible distractor, but it conflicts with the required combining gc percentage with hydrogen-bond counting and therefore cannot be the single correct response. B: 200 is a plausible distractor, but it conflicts with the required combining gc percentage with hydrogen-bond counting and therefore cannot be the single correct response. D: 300 is a plausible distractor, but it conflicts with the required combining gc percentage with hydrogen-bond counting and therefore cannot be the single correct response.

BIO-11-CH01-0116 | Challenging | Factors affecting activity

Correct answer: D. II and III only

Solution: A specific inhibitor lowers activity and heating to 65 degree C can denature the enzyme. More enzyme with abundant substrate increases capacity, while more substrate at saturation causes little change rather than a decrease.

Distractor review: A: I and IV only is a plausible distractor, but it conflicts with the required separating effects of concentration, inhibitor and denaturation and therefore cannot be the single correct response. B: I, II and III only is a plausible distractor, but it conflicts with the required separating effects of concentration, inhibitor and denaturation and therefore cannot be the single correct response. C: II and IV only is a plausible distractor, but it conflicts with the required separating effects of concentration, inhibitor and denaturation and therefore cannot be the single correct response.

Standard 11 | Chapter 02

Respiration and Energy Transfer

BIO-11-CH02-0001 | Easy | Modes of phosphorylation

Correct answer: A. Photophosphorylation, substrate-level phosphorylation and oxidative phosphorylation

Solution: The three ways of forming ATP by phosphorylation are photophosphorylation, which occurs during photosynthesis, substrate-level phosphorylation, and oxidative phosphorylation.

Distractor review: B: Chemiosmosis is the name given to proton transfer across the inner mitochondrial membrane accompanied by ATP synthesis; it is not listed as a separate mode of phosphorylation. C: Terminal oxidation is the step in which reduced oxygen combines with protons to give metabolic water; it is a step of aerobic respiration, not a mode of ATP formation. D: Dehydrogenation is the removal of hydrogen from a respiratory material by dehydrogenases; no phosphate group is transferred to ADP in it.

BIO-11-CH02-0002 | Easy | Substrate-level phosphorylation

Correct answer: A. Direct phosphorylation of ADP by transfer of a phosphate group from a suitable substrate

Solution: In substrate-level phosphorylation the phosphate group comes straight from a phosphorylated substrate and is handed to ADP, so ATP is formed without any coenzyme oxidation.

Distractor review: B: This is the definition of oxidative phosphorylation, in which the phosphate is not handed over by the substrate itself but is added using energy from coenzyme oxidation. C: This describes photophosphorylation, the mode that operates in photosynthesis and not in respiration. D: This is the hexokinase reaction of glycolysis; here the substrate is phosphorylated and ATP is consumed, which is the reverse of ATP formation.

BIO-11-CH02-0003 | Medium | Substrate-level versus oxidative phosphorylation

Correct answer: A. Oxidative phosphorylation uses energy released when NADH+H⁺ and FADH₂ are oxidised, and occurs on the inner mitochondrial membrane

Solution: The defining features of oxidative phosphorylation are its energy source, the oxidation of the reduced coenzymes NADH+H⁺ and FADH₂, and its location on the inner mitochondrial membrane.

Distractor review: B: The direct transfer of a phosphate group from a substrate is substrate-level phosphorylation; this option attaches that mechanism to the wrong mode. C: The sites are interchanged; the matrix is the site of the Krebs cycle and the inner mitochondrial membrane carries oxidative phosphorylation. D: In both modes it is ADP that receives the phosphate group; molecular oxygen serves as the final electron acceptor and does not phosphorylate the substrate.

BIO-11-CH02-0004 | Medium | Substrate-level phosphorylation in the balance sheet

Correct answer: C. 6

Solution: Glycolysis contributes 4 ATP by substrate-level phosphorylation and each of the two turns of the Krebs cycle contributes 1, so $4 + 2 = 6$ ATP arise by substrate-level phosphorylation per glucose.

Distractor review: A: This counts only the glycolytic contribution and omits the one GTP (ATP) formed in each of the two turns of the Krebs cycle. B: This counts only the two turns of the Krebs cycle and omits the four ATP formed at the two ATP-yielding conversions of glycolysis. D: 8 is the net benefit entered for glycolysis alone, which already includes the ATP credited to oxidative phosphorylation of the two NADH+H⁺.

BIO-11-CH02-0005 | Challenging | Oxidative phosphorylation in the balance sheet

Correct answer: B. 34 by oxidative phosphorylation, gross total 40

Solution: The 10 NADH+H⁺ account for $10 \times 3 = 30$ ATP and the 2 FADH₂ for $2 \times 2 = 4$ ATP, giving 34 by oxidative phosphorylation; adding the 6 ATP from substrate-level phosphorylation gives a gross total of 40 ATP.

Distractor review: A: This values only the NADH+H⁺ and leaves out the 2 FADH₂, and it also mistakes the net figure for the gross total. C: The oxidative contribution is right but 38 is the net figure obtained after subtracting the 2 ATP consumed in glycolysis, not the gross total. D: This values every reduced coenzyme at 2 ATP, applying the FADH₂ factor to NADH+H⁺ as well.

BIO-11-CH02-0006 | Hard | Identifying the mode of phosphorylation from the reaction

Correct answer: D. Substrate-level phosphorylation

Solution: In the succinyl-CoA to succinate conversion the phosphate group reaches the nucleotide directly from the substrate, so the ATP is formed by substrate-level phosphorylation even though the reaction takes place inside the mitochondrion.

Distractor review: A: The reaction is placed inside the mitochondrion, but no reduced coenzyme is oxidised on the inner membrane to drive it, so it does not meet the stated definition. B: Photophosphorylation occurs during photosynthesis and has no role in any step of the Krebs cycle. C: Chemiosmosis is the synthesis of ATP that accompanies the channelling of protons through the F₀ part of the oxysome, which is not involved in this cycle step.

BIO-11-CH02-0007 | Easy | Site of glycolysis**Correct answer: B. Cytoplasm**

Solution: All ten steps of glycolysis, from glucose to pyruvic acid, take place in the cytoplasm of the cell.

Distractor review: A: The mitochondrial matrix is the site of the Krebs cycle, which follows glycolysis and the connecting link reaction. C: The inner mitochondrial membrane carries the electron transport complexes and oxidative phosphorylation, not the glycolytic enzymes. D: The outer chamber is where protons are released during oxidation of NADH+H⁺ and FADH₂; no reaction of glycolysis occurs there.

BIO-11-CH02-0008 | Easy | Definition and extent of glycolysis**Correct answer: A. Ten steps; two pyruvic acid molecules**

Solution: Glycolysis runs in ten steps and splits one six-carbon glucose into two three-carbon pyruvic acid molecules.

Distractor review: B: One six-carbon glucose is split into two three-carbon units, so the yield is two pyruvic acid molecules, not one. C: The flow chart of the pathway shows ten enzyme-catalysed conversions from glucose to pyruvic acid. D: Three three-carbon products cannot arise from a single six-carbon glucose, and the pathway is not twelve steps long.

BIO-11-CH02-0009 | Easy | Preparatory phase sequence**Correct answer: D. II, I, III, IV**

Solution: Glucose is phosphorylated to glucose-6-phosphate, isomerised to fructose-6-phosphate, phosphorylated again to fructose-1,6-diphosphate, and this is then cleaved to give glyceraldehyde-3-phosphate.

Distractor review: A: This starts with fructose-6-phosphate, but glucose is first phosphorylated at carbon 6 to glucose-6-phosphate before any isomerisation to a fructose derivative. B: This places the three-carbon triose before fructose-1,6-diphosphate, although the triose arises only when fructose-1,6-diphosphate is cleaved. C: This begins with fructose-1,6-diphosphate, which is the product of the third step and not the entry compound of the pathway.

BIO-11-CH02-0010 | Medium | Pay-off phase sequence**Correct answer: C. II, IV, I, III**

Solution: 1,3-Bisphosphoglycerate gives 3-phosphoglycerate with formation of ATP, the phosphate is then shifted to give 2-phosphoglycerate, and removal of water gives phosphoenol pyruvate.

Distractor review: A: This places 2-phosphoglycerate before 3-phosphoglycerate, although the mutase acts on 3-phosphoglycerate to give 2-phosphoglycerate. B: This starts with 3-phosphoglycerate, but 3-phosphoglycerate is itself formed from 1,3-bisphosphoglycerate with the release of ATP. D: This places phosphoenol pyruvate before 2-phosphoglycerate, although phosphoenol pyruvate is formed from 2-phosphoglycerate by removal of water.

BIO-11-CH02-0011 | Easy | Enzymes of glycolysis**Correct answer: A. Hexokinase**

Solution: Hexokinase phosphorylates glucose at the cost of one ATP to give glucose-6-phosphate, the first intermediate of the pathway.

Distractor review: B: Phosphofructokinase phosphorylates fructose-6-phosphate to fructose-1,6-diphosphate; it acts on one hexose derivative later. C: Phosphohexose isomerase converts glucose-6-phosphate to fructose-6-phosphate; it isomerises and does not transfer a phosphate group. D: Phosphoglycerate kinase acts in the pay-off phase on 1,3-bisphosphoglycerate, where it produces ATP instead of consuming it.

BIO-11-CH02-0012 | Medium | Enzymes of glycolysis**Correct answer: C. I-Aldolase, II-Triose phosphate isomerase, III-Enolase, IV-Pyruvate kinase**

Solution: Aldolase splits fructose-1,6-diphosphate, triose phosphate isomerase converts dihydroxyacetone phosphate to a second glyceraldehyde-3-phosphate, enolase removes water from 2-phosphoglycerate, and pyruvate kinase converts phosphoenol pyruvate to pyruvic acid.

Distractor review: A: The last two are interchanged; the removal of water from 2-phosphoglycerate is catalysed by enolase and the final ATP-yielding step by pyruvate kinase. B: The first two are interchanged; aldolase cleaves the six-carbon diphosphate and the isomerase interconverts the two three-carbon trioses. D: Phosphohexose isomerase acts on a six-carbon phosphorylated hexose, not on the three-carbon dihydroxyacetone phosphate.

BIO-11-CH02-0013 | Medium | Cofactors of glycolysis**Correct answer: B. Glyceraldehyde-3-phosphate to 1,3-bisphosphoglycerate, with 2 NADH+H⁺**

Solution: The only Zn⁺⁺-requiring conversion of the pathway is the oxidation of glyceraldehyde-3-phosphate to 1,3-bisphosphoglycerate, in which 2 NAD⁺ are reduced to 2 NADH+H⁺.

Distractor review: A: This first step is shown with Mg⁺⁺ as its cofactor, and although ADP is released the metal ion named for it is not Zn⁺⁺. C: This step is catalysed by phosphofructokinase with Mg⁺⁺, and no coenzyme is reduced during it. D: Water is indeed released here, but the enolase step is shown with Mg⁺⁺ and not with Zn⁺⁺.

BIO-11-CH02-0014 | Hard | Pay-off phase chemistry**Correct answer: D. Inorganic phosphate, so no further ATP is spent**

Solution: Both molecules of glyceraldehyde-3-phosphate are converted to 1,3-bisphosphoglycerate using inorganic phosphate as the phosphate source, which is why the pathway invests only the two ATP of the preparatory phase.

Distractor review: A: Treating this addition like the two preparatory-phase phosphorylations would wrongly raise the ATP investment of the pathway from two to four. B: GTP appears in the Krebs cycle as a product of substrate-level phosphorylation and is not a phosphate donor anywhere in glycolysis. C: ADP acts in glycolysis as the acceptor of phosphate groups, and no conversion of ADP to AMP occurs in the pathway.

BIO-11-CH02-0015 | Medium | ATP investment**Correct answer: C. Two ATP; hexokinase and phosphofructokinase**

Solution: The preparatory phase spends one ATP converting glucose to glucose-6-phosphate and a second converting fructose-6-phosphate to fructose-1,6-diphosphate, an investment of two ATP per glucose.

Distractor review: A: Phosphoglycerate kinase acts in the pay-off phase, where it produces ATP rather than consuming it. B: Four is the number of ATP produced by substrate-level phosphorylation in the pay-off phase, not the number invested. D: This counts only the first phosphorylation and misses the second, at which fructose-6-phosphate is converted to fructose-1,6-diphosphate.

BIO-11-CH02-0016 | Medium | ATP production**Correct answer: D. Four; at 1,3-bisphosphoglycerate to 3-phosphoglycerate and at phosphoenol pyruvate to pyruvic acid**

Solution: Since 2 ATP are consumed and the net gain is 2, the gross output is 4 ATP, produced two at a time at the conversion of 1,3-bisphosphoglycerate to 3-phosphoglycerate and at the conversion of phosphoenol pyruvate to pyruvic acid.

Distractor review: A: Each of these two conversions occurs twice per glucose because two trioses are processed, so each yields 2 ATP and the total is four. B: These two conversions consume ATP in the preparatory phase; they are the investment of the pathway, not its yield. C: The enolase step releases water and forms no ATP, so counting it adds two ATP that are never made.

BIO-11-CH02-0017 | Hard | Net gain of glycolysis**Correct answer: B. 10 pyruvic acid and 10 net ATP**

Solution: Each glucose gives 2 pyruvic acid and a net of $4 - 2 = 2$ ATP, so 5 glucose molecules give 10 pyruvic acid and 10 net ATP.

Distractor review: A: This uses the gross output of 4 ATP per glucose and never subtracts the 2 ATP invested in the preparatory phase. C: This assumes one pyruvic acid per glucose, whereas the six-carbon sugar is split into two three-carbon products. D: This adds the invested ATP to the output instead of subtracting it, giving 6 per glucose.

BIO-11-CH02-0018 | Hard | NADH+H⁺ of glycolysis**Correct answer: B. 8 NADH+H⁺ and 24 ATP**

Solution: Four glucose molecules give $4 \times 2 = 8$ NADH+H⁺, and at 3 ATP each in the balance sheet these account for $8 \times 3 = 24$ ATP.

Distractor review: A: This applies the FADH₂ factor of 2 ATP to NADH+H⁺, which the balance sheet values at 3 ATP each. C: This credits only one NADH+H⁺ per glucose, forgetting that both trioses are oxidised at the triose phosphate dehydrogenase step. D: This values each NADH+H⁺ at 4 ATP, a factor that appears nowhere in the balance sheet.

BIO-11-CH02-0019 | Medium | Glycolysis in the ATP balance sheet**Correct answer: B. Total 10, net benefit 8**

Solution: Glycolysis totals $4 + 6 = 10$ ATP, and after subtracting the 2 ATP consumed the net benefit entered for glycolysis is 8 ATP.

Distractor review: A: This copies the total into the net benefit column and ignores the 2 ATP consumed in the preparatory phase. C: This subtracts the consumption before entering the total, so both columns are reduced by two. D: This adds the 2 consumed ATP to the total instead of treating them as an expenditure.

BIO-11-CH02-0020 | Medium | Summary equation of glycolysis**Correct answer: C. 2 Pyruvate + 2 ADP + 4 ATP + 2 NADH+H⁺ + 2 H₂O**

Solution: The products of glycolysis are two pyruvate, the two ADP left after the invested ATP are used, the four ATP formed by substrate-level phosphorylation, two NADH+H⁺ and two water molecules.

Distractor review: A: The ADP and ATP coefficients are interchanged; four ADP are phosphorylated to four ATP, while the two ATP fed in are released as two ADP. B: NAD⁺ is a reactant of the pathway; on the product side it appears in its reduced form as NADH+H⁺. D: No carbon dioxide is released in glycolysis; the water shown is the water removed at the enolase step.

BIO-11-CH02-0021 | Easy | Fate of pyruvic acid**Correct answer: C. Lactic acid, ethanol or acetyl CoA**

Solution: Depending on the conditions, pyruvic acid is reduced to lactic acid in muscle, converted to ethanol in yeast, or oxidatively decarboxylated to acetyl CoA when respiration is aerobic.

Distractor review: A: Citric acid is formed only when acetyl CoA condenses with oxaloacetate in the Krebs cycle, not directly from pyruvic acid. B: Oxaloacetate is regenerated within the Krebs cycle and is not listed as a product formed from pyruvic acid. D: Glutamic acid is synthesised from the Krebs cycle intermediate alpha-ketoglutarate, several steps beyond pyruvic acid.

BIO-11-CH02-0022 | Medium | Connecting link reaction**Correct answer: A. An oxidative decarboxylation by the pyruvate dehydrogenase complex, giving acetyl CoA, CO₂ and NADH+H⁺**

Solution: Pyruvic acid is oxidatively decarboxylated by the pyruvate dehydrogenase complex, so one carbon leaves as CO₂, NAD⁺ is reduced to NADH+H⁺ and the remaining two-carbon unit is carried by CoA-SH as acetyl CoA.

Distractor review: B: This is the first reaction of the Krebs cycle, in which acetyl CoA that has already been formed condenses with oxaloacetate. C: This is the first step of alcoholic fermentation in yeast, which does not produce acetyl CoA and reduces no coenzyme. D: Succinate dehydrogenase acts within the Krebs cycle on succinate and has no role in processing pyruvic acid.

BIO-11-CH02-0023 | Easy | Tissues dependent on glycolysis**Correct answer: A. Erythrocytes, renal medulla, brain and sperm**

Solution: The four cells and tissues that depend entirely on glycolysis for their energy are erythrocytes, the renal medulla, the brain and sperm.

Distractor review: B: Liver is not among the tissues listed as depending wholly on glycolysis; the renal medulla is the kidney region named. C: Cardiac muscle and the renal cortex are not named; the kidney tissue listed is the medulla and not the cortex. D: Skeletal muscle carries out fermentation only when it runs short of oxygen, and liver is not on the list of wholly glycolysis-dependent tissues.

BIO-11-CH02-0024 | Hard | Muscle fibre types**Correct answer: B. Red fibres contain more myoglobin and sustain aerobic oxidation, suiting endurance events such as marathon and swimming**

Solution: Red (dark) fibres owe their colour to a higher myoglobin content and can sustain aerobic oxidation, which suits them to endurance events such as marathon running and swimming.

Distractor review: A: The higher myoglobin content and the capacity for sustained aerobic oxidation belong to the red fibres, not the white ones. C: Rapid lactate accumulation and quick fatigue are features of white fibres, which is why sprinters have a higher proportion of them. D: White fibres do fatigue quickly through lactate accumulation, but they are the fibres with less myoglobin, which is why they are pale.

BIO-11-CH02-0025 | Medium | Regulation of glycolysis

Correct answer: D. ATP consumption, NADH₂ regeneration and regulation of glycolytic enzymes, with glucagon, epinephrine and insulin

Solution: Control of glycolysis rests on how fast ATP is consumed, how fast NADH₂ is regenerated and how the glycolytic enzymes are regulated, with glucagon, epinephrine and insulin acting as hormonal regulators.

Distractor review: A: The intracellular factors are right, but thyroxine is not among the hormones named as regulators of glycolysis. B: Glycolysis itself neither uses oxygen nor releases carbon dioxide, so these gases are not given as controllers of its rate. C: FADH₂ is produced in the Krebs cycle and not in glycolysis, and cortisol is not named among the regulating hormones.

BIO-11-CH02-0026 | Challenging | Pay-off phase accounting

Correct answer: D. 12 ATP and 6 NADH+H⁺, with no ATP invested

Solution: Each glyceraldehyde-3-phosphate yields 2 ATP and 1 NADH+H⁺, so 6 of them give 12 ATP and 6 NADH+H⁺; because the two ATP-consuming phosphorylations happen before this intermediate is reached, no ATP is invested.

Distractor review: A: The two ATP-consuming steps lie in the preparatory phase, before glyceraldehyde-3-phosphate is formed, so they are bypassed entirely here. B: This credits only one ATP to each triose, although each triose passes through two ATP-yielding conversions. C: This treats each triose as if it were a whole glucose and so doubles both the ATP and the reduced coenzyme.

BIO-11-CH02-0027 | Easy | Definition of anaerobic respiration

Correct answer: D. Cellular respiration that does not involve oxygen at all, also called fermentation

Solution: Anaerobic respiration proceeds without any involvement of oxygen and is the process also referred to as fermentation.

Distractor review: A: Molecular oxygen as final electron acceptor is the defining feature of aerobic respiration. B: Complete oxidation of glucose belongs to aerobic respiration; in fermentation the carbon ends up in lactic acid or ethanol. C: The matrix houses the Krebs cycle of aerobic respiration, whereas the reactions of fermentation follow glycolysis in the cytoplasm.

BIO-11-CH02-0028 | Medium | Lactic acid fermentation

Correct answer: C. Pyruvic acid is the acceptor and lactic acid is the product

Solution: In muscle the NADH+H⁺ from glycolysis reduces pyruvic acid to lactic acid, and the NAD⁺ released in the process becomes available for glycolysis again.

Distractor review: A: Acetaldehyde accepts hydrogen only in yeast, after pyruvate has first been decarboxylated; muscle does not carry out that decarboxylation. B: Handing hydrogen to molecular oxygen to give water is terminal oxidation in aerobic respiration, which cannot occur in the absence of oxygen. D: The malate to oxaloacetate conversion belongs to the Krebs cycle in the mitochondrial matrix and is not part of fermentation.

BIO-11-CH02-0029 | Medium | Alcoholic fermentation

Correct answer: D. Acetaldehyde is formed first and ethanol finally

Solution: Yeast decarboxylates pyruvate to acetaldehyde, releasing carbon dioxide, and the acetaldehyde is then reduced by NADH+H⁺ to ethanol.

Distractor review: A: Acetyl CoA is the product of the connecting link reaction of aerobic respiration and is not an intermediate of alcoholic fermentation. B: Lactic acid is the end product of the muscle pathway and is not an intermediate on the way to ethanol. C: The acetaldehyde is reduced by NADH+H⁺ rather than oxidised, so the alcohol and not the acid is obtained.

BIO-11-CH02-0030 | Medium | Comparison of the two fermentations

Correct answer: A. Carbon dioxide is released in alcoholic fermentation but not in lactic acid fermentation

Solution: Alcoholic fermentation decarboxylates pyruvate to acetaldehyde and so gives off carbon dioxide, whereas lactic acid fermentation simply reduces pyruvic acid and releases no gas.

Distractor review: B: The comparison is reversed; pyruvic acid is reduced without losing a carbon in muscle, while yeast decarboxylates it. C: Both fermentations reoxidise NADH+H⁺ to NAD⁺; that is exactly why either pathway allows glycolysis to continue. D: Both pathways begin from the pyruvic acid delivered by glycolysis, and both are preceded by the same glycolytic steps.

BIO-11-CH02-0031 | Hard | Role of coenzyme reoxidation

Correct answer: A. Regeneration of NAD⁺ so that glycolysis can continue, with no ATP formed at this step

Solution: Fermentation oxidises NADH+H⁺ only to return NAD⁺ to glycolysis; because the hydrogen goes to pyruvic acid or acetaldehyde and not into the electron transport system, no ATP is formed at this step.

Distractor review: B: That conversion belongs to oxidative phosphorylation on the inner mitochondrial membrane, which does not operate when oxygen is absent. C: Formation of metabolic water is terminal oxidation of aerobic respiration and requires the molecular oxygen that fermentation lacks. D: Pyruvic acid is reduced rather than oxidised in fermentation, and its carbon is retained in lactic acid or ethanol.

BIO-11-CH02-0032 | Medium | ATP yield of fermentation**Correct answer: B. 2 ATP (net per glucose)**

Solution: Only the substrate-level phosphorylations of glycolysis yield ATP in fermentation, so the net gain is $4 - 2 = 2$ ATP per glucose.

Distractor review: A: This is the gross output of the pay-off phase and ignores the 2 ATP invested in the preparatory phase. C: This adds the ATP that the balance sheet credits to oxidative phosphorylation of the glycolytic $\text{NADH} + \text{H}^+$, which cannot occur without oxygen. D: 8 is the net benefit entered for glycolysis in the aerobic balance sheet, which already credits oxidative phosphorylation of the two $\text{NADH} + \text{H}^+$.

BIO-11-CH02-0033 | Challenging | Stoichiometry of alcoholic fermentation**Correct answer: A. 8 net ATP and 8 CO_2**

Solution: Each glucose gives a net of 2 ATP and 2 carbon dioxide molecules, so 4 glucose molecules give 8 net ATP and 8 carbon dioxide molecules.

Distractor review: B: This uses the gross figure of 4 ATP per glucose without subtracting the 2 ATP invested. C: This releases one carbon dioxide per glucose, forgetting that each glucose gives two pyruvic acid molecules and each loses a carbon. D: This assumes the carbon skeleton of pyruvic acid is fully oxidised to three carbon dioxide molecules, which happens only in aerobic respiration.

BIO-11-CH02-0034 | Challenging | Comparison with aerobic respiration**Correct answer: B. Glucose is not completely oxidised, and the reduced coenzyme is spent on lactic acid or ethanol**

Solution: In fermentation the carbon of glucose is retained in lactic acid or ethanol instead of being fully oxidised, and the $\text{NADH} + \text{H}^+$ of glycolysis is used to reduce pyruvic acid rather than being oxidised through the electron transport system, so only the substrate-level phosphorylations of glycolysis contribute ATP.

Distractor review: A: Fermentation leaves the carbon of glucose locked in lactic acid or ethanol, so the oxidation is incomplete, and no such wholesale hydrolysis of the ATP formed is described. C: The shortfall is in ATP obtained per glucose, not in the number of glucose molecules entering the pathway; glycolysis proceeds identically either way. D: Glycolysis is common to both, and anaerobic respiration consists of glycolysis followed by the conversion of pyruvic acid to lactic acid or ethanol.

BIO-11-CH02-0035 | Medium | Try this: fermentation by yeast**Correct answer: A. Lime water gradually turns milky and a smell of alcohol develops**

Solution: The lime water turns milky because carbon dioxide is evolved, and the smell of alcohol shows that ethanol has been formed, the two products of alcoholic fermentation.

Distractor review: B: A decline in gas volume is expressly not observed; the gas evolved continues to collect. C: The carbon dioxide given off during fermentation does turn lime water milky, so the lime water cannot stay clear. D: Yeast carries out alcoholic and not lactic acid fermentation, and the odour recorded is that of alcohol.

BIO-11-CH02-0036 | Hard | Try this: germinating seeds in mercury**Correct answer: B. The gas is carbon dioxide, and the seeds respire anaerobically in the absence of oxygen**

Solution: The gas that collects is carbon dioxide, shown by its complete absorption by the potassium hydroxide pellet, and because the mercury excludes air the seeds must have produced it by respiring anaerobically.

Distractor review: A: Potassium hydroxide absorbs carbon dioxide and not oxygen, and germinating seeds do not evolve oxygen. C: The mercury seals the seeds off from air, so no oxygen is available and the respiration demonstrated is anaerobic. D: The gas collected is a product of respiration that potassium hydroxide absorbs chemically; water vapour is not what the activity is designed to show.

BIO-11-CH02-0037 | Easy | Nature of the connecting link reaction**Correct answer: A. Oxidative decarboxylation**

Solution: The step is described as the connecting link reaction between glycolysis and the Krebs cycle, and its reaction type is oxidative decarboxylation because oxidation and loss of carbon dioxide occur together.

Distractor review: B: Substrate level phosphorylation is the direct transfer of a phosphate group from a substrate to ADP; no phosphate is transferred when pyruvic acid becomes acetyl CoA. C: Oxidative phosphorylation is the phosphorylation of ADP using energy from the oxidation of $\text{NADH} + \text{H}^+$ and FADH_2 on the inner mitochondrial membrane, not a reaction of the substrate itself. D: Condensation is the joining of two molecules with loss of a small molecule, which is what happens when acetyl CoA meets oxaloacetate, not when pyruvic acid is broken down.

BIO-11-CH02-0038 | Easy | Products of the connecting link reaction**Correct answer: D. Acetyl CoA, carbon dioxide and NADH+H⁺**

Solution: The reaction removes one carbon as carbon dioxide, hands two hydrogens to NAD⁺ to give NADH+H⁺, and attaches the remaining acetyl group to coenzyme A to give acetyl CoA.

Distractor review: A: FADH₂ is formed only where FAD⁺ is the hydrogen acceptor, which in this chapter happens at the succinate to fumarate step, not in the connecting link reaction. B: No ATP is generated in this step; the energy of the oxidation is carried away as reduced coenzyme rather than as a phosphate bond. C: Water is not a product here; the carbon removed from pyruvic acid leaves as carbon dioxide, which is why the step is a decarboxylation.

BIO-11-CH02-0039 | Medium | Enzyme and coenzymes of the link reaction**Correct answer: B. Pyruvate dehydrogenase complex, with NAD⁺ and CoA-SH**

Solution: The pyruvate dehydrogenase complex oxidises pyruvic acid, passing hydrogen to NAD⁺, and transfers the resulting acetyl group to CoA-SH, so both NAD⁺ and CoA-SH are required.

Distractor review: A: The enzyme is right but the hydrogen acceptor is wrong; this step reduces NAD⁺ to NADH+H⁺, and FAD⁺ is not a coenzyme of the link reaction. C: Citrate synthase acts one step later, joining acetyl CoA to oxaloacetate, and it releases CoA-SH rather than oxidising pyruvic acid. D: Isocitrate dehydrogenase works inside the Krebs cycle on isocitrate, and it uses NAD⁺ alone rather than a pair of hydrogen acceptors.

BIO-11-CH02-0040 | Medium | Site of the connecting link reaction**Correct answer: B. In the mitochondria in eukaryotes and in the cytosol in prokaryotes**

Solution: The link reaction follows glycolysis, which ends in the cytoplasm. In a eukaryotic cell the pyruvic acid is taken into the mitochondria for this step, while a prokaryotic cell, having no mitochondria, carries it out in the cytosol.

Distractor review: A: This reverses the two cases; prokaryotic cells have no mitochondria at all, so the conversion cannot be placed there. C: Prokaryotic cells lack mitochondria, so the same compartment cannot serve both cell types for this reaction. D: In eukaryotes only glycolysis is completed in the cytoplasm; the link reaction is placed in the mitochondria.

BIO-11-CH02-0041 | Medium | Vitamin requirement of the link reaction**Correct answer: D. Pyruvic acid can no longer be converted into acetyl CoA**

Solution: Without thiamin the pyruvate dehydrogenase complex cannot work, so pyruvic acid is not converted to acetyl CoA and accumulates instead. The chapter links this to pyruvic acidosis and lactic acidosis.

Distractor review: A: The decarboxylation and the oxidation are parts of the same catalysed step, so they cannot be separated by removing the vitamin the enzyme needs. B: The reduction of NAD⁺ is carried out by the same complex, so a deficiency that blocks the complex blocks the whole conversion rather than one of its outputs. C: Only acetyl CoA condenses with oxaloacetate to start the cycle; pyruvic acid has no route into the cycle that bypasses this step.

BIO-11-CH02-0042 | Medium | ATP accounting for the link reaction**Correct answer: C. 6**

Solution: Two pyruvic acid molecules each give one NADH+H⁺, so 2 NADH+H⁺ are formed. At 3 ATP per NADH+H⁺ this is 2 x 3 = 6 ATP, and nothing is consumed at this step, so the net contribution is 6 ATP.

Distractor review: A: This counts only one pyruvic acid; each glucose gives two pyruvic acid molecules, so the link reaction runs twice per glucose. B: This counts the two NADH+H⁺ molecules themselves rather than converting each of them into 3 ATP. D: This applies a value of 2 ATP per reduced coenzyme, which is the figure for FADH₂; the link reaction produces NADH+H⁺, worth 3 ATP each.

BIO-11-CH02-0043 | Easy | Site of the Krebs cycle**Correct answer: D. Mitochondrial matrix — Krebs-cycle site**

Solution: All the enzymes of the Krebs cycle act in the mitochondrial matrix, which is why acetyl CoA must first be brought into the mitochondrion.

Distractor review: A: The inner mitochondrial membrane carries the electron carriers and the site of oxidative phosphorylation, not the enzymes of the cycle itself. B: The outer chamber is where protons collect during the oxidation of reduced coenzymes; no reaction of the cycle is placed there. C: The cytoplasm is the site of glycolysis; the cycle lies inside the mitochondrion.

BIO-11-CH02-0044 | Easy | Entry of acetyl CoA**Correct answer: C. Oxaloacetate**

Solution: The first reaction of the cycle is a condensation of acetyl CoA with oxaloacetate to form citrate. This is why oxaloacetate must be available at the start of every turn.

Distractor review: A: Alpha-ketoglutarate is formed later in the cycle and is itself oxidatively decarboxylated; it does not accept the incoming acetyl group. B: Succinate arises after the substrate level phosphorylation step and is oxidised to fumarate; it plays no part in accepting acetyl CoA. D: Malate is the compound oxidised in the last step of the cycle, and its product rather than malate itself accepts the acetyl group.

BIO-11-CH02-0045 | Easy | Order of intermediates in the first half**Correct answer: A. Citrate --> cis-aconitate --> isocitrate --> oxalosuccinic acid**

Solution: Citrate is dehydrated by aconitase to cis-aconitate, which is then hydrated by the same enzyme to isocitrate. Isocitrate is then oxidatively decarboxylated through oxalosuccinic acid.

Distractor review: B: This swaps the dehydration and hydration products; citrate first loses water to give cis-aconitate, which then takes up water to give isocitrate. C: Citrate is the condensation product that begins the cycle, so nothing of the cycle proper can precede it. D: Oxalosuccinic acid is formed from isocitrate during its oxidative decarboxylation, so it cannot come before isocitrate.

BIO-11-CH02-0046 | Easy | Substrate level phosphorylation in the cycle**Correct answer: D. Succinyl-CoA to succinate**

Solution: The energy of the thioester bond of succinyl-CoA is used to phosphorylate directly, giving one GTP (ATP) for every molecule of citric acid oxidised.

Distractor review: A: This step is an oxidative decarboxylation that yields NADH+H⁺ and carbon dioxide; the phosphate-forming energy is still locked in the thioester bond of succinyl-CoA. B: This is a dehydrogenation in which FAD⁺ is reduced to FADH₂; no phosphate group is transferred to ADP. C: This is a hydration catalysed by fumarase, involving neither oxidation nor phosphorylation.

BIO-11-CH02-0047 | Easy | Regeneration of oxaloacetate**Correct answer: C. Malate to oxaloacetate**

Solution: Oxaloacetate is consumed in the opening condensation and regenerated in the final dehydrogenation of malate, which is what makes the pathway a cycle.

Distractor review: A: This hydration produces malate, which must still be oxidised before the acceptor is available. B: This dehydrogenation lies two steps earlier and yields FADH₂ rather than the acceptor of acetyl CoA. D: This dehydration belongs to the first half of the cycle and consumes the condensation product rather than regenerating the acceptor.

BIO-11-CH02-0048 | Medium | Order of intermediates in the second half**Correct answer: B. Succinate --> fumarate --> malate --> oxaloacetate**

Solution: Succinyl-CoA synthetase gives succinate, succinate dehydrogenase gives fumarate, fumarase adds water to give malate, and malate dehydrogenase gives oxaloacetate.

Distractor review: A: This exchanges the dehydrogenation and the hydration; succinate is first oxidised to fumarate, and only then is water added to give malate. C: Succinyl-CoA gives succinate directly, so succinate must be the first of this group. D: Oxaloacetate is the product of the oxidation of malate, so it cannot precede malate.

BIO-11-CH02-0049 | Medium | Adjacency of intermediates**Correct answer: C. Oxalosuccinic acid and succinate**

Solution: Isocitrate is oxidatively decarboxylated through oxalosuccinic acid to alpha-ketoglutarate, so oxalosuccinic acid is the species immediately before it. Succinyl-CoA is converted to succinate at the substrate level phosphorylation step.

Distractor review: A: Isocitrate does precede alpha-ketoglutarate, but not immediately; oxalosuccinic acid lies between them in the printed scheme. B: Fumarate comes one step too late; succinyl-CoA gives succinate first, and only then is succinate oxidised to fumarate. D: cis-Aconitate lies before isocitrate and is separated from alpha-ketoglutarate by two further conversions.

BIO-11-CH02-0050 | Medium | Enzymes of the Krebs cycle**Correct answer: B. Fumarase with the conversion of fumarate to malate**

Solution: Fumarase catalyses the hydration of fumarate to malate, adding a molecule of water without any oxidation.

Distractor review: A: Citrate synthase makes citrate from acetyl CoA and oxaloacetate; the removal of water from citrate is the work of aconitase. C: Succinate dehydrogenase oxidises succinate to fumarate; the oxidation of malate is catalysed by malate dehydrogenase. D: Aconitase acts on citrate and cis-aconitate; succinyl-CoA synthetase handles the substrate level phosphorylation step.

BIO-11-CH02-0051 | Medium | Decarboxylation steps of the cycle**Correct answer: B. Isocitrate to alpha-ketoglutarate and alpha-ketoglutarate to succinyl-CoA**

Solution: Isocitrate is oxidatively decarboxylated through oxalosuccinic acid to alpha-ketoglutarate, and alpha-ketoglutarate is oxidatively decarboxylated to succinyl-CoA. These are the only carbon dioxide releasing steps of the cycle.

Distractor review: A: These two conversions are a dehydration followed by a hydration, in which water is removed and then added; no carbon leaves the molecule. C: These are a dehydrogenation and a hydration respectively, so one yields FADH₂ and the other adds water, with no loss of carbon. D: Only the first of this pair releases carbon dioxide; the second is a substrate level phosphorylation that yields GTP (ATP).

BIO-11-CH02-0052 | Hard | ATP yield per turn**Correct answer: D. 12**

Solution: The three NADH+H⁺ give $3 \times 3 = 9$ ATP, the single FADH₂ gives $1 \times 2 = 2$ ATP, and the GTP counts as 1 ATP. The total for one turn is $9 + 2 + 1 = 12$ ATP.

Distractor review: A: This omits the GTP formed by substrate level phosphorylation and counts only the reduced coenzymes. B: This applies 2 ATP to every reduced coenzyme, whereas only FADH₂ is worth 2 ATP and each NADH+H⁺ is worth 3. C: This is the figure for two turns, that is for a whole glucose, rather than for a single turn.

BIO-11-CH02-0053 | Medium | Carbon dioxide yield of the cycle**Correct answer: C. 4 CO₂ molecules per glucose**

Solution: Two molecules of carbon dioxide leave in each turn, at the isocitrate and alpha-ketoglutarate oxidative decarboxylations. With two turns per glucose this gives $2 \times 2 = 4$ molecules of carbon dioxide.

Distractor review: A: This counts a single turn only; a glucose molecule gives two acetyl CoA and therefore drives two turns. B: This adds the two carbon dioxide molecules of the connecting link reaction, which are released before the cycle begins. D: This assumes four carbon dioxide molecules per turn, whereas only the two oxidative decarboxylations release carbon.

BIO-11-CH02-0054 | Hard | Split of the Krebs contribution by mechanism**Correct answer: C. 22**

Solution: Substrate level phosphorylation gives 1 ATP per turn, so $1 \times 2 = 2$ ATP per glucose. Subtracting from the total of 24 leaves $24 - 2 = 22$ ATP arising by oxidative phosphorylation.

Distractor review: A: This subtracts the GTP of a single turn; two turns occur per glucose, so 2 ATP come from substrate level phosphorylation. B: This subtracts four rather than two, treating each turn as though it carried out two substrate level phosphorylations. D: This is the share contributed by the NADH+H⁺ alone and leaves out the 4 ATP that come from the FADH₂.

BIO-11-CH02-0055 | Hard | Reading the labels on a cycle step**Correct answer: A. Alpha-ketoglutarate to succinyl-CoA**

Solution: Only the alpha-ketoglutarate dehydrogenase complex step takes in both NAD⁺ and CoA-SH while releasing NADH+H⁺ and carbon dioxide, and its product succinyl-CoA is the substrate of the substrate level phosphorylation.

Distractor review: B: This conversion is an oxidative decarboxylation with NAD⁺, but no CoA-SH enters it and its product is not the substrate of the phosphorylating step. C: This is the substrate level phosphorylation itself, not the conversion that produces its substrate, and it releases no carbon dioxide. D: This is a simple dehydrogenation with NAD⁺; no CoA-SH is involved and no carbon dioxide is released.

BIO-11-CH02-0056 | Medium | Reduced coenzyme yield of the cycle**Correct answer: D. 8**

Solution: Each turn gives 3 NADH+H⁺ and 1 FADH₂, that is 4 reduced coenzyme molecules. Over two turns this is $4 \times 2 = 8$, made up of 6 NADH+H⁺ and 2 FADH₂.

Distractor review: A: This is the yield of a single turn; two turns take place for each glucose molecule. B: This counts only the NADH+H⁺ molecules of the two turns and leaves out the two FADH₂. C: This adds the two NADH+H⁺ of glycolysis, which are produced in the cytoplasm and not by the Krebs cycle.

BIO-11-CH02-0057 | Challenging | Carbon balance of glucose oxidation**Correct answer: C. 6 carbon atoms**

Solution: The two connecting link reactions release 2 molecules of carbon dioxide and the two turns of the cycle release $2 \times 2 = 4$ more, giving 6 in all. That equals the six carbon atoms of one glucose, so all of the carbon of the glucose has been released.

Distractor review: A: This counts only the carbon released by the two turns of the cycle and ignores the two carbon atoms lost in the connecting link reaction. B: This counts only the connecting link reaction and leaves the four carbon atoms released within the cycle unaccounted for. D: This follows a single pyruvic acid through to completion and forgets that glycolysis produces two of them per glucose.

BIO-11-CH02-0058 | Hard | Substrate level phosphorylation across the pathway**Correct answer: A. 6**

Solution: Glycolysis contributes 4 ATP by direct phosphate transfer and the Krebs cycle contributes $1 \times 2 = 2$, giving $4 + 2 = 6$ ATP by substrate level phosphorylation for one glucose.

Distractor review: B: This counts the GTP of one turn only, although two turns of the cycle run for every glucose molecule. C: This is the glycolytic contribution alone and omits the GTP of the Krebs cycle turns. D: This is the net gain of glycolysis after the two ATP invested in the preparatory phase, not the amount produced by direct phosphate transfer.

BIO-11-CH02-0059 | Challenging | Working back from a measured ATP total**Correct answer: B. 5 turns and 10 molecules of carbon dioxide**

Solution: One turn yields $9 + 2 + 1 = 12$ ATP, so 60 ATP corresponds to $60 / 12 = 5$ turns. Each turn releases 2 molecules of carbon dioxide, so $5 \times 2 = 10$ molecules were released.

Distractor review: A: The number of turns is right but each turn releases two molecules of carbon dioxide, not one. C: This takes each turn to be worth 10 ATP, which follows from valuing every reduced coenzyme at 2 ATP and dropping the GTP. D: This values each turn at 6 ATP, counting the reduced coenzymes as single ATP molecules rather than applying the stated conversion factors.

BIO-11-CH02-0060 | Medium | Hydrogen acceptor at the oxidation steps**Correct answer: A. Succinate to fumarate**

Solution: Succinate dehydrogenase is the one enzyme of the cycle that passes hydrogen to FAD^+ , giving FADH_2 . This is why one turn produces 3 $\text{NADH} + \text{H}^+$ but only 1 FADH_2 .

Distractor review: B: This oxidation follows the rule; isocitrate dehydrogenase reduces NAD^+ and also releases carbon dioxide. C: Malate dehydrogenase also reduces NAD^+ , so this closing oxidation conforms to the pattern rather than breaking it. D: The α -ketoglutarate dehydrogenase complex reduces NAD^+ as well, with CoA-SH taken up and carbon dioxide released.

BIO-11-CH02-0061 | Hard | An enzyme acting twice in one turn**Correct answer: D. Aconitase**

Solution: The conversion of citrate to isocitrate is shown as two steps through cis-aconitate, and the same enzyme, aconitase, catalyses both the removal of water and its addition.

Distractor review: A: Citrate synthase acts once, joining acetyl CoA to oxaloacetate to give citrate, and takes no part in the conversions that follow. B: Fumarase acts once, adding water to fumarate to give malate, and it does not catalyse the oxidation that follows. C: Malate dehydrogenase acts once, oxidising malate to oxaloacetate at the close of the turn.

BIO-11-CH02-0062 | Medium | Dehydrogenases of the cycle**Correct answer: A. Fumarase**

Solution: Fumarase converts fumarate to malate by adding water. No hydrogen is removed and no coenzyme is reduced, so it is not one of the four dehydrogenase-catalysed oxidations.

Distractor review: B: This enzyme carries out one of the four oxidations, removing hydrogen from succinate and reducing FAD^+ . C: This enzyme joins acetyl CoA to oxaloacetic acid to give citric acid, a condensation, and no water is added across a double bond. D: This enzyme carries out the substrate level phosphorylation of the turn, converting succinyl CoA to succinic acid, which is neither an oxidation nor a hydration.

BIO-11-CH02-0063 | Challenging | Order of intermediates across the whole cycle**Correct answer: C. cis-Aconitate --> alpha-ketoglutarate --> succinyl-CoA --> malate**

Solution: cis-Aconitate lies early, between citrate and isocitrate. Alpha-ketoglutarate follows oxalosuccinic acid, succinyl-CoA follows alpha-ketoglutarate, and malate lies late, between fumarate and oxaloacetate.

Distractor review: A: This places succinyl-CoA before alpha-ketoglutarate, whereas succinyl-CoA is formed by the oxidative decarboxylation of alpha-ketoglutarate. B: cis-Aconitate belongs to the first half of the turn, between citrate and isocitrate, so it precedes alpha-ketoglutarate. D: Malate lies near the close of the turn, after succinate and fumarate, so it cannot precede succinyl-CoA.

BIO-11-CH02-0064 | Hard | Net fate of oxaloacetate**Correct answer: C. It is re-formed at the last step of the same turn by the oxidation of malate**

Solution: Because malate dehydrogenase restores oxaloacetate at the end of every turn, the same molecule can accept the next acetyl CoA. This regeneration is what makes the pathway a cycle rather than a linear sequence.

Distractor review: A: The condensation consumes both acetyl CoA and oxaloacetate to make citrate; it cannot also produce the oxaloacetate it uses. B: The pyruvate dehydrogenase complex produces acetyl CoA, not oxaloacetate, so this reaction cannot replenish the acceptor. D: Those steps yield alpha-ketoglutarate and succinyl-CoA together with carbon dioxide and NADH+H⁺, and neither of them yields oxaloacetate.

BIO-11-CH02-0065 | Easy | Meaning of amphibolic**Correct answer: B. A pathway involving catabolism as well as anabolism**

Solution: The Krebs cycle oxidises the acetyl part of acetyl CoA with release of energy and carbon dioxide, and at the same time supplies intermediates that are used as precursors for synthesis, so it is both catabolic and anabolic.

Distractor review: A: That describes a purely catabolic pathway, which is exactly the description the chapter says is not entirely correct for the Krebs cycle. C: That describes a purely anabolic pathway; the Krebs cycle plainly also oxidises the acetyl part of acetyl CoA and releases carbon dioxide. D: The term refers to the chemistry of the pathway rather than to its location, and the reactions of the cycle are confined to the mitochondrial matrix.

BIO-11-CH02-0066 | Easy | Precursor of aspartic acid**Correct answer: D. Oxaloacetate**

Solution: Oxaloacetate withdrawn from the cycle is used as the precursor for the synthesis of aspartic acid, which is one of the anabolic uses that make the cycle amphibolic.

Distractor review: A: Alpha-ketoglutarate is the precursor named for glutamic acid, not for aspartic acid, so the two amino acids are being swapped. B: Succinate is named only as an intermediate of the oxidative sequence; the chapter gives it no role as a precursor for amino acid synthesis. C: Fumarate likewise appears only in the oxidative sequence and is not listed among the precursors used for synthesis.

BIO-11-CH02-0067 | Medium | Precursor and product pairings**Correct answer: B. Alpha-ketoglutarate with glutamic acid**

Solution: Alpha-ketoglutarate withdrawn from the cycle serves as the precursor for glutamic acid, one of the three precursor to product relationships that make the cycle amphibolic.

Distractor review: A: Aspartic acid is made from oxaloacetate; this pairing attaches the wrong amino acid to alpha-ketoglutarate. C: Fatty acids are synthesised from acetyl CoA; oxaloacetate is the precursor named for aspartic acid. D: Acetyl CoA is the precursor named for fatty acids, and aspartic acid comes from oxaloacetate instead.

BIO-11-CH02-0068 | Medium | Entry of fats and proteins**Correct answer: D. Acetyl CoA**

Solution: Acetyl CoA is the common entry compound. Whether the substrate is a carbohydrate, a fat or a protein, it is fed into the cycle as acetyl CoA, which is why the cycle is the common oxidative pathway for all three.

Distractor review: A: Oxaloacetate is the acceptor already present in the cycle and is named as a precursor for aspartic acid, not as the port of entry for these substrates. B: Alpha-ketoglutarate is an intermediate of the cycle and a precursor for glutamic acid; substrates do not join the cycle at this point. C: Succinyl-CoA carries coenzyme A but is produced within the cycle by oxidative decarboxylation rather than serving as an entry point.

BIO-11-CH02-0069 | Medium | Consequence of a non-carbohydrate substrate**Correct answer: A. Glycolysis**

Solution: Fatty acids enter the pathway as acetyl CoA, past the point at which glucose is broken down. Glycolysis is therefore the step that is left out, while the cycle, the electron transfer chain and terminal oxidation all still operate.

Distractor review: B: The Krebs cycle is precisely the stage that fatty acid derived acetyl CoA feeds into, so it is the one stage that certainly still runs. C: The reduced coenzymes produced from fatty acid oxidation still have to be oxidised, so the electron transfer chain is still needed. D: Terminal oxidation is where oxygen finally accepts the electrons; it is required whatever the respiratory substrate.

BIO-11-CH02-0070 | Hard | Why respiration is not purely catabolic**Correct answer: D. Some of its intermediates are withdrawn as precursors for the synthesis of other biomolecules**

Solution: Acetyl CoA and intermediates such as alpha-ketoglutarate and oxaloacetate are drawn off as needed for the synthesis of fatty acids, glutamic acid and aspartic acid. Because synthesis as well as breakdown depends on the cycle, it is correctly called amphibolic.

Distractor review: A: Regeneration of coenzymes is part of the oxidative machinery itself and does not amount to the building up of new biomolecules. B: Stepwise release is a feature of how catabolism is organised and does not make the pathway anabolic in any part. C: That makes it a common oxidative pathway, but every one of those substrates is still being broken down rather than built up.

BIO-11-CH02-0071 | Hard | Common oxidative pathway**Correct answer: B. The cycle still operates, because these substrates enter it through acetyl CoA**

Solution: Amino acids from protein metabolism and fatty acids from lipid metabolism participate in the cycle through acetyl CoA, so the cycle runs normally in the absence of carbohydrate and continues to release carbon dioxide and reduced coenzymes.

Distractor review: A: Acetyl CoA is the common entry compound for carbohydrates, fats and proteins alike, so its supply does not depend on pyruvic acid. C: The carbon dioxide of the cycle comes from the acetyl carbon delivered by acetyl CoA, whatever substrate that acetyl CoA was derived from. D: The chapter routes amino acids and fatty acids into the cycle through acetyl CoA directly, with no conversion into glucose.

BIO-11-CH02-0072 | Challenging | Cost of withdrawing intermediates**Correct answer: D. Oxaloacetate must be present at the start of each turn to accept acetyl CoA, so its removal limits the number of turns**

Solution: Every turn begins with the condensation of acetyl CoA with oxaloacetate and ends with the regeneration of oxaloacetate from malate. If oxaloacetate is drawn off for aspartic acid synthesis, fewer molecules are left to accept acetyl CoA, so fewer turns occur and less reduced coenzyme and carbon dioxide are produced even though acetyl CoA is plentiful.

Distractor review: A: The acetyl group is carried into the cycle by coenzyme A and is handed to oxaloacetate at the condensation step; alpha-ketoglutarate has no such carrying role. B: The hydrogen acceptors at those steps are NAD⁺ and FAD⁺, not the acids of the cycle themselves. C: CoA-SH is released at the condensation step rather than being supplied by the intermediates of the cycle.

BIO-11-CH02-0073 | Easy | Site of the electron transport system**Correct answer: C. Inner mitochondrial membrane — ETC location**

Solution: The chapter places the carriers of the electron transport system, arranged in the form of complexes, on the inner mitochondrial membrane, the same membrane on which oxidative phosphorylation occurs.

Distractor review: A: The matrix houses the reactions of the Krebs cycle; the carriers of the transport system are embedded in the inner membrane rather than dissolved in the matrix. B: The outer chamber receives the protons released during oxidation of the coenzymes, but the carriers themselves are set in the membrane bounding it. D: The cytoplasm is the site of glycolysis; the oxidation of the reduced coenzymes by the carrier complexes is mitochondrial.

BIO-11-CH02-0074 | Easy | Terminal electron acceptor**Correct answer: A. Molecular oxygen**

Solution: Aerobic respiration is defined by the involvement of molecular oxygen as the final electron acceptor, and cytochrome C oxidase passes the electrons to it during terminal oxidation.

Distractor review: B: Ubiquinone is a mobile carrier that accepts electrons midway in the chain and passes them on to the cytochrome bc₁ complex. C: Cytochrome C carries electrons between the cytochrome bc₁ complex and cytochrome C oxidase, so it hands them on rather than retaining them. D: NAD⁺ collects hydrogen during glycolysis, the connecting link reaction and the Krebs cycle, and is regenerated by the transport system rather than terminating it.

BIO-11-CH02-0075 | Easy | Terminal oxidation and metabolic water**Correct answer: A. Reduced molecular oxygen combines with protons**

Solution: The chapter states that reduced molecular oxygen reacts with protons to form a water molecule called metabolic water, which completes terminal oxidation.

Distractor review: B: Oxygen becomes reduced precisely by accepting the electrons, so naming electrons again counts the same step twice and leaves the hydrogen unaccounted for. C: Ubiquinol surrenders its electrons to the cytochrome bc1 complex; it is not the species reduced at the end of the chain. D: Cytochrome C is a mobile electron carrier between the cytochrome bc1 complex and cytochrome C oxidase and takes no part in forming water.

BIO-11-CH02-0076 | Easy | Chemiosmosis**Correct answer: C. Chemiosmosis**

Solution: As the transfer of protons is accompanied by the synthesis of ATP, the process was named chemiosmosis by Peter Mitchell.

Distractor review: A: Substrate-level phosphorylation is the direct transfer of a phosphate group from a suitable substrate to ADP and involves no proton movement across a membrane. B: Photophosphorylation is the formation of ATP during photosynthesis and is not the process coupled to proton transfer in the mitochondrion. D: Oxidative decarboxylation describes the conversion of pyruvic acid to acetyl CoA and steps of the Krebs cycle, in which carbon dioxide is released.

BIO-11-CH02-0077 | Easy | Oxysome**Correct answer: B. Oxysome**

Solution: The chapter names the ATP-forming particle of the inner mitochondrial membrane the oxysome and describes it as having an F0 part and an F1 part.

Distractor review: A: NADH dehydrogenase oxidises $\text{NADH} + \text{H}^+$ and passes its electrons to ubiquinone; it has no F0 or F1 part. C: The cytochrome bc1 complex oxidises ubiquinol and forwards the electrons to cytochrome C rather than making ATP. D: Cytochrome C oxidase contains cytochrome a and a3 and carries out terminal oxidation by delivering electrons to molecular oxygen.

BIO-11-CH02-0078 | Medium | Sequence of electron carriers**Correct answer: A. II, IV, V, I, III**

Solution: $\text{NADH} + \text{H}^+$ is oxidised by NADH dehydrogenase, which passes electrons to ubiquinone; the ubiquinol so formed is oxidised by the cytochrome bc1 complex, which reduces cytochrome C, and cytochrome C delivers the electrons to cytochrome C oxidase.

Distractor review: B: This puts cytochrome C ahead of the cytochrome bc1 complex, whereas cytochrome C receives its electrons from that complex. C: This puts the cytochrome bc1 complex ahead of ubiquinone, but the complex acts on ubiquinol and therefore follows it. D: This begins at ubiquinone, whereas $\text{NADH} + \text{H}^+$ is first oxidised by NADH dehydrogenase, which then reduces ubiquinone.

BIO-11-CH02-0079 | Medium | Sequence of electron carriers**Correct answer: C. III, I, IV, V, II**

Solution: FADH_2 is oxidised by succinate dehydrogenase, whose electrons reach ubiquinone; from ubiquinol they pass to the cytochrome bc1 complex, then to cytochrome C, and finally to cytochrome C oxidase.

Distractor review: A: This starts at ubiquinone, but FADH_2 is oxidised first by succinate dehydrogenase, which then reduces ubiquinone. B: This puts cytochrome C ahead of the cytochrome bc1 complex, whereas cytochrome C is reduced by that complex. D: This puts the cytochrome bc1 complex ahead of ubiquinone, but the complex oxidises ubiquinol and therefore comes after it.

BIO-11-CH02-0080 | Hard | Sequence of electron carriers**Correct answer: D. III, V, II, IV, I**

Solution: Ubiquinol is oxidised by the cytochrome bc1 complex, which reduces cytochrome C; cytochrome C passes the electrons to cytochrome C oxidase, which reduces molecular oxygen, and the reduced oxygen then reacts with protons to give metabolic water.

Distractor review: A: This puts cytochrome C oxidase before cytochrome C, whereas cytochrome C is the carrier that delivers electrons to the oxidase. B: This begins with cytochrome C, but ubiquinol is oxidised by the cytochrome bc1 complex, which then reduces cytochrome C. C: This forms the water before oxygen has been involved, whereas the water appears only after reduced oxygen has reacted with protons.

BIO-11-CH02-0081 | Hard | Sequence of electron carriers**Correct answer: B. II, IV, V, I, III**

Solution: Electrons flow from NADH dehydrogenase to ubiquinone, then to the cytochrome bc1 complex, then to cytochrome C and finally to cytochrome C oxidase, so reading backwards from the oxidase gives cytochrome C oxidase, cytochrome C, cytochrome bc1 complex, ubiquinone, NADH dehydrogenase.

Distractor review: A: This is the sequence read in the direction of electron flow, whereas the stem asks for it against that direction. C: This interchanges cytochrome C with the cytochrome bc1 complex; going backwards from the oxidase, cytochrome C is met before that complex. D: This interchanges ubiquinone with the cytochrome bc1 complex; going backwards, the complex is met before ubiquinone.

BIO-11-CH02-0082 | Medium | Entry of the reduced coenzymes**Correct answer: B. NADH+H⁺ is oxidised by NADH dehydrogenase and FADH₂ by succinate dehydrogenase**

Solution: NADH+H⁺ is oxidised by NADH dehydrogenase while FADH₂ is oxidised by succinate dehydrogenase, and both complexes deliver their electrons to ubiquinone.

Distractor review: A: The two complexes are interchanged; succinate dehydrogenase is the complex that handles FADH₂, the coenzyme reduced when succinate is converted to fumarate. C: The cytochrome bc1 complex acts on ubiquinol and receives no coenzyme directly. D: Ubiquinone receives electrons from the two dehydrogenase complexes and does not itself oxidise a coenzyme.

BIO-11-CH02-0083 | Medium | Ubiquinone as a mobile carrier**Correct answer: A. Ubiquinol, oxidised by the cytochrome bc1 complex**

Solution: Reduced ubiquinone is called ubiquinol, and it is oxidised by the cytochrome bc1 complex, which then reduces cytochrome C.

Distractor review: B: Cytochrome C oxidase acts on reduced cytochrome C and carries out terminal oxidation; it is two steps beyond ubiquinol. C: Succinate dehydrogenase donates electrons to ubiquinone; it does not receive them back from ubiquinol. D: The name ubiquinone belongs to the oxidised form of the carrier, so it cannot also name the reduced form that the complex oxidises.

BIO-11-CH02-0084 | Hard | Mobile carriers of the chain**Correct answer: C. Cytochrome C**

Solution: Cytochrome C is described as a small, iron-containing protein that acts as a mobile electron carrier between the cytochrome bc1 complex and cytochrome C oxidase, unlike the carriers fixed in the membrane as complexes.

Distractor review: A: Ubiquinone is also a mobile carrier, but it is a quinone and not the small iron-containing protein described in the stem. B: Cytochrome C oxidase is itself one of the fixed complexes and contains cytochrome a and a₃. D: The cytochrome bc1 complex is a fixed complex of the membrane and is the carrier that reduces the mobile protein rather than being it.

BIO-11-CH02-0085 | Medium | Cytochrome C oxidase**Correct answer: B. Cytochrome C oxidase**

Solution: Cytochrome C oxidase is the complex that contains cytochrome a and a₃ and passes the electrons of the chain to molecular oxygen in terminal oxidation.

Distractor review: A: The cytochrome bc1 complex oxidises ubiquinol and hands electrons to cytochrome C, well before oxygen is reached. C: Cytochrome C is a single small iron-containing protein that ferries electrons to the oxidase and does not itself contain cytochrome a and a₃. D: NADH dehydrogenase is the entry complex for NADH+H⁺ and delivers electrons to ubiquinone, not to oxygen.

BIO-11-CH02-0086 | Medium | ATP from reduced coenzymes**Correct answer: D. 19**

Solution: Five molecules of NADH+H⁺ give $5 \times 3 = 15$ ATP and two molecules of FADH₂ give $2 \times 2 = 4$ ATP, so the total is $15 + 4 = 19$ ATP.

Distractor review: A: This credits each FADH₂ with 3 ATP, giving 15 plus 6; the balance sheet allows only 2 ATP for each FADH₂. B: This interchanges the two factors, taking 2 ATP for each NADH+H⁺ and 3 for each FADH₂, giving 10 plus 6. C: This applies 2 ATP to both coenzymes, giving 10 plus 4, and so undervalues every NADH+H⁺.

BIO-11-CH02-0087 | Hard | ATP accounting for the Krebs cycle**Correct answer: C. 24**

Solution: Six molecules of $\text{NADH}+\text{H}^+$ give $6 \times 3 = 18$ ATP, two molecules of FADH_2 give $2 \times 2 = 4$ ATP, and substrate-level phosphorylation contributes 2 ATP, so the row total is $18 + 4 + 2 = 24$ ATP.

Distractor review: A: This leaves out the 2 ATP formed by substrate-level phosphorylation in the two turns of the cycle. B: This credits each FADH_2 with 3 ATP, adding 6 in place of 4. D: This is the yield of a single turn of the cycle, whereas the balance sheet is drawn for one glucose molecule and therefore covers two turns.

BIO-11-CH02-0088 | Hard | ATP from reduced coenzymes**Correct answer: A. 10 $\text{NADH}+\text{H}^+$ and 2 FADH_2**

Solution: Four ATP from FADH_2 at 2 ATP each means 2 molecules of FADH_2 ; the remaining $34 - 4 = 30$ ATP at 3 ATP each means 10 molecules of $\text{NADH}+\text{H}^+$.

Distractor review: B: This divides the remaining 30 ATP by 2 instead of by 3 and so overstates the number of $\text{NADH}+\text{H}^+$ molecules. C: This reads the 4 ATP as 4 molecules of FADH_2 , which would mean only 1 ATP for each FADH_2 . D: This adds the 2 ATP consumed during glycolysis to the 30 before dividing by 3, although those 2 ATP are a consumption entry and not a product of oxidative phosphorylation.

BIO-11-CH02-0089 | Challenging | ATP yield per reduced coenzyme**Correct answer: B. FADH_2 , credited with 2 ATP for each molecule**

Solution: The Krebs cycle row of the balance sheet enters 2 FADH_2 as $2 \times 2 = 4$ ATP, so FADH_2 alone is credited with 2 ATP for each molecule while every $\text{NADH}+\text{H}^+$ is credited with 3.

Distractor review: A: The Krebs cycle row shows 2 FADH_2 accounting for a total of 4 ATP, so the credit is 2 ATP for each molecule and not 4. C: The glycolysis row shows 2 $\text{NADH}+\text{H}^+$ accounting for 6 ATP, which is the same 3 ATP for each molecule as elsewhere. D: The row for pyruvic acid to acetyl CoA shows 2 $\text{NADH}+\text{H}^+$ accounting for 6 ATP, again 3 ATP for each molecule.

BIO-11-CH02-0090 | Medium | Proton movement and the oxysome**Correct answer: C. Released into the outer chamber, they pass through the F_0 part**

Solution: Oxidation of the reduced coenzymes releases protons into the outer chamber of the mitochondrion, and these protons are channelled into the inner chamber by the F_0 part of the oxysome, the passage being coupled to the catalytic site of F_1 .

Distractor review: A: The F_1 part carries the catalytic site at which ATP is synthesised; the passage for the protons is the F_0 part. B: The protons are released into the outer chamber and are then channelled inwards, so naming the inner chamber reverses the direction of the movement. D: This reverses both the compartment of release and the part of the oxysome that provides the channel.

BIO-11-CH02-0091 | Easy | Regeneration of oxidised coenzymes**Correct answer: C. NAD^+ and FAD^+**

Solution: The electron transport system oxidises $\text{NADH}+\text{H}^+$ and FADH_2 and thereby regenerates the oxidised coenzymes NAD^+ and FAD^+ for recycling in the earlier steps of respiration.

Distractor review: A: These are the reduced forms that the system consumes; regeneration returns them to their oxidised state. B: CoA-SH is released within the Krebs cycle when citrate is formed and is not a coenzyme regenerated by the electron transport system. D: This pairs the correct oxidised flavin coenzyme with CoA-SH, which is not regenerated by the transport system, and omits NAD^+ altogether.

BIO-11-CH02-0092 | Medium | Water supplied to the Krebs cycle**Correct answer: D. Water molecules, required by the Krebs cycle**

Solution: The electron transport system provides water molecules that are necessary for the Krebs cycle, in which hydration steps consume water.

Distractor review: A: The chapter attaches the water provided by the transport system to the requirement of the Krebs cycle, not of glycolysis. B: Carbon dioxide is released by the connecting link reaction and the Krebs cycle rather than being supplied to the cycle. C: Molecular oxygen is consumed by the transport system as the final electron acceptor and is not a product handed on to the cycle.

BIO-11-CH02-0093 | Medium | Stepwise release of energy

Correct answer: A. Energy is released in a stepwise manner, and this prevents damage of cells

Solution: Because electrons pass through a series of carriers, the energy of the reduced coenzymes is released in a stepwise manner, which the chapter states prevents damage of cells.

Distractor review: B: The transport system passes electrons through a series of carriers, so the release is graded and not a single event. C: The stated purpose of the graded release is the protection of the cell, and the chapter treats ATP rather than stored heat as the useful outcome. D: Molecular oxygen remains indispensable as the final electron acceptor however gradually the energy is released.

BIO-11-CH02-0094 | Easy | Share of the total ATP

Correct answer: D. 4 ATP molecules from non-ETS steps

Solution: If 34 of the net 38 ATP molecules come through the electron transport system, then $38 - 34 = 4$ ATP arise by other means, these being the net gain from substrate-level phosphorylation.

Distractor review: A: 6 is the gross substrate-level phosphorylation total of the balance sheet; 2 of those ATP are consumed during glycolysis, so the net figure is smaller. B: 2 is only the substrate-level phosphorylation credited to the Krebs cycle, and the glycolysis row contributes as well. C: 8 is the net benefit entered for the glycolysis row, and that figure already includes ATP produced through the transport system.

BIO-11-CH02-0095 | Easy | Definition of respiratory quotient

Correct answer: A. The ratio of the volume of CO₂ released to the volume of O₂ consumed in respiration, also called the respiratory ratio

Solution: The respiratory quotient, also called the respiratory ratio, is defined as the ratio of the volume of carbon dioxide released to the volume of oxygen consumed in respiration.

Distractor review: B: The ratio is inverted; carbon dioxide released is the numerator and oxygen consumed the denominator. C: The ratio is stated correctly but the alternative name printed for it is respiratory ratio. D: A difference of volumes would carry units, whereas the quantity defined is a ratio and is therefore a pure number.

BIO-11-CH02-0096 | Easy | Formula for respiratory quotient

Correct answer: C. V₁ divided by V₂

Solution: The respiratory quotient is the volume of carbon dioxide released divided by the volume of oxygen consumed, so it is V₁ divided by V₂.

Distractor review: A: This inverts the ratio and would give a value above 1 for fats, whereas fats give a value below 1. B: A product of two volumes is not a ratio and would change if the same tissue were simply observed for twice as long. D: A difference of volumes is not the defined quantity, which is a ratio and therefore independent of the units in which the volumes are measured.

BIO-11-CH02-0097 | Medium | RQ of carbohydrate

Correct answer: B. A completely oxidised carbohydrate; the volume of CO₂ evolved equals the volume of O₂ consumed

Solution: When a carbohydrate is completely oxidised the volume of carbon dioxide evolved equals the volume of oxygen consumed, so the respiratory quotient works out to 1.

Distractor review: A: Fats give a value of about 0.7, for which the volume of carbon dioxide evolved is less than the volume of oxygen consumed. C: Proteins give a value of about 0.9, so the volume of carbon dioxide evolved falls short of the volume of oxygen consumed. D: A twofold excess of carbon dioxide would make the ratio 2; equality of the two volumes is what gives the value 1.

BIO-11-CH02-0098 | Medium | RQ from measured gas volumes

Correct answer: B. 0.7, indicating fat

Solution: Dividing 14 mL of carbon dioxide by 20 mL of oxygen gives 0.7, and about 0.7 is the value the chapter gives for fats.

Distractor review: A: The arithmetic is right, but a value of about 0.7 is the printed value for fats while proteins give about 0.9. C: A value of 0.9 would require 18 mL of carbon dioxide against 20 mL of oxygen; dividing 14 by 20 gives 0.7. D: A value of 1 requires the volume of carbon dioxide evolved to equal the volume of oxygen consumed, which is not the case here.

BIO-11-CH02-0099 | Medium | Gas volume predicted from a known RQ**Correct answer: D. 27 mL**

Solution: Rearranging the definition gives volume of carbon dioxide = $0.9 \times 30 \text{ mL} = 27 \text{ mL}$.

Distractor review: A: Equal volumes would give a respiratory quotient of 1, which is the value for a completely oxidised carbohydrate rather than for protein. B: This applies the value 0.7, which the chapter gives for fats, instead of the 0.9 stated for protein. C: This is the difference between the two volumes rather than the volume of carbon dioxide itself.

BIO-11-CH02-0100 | Hard | RQ from measured gas volumes**Correct answer: D. P is respiring on carbohydrate and Q on fat**

Solution: Sample P gives 25 divided by 25, that is 1, the value for a completely oxidised carbohydrate; sample Q gives 35 divided by 50, that is 0.7, the value the chapter gives for fats.

Distractor review: A: The two samples are interchanged; equal gas volumes give a value of 1, which belongs to a completely oxidised carbohydrate. B: Sample Q gives 35 divided by 50, which is 0.7 and the value for fats; protein would give about 0.9. C: Sample P gives 25 divided by 25, which is exactly 1; a value of about 0.9 would be needed for protein.

BIO-11-CH02-0101 | Hard | RQ of anaerobic respiration**Correct answer: A. Anaerobic respiration, for which the respiratory quotient is infinity**

Solution: In anaerobic respiration carbon dioxide is evolved without oxygen being consumed, so the denominator of the ratio is zero and the respiratory quotient is always infinity.

Distractor review: B: Zero would require no carbon dioxide to be evolved; in fermentation carbon dioxide is evolved while no oxygen is taken up, so it is the denominator that vanishes. C: A value of 1 needs equal volumes of the two gases, but no oxygen at all is consumed in anaerobic respiration. D: Protein oxidised aerobically consumes oxygen and gives a finite value of about 0.9.

BIO-11-CH02-0102 | Challenging | RQ under aerobic and anaerobic conditions**Correct answer: D. 0.9 and infinity**

Solution: With oxygen available the ratio is 36 divided by 40, that is 0.9, the value given for proteins; without oxygen the denominator is zero while carbon dioxide is still evolved, so the ratio is infinity.

Distractor review: A: The second value is obtained by putting the zero oxygen figure in the numerator; the oxygen volume is the denominator, so the ratio grows without limit instead of vanishing. B: The first value is misread as the fat figure; 36 divided by 40 is 0.9, the value the chapter gives for proteins. C: A first value of 1 would need the two volumes to be equal, but 36 mL of carbon dioxide is less than 40 mL of oxygen.

BIO-11-CH02-0103 | Easy | Utility of stepwise oxidation**Correct answer: A. The utilization of a relatively higher proportion of that energy in ATP synthesis**

Solution: Releasing the bond energy of the substrate in many small steps allows a relatively higher proportion of it to be used for ATP synthesis than a single large release would.

Distractor review: B: The claim made is that a relatively higher proportion of the energy is captured, not that the capture is complete. C: The purpose of the graded release is to improve the capture of energy as ATP, not to divert it into heat. D: Each of the many steps is catalysed by its own enzyme, and it is precisely the enzymes of those steps that allow the pathway to be regulated.

BIO-11-CH02-0104 | Medium | Utility of stepwise oxidation**Correct answer: B. Control of the rate of the pathway and of the energy output according to the need of the cell**

Solution: Each step of a many-stepped pathway is catalysed by its own enzyme, and since specific compounds can enhance or inhibit those enzymes, the rate of the pathway and the resulting energy output can be controlled according to the need of the cell.

Distractor review: A: The steps of the pathway are fixed; what varies is the rate at which they proceed. C: Intermediates are formed at every step, and some of them are drawn off for the synthesis of other biomolecules. D: A constant rate is the opposite of what such regulation achieves, since the point of enhancing or inhibiting the enzymes is to match supply with demand.

BIO-11-CH02-0105 | Medium | Utility of stepwise oxidation

Correct answer: D. The same pathway supplies intermediates used in the synthesis of other biomolecules such as amino acids

Solution: Because a many-stepped pathway generates a series of intermediates, those intermediates can be used for forming other biomolecules such as amino acids.

Distractor review: A: Molecular oxygen is consumed in respiration as the final electron acceptor and is not synthesised from any intermediate. B: The intermediates are in fact drawn upon for synthesis, which is why the Krebs cycle is described as an amphibolic pathway. C: Regeneration of the oxidised coenzymes is credited to the electron transport system, while the intermediates themselves act as precursors for synthesis.

BIO-11-CH02-0106 | Medium | Industrial use of fermentation

Correct answer: C. Dairies, bakeries, distilleries, leather and paper industries; alcohol, organic acids, vitamins and antibiotics

Solution: Fermentation is used in dairies, bakeries, distilleries, leather industries and paper industries, and commercially for the production of alcohol, organic acids, vitamins and antibiotics.

Distractor review: A: Pigments are named among the cellular materials built with the energy of respiration, whereas antibiotics are the product listed for fermentation industries. B: The list of industries is incomplete, since leather and paper industries are also named. D: Water is a product of terminal oxidation in aerobic respiration and is not a commercial fermentation product.

BIO-11-CH02-0107 | Hard | Utility of stepwise oxidation

Correct answer: C. I, II and III

Solution: Better capture of the chemical bond energy as ATP, regulation of the rate and energy output through the enzymes of the individual steps, and supply of intermediates for the synthesis of other biomolecules are the purposes served; a pathway cannot proceed without its enzymes, since it is those enzymes that make regulation possible.

Distractor review: A: This omits the supply of intermediates for the synthesis of other biomolecules, which is one of the stated purposes. B: This admits the claim that the pathway can run without enzymes, whereas each step needs its own enzyme, and it drops the improved capture of bond energy. D: This admits the enzyme-free claim and drops the regulation of rate and energy output, which is one of the stated purposes.

BIO-11-CH02-0108 | Challenging | Significance of respiration

Correct answer: C. Building blocks for the synthesis of complex compounds are provided by the intermediates of the Krebs cycle

Solution: The chapter states that various intermediates of the Krebs cycle are used as building blocks for the synthesis of other complex compounds, which is why the cycle is also described as an amphibolic pathway.

Distractor review: A: The water formed in terminal oxidation is stated to meet the requirement of the Krebs cycle, while the building blocks are the intermediates of that cycle. B: This balance is stated to be maintained by respiration coupled with photosynthesis, since respiration alone would only add carbon dioxide. D: That conversion is attributed to the energy released by respiration, whereas the regenerated coenzymes are recycled for further oxidation.

BIO-11-CH02-0109 | Easy | Cellular site

Correct answer: A. Cytosol

Solution: Glycolysis occurs in the cytosol. The link reaction and Krebs cycle occur in the mitochondrial matrix, while the electron transport system is associated with the inner membrane.

Distractor review: B: Mitochondrial matrix is a plausible distractor, but it conflicts with the required locating glycolysis within the cell and therefore cannot be the single correct response. C: Inner mitochondrial membrane is a plausible distractor, but it conflicts with the required locating glycolysis within the cell and therefore cannot be the single correct response. D: Outer mitochondrial membrane is a plausible distractor, but it conflicts with the required locating glycolysis within the cell and therefore cannot be the single correct response.

BIO-11-CH02-0110 | Medium | Oxidative decarboxylation

Correct answer: B. Succinyl-CoA

Solution: The oxidative decarboxylation of alpha-ketoglutarate uses CoA-SH and NAD⁺, releases carbon dioxide and NADH+H⁺, and forms succinyl-CoA.

Distractor review: A: Citrate is a plausible distractor, but it conflicts with the required reading the alpha-ketoglutarate conversion from a pathway segment and therefore cannot be the single correct response. C: Fumarate is a plausible distractor, but it conflicts with the required reading the alpha-ketoglutarate conversion from a pathway segment and therefore cannot be the single correct response. D: Oxaloacetate is a plausible distractor, but it conflicts with the required reading the alpha-ketoglutarate conversion from a pathway segment and therefore cannot be the single correct response.

BIO-11-CH02-0111 | Hard | Experimental interpretation

Correct answer: C. 0.7; fat

Solution: $RQ = \text{carbon dioxide released} / \text{oxygen consumed} = 35/50 = 0.7$. A value near 0.7 is characteristic of fat as the main respiratory substrate.

Distractor review: A: 1.0; carbohydrate is a plausible distractor, but it conflicts with the required calculating and interpreting respiratory quotient and therefore cannot be the single correct response. B: 0.5; organic acid is a plausible distractor, but it conflicts with the required calculating and interpreting respiratory quotient and therefore cannot be the single correct response. D: 1.3; protein is a plausible distractor, but it conflicts with the required calculating and interpreting respiratory quotient and therefore cannot be the single correct response.

BIO-11-CH02-0112 | Challenging | Mixed substrates

Correct answer: D. 0.85

Solution: P releases 100 ml carbon dioxide and Q releases 70 ml. Combined $RQ = (100 + 70)/(100 + 100) = 170/200 = 0.85$.

Distractor review: A: 0.35 is a plausible distractor, but it conflicts with the required combining gas exchange from two respiratory substrates and therefore cannot be the single correct response. B: 0.70 is a plausible distractor, but it conflicts with the required combining gas exchange from two respiratory substrates and therefore cannot be the single correct response. C: 1.70 is a plausible distractor, but it conflicts with the required combining gas exchange from two respiratory substrates and therefore cannot be the single correct response.

Standard 11 | Chapter 03

Human Nutrition

BIO-11-CH03-0001 | Easy | Alimentary canal, general plan

Correct answer: A. About 8 to 10 metres

Solution: The alimentary canal is a continuous tube running from the mouth to the anus and is about 8 to 10 metres long.

Distractor review: B: About 6 metres is the length of the small intestine alone; the student mistakes the longest single region for the whole canal. C: No region or total of this size is given for the human canal; the figure is an inflated guess not supported by any stated dimension. D: 1.5 metres is the length of the large intestine; the student substitutes one terminal region for the entire canal.

BIO-11-CH03-0002 | Easy | Buccal cavity

Correct answer: C. Palate

Solution: The roof of the buccal cavity is formed by the palate, while the tongue forms its floor, the cheeks its side walls and the fleshy lips its front boundary.

Distractor review: A: The tongue lies along the floor of the buccal cavity, not its roof; the student inverts the two horizontal boundaries. B: The cheeks form the side walls of the buccal cavity; the student assigns a lateral boundary to the upper one. D: The fleshy lips bound the mouth at its opening in front; they do not roof the cavity.

BIO-11-CH03-0003 | Easy | Dentition

Correct answer: C. Dentition

Solution: Dentition is the study of teeth with respect to their number, arrangement, development and related features.

Distractor review: A: Thecodont describes the fixing of each tooth in a separate socket of the jaw bone; it names a mode of attachment, not a field of study. B: Diphyodont describes the appearance of only two sets of teeth in a lifetime; it names a pattern of replacement, not a field of study. D: Heterodont describes the presence of four different types of teeth; it names variety of form, not a field of study.

BIO-11-CH03-0004 | Easy | Tongue

Correct answer: D. Papillae, some of which bear taste buds

Solution: The upper surface of the tongue bears numerous projections called papillae, and some papillae bear sensory receptors called taste buds.

Distractor review: A: Villi are the finger like foldings of the mucosa of the small intestine; they are not present on the tongue. B: The papillae carry taste buds; the ducts of the salivary glands open into the buccal cavity independently of the papillae. C: Rugae are the irregular folds into which the mucosa of the stomach is thrown; they are not surface projections of the tongue.

BIO-11-CH03-0005 | Easy | Oesophagus

Correct answer: D. About 25 cm

Solution: The oesophagus is about 25 cm long and runs from the neck, through the central aspect of the rib cage, through the diaphragm, to join the stomach.

Distractor review: A: About 26 cm is the stated length of the duodenum; the student attaches an intestinal dimension to the oesophagus. B: 6 cm is the stated length of the caecum; the student transfers a large intestinal dimension to the oesophagus. C: No region of the alimentary canal is given this length; the figure is an overestimate not supported by any stated dimension.

BIO-11-CH03-0006 | Easy | Stomach

Correct answer: A. A muscular sac-like 'J' shaped organ in the upper left portion of the abdominal cavity

Solution: The stomach is a muscular sac-like organ shaped like the letter 'J' and is located in the upper left portion of the abdominal cavity.

Distractor review: B: The 'U' shape belongs to the duodenum, the first part of the small intestine, not to the stomach. C: The shape is right but the position is wrong; the stomach lies in the upper left portion of the abdominal cavity. D: Both shape and side are wrong; no region of the canal is described as 'C' shaped and the stomach is not a right-sided organ.

BIO-11-CH03-0007 | Easy | Large intestine**Correct answer: D. Caecum**

Solution: The vermiform appendix arises from the caecum, the small blind sac 6 cm in length present at the junction of the ileum and the colon, and it is a vestigial organ in human beings.

Distractor review: A: The ascending colon is the first part of the colon and follows the caecum; the appendix arises from the caecum itself, not from the colon. B: The transverse colon is the middle part of the colon and lies far from the junction of the ileum and colon where the appendix is situated. C: The rectum is the posterior storage region of the large intestine; the appendix is a proximal outgrowth, not a terminal one.

BIO-11-CH03-0008 | Easy | Large intestine**Correct answer: A. Rectum**

Solution: The rectum is the posterior region of the large intestine and temporarily stores the undigested waste before it leaves through the anus.

Distractor review: B: The caecum is the small blind sac at the junction of the ileum and colon at the beginning of the large intestine; it is not a storage region for waste. C: The descending colon is the third part of the colon and conveys material onward; the temporary store of undigested waste lies beyond it. D: The anus is the terminal opening of the alimentary canal, an aperture rather than a region capable of storing waste.

BIO-11-CH03-0009 | Easy | Buccal cavity**Correct answer: D. Buccal cavity**

Solution: The salivary glands open into the buccal cavity, so their secretion is added to food at the very first region of the alimentary canal.

Distractor review: A: The pharynx is the short common passage for food and air that follows the buccal cavity; the salivary glands discharge before this point. B: The oesophagus is a conducting tube joining the pharynx to the stomach and receives no salivary secretion. C: The stomach receives the secretion of its own gastric glands; salivary secretion reaches it only along with swallowed food.

BIO-11-CH03-0010 | Medium | Alimentary canal, general plan**Correct answer: D. IV, II, I, III**

Solution: Food taken into the buccal cavity passes into the short pharynx, then through the oesophagus, which pierces the diaphragm, and finally into the stomach.

Distractor review: A: This places the oesophagus before the pharynx; food leaves the buccal cavity into the short pharynx first and only then enters the oesophagus. B: This begins at the pharynx; the buccal cavity is the first region and it leads to the pharynx, not the reverse. C: This puts the stomach before the oesophagus; the oesophagus pierces the diaphragm and joins the stomach, so it must precede it.

BIO-11-CH03-0011 | Medium | Stomach**Correct answer: A. III, IV, I, II**

Solution: The cardia is the first part, into which the oesophagus opens; the fundus is the dome shaped region above and left of the cardia; the body is the large central portion that stores food; and the pylorus is the narrow posterior region that opens into the duodenum.

Distractor review: B: This places the body before the fundus; the fundus is the dome shaped region lying immediately above and to the left of the cardia, so it is named before the large central body. C: This begins at the fundus; the part at which the oesophagus opens is the cardia, which therefore comes first. D: This ends with the body; the narrow posterior part that opens into the duodenum is the pylorus, so the pylorus must be last.

BIO-11-CH03-0012 | Medium | Small intestine**Correct answer: C. Order: II, IV, I, III**

Solution: Chyme passes from the stomach into the duodenum, then into the coiled middle jejunum, then into the highly coiled ileum, and reaches the caecum at the junction of the ileum and the colon.

Distractor review: A: This places the ileum before the jejunum; the jejunum is the coiled middle portion and lies between the duodenum and the ileum. B: This begins at the jejunum; the duodenum is the first part of the small intestine and receives chyme directly from the stomach. D: This places the caecum before the ileum; the caecum lies at the junction of the ileum and the colon and so follows the ileum.

BIO-11-CH03-0013 | Medium | Large intestine**Correct answer: A. II, III, IV, I**

Solution: Material entering the large intestine reaches the caecum first, then passes through the ascending colon, the transverse colon and the descending colon in that order.

Distractor review: B: This places the descending colon immediately after the caecum; the colon is traversed in the order ascending, transverse, descending. C: This begins at the ascending colon; the caecum lies at the junction of the ileum and colon and is reached before any part of the colon. D: This places the transverse colon before the ascending colon, reversing the first two parts of the colon.

BIO-11-CH03-0014 | Medium | Alimentary canal, general plan**Correct answer: B. II, I, IV, III**

Solution: The capsule passes the cardiac sphincter at the junction of the oesophagus and stomach, then the pyloric sphincter guarding the exit of the stomach into the duodenum, then travels through the jejunum, and finally reaches the rectum.

Distractor review: A: This places the pyloric sphincter before the cardiac sphincter; the cardiac sphincter lies at the entry to the stomach and the pyloric sphincter at its exit. C: This places the rectum before the jejunum; the jejunum is a part of the small intestine and is reached long before the rectum. D: This places the jejunum before the pyloric sphincter; material can reach the small intestine only after passing the pyloric sphincter.

BIO-11-CH03-0015 | Medium | Dentition**Correct answer: A. 8**

Solution: The formula gives 2 premolars in one half of the upper jaw and 2 in one half of the lower jaw. Each jaw has two symmetrical halves, so the total is 2×2 for the upper jaw plus 2×2 for the lower jaw, that is $4 + 4 = 8$ premolars.

Distractor review: B: 4 is obtained by counting the two premolars of one half of the upper jaw and the two of one half of the lower jaw only, forgetting that each jaw has two halves. C: 12 is the number of molars, obtained by reading the fourth figure of the formula instead of the third. D: 16 is the total number of teeth in one whole jaw, obtained by using the jaw total in place of the premolar count.

BIO-11-CH03-0016 | Hard | Dentition**Correct answer: B. 37.5 per cent**

Solution: The formula gives 3 molars in one half of the upper jaw and 3 in one half of the lower jaw. Each jaw has two halves, so the total is $3 \times 2 + 3 \times 2 = 12$ molars. As a percentage of the 32 teeth this is $(12 / 32) \times 100 = 37.5$ per cent.

Distractor review: A: 25 per cent corresponds to 8 teeth out of 32, which is the count of incisors or of premolars, not of molars. C: 12.5 per cent corresponds to 4 teeth out of 32, which is the count of canines. D: 50 per cent corresponds to 16 teeth, which is the number of teeth in one whole jaw rather than the number of molars.

BIO-11-CH03-0017 | Hard | Small intestine**Correct answer: C. 626 cm**

Solution: Converting to centimetres, the jejunum is $2.5 \times 100 = 250$ cm and the ileum is $3.5 \times 100 = 350$ cm. Adding the duodenum gives $26 + 250 + 350 = 626$ cm.

Distractor review: A: 600 cm is obtained by adding only the jejunum and the ileum and dropping the duodenum from the sum. B: 656 cm results from converting 2.5 metres to 280 cm by an arithmetic slip while converting the metre values. D: 326 cm results from converting each of 2.5 metres and 3.5 metres to 150 cm and 150 cm, that is from halving the metre figures before conversion.

BIO-11-CH03-0018 | Hard | Alimentary canal, general plan**Correct answer: C. 3/4**

Solution: The two intestines together measure $6 + 1.5 = 7.5$ metres. As a fraction of the 10 metre canal this is $7.5 / 10 = 3/4$.

Distractor review: A: $3/5$ corresponds to 6 metres out of 10, that is the small intestine alone, with the large intestine left out of the sum. B: $2/3$ does not correspond to any combination of the stated lengths and arises from rounding 7.5 metres to about 6.7 metres. D: $4/5$ corresponds to 8 metres out of 10, obtained by taking the large intestine as 2 metres instead of its stated 1.5 metres.

BIO-11-CH03-0019 | Medium | Small intestine**Correct answer: D. 4:1**

Solution: Both lengths are in metres, so the ratio is $6 : 1.5$. Dividing both terms by 1.5 gives $4 : 1$.

Distractor review: A: $3:1$ arises from taking the large intestine as 2 metres rather than its stated 1.5 metres. B: $6:1$ arises from dividing the small intestinal length by 1 metre, that is from treating the large intestine as 1 metre long. C: $1:4$ is the reciprocal of the required ratio and results from writing the large intestine first even though the question asks for small to large.

BIO-11-CH03-0020 | Medium | Alimentary canal, general plan**Correct answer: A. I-P, II-Q, III-R, IV-P**

Solution: The fundus and the pylorus are named regions of the stomach, the jejunum is the coiled middle portion of the small intestine, and the caecum is the blind sac of the large intestine at the junction of the ileum and colon.

Distractor review: B: This places the caecum in the small intestine and the pylorus in the large intestine; the caecum belongs to the large intestine and the pylorus is the posterior part of the stomach. C: This places the fundus in the large intestine; the fundus is the dome shaped region of the stomach above and left of the cardia. D: This places the jejunum in the large intestine and the pylorus in the small intestine; the jejunum is the coiled middle part of the small intestine and the pylorus is a stomach region.

BIO-11-CH03-0021 | Medium | Stomach**Correct answer: B. The cardiac sphincter is a band of circular muscles at the junction of the oesophagus and the stomach that prevents regurgitation of food, while the pyloric sphincter regulates passage of material from the stomach to the small intestine**

Solution: The cardiac sphincter is a band of circular muscles present at the junction of the oesophagus and the stomach and prevents regurgitation of food, whereas the pyloric sphincter is a set of muscles at the pylorus that regulates the passage of material from the stomach into the small intestine.

Distractor review: A: The positions and functions of the two sphincters are interchanged; the cardiac sphincter belongs to the cardia at the oesophageal end. C: The two sphincters guard opposite ends of the stomach; only the cardiac sphincter lies at the oesophageal junction. D: The opening of the hepato-pancreatic duct is guarded by the sphincter of Oddi; neither sphincter of the stomach is concerned with it.

BIO-11-CH03-0022 | Medium | Small intestine**Correct answer: B. The jejunum is about 2.5 metres long and is the coiled middle portion, while the ileum is about 3.5 metres long and is highly coiled**

Solution: The jejunum is about 2.5 metres long and forms the coiled middle portion of the small intestine, while the ileum is about 3.5 metres long and is highly coiled.

Distractor review: A: The two lengths and the two descriptions are interchanged; the middle portion is the shorter of the two. C: The first part of the small intestine is the duodenum; the jejunum is the middle portion and the ileum is the last part. D: The length of about 26 cm and the 'U' shape belong to the duodenum, not to the jejunum.

BIO-11-CH03-0023 | Medium | Dentition**Correct answer: B. I-Q, II-S, III-P, IV-R**

Solution: Thecodont means each tooth is fixed in a separate socket present in the jaw bones, and the joint by which it is so fixed is called a gomphosis. Diphyodont means only two sets of teeth, milk teeth and permanent teeth, appear in a lifetime. Heterodont means four different types of teeth are present.

Distractor review: A: This makes diphyodont mean four types of teeth and heterodont mean two sets of teeth; the two terms are interchanged. C: This makes thecodont mean two sets of teeth and diphyodont mean fixing in separate sockets; the terms for attachment and replacement are interchanged. D: This makes thecodont the name of a joint and gomphosis the fixing in a socket; gomphosis is the joint and thecodont is the condition of being socketed.

BIO-11-CH03-0024 | Hard | Large intestine**Correct answer: D. Vermiform appendix**

Solution: The vermiform appendix is an elongated worm like outgrowth arising from the caecum. It leads on to no further region and is a vestigial organ in human beings, so it is the exception to the through-passage rule.

Distractor review: A: The caecum is described as a blind sac, but it lies at the junction of the ileum and colon and material passes on from it into the ascending colon. B: The fundus is a dome shaped region of the stomach continuous with the cardia and the body, so its contents pass on towards the pylorus. C: The rectum temporarily stores undigested waste but opens onward at the anus, the terminal opening of the alimentary canal.

BIO-11-CH03-0025 | Medium | Large intestine**Correct answer: C. Ascending colon, transverse colon and descending colon**

Solution: The colon consists of the ascending, transverse and descending colon only. The caecum is a region of the large intestine lying at the junction of the ileum and colon, but it is not a part of the colon.

Distractor review: A: The caecum is a region of the large intestine lying at the junction of the ileum and colon, but it is not a part of the colon itself. B: The rectum is the region that follows the colon and stores the waste temporarily; it is not counted among the parts of the colon. D: Two of the three named parts are right, but the caecum belongs to the large intestine as a whole and the ascending colon has been omitted.

BIO-11-CH03-0026 | Hard | Dentition

Correct answer: C. The presence of incisors, canines, premolars and molars makes the dentition heterodont

Solution: Heterodont dentition means four different types of teeth are present, and in an adult human these are the incisors, canines, premolars and molars. Two sets of teeth in a lifetime is the diphyodont feature and socketed attachment by a gomphosis joint is the thecodont feature.

Distractor review: A: Two successive sets of teeth is a feature of replacement and makes the dentition diphyodont, not heterodont. B: Fixing in a separate socket is a feature of attachment and makes the dentition thecodont, not diphyodont. D: The gomphosis joint concerns attachment, so it supports the term thecodont; heterodont refers only to the variety of tooth types.

BIO-11-CH03-0027 | Medium | Stomach

Correct answer: B. Fundus

Solution: The described organ is the stomach. The dome shaped region lying above and to the left of the point where the oesophagus opens in is the fundus, while the large central storing portion is the body and the narrow outlet into the duodenum is the pylorus.

Distractor review: A: The cardia is the region at which the tube from the diaphragm opens in, and the dome lies above and to the left of it rather than at it. C: The body is the large central portion below the dome that stores food, not the dome itself. D: The pylorus is the narrow far region that opens into the 'U' shaped tube 26 cm long, at the opposite end from the dome.

BIO-11-CH03-0028 | Easy | Large intestine

Correct answer: C. Ascending colon

Solution: The highly coiled 3.5 metre tube is the ileum and the wider tube is the large intestine. The portion running upwards from the junction is the ascending colon; it is followed by the transverse colon and then the descending colon, which leads to the rectum.

Distractor review: A: The transverse colon is the portion that runs across, after the upward running portion. B: The descending colon is the portion that runs downwards, after the across running portion. D: The rectum is the region that temporarily stores waste before the terminal opening, not a part of the colon.

BIO-11-CH03-0029 | Hard | Alimentary canal, general plan

Correct answer: B. 5

Solution: The short common passage for food and air is the pharynx and the small blind sac 6 cm long is the caecum. Strictly between them lie the oesophagus, the stomach, the duodenum, the jejunum and the ileum, that is 5 regions.

Distractor review: A: 4 is obtained by leaving out one of the three parts of the small intestine, most often the 'U' shaped duodenum, from the count. C: 6 is obtained by including one of the two boundary regions, that is by counting inclusively at one end. D: 7 is obtained by counting inclusively at both ends, so that both boundary regions are added to the five intervening regions.

BIO-11-CH03-0030 | Medium | Pharynx

Correct answer: B. It is a common passage for food and air

Solution: The pharynx is a short region following the buccal cavity and is a common passage for both food and air.

Distractor review: A: The pharynx is shared by both food and air; a passage exclusive to food begins only at the oesophagus. C: The tube about 25 cm long that pierces the diaphragm and joins the stomach is the oesophagus; the pharynx is described as short. D: Rugae are the irregular folds of the mucosa of the stomach; they are not a feature of the pharynx.

BIO-11-CH03-0031 | Hard | Oesophagus

Correct answer: C. Rhythmic waves of contraction and relaxation of its muscles push the food along the tube

Solution: Food is moved through the oesophagus by peristalsis, the rhythmic wave of contraction and relaxation of the muscles of its wall, which propels the food regardless of body posture.

Distractor review: A: Weight would act only when the tube is vertical; the morsel reaches the stomach even when the tube is horizontal, so gravity cannot be the mechanism. B: The cardiac sphincter is a band of circular muscles that prevents regurgitation; it guards the junction and does not pull food along the oesophagus. D: Churning is the mechanical process described for the stomach; the oesophagus conducts food without grinding it.

BIO-11-CH03-0032 | Hard | Small intestine**Correct answer: B. The ileum, which normally leads on to the caecum**

Solution: The terminal part of the small intestine is the ileum, which is about 3.5 metres long, so a terminal 1.5 metre segment lies wholly within it. The ileum normally leads on to the junction with the colon, at which the caecum, a small blind sac 6 cm long, is present.

Distractor review: A: The jejunum is the middle portion of the small intestine, so it is not the terminal segment; the terminal 1.5 metres lie within the ileum. C: The duodenum is only about 26 cm long and is the first part of the small intestine, so it cannot supply a terminal 1.5 metre segment. D: The correct segment is named but its continuity is reversed; the duodenum precedes the ileum and does not follow it.

BIO-11-CH03-0033 | Hard | Alimentary canal, general plan**Correct answer: C. I-Q, II-R, III-S, IV-P**

Solution: The caecum is 6 cm long, the duodenum about 26 cm, the large intestine 1.5 metres and the ileum about 3.5 metres.

Distractor review: A: This gives the ileum 1.5 metres and the large intestine 3.5 metres; the values for these two structures are interchanged. B: This gives the duodenum 6 cm and the caecum 26 cm; the two short dimensions are interchanged. D: This gives the large intestine 26 cm and the duodenum 1.5 metres, values far removed from the stated ones for both.

BIO-11-CH03-0034 | Challenging | Alimentary canal, general plan**Correct answer: D. 95 cm**

Solution: Converting to centimetres, the canal is $9 \times 100 = 900$ cm, the small intestine is 600 cm and the large intestine is 150 cm. The four regions together measure $25 + 30 + 600 + 150 = 805$ cm. The excess is $900 - 805 = 95$ cm.

Distractor review: A: 125 cm results from omitting the 30 cm stomach from the combined length before subtracting. B: 120 cm results from omitting the 25 cm oesophagus from the combined length before subtracting. C: 295 cm results from taking the small intestine as 4 metres instead of the stated 6 metres.

BIO-11-CH03-0035 | Challenging | Stomach**Correct answer: A. I, II and IV only**

Solution: The body is the large central portion that stores food, the fundus is the dome shaped region above and to the left of the cardia, and a band of circular muscles at the oesophageal junction prevents regurgitation. The pylorus, however, opens into the duodenum and not into the jejunum, so that statement is faulty.

Distractor review: B: Statement III is faulty because the pylorus opens into the duodenum, the first part of the small intestine, and not into the jejunum. C: This accepts the faulty statement III about the pylorus opening into the jejunum and rejects the correct statement I about the body. D: This rejects the correct statements II and IV about the fundus and the cardiac sphincter while accepting the faulty statement III.

BIO-11-CH03-0036 | Challenging | Dentition**Correct answer: C. 26 teeth and 26 joints**

Solution: The formula gives 3 molars in one half of the lower jaw, so the lower jaw carries $3 \times 2 = 6$ molars. Removing them leaves $32 - 6 = 26$ teeth. Because each tooth is fixed in its own socket by one gomphosis joint, 26 such joints remain.

Distractor review: A: The tooth count is right but the joint count is not reduced; each lost tooth also removes its own socket joint, since there is one gomphosis joint per tooth. B: 20 remains after removing all 12 molars of both jaws; the question removes only the molars of the lower jaw. D: 29 results from removing only 3 molars, that is from treating the formula figure as a whole-jaw count rather than a half-jaw count.

BIO-11-CH03-0037 | Easy | Layers of the gut wall**Correct answer: C. Serosa, muscularis, submucosa, mucosa**

Solution: Moving from the outer surface of the gut towards the lumen, the first layer met is the serosa, then the smooth muscle of the muscularis, then the loose connective tissue of the submucosa, and finally the mucosa, which lines the lumen.

Distractor review: A: This is the same set of layers read from the lumen outwards, so the outermost and innermost layers have been interchanged. B: The two middle layers have been interchanged; the muscle coat lies outside the loose connective tissue of the submucosa, not inside it. D: This places the smooth muscle coat outside the serosa, whereas the serosa is the outermost covering of the canal.

BIO-11-CH03-0038 | Easy | Serosa**Correct answer: D. Mesothelium**

Solution: The squamous epithelium forming the outer face of the serosa is called mesothelium, and connective tissue lies internal to it.

Distractor review: A: The mucosa is the innermost layer lining the lumen and bears goblet cells; it is not the squamous epithelium of the outer covering. B: Brush border is the name given to the microvilli-bearing epithelial cells lining the villi of the small intestine, a part of the mucosa. C: Glisson's capsule is the thin covering of each lobe of the liver, not a layer of the wall of the alimentary canal.

BIO-11-CH03-0039 | Medium | Submucosa and mucosa**Correct answer: A. Submucosa carries the vessels and nerves; mucosa carries the goblet cells**

Solution: Blood vessels, lymph vessels and nerves run in the loose connective tissue of the submucosa, and the goblet cells that secrete mucus lie in the mucosa, which borders the lumen.

Distractor review: B: This reverses the two layers; the goblet cells lie in the layer that borders the lumen, which is the mucosa, not the submucosa. C: The serosa is the outermost covering of mesothelium and connective tissue; the vessels and nerves are described in the submucosa. D: The muscularis is formed of smooth muscle only; mucus-secreting goblet cells belong to the mucosa.

BIO-11-CH03-0040 | Medium | Muscularis**Correct answer: B. The intestine, where the oblique layer is absent**

Solution: The three-layered plan holds generally, and the single stated exception is the intestine, from which the innermost oblique muscle layer is absent.

Distractor review: A: The circular layer is retained throughout; it is the innermost oblique layer that is missing from the intestine. C: The stomach retains the full three-layered arrangement, including the innermost oblique muscles. D: The outermost longitudinal muscles are present in the stomach; no region is described as lacking the longitudinal layer.

BIO-11-CH03-0041 | Medium | Layers of the gut wall**Correct answer: B. Submucosa**

Solution: The layer bearing mucus-secreting cells and bordering the lumen is the mucosa. The layer described as loose connective tissue with blood vessels, lymph vessels and nerves is the submucosa, and it lies immediately outside the mucosa.

Distractor review: A: The smooth muscle coat is separated from the lining layer by the loose connective tissue layer, so it is not the layer immediately outside it. C: The serosa is the outermost covering of the whole wall and is separated from the lining layer by two further layers. D: Mesothelium is the squamous epithelium of the serosa itself, so it lies at the outer surface and not next to the lining layer.

BIO-11-CH03-0042 | Medium | Mucosa of the small intestine**Correct answer: C. Lacteals**

Solution: The lymph vessels running in the villi of the small intestine are named lacteals; the capillary network in the same villus carries blood.

Distractor review: A: Sinusoids are the blood-filled spaces lying between the cords of hepatic cells in a liver lobule, not vessels of the intestinal villi. B: Crypts of Lieberkuhn are the depressions formed by the mucosa between the bases of the villi, not lymph vessels. D: Microvilli are the projections of the epithelial cells that give the villus its brush border; they are cell surface structures, not vessels.

BIO-11-CH03-0043 | Medium | Mucosa of the small intestine**Correct answer: B. Mucosa, and they are called crypts of Lieberkuhn**

Solution: The same lining layer that raises the villi also forms the depressions between their bases, so the crypts of Lieberkuhn are a modification of the mucosa.

Distractor review: A: The crypts are named correctly but assigned to the wrong layer; they are formed by the same lining layer that raises the villi. C: The layer is right but the name is wrong; Brunner's glands are a separate set of glands that assist the goblet cells. D: Both parts are wrong for the crypts between the villus bases, since these depressions are formed by the lining layer of the intestine.

BIO-11-CH03-0044 | Hard | Muscularis**Correct answer: D. Chyme and the digestive juices would remain poorly combined**

Solution: The stated role of the muscularis in the small intestine is mechanical mixing, so abolishing its contractions leaves secretion intact but prevents the juices from being properly mixed with the chyme.

Distractor review: A: Enzyme production by the glands is a secretory activity of the mucosa and submucosa and does not depend on contraction of the muscle coat. B: Mucus is secreted by the goblet cells of the mucosa, which are not the contractile cells of the muscularis. C: Bile is secreted by the liver and delivered through its own duct system, so it is unaffected by the muscle coat of the intestine.

BIO-11-CH03-0045 | Hard | Regional modifications of the mucosa**Correct answer: A. Mucosa, showing rugae in the stomach and villi in the small intestine**

Solution: Both the irregular folds called rugae and the finger like foldings called villi are modifications of the mucosa, the rugae belonging to the stomach and the villi to the small intestine.

Distractor review: B: The layer is right but the two modifications are interchanged; the irregular folds belong to the stomach and the finger like foldings to the small intestine. C: The submucosa is loose connective tissue carrying vessels and nerves; the folds and foldings are raised by the lining layer. D: The muscularis is the smooth muscle coat and is not the layer thrown into rugae or raised into villi.

BIO-11-CH03-0046 | Hard | Regional structure of the gut wall**Correct answer: A. Stomach; mucosa; oblique muscles**

Solution: A lining thrown into irregular folds that forms glands secreting a digestive juice identifies the stomach, those glands are formed by the mucosa, and the innermost of the three muscle layers present there is the oblique layer.

Distractor review: B: The region and the muscle layer are right, but glands raised from the folded lining are formed by the mucosa, not by the connective tissue layer beneath it. C: The circular muscles form the middle of the three layers; the innermost of the three is the oblique layer. D: The small intestine has no oblique muscle layer and its mucosa forms villi rather than irregular folds bearing gastric glands.

BIO-11-CH03-0047 | Easy | Salivary glands**Correct answer: D. Parotid in front of the ear, submandibular below the lower jaw, sublingual below the tongue**

Solution: The parotid glands lie in front of the ear, the submandibular glands lie below the lower jaw and the sublingual glands lie below the tongue, and all three pairs open into the buccal cavity.

Distractor review: A: The positions of the parotid and submandibular glands have been interchanged; the gland lying in front of the ear is the parotid. B: The parotid and sublingual positions have been interchanged; the name sublingual refers to the position below the tongue. C: The submandibular and sublingual positions have been interchanged; the gland below the lower jaw is the submandibular.

BIO-11-CH03-0048 | Easy | Salivary glands**Correct answer: A. Six**

Solution: Three pairs means three multiplied by two, that is six individual salivary glands, all of which open into the buccal cavity.

Distractor review: B: This counts each pair as a single gland and so reports the number of types rather than the number of glands. C: This treats the glands as a single pair and ignores that three different pairs are named. D: This doubles the six glands a second time, as though each named gland were itself paired.

BIO-11-CH03-0049 | Medium | Salivary glands**Correct answer: A. Serous cells produce salivary amylase; mucous cells produce mucus**

Solution: The enzyme-containing fluid of saliva comes from the serous cells and the lubricating mucus comes from the mucous cells.

Distractor review: B: The two cell types have been interchanged; the enzyme-containing fluid comes from the serous cells. C: Lysozyme is listed as a constituent of saliva as a whole and is not assigned to the serous cells, which are named for the amylase. D: The secretion of the mucous cells is the lubricating mucus; lysozyme is not assigned to these cells.

BIO-11-CH03-0050 | Medium | Composition of saliva**Correct answer: D. 49:1**

Solution: Water forms 98 parts and all other constituents together form 2 parts of every 100 parts of saliva, and 98 divided by 2 gives 49, so the ratio is 49 to 1.

Distractor review: A: This uses the percentage of water directly against unity and forgets that the other constituents amount to 2 per cent, not 1 per cent. B: This compares the whole sample of 100 parts with the 2 parts of other constituents instead of comparing water with them. C: This divides the water percentage by 4 instead of by the 2 per cent of other constituents.

BIO-11-CH03-0051 | Easy | Gastric glands**Correct answer: B. Intrinsic factor**

Solution: The constituent of gastric juice named as essential for absorption of vitamin B12 is the intrinsic factor, which is secreted along with hydrochloric acid by the same cells.

Distractor review: A: Pepsinogen is the inactive enzyme of gastric juice that is converted to pepsin and acts on proteins, not on vitamin absorption. C: The mucus of gastric juice protects the inner lining of the stomach from hydrochloric acid; it is not linked to vitamin absorption. D: Hydrochloric acid makes the food acidic, activates pepsinogen and kills germs, but the factor needed for vitamin B12 absorption is a separate constituent.

BIO-11-CH03-0052 | Easy | Gastric glands**Correct answer: C. Mucus cells secrete mucus; peptic cells secrete pepsinogen; parietal cells secrete hydrochloric acid and intrinsic factor**

Solution: Mucus comes from the mucus cells, the inactive pepsinogen from the peptic or chief cells, and hydrochloric acid together with intrinsic factor from the parietal or oxyntic cells.

Distractor review: A: The peptic and parietal cells have been interchanged; the peptic or chief cells are the source of the proenzyme. B: The mucus and peptic cells have been interchanged, although each cell type is named after what it secretes. D: Intrinsic factor and mucus have been exchanged between the cell types; intrinsic factor accompanies the acid from the parietal cells.

BIO-11-CH03-0053 | Medium | Gastric glands**Correct answer: A. It makes the food acidic and thereby stops the action of salivary amylase**

Solution: Salivary amylase continues to act on starch after swallowing until the gastric juice mixes with the food; the hydrochloric acid then makes the food acidic, which stops the amylase.

Distractor review: B: Neutralising the contents and making them alkaline is the action of bile salts in the small intestine, not of gastric hydrochloric acid. C: Hydrochloric acid does activate pepsinogen, but pepsin is described as acting on proteins to give peptones and proteoses, and the stated reason for amylase stopping is the acidity. D: Lubrication of food is the role of mucus, and no separation of enzyme from substrate is described.

BIO-11-CH03-0054 | Hard | Gastric glands**Correct answer: C. Pepsinogen is still secreted but is not converted into pepsin**

Solution: The peptic cells continue to supply pepsinogen, but the hydrochloric acid that converts pepsinogen into pepsin comes from the parietal cells, so with those cells destroyed the proenzyme remains unconverted.

Distractor review: A: Pepsinogen comes from the peptic or chief cells, which are stated to be working normally, so its secretion continues. B: The gastric gland secretes the enzyme only in the inactive form pepsinogen; active pepsin is never secreted as such. D: Mucus protects the stomach lining from acid; the conversion of pepsinogen to pepsin is brought about by hydrochloric acid.

BIO-11-CH03-0055 | Medium | Gastric glands**Correct answer: D. Rennin, in infants**

Solution: In addition to the standard constituents of gastric juice, the stomach also secretes rennin in infants.

Distractor review: A: The enzyme is named correctly but the group is wrong; this secretion is stated for infants. B: Pepsin is formed in the stomach at every age from pepsinogen and is not confined to one group. C: Trypsin belongs to the pancreatic juice acting in the small intestine and is not a gastric secretion at all.

BIO-11-CH03-0056 | Easy | Liver**Correct answer: C. Glisson's capsule**

Solution: The thin covering over each lobe of the bilobed liver is Glisson's capsule.

Distractor review: A: Mesothelium is the squamous epithelium forming the outer part of the serosa of the alimentary canal wall. B: The serosa is the outermost layer of the wall of the alimentary canal, not the named covering of a liver lobe. D: The mucosa is the layer lining the lumen of the alimentary canal and has no part in covering the liver.

BIO-11-CH03-0057 | Medium | Liver**Correct answer: A. About 1.2 to 1.5 kg**

Solution: The mass given for the adult human liver is about 1.2 to 1.5 kg.

Distractor review: B: This range is an order of magnitude too small for the largest gland of the body as described. C: This adds a kilogram to each end of the stated range. D: This is ten times the stated range and would exceed a large part of the body mass.

BIO-11-CH03-0058 | Medium | Histology of the liver**Correct answer: B. Central vein; sinusoids; Kupffer cells**

Solution: The vein at the centre of a hepatic lobule is the central vein, the blood-carrying spaces between the cords of hepatic cells are the sinusoids, and the phagocytic cells within them are the Kupffer cells. The triangular portal area at the junction of adjacent lobules carries a branch of the hepatic artery, of the hepatic portal vein and of the bile duct.

Distractor review: A: A branch of the hepatic portal vein lies in the triangular portal area at the junction of adjacent lobules, not at the centre of a lobule. C: The portal area is the triangular region between adjacent lobules; the blood-filled spaces between the cords have a different name. D: Acinar cells are the secretory cells of the exocrine part of the pancreas and are not the phagocytic cells of the liver.

BIO-11-CH03-0059 | Easy | Intestinal glands**Correct answer: D. Succus entericus**

Solution: The intestinal juice, made of mucus together with the intestinal enzymes, is also known as succus entericus.

Distractor review: A: Chyme is the semifluid acidic mass of partially digested food formed at the end of gastric digestion, not a secretion of glands. B: Chyle is a name used for the intestinal contents after digestion, not for the juice secreted by the intestinal glands. C: Bolus is the mass of food bound together by mucus in the buccal cavity before swallowing.

BIO-11-CH03-0060 | Medium | Intestinal glands**Correct answer: C. Brunner's glands, lying in the submucosa**

Solution: The glands that help in the action of the goblet cells are Brunner's glands, and they lie in the submucosa of the intestinal wall.

Distractor review: A: The glands are named correctly but placed in the wrong layer; they are described as sub-mucosal. B: The crypts of Lieberkuhn are formed by the mucosa between the bases of the villi and are not the glands said to help the goblet cells. D: The layer is right for these crypts, but they are not the glands named as assisting the action of the goblet cells.

BIO-11-CH03-0061 | Hard | Gall bladder and bile ducts**Correct answer: C. Duct of the gall bladder with hepatic duct --> common bile duct --> joined by pancreatic duct --> hepato-pancreatic duct**

Solution: Bile leaves by the hepatic duct, which unites with the duct of the gall bladder to form the common bile duct; this joins the pancreatic duct to form the hepato-pancreatic duct, whose opening into the duodenum is guarded by the sphincter of Oddi.

Distractor review: A: The pancreatic duct and the hepatic duct have exchanged places; the common bile duct is formed before the pancreatic duct joins. B: This reverses the order of formation, making the common bile duct the last duct instead of the one that receives the pancreatic duct. D: The names of the two composite ducts have been interchanged; the union of the gall bladder duct and the hepatic duct gives the common bile duct.

BIO-11-CH03-0062 | Hard | Gall bladder**Correct answer: A. Bile continues to be produced by the liver but is no longer stored before reaching the duodenum**

Solution: Bile is a secretion of the liver, so its production continues after the gall bladder is removed; what is lost is the sac in which the bile was stored before reaching the duodenum.

Distractor review: B: Bile is secreted by the liver; the gall bladder is the sac in which the bile is stored. C: The pancreas secretes pancreatic juice into its own duct and is nowhere described as storing bile. D: The duct system opens into the duodenum, so removal of the storage sac cannot redirect bile into the stomach.

BIO-11-CH03-0063 | Medium | Bile

Correct answer: D. Bile juice, whose salts neutralise the acidity of the chyme and emulsify fats

Solution: Bile juice is the exception, since it is stated to contain no digestive enzyme; its salts make the chyme alkaline and emulsify fats.

Distractor review: A: The juice is identified correctly, but hydrolysis of fats is carried out by lipases, while bile salts only emulsify the fats. B: Pancreatic juice is alkaline but does contain enzymes such as amylases, lipases and nucleases, so it is not the enzyme-free juice. C: Intestinal juice is defined as mucus together with intestinal enzymes, so it also carries enzymes.

BIO-11-CH03-0064 | Medium | Bile

Correct answer: B. Bilirubin and biliverdin are the pigments; Na-glycocholate and Na-taurocholate are the salts

Solution: The pigments of bile are bilirubin and biliverdin, while its salts are Na-glycocholate and Na-taurocholate; cholesterol and phospholipid form a third group of constituents.

Distractor review: A: The two classes have been interchanged; the coloured compounds of bile are bilirubin and biliverdin. C: Cholesterol and phospholipid are listed as separate constituents of bile and are not its salts. D: Cholesterol and phospholipid are colourless constituents listed apart from the pigments of bile.

BIO-11-CH03-0065 | Hard | Pancreas

Correct answer: D. Produced by acinar cells, released into the duodenum through the pancreatic duct, acting on food in the small intestine

Solution: Pancreatic juice is produced by the acinar cells of the exocrine part of the pancreas, is collected and carried to the duodenum by the pancreatic duct, and its enzymes act on the food in the small intestine.

Distractor review: A: The islets of Langerhans form the endocrine part of the pancreas and secrete hormones; the juice is produced by the acinar cells. B: The pancreatic duct carries the juice to the duodenum, and the enzymes of pancreatic juice are described as acting in the small intestine. C: The gall bladder stores bile, not pancreatic juice, which is carried directly by the pancreatic duct.

BIO-11-CH03-0066 | Medium | Pancreas

Correct answer: B. alpha-cells secrete glucagon, beta-cells secrete insulin, delta-cells secrete somatostatin

Solution: In the islets of Langerhans the alpha-cells secrete glucagon, the beta-cells secrete insulin and the delta-cells secrete somatostatin, and glucagon together with insulin controls the blood-sugar level.

Distractor review: A: The hormones of the alpha and beta cells have been interchanged; insulin is the secretion of the beta-cells. C: The hormones of the alpha and delta cells have been interchanged; glucagon belongs to the alpha-cells. D: The hormones of the beta and delta cells have been interchanged; somatostatin is the secretion of the delta-cells.

BIO-11-CH03-0067 | Hard | Gall bladder and bile ducts

Correct answer: D. Bile juice and pancreatic juice

Solution: Bile reaches the duodenum through the common bile duct and pancreatic juice through the pancreatic duct, and both of these unite to form the hepato-pancreatic duct, so a block just before its opening holds back both juices. Intestinal juice is made in the intestinal wall and is unaffected.

Distractor review: A: Intestinal juice is produced in the wall of the intestine itself and does not travel through this duct. B: Intestinal juice does not pass through the hepato-pancreatic duct, while bile does and would also be held back. C: Gastric juice is poured into the stomach by the gastric glands and never enters this duct.

BIO-11-CH03-0068 | Challenging | Intestinal glands

Correct answer: A. Trypsinogen would arrive but would not be converted into trypsin

Solution: The pancreas contributes trypsinogen in the inactive form, and the conversion to trypsin is brought about by enterokinase supplied by the intestinal juice, so without enterokinase the trypsinogen reaches the intestine but remains unconverted.

Distractor review: B: Secretion by the pancreatic acinar cells is stated to continue; only the intestinal activator is missing. C: Bile salts neutralise the acidity of the chyme and emulsify fats; no activating role for them is described. D: The pancreas secretes trypsinogen and chymotrypsinogen as inactive enzymes, not the active forms.

BIO-11-CH03-0069 | Easy | Digestion in the buccal cavity**Correct answer: C. Lysozyme**

Solution: Among the constituents of saliva, lysozyme is the one described as an antibacterial agent that prevents infections, while amylase digests and mucus lubricates.

Distractor review: A: Salivary amylase is the digestive enzyme of saliva that hydrolyses starch; it has no antibacterial role. B: Mucus of saliva lubricates the food and binds the food particles into a bolus rather than acting against bacteria. D: Bicarbonates are listed among the electrolytes of saliva along with sodium, potassium, calcium and chloride, and no antibacterial action is attributed to them.

BIO-11-CH03-0070 | Easy | Digestion in the buccal cavity**Correct answer: A. Maltose**

Solution: Salivary amylase converts starch, a polysaccharide, into maltose, a disaccharide, and this is the only chemical digestion occurring in the buccal cavity.

Distractor review: B: Glucose is obtained from maltose only later, by the maltase of intestinal juice, and not by salivary amylase in the mouth. C: Fructose appears when sucrase splits sucrose in the small intestine; it is not a product of starch hydrolysis. D: Galactose is released when lactase splits lactose in the small intestine and has no connection with starch.

BIO-11-CH03-0071 | Medium | Digestion in the buccal cavity**Correct answer: B. No chemical digestion of food occurs in the buccal cavity**

Solution: Since the only chemical digestion in the mouth is by the action of salivary amylase, saliva lacking that enzyme leaves the buccal cavity with mechanical digestion alone.

Distractor review: A: Binding of food into a bolus is done by the mucus of saliva, which the stem leaves unaltered. C: Action against bacteria is due to lysozyme, which is still present under the conditions stated. D: Mastication is mechanical digestion performed by the teeth and is independent of any salivary enzyme.

BIO-11-CH03-0072 | Medium | Digestion in the stomach**Correct answer: D. Peptic cells produce pepsinogen and parietal cells produce intrinsic factor**

Solution: The peptic cells, also called chief cells, are the source of the proenzyme pepsinogen, while the parietal or oxyntic cells produce both HCl and intrinsic factor.

Distractor review: A: The two secretions are interchanged; the oxyntic or parietal cells are the source of HCl and intrinsic factor, not of the proenzyme. B: Mucus cells produce only mucus; the proenzyme comes from the peptic or chief cells. C: Intrinsic factor is a parietal cell product; mucus cells contribute only the mucus of gastric juice.

BIO-11-CH03-0073 | Easy | Digestion in the stomach**Correct answer: A. Absorption of vitamin B12**

Solution: Intrinsic factor, released along with HCl by the parietal cells, is essential for the absorption of vitamin B12.

Distractor review: B: This conversion is brought about by HCl, a separate constituent of gastric juice. C: Emulsification of fats is brought about by the bile salts of bile juice, which acts in the small intestine. D: The stomach lining is protected by the mucus of gastric juice, not by intrinsic factor.

BIO-11-CH03-0074 | Medium | Digestion in the stomach**Correct answer: B. Pepsinogen remains unconverted to pepsin**

Solution: HCl converts the inactive pepsinogen into active pepsin, so a stomach receiving pepsinogen but no acid retains the proenzyme in its inactive form.

Distractor review: A: Intrinsic factor is a separate parietal cell product and the stem states that its release continues. C: Mucus is produced by the mucus cells of the gastric gland and its release is unaffected by the loss of acid. D: Polypeptides are formed by trypsin in the small intestine; gastric digestion of protein cannot even begin without active pepsin.

BIO-11-CH03-0075 | Easy | Digestion in the stomach**Correct answer: C. Peptones and proteoses**

Solution: Pepsin, the active form derived from pepsinogen, hydrolyses proteins to peptones and proteoses within the stomach.

Distractor review: A: Polypeptides are formed by trypsin and dipeptides by chymotrypsin, both of which act in the small intestine and not in the stomach. B: Dipeptides come from the action of chymotrypsin and amino acids from that of dipeptidase, both belonging to intestinal digestion. D: Ammonia arises from the deamination of excess amino acids in the liver and is not a product of any gastric enzyme.

BIO-11-CH03-0076 | Medium | Digestion in the stomach**Correct answer: B. Rennin, which the stomach secretes in infants**

Solution: Besides the four usual constituents of gastric juice, the stomach secretes rennin in infants, and this is the only age-restricted gastric secretion stated.

Distractor review: A: Pepsin is not secreted as such at any age; the stomach releases the proenzyme pepsinogen, which HCl activates. C: The age group is reversed; rennin is stated to be secreted by the stomach in infants. D: Lysozyme is a constituent of saliva, not of gastric juice, and is not linked to any age group.

BIO-11-CH03-0077 | Medium | Digestion in the stomach**Correct answer: D. II, III, I, IV**

Solution: Churning begins gastric digestion, HCl then converts pepsinogen to pepsin, pepsin hydrolyses proteins to peptones and proteoses, and the semifluid acidic mass called chyme is the end result.

Distractor review: A: Activation of the proenzyme is placed before the food is churned, although churning is the mechanical step that begins gastric digestion. B: Protein hydrolysis is placed before pepsinogen has been activated, but peptones and proteoses can appear only after pepsin exists. C: Chyme is placed before protein hydrolysis, although chyme is the partially digested mass formed at the end of gastric digestion.

BIO-11-CH03-0078 | Medium | Digestion in the small intestine**Correct answer: D. Bile juice**

Solution: Bile juice contains bile pigments, bile salts, cholesterol and phospholipid but no digestive enzyme, and it acts by neutralising the acidity of the chyme and emulsifying fats.

Distractor review: A: Pancreatic juice carries amylases, lipases, nucleases and the proenzymes trypsinogen and chymotrypsinogen. B: Intestinal juice is mucus together with enzymes such as dipeptidases, lipases and disaccharidases. C: Gastric juice carries the proenzyme pepsinogen, which HCl converts into the active enzyme pepsin.

BIO-11-CH03-0079 | Hard | Digestion in the small intestine**Correct answer: B. Trypsinogen and chymotrypsinogen**

Solution: Of the pancreatic enzymes, trypsinogen and chymotrypsinogen are the two named as inactive enzymes, the rest reaching the duodenum ready to act.

Distractor review: A: These two are carried in pancreatic juice in their working form and need no activating agent in the duodenum. C: Nucleases act on nucleic acids and lipases on fat molecules; neither is listed among the inactive enzymes of pancreatic juice. D: Only one member of this pair is a proenzyme; the amylase of pancreatic juice is not released in an inactive form.

BIO-11-CH03-0080 | Hard | Digestion in the small intestine**Correct answer: D. Trypsin converts proteoses and peptones into polypeptides, and chymotrypsin converts polypeptides into dipeptides**

Solution: Trypsin takes proteins, proteoses and peptones down to polypeptides, and chymotrypsin then reduces those polypeptides to dipeptides.

Distractor review: A: The two enzymes are interchanged; trypsin acts on the gastric products and hands polypeptides on to chymotrypsin. B: Peptones and proteoses are gastric products of pepsin, and amino acids are released by dipeptidase, not by chymotrypsin. C: Dipeptides are hydrolysed to amino acids by the dipeptidase of intestinal juice, a step that belongs to neither pancreatic enzyme.

BIO-11-CH03-0081 | Easy | Digestion in the small intestine**Correct answer: D. Mucus together with the intestinal enzymes**

Solution: Mucus and the intestinal enzymes together constitute the intestinal juice or succus entericus, the mucus being produced by the goblet cells of the mucosa.

Distractor review: A: Bile salts belong to bile juice, which is a separate secretion reaching the duodenum through the hepato-pancreatic duct. B: Hydrochloric acid is a constituent of gastric juice and takes no part in the composition of intestinal juice. C: The pancreatic enzymes are carried in pancreatic juice; the enzymes of succus entericus are the intestinal enzymes such as dipeptidases and disaccharidases.

BIO-11-CH03-0082 | Hard | Digestion in the small intestine**Correct answer: D. Trypsinogen arriving in the duodenum stays unconverted to trypsin**

Solution: Trypsinogen is activated to trypsin by the enterokinase of intestinal juice, so intestinal juice lacking enterokinase leaves the pancreatic proenzyme in its inactive form.

Distractor review: A: Pepsinogen is converted to pepsin by HCl within the stomach, so this step does not depend on intestinal juice. B: Pancreatic amylase reaches the duodenum in working form and needs no activating enzyme. C: Emulsification is a property of the bile salts themselves and is independent of any enzyme of intestinal juice.

BIO-11-CH03-0083 | Hard | Digestion in the small intestine**Correct answer: B. Bile juice and pancreatic juice**

Solution: Because the common bile duct and the pancreatic duct join to form the hepato-pancreatic duct, a block in that duct stops both bile juice and pancreatic juice from entering the duodenum, while intestinal juice from the intestinal wall is unaffected.

Distractor review: A: Intestinal juice is released by the mucosa of the intestinal wall itself and does not travel through the hepato-pancreatic duct. C: Only bile of this pair uses the blocked route; the intestinal enzymes and mucus are added directly from the intestinal lining. D: Gastric juice is poured into the stomach by the gastric glands and never passes through the hepato-pancreatic duct.

BIO-11-CH03-0084 | Medium | Digestion in the small intestine**Correct answer: C. Lactose to glucose and galactose**

Solution: Lactase is the intestinal enzyme that splits lactose into glucose and galactose, so its absence blocks only that conversion while maltase, sucrase and dipeptidase continue to work.

Distractor review: A: Maltose is split by maltase, an enzyme the stem leaves present in the intestinal juice. B: Sucrose is split by sucrase, which is unaffected by the absence of lactase. D: Dipeptides are hydrolysed by dipeptidase, an enzyme acting on peptides and not on any disaccharide.

BIO-11-CH03-0085 | Medium | Neurohormonal control of digestion**Correct answer: A. Inhibition of gastric juice secretion, together with stimulation of bile, pancreatic and intestinal juice secretion**

Solution: Secretin inhibits the secretion of gastric juice while stimulating the secretion of bile juice from the liver, of pancreatic juice and of intestinal juice.

Distractor review: B: Both halves are reversed; secretin holds back the gastric secretion and promotes the three secretions that act beyond the stomach. C: The stimulating action of secretin is not confined to bile; pancreatic juice and intestinal juice are stimulated as well. D: The feeling of fullness is the action attributed to cholecystokinin, not to secretin.

BIO-11-CH03-0086 | Challenging | Neurohormonal control of digestion**Correct answer: A. Gastrin stimulates the secretion of gastric juice, while gastric inhibiting peptide inhibits it**

Solution: Gastrin brings about the same effect as the tenth cranial nerve and stimulates gastric juice secretion, whereas gastric inhibiting peptide inhibits gastric secretion, as do secretin and cholecystokinin.

Distractor review: B: Secretin inhibits rather than stimulates gastric secretion, and the only stated action of gastric inhibiting peptide is inhibition of gastric secretion. C: Cholecystokinin brings about an action similar to that of secretin, so it holds back gastric secretion, and it induces satiety instead of suppressing it. D: The tenth cranial nerve does stimulate gastric secretion, but gastrin brings about the same effect rather than the opposite one.

BIO-11-CH03-0087 | Easy | Sites of absorption**Correct answer: C. Small intestine**

Solution: About 90 per cent of absorption takes place in the small intestine, the mouth, stomach and large intestine together accounting for the rest.

Distractor review: A: The gastric mucosa is impermeable to most substances, so only a small share of absorption occurs there. B: The large intestine absorbs water, electrolytes, drugs and some vitamins, but this is part of the remaining share only. D: Absorption through the mucosa of the mouth and the lower side of the tongue is a minor contribution compared with the intestinal villi.

BIO-11-CH03-0088 | Medium | Sites of absorption

Correct answer: A. Glucose, amino acids and water soluble vitamins into the blood capillaries; lipids and fat soluble vitamins into the lacteals

Solution: Monosaccharides, amino acids, minerals and water soluble vitamins are absorbed into the blood capillaries of the villi, whereas lipids and the fat soluble vitamins A, D, E and K are absorbed into the lacteals.

Distractor review: B: The two routes are interchanged; the lacteals take up the fatty material, not the sugars and amino acids. C: Amino acids follow the same route as the monosaccharides, into the blood capillaries of the villi. D: The fat soluble vitamins accompany the lipids into the lacteals, and amino acids enter the blood capillaries.

BIO-11-CH03-0089 | Hard | Sites of absorption

Correct answer: D. A little water, some electrolytes, alcohol and certain drugs like aspirin

Solution: In the stomach only a little water, some electrolytes, alcohol and certain drugs like aspirin are absorbed, since the gastric mucosa is impermeable to most substances.

Distractor review: A: These are the end products absorbed in the blood capillaries of the intestinal villi, not through the gastric mucosa. B: This is the material taken up by the lacteals of the intestinal villi. C: This combination describes absorption in the large intestine rather than in the stomach.

BIO-11-CH03-0090 | Hard | Mechanisms of absorption

Correct answer: B. I-R, II-S, III-P, IV-Q

Solution: Part of the glucose and chloride ions are absorbed by simple diffusion along the gradient, fructose by facilitated transport with carrier ions like Na⁺, sodium by active transport against the gradient with expenditure of energy, and water by osmosis along the gradient.

Distractor review: A: The carrier-assisted uptake of fructose is placed under simple diffusion, although simple diffusion needs no carrier. C: Water movement along the gradient is labelled active transport, although active transport is defined by movement against the gradient with expenditure of energy. D: Water is assigned to simple diffusion and the gradient-dependent uptake of glucose and chloride to osmosis, reversing the two.

BIO-11-CH03-0091 | Medium | Absorption of fats

Correct answer: C. Micelles are spherical aggregates formed by bile salts in which monoglycerides and fatty acids dissolve, and chylomicrons are small protein coated fat globules

Solution: Micelles are the spherical aggregates formed by bile salts in which the monoglycerides and fatty acids dissolve, while chylomicrons are small protein coated fat globules into which the fat is reformed.

Distractor review: A: The two names are interchanged; the aggregates put together by bile salts are the micelles. B: Micelles are formed by bile salts and not by the bile pigments, and the coat of a chylomicron is protein. D: Both descriptions are misassigned; the aggregates of bile salts are micelles, and a chylomicron carries a protein coat.

BIO-11-CH03-0092 | Easy | Assimilation

Correct answer: B. Assimilation

Solution: Assimilation is the process in which the absorbed food material reaches the tissue and becomes a part of the protoplasm.

Distractor review: A: Absorption is the earlier step, the passage of the end products of digestion through the mucosal lining into blood and lymph. C: Egestion is the removal of the undigested waste as faeces and involves no incorporation into tissue. D: Digestion is the conversion of complex non-diffusible food substances into simple diffusible and assimilable ones.

BIO-11-CH03-0093 | Medium | Egestion

Correct answer: A. Water, inorganic salts, sloughed mucosal cells, bacteria and undigested food

Solution: Faeces contain water, inorganic salts, sloughed mucosal cells, bacteria and undigested food, and are formed in the colon before being stored temporarily in the rectum.

Distractor review: B: Bile pigments are named as constituents of bile juice and are not listed among the contents of faeces. C: No digestive enzyme is listed in the faeces, and the mucosa of the large intestine produces mucus but no enzymes. D: Vitamins that have been absorbed have already left the canal, so they cannot form part of the material egested.

BIO-11-CH03-0094 | Challenging | Mechanisms of absorption**Correct answer: D. Fructose is absorbed by facilitated transport with the help of carrier ions like Na⁺**

Solution: Fructose, along with some amino acids, is absorbed by facilitated transport, a method in which carrier ions like Na⁺ bring about the absorption.

Distractor review: A: Movement against the gradient with expenditure of energy defines active transport, whereas water is absorbed by osmosis along the gradient. B: Sodium is the mineral named as an example of active transport, absorbed against the gradient with expenditure of energy. C: Chloride ions are named among the electrolytes absorbed by simple diffusion depending on the concentration gradient.

BIO-11-CH03-0095 | Easy | Protein energy malnutrition**Correct answer: C. Protruding belly, slender legs, bulging eye and oedema of the lower legs and face**

Solution: Kwashiorkor shows a protruding belly, slender legs, bulging eye and oedema of the lower legs and face, along with stunted growth, poor brain development, loss of appetite and anaemia.

Distractor review: A: These are the stated symptoms of jaundice, a sign of abnormal bilirubin metabolism and excretion. B: This describes diarrhoea, a disorder of the digestive system and not a protein deficiency disorder. D: These are the symptoms listed for indigestion, which follows overeating, spicy food or anxiety.

BIO-11-CH03-0096 | Medium | Protein energy malnutrition**Correct answer: B. Marasmus is found in infants under one year of age and shows no oedema**

Solution: Marasmus occurs in infants under one year of age and shows no oedema, whereas kwashiorkor is found generally in children between one and three years of age and does show oedema of the lower legs and face.

Distractor review: A: Both halves belong to kwashiorkor; marasmus affects a younger age group and lacks oedema altogether. C: The age group is right but the swelling is not; oedema is absent in marasmus. D: The absence of oedema is right but the age group is not; the one to three year range belongs to kwashiorkor.

BIO-11-CH03-0097 | Medium | Disorders of the digestive system**Correct answer: B. Overeating, inadequate enzyme secretion, spicy food and anxiety**

Solution: Overeating, inadequate enzyme secretion, spicy food and anxiety are the causes given for indigestion, which then produces discomfort with acidity, heart burn and upper abdominal pain.

Distractor review: A: This is given as the major cause of the protein energy malnutrition disorders kwashiorkor and marasmus. C: This is the defining feature of constipation, and is a description of a condition rather than a cause of indigestion. D: Reverse peristaltic movements of the gastric wall are the mechanism of vomiting.

BIO-11-CH03-0098 | Hard | Disorders of the digestive system**Correct answer: A. I-R, II-P, III-S, IV-Q**

Solution: Constipation is defined by a defaecation frequency of less than once per week, diarrhoea by loose watery stools more than three times a day, vomiting by expulsion of stomach contents through reverse peristaltic movements, and jaundice by yellowness of the conjunctiva and skin with whitish stool.

Distractor review: B: The two stool disorders are interchanged; reduced frequency defines constipation and the frequent watery stool defines diarrhoea. C: The reverse peristaltic mechanism is assigned to jaundice and the yellowing of the eyes to vomiting, exchanging the two. D: Constipation is given the mechanism of vomiting, although constipation is defined by the frequency of defaecation.

BIO-11-CH03-0099 | Medium | Calorific value of food**Correct answer: C. 165 Kcal**

Solution: Using the physiological values, the carbohydrate gives $20 \times 4.0 = 80$ Kcal, the protein gives $10 \times 4.0 = 40$ Kcal and the fat gives $5 \times 9.0 = 45$ Kcal, so the meal produces $80 + 40 + 45 = 165$ Kcal.

Distractor review: A: The gross calorific value 4.1 Kcal/g has been used for the carbohydrate instead of its physiological value 4.0 Kcal/g. B: The gross calorific value 5.65 Kcal/g has been used for the protein although the question asks for the energy actually produced. D: The gross calorific values 4.1, 5.65 and 9.45 Kcal/g have been used throughout, which gives the heat of combustion and not the energy actually produced.

BIO-11-CH03-0100 | Challenging | Calorific value of food**Correct answer: A. 42.5 Kcal**

Solution: The per gram differences are $4.1 - 4.0 = 0.1$ Kcal/g for carbohydrate, $5.65 - 4.0 = 1.65$ Kcal/g for protein and $9.45 - 9.0 = 0.45$ Kcal/g for fat. The excess is therefore $50 \times 0.1 = 5$ Kcal plus $20 \times 1.65 = 33$ Kcal plus $10 \times 0.45 = 4.5$ Kcal, giving 42.5 Kcal.

Distractor review: B: The difference of 0.45 Kcal/g for the 10 g of fat has been left out of the total. C: The difference of 0.1 Kcal/g for the 50 g of carbohydrate has been dismissed as negligible and omitted. D: Only the protein difference has been counted, on the assumption that carbohydrate and fat show no gap between the two values.

BIO-11-CH03-0101 | Easy | Stomach regions**Correct answer: A. Pylorus (region Y)**

Solution: The narrow terminal region that opens into the duodenum is the pylorus.

Distractor review: B: Fundus is a plausible distractor, but it conflicts with the required identifying the pyloric region on a stomach diagram and therefore cannot be the single correct response. C: Cardia is a plausible distractor, but it conflicts with the required identifying the pyloric region on a stomach diagram and therefore cannot be the single correct response. D: Body is a plausible distractor, but it conflicts with the required identifying the pyloric region on a stomach diagram and therefore cannot be the single correct response.

BIO-11-CH03-0102 | Medium | Liver histology**Correct answer: B. Sinusoid**

Solution: The spaces between the radiating cords of hepatic cells are sinusoids through which blood flows.

Distractor review: A: Bile canaliculus is a plausible distractor, but it conflicts with the required identifying a sinusoid in a liver lobule diagram and therefore cannot be the single correct response. C: Central vein is a plausible distractor, but it conflicts with the required identifying a sinusoid in a liver lobule diagram and therefore cannot be the single correct response. D: Hepatic duct is a plausible distractor, but it conflicts with the required identifying a sinusoid in a liver lobule diagram and therefore cannot be the single correct response.

BIO-11-CH03-0103 | Hard | Wall layers**Correct answer: C. Submucosa**

Solution: The loose connective-tissue layer containing blood vessels, lymph vessels and nerves is the submucosa.

Distractor review: A: Serosa is a plausible distractor, but it conflicts with the required using location and contents to identify the submucosa and therefore cannot be the single correct response. B: Muscularis is a plausible distractor, but it conflicts with the required using location and contents to identify the submucosa and therefore cannot be the single correct response. D: Mucosal epithelium is a plausible distractor, but it conflicts with the required using location and contents to identify the submucosa and therefore cannot be the single correct response.

BIO-11-CH03-0104 | Challenging | Calorific values**Correct answer: D. 54 Kcal**

Solution: The excess is $60(0.1) + 25(1.65) + 15(0.45) = 6 + 41.25 + 6.75 = 54$ Kcal.

Distractor review: A: 40 Kcal is a plausible distractor, but it conflicts with the required comparing gross and physiological energy for a mixed meal and therefore cannot be the single correct response. B: 46.5 Kcal is a plausible distractor, but it conflicts with the required comparing gross and physiological energy for a mixed meal and therefore cannot be the single correct response. C: 50 Kcal is a plausible distractor, but it conflicts with the required comparing gross and physiological energy for a mixed meal and therefore cannot be the single correct response.

Standard 11 | Chapter 04

Excretion and Osmoregulation

BIO-11-CH04-0001 | Easy | Excretion, secretion and egestion

Correct answer: C. Elimination of metabolic waste products from the body

Solution: Excretion is the elimination of metabolic waste products from the body. The wastes concerned are formed inside body cells, which separates excretion from the removal of digestive wastes and from the release of useful secretions.

Distractor review: A: This describes secretion, in which the substance released is put to use by the body; the student treats every outward movement of a substance as excretion. B: This describes digestive waste, which is made of substances that have never entered the body cells, so it is not a metabolic waste product. D: This describes the catabolic side of metabolism, which produces the wastes; it is not their elimination.

BIO-11-CH04-0002 | Easy | Nitrogenous waste products

Correct answer: C. The amino group is separated and ammonia is formed

Solution: Excess amino acids cannot be stored, so they are deaminated. The amino group is separated from the amino acid and ammonia is formed; this ammonia is then either excreted as such or converted into urea or uric acid.

Distractor review: A: Loss of a carboxyl group is decarboxylation; deamination is named for the removal of the amino group, which is the nitrogen bearing part. B: Urea is formed only later, when the ammonia released by deamination is converted in the liver; it is not the immediate product of deamination. D: Uric acid is a downstream conversion product of ammonia, and the group removed in deamination is the amino group, not the carboxyl group.

BIO-11-CH04-0003 | Easy | Relative toxicity of nitrogenous wastes

Correct answer: B. Uric acid

Solution: Uric acid is the least toxic nitrogenous waste, which is why uricotelic animals can retain it in the body for some time in a concentrated form before eliminating it.

Distractor review: A: Ammonia is the most toxic of the three; it is basic and even a slight rise in pH would disturb enzyme catalyzed reactions, so it must be flushed out at once. C: Urea is less toxic than ammonia and can be concentrated to some extent, but it is not the least toxic of the three wastes. D: Guanine is the waste of guanotelic animals such as spiders, scorpions and penguins; its toxicity is not placed in the three way ranking of ammonia, urea and uric acid.

BIO-11-CH04-0004 | Easy | Ureotelism

Correct answer: D. In the liver, by the ornithine cycle

Solution: Ammonia formed by deamination is carried to the liver, where the ornithine cycle, also called the urea cycle, converts it into urea. The kidney then removes the urea from the blood.

Distractor review: A: The kidney eliminates the urea that reaches it in the blood; the conversion of ammonia into urea takes place before that, in the liver. B: The organ is right but the pathway is not; the inosinic acid pathway yields uric acid, not urea. C: Both the organ and the pathway are misplaced; the kidney only excretes the waste, and the inosinic acid pathway forms uric acid.

BIO-11-CH04-0005 | Easy | Excretory organs of animal groups

Correct answer: C. Metanephridia

Solution: Earthworms excrete through metanephridia, unbranched coiled tubes that connect with the body cavity through funnel like nephrostomes and discharge through the nephridiopore.

Distractor review: A: Protonephridia, the dead end tubes ending in flame cells, belong to platyhelminthes, rotifers, some annelids and Amphioxus, not to the earthworm. B: Malpighian tubules are the blind ended tubules of insects; the student assigns an arthropod organ to an annelid. D: Green glands are the excretory organs of crustaceans; the student transfers a crustacean organ to the earthworm.

BIO-11-CH04-0006 | Easy | Guanotelism

Correct answer: D. Spiders, scorpions and penguins

Solution: Spiders, scorpions and penguins excrete guanine, and this mode of excretion is called guanotelism.

Distractor review: A: Land snails are uricotelic and the albatross is named for its salt glands, not for guanine; only the spider in this set is guanotelic. B: Sharks and rays are cartilaginous fishes and are ureotelic; grouping them with the scorpion confuses guanotelism with ureotelism. C: Camel and kangaroo rat are quoted as mammals that economise water while excreting urea, so they are ureotelic, not guanotelic.

BIO-11-CH04-0007 | Easy | Osmoregulation

Correct answer: A. The process of controlling solute concentrations and water balance in the body

Solution: Osmoregulation is the control of solute concentrations and of water balance. It is why the composition of blood is said to be determined not by what the mouth ingests but by what the excretory organs retain.

Distractor review: B: This is excretion; osmoregulation is concerned with the concentration of solutes and with water balance, not only with nitrogenous waste. C: This is deamination, a metabolic step that generates a waste; it does not describe the regulation of body fluid composition. D: This is storage of surplus nutrients, which the body can do; it has nothing to do with regulating solute concentration and water balance.

BIO-11-CH04-0008 | Easy | Nitrogenous waste products

Correct answer: B. Urea

Solution: The molecule $\text{H}_2\text{N}-\text{CO}-\text{NH}_2$ has one carbon flanked by two amino groups and is urea, the waste of ureotelic animals.

Distractor review: A: Uric acid is represented as $\text{C}_5\text{H}_4\text{O}_3\text{N}_4$, a much larger ring containing molecule; the student ignores the number of nitrogen and carbon atoms shown. C: Ammonia carries a single nitrogen and no carbon, so it cannot be the molecule shown, which has one carbon between two amino groups. D: Creatinine is a nitrogenous waste produced from creatinine phosphate of skeletal muscle, and no such formula is given for it.

BIO-11-CH04-0009 | Easy | Pigment wastes

Correct answer: D. Haemoglobin; bilirubin through the faeces and urochrome through the urine

Solution: Both pigments come from the breakdown of haemoglobin. Bilirubin is excreted through the faeces while urochrome is eliminated through the urine.

Distractor review: A: The source is right but the routes are interchanged; it is urochrome that leaves in the urine and bilirubin that leaves in the faeces. B: Creatinine phosphate catabolism yields creatinine in skeletal muscle; it does not yield these two pigments. C: Breakdown of excess amino acids yields ammonia, and both the source and the two routes are misassigned here.

BIO-11-CH04-0010 | Medium | Relative properties of nitrogenous wastes

Correct answer: B. Ammonia, urea, uric acid

Solution: Ammonia is readily soluble in water. Urea is less water soluble than ammonia, so it can be concentrated to some extent, and uric acid is the least soluble of the three.

Distractor review: A: This exactly reverses the order; uric acid is the least soluble of the three and ammonia is readily soluble in water. C: Urea is placed at the top, but urea is stated to be less soluble in water than ammonia, which is why it can be concentrated to some extent. D: This puts the least soluble waste first and separates ammonia from urea wrongly; the student ranks by molecular size instead of by the stated solubility.

BIO-11-CH04-0011 | Medium | Energy cost of excretory modes

Correct answer: A. Excretion of ammonia is an energy saving mechanism, while animals excreting uric acid have to spend more energy

Solution: Ammonia is excreted without any conversion, so ammonotelism is described as an energy saving mechanism. Uricotelic animals must convert ammonia into uric acid and therefore have to spend more energy.

Distractor review: B: The two modes are interchanged; ammonia needs no conversion and so saves energy, whereas the manufacture of uric acid is expensive. C: Urea formation does cost energy, but so does uric acid formation; treating uricotelism as cheap ignores the extra energy those animals must spend. D: Ammonia is excreted as it is formed, without conversion, so ammonotelism is the energy saving mode rather than a costly one.

BIO-11-CH04-0012 | Medium | Habitat and excretory product

Correct answer: D. Ammonia, because abundant water can dilute it and so reduce its toxicity

Solution: Bony fishes are ammonotelic. Where water is freely available, ammonia can be diluted enough to reduce its toxicity, and excreting it saves the animal the energy of conversion.

Distractor review: A: The conversion of ammonia into urea is a device for saving water; an animal surrounded by water gains nothing from paying that energy cost. B: Uric acid is the least soluble waste and is the choice of animals that must conserve water, not of animals surrounded by it. C: Guanine is the waste of spiders, scorpions and penguins; bony fishes are not placed in the guanotelic group.

BIO-11-CH04-0013 | Medium | Habitat and excretory product

Correct answer: C. Uric acid, because it is the least toxic waste and can be retained in a concentrated form with very little water

Solution: Land snails are uricotelic. Uric acid is the least toxic waste, so it can be held in the body in concentrated form and eliminated with a minimum of water, which suits a dry habitat.

Distractor review: A: Ready solubility does not reduce the water bill; ammonia is so toxic that about 300 to 500 ml of water is needed for each gram eliminated. B: Urea is less toxic than ammonia but not the least toxic, and its elimination still needs about 50 ml of water per gram. D: Guanine is the waste of spiders, scorpions and penguins, and land snails are placed among the uricotelic animals.

BIO-11-CH04-0014 | Medium | Energy cost of urea formation

Correct answer: B. 12 ATP molecules

Solution: Each molecule of urea costs 3 ATP, so four molecules cost 3 multiplied by 4, that is 12 ATP molecules.

Distractor review: A: This is the cost of a single molecule of urea; the student quotes the unit value without multiplying by the number of molecules. C: This comes from adding 3 and 4 instead of multiplying them, treating the number of molecules as though it were an extra energy cost. D: This multiplies 3 by 3, so one of the four urea molecules is left out of the calculation.

BIO-11-CH04-0015 | Medium | Water requirement of nitrogenous wastes

Correct answer: B. 300 ml

Solution: At about 50 ml of water per gram of nitrogen, removal of 6 g in the form of urea needs $50 \times 6 = 300$ ml of water.

Distractor review: A: This is the requirement for a single gram; the student quotes the unit value and never multiplies by the given mass. C: This corresponds to 5 gm rather than 6 gm, so one gram of the given mass is dropped from the calculation. D: This doubles the correct answer, as though 100 ml were needed per gram; the stated unit requirement for urea has been misread.

BIO-11-CH04-0016 | Medium | Uricotelism

Correct answer: B. Birds

Solution: Uric acid is produced from ammonia by the inosinic acid pathway, and this pathway is placed in the liver of birds.

Distractor review: A: Land snails are uricotelic, but the inosinic acid pathway of the liver is described for birds; the student assumes one named pathway covers every uricotelic animal. C: Many reptiles are indeed uricotelic, yet the liver pathway named is that of birds; membership of the same excretory class does not imply the same named pathway. D: Some insects are uricotelic and excrete through Malpighian tubules, but the inosinic acid pathway of the liver is not attributed to them.

BIO-11-CH04-0017 | Medium | Excretory organs of animal groups

Correct answer: A. P-III, Q-I, R-IV, S-II

Solution: Insects excrete through Malpighian tubules, crustaceans through green glands, platyhelminthes through protonephridia bearing flame cells, and unicellular forms through contractile vacuoles.

Distractor review: B: The organs of insects and crustaceans are interchanged; Malpighian tubules are the insect organ and green glands the crustacean one. C: Platyhelminthes and unicellular forms are interchanged; contractile vacuoles belong to unicellular forms and protonephridia to platyhelminthes. D: Every pairing is displaced, so protonephridia are given to insects and Malpighian tubules to platyhelminthes.

BIO-11-CH04-0018 | Medium | Ammonotelism

Correct answer: D. It is basic in nature, and a slight rise in pH would disturb all enzyme catalyzed reactions and make the plasma membrane unstable

Solution: Ammonia is basic. If it were retained it would raise the pH of the body, and even a slight rise would disturb all enzyme catalyzed reactions and make the plasma membrane unstable.

Distractor review: A: The consequences are correctly stated but the chemical nature is inverted; ammonia is basic, so the danger is a rise in pH, not a fall. B: Deposition in the joints is associated with the accumulation of uric acid, not with ammonia; the student transfers the effect of one waste to another. C: Ammonia is in fact readily soluble in water; it is the toxicity of ammonia rather than any insolubility that forces its immediate removal.

BIO-11-CH04-0019 | Medium | Creatinine

Correct answer: C. Catabolism of creatinine phosphate during skeletal muscle contraction

Solution: Creatinine phosphate provides a ready source of high energy phosphate in skeletal muscle. Its catabolism during muscle contraction produces the plasma creatinine that the kidney excretes.

Distractor review: A: Deamination of amino acids yields ammonia, which is then converted into urea; it is not the source of plasma creatinine. B: The breakdown of haemoglobin yields the pigments bilirubin and urobilinogen, not creatinine. D: The ornithine cycle produces urea; creatinine arises independently of it, in contracting skeletal muscle.

BIO-11-CH04-0020 | Medium | Osmoregulation in marine animals**Correct answer: C. Near the nostrils, by active transport**

Solution: The salt glands of marine birds such as the albatross lie near the nostrils and secrete salts by active transport, so that osmotic balance is maintained. Sea turtles and marine iguanas have comparable salt excreting glands.

Distractor review: A: The position is right but the process is not; salts are moved against their concentration difference, which requires active transport. B: The process is right but the site is not; the salt glands of marine birds are placed near the nostrils. D: Both the site and the process are misplaced; the cloaca is not the site of these glands and passive diffusion would not remove excess salt.

BIO-11-CH04-0021 | Medium | Excretion, secretion and egestion**Correct answer: A. Food residue that has passed the length of the alimentary canal without crossing the gut wall**

Solution: Undigested food residues never enter the body cells, so their removal is not excretion. Bilirubin also leaves in the faeces, but it is formed within the body and is therefore a true excretory product.

Distractor review: B: Bilirubin is formed inside the body from the breakdown of haemoglobin, so its removal is excretion even though it leaves by the same route as digestive waste. C: Urochrome is a pigment produced within the body from haemoglobin breakdown, which makes its elimination a genuine excretory act. D: These substances are absorbed into the body and then eliminated through the lungs, so they are listed among the excretory products.

BIO-11-CH04-0022 | Medium | Osmoregulation**Correct answer: D. They are euryhaline, whereas stenohaline organisms handle only a narrow range of salt concentrations**

Solution: Barnacles and clams are euryhaline, that is, capable of handling wide changes in salinity. Stenohaline organisms, in contrast, can handle only a narrow range of salt concentrations.

Distractor review: A: The two terms are interchanged; the prefix steno means narrow, so it is the stenohaline organisms that are restricted. B: The first term is right, but stenohaline organisms are the restricted type and cannot exceed the tolerance of euryhaline forms. C: Both halves are inverted; euryhaline organisms are precisely those that tolerate wide changes in salinity.

BIO-11-CH04-0023 | Medium | Ammonotelism**Correct answer: A. General body surface, gills and kidneys**

Solution: Ammonia leaves ammonotelic animals through the general body surface or skin, through the gills and through the kidneys.

Distractor review: B: The liver manufactures wastes rather than releasing them outward, and lungs are not the route named for ammonia in ammonotelic animals. C: Malpighian tubules belong to insects, which are uricotelic; they are not a route of ammonia loss in ammonotelic animals. D: Green glands are crustacean organs listed under excretory organs of groups, and lungs are not named as a route for ammonia.

BIO-11-CH04-0024 | Medium | Water requirement of nitrogenous wastes**Correct answer: D. 20 ml to 40 ml**

Solution: Both limits must be multiplied by the mass. Four grams of uric acid therefore need between 5 multiplied by 4 and 10 multiplied by 4, that is between 20 ml and 40 ml of water.

Distractor review: A: This is the requirement for a single gram; the student quotes the unit range without scaling it to the given mass. B: This scales the unit range by 8 instead of 4, doubling both limits of the correct answer. C: This adds 4 to each limit instead of multiplying, treating the mass as an increment rather than a multiplier.

BIO-11-CH04-0025 | Hard | Creatinine**Correct answer: A. 0.52 mg/dl above the upper limit, indicating poor kidney function**

Solution: The upper limit for males is 1.4 mg/dl, so the excess is 1.92 minus 1.40, that is 0.52 mg/dl. Because creatinine is normally produced and excreted at matching rates, a level above the normal range indicates poor kidney function.

Distractor review: B: This subtracts 1.2 mg/dl, the upper limit of the normal range quoted for females, from a male patient's value. C: This subtracts 0.6 mg/dl, the lower limit of the normal range, instead of the upper limit. D: The arithmetic is right but the interpretation is inverted; a creatinine level above the normal range indicates poor renal function.

BIO-11-CH04-0026 | Hard | Water requirement of nitrogenous wastes**Correct answer: B. Uric acid, about 5 to 10 ml; urea, about 50 ml; ammonia, about 300 to 500 ml**

Solution: Uric acid needs the least water, about 5 to 10 ml per gram or none at all; urea needs about 50 ml per gram; ammonia, the most toxic, needs about 300 to 500 ml per gram to dilute it.

Distractor review: A: The two extremes are interchanged; it is ammonia that needs the largest volume and uric acid the smallest. C: Urea and uric acid are interchanged; about 50 ml per gram is the figure for urea and 5 to 10 ml the figure for uric acid. D: Ammonia and urea are interchanged; ammonia is the waste that must be heavily diluted and therefore takes the largest volume.

BIO-11-CH04-0027 | Hard | Relative properties of nitrogenous wastes**Correct answer: C. It is the least toxic waste and can be retained in concentrated form, so about 5 to 10 ml of water per gram suffices**

Solution: It is toxicity, not solubility, that sets the water bill. Uric acid is the least toxic waste, so it can be held in the body in concentrated form and eliminated with only about 5 to 10 ml of water per gram, or with no water at all.

Distractor review: A: This repeats the very ranking that is being questioned; ammonia, not uric acid, is the most toxic waste and the one needing 300 to 500 ml per gram. B: About 50 ml per gram is the figure quoted for urea; the student transfers the water cost of ureotelism to uricotelism. D: Uric acid is stated to be the least soluble of the three wastes, so it cannot dissolve as readily as ammonia anywhere in the body.

BIO-11-CH04-0028 | Hard | Comparative water economy and habitat**Correct answer: C. I and II only (excretion statements)**

Solution: Removing nitrogen as urea saves water, about 100 times less in human beings and several hundred times less in camel, kangaroo rat and shark. However, the major excretory product shows no clear correlation with phylogenetic relationship; it is the habitat that correlates with it.

Distractor review: A: Statement III inverts the stated correlation; there is no clear correlation between phylogenetic relationship and the major excretory product, but habitat does correlate with it. B: Statement II is correct but statement III reverses habitat and phylogeny, so this combination cannot stand while statement I is left out. D: Statements I and II are correct, but including statement III accepts a correlation with phylogeny that is explicitly denied.

BIO-11-CH04-0029 | Hard | Nitrogenous waste products**Correct answer: C. The surplus amino acids are deaminated, and the ammonia formed is either excreted as such or converted into urea or uric acid**

Solution: Carbohydrate and fat can be stored, but surplus amino acids cannot. They are therefore deaminated, the amino group being separated to yield ammonia, and that ammonia is either excreted directly or converted into urea or uric acid.

Distractor review: A: This is the very storage that the body cannot perform for amino acids, although it can store surplus carbohydrate and fat. B: Excretion of the intact amino acid is not the route described; the amino group is first separated by deamination and appears as ammonia. D: The nitrogen of the amino group cannot be absorbed into a fat store; it must leave the body as ammonia, urea or uric acid.

BIO-11-CH04-0030 | Hard | Excretory organs of animal groups**Correct answer: B. Metanephridia are unbranched coiled tubes connected to the body cavity through funnel like nephrostomes, whereas protonephridia are networks of dead end tubes called flame cells**

Solution: A protonephridium is a network of dead end tubes called flame cells. A metanephridium is an unbranched coiled tube that connects with the body cavity through funnel like nephrostomes and discharges through the nephridiopore.

Distractor review: A: The two descriptions are interchanged; it is the protonephridium that ends blindly in flame cells and the metanephridium that opens into the body cavity. C: Blind ended tubules opening into the gut describes the Malpighian tubules of insects, and the nephridiopore is the outlet of the metanephridium, not of the protonephridium. D: Green glands are separate excretory organs of crustaceans and are not a part of any nephridium, and the nephrostome belongs to the metanephridium.

BIO-11-CH04-0031 | Hard | Excretion, secretion and egestion**Correct answer: A. Bilirubin leaving in the faeces is excretion, while undigested residue leaving in the faeces is egestion**

Solution: Bilirubin is formed within the body from the breakdown of haemoglobin, so its removal is excretion even though it leaves in the faeces. Undigested residue has never entered the body cells, so its removal is egestion.

Distractor review: B: Bilirubin is a waste of haemoglobin breakdown and is not put to use by the body, and undigested residue never entered the cells at all. C: The two are exactly interchanged; bilirubin is formed inside the body while the residue has never entered any cell. D: This classifies by route rather than by origin, which is the distinction the definitions actually rest on.

BIO-11-CH04-0032 | Hard | Osmoregulation in marine animals

Correct answer: D. It makes their blood isotonic to the surrounding sea water and so prevents the loss of water by exosmosis

Solution: By retaining urea, sharks keep their body fluid isotonic to the surrounding marine water. Since there is then no osmotic difference across the body surface, water is not lost by exosmosis.

Distractor review: A: A body fluid more dilute than sea water would lose water to the surroundings, which is the very outcome the retained urea prevents. B: The retained urea is described as matching the blood to sea water, not as exceeding it, and salt removal is not the stated purpose. C: Sharks are marine, so the reference fluid is sea water; matching the blood to fresh water would not solve their osmotic problem.

BIO-11-CH04-0033 | Challenging | Habitat and excretory product

Correct answer: A. The land snail

Solution: Land snails are uricotelic, and uric acid needs only about 5 to 10 ml of water per gram, or none at all. The shark and the camel are ureotelic, needing about 50 ml per gram, and the bony fish is ammonotelic, needing about 300 to 500 ml per gram.

Distractor review: B: The shark is ureotelic, so about 50 ml of water per gram is needed; it is economical compared with an ammonotelic animal but not the most economical here. C: Bony fishes are ammonotelic, and ammonia is the costliest waste in water terms, needing about 300 to 500 ml per gram. D: The camel is quoted as a ureotelic animal that saves a great deal of water, but urea still needs about 50 ml per gram against the far smaller need for uric acid.

BIO-11-CH04-0034 | Challenging | Energy cost of excretory modes

Correct answer: D. P spends more energy in producing its waste but needs less water to eliminate it than Q

Solution: Uricotelic animals have to spend more energy to make uric acid, but uric acid is the least toxic waste and needs only about 5 to 10 ml of water per gram. Urea needs about 50 ml per gram, so animal P saves water at the cost of energy.

Distractor review: A: Uric acid is the cheaper waste in water but not in energy; uricotelic animals have to spend more energy, not less. B: The energy claim is right but the water claim is not; uric acid needs about 5 to 10 ml per gram against about 50 ml per gram for urea. C: Both halves are inverted, which amounts to treating uricotelism as the cheap and wasteful option instead of the costly and thrifty one.

BIO-11-CH04-0035 | Challenging | Habitat and excretory product

Correct answer: D. The tadpole of a frog excretes ammonia while adult frogs are mostly ureotelic, and some terrestrial turtles excrete uric acid while others excrete urea or ammonia

Solution: The frog changes its excretory product with its life stage, the tadpole being ammonotelic and the adult mostly ureotelic. Among terrestrial turtles some excrete uric acid while others excrete urea or ammonia. Both observations break the proposed rule.

Distractor review: A: These pairings show groups agreeing with one another, so they support the student's rule instead of breaking it. B: Here again each pair shares a single mode of excretion, which is exactly what the rule predicts. C: Both pairs are internally consistent, so they illustrate uniformity rather than the variation that would disprove the rule.

BIO-11-CH04-0036 | Challenging | Osmoregulation in marine animals

Correct answer: D. The bird secretes salts by active transport through glands near its nostrils, while the shark retains urea in its body fluid to stay isotonic to sea water

Solution: Marine birds such as the albatross have salt glands near the nostrils that secrete salts by active transport. Sharks instead retain urea in their blood so that it becomes isotonic to the surrounding marine water and water is not lost by exosmosis.

Distractor review: A: The two strategies are interchanged; salt glands near the nostrils are described for marine birds, and urea retention for sharks. B: Salt glands near the nostrils are attributed to marine birds, sea turtles and marine iguanas, and the shark is described instead as retaining urea. C: Urea retention is the strategy described for sharks alone; the marine bird is described as handling salts through its salt glands.

BIO-11-CH04-0037 | Easy | Organs of the excretory system

Correct answer: B. One pair of kidneys, one pair of ureters, one urinary bladder and one urethra

Solution: Urine is formed in the two kidneys, carried by the two ureters to the single median urinary bladder, and discharged to the exterior through the single urethra arising from that bladder.

Distractor review: A: Each kidney has its own ureter, so the ureters are paired; the urinary bladder is a single median sac, not a pair. C: The kidneys are a pair lying on either side of the vertebral column, and a single urethra arises from the bladder. D: The student pairs every organ of the system; only the kidneys and the ureters are paired, the bladder and the urethra being single.

BIO-11-CH04-0038 | Easy | Position of the kidneys**Correct answer: A. They lie behind the peritoneum, extending from the 12th thoracic to the 3rd lumbar vertebra**

Solution: The kidneys are placed behind the peritoneum, which is why they are called retroperitoneal, and each one spans the levels of the 12th thoracic vertebra down to the 3rd lumbar vertebra.

Distractor review: B: The vertebral extent is right but 'retroperitoneal' means behind the peritoneum, not in front of it. C: The two vertebral numbers have been exchanged between the thoracic and lumbar regions, giving an extent far longer than the kidney. D: Both the relation to the peritoneum and the vertebral numbers are reversed from the described position.

BIO-11-CH04-0039 | Easy | Gross external form of the kidney**Correct answer: D. The hilum; the renal artery enters through it**

Solution: The described notch on the concave inner surface is the hilum. Blood is brought in through the renal artery at this point, while the renal vein and the ureter pass outwards through it.

Distractor review: A: The notch is named correctly, but the renal vein leaves the kidney along with the ureter; it is the renal artery that enters. B: The renal pelvis is the funnel inside the kidney that collects urine and continues as the ureter, not the notch on the concave border. C: Both the name of the notch and the direction of flow in the renal vein have been confused with the collecting funnel of the kidney.

BIO-11-CH04-0040 | Easy | Internal regions of the kidney**Correct answer: B. Renal cortex**

Solution: The outer peripheral region of the kidney with a red, granular appearance is the renal cortex; it holds the Malpighian bodies, the convoluted tubules and blood vessels.

Distractor review: A: The medulla is the inner region and is pale red and striated, not outer, red and granular. C: The renal pelvis is the funnel shaped urine collecting space continuous with the ureter, not a tissue region of the kidney substance. D: The renal papilla is only the narrow tip of a pyramid and is not one of the two broad internal regions.

BIO-11-CH04-0041 | Medium | Tissue layers around the kidney**Correct answer: B. Adipose capsule -- shock absorption**

Solution: The mass of fatty tissue surrounding the kidney is the adipose capsule, and cushioning the organ against mechanical shock is its function.

Distractor review: A: The renal fascia is thin fibrous connective tissue that anchors the kidney to the abdominal wall; the fatty layer is the one that absorbs shock. C: The renal capsule is the smooth transparent fibrous membrane immediately on the kidney; anchoring is the role of the outermost fascia. D: Acting as a barrier against the spread of infection is the stated role of the innermost renal capsule, not of the fatty layer.

BIO-11-CH04-0042 | Medium | Urine collecting spaces of the kidney**Correct answer: C. Renal papilla --> minor calyx --> major calyx --> renal pelvis --> ureter**

Solution: The narrow tip of a pyramid is the renal papilla. From there urine is received by a minor calyx, which drains into a major calyx, then into the renal pelvis, and the pelvis continues as the ureter.

Distractor review: A: The two calyces are interchanged; several minor calyces open into a major calyx, so the minor one is met first. B: This places the calyx before the papilla, whereas the papilla is the tip of the pyramid from which urine leaves the kidney substance. D: The renal pelvis is put before the major calyx; the pelvis is the last wide space in the kidney and continues directly as the ureter.

BIO-11-CH04-0043 | Hard | Renal pyramids and renal columns**Correct answer: B. Strips of cortical tissue run inwards between adjacent pyramids as the columns of Bertini**

Solution: Cortical tissue is not restricted to the outer rim of the kidney. It extends inwards between adjacent medullary pyramids as the renal columns, so tissue lying deep in the kidney can still be cortex.

Distractor review: A: This reverses the rule; the pyramids are the medullary masses and the strips between them are cortical. C: The renal capsule is a covering membrane on the outside of the kidney and does not form the tissue strips between pyramids. D: The calyces are urine collecting spaces that receive the papillae; they do not extend outwards as tissue between the pyramids.

BIO-11-CH04-0044 | Easy | Ureters**Correct answer: D. From the renal pelvis; 25 to 30 cm**

Solution: The ureter is the continuation of the renal pelvis and measures 25 to 30 cm, conveying urine from the renal pelvis to the urinary bladder.

Distractor review: A: The origin is right, but 4 to 6 cm is the length of a single renal tubule of a nephron, not of the ureter. B: The length is right, but the ureter continues from the renal pelvis; the cortex is a tissue region and gives rise to no duct. C: Both the site of origin and the length have been taken from the nephron rather than from the ureter.

BIO-11-CH04-0045 | Hard | Ureteric openings into the bladder

Correct answer: A. The ureters pass obliquely through the wall of the bladder, so compression closes them

Solution: Each ureter runs obliquely through the thickness of the bladder wall. When the bladder is compressed, the pressure squeezes this oblique stretch shut, so urine cannot be driven back up the ureter even though no valve guards the opening.

Distractor review: B: The openings form the two basal corners of the trigone and are not beyond the reach of urine; their separation does not stop backflow. C: Skeletal muscle in this region forms the external urethral sphincter at the outlet of the bladder, not rings around the ureteric openings. D: The transitional epithelium allows the bladder wall to stretch; it is a lining tissue and does not act as a closing mechanism.

BIO-11-CH04-0046 | Easy | Trigone of the urinary bladder

Correct answer: C. Trigone; the opening of the urethra lies at the apex

Solution: The described area is the trigone. A triangle has one apex and two basal corners, so the single opening at the apex must be the one leading out of the bladder, that is the urethral opening, while the paired ureteric openings lie along the base.

Distractor review: A: The area is named correctly, but the two ureteric openings form the base of the triangle, not its single apex. B: Detrusor is the name of the smooth muscle coat of the bladder wall, not of the triangular area at its base. D: Both the name of the area and the position of the paired openings have been confused with the muscle layer of the bladder.

BIO-11-CH04-0047 | Easy | Urinary bladder

Correct answer: C. In the pelvic cavity, posterior to the pubic symphysis; lined by transitional epithelium

Solution: The bladder is a median sac in the pelvic cavity, placed behind the pubic symphysis, and its innermost lining of transitional epithelium is what allows the organ to stretch as urine accumulates.

Distractor review: A: The position is right, but simple squamous epithelium is the thin lining of the descending limb of the loop of Henle and cannot provide the stretch of the bladder. B: The lining is right, but the bladder is a pelvic organ lying behind the pubic symphysis, not in front of it. D: Both the position of the bladder and the nature of its stretchable lining have been misplaced.

BIO-11-CH04-0048 | Medium | Urethral sphincters

Correct answer: C. The internal sphincter is of smooth muscle and is involuntary, while the external sphincter is of skeletal muscle and is voluntary

Solution: Smooth muscle is not under conscious control, so the internal sphincter formed of it is involuntary, whereas the external sphincter is formed of skeletal muscle and can be controlled voluntarily.

Distractor review: A: The two sphincters have been interchanged; the deeper, internal one is the smooth involuntary sphincter. B: Voluntary control requires skeletal muscle, so the external sphincter cannot be a smooth muscle structure. D: The internal sphincter is of smooth muscle and is not under conscious control, so it cannot be skeletal and voluntary.

BIO-11-CH04-0049 | Easy | Nephron as the unit of the kidney

Correct answer: A. Almost one million

Solution: The kidney is built of almost one million nephrons, each being a structural as well as a functional unit of the organ.

Distractor review: B: This underestimates the count by two orders of magnitude; a kidney with only ten thousand units could not filter the blood volume it receives. C: This overstates the printed count tenfold, exaggerating the number of units in a single kidney. D: A count of this order would describe a few tubules rather than the filtering units of an entire kidney.

BIO-11-CH04-0050 | Medium | Overall build of a nephron

Correct answer: A. 4 to 6 cm long; the capillary bunch is the glomerulus

Solution: The renal tubule of a nephron measures 4 to 6 cm, and the bunch of capillaries associated with it at its beginning is the glomerulus.

Distractor review: B: The length is right, but the vasa recta are the capillaries around the loop of Henle of juxtamedullary nephrons, not the bunch that forms part of the nephron unit itself. C: The capillary bunch is named correctly, but 25 to 30 cm is the length of a ureter, not of a renal tubule. D: Both the length quoted for the ureter and the name of the peritubular capillaries of the medulla have been transferred to the nephron.

BIO-11-CH04-0051 | Hard | Path of the filtrate through the nephron

Correct answer: C. Capsular space --> neck --> proximal convoluted tubule --> descending limb --> ascending limb

Solution: Filtrate formed at the glomerulus enters the capsular space of Bowman's capsule, leaves it at the urinary pole through the ciliated neck, passes into the proximal convoluted tubule, and then runs down the descending limb of the loop of Henle before turning up the ascending limb.

Distractor review: A: The neck is the short ciliated connection between the capsule and the proximal convoluted tubule, so it cannot follow that tubule. B: Filtration delivers fluid into the capsular space first, and the loop of Henle is descended before it is ascended. D: This places the loop of Henle before the proximal convoluted tubule, whereas the loop follows that tubule.

BIO-11-CH04-0052 | Medium | Path of the filtrate through the nephron

Correct answer: B. Ascending limb --> distal convoluted tubule --> collecting tubule --> collecting duct

Solution: The ascending limb of the loop of Henle continues as the distal convoluted tubule; the short straight collecting tubule is the continuation of that tubule, and it opens into the collecting duct.

Distractor review: A: The collecting tubule is the short straight continuation of the distal convoluted tubule and therefore follows it. C: The distal convoluted tubule begins where the ascending limb ends, so it cannot come before that limb. D: The collecting tubule of a nephron opens into the collecting duct, so the duct is reached last.

BIO-11-CH04-0053 | Medium | Bowman's capsule and podocytes

Correct answer: B. The visceral wall; the gaps are the filtration slits

Solution: The inner wall of the double walled capsule is the visceral wall, and it is made of podocytes. Their foot like pedicels do not meet completely, and the gaps left between the pedicels of neighbouring cells are the filtration slits.

Distractor review: A: The gaps are named correctly, but the parietal wall is the outer wall of the capsule and carries no podocytes. C: The wall is right, but the capsular space is the cavity between the two walls of the capsule, not the narrow gaps between neighbouring foot like processes. D: Both the wall bearing the special cells and the identity of the narrow gaps between the processes have been confused.

BIO-11-CH04-0054 | Medium | Malpighian body

Correct answer: A. Malpighian body, made of the glomerulus together with Bowman's capsule

Solution: A capillary bunch enclosed within a double walled cup is the glomerulus inside Bowman's capsule, and the two together constitute the Malpighian body or renal corpuscle, which measures about 200 micrometres and lies in the cortex.

Distractor review: B: The name is right, but the arterioles carry blood to and from the capillary bunch; the cup enclosing it is Bowman's capsule. C: The composition is right, but the juxtaglomerular apparatus is a small region formed where the distal convoluted tubule contacts the afferent arteriole. D: Neither the name nor the composition matches a capillary bunch enclosed within a double walled cup.

BIO-11-CH04-0055 | Medium | Loop of Henle

Correct answer: A. The descending limb is thin walled with simple squamous epithelium and is permeable to water, while the ascending limb is thick walled with simple cuboidal epithelium and is impermeable to water

Solution: Water leaves the tubular fluid in the descending limb because that limb is thin walled, lined by simple squamous epithelium and permeable to water, whereas the thick walled ascending limb with its simple cuboidal epithelium is impermeable to water.

Distractor review: B: The two limbs have been interchanged completely; it is the descending limb that is thin, squamous and water permeable. C: The epithelium and the permeability have been swapped while the wall thickness is kept, giving a thin cuboidal and a thick squamous limb. D: The ascending limb is thick walled with cuboidal epithelium and is impermeable to water, so the limbs cannot be alike in structure.

BIO-11-CH04-0056 | Easy | Neck of the nephron

Correct answer: B. Neck

Solution: The ciliated segment at the urinary pole through which filtrate leaves Bowman's capsule on its way to the proximal convoluted tubule is the neck of the nephron.

Distractor review: A: The macula densa is the densely packed patch of cells at the beginning of the distal convoluted tubule, at the opposite end of the tubule. C: The collecting tubule is the short straight continuation of the distal convoluted tubule and lies at the end of the nephron. D: The renal papilla is the narrow tip of a medullary pyramid and is not a part of the nephron tubule.

BIO-11-CH04-0057 | Challenging | Identification of nephron parts**Correct answer: A. Distal convoluted tubule**

Solution: Only one part of the nephron combines a coiled form, a simple cuboidal lining without a brush border, a cortical position and the three functions of water reabsorption, pH regulation and tubular secretion, and that is the distal convoluted tubule.

Distractor review: B: The proximal convoluted tubule is coiled and cortical but its cells bear a brush border of microvilli and it is the site of selective reabsorption rather than of tubular secretion and pH regulation. C: The collecting tubule is a short straight continuation of the distal convoluted tubule, so it is not a coiled part. D: The ascending limb is lined by simple cuboidal epithelium but it is a straight limb of the loop lying in the medulla, and it is impermeable to water.

BIO-11-CH04-0058 | Medium | Nephron parts in cortex and medulla**Correct answer: A. Cortex: Malpighian bodies, convoluted tubules and blood vessels; Medulla: loops of Henle and collecting ducts**

Solution: The granular appearance of the cortex comes from the Malpighian bodies and convoluted tubules it contains along with blood vessels, while the striated appearance of the medulla comes from the straight loops of Henle and collecting ducts it holds.

Distractor review: B: The two sets have been exchanged; the Malpighian bodies and convoluted tubules belong to the outer region. C: The loops of Henle are placed in the cortex and the convoluted tubules in the medulla, reversing these two structures. D: The collecting ducts are put in the cortex and the Malpighian bodies in the medulla, contrary to the stated regional contents.

BIO-11-CH04-0059 | Hard | Nephron parts in cortex and medulla**Correct answer: D. Distal convoluted tubule**

Solution: The distal convoluted tubule is a convoluted tubule and therefore lies in the cortex, yet it follows the loop of Henle, which lies in the medulla. The tubule must therefore come back into the cortex after its medullary loop.

Distractor review: A: This tubule is indeed cortical, but it is traversed before the tubule enters the medulla, so it says nothing about a return. B: The collecting duct is one of the medullary structures, so it supports rather than refutes the idea of going deeper. C: The descending limb is the part that carries the tubule into the medulla in the first place.

BIO-11-CH04-0060 | Medium | Types of nephron**Correct answer: B. Cortical nephrons have a shorter loop of Henle that extends very little into the medulla, and they form most of the nephrons**

Solution: The type whose loop of Henle barely enters the medulla is the cortical nephron, and most of the nephrons of a kidney are of this type; the juxtamedullary nephrons with their long loops are few.

Distractor review: A: The abundance is right, but a loop running deep into the medulla is the feature of the other type of nephron. C: Both features have been transferred to the wrong type; juxtamedullary nephrons are few and have long loops. D: The loop length is right for this type, but juxtamedullary nephrons are only a few of the nephrons of the kidney.

BIO-11-CH04-0061 | Hard | Vasa recta and nephron type**Correct answer: D. Vasa recta are associated with only a few of the nephrons of the kidney**

Solution: Since the vasa recta occur only around the loops of Henle of juxtamedullary nephrons, and such nephrons are few compared with the cortical ones, the vasa recta can be associated with only a small proportion of the nephrons of the kidney.

Distractor review: A: This ignores the second statement; since juxtamedullary nephrons are few, so are the loops that the vasa recta can surround. B: The capillaries around the proximal convoluted tubule belong to the peritubular network; the vasa recta are placed around the loop of Henle. C: Cortical nephrons are supplied by the peritubular capillary network and are not the nephrons around which the vasa recta lie.

BIO-11-CH04-0062 | Hard | Path of blood through the kidney**Correct answer: C. Renal artery --> afferent arteriole --> glomerulus --> efferent arteriole --> peritubular capillary network --> renal venule**

Solution: Blood is brought into the kidney by the renal artery and reaches the glomerulus through the afferent arteriole. It leaves the glomerulus by the efferent arteriole, which supplies the peritubular capillary network around the tubules, and this drains into the renal venule.

Distractor review: A: The two arterioles are interchanged; the afferent arteriole is the one that brings blood towards the glomerulus. B: This puts the peritubular network before the efferent arteriole, whereas that arteriole is what delivers blood to the network. D: The glomerulus is placed after the efferent arteriole, although the efferent arteriole drains the glomerulus.

BIO-11-CH04-0063 | Medium | Path of blood through the kidney

Correct answer: D. Renal vein --> renal venule --> peritubular capillary network --> efferent arteriole --> glomerulus

Solution: Going upstream from the renal vein, blood is met first in the renal venule, then in the peritubular capillary network around the tubules, then in the efferent arteriole that supplied that network, and finally in the glomerulus that the efferent arteriole drains.

Distractor review: A: Working backwards from the tubules reaches the glomerulus through the efferent arteriole, since the afferent arteriole lies on the upstream side of the glomerulus. B: This puts the efferent arteriole downstream of the peritubular network, although that arteriole supplies the network. C: The renal venule lies between the peritubular network and the renal vein, so it cannot be reached after the network when moving backwards.

BIO-11-CH04-0064 | Hard | From renal blood to glomerular filtrate

Correct answer: C. Renal artery --> afferent arteriole --> glomerulus --> capsular space --> neck --> proximal convoluted tubule

Solution: The molecule travels in blood through the renal artery and the afferent arteriole into the glomerulus, is filtered from there into the capsular space of Bowman's capsule, and then leaves that space through the neck to enter the proximal convoluted tubule.

Distractor review: A: Filtration occurs from the glomerulus into the capsular space, so the capillary bunch must be reached before that space. B: The efferent arteriole carries blood away from the glomerulus and cannot deliver blood to it. D: The neck leads out of the capsular space towards the proximal convoluted tubule, so it is reached after that space.

BIO-11-CH04-0065 | Challenging | Blood supply of a juxtamedullary nephron

Correct answer: A. Afferent arteriole --> glomerulus --> efferent arteriole --> vasa recta --> renal venule

Solution: The nephron described is a juxtamedullary one, and the loop shaped capillaries around its loop of Henle are the vasa recta. These capillaries belong to the bed supplied by the efferent arteriole, so blood reaches them after leaving the glomerulus and drains from them into the renal venule.

Distractor review: B: The vasa recta form part of the capillary bed supplied by the efferent arteriole, so they are reached after that arteriole and not before it. C: This places the capillaries of the loop before the glomerulus, although the afferent arteriole opens directly into the glomerulus. D: The two arterioles are interchanged, so blood is shown entering the glomerulus by the vessel that in fact drains it.

BIO-11-CH04-0066 | Easy | Juxtaglomerular apparatus

Correct answer: C. The macula densa and the JG cells

Solution: The juxtaglomerular apparatus is formed where the densely packed cells of the macula densa lie against the JG cells of the afferent arteriole of the same nephron.

Distractor review: A: Podocytes are the special squamous cells of the visceral wall of Bowman's capsule and take no part in this apparatus. B: One component is right, but the second component is a patch of densely packed cells of the distal convoluted tubule, not the podocytes. D: The peritubular capillaries form the network around the tubules and are not a component of this apparatus.

BIO-11-CH04-0067 | Medium | Diameters of the glomerular vessels

Correct answer: C. The afferent arteriole has the largest diameter, the efferent arteriole is narrower, and the capillaries are narrower than both arterioles

Solution: Of the three vessels the afferent arteriole is the widest, the efferent arteriole is narrower than it, and the glomerular capillaries are narrower than both arterioles.

Distractor review: A: The two arterioles have been interchanged; the vessel bringing blood to the glomerulus is the wider of the two. B: The capillary diameter is smaller than that of both arterioles, so the capillaries cannot be the widest vessels. D: The two arterioles differ in diameter, and the capillaries are the narrowest of the three.

BIO-11-CH04-0068 | Challenging | Integration of nephron type, blood supply and juxtaglomerular apparatus

Correct answer: C. It is a juxtamedullary nephron, the capillaries around its loop are the vasa recta, and the densely packed tubule cells form the macula densa

Solution: The long medullary loop identifies the nephron as juxtamedullary, and only such nephrons have vasa recta around their loops. The densely packed cells belong to the wall of the distal convoluted tubule where it meets the afferent arteriole, and these constitute the macula densa, the JG cells being the modified smooth muscle cells of the arteriole itself.

Distractor review: A: A loop running deep into the medulla belongs to a juxtamedullary nephron; the cortical type has a loop that extends very little into the medulla. B: The glomerular capillaries lie within Bowman's capsule, and the JG cells are modified smooth muscle cells of the afferent arteriole wall, not tubule cells. D: Every part of this description is misassigned, from the type of nephron to the identity of the capillaries and of the densely packed cells.

BIO-11-CH04-0069 | Easy | The three steps of urine formation**Correct answer: A. Tubular reabsorption**

Solution: Movement of substances from the tubular fluid back into the blood is the defining statement of tubular reabsorption, the second of the three successive steps of urine formation.

Distractor review: B: Tubular secretion is defined in the opposite direction, as the movement of substances from the blood into the tubular fluid. C: Glomerular filtration is the movement of substances from the blood within the glomerulus into the capsular space, not out of the tubular fluid. D: Counter current multiplication is the operation of the limbs of Henle's loop that builds the medullary osmotic gradient; it is not one of the three defined steps of urine formation.

BIO-11-CH04-0070 | Medium | Sites of the three steps**Correct answer: D. Distal convoluted tubule and collecting tubule**

Solution: Tubular secretion, or augmentation, takes place in the distal convoluted tubule and the collecting tubule by active transport, just as selective reabsorption takes place mainly in the proximal convoluted tubule.

Distractor review: A: The glomerulus and the capsular space are the site of glomerular filtration, the first step, and no secretion into the tubular fluid is described there. B: The limbs of Henle's loop are described in terms of water and NaCl movement for the counter current mechanism; no tubular secretion is placed there. C: These are blood vessels supplying the glomerulus and surrounding the tubule; they are not parts of the renal tubule and cannot secrete into the tubular fluid.

BIO-11-CH04-0071 | Medium | Effective filtration pressure**Correct answer: B. 10 mm Hg**

Solution: The two opposing pressures total $30 + 15 = 45$ mm Hg. Subtracting this from the glomerular hydrostatic pressure of 55 mm Hg gives an effective filtration pressure of 10 mm Hg.

Distractor review: A: This is obtained by subtracting only the osmotic pressure of the blood and ignoring the hydrostatic pressure of the filtrate in the capsular space. C: This is obtained by subtracting only the filtrate hydrostatic pressure and ignoring the osmotic pressure of the blood. D: This is the sum of all three pressures; the two opposing pressures must be subtracted from the glomerular hydrostatic pressure, not added to it.

BIO-11-CH04-0072 | Hard | Effective filtration pressure**Correct answer: D. 15 mm Hg**

Solution: The two opposing pressures together must be $55 - 10 = 45$ mm Hg. Since the osmotic pressure of the blood is 30 mm Hg, the hydrostatic pressure of the filtrate is $45 - 30 = 15$ mm Hg.

Distractor review: A: This is the total of both opposing pressures; the osmotic pressure of the blood still has to be removed from it to leave the filtrate hydrostatic pressure. B: This comes from subtracting the effective filtration pressure from the osmotic pressure of the blood, which reverses the roles of the two quantities. C: This treats the effective filtration pressure as the difference between the osmotic pressure of blood and the filtrate pressure, which is not the stated relationship.

BIO-11-CH04-0073 | Hard | Glomerular filtration rate**Correct answer: D. 180 litres**

Solution: In one day the filtrate volume is $125 \times 60 \times 24 = 180000$ ml, which is 180 litres, exactly the daily figure quoted alongside the rate of 125 ml per minute.

Distractor review: A: This is a tenfold underestimate arising from a slipped decimal point while converting 180000 ml into litres. B: This is the volume formed in 10 hours only and treats part of the day rather than the whole 24 hours. C: This is a tenfold overestimate arising from converting millilitres to litres by dividing by 100 instead of 1000.

BIO-11-CH04-0074 | Hard | Selective reabsorption**Correct answer: B. 123.75 ml**

Solution: Ninety nine per cent of 125 ml is 123.75 ml, so about 123.75 ml of the filtrate formed each minute is returned to the blood in the proximal and distal convoluted tubules.

Distractor review: A: This is obtained by subtracting 1 per cent as a flat 0.25 ml rather than as 1 per cent of 125 ml, which is 1.25 ml. C: This is 125 ml reduced by 5 per cent instead of by 1 per cent, misreading the reabsorbed fraction. D: This adds 1 per cent of the filtration rate instead of taking 99 per cent of it, so it exceeds the volume actually filtered.

BIO-11-CH04-0075 | Medium | High threshold and low threshold substances**Correct answer: C. Sulphates, absorbed passively as a low threshold substance**

Solution: Sulphates belong to the low threshold group, which also includes water and nitrates, and this whole group is taken back passively rather than by active transport.

Distractor review: A: Glucose is listed among the high threshold substances of considerable importance that are absorbed actively, not passively. B: Vitamin C appears in the list of high threshold substances taken back actively along with glucose, amino acids and the major ions. D: Water heads the list of low threshold substances that are absorbed passively, so its mode of absorption is misassigned here.

BIO-11-CH04-0076 | Medium | Composition of the glomerular filtrate**Correct answer: A. Albumin**

Solution: Albumin is a plasma protein, and proteins are the macromolecules that the glomerular capillary wall holds back, which is why the filtrate is called deproteinized plasma.

Distractor review: B: Glucose is a small solute that passes freely into the capsular space and is named among the contents of the glomerular filtrate. C: Urea is listed among the substances present in the glomerular filtrate, which is described as deproteinized plasma containing urea. D: Inorganic ions cross the glomerular capillary wall readily and are named among the constituents of the filtrate.

BIO-11-CH04-0077 | Challenging | Reabsorption contrasted with secretion**Correct answer: A. Distal convoluted tubule and collecting tubule**

Solution: K⁺ together with H⁺ and wastes such as creatinine is actively secreted from the blood into the tubular fluid in the distal convoluted tubule and the collecting tubule, even though the same ion is reabsorbed earlier in the proximal convoluted tubule.

Distractor review: B: The PCT is the main site of selective reabsorption, where K⁺ moves in the opposite direction, from the tubular fluid into the blood. C: The limbs of the loop are described in terms of water and NaCl movement for the counter current mechanism, not active movement of K⁺ into the tubular fluid. D: Bowman's capsule receives the glomerular filtrate, so substances arrive there by filtration rather than by active secretion.

BIO-11-CH04-0078 | Easy | Juxta glomerular apparatus**Correct answer: D. Macula densa**

Solution: The densely packed cells of the wall of the beginning of the distal convoluted tubule where it touches the afferent arteriole are called the macula densa, which together with the JG cells forms the juxta glomerular apparatus.

Distractor review: A: JG cells are the modified smooth muscle cells of the wall of the afferent arteriole with granular sarcoplasm, not cells of the distal convoluted tubule. B: Podocytes are the special squamous cells of the visceral wall of Bowman's capsule bearing pedicels, and lie nowhere near the distal convoluted tubule. C: Peritubular capillaries are the network of blood vessels surrounding the tubule and are not a densely packed region of the tubular wall.

BIO-11-CH04-0079 | Medium | Path of the tubular fluid**Correct answer: D. Proximal convoluted tubule --> Henle's loop --> distal convoluted tubule --> collecting tubule**

Solution: The tubular fluid leaves Bowman's capsule and the neck, passes through the proximal convoluted tubule, then the U shaped Henle's loop, then the distal convoluted tubule, and finally the collecting tubule which opens into the collecting duct.

Distractor review: A: This places the two convoluted tubules together, whereas Henle's loop lies between them and connects the proximal to the distal tubule. B: This puts Henle's loop first, but the filtrate leaves the neck and enters the proximal convoluted tubule before reaching the loop. C: This reverses the last two regions; the collecting tubule is a short straight continuation of the distal convoluted tubule and therefore follows it.

BIO-11-CH04-0080 | Medium | Structure and function of the PCT**Correct answer: A. Selective reabsorption of useful molecules from the tubular fluid**

Solution: The PCT is the place of selective reabsorption, and its microvillous brush border together with its convolutions is what allows maximum reabsorption of useful molecules, so losing the brush border reduces reabsorption specifically.

Distractor review: B: Filtration depends on the glomerular capillary wall and the balance of pressures at the Malpighian body, neither of which involves the brush border of the PCT. C: Secretion of these ions occurs by active transport in the distal convoluted tubule and collecting tubule, regions unaffected by the change described. D: The medullary gradient is built by the two limbs of Henle's loop carrying fluid in opposite directions, and does not depend on the surface of the PCT cells.

BIO-11-CH04-0081 | Easy | Counter current mechanism**Correct answer: C. Vasa recta**

Solution: The vasa recta are the capillaries surrounding Henle's loop of juxtamedullary nephrons, and they maintain the medullary osmotic gradient by operating the counter current exchange system.

Distractor review: A: The peritubular capillary network surrounds the convoluted tubules for reabsorption but is not the loop shaped set described as the counter current exchanger. B: The glomerular capillaries form the bunch where filtration occurs, and they lie in the Malpighian body rather than around Henle's loop in the medulla. D: The afferent arteriole is a single vessel bringing blood to the glomerulus and is not a capillary network operating an exchange system.

BIO-11-CH04-0082 | Medium | Limbs of Henle's loop**Correct answer: D. Descending limb, thin walled, permeable to water, simple squamous epithelium**

Solution: The descending limb of Henle's loop is the thin walled limb of simple squamous epithelium that is permeable to water, so water moves out of it by osmosis into the tissue fluid.

Distractor review: A: These are the features of the ascending limb; the descending limb is the thin walled limb through which water leaves by osmosis. B: The ascending limb is thick walled and impermeable to water, which is why NaCl rather than water leaves it. C: The wall thickness and epithelium are right for the ascending limb but its impermeability to water is the property that makes the counter current mechanism work.

BIO-11-CH04-0083 | Medium | Medullary osmotic gradient**Correct answer: B. 4 times**

Solution: Dividing 1200 mOsmol L⁻¹ by 300 mOsmol L⁻¹ gives 4, which is why the urine can be almost four times as concentrated as the blood.

Distractor review: A: This comes from taking the difference of 900 mOsmol L⁻¹ and dividing it by 300, which gives the increase rather than the ratio. C: This corresponds to the outer medullary value of 600 mOsmol L⁻¹, not to the deep inner medullary value given in the stem. D: This would need a deep inner medullary value of 1800 mOsmol L⁻¹, which is above the highest osmolarity reached in the medulla.

BIO-11-CH04-0084 | Hard | Urea recycling**Correct answer: B. Thick ascending limb segment**

Solution: Urea cannot pass out through the thick ascending limb segment, and it is this restriction that traps urea in the medullary tissue fluid and allows more water to be reabsorbed so that a small volume of concentrated urine is excreted.

Distractor review: A: The proximal convoluted tubule is described as the site of bulk reabsorption of the filtrate and is not named as the barrier that traps urea in the medulla. C: The neck is the short segment joining Bowman's capsule to the proximal convoluted tubule and takes no part in the urea recycling described here. D: The descending limb is described in terms of its permeability to water, and it is not named as the barrier to urea movement.

BIO-11-CH04-0085 | Medium | Hormonal regulation, sources**Correct answer: C. Aldosterone, produced by the adrenal cortex**

Solution: Aldosterone is released from the adrenal cortex, from where it travels to the distal convoluted tubule and collecting ducts to increase Na⁺ and water reabsorption.

Distractor review: A: Renin is released by the JG cells, the modified smooth muscle cells of the wall of the afferent arteriole, and not by the macula densa. B: The collecting duct is the target on which atrial natriuretic peptide acts; the hormone itself is produced by the atrial wall. D: Hypothalamic osmoreceptors detect the low water concentration of the blood, but the hormone is secreted by the neurohypophysis.

BIO-11-CH04-0086 | Medium | Renin angiotensin aldosterone system**Correct answer: B. Angiotensin I**

Solution: Renin acting on angiotensinogen produces angiotensin I, which is then converted by angiotensin converting enzyme into the active angiotensin II.

Distractor review: A: Angiotensin II is the active form produced one step later, when angiotensin converting enzyme acts on angiotensin I. C: Aldosterone is released from the adrenal cortex only after angiotensin II has stimulated it, so it is not the direct product of renin. D: This enzyme acts on angiotensin I to make angiotensin II; it is a catalyst in the pathway and not a product of renin.

BIO-11-CH04-0087 | Hard | Hormonal regulation, effects**Correct answer: A. Atrial natriuretic peptide**

Solution: Atrial natriuretic peptide, secreted by the atrial wall when blood volume and pressure increase greatly, inhibits Na⁺ and Cl⁻ reabsorption from the collecting ducts and produces natriuresis and diuresis.

Distractor review: B: Antidiuretic hormone also acts on the collecting ducts but increases their permeability to water and makes the urine concentrated, which is the opposite of diuresis. C: Angiotensin II acts on the PCT cells to enhance reabsorption of Na⁺, Cl⁻ and water, so it works in the same direction as aldosterone rather than against it. D: This enzyme acts on angiotensin I in the blood to make angiotensin II and has no direct action on the tubular wall of the collecting ducts.

BIO-11-CH04-0088 | Challenging | Antidiuretic hormone**Correct answer: D. Permeability of the target cells to water increases, water moves from the tubular fluid into the tissue fluid and concentrated urine is formed**

Solution: Antidiuretic hormone makes the cells in the wall of the collecting ducts permeable to water, so water moves from the tubular fluid into the tissue fluid and the urine that is passed is concentrated.

Distractor review: A: This is what happens when antidiuretic hormone secretion falls, giving diuresis, and it reverses the effect of the hormone being released. B: Increased Na⁺ reabsorption is the effect of aldosterone, and in any case Na⁺ reabsorption raises blood volume rather than lowering it. C: Movement of urea into the medullary tissue fluid belongs to urea recycling at the deep medullary collecting duct and is not the action attributed to this hormone.

BIO-11-CH04-0089 | Easy | Colour of normal urine**Correct answer: B. Urochrome**

Solution: The pale yellow colour of normal transparent urine is due to the pigment urochrome, which is eliminated through the urine.

Distractor review: A: Bilirubin is the other pigment formed from the breakdown of haemoglobin, but it is excreted through the faeces rather than colouring the urine. C: Creatinine is a nitrogenous waste produced from catabolism of creatinine phosphate in skeletal muscle and is not a pigment. D: Uric acid is a nitrogenous excretory product of uricotelic animals and contributes no colour to urine.

BIO-11-CH04-0090 | Easy | Urine report**Correct answer: C. Typically slightly acidic**

Solution: The prescribed report describes normal urine as typically acidic. Physiologically, urine pH can vary across an acidic-to-alkaline reference range, so the qualified wording is important.

Distractor review: A: The tubular fluid is alkaline only as it leaves the glomerulus; tubular secretion changes it before the urine is voided. B: The report records a definite character rather than a neutral one, and tubular secretion shifts the reaction away from neutrality. D: Amphoterics describe substances able to act as an acid or a base, and it is not a term used to record the reaction of urine in the report.

BIO-11-CH04-0091 | Hard | Reaction of urine**Correct answer: D. Tubular secretion in the distal convoluted tubule and collecting tubule**

Solution: Tubular secretion, or augmentation, occurring in the distal convoluted tubule and collecting tubule adds H⁺ and K⁺ and wastes such as creatinine to the tubular fluid, and this is the step that changes the reaction from alkaline to acidic.

Distractor review: A: Filtration produces the alkaline filtrate in the first place, so it cannot be the step that turns the fluid acidic. B: Selective reabsorption returns useful high and low threshold substances to the blood; the change of reaction is not attributed to it. C: The limbs of Henle's loop build the medullary osmotic gradient that concentrates the urine, which is a change in osmolarity and not in reaction.

BIO-11-CH04-0092 | Medium | Specific gravity of urine**Correct answer: B. It would increase**

Solution: A rise in antidiuretic hormone increases the permeability of the collecting duct wall to water, more water is reabsorbed, the urine becomes more concentrated and its specific gravity rises above the recorded value.

Distractor review: A: A fall in specific gravity accompanies diuresis, which follows a decrease and not an increase in antidiuretic hormone. C: Antidiuretic hormone alters the water content of the urine, so a change in its level must be reflected in the specific gravity. D: No transient fall is described; the hormone acts in one direction by making the collecting duct wall permeable to water.

BIO-11-CH04-0093 | Easy | Skin**Correct answer: A. Thermoregulation of the body**

Solution: Although sweat carries out several waste substances, it is primarily produced for thermoregulation, and its excretory role is secondary to that.

Distractor review: B: Lubrication and protection of the skin are provided by sebum from the sebaceous glands, not by sweat. C: Control of solute concentration and water balance in the blood is carried out by the kidneys, and sweat is not described as serving that purpose. D: Volatile substances present in spices are excreted through the lungs rather than through the sweat glands.

BIO-11-CH04-0094 | Medium | Skin and lungs**Correct answer: B. Sweat glands, giving out water, NaCl, lactic acid and urea**

Solution: Sweat glands are the source of sweat, which carries out water, NaCl, lactic acid and urea in addition to its thermoregulatory role.

Distractor review: A: Sebaceous glands lie at the necks of hair follicles and secrete sebum; the watery secretion carrying these solutes comes from the sweat glands. C: Sweat glands are simple, unbranched, coiled tubular glands opening through sweat pores, and sebum is not their secretion. D: The lungs give out volatile substances such as CO₂ and water vapour produced during cellular respiration, not an oily skin secretion.

BIO-11-CH04-0095 | Easy | Haemodialysis**Correct answer: A. Cellophane**

Solution: The tube through which the blood passes during haemodialysis is made of cellophane, which acts as the semipermeable membrane allowing excess salts and wastes to move into the dialysate.

Distractor review: B: The peritoneal membrane serves as the dialyzing membrane in peritoneal dialysis, where fluid is introduced into the abdominal cavity, not in haemodialysis. C: The renal capsule is the fibrous covering of the kidney itself and has no part in the apparatus used outside the body. D: The glomerular basement membrane is part of the filtration barrier inside the nephron, not a material from which the dialysis tube is made.

BIO-11-CH04-0096 | Easy | Kidney stones**Correct answer: B. Struvite stones**

Solution: Struvite stones are the type formed in response to bacterial infection caused by urea splitting bacteria, and they are the type that grows quickly and becomes quite large.

Distractor review: A: Calcium stones are calcium oxalate or calcium phosphate stones and are not attributed to infection by urea splitting bacteria. C: Uric acid stones form in people who do not drink enough water or who consume a high protein diet, not as a response to bacterial infection. D: Cystine stones arise from a genetic disorder that makes the kidney excrete too much of a certain amino acid.

BIO-11-CH04-0097 | Hard | Uremia**Correct answer: A. 2 times**

Solution: Dividing the measured 0.06 per cent by the upper normal limit of 0.03 per cent gives 2, so the patient's blood urea is twice the upper limit of normal and also above the uremia threshold of 0.05 per cent.

Distractor review: B: This compares the value with the uremia threshold of 0.05 per cent rather than with the upper limit of the normal range. C: This would need a blood urea of 0.09 per cent, which is higher than the value given in the stem. D: This compares the value with the lower limit of the normal range, 0.01 per cent, instead of the upper limit.

BIO-11-CH04-0098 | Medium | Acute renal failure**Correct answer: D. 300 ml per day, which satisfies the criterion**

Solution: At 150 ml per 12 hours the output for 24 hours is 300 ml per day, which is less than 400 ml per day and therefore satisfies the adult oligouria criterion.

Distractor review: A: The daily volume is worked out correctly but is then wrongly judged, since 300 ml per day is below the stated limit of 400 ml per day. B: This quotes the 12 hour volume as though it were the whole day, without doubling it to cover the full 24 hours. C: This multiplies the 12 hour volume by four instead of by two, treating the measurement period as six hours.

BIO-11-CH04-0099 | Medium | Acute renal failure**Correct answer: C. 10 ml per hour**

Solution: For a 20 kg child the threshold is $0.5 \times 20 = 10$ ml per hour, so an output below 10 ml per hour satisfies the paediatric oligouria criterion.

Distractor review: A: This halves the correct threshold by treating the child's weight as 10 kg instead of the 20 kg given. B: This uses 1 ml per kilogram per hour instead of the stated 0.5 ml per kilogram per hour. D: This multiplies the weight by 2 instead of by 0.5, inverting the rate given in the criterion.

BIO-11-CH04-0100 | Challenging | Dialysis and transplant**Correct answer: C. Secretion of erythropoietin, renin and calcitriol**

Solution: Dialysis machines can clear wastes and excess salts, but they cannot replace the kidney's secretions of erythropoietin, renin and calcitriol, so dialysis serves only as a holding measure until a renal transplant can be performed.

Distractor review: A: Waste substances move from the blood into the dialyzing fluid across the membrane, so this function is performed by dialysis. B: Excess salts pass through the membrane into the dialysate, which is isosmotic to normal blood plasma, so dialysis handles this. D: Movement of water and solutes across the dialyzing membrane allows dialysis to act on the water content of the blood.

BIO-11-CH04-0101 | Easy | Kidney hilum**Correct answer: A. Renal artery**

Solution: The renal artery enters the kidney through the hilum; the renal vein and ureter leave through the same region.

Distractor review: B: Renal vein is a plausible distractor, but it conflicts with the required identifying the vessel entering the kidney at the hilum and therefore cannot be the single correct response. C: Ureter is a plausible distractor, but it conflicts with the required identifying the vessel entering the kidney at the hilum and therefore cannot be the single correct response. D: Inferior vena cava is a plausible distractor, but it conflicts with the required identifying the vessel entering the kidney at the hilum and therefore cannot be the single correct response.

BIO-11-CH04-0102 | Medium | Malpighian body**Correct answer: B. Capsular space**

Solution: Filtrate enters the capsular space between the parietal and visceral walls of Bowman's capsule.

Distractor review: A: Glomerular lumen is a plausible distractor, but it conflicts with the required identifying the capsular space in a renal corpuscle and therefore cannot be the single correct response. C: Collecting duct is a plausible distractor, but it conflicts with the required identifying the capsular space in a renal corpuscle and therefore cannot be the single correct response. D: Renal pelvis is a plausible distractor, but it conflicts with the required identifying the capsular space in a renal corpuscle and therefore cannot be the single correct response.

BIO-11-CH04-0103 | Hard | Ultrafiltration**Correct answer: C. 10 mm Hg**

Solution: Effective filtration pressure = $60 - (32 + 18) = 10$ mm Hg.

Distractor review: A: 6 mm Hg is a plausible distractor, but it conflicts with the required calculating effective filtration pressure from three pressures and therefore cannot be the single correct response. B: 8 mm Hg is a plausible distractor, but it conflicts with the required calculating effective filtration pressure from three pressures and therefore cannot be the single correct response. D: 14 mm Hg is a plausible distractor, but it conflicts with the required calculating effective filtration pressure from three pressures and therefore cannot be the single correct response.

BIO-11-CH04-0104 | Challenging | Dialysis and transplantation**Correct answer: D. Erythropoietin, renin and calcitriol**

Solution: Dialysis can remove small wastes and adjust fluid and electrolyte composition, but it does not reproduce renal endocrine secretion of erythropoietin, renin and calcitriol.

Distractor review: A: Urea, uric acid and creatinine is a plausible distractor, but it conflicts with the required distinguishing filtration functions from endocrine functions of the kidney and therefore cannot be the single correct response. B: Water, sodium ions and urea is a plausible distractor, but it conflicts with the required distinguishing filtration functions from endocrine functions of the kidney and therefore cannot be the single correct response. C: Glucose, amino acids and bicarbonate is a plausible distractor, but it conflicts with the required distinguishing filtration functions from endocrine functions of the kidney and therefore cannot be the single correct response.

Standard 12 | Chapter 01

Reproduction in Lower and Higher Plants

BIO-12-CH01-0001 | Medium | Double fertilization in an angiosperm

Correct answer: D. Male gamete with the two polar nuclei

Solution: Triple fusion joins one male gamete with the two polar nuclei, producing the triploid primary endosperm nucleus.

Distractor review: A: The alternative 'Male gamete with the egg' conflicts with the labelled relationship, scale or mechanism in the figure. Triple fusion joins one male gamete with the two polar nuclei, producing the triploid primary endosperm nucleus. B: The alternative 'Egg with a synergid' conflicts with the labelled relationship, scale or mechanism in the figure. Triple fusion joins one male gamete with the two polar nuclei, producing the triploid primary endosperm nucleus. C: The alternative 'Two male gametes with each other' conflicts with the labelled relationship, scale or mechanism in the figure. Triple fusion joins one male gamete with the two polar nuclei, producing the triploid primary endosperm nucleus.

BIO-12-CH01-0002 | Hard | Pollination pathways

Correct answer: B. Pathway C: xenogamy

Solution: Xenogamy transfers pollen between different plants of the same species and therefore most directly promotes outcrossing.

Distractor review: A: The alternative 'Pathway A: autogamy' conflicts with the labelled relationship, scale or mechanism in the figure. Xenogamy transfers pollen between different plants of the same species and therefore most directly promotes outcrossing. C: The alternative 'Pathway B: geitonogamy' conflicts with the labelled relationship, scale or mechanism in the figure. Xenogamy transfers pollen between different plants of the same species and therefore most directly promotes outcrossing. D: The alternative 'All three pathways equally' conflicts with the labelled relationship, scale or mechanism in the figure. Xenogamy transfers pollen between different plants of the same species and therefore most directly promotes outcrossing.

BIO-12-CH01-0003 | Hard | Fertilization

Correct answer: B. Fusion of the second male gamete with the two polar nuclei - Sporopollenin

Solution: 'Fusion of the second male gamete with the two polar nuclei' should be paired with Triple fusion, not Sporopollenin. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Transfer between flowers on different plants of the same species - Xenogamy. C: This is a valid association: Pollination carried out by wind - Anemophily. D: This is a valid association: Embryos arising directly from nucellus or integuments - Adventive embryony.

BIO-12-CH01-0004 | Hard | Pollination

Correct answer: C. Transfer of pollen to the stigma of the same flower - Sporopollenin

Solution: 'Transfer of pollen to the stigma of the same flower' should be paired with Autogamy, not Sporopollenin. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Syngamy and triple fusion occurring in one embryo sac - Double fertilization. B: This is a valid association: Gemma formation - Marchantia. D: This is a valid association: Anther wall layer that nourishes developing pollen - Tapetum.

BIO-12-CH01-0005 | Medium | Ovule and embryo sac

Correct answer: C. Junction between the ovule body and its stalk - Hilum

Solution: The accurate entry is 'Junction between the ovule body and its stalk - Hilum'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Fusion of the second male gamete with the two polar nuclei' should be paired with Triple fusion, not Porogamy. B: 'Embryonic axis below the cotyledons' should be paired with Hypocotyl, not Hilum. D: 'Syngamy and triple fusion occurring in one embryo sac' should be paired with Double fertilization, not Seed.

BIO-12-CH01-0006 | Medium | Asexual reproduction

Correct answer: B. Yeast

Solution: The chapter links Budding under favourable conditions with Yeast, so Yeast completes the statement.

Distractor review: A: Marchantia belongs with 'Gemma formation', whereas the tested description is 'Budding under favourable conditions'. C: Endosperm belongs with 'Primary endosperm nucleus generally develops into', whereas the tested description is 'Budding under favourable conditions'. D: Spirogyra belongs with 'Fragmentation in a filamentous alga', whereas the tested description is 'Budding under favourable conditions'.

BIO-12-CH01-0007 | Medium | Asexual reproduction**Correct answer: A. Motile zoospore formation - Chlamydomonas**

Solution: The accurate entry is 'Motile zoospore formation - Chlamydomonas'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Transfer of pollen to the stigma of the same flower' should be paired with Autogamy, not Chlamydomonas. C: 'Embryonic axis below the cotyledons' should be paired with Hypocotyl, not Autogamy. D: 'Primary endosperm nucleus generally develops into' should be paired with Endosperm, not Geitonogamy.

BIO-12-CH01-0008 | Medium | Post-fertilization development**Correct answer: C. Ovule after fertilization - Seed**

Solution: The accurate entry is 'Ovule after fertilization - Seed'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Stalk attaching an ovule to the placenta' should be paired with Funicle, not Syngamy. B: 'Fusion of one male gamete with the egg' should be paired with Syngamy, not A tetrad of haploid microspores. D: 'Fruit formation without fertilization' should be paired with Parthenocarpy, not Seed.

BIO-12-CH01-0009 | Easy | Pollination pathways**Correct answer: A. Autogamy**

Solution: Transfer of pollen to the stigma of the same flower is autogamy.

Distractor review: B: The alternative 'Geitonogamy' conflicts with the labelled relationship, scale or mechanism in the figure. Transfer of pollen to the stigma of the same flower is autogamy. C: The alternative 'Xenogamy' conflicts with the labelled relationship, scale or mechanism in the figure. Transfer of pollen to the stigma of the same flower is autogamy. D: The alternative 'Anemophily' conflicts with the labelled relationship, scale or mechanism in the figure. Transfer of pollen to the stigma of the same flower is autogamy.

BIO-12-CH01-0010 | Easy | Pollination**Correct answer: A. Anemophily**

Solution: 'Pollination carried out by wind' is the defining description for Anemophily. The other choices belong to different chapter concepts.

Distractor review: B: Funicle belongs with 'Stalk attaching an ovule to the placenta', whereas the tested description is 'Pollination carried out by wind'. C: Porogamy belongs with 'Pollen-tube entry into an ovule through the micropyle', whereas the tested description is 'Pollination carried out by wind'. D: Seven-celled and eight-nucleate belongs with 'Typical mature angiosperm embryo sac', whereas the tested description is 'Pollination carried out by wind'.

BIO-12-CH01-0011 | Medium | Microsporogenesis and pollen**Correct answer: C. Highly resistant material of the pollen exine - Sporopollenin**

Solution: The accurate entry is 'Highly resistant material of the pollen exine - Sporopollenin'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Transfer of pollen to the stigma of the same flower' should be paired with Autogamy, not Hypocotyl. B: 'Embryonic axis below the cotyledons' should be paired with Hypocotyl, not Sporopollenin. D: 'Seed formation without fertilization' should be paired with Apomixis, not Porogamy.

BIO-12-CH01-0012 | Medium | Microsporogenesis and pollen**Correct answer: A. Tapetum**

Solution: The observation 'Anther wall layer that nourishes developing pollen' points to Tapetum; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Seven-celled and eight-nucleate belongs with 'Typical mature angiosperm embryo sac', whereas the tested description is 'Anther wall layer that nourishes developing pollen'. C: Micropyle belongs with 'Opening left by the integuments at the ovule apex', whereas the tested description is 'Anther wall layer that nourishes developing pollen'. D: Endosperm belongs with 'Primary endosperm nucleus generally develops into', whereas the tested description is 'Anther wall layer that nourishes developing pollen'.

BIO-12-CH01-0013 | Challenging | Double fertilization in an angiosperm**Correct answer: B. A zygote may form, but normal endosperm initiation is absent**

Solution: Syngamy can form the diploid zygote, but blocking triple fusion prevents formation of the usual triploid primary endosperm nucleus.

Distractor review: A: The alternative 'Neither zygote nor endosperm can form' conflicts with the labelled relationship, scale or mechanism in the figure. Syngamy can form the diploid zygote, but blocking triple fusion prevents formation of the usual triploid primary endosperm nucleus. C: The alternative 'Endosperm forms normally without a zygote' conflicts with the labelled relationship, scale or mechanism in the figure. Syngamy can form the diploid zygote, but blocking triple fusion prevents formation of the usual triploid primary endosperm nucleus. D: The alternative 'Two diploid embryos form in the embryo sac' conflicts with the labelled relationship, scale or mechanism in the figure. Syngamy can form the diploid zygote, but blocking triple fusion prevents formation of the usual triploid primary endosperm nucleus.

BIO-12-CH01-0014 | Easy | Ovule and embryo sac**Correct answer: D. Seven-celled and eight-nucleate**

Solution: 'Typical mature angiosperm embryo sac' is the defining description for Seven-celled and eight-nucleate. The other choices belong to different chapter concepts.

Distractor review: A: Fruit belongs with 'Ovary after fertilization', whereas the tested description is 'Typical mature angiosperm embryo sac'. B: Yeast belongs with 'Budding under favourable conditions', whereas the tested description is 'Typical mature angiosperm embryo sac'. C: Micropyle belongs with 'Opening left by the integuments at the ovule apex', whereas the tested description is 'Typical mature angiosperm embryo sac'.

BIO-12-CH01-0015 | Medium | Special reproductive phenomena**Correct answer: B. Parthenocarp**

Solution: The observation 'Fruit formation without fertilization' points to Parthenocarp; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Polyembryony belongs with 'Occurrence of more than one embryo in a seed', whereas the tested description is 'Fruit formation without fertilization'. C: Tapetum belongs with 'Anther wall layer that nourishes developing pollen', whereas the tested description is 'Fruit formation without fertilization'. D: Autogamy belongs with 'Transfer of pollen to the stigma of the same flower', whereas the tested description is 'Fruit formation without fertilization'.

BIO-12-CH01-0016 | Challenging | Ovule and embryo sac**Correct answer: B. I, II and III only**

Solution: Statement I is correct because 'Junction between the ovule body and its stalk' corresponds to Hilum. Statement II is correct because 'Opening left by the integuments at the ovule apex' corresponds to Micropyle. Statement III is correct because 'Typical mature angiosperm embryo sac' corresponds to Seven-celled and eight-nucleate. Statement IV is incorrect: 'Product of meiosis in a microspore mother cell' corresponds to A tetrad of haploid microspores, not Hilum. Therefore, the valid combination is I, II and III only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) III. C: This combination omits true statement(s) II. D: This combination includes false statement(s) IV and omits true statement(s) I, III.

BIO-12-CH01-0017 | Medium | Fertilization**Correct answer: D. Syngamy**

Solution: The chapter links Fusion of one male gamete with the egg with Syngamy, so Syngamy completes the statement.

Distractor review: A: Porogamy belongs with 'Pollen-tube entry into an ovule through the micropyle', whereas the tested description is 'Fusion of one male gamete with the egg'. B: Four belongs with 'Number of pollen sacs in a typical dithecous anther', whereas the tested description is 'Fusion of one male gamete with the egg'. C: Autogamy belongs with 'Transfer of pollen to the stigma of the same flower', whereas the tested description is 'Fusion of one male gamete with the egg'.

BIO-12-CH01-0018 | Medium | Special reproductive phenomena**Correct answer: D. Adventive embryony**

Solution: The chapter links Embryos arising directly from nucellus or integuments with Adventive embryony, so Adventive embryony completes the statement.

Distractor review: A: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Embryos arising directly from nucellus or integuments'. B: Polyembryony belongs with 'Occurrence of more than one embryo in a seed', whereas the tested description is 'Embryos arising directly from nucellus or integuments'. C: Micropyle belongs with 'Opening left by the integuments at the ovule apex', whereas the tested description is 'Embryos arising directly from nucellus or integuments'.

BIO-12-CH01-0019 | Hard | Post-fertilization development

Correct answer: B. Ovary after fertilization - Fruit; Embryos arising directly from nucellus or integuments - Adventive embryony

Solution: Both parts of 'Ovary after fertilization - Fruit; Embryos arising directly from nucellus or integuments - Adventive embryony' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Fusion of one male gamete with the egg' requires Syngamy, and 'Seed formation without fertilization' requires Apomixis. C: The answers have been interchanged: 'Typical mature angiosperm embryo sac' requires Seven-celled and eight-nucleate, and 'Opening left by the integuments at the ovule apex' requires Micropyle. D: The answers have been interchanged: 'Pollen-tube entry into an ovule through the micropyle' requires Porogamy, and 'Fusion of one male gamete with the egg' requires Syngamy.

BIO-12-CH01-0020 | Hard | Pollination

Correct answer: A. Transfer between flowers on different plants of the same species - Xenogamy; Junction between the ovule body and its stalk - Hilum

Solution: Both parts of 'Transfer between flowers on different plants of the same species - Xenogamy; Junction between the ovule body and its stalk - Hilum' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Fruit formation without fertilization' requires Parthenocarpy, and 'Typical mature angiosperm embryo sac' requires Seven-celled and eight-nucleate. C: The answers have been interchanged: 'Embryos arising directly from nucellus or integuments' requires Adventive embryony, and 'Primary endosperm nucleus generally develops into' requires Endosperm. D: The answers have been interchanged: 'Ovule after fertilization' requires Seed, and 'Embryos arising directly from nucellus or integuments' requires Adventive embryony.

BIO-12-CH01-0021 | Easy | Fertilization

Correct answer: B. Porogamy

Solution: 'Pollen-tube entry into an ovule through the micropyle' is the defining description for Porogamy. The other choices belong to different chapter concepts.

Distractor review: A: Hilum belongs with 'Junction between the ovule body and its stalk', whereas the tested description is 'Pollen-tube entry into an ovule through the micropyle'. C: Sporopollenin belongs with 'Highly resistant material of the pollen exine', whereas the tested description is 'Pollen-tube entry into an ovule through the micropyle'. D: Seed belongs with 'Ovule after fertilization', whereas the tested description is 'Pollen-tube entry into an ovule through the micropyle'.

BIO-12-CH01-0022 | Medium | Pollination

Correct answer: B. Anemophily

Solution: The observation 'Pollination carried out by wind' points to Anemophily; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Funicle belongs with 'Stalk attaching an ovule to the placenta', whereas the tested description is 'Pollination carried out by wind'. C: Seven-celled and eight-nucleate belongs with 'Typical mature angiosperm embryo sac', whereas the tested description is 'Pollination carried out by wind'. D: Double fertilization belongs with 'Syngamy and triple fusion occurring in one embryo sac', whereas the tested description is 'Pollination carried out by wind'.

BIO-12-CH01-0023 | Hard | Pollination

Correct answer: A. Transfer between flowers on different plants of the same species - Hilum

Solution: 'Transfer between flowers on different plants of the same species' should be paired with Xenogamy, not Hilum. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Embryos arising directly from nucellus or integuments - Adventive embryony. C: This is a valid association: Typical mature angiosperm embryo sac - Seven-celled and eight-nucleate. D: This is a valid association: Pollen-tube entry into an ovule through the micropyle - Porogamy.

BIO-12-CH01-0024 | Easy | Ovule and embryo sac

Correct answer: A. Hilum

Solution: 'Junction between the ovule body and its stalk' is the defining description for Hilum. The other choices belong to different chapter concepts.

Distractor review: B: Double fertilization belongs with 'Syngamy and triple fusion occurring in one embryo sac', whereas the tested description is 'Junction between the ovule body and its stalk'. C: Micropyle belongs with 'Opening left by the integuments at the ovule apex', whereas the tested description is 'Junction between the ovule body and its stalk'. D: Seed belongs with 'Ovule after fertilization', whereas the tested description is 'Junction between the ovule body and its stalk'.

BIO-12-CH01-0025 | Challenging | Microsporogenesis and pollen**Correct answer: B. III only**

Solution: Statement I is incorrect: 'Anther wall layer that nourishes developing pollen' corresponds to Tapetum, not Sporopollenin. Statement II is incorrect: 'Highly resistant material of the pollen exine' corresponds to Sporopollenin, not Four. Statement III is correct because 'Number of pollen sacs in a typical dithecal anther' corresponds to Four. Statement IV is incorrect: 'Embryos arising directly from nucellus or integuments' corresponds to Adventive embryony, not Tapetum. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) III. C: This combination includes false statement(s) I, IV and omits true statement(s) III. D: This combination includes false statement(s) I.

BIO-12-CH01-0026 | Easy | Asexual reproduction**Correct answer: D. Fragmentation in a filamentous alga**

Solution: Spirogyra represents 'Fragmentation in a filamentous alga'. Each alternative describes a different term or result.

Distractor review: A: 'Fusion of one male gamete with the egg' identifies Syngamy, not Spirogyra. B: 'Gemma formation' identifies Marchantia, not Spirogyra. C: 'Product of meiosis in a microspore mother cell' identifies A tetrad of haploid microspores, not Spirogyra.

BIO-12-CH01-0027 | Medium | Pollination**Correct answer: B. Xenogamy**

Solution: The observation 'Transfer between flowers on different plants of the same species' points to Xenogamy; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Seven-celled and eight-nucleate belongs with 'Typical mature angiosperm embryo sac', whereas the tested description is 'Transfer between flowers on different plants of the same species'. C: Parthenocarpy belongs with 'Fruit formation without fertilization', whereas the tested description is 'Transfer between flowers on different plants of the same species'. D: Porogamy belongs with 'Pollen-tube entry into an ovule through the micropyle', whereas the tested description is 'Transfer between flowers on different plants of the same species'.

BIO-12-CH01-0028 | Hard | Ovule and embryo sac**Correct answer: D. Junction between the ovule body and its stalk - Double fertilization**

Solution: 'Junction between the ovule body and its stalk' should be paired with Hilum, not Double fertilization. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Ovule after fertilization - Seed. B: This is a valid association: Budding under favourable conditions - Yeast. C: This is a valid association: Fusion of the second male gamete with the two polar nuclei - Triple fusion.

BIO-12-CH01-0029 | Medium | Fertilization**Correct answer: A. Syngamy and triple fusion occurring in one embryo sac - Double fertilization**

Solution: The accurate entry is 'Syngamy and triple fusion occurring in one embryo sac - Double fertilization'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Stalk attaching an ovule to the placenta' should be paired with Funicle, not Double fertilization. C: 'Fragmentation in a filamentous alga' should be paired with Spirogyra, not Geitonogamy. D: 'Transfer between different flowers of the same plant' should be paired with Geitonogamy, not Chlamydomonas.

BIO-12-CH01-0030 | Hard | Asexual reproduction**Correct answer: C. Budding under favourable conditions - Triple fusion**

Solution: 'Budding under favourable conditions' should be paired with Yeast, not Triple fusion. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Pollen-tube entry into an ovule through the micropyle - Porogamy. B: This is a valid association: Seed formation without fertilization - Apomixis. D: This is a valid association: Junction between the ovule body and its stalk - Hilum.

BIO-12-CH01-0031 | Medium | Special reproductive phenomena**Correct answer: B. Parthenocarp**

Solution: The chapter links Fruit formation without fertilization with Parthenocarp, so Parthenocarp completes the statement.

Distractor review: A: Fruit belongs with 'Ovary after fertilization', whereas the tested description is 'Fruit formation without fertilization'. C: Autogamy belongs with 'Transfer of pollen to the stigma of the same flower', whereas the tested description is 'Fruit formation without fertilization'. D: Adventive embryony belongs with 'Embryos arising directly from nucellus or integuments', whereas the tested description is 'Fruit formation without fertilization'.

BIO-12-CH01-0032 | Medium | Fertilization**Correct answer: D. Porogamy**

Solution: The chapter links Pollen-tube entry into an ovule through the micropyle with Porogamy, so Porogamy completes the statement.

Distractor review: A: Polyembryony belongs with 'Occurrence of more than one embryo in a seed', whereas the tested description is 'Pollen-tube entry into an ovule through the micropyle'. B: Hilum belongs with 'Junction between the ovule body and its stalk', whereas the tested description is 'Pollen-tube entry into an ovule through the micropyle'. C: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Pollen-tube entry into an ovule through the micropyle'.

BIO-12-CH01-0033 | Medium | Ovule and embryo sac**Correct answer: D. Seven-celled and eight-nucleate**

Solution: The chapter links Typical mature angiosperm embryo sac with Seven-celled and eight-nucleate, so Seven-celled and eight-nucleate completes the statement.

Distractor review: A: Four belongs with 'Number of pollen sacs in a typical dithecous anther', whereas the tested description is 'Typical mature angiosperm embryo sac'. B: Anemophily belongs with 'Pollination carried out by wind', whereas the tested description is 'Typical mature angiosperm embryo sac'. C: Yeast belongs with 'Budding under favourable conditions', whereas the tested description is 'Typical mature angiosperm embryo sac'.

BIO-12-CH01-0034 | Easy | Fertilization**Correct answer: C. Syngamy and triple fusion occurring in one embryo sac**

Solution: Double fertilization represents 'Syngamy and triple fusion occurring in one embryo sac'. Each alternative describes a different term or result.

Distractor review: A: 'Transfer between different flowers of the same plant' identifies Geitonogamy, not Double fertilization. B: 'Occurrence of more than one embryo in a seed' identifies Polyembryony, not Double fertilization. D: 'Motile zoospore formation' identifies Chlamydomonas, not Double fertilization.

BIO-12-CH01-0035 | Hard | Ovule and embryo sac**Correct answer: A. Opening left by the integuments at the ovule apex - Fruit**

Solution: 'Opening left by the integuments at the ovule apex' should be paired with Micropyle, not Fruit. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Anther wall layer that nourishes developing pollen - Tapetum. C: This is a valid association: Embryonic axis below the cotyledons - Hypocotyl. D: This is a valid association: Gemma formation - Marchantia.

BIO-12-CH01-0036 | Hard | Asexual reproduction**Correct answer: A. Gemma formation - Hypocotyl**

Solution: 'Gemma formation' should be paired with Marchantia, not Hypocotyl. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Highly resistant material of the pollen exine - Sporopollenin. C: This is a valid association: Occurrence of more than one embryo in a seed - Polyembryony. D: This is a valid association: Fruit formation without fertilization - Parthenocarp.

BIO-12-CH01-0037 | Hard | Special reproductive phenomena

Correct answer: D. Embryos arising directly from nucellus or integuments - Adventive embryony; Fusion of one male gamete with the egg - Syngamy

Solution: Both parts of 'Embryos arising directly from nucellus or integuments - Adventive embryony; Fusion of one male gamete with the egg - Syngamy' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Transfer between flowers on different plants of the same species' requires Xenogamy, and 'Primary endosperm nucleus generally develops into' requires Endosperm. B: The answers have been interchanged: 'Motile zoospore formation' requires Chlamydomonas, and 'Transfer between flowers on different plants of the same species' requires Xenogamy. C: The answers have been interchanged: 'Fusion of the second male gamete with the two polar nuclei' requires Triple fusion, and 'Embryonic axis below the cotyledons' requires Hypocotyl.

BIO-12-CH01-0038 | Medium | Asexual reproduction

Correct answer: C. Spirogyra

Solution: The chapter links Fragmentation in a filamentous alga with Spirogyra, so Spirogyra completes the statement.

Distractor review: A: Double fertilization belongs with 'Syngamy and triple fusion occurring in one embryo sac', whereas the tested description is 'Fragmentation in a filamentous alga'. B: A tetrad of haploid microspores belongs with 'Product of meiosis in a microspore mother cell', whereas the tested description is 'Fragmentation in a filamentous alga'. D: Yeast belongs with 'Budding under favourable conditions', whereas the tested description is 'Fragmentation in a filamentous alga'.

BIO-12-CH01-0039 | Medium | Post-fertilization development

Correct answer: C. Seed

Solution: The observation 'Ovule after fertilization' points to Seed; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Ovule after fertilization'. B: Porogamy belongs with 'Pollen-tube entry into an ovule through the micropyle', whereas the tested description is 'Ovule after fertilization'. D: Anemophily belongs with 'Pollination carried out by wind', whereas the tested description is 'Ovule after fertilization'.

BIO-12-CH01-0040 | Medium | Special reproductive phenomena

Correct answer: D. Embryos arising directly from nucellus or integuments - Adventive embryony

Solution: The accurate entry is 'Embryos arising directly from nucellus or integuments - Adventive embryony'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Primary endosperm nucleus generally develops into' should be paired with Endosperm, not Triple fusion. B: 'Opening left by the integuments at the ovule apex' should be paired with Micropyle, not Hypocotyl. C: 'Fusion of the second male gamete with the two polar nuclei' should be paired with Triple fusion, not Adventive embryony.

BIO-12-CH01-0041 | Easy | Post-fertilization development

Correct answer: A. Ovary after fertilization

Solution: Fruit represents 'Ovary after fertilization'. Each alternative describes a different term or result.

Distractor review: B: 'Fusion of one male gamete with the egg' identifies Syngamy, not Fruit. C: 'Primary endosperm nucleus generally develops into' identifies Endosperm, not Fruit. D: 'Highly resistant material of the pollen exine' identifies Sporopollenin, not Fruit.

BIO-12-CH01-0042 | Hard | Special reproductive phenomena

Correct answer: A. Fruit formation without fertilization - Parthenocarpy; Fusion of the second male gamete with the two polar nuclei - Triple fusion

Solution: Both parts of 'Fruit formation without fertilization - Parthenocarpy; Fusion of the second male gamete with the two polar nuclei - Triple fusion' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Transfer of pollen to the stigma of the same flower' requires Autogamy, and 'Budding under favourable conditions' requires Yeast. C: The answers have been interchanged: 'Anther wall layer that nourishes developing pollen' requires Tapetum, and 'Transfer of pollen to the stigma of the same flower' requires Autogamy. D: The answers have been interchanged: 'Opening left by the integuments at the ovule apex' requires Micropyle, and 'Product of meiosis in a microspore mother cell' requires A tetrad of haploid microspores.

BIO-12-CH01-0043 | Easy | Ovule and embryo sac**Correct answer: B. Opening left by the integuments at the ovule apex**

Solution: Micropyle represents 'Opening left by the integuments at the ovule apex'. Each alternative describes a different term or result.

Distractor review: A: 'Ovary after fertilization' identifies Fruit, not Micropyle. C: 'Gemma formation' identifies Marchantia, not Micropyle. D: 'Typical mature angiosperm embryo sac' identifies Seven-celled and eight-nucleate, not Micropyle.

BIO-12-CH01-0044 | Easy | Fertilization**Correct answer: A. Triple fusion**

Solution: 'Fusion of the second male gamete with the two polar nuclei' is the defining description for Triple fusion. The other choices belong to different chapter concepts.

Distractor review: B: Xenogamy belongs with 'Transfer between flowers on different plants of the same species', whereas the tested description is 'Fusion of the second male gamete with the two polar nuclei'. C: Anemophily belongs with 'Pollination carried out by wind', whereas the tested description is 'Fusion of the second male gamete with the two polar nuclei'. D: Double fertilization belongs with 'Syngamy and triple fusion occurring in one embryo sac', whereas the tested description is 'Fusion of the second male gamete with the two polar nuclei'.

BIO-12-CH01-0045 | Medium | Ovule and embryo sac**Correct answer: A. Hilum**

Solution: The chapter links Junction between the ovule body and its stalk with Hilum, so Hilum completes the statement.

Distractor review: B: Triple fusion belongs with 'Fusion of the second male gamete with the two polar nuclei', whereas the tested description is 'Junction between the ovule body and its stalk'. C: Yeast belongs with 'Budding under favourable conditions', whereas the tested description is 'Junction between the ovule body and its stalk'. D: Geitonogamy belongs with 'Transfer between different flowers of the same plant', whereas the tested description is 'Junction between the ovule body and its stalk'.

BIO-12-CH01-0046 | Easy | Asexual reproduction**Correct answer: C. Yeast**

Solution: 'Budding under favourable conditions' is the defining description for Yeast. The other choices belong to different chapter concepts.

Distractor review: A: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Budding under favourable conditions'. B: Seed belongs with 'Ovule after fertilization', whereas the tested description is 'Budding under favourable conditions'. D: Endosperm belongs with 'Primary endosperm nucleus generally develops into', whereas the tested description is 'Budding under favourable conditions'.

BIO-12-CH01-0047 | Challenging | Microsporogenesis and pollen**Correct answer: C. III and IV only**

Solution: Statement I is incorrect: 'Number of pollen sacs in a typical dithecous anther' corresponds to Four, not Sporopollenin. Statement II is incorrect: 'Highly resistant material of the pollen exine' corresponds to Sporopollenin, not A tetrad of haploid microspores. Statement III is correct because 'Product of meiosis in a microspore mother cell' corresponds to A tetrad of haploid microspores. Statement IV is correct because 'Ovule after fertilization' corresponds to Seed. Therefore, the valid combination is III and IV only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) IV. B: This combination includes false statement(s) I and omits true statement(s) III, IV. D: This combination includes false statement(s) I and omits true statement(s) IV.

BIO-12-CH01-0048 | Challenging | Asexual reproduction**Correct answer: B. II only**

Solution: Statement I is incorrect: 'Motile zoospore formation' corresponds to Chlamydomonas, not Marchantia. Statement II is correct because 'Gemma formation' corresponds to Marchantia. Statement III is incorrect: 'Budding under favourable conditions' corresponds to Yeast, not Xenogamy. Statement IV is incorrect: 'Transfer between flowers on different plants of the same species' corresponds to Xenogamy, not Chlamydomonas. Therefore, the valid combination is II only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) II. C: This combination includes false statement(s) III. D: This combination includes false statement(s) I, IV and omits true statement(s) II.

BIO-12-CH01-0049 | Hard | Asexual reproduction**Correct answer: D. Motile zoospore formation - Chlamydomonas; Seed formation without fertilization - Apomixis**

Solution: Both parts of 'Motile zoospore formation - Chlamydomonas; Seed formation without fertilization - Apomixis' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Transfer of pollen to the stigma of the same flower' requires Autogamy, and 'Number of pollen sacs in a typical ditheous anther' requires Four. B: The answers have been interchanged: 'Transfer between different flowers of the same plant' requires Geitonogamy, and 'Transfer of pollen to the stigma of the same flower' requires Autogamy. C: The answers have been interchanged: 'Embryonic axis below the cotyledons' requires Hypocotyl, and 'Transfer between different flowers of the same plant' requires Geitonogamy.

BIO-12-CH01-0050 | Challenging | Asexual reproduction**Correct answer: C. II and IV only**

Solution: Statement I is incorrect: 'Gemma formation' corresponds to Marchantia, not Chlamydomonas. Statement II is correct because 'Motile zoospore formation' corresponds to Chlamydomonas. Statement III is incorrect: 'Fragmentation in a filamentous alga' corresponds to Spirogyra, not Porogamy. Statement IV is correct because 'Pollen-tube entry into an ovule through the micropyle' corresponds to Porogamy. Therefore, the valid combination is II and IV only.

Distractor review: A: This combination includes false statement(s) I, III and omits true statement(s) II, IV. B: This combination includes false statement(s) I, III and omits true statement(s) II. D: This combination includes false statement(s) I and omits true statement(s) IV.

BIO-12-CH01-0051 | Medium | Fertilization**Correct answer: C. Syngamy**

Solution: The observation 'Fusion of one male gamete with the egg' points to Syngamy; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Adventive embryony belongs with 'Embryos arising directly from nucellus or integuments', whereas the tested description is 'Fusion of one male gamete with the egg'. B: Autogamy belongs with 'Transfer of pollen to the stigma of the same flower', whereas the tested description is 'Fusion of one male gamete with the egg'. D: Seed belongs with 'Ovule after fertilization', whereas the tested description is 'Fusion of one male gamete with the egg'.

BIO-12-CH01-0052 | Hard | Special reproductive phenomena**Correct answer: A. Occurrence of more than one embryo in a seed - Polyembryony; Transfer between different flowers of the same plant - Geitonogamy**

Solution: Both parts of 'Occurrence of more than one embryo in a seed - Polyembryony; Transfer between different flowers of the same plant - Geitonogamy' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Pollination carried out by wind' requires Anemophily, and 'Number of pollen sacs in a typical ditheous anther' requires Four. C: The answers have been interchanged: 'Opening left by the integuments at the ovule apex' requires Micropyle, and 'Fusion of one male gamete with the egg' requires Syngamy. D: The answers have been interchanged: 'Fusion of one male gamete with the egg' requires Syngamy, and 'Pollination carried out by wind' requires Anemophily.

BIO-12-CH01-0053 | Medium | Asexual reproduction**Correct answer: B. Yeast**

Solution: The observation 'Budding under favourable conditions' points to Yeast; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Triple fusion belongs with 'Fusion of the second male gamete with the two polar nuclei', whereas the tested description is 'Budding under favourable conditions'. C: Marchantia belongs with 'Gemma formation', whereas the tested description is 'Budding under favourable conditions'. D: Spirogyra belongs with 'Fragmentation in a filamentous alga', whereas the tested description is 'Budding under favourable conditions'.

BIO-12-CH01-0054 | Easy | Ovule and embryo sac**Correct answer: D. Stalk attaching an ovule to the placenta**

Solution: Funicle represents 'Stalk attaching an ovule to the placenta'. Each alternative describes a different term or result.

Distractor review: A: 'Fusion of one male gamete with the egg' identifies Syngamy, not Funicle. B: 'Fruit formation without fertilization' identifies Parthenocarp, not Funicle. C: 'Pollination carried out by wind' identifies Anemophily, not Funicle.

BIO-12-CH01-0055 | Easy | Post-fertilization development**Correct answer: A. Hypocotyl**

Solution: 'Embryonic axis below the cotyledons' is the defining description for Hypocotyl. The other choices belong to different chapter concepts.

Distractor review: B: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Embryonic axis below the cotyledons'. C: Funicle belongs with 'Stalk attaching an ovule to the placenta', whereas the tested description is 'Embryonic axis below the cotyledons'. D: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Embryonic axis below the cotyledons'.

BIO-12-CH01-0056 | Easy | Special reproductive phenomena**Correct answer: C. Occurrence of more than one embryo in a seed**

Solution: Polyembryony represents 'Occurrence of more than one embryo in a seed'. Each alternative describes a different term or result.

Distractor review: A: 'Transfer between different flowers of the same plant' identifies Geitonogamy, not Polyembryony. B: 'Opening left by the integuments at the ovule apex' identifies Micropyle, not Polyembryony. D: 'Fusion of one male gamete with the egg' identifies Syngamy, not Polyembryony.

BIO-12-CH01-0057 | Hard | Microsporogenesis and pollen**Correct answer: D. Highly resistant material of the pollen exine - Sporopollenin; Fusion of one male gamete with the egg - Syngamy**

Solution: Both parts of 'Highly resistant material of the pollen exine - Sporopollenin; Fusion of one male gamete with the egg - Syngamy' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Transfer between different flowers of the same plant' requires Geitonogamy, and 'Seed formation without fertilization' requires Apomixis. B: The answers have been interchanged: 'Pollen-tube entry into an ovule through the micropyle' requires Porogamy, and 'Embryonic axis below the cotyledons' requires Hypocotyl. C: The answers have been interchanged: 'Embryonic axis below the cotyledons' requires Hypocotyl, and 'Syngamy and triple fusion occurring in one embryo sac' requires Double fertilization.

BIO-12-CH01-0058 | Challenging | Pollination**Correct answer: A. I only**

Solution: Statement I is correct because 'Transfer between different flowers of the same plant' corresponds to Geitonogamy. Statement II is incorrect: 'Transfer of pollen to the stigma of the same flower' corresponds to Autogamy, not Anemophily. Statement III is incorrect: 'Pollination carried out by wind' corresponds to Anemophily, not Double fertilization. Statement IV is incorrect: 'Syngamy and triple fusion occurring in one embryo sac' corresponds to Double fertilization, not Geitonogamy. Therefore, the valid combination is I only.

Distractor review: B: This combination includes false statement(s) III. C: This combination includes false statement(s) III, IV and omits true statement(s) I. D: This combination includes false statement(s) II, IV.

BIO-12-CH01-0059 | Hard | Asexual reproduction**Correct answer: A. Fragmentation in a filamentous alga - Spirogyra; Product of meiosis in a microspore mother cell - A tetrad of haploid microspores**

Solution: Both parts of 'Fragmentation in a filamentous alga - Spirogyra; Product of meiosis in a microspore mother cell - A tetrad of haploid microspores' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Occurrence of more than one embryo in a seed' requires Polyembryony, and 'Highly resistant material of the pollen exine' requires Sporopollenin. C: The answers have been interchanged: 'Syngamy and triple fusion occurring in one embryo sac' requires Double fertilization, and 'Fruit formation without fertilization' requires Parthenocarpy. D: The answers have been interchanged: 'Anther wall layer that nourishes developing pollen' requires Tapetum, and 'Occurrence of more than one embryo in a seed' requires Polyembryony.

BIO-12-CH01-0060 | Hard | Microsporogenesis and pollen**Correct answer: C. Number of pollen sacs in a typical dithecous anther - Geitonogamy**

Solution: 'Number of pollen sacs in a typical dithecous anther' should be paired with Four, not Geitonogamy. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Opening left by the integuments at the ovule apex - Micropyle. B: This is a valid association: Stalk attaching an ovule to the placenta - Funicle. D: This is a valid association: Seed formation without fertilization - Apomixis.

BIO-12-CH01-0061 | Easy | Special reproductive phenomena**Correct answer: B. Embryos arising directly from nucellus or integuments**

Solution: Adventive embryony represents 'Embryos arising directly from nucellus or integuments'. Each alternative describes a different term or result.

Distractor review: A: 'Fusion of one male gamete with the egg' identifies Syngamy, not Adventive embryony. C: 'Transfer between flowers on different plants of the same species' identifies Xenogamy, not Adventive embryony. D: 'Primary endosperm nucleus generally develops into' identifies Endosperm, not Adventive embryony.

BIO-12-CH01-0062 | Hard | Fertilization**Correct answer: D. Fusion of the second male gamete with the two polar nuclei - Triple fusion; Highly resistant material of the pollen exine - Sporopollenin**

Solution: Both parts of 'Fusion of the second male gamete with the two polar nuclei - Triple fusion; Highly resistant material of the pollen exine - Sporopollenin' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Transfer between flowers on different plants of the same species' requires Xenogamy, and 'Embryos arising directly from nucellus or integuments' requires Adventive embryony. B: The answers have been interchanged: 'Seed formation without fertilization' requires Apomixis, and 'Motile zoospore formation' requires Chlamydomonas. C: The answers have been interchanged: 'Motile zoospore formation' requires Chlamydomonas, and 'Pollination carried out by wind' requires Anemophily.

BIO-12-CH01-0063 | Hard | Ovule and embryo sac**Correct answer: A. Opening left by the integuments at the ovule apex - Micropyle; Ovary after fertilization - Fruit**

Solution: Both parts of 'Opening left by the integuments at the ovule apex - Micropyle; Ovary after fertilization - Fruit' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Anther wall layer that nourishes developing pollen' requires Tapetum, and 'Embryonic axis below the cotyledons' requires Hypocotyl. C: The answers have been interchanged: 'Gemma formation' requires Marchantia, and 'Number of pollen sacs in a typical ditheous anther' requires Four. D: The answers have been interchanged: 'Number of pollen sacs in a typical ditheous anther' requires Four, and 'Pollen-tube entry into an ovule through the micropyle' requires Porogamy.

BIO-12-CH01-0064 | Hard | Post-fertilization development**Correct answer: A. Ovary after fertilization - Adventive embryony**

Solution: 'Ovary after fertilization' should be paired with Fruit, not Adventive embryony. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Seed formation without fertilization - Apomixis. C: This is a valid association: Typical mature angiosperm embryo sac - Seven-celled and eight-nucleate. D: This is a valid association: Pollen-tube entry into an ovule through the micropyle - Porogamy.

BIO-12-CH01-0065 | Hard | Post-fertilization development**Correct answer: D. Ovule after fertilization - Seed; Stalk attaching an ovule to the placenta - Funicle**

Solution: Both parts of 'Ovule after fertilization - Seed; Stalk attaching an ovule to the placenta - Funicle' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Fusion of one male gamete with the egg' requires Syngamy, and 'Pollination carried out by wind' requires Anemophily. B: The answers have been interchanged: 'Pollination carried out by wind' requires Anemophily, and 'Product of meiosis in a microspore mother cell' requires A tetrad of haploid microspores. C: The answers have been interchanged: 'Product of meiosis in a microspore mother cell' requires A tetrad of haploid microspores, and 'Fruit formation without fertilization' requires Parthenocarp.

BIO-12-CH01-0066 | Challenging | Post-fertilization development**Correct answer: A. II and III only**

Solution: Statement I is incorrect: 'Ovary after fertilization' corresponds to Fruit, not Seed. Statement II is correct because 'Ovule after fertilization' corresponds to Seed. Statement III is correct because 'Primary endosperm nucleus generally develops into' corresponds to Endosperm. Statement IV is incorrect: 'Number of pollen sacs in a typical ditheous anther' corresponds to Four, not Fruit. Therefore, the valid combination is II and III only.

Distractor review: B: This combination omits true statement(s) II. C: This combination omits true statement(s) III. D: This combination includes false statement(s) IV and omits true statement(s) II, III.

BIO-12-CH01-0067 | Medium | Microsporogenesis and pollen**Correct answer: D. Tapetum**

Solution: The chapter links Anther wall layer that nourishes developing pollen with Tapetum, so Tapetum completes the statement.

Distractor review: A: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Anther wall layer that nourishes developing pollen'. B: Four belongs with 'Number of pollen sacs in a typical dithecous anther', whereas the tested description is 'Anther wall layer that nourishes developing pollen'. C: Sporopollenin belongs with 'Highly resistant material of the pollen exine', whereas the tested description is 'Anther wall layer that nourishes developing pollen'.

BIO-12-CH01-0068 | Medium | Post-fertilization development**Correct answer: D. Endosperm**

Solution: The chapter links Primary endosperm nucleus generally develops into with Endosperm, so Endosperm completes the statement.

Distractor review: A: Spirogyra belongs with 'Fragmentation in a filamentous alga', whereas the tested description is 'Primary endosperm nucleus generally develops into'. B: Funicle belongs with 'Stalk attaching an ovule to the placenta', whereas the tested description is 'Primary endosperm nucleus generally develops into'. C: Hilum belongs with 'Junction between the ovule body and its stalk', whereas the tested description is 'Primary endosperm nucleus generally develops into'.

BIO-12-CH01-0069 | Challenging | Pollination**Correct answer: C. I, II, III and IV**

Solution: Statement I is correct because 'Transfer of pollen to the stigma of the same flower' corresponds to Autogamy. Statement II is correct because 'Transfer between flowers on different plants of the same species' corresponds to Xenogamy. Statement III is correct because 'Transfer between different flowers of the same plant' corresponds to Geitonogamy. Statement IV is correct because 'Fragmentation in a filamentous alga' corresponds to Spirogyra. Therefore, the valid combination is I, II, III and IV.

Distractor review: A: This combination omits true statement(s) I. B: This combination omits true statement(s) I, III. D: This combination omits true statement(s) II, IV.

BIO-12-CH01-0070 | Easy | Pollination**Correct answer: C. Transfer between flowers on different plants of the same species**

Solution: Xenogamy represents 'Transfer between flowers on different plants of the same species'. Each alternative describes a different term or result.

Distractor review: A: 'Junction between the ovule body and its stalk' identifies Hilum, not Xenogamy. B: 'Transfer between different flowers of the same plant' identifies Geitonogamy, not Xenogamy. D: 'Typical mature angiosperm embryo sac' identifies Seven-celled and eight-nucleate, not Xenogamy.

BIO-12-CH01-0071 | Medium | Microsporogenesis and pollen**Correct answer: D. Four**

Solution: The chapter links Number of pollen sacs in a typical dithecous anther with Four, so Four completes the statement.

Distractor review: A: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Number of pollen sacs in a typical dithecous anther'. B: Geitonogamy belongs with 'Transfer between different flowers of the same plant', whereas the tested description is 'Number of pollen sacs in a typical dithecous anther'. C: Tapetum belongs with 'Anther wall layer that nourishes developing pollen', whereas the tested description is 'Number of pollen sacs in a typical dithecous anther'.

BIO-12-CH01-0072 | Hard | Microsporogenesis and pollen**Correct answer: B. Number of pollen sacs in a typical dithecous anther - Four; Transfer between different flowers of the same plant - Geitonogamy**

Solution: Both parts of 'Number of pollen sacs in a typical dithecous anther - Four; Transfer between different flowers of the same plant - Geitonogamy' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Seed formation without fertilization' requires Apomixis, and 'Opening left by the integuments at the ovule apex' requires Micropyle. C: The answers have been interchanged: 'Typical mature angiosperm embryo sac' requires Seven-celled and eight-nucleate, and 'Seed formation without fertilization' requires Apomixis. D: The answers have been interchanged: 'Stalk attaching an ovule to the placenta' requires Funicle, and 'Embryos arising directly from nucellus or integuments' requires Adventive embryony.

BIO-12-CH01-0073 | Hard | Post-fertilization development**Correct answer: C. Embryonic axis below the cotyledons - Funicle**

Solution: 'Embryonic axis below the cotyledons' should be paired with Hypocotyl, not Funicle. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Budding under favourable conditions - Yeast. B: This is a valid association: Transfer of pollen to the stigma of the same flower - Autogamy. D: This is a valid association: Fusion of one male gamete with the egg - Syngamy.

BIO-12-CH01-0074 | Medium | Post-fertilization development**Correct answer: C. Primary endosperm nucleus generally develops into - Endosperm**

Solution: The accurate entry is 'Primary endosperm nucleus generally develops into - Endosperm'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Transfer between different flowers of the same plant' should be paired with Geitonogamy, not Four. B: 'Fragmentation in a filamentous alga' should be paired with Spirogyra, not Yeast. D: 'Budding under favourable conditions' should be paired with Yeast, not Funicle.

BIO-12-CH01-0075 | Medium | Fertilization**Correct answer: A. Triple fusion**

Solution: The observation 'Fusion of the second male gamete with the two polar nuclei' points to Triple fusion; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Anemophily belongs with 'Pollination carried out by wind', whereas the tested description is 'Fusion of the second male gamete with the two polar nuclei'. C: Chlamydomonas belongs with 'Motile zoospore formation', whereas the tested description is 'Fusion of the second male gamete with the two polar nuclei'. D: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Fusion of the second male gamete with the two polar nuclei'.

BIO-12-CH01-0076 | Hard | Pollination**Correct answer: C. Transfer of pollen to the stigma of the same flower - Autogamy; Highly resistant material of the pollen exine - Sporopollenin**

Solution: Both parts of 'Transfer of pollen to the stigma of the same flower - Autogamy; Highly resistant material of the pollen exine - Sporopollenin' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Syngamy and triple fusion occurring in one embryo sac' requires Double fertilization, and 'Anther wall layer that nourishes developing pollen' requires Tapetum. B: The answers have been interchanged: 'Gemma formation' requires Marchantia, and 'Seed formation without fertilization' requires Apomixis. D: The answers have been interchanged: 'Fragmentation in a filamentous alga' requires Spirogyra, and 'Syngamy and triple fusion occurring in one embryo sac' requires Double fertilization.

BIO-12-CH01-0077 | Medium | Special reproductive phenomena**Correct answer: A. Polyembryony**

Solution: The observation 'Occurrence of more than one embryo in a seed' points to Polyembryony; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Occurrence of more than one embryo in a seed'. C: Micropyle belongs with 'Opening left by the integuments at the ovule apex', whereas the tested description is 'Occurrence of more than one embryo in a seed'. D: Geitonogamy belongs with 'Transfer between different flowers of the same plant', whereas the tested description is 'Occurrence of more than one embryo in a seed'.

BIO-12-CH01-0078 | Easy | Asexual reproduction**Correct answer: B. Marchantia**

Solution: 'Gemma formation' is the defining description for Marchantia. The other choices belong to different chapter concepts.

Distractor review: A: Sporopollenin belongs with 'Highly resistant material of the pollen exine', whereas the tested description is 'Gemma formation'. C: Spirogyra belongs with 'Fragmentation in a filamentous alga', whereas the tested description is 'Gemma formation'. D: Autogamy belongs with 'Transfer of pollen to the stigma of the same flower', whereas the tested description is 'Gemma formation'.

BIO-12-CH01-0079 | Medium | Ovule and embryo sac**Correct answer: D. Micropyle**

Solution: The chapter links Opening left by the integuments at the ovule apex with Micropyle, so Micropyle completes the statement.

Distractor review: A: Seven-celled and eight-nucleate belongs with 'Typical mature angiosperm embryo sac', whereas the tested description is 'Opening left by the integuments at the ovule apex'. B: Anemophily belongs with 'Pollination carried out by wind', whereas the tested description is 'Opening left by the integuments at the ovule apex'. C: Tapetum belongs with 'Anther wall layer that nourishes developing pollen', whereas the tested description is 'Opening left by the integuments at the ovule apex'.

BIO-12-CH01-0080 | Medium | Microsporogenesis and pollen**Correct answer: C. Four**

Solution: The observation 'Number of pollen sacs in a typical ditheous anther' points to Four; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Number of pollen sacs in a typical ditheous anther'. B: Tapetum belongs with 'Anther wall layer that nourishes developing pollen', whereas the tested description is 'Number of pollen sacs in a typical ditheous anther'. D: Seven-celled and eight-nucleate belongs with 'Typical mature angiosperm embryo sac', whereas the tested description is 'Number of pollen sacs in a typical ditheous anther'.

BIO-12-CH01-0081 | Easy | Microsporogenesis and pollen**Correct answer: C. Four**

Solution: 'Number of pollen sacs in a typical ditheous anther' is the defining description for Four. The other choices belong to different chapter concepts.

Distractor review: A: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Number of pollen sacs in a typical ditheous anther'. B: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Number of pollen sacs in a typical ditheous anther'. D: Tapetum belongs with 'Anther wall layer that nourishes developing pollen', whereas the tested description is 'Number of pollen sacs in a typical ditheous anther'.

BIO-12-CH01-0082 | Medium | Pollination**Correct answer: D. Autogamy**

Solution: The observation 'Transfer of pollen to the stigma of the same flower' points to Autogamy; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Xenogamy belongs with 'Transfer between flowers on different plants of the same species', whereas the tested description is 'Transfer of pollen to the stigma of the same flower'. B: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Transfer of pollen to the stigma of the same flower'. C: Spirogyra belongs with 'Fragmentation in a filamentous alga', whereas the tested description is 'Transfer of pollen to the stigma of the same flower'.

BIO-12-CH01-0083 | Hard | Ovule and embryo sac**Correct answer: A. Stalk attaching an ovule to the placenta - Funicle; Embryos arising directly from nucellus or integuments - Adventive embryony**

Solution: Both parts of 'Stalk attaching an ovule to the placenta - Funicle; Embryos arising directly from nucellus or integuments - Adventive embryony' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Fusion of one male gamete with the egg' requires Syngamy, and 'Fruit formation without fertilization' requires Parthenocarp. C: The answers have been interchanged: 'Fruit formation without fertilization' requires Parthenocarp, and 'Ovary after fertilization' requires Fruit. D: The answers have been interchanged: 'Ovule after fertilization' requires Seed, and 'Transfer between different flowers of the same plant' requires Geitonogamy.

BIO-12-CH01-0084 | Medium | Ovule and embryo sac**Correct answer: C. Seven-celled and eight-nucleate**

Solution: The observation 'Typical mature angiosperm embryo sac' points to Seven-celled and eight-nucleate; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Micropyle belongs with 'Opening left by the integuments at the ovule apex', whereas the tested description is 'Typical mature angiosperm embryo sac'. B: Parthenocarp belongs with 'Fruit formation without fertilization', whereas the tested description is 'Typical mature angiosperm embryo sac'. D: Anemophily belongs with 'Pollination carried out by wind', whereas the tested description is 'Typical mature angiosperm embryo sac'.

BIO-12-CH01-0085 | Medium | Fertilization**Correct answer: D. Double fertilization**

Solution: The observation 'Syngamy and triple fusion occurring in one embryo sac' points to Double fertilization; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Hilum belongs with 'Junction between the ovule body and its stalk', whereas the tested description is 'Syngamy and triple fusion occurring in one embryo sac'. B: Polyembryony belongs with 'Occurrence of more than one embryo in a seed', whereas the tested description is 'Syngamy and triple fusion occurring in one embryo sac'. C: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Syngamy and triple fusion occurring in one embryo sac'.

BIO-12-CH01-0086 | Easy | Fertilization**Correct answer: B. Fusion of the second male gamete with the two polar nuclei**

Solution: Triple fusion represents 'Fusion of the second male gamete with the two polar nuclei'. Each alternative describes a different term or result.

Distractor review: A: 'Transfer between flowers on different plants of the same species' identifies Xenogamy, not Triple fusion. C: 'Motile zoospore formation' identifies Chlamydomonas, not Triple fusion. D: 'Pollination carried out by wind' identifies Anemophily, not Triple fusion.

BIO-12-CH01-0087 | Medium | Special reproductive phenomena**Correct answer: C. Adventive embryony**

Solution: The observation 'Embryos arising directly from nucellus or integuments' points to Adventive embryony; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Embryos arising directly from nucellus or integuments'. B: Xenogamy belongs with 'Transfer between flowers on different plants of the same species', whereas the tested description is 'Embryos arising directly from nucellus or integuments'. D: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Embryos arising directly from nucellus or integuments'.

BIO-12-CH01-0088 | Medium | Ovule and embryo sac**Correct answer: B. Typical mature angiosperm embryo sac - Seven-celled and eight-nucleate**

Solution: The accurate entry is 'Typical mature angiosperm embryo sac - Seven-celled and eight-nucleate'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Pollination carried out by wind' should be paired with Anemophily, not Geitonogamy. C: 'Ovary after fertilization' should be paired with Fruit, not Xenogamy. D: 'Transfer between different flowers of the same plant' should be paired with Geitonogamy, not Seven-celled and eight-nucleate.

BIO-12-CH01-0089 | Hard | Ovule and embryo sac**Correct answer: A. Typical mature angiosperm embryo sac - Yeast**

Solution: 'Typical mature angiosperm embryo sac' should be paired with Seven-celled and eight-nucleate, not Yeast. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Ovary after fertilization - Fruit. C: This is a valid association: Transfer between different flowers of the same plant - Geitonogamy. D: This is a valid association: Number of pollen sacs in a typical ditheous anther - Four.

BIO-12-CH01-0090 | Medium | Asexual reproduction**Correct answer: A. Budding under favourable conditions - Yeast**

Solution: The accurate entry is 'Budding under favourable conditions - Yeast'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Primary endosperm nucleus generally develops into' should be paired with Endosperm, not Porogamy. C: 'Ovule after fertilization' should be paired with Seed, not Apomixis. D: 'Junction between the ovule body and its stalk' should be paired with Hilum, not Endosperm.

BIO-12-CH01-0091 | Medium | Post-fertilization development**Correct answer: D. Embryonic axis below the cotyledons - Hypocotyl**

Solution: The accurate entry is 'Embryonic axis below the cotyledons - Hypocotyl'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Highly resistant material of the pollen exine' should be paired with Sporopollenin, not Yeast. B: 'Budding under favourable conditions' should be paired with Yeast, not Hypocotyl. C: 'Seed formation without fertilization' should be paired with Apomixis, not Tapetum.

BIO-12-CH01-0092 | Medium | Ovule and embryo sac**Correct answer: B. Funicle**

Solution: The chapter links Stalk attaching an ovule to the placenta with Funicle, so Funicle completes the statement.

Distractor review: A: Parthenocarpy belongs with 'Fruit formation without fertilization', whereas the tested description is 'Stalk attaching an ovule to the placenta'. C: Micropyle belongs with 'Opening left by the integuments at the ovule apex', whereas the tested description is 'Stalk attaching an ovule to the placenta'. D: Fruit belongs with 'Ovary after fertilization', whereas the tested description is 'Stalk attaching an ovule to the placenta'.

BIO-12-CH01-0093 | Medium | Pollination**Correct answer: B. Geitonogamy**

Solution: The chapter links Transfer between different flowers of the same plant with Geitonogamy, so Geitonogamy completes the statement.

Distractor review: A: Spirogyra belongs with 'Fragmentation in a filamentous alga', whereas the tested description is 'Transfer between different flowers of the same plant'. C: Adventive embryony belongs with 'Embryos arising directly from nucellus or integuments', whereas the tested description is 'Transfer between different flowers of the same plant'. D: Autogamy belongs with 'Transfer of pollen to the stigma of the same flower', whereas the tested description is 'Transfer between different flowers of the same plant'.

BIO-12-CH01-0094 | Medium | Microsporogenesis and pollen**Correct answer: D. Anther wall layer that nourishes developing pollen - Tapetum**

Solution: The accurate entry is 'Anther wall layer that nourishes developing pollen - Tapetum'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Primary endosperm nucleus generally develops into' should be paired with Endosperm, not Autogamy. B: 'Typical mature angiosperm embryo sac' should be paired with Seven-celled and eight-nucleate, not Seed. C: 'Fusion of one male gamete with the egg' should be paired with Syngamy, not Endosperm.

BIO-12-CH01-0095 | Medium | Pollination**Correct answer: C. Geitonogamy**

Solution: The observation 'Transfer between different flowers of the same plant' points to Geitonogamy; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Parthenocarpy belongs with 'Fruit formation without fertilization', whereas the tested description is 'Transfer between different flowers of the same plant'. B: Adventive embryony belongs with 'Embryos arising directly from nucellus or integuments', whereas the tested description is 'Transfer between different flowers of the same plant'. D: Anemophily belongs with 'Pollination carried out by wind', whereas the tested description is 'Transfer between different flowers of the same plant'.

BIO-12-CH01-0096 | Hard | Special reproductive phenomena**Correct answer: D. Seed formation without fertilization - Apomixis; Product of meiosis in a microspore mother cell - A tetrad of haploid microspores**

Solution: Both parts of 'Seed formation without fertilization - Apomixis; Product of meiosis in a microspore mother cell - A tetrad of haploid microspores' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Number of pollen sacs in a typical dithecous anther' requires Four, and 'Pollination carried out by wind' requires Anemophily. B: The answers have been interchanged: 'Typical mature angiosperm embryo sac' requires Seven-celled and eight-nucleate, and 'Number of pollen sacs in a typical dithecous anther' requires Four. C: The answers have been interchanged: 'Motile zoospore formation' requires Chlamydomonas, and 'Typical mature angiosperm embryo sac' requires Seven-celled and eight-nucleate.

BIO-12-CH01-0097 | Medium | Post-fertilization development**Correct answer: B. Fruit**

Solution: The chapter links Ovary after fertilization with Fruit, so Fruit completes the statement.

Distractor review: A: Seed belongs with 'Ovule after fertilization', whereas the tested description is 'Ovary after fertilization'. C: Endosperm belongs with 'Primary endosperm nucleus generally develops into', whereas the tested description is 'Ovary after fertilization'. D: Sporopollenin belongs with 'Highly resistant material of the pollen exine', whereas the tested description is 'Ovary after fertilization'.

BIO-12-CH01-0098 | Medium | Pollination**Correct answer: D. Autogamy**

Solution: The chapter links Transfer of pollen to the stigma of the same flower with Autogamy, so Autogamy completes the statement.

Distractor review: A: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Transfer of pollen to the stigma of the same flower'. B: Double fertilization belongs with 'Syngamy and triple fusion occurring in one embryo sac', whereas the tested description is 'Transfer of pollen to the stigma of the same flower'. C: Marchantia belongs with 'Gemma formation', whereas the tested description is 'Transfer of pollen to the stigma of the same flower'.

BIO-12-CH01-0099 | Medium | Special reproductive phenomena**Correct answer: D. Seed formation without fertilization - Apomixis**

Solution: The accurate entry is 'Seed formation without fertilization - Apomixis'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Number of pollen sacs in a typical dithecous anther' should be paired with Four, not Apomixis. B: 'Transfer of pollen to the stigma of the same flower' should be paired with Autogamy, not Chlamydomonas. C: 'Product of meiosis in a microspore mother cell' should be paired with A tetrad of haploid microspores, not Hypocotyl.

BIO-12-CH01-0100 | Challenging | Fertilization**Correct answer: B. I and IV only**

Solution: Statement I is correct because 'Syngamy and triple fusion occurring in one embryo sac' corresponds to Double fertilization. Statement II is incorrect: 'Fusion of the second male gamete with the two polar nuclei' corresponds to Triple fusion, not Porogamy. Statement III is incorrect: 'Pollen-tube entry into an ovule through the micropyle' corresponds to Porogamy, not A tetrad of haploid microspores. Statement IV is correct because 'Product of meiosis in a microspore mother cell' corresponds to A tetrad of haploid microspores. Therefore, the valid combination is I and IV only.

Distractor review: A: This combination includes false statement(s) III. C: This combination includes false statement(s) III and omits true statement(s) I, IV. D: This combination includes false statement(s) II.

BIO-12-CH01-0101 | Medium | Asexual reproduction**Correct answer: A. Spirogyra**

Solution: The observation 'Fragmentation in a filamentous alga' points to Spirogyra; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: A tetrad of haploid microspores belongs with 'Product of meiosis in a microspore mother cell', whereas the tested description is 'Fragmentation in a filamentous alga'. C: Syngamy belongs with 'Fusion of one male gamete with the egg', whereas the tested description is 'Fragmentation in a filamentous alga'. D: Double fertilization belongs with 'Syngamy and triple fusion occurring in one embryo sac', whereas the tested description is 'Fragmentation in a filamentous alga'.

BIO-12-CH01-0102 | Easy | Post-fertilization development**Correct answer: D. Ovule after fertilization**

Solution: Seed represents 'Ovule after fertilization'. Each alternative describes a different term or result.

Distractor review: A: 'Pollination carried out by wind' identifies Anemophily, not Seed. B: 'Primary endosperm nucleus generally develops into' identifies Endosperm, not Seed. C: 'Fusion of one male gamete with the egg' identifies Syngamy, not Seed.

BIO-12-CH01-0103 | Hard | Fertilization**Correct answer: B. Syngamy and triple fusion occurring in one embryo sac - Double fertilization; Fragmentation in a filamentous alga - Spirogyra**

Solution: Both parts of 'Syngamy and triple fusion occurring in one embryo sac - Double fertilization; Fragmentation in a filamentous alga - Spirogyra' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Stalk attaching an ovule to the placenta' requires Funicle, and 'Transfer between flowers on different plants of the same species' requires Xenogamy. C: The answers have been interchanged: 'Transfer between different flowers of the same plant' requires Geitonogamy, and 'Junction between the ovule body and its stalk' requires Hilum. D: The answers have been interchanged: 'Junction between the ovule body and its stalk' requires Hilum, and 'Motile zoospore formation' requires Chlamydomonas.

BIO-12-CH01-0104 | Hard | Pollination**Correct answer: A. Pollination carried out by wind - Seven-celled and eight-nucleate**

Solution: 'Pollination carried out by wind' should be paired with Anemophily, not Seven-celled and eight-nucleate. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Pollen-tube entry into an ovule through the micropyle - Porogamy. C: This is a valid association: Stalk attaching an ovule to the placenta - Funicle. D: This is a valid association: Anther wall layer that nourishes developing pollen - Tapetum.

BIO-12-CH01-0105 | Easy | Pollination**Correct answer: C. Xenogamy**

Solution: 'Transfer between flowers on different plants of the same species' is the defining description for Xenogamy. The other choices belong to different chapter concepts.

Distractor review: A: Adventive embryony belongs with 'Embryos arising directly from nucellus or integuments', whereas the tested description is 'Transfer between flowers on different plants of the same species'. B: Sporopollenin belongs with 'Highly resistant material of the pollen exine', whereas the tested description is 'Transfer between flowers on different plants of the same species'. D: Porogamy belongs with 'Pollen-tube entry into an ovule through the micropyle', whereas the tested description is 'Transfer between flowers on different plants of the same species'.

BIO-12-CH01-0106 | Medium | Fertilization**Correct answer: B. Triple fusion**

Solution: The chapter links Fusion of the second male gamete with the two polar nuclei with Triple fusion, so Triple fusion completes the statement.

Distractor review: A: Apomixis belongs with 'Seed formation without fertilization', whereas the tested description is 'Fusion of the second male gamete with the two polar nuclei'. C: Chlamydomonas belongs with 'Motile zoospore formation', whereas the tested description is 'Fusion of the second male gamete with the two polar nuclei'. D: Anemophily belongs with 'Pollination carried out by wind', whereas the tested description is 'Fusion of the second male gamete with the two polar nuclei'.

BIO-12-CH01-0107 | Hard | Fertilization**Correct answer: D. Pollen-tube entry into an ovule through the micropyle - Hilum**

Solution: 'Pollen-tube entry into an ovule through the micropyle' should be paired with Porogamy, not Hilum. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Ovule after fertilization - Seed. B: This is a valid association: Highly resistant material of the pollen exine - Sporopollenin. C: This is a valid association: Ovary after fertilization - Fruit.

BIO-12-CH01-0108 | Easy | Special reproductive phenomena**Correct answer: C. Seed formation without fertilization**

Solution: Apomixis represents 'Seed formation without fertilization'. Each alternative describes a different term or result.

Distractor review: A: 'Typical mature angiosperm embryo sac' identifies Seven-celled and eight-nucleate, not Apomixis. B: 'Occurrence of more than one embryo in a seed' identifies Polyembryony, not Apomixis. D: 'Embryos arising directly from nucellus or integuments' identifies Adventive embryony, not Apomixis.

BIO-12-CH01-0109 | Medium | Microsporogenesis and pollen**Correct answer: A. Number of pollen sacs in a typical dithecal anther - Four**

Solution: The accurate entry is 'Number of pollen sacs in a typical dithecal anther - Four'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Transfer between different flowers of the same plant' should be paired with Geitonogamy, not Apomixis. C: 'Typical mature angiosperm embryo sac' should be paired with Seven-celled and eight-nucleate, not Micropyle. D: 'Stalk attaching an ovule to the placenta' should be paired with Funicle, not Four.

BIO-12-CH01-0110 | Easy | Microsporogenesis and pollen**Correct answer: C. A tetrad of haploid microspores**

Solution: 'Product of meiosis in a microspore mother cell' is the defining description for A tetrad of haploid microspores. The other choices belong to different chapter concepts.

Distractor review: A: Autogamy belongs with 'Transfer of pollen to the stigma of the same flower', whereas the tested description is 'Product of meiosis in a microspore mother cell'. B: Hilum belongs with 'Junction between the ovule body and its stalk', whereas the tested description is 'Product of meiosis in a microspore mother cell'. D: Double fertilization belongs with 'Syngamy and triple fusion occurring in one embryo sac', whereas the tested description is 'Product of meiosis in a microspore mother cell'.

BIO-12-CH01-0111 | Challenging | Special reproductive phenomena**Correct answer: C. II, III and IV only**

Solution: Statement I is incorrect: 'Fruit formation without fertilization' corresponds to Parthenocarpy, not Adventive embryony. Statement II is correct because 'Embryos arising directly from nucellus or integuments' corresponds to Adventive embryony. Statement III is correct because 'Seed formation without fertilization' corresponds to Apomixis. Statement IV is correct because 'Pollination carried out by wind' corresponds to Anemophily. Therefore, the valid combination is II, III and IV only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) II, III. B: This combination includes false statement(s) I and omits true statement(s) IV. D: This combination includes false statement(s) I and omits true statement(s) II, III, IV.

BIO-12-CH01-0112 | Easy | Microsporogenesis and pollen**Correct answer: B. Number of pollen sacs in a typical dithecal anther**

Solution: Four represents 'Number of pollen sacs in a typical dithecal anther'. Each alternative describes a different term or result.

Distractor review: A: 'Transfer between different flowers of the same plant' identifies Geitonogamy, not Four. C: 'Highly resistant material of the pollen exine' identifies Sporopollenin, not Four. D: 'Opening left by the integuments at the ovule apex' identifies Micropyle, not Four.

BIO-12-CH01-0113 | Easy | Pollination**Correct answer: B. Autogamy**

Solution: 'Transfer of pollen to the stigma of the same flower' is the defining description for Autogamy. The other choices belong to different chapter concepts.

Distractor review: A: Double fertilization belongs with 'Syngamy and triple fusion occurring in one embryo sac', whereas the tested description is 'Transfer of pollen to the stigma of the same flower'. C: Spirogyra belongs with 'Fragmentation in a filamentous alga', whereas the tested description is 'Transfer of pollen to the stigma of the same flower'. D: Geitonogamy belongs with 'Transfer between different flowers of the same plant', whereas the tested description is 'Transfer of pollen to the stigma of the same flower'.

BIO-12-CH01-0114 | Easy | Ovule and embryo sac**Correct answer: C. Micropyle**

Solution: 'Opening left by the integuments at the ovule apex' is the defining description for Micropyle. The other choices belong to different chapter concepts.

Distractor review: A: Fruit belongs with 'Ovary after fertilization', whereas the tested description is 'Opening left by the integuments at the ovule apex'. B: Tapetum belongs with 'Anther wall layer that nourishes developing pollen', whereas the tested description is 'Opening left by the integuments at the ovule apex'. D: Hypocotyl belongs with 'Embryonic axis below the cotyledons', whereas the tested description is 'Opening left by the integuments at the ovule apex'.

BIO-12-CH01-0115 | Easy | Microsporogenesis and pollen**Correct answer: D. Highly resistant material of the pollen exine**

Solution: Sporopollenin represents 'Highly resistant material of the pollen exine'. Each alternative describes a different term or result.

Distractor review: A: 'Fusion of one male gamete with the egg' identifies Syngamy, not Sporopollenin. B: 'Pollen-tube entry into an ovule through the micropyle' identifies Porogamy, not Sporopollenin. C: 'Transfer between different flowers of the same plant' identifies Geitonogamy, not Sporopollenin.

BIO-12-CH01-0116 | Easy | Pollination**Correct answer: B. Transfer of pollen to the stigma of the same flower**

Solution: Autogamy represents 'Transfer of pollen to the stigma of the same flower'. Each alternative describes a different term or result.

Distractor review: A: 'Syngamy and triple fusion occurring in one embryo sac' identifies Double fertilization, not Autogamy. C: 'Highly resistant material of the pollen exine' identifies Sporopollenin, not Autogamy. D: 'Gemma formation' identifies Marchantia, not Autogamy.

BIO-12-CH01-0117 | Easy | Post-fertilization development**Correct answer: B. Fruit**

Solution: 'Ovary after fertilization' is the defining description for Fruit. The other choices belong to different chapter concepts.

Distractor review: A: Seed belongs with 'Ovule after fertilization', whereas the tested description is 'Ovary after fertilization'. C: Adventive embryony belongs with 'Embryos arising directly from nucellus or integuments', whereas the tested description is 'Ovary after fertilization'. D: Sporopollenin belongs with 'Highly resistant material of the pollen exine', whereas the tested description is 'Ovary after fertilization'.

BIO-12-CH01-0118 | Medium | Asexual reproduction**Correct answer: D. Fragmentation in a filamentous alga - Spirogyra**

Solution: The accurate entry is 'Fragmentation in a filamentous alga - Spirogyra'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Syngamy and triple fusion occurring in one embryo sac' should be paired with Double fertilization, not Tapetum. B: 'Fusion of one male gamete with the egg' should be paired with Syngamy, not Parthenocarp. C: 'Product of meiosis in a microspore mother cell' should be paired with A tetrad of haploid microspores, not Double fertilization.

BIO-12-CH01-0119 | Challenging | Post-fertilization development**Correct answer: A. I and III only**

Solution: Statement I is correct because 'Embryonic axis below the cotyledons' corresponds to Hypocotyl. Statement II is incorrect: 'Ovary after fertilization' corresponds to Fruit, not Seed. Statement III is correct because 'Ovule after fertilization' corresponds to Seed. Statement IV is incorrect: 'Pollen-tube entry into an ovule through the micropyle' corresponds to Porogamy, not Hypocotyl. Therefore, the valid combination is I and III only.

Distractor review: B: This combination includes false statement(s) II and omits true statement(s) I, III. C: This combination includes false statement(s) IV and omits true statement(s) I, III. D: This combination includes false statement(s) II, IV and omits true statement(s) III.

BIO-12-CH01-0120 | Easy | Special reproductive phenomena**Correct answer: C. Fruit formation without fertilization**

Solution: Parthenocarp represents 'Fruit formation without fertilization'. Each alternative describes a different term or result.

Distractor review: A: 'Fusion of the second male gamete with the two polar nuclei' identifies Triple fusion, not Parthenocarp. B: 'Transfer of pollen to the stigma of the same flower' identifies Autogamy, not Parthenocarp. D: 'Occurrence of more than one embryo in a seed' identifies Polyembryony, not Parthenocarp.

Standard 12 | Chapter 02

Reproduction in Lower and Higher Animals

BIO-12-CH02-0001 | Easy | Hormonal pattern across a menstrual cycle

Correct answer: A. LH surge

Solution: A sharp rise in luteinizing hormone near day 14 triggers ovulation.

Distractor review: B: The alternative 'Progesterone fall' conflicts with the labelled relationship, scale or mechanism in the figure. A sharp rise in luteinizing hormone near day 14 triggers ovulation. C: The alternative 'FSH absence' conflicts with the labelled relationship, scale or mechanism in the figure. A sharp rise in luteinizing hormone near day 14 triggers ovulation. D: The alternative 'Prolactin surge' conflicts with the labelled relationship, scale or mechanism in the figure. A sharp rise in luteinizing hormone near day 14 triggers ovulation.

BIO-12-CH02-0002 | Medium | Early human development

Correct answer: D. Blastocyst

Solution: The blastocyst, with trophoblast and inner cell mass, is the stage that initiates implantation.

Distractor review: A: The alternative 'Zygote' conflicts with the labelled relationship, scale or mechanism in the figure. The blastocyst, with trophoblast and inner cell mass, is the stage that initiates implantation. B: The alternative 'Two-cell embryo' conflicts with the labelled relationship, scale or mechanism in the figure. The blastocyst, with trophoblast and inner cell mass, is the stage that initiates implantation. C: The alternative 'Morula before cavitation' conflicts with the labelled relationship, scale or mechanism in the figure. The blastocyst, with trophoblast and inner cell mass, is the stage that initiates implantation.

BIO-12-CH02-0003 | Medium | Male reproductive system

Correct answer: C. Seminiferous tubules

Solution: The chapter links Site of sperm formation with Seminiferous tubules, so Seminiferous tubules completes the statement.

Distractor review: A: Progesterone belongs with 'Hormone chiefly maintaining the secretory endometrium', whereas the tested description is 'Site of sperm formation'. B: Spermatogonia belongs with 'Diploid male germ cells multiplying by mitosis', whereas the tested description is 'Site of sperm formation'. D: Gastrulation belongs with 'Process producing the three primary germ layers', whereas the tested description is 'Site of sperm formation'.

BIO-12-CH02-0004 | Challenging | Early human development

Correct answer: A. Formation of a morula and blastocyst

Solution: The morula and blastocyst require repeated cleavage divisions after fertilization.

Distractor review: B: The alternative 'Fusion of sperm and ovum' conflicts with the labelled relationship, scale or mechanism in the figure. The morula and blastocyst require repeated cleavage divisions after fertilization. C: The alternative 'Completion of ovulation' conflicts with the labelled relationship, scale or mechanism in the figure. The morula and blastocyst require repeated cleavage divisions after fertilization. D: The alternative 'Formation of the corpus luteum' conflicts with the labelled relationship, scale or mechanism in the figure. The morula and blastocyst require repeated cleavage divisions after fertilization.

BIO-12-CH02-0005 | Medium | Menstrual cycle

Correct answer: D. Progesterone

Solution: The observation 'Hormone chiefly maintaining the secretory endometrium' points to Progesterone; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Ovary belongs with 'Organ producing ova, estrogen and progesterone', whereas the tested description is 'Hormone chiefly maintaining the secretory endometrium'. B: Freshwater sponge belongs with 'Gemmule formation', whereas the tested description is 'Hormone chiefly maintaining the secretory endometrium'. C: LH surge belongs with 'Immediate endocrine trigger for ovulation', whereas the tested description is 'Hormone chiefly maintaining the secretory endometrium'.

BIO-12-CH02-0006 | Medium | Female reproductive system

Correct answer: B. Ovary

Solution: The chapter links Organ producing ova, estrogen and progesterone with Ovary, so Ovary completes the statement.

Distractor review: A: Tubectomy belongs with 'Surgical interruption of the fallopian tubes', whereas the tested description is 'Organ producing ova, estrogen and progesterone'. C: Ampulla of the fallopian tube belongs with 'Usual site of fertilization', whereas the tested description is 'Organ producing ova, estrogen and progesterone'. D: FSH belongs with 'Pituitary hormone promoting follicular growth', whereas the tested description is 'Organ producing ova, estrogen and progesterone'.

BIO-12-CH02-0007 | Medium | Asexual reproduction**Correct answer: B. Hydra**

Solution: The observation 'Budding' points to Hydra; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Freshwater sponge belongs with 'Gemmule formation', whereas the tested description is 'Budding'. C: Sertoli cells belongs with 'Cells nourishing developing germ cells', whereas the tested description is 'Budding'. D: Vasectomy belongs with 'Surgical interruption of the vas deferens', whereas the tested description is 'Budding'.

BIO-12-CH02-0008 | Medium | Birth control and reproductive health**Correct answer: C. Surgical interruption of the fallopian tubes - Tubectomy**

Solution: The accurate entry is 'Surgical interruption of the fallopian tubes - Tubectomy'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Organ producing ova, estrogen and progesterone' should be paired with Ovary, not Tubectomy. B: 'Embryonic stage with trophoblast and inner cell mass' should be paired with Blastocyst, not Primary spermatocyte. D: 'Bacterial sexually transmitted infection caused by Neisseria' should be paired with Gonorrhoea, not Ovary.

BIO-12-CH02-0009 | Challenging | Male reproductive system**Correct answer: C. II and III only**

Solution: Statement I is incorrect: 'Cells secreting testicular androgens' corresponds to Leydig cells, not Seminiferous tubules. Statement II is correct because 'Site of sperm formation' corresponds to Seminiferous tubules. Statement III is correct because 'Duct carrying sperm away from the epididymis' corresponds to Vas deferens. Statement IV is incorrect: 'Barrier method that also lowers STI transmission' corresponds to Condom, not Leydig cells. Therefore, the valid combination is II and III only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) II. B: This combination omits true statement(s) II. D: This combination includes false statement(s) I, IV.

BIO-12-CH02-0010 | Hard | Hormonal pattern across a menstrual cycle**Correct answer: A. Progesterone**

Solution: Progesterone secreted mainly by the corpus luteum maintains the secretory endometrium after ovulation.

Distractor review: B: The alternative 'LH' conflicts with the labelled relationship, scale or mechanism in the figure. Progesterone secreted mainly by the corpus luteum maintains the secretory endometrium after ovulation. C: The alternative 'FSH' conflicts with the labelled relationship, scale or mechanism in the figure. Progesterone secreted mainly by the corpus luteum maintains the secretory endometrium after ovulation. D: The alternative 'Oxytocin' conflicts with the labelled relationship, scale or mechanism in the figure. Progesterone secreted mainly by the corpus luteum maintains the secretory endometrium after ovulation.

BIO-12-CH02-0011 | Medium | Gametogenesis**Correct answer: D. Acrosome**

Solution: The chapter links Cap-like structure over the sperm nucleus with Acrosome, so Acrosome completes the statement.

Distractor review: A: Amoeba belongs with 'Binary fission', whereas the tested description is 'Cap-like structure over the sperm nucleus'. B: Blastocyst belongs with 'Embryonic stage with trophoblast and inner cell mass', whereas the tested description is 'Cap-like structure over the sperm nucleus'. C: Ampulla of the fallopian tube belongs with 'Usual site of fertilization', whereas the tested description is 'Cap-like structure over the sperm nucleus'.

BIO-12-CH02-0012 | Medium | Birth control and reproductive health**Correct answer: A. Surgical interruption of the vas deferens - Vasectomy**

Solution: The accurate entry is 'Surgical interruption of the vas deferens - Vasectomy'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Process producing the three primary germ layers' should be paired with Gastrulation, not Amoeba. C: 'Binary fission' should be paired with Amoeba, not Spermatogonia. D: 'Usual site of fertilization' should be paired with Ampulla of the fallopian tube, not Freshwater sponge.

BIO-12-CH02-0013 | Medium | Male reproductive system**Correct answer: B. Duct carrying sperm away from the epididymis - Vas deferens**

Solution: The accurate entry is 'Duct carrying sperm away from the epididymis - Vas deferens'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Process producing the three primary germ layers' should be paired with Gastrulation, not Infundibulum. C: 'Hormone chiefly maintaining the secretory endometrium' should be paired with Progesterone, not Vas deferens. D: 'Transfer of an embryo with more than eight blastomeres into the uterus' should be paired with IUT, not Ovary.

BIO-12-CH02-0014 | Medium | Early development**Correct answer: D. Cleavage**

Solution: The observation 'Rapid mitotic divisions of the zygote without overall growth' points to Cleavage; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Hydra belongs with 'Budding', whereas the tested description is 'Rapid mitotic divisions of the zygote without overall growth'. B: FSH belongs with 'Pituitary hormone promoting follicular growth', whereas the tested description is 'Rapid mitotic divisions of the zygote without overall growth'. C: Blastocyst belongs with 'Embryonic stage with trophoblast and inner cell mass', whereas the tested description is 'Rapid mitotic divisions of the zygote without overall growth'.

BIO-12-CH02-0015 | Easy | Female reproductive system**Correct answer: A. Endometrium**

Solution: 'Uterine layer receiving the implanting blastocyst' is the defining description for Endometrium. The other choices belong to different chapter concepts.

Distractor review: B: Corpus luteum belongs with 'Temporary ovarian body formed after ovulation', whereas the tested description is 'Uterine layer receiving the implanting blastocyst'. C: Infundibulum belongs with 'Funnel-like oviduct part bearing fimbriae', whereas the tested description is 'Uterine layer receiving the implanting blastocyst'. D: Ovary belongs with 'Organ producing ova, estrogen and progesterone', whereas the tested description is 'Uterine layer receiving the implanting blastocyst'.

BIO-12-CH02-0016 | Easy | Female reproductive system**Correct answer: A. Uterine layer receiving the implanting blastocyst**

Solution: Endometrium represents 'Uterine layer receiving the implanting blastocyst'. Each alternative describes a different term or result.

Distractor review: B: 'Immediate endocrine trigger for ovulation' identifies LH surge, not Endometrium. C: 'Regeneration producing complete individuals' identifies Planaria, not Endometrium. D: 'Bacterial sexually transmitted infection caused by Neisseria' identifies Gonorrhoea, not Endometrium.

BIO-12-CH02-0017 | Medium | Male reproductive system**Correct answer: C. Cells secreting testicular androgens - Leydig cells**

Solution: The accurate entry is 'Cells secreting testicular androgens - Leydig cells'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Budding' should be paired with Hydra, not Spermiogenesis. B: 'Transformation of spermatids into spermatozoa' should be paired with Spermiogenesis, not Infundibulum. D: 'Surgical interruption of the fallopian tubes' should be paired with Tubectomy, not LH surge.

BIO-12-CH02-0018 | Easy | Birth control and reproductive health**Correct answer: C. Copper-T**

Solution: 'Copper-releasing device placed in the uterus' is the defining description for Copper-T. The other choices belong to different chapter concepts.

Distractor review: A: Seminiferous tubules belongs with 'Site of sperm formation', whereas the tested description is 'Copper-releasing device placed in the uterus'. B: Primary spermatocyte belongs with 'Cell completing meiosis I to form secondary spermatocytes', whereas the tested description is 'Copper-releasing device placed in the uterus'. D: Condom belongs with 'Barrier method that also lowers STI transmission', whereas the tested description is 'Copper-releasing device placed in the uterus'.

BIO-12-CH02-0019 | Easy | Male reproductive system**Correct answer: A. Leydig cells**

Solution: 'Cells secreting testicular androgens' is the defining description for Leydig cells. The other choices belong to different chapter concepts.

Distractor review: B: IUT belongs with 'Transfer of an embryo with more than eight blastomeres into the uterus', whereas the tested description is 'Cells secreting testicular androgens'. C: Hydra belongs with 'Budding', whereas the tested description is 'Cells secreting testicular androgens'. D: Vas deferens belongs with 'Duct carrying sperm away from the epididymis', whereas the tested description is 'Cells secreting testicular androgens'.

BIO-12-CH02-0020 | Medium | Male reproductive system**Correct answer: A. Sertoli cells**

Solution: The observation 'Cells nourishing developing germ cells' points to Sertoli cells; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: ZIFT belongs with 'Transfer of an early embryo with up to eight blastomeres into the oviduct', whereas the tested description is 'Cells nourishing developing germ cells'. C: Gastrulation belongs with 'Process producing the three primary germ layers', whereas the tested description is 'Cells nourishing developing germ cells'. D: Vas deferens belongs with 'Duct carrying sperm away from the epididymis', whereas the tested description is 'Cells nourishing developing germ cells'.

BIO-12-CH02-0021 | Easy | Gametogenesis**Correct answer: B. Primary spermatocyte**

Solution: 'Cell completing meiosis I to form secondary spermatocytes' is the defining description for Primary spermatocyte. The other choices belong to different chapter concepts.

Distractor review: A: Seminiferous tubules belongs with 'Site of sperm formation', whereas the tested description is 'Cell completing meiosis I to form secondary spermatocytes'. C: Gastrulation belongs with 'Process producing the three primary germ layers', whereas the tested description is 'Cell completing meiosis I to form secondary spermatocytes'. D: Spermiogenesis belongs with 'Transformation of spermatids into spermatozoa', whereas the tested description is 'Cell completing meiosis I to form secondary spermatocytes'.

BIO-12-CH02-0022 | Hard | Assisted reproduction and infection**Correct answer: D. Viral syndrome caused by HIV - AIDS; Solid ball of about 8 to 16 blastomeres - Morula**

Solution: Both parts of 'Viral syndrome caused by HIV - AIDS; Solid ball of about 8 to 16 blastomeres - Morula' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Funnel-like oviduct part bearing fimbriae' requires Infundibulum, and 'Copper-releasing device placed in the uterus' requires Copper-T. B: The answers have been interchanged: 'Hormone chiefly maintaining the secretory endometrium' requires Progesterone, and 'Funnel-like oviduct part bearing fimbriae' requires Infundibulum. C: The answers have been interchanged: 'Site of sperm formation' requires Seminiferous tubules, and 'Regeneration producing complete individuals' requires Planaria.

BIO-12-CH02-0023 | Medium | Assisted reproduction and infection**Correct answer: A. Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT**

Solution: The accurate entry is 'Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Solid ball of about 8 to 16 blastomeres' should be paired with Morula, not Acrosome. C: 'Transformation of spermatids into spermatozoa' should be paired with Spermiogenesis, not Seminiferous tubules. D: 'Site of sperm formation' should be paired with Seminiferous tubules, not Sertoli cells.

BIO-12-CH02-0024 | Easy | Gametogenesis**Correct answer: A. Cap-like structure over the sperm nucleus**

Solution: Acrosome represents 'Cap-like structure over the sperm nucleus'. Each alternative describes a different term or result.

Distractor review: B: 'Process producing the three primary germ layers' identifies Gastrulation, not Acrosome. C: 'Binary fission' identifies Amoeba, not Acrosome. D: 'Usual site of fertilization' identifies Ampulla of the fallopian tube, not Acrosome.

BIO-12-CH02-0025 | Medium | Early development**Correct answer: D. Blastocyst**

Solution: The observation 'Embryonic stage with trophoblast and inner cell mass' points to Blastocyst; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Cleavage belongs with 'Rapid mitotic divisions of the zygote without overall growth', whereas the tested description is 'Embryonic stage with trophoblast and inner cell mass'. B: IUT belongs with 'Transfer of an embryo with more than eight blastomeres into the uterus', whereas the tested description is 'Embryonic stage with trophoblast and inner cell mass'. C: Tubectomy belongs with 'Surgical interruption of the fallopian tubes', whereas the tested description is 'Embryonic stage with trophoblast and inner cell mass'.

BIO-12-CH02-0026 | Hard | Gametogenesis**Correct answer: B. Cap-like structure over the sperm nucleus - Acrosome; Binary fission - Amoeba**

Solution: Both parts of 'Cap-like structure over the sperm nucleus - Acrosome; Binary fission - Amoeba' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Surgical interruption of the vas deferens' requires Vasectomy, and 'Duct carrying sperm away from the epididymis' requires Vas deferens. C: The answers have been interchanged: 'Immediate endocrine trigger for ovulation' requires LH surge, and 'Process producing the three primary germ layers' requires Gastrulation. D: The answers have been interchanged: 'Process producing the three primary germ layers' requires Gastrulation, and 'Embryonic stage with trophoblast and inner cell mass' requires Blastocyst.

BIO-12-CH02-0027 | Easy | Female reproductive system**Correct answer: D. Funnel-like oviduct part bearing fimbriae**

Solution: Infundibulum represents 'Funnel-like oviduct part bearing fimbriae'. Each alternative describes a different term or result.

Distractor review: A: 'Uterine layer receiving the implanting blastocyst' identifies Endometrium, not Infundibulum. B: 'Surgical interruption of the vas deferens' identifies Vasectomy, not Infundibulum. C: 'Transfer of an embryo with more than eight blastomeres into the uterus' identifies IUT, not Infundibulum.

BIO-12-CH02-0028 | Hard | Assisted reproduction and infection**Correct answer: C. Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT; Funnel-like oviduct part bearing fimbriae - Infundibulum**

Solution: Both parts of 'Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT; Funnel-like oviduct part bearing fimbriae - Infundibulum' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Solid ball of about 8 to 16 blastomeres' requires Morula, and 'Site of sperm formation' requires Seminiferous tubules. B: The answers have been interchanged: 'Binary fission' requires Amoeba, and 'Transformation of spermatids into spermatozoa' requires Spermiogenesis. D: The answers have been interchanged: 'Transformation of spermatids into spermatozoa' requires Spermiogenesis, and 'Solid ball of about 8 to 16 blastomeres' requires Morula.

BIO-12-CH02-0029 | Medium | Menstrual cycle**Correct answer: D. LH surge**

Solution: The observation 'Immediate endocrine trigger for ovulation' points to LH surge; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Vasectomy belongs with 'Surgical interruption of the vas deferens', whereas the tested description is 'Immediate endocrine trigger for ovulation'. B: Endometrium belongs with 'Uterine layer receiving the implanting blastocyst', whereas the tested description is 'Immediate endocrine trigger for ovulation'. C: Progesterone belongs with 'Hormone chiefly maintaining the secretory endometrium', whereas the tested description is 'Immediate endocrine trigger for ovulation'.

BIO-12-CH02-0030 | Challenging | Assisted reproduction and infection**Correct answer: A. I, II, III and IV**

Solution: Statement I is correct because 'Transfer of an early embryo with up to eight blastomeres into the oviduct' corresponds to ZIFT. Statement II is correct because 'Bacterial sexually transmitted infection caused by Neisseria' corresponds to Gonorrhoea. Statement III is correct because 'Viral syndrome caused by HIV' corresponds to AIDS. Statement IV is correct because 'Process producing the three primary germ layers' corresponds to Gastrulation. Therefore, the valid combination is I, II, III and IV.

Distractor review: B: This combination omits true statement(s) I. C: This combination omits true statement(s) II, III. D: This combination omits true statement(s) III.

BIO-12-CH02-0031 | Hard | Menstrual cycle**Correct answer: D. Pituitary hormone promoting follicular growth - Primary spermatocyte**

Solution: 'Pituitary hormone promoting follicular growth' should be paired with FSH, not Primary spermatocyte. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Surgical interruption of the fallopian tubes - Tubectomy. B: This is a valid association: Organ producing ova, estrogen and progesterone - Ovary. C: This is a valid association: Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT.

BIO-12-CH02-0032 | Easy | Birth control and reproductive health**Correct answer: B. Vasectomy**

Solution: 'Surgical interruption of the vas deferens' is the defining description for Vasectomy. The other choices belong to different chapter concepts.

Distractor review: A: Condom belongs with 'Barrier method that also lowers STI transmission', whereas the tested description is 'Surgical interruption of the vas deferens'. C: Amoeba belongs with 'Binary fission', whereas the tested description is 'Surgical interruption of the vas deferens'. D: Gastrulation belongs with 'Process producing the three primary germ layers', whereas the tested description is 'Surgical interruption of the vas deferens'.

BIO-12-CH02-0033 | Medium | Birth control and reproductive health**Correct answer: C. Condom**

Solution: The chapter links Barrier method that also lowers STI transmission with Condom, so Condom completes the statement.

Distractor review: A: Vasectomy belongs with 'Surgical interruption of the vas deferens', whereas the tested description is 'Barrier method that also lowers STI transmission'. B: Cleavage belongs with 'Rapid mitotic divisions of the zygote without overall growth', whereas the tested description is 'Barrier method that also lowers STI transmission'. D: Ampulla of the fallopian tube belongs with 'Usual site of fertilization', whereas the tested description is 'Barrier method that also lowers STI transmission'.

BIO-12-CH02-0034 | Medium | Birth control and reproductive health**Correct answer: B. Condom**

Solution: The observation 'Barrier method that also lowers STI transmission' points to Condom; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Blastocyst belongs with 'Embryonic stage with trophoblast and inner cell mass', whereas the tested description is 'Barrier method that also lowers STI transmission'. C: IUT belongs with 'Transfer of an embryo with more than eight blastomeres into the uterus', whereas the tested description is 'Barrier method that also lowers STI transmission'. D: Vasectomy belongs with 'Surgical interruption of the vas deferens', whereas the tested description is 'Barrier method that also lowers STI transmission'.

BIO-12-CH02-0035 | Hard | Male reproductive system**Correct answer: A. Cells secreting testicular androgens - IUT**

Solution: 'Cells secreting testicular androgens' should be paired with Leydig cells, not IUT. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Solid ball of about 8 to 16 blastomeres - Morula. C: This is a valid association: Surgical interruption of the fallopian tubes - Tubectomy. D: This is a valid association: Transformation of spermatids into spermatozoa - Spermiogenesis.

BIO-12-CH02-0036 | Medium | Early development**Correct answer: B. Rapid mitotic divisions of the zygote without overall growth - Cleavage**

Solution: The accurate entry is 'Rapid mitotic divisions of the zygote without overall growth - Cleavage'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Cap-like structure over the sperm nucleus' should be paired with Acrosome, not FSH. C: 'Cell completing meiosis I to form secondary spermatocytes' should be paired with Primary spermatocyte, not Cleavage. D: 'Pituitary hormone promoting follicular growth' should be paired with FSH, not Vas deferens.

BIO-12-CH02-0037 | Easy | Assisted reproduction and infection**Correct answer: D. ZIFT**

Solution: 'Transfer of an early embryo with up to eight blastomeres into the oviduct' is the defining description for ZIFT. The other choices belong to different chapter concepts.

Distractor review: A: Seminiferous tubules belongs with 'Site of sperm formation', whereas the tested description is 'Transfer of an early embryo with up to eight blastomeres into the oviduct'. B: Infundibulum belongs with 'Funnel-like oviduct part bearing fimbriae', whereas the tested description is 'Transfer of an early embryo with up to eight blastomeres into the oviduct'. C: AIDS belongs with 'Viral syndrome caused by HIV', whereas the tested description is 'Transfer of an early embryo with up to eight blastomeres into the oviduct'.

BIO-12-CH02-0038 | Medium | Early development**Correct answer: B. Morula**

Solution: The observation 'Solid ball of about 8 to 16 blastomeres' points to Morula; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: ZIFT belongs with 'Transfer of an early embryo with up to eight blastomeres into the oviduct', whereas the tested description is 'Solid ball of about 8 to 16 blastomeres'. C: Progesterone belongs with 'Hormone chiefly maintaining the secretory endometrium', whereas the tested description is 'Solid ball of about 8 to 16 blastomeres'. D: Cleavage belongs with 'Rapid mitotic divisions of the zygote without overall growth', whereas the tested description is 'Solid ball of about 8 to 16 blastomeres'.

BIO-12-CH02-0039 | Hard | Female reproductive system**Correct answer: B. Uterine layer receiving the implanting blastocyst - Endometrium; Regeneration producing complete individuals - Planaria**

Solution: Both parts of 'Uterine layer receiving the implanting blastocyst - Endometrium; Regeneration producing complete individuals - Planaria' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Bacterial sexually transmitted infection caused by Neisseria' requires Gonorrhoea, and 'Temporary ovarian body formed after ovulation' requires Corpus luteum. C: The answers have been interchanged: 'Immediate endocrine trigger for ovulation' requires LH surge, and 'Cells nourishing developing germ cells' requires Sertoli cells. D: The answers have been interchanged: 'Temporary ovarian body formed after ovulation' requires Corpus luteum, and 'Cells secreting testicular androgens' requires Leydig cells.

BIO-12-CH02-0040 | Medium | Gametogenesis**Correct answer: C. Cap-like structure over the sperm nucleus - Acrosome**

Solution: The accurate entry is 'Cap-like structure over the sperm nucleus - Acrosome'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Embryonic stage with trophoblast and inner cell mass' should be paired with Blastocyst, not Vas deferens. B: 'Immediate endocrine trigger for ovulation' should be paired with LH surge, not Blastocyst. D: 'Usual site of fertilization' should be paired with Ampulla of the fallopian tube, not Gastrulation.

BIO-12-CH02-0041 | Easy | Birth control and reproductive health**Correct answer: C. Copper-releasing device placed in the uterus**

Solution: Copper-T represents 'Copper-releasing device placed in the uterus'. Each alternative describes a different term or result.

Distractor review: A: 'Duct carrying sperm away from the epididymis' identifies Vas deferens, not Copper-T. B: 'Pituitary hormone promoting follicular growth' identifies FSH, not Copper-T. D: 'Surgical interruption of the fallopian tubes' identifies Tubectomy, not Copper-T.

BIO-12-CH02-0042 | Medium | Birth control and reproductive health**Correct answer: B. Tubectomy**

Solution: The observation 'Surgical interruption of the fallopian tubes' points to Tubectomy; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Condom belongs with 'Barrier method that also lowers STI transmission', whereas the tested description is 'Surgical interruption of the fallopian tubes'. C: Blastocyst belongs with 'Embryonic stage with trophoblast and inner cell mass', whereas the tested description is 'Surgical interruption of the fallopian tubes'. D: Gonorrhoea belongs with 'Bacterial sexually transmitted infection caused by Neisseria', whereas the tested description is 'Surgical interruption of the fallopian tubes'.

BIO-12-CH02-0043 | Medium | Asexual reproduction**Correct answer: B. Budding - Hydra**

Solution: The accurate entry is 'Budding - Hydra'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Surgical interruption of the fallopian tubes' should be paired with Tubectomy, not Hydra. C: 'Surgical interruption of the vas deferens' should be paired with Vasectomy, not Gastrulation. D: 'Uterine layer receiving the implanting blastocyst' should be paired with Endometrium, not Vasectomy.

BIO-12-CH02-0044 | Easy | Asexual reproduction**Correct answer: A. Budding**

Solution: Hydra represents 'Budding'. Each alternative describes a different term or result.

Distractor review: B: 'Gemmule formation' identifies Freshwater sponge, not Hydra. C: 'Binary fission' identifies Amoeba, not Hydra. D: 'Process producing the three primary germ layers' identifies Gastrulation, not Hydra.

BIO-12-CH02-0045 | Hard | Male reproductive system**Correct answer: B. Duct carrying sperm away from the epididymis - Vas deferens; Transfer of an embryo with more than eight blastomeres into the uterus - IUT**

Solution: Both parts of 'Duct carrying sperm away from the epididymis - Vas deferens; Transfer of an embryo with more than eight blastomeres into the uterus - IUT' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Hormone chiefly maintaining the secretory endometrium' requires Progesterone, and 'Funnel-like oviduct part bearing fimbriae' requires Infundibulum. C: The answers have been interchanged: 'Viral syndrome caused by HIV' requires AIDS, and 'Process producing the three primary germ layers' requires Gastrulation. D: The answers have been interchanged: 'Process producing the three primary germ layers' requires Gastrulation, and 'Hormone chiefly maintaining the secretory endometrium' requires Progesterone.

BIO-12-CH02-0046 | Challenging | Gametogenesis**Correct answer: B. I and IV only**

Solution: Statement I is correct because 'Transformation of spermatids into spermatozoa' corresponds to Spermiogenesis. Statement II is incorrect: 'Diploid male germ cells multiplying by mitosis' corresponds to Spermatogonia, not Primary spermatocyte. Statement III is incorrect: 'Cell completing meiosis I to form secondary spermatocytes' corresponds to Primary spermatocyte, not Tubectomy. Statement IV is correct because 'Surgical interruption of the fallopian tubes' corresponds to Tubectomy. Therefore, the valid combination is I and IV only.

Distractor review: A: This combination includes false statement(s) II, III and omits true statement(s) IV. C: This combination omits true statement(s) IV. D: This combination omits true statement(s) I.

BIO-12-CH02-0047 | Challenging | Birth control and reproductive health**Correct answer: D. II, III and IV only**

Solution: Statement I is incorrect: 'Barrier method that also lowers STI transmission' corresponds to Condom, not Vasectomy. Statement II is correct because 'Surgical interruption of the vas deferens' corresponds to Vasectomy. Statement III is correct because 'Copper-releasing device placed in the uterus' corresponds to Copper-T. Statement IV is correct because 'Viral syndrome caused by HIV' corresponds to AIDS. Therefore, the valid combination is II, III and IV only.

Distractor review: A: This combination omits true statement(s) III, IV. B: This combination includes false statement(s) I and omits true statement(s) III, IV. C: This combination includes false statement(s) I and omits true statement(s) II, IV.

BIO-12-CH02-0048 | Hard | Female reproductive system**Correct answer: C. Funnel-like oviduct part bearing fimbriae - Infundibulum; Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea**

Solution: Both parts of 'Funnel-like oviduct part bearing fimbriae - Infundibulum; Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Duct carrying sperm away from the epididymis' requires Vas deferens, and 'Binary fission' requires Amoeba. B: The answers have been interchanged: 'Binary fission' requires Amoeba, and 'Transfer of an embryo with more than eight blastomeres into the uterus' requires IUT. D: The answers have been interchanged: 'Cap-like structure over the sperm nucleus' requires Acrosome, and 'Transfer of an early embryo with up to eight blastomeres into the oviduct' requires ZIFT.

BIO-12-CH02-0049 | Easy | Menstrual cycle**Correct answer: D. LH surge**

Solution: 'Immediate endocrine trigger for ovulation' is the defining description for LH surge. The other choices belong to different chapter concepts.

Distractor review: A: Ovary belongs with 'Organ producing ova, estrogen and progesterone', whereas the tested description is 'Immediate endocrine trigger for ovulation'. B: Seminiferous tubules belongs with 'Site of sperm formation', whereas the tested description is 'Immediate endocrine trigger for ovulation'. C: FSH belongs with 'Pituitary hormone promoting follicular growth', whereas the tested description is 'Immediate endocrine trigger for ovulation'.

BIO-12-CH02-0050 | Hard | Asexual reproduction**Correct answer: D. Budding - Hydra; Uterine layer receiving the implanting blastocyst - Endometrium**

Solution: Both parts of 'Budding - Hydra; Uterine layer receiving the implanting blastocyst - Endometrium' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Surgical interruption of the vas deferens' requires Vasectomy, and 'Cells nourishing developing germ cells' requires Sertoli cells. B: The answers have been interchanged: 'Rapid mitotic divisions of the zygote without overall growth' requires Cleavage, and 'Surgical interruption of the vas deferens' requires Vasectomy. C: The answers have been interchanged: 'Cells nourishing developing germ cells' requires Sertoli cells, and 'Process producing the three primary germ layers' requires Gastrulation.

BIO-12-CH02-0051 | Medium | Female reproductive system**Correct answer: B. Infundibulum**

Solution: The observation 'Funnel-like oviduct part bearing fimbriae' points to Infundibulum; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Vas deferens belongs with 'Duct carrying sperm away from the epididymis', whereas the tested description is 'Funnel-like oviduct part bearing fimbriae'. C: Vasectomy belongs with 'Surgical interruption of the vas deferens', whereas the tested description is 'Funnel-like oviduct part bearing fimbriae'. D: IUT belongs with 'Transfer of an embryo with more than eight blastomeres into the uterus', whereas the tested description is 'Funnel-like oviduct part bearing fimbriae'.

BIO-12-CH02-0052 | Hard | Gametogenesis**Correct answer: A. Cap-like structure over the sperm nucleus - Amoeba**

Solution: 'Cap-like structure over the sperm nucleus' should be paired with Acrosome, not Amoeba. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Process producing the three primary germ layers - Gastrulation. C: This is a valid association: Immediate endocrine trigger for ovulation - LH surge. D: This is a valid association: Embryonic stage with trophoblast and inner cell mass - Blastocyst.

BIO-12-CH02-0053 | Hard | Female reproductive system**Correct answer: B. Uterine layer receiving the implanting blastocyst - Planaria**

Solution: 'Uterine layer receiving the implanting blastocyst' should be paired with Endometrium, not Planaria. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea. C: This is a valid association: Immediate endocrine trigger for ovulation - LH surge. D: This is a valid association: Cells secreting testicular androgens - Leydig cells.

BIO-12-CH02-0054 | Challenging | Menstrual cycle**Correct answer: A. I and III only**

Solution: Statement I is correct because 'Hormone chiefly maintaining the secretory endometrium' corresponds to Progesterone. Statement II is incorrect: 'Immediate endocrine trigger for ovulation' corresponds to LH surge, not Corpus luteum. Statement III is correct because 'Temporary ovarian body formed after ovulation' corresponds to Corpus luteum. Statement IV is incorrect: 'Budding' corresponds to Hydra, not Progesterone. Therefore, the valid combination is I and III only.

Distractor review: B: This combination omits true statement(s) III. C: This combination includes false statement(s) II and omits true statement(s) I, III. D: This combination includes false statement(s) II and omits true statement(s) I.

BIO-12-CH02-0055 | Hard | Asexual reproduction

Correct answer: B. Regeneration producing complete individuals - Planaria; Copper-releasing device placed in the uterus - Copper-T

Solution: Both parts of 'Regeneration producing complete individuals - Planaria; Copper-releasing device placed in the uterus - Copper-T' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Embryonic stage with trophoblast and inner cell mass' requires Blastocyst, and 'Cap-like structure over the sperm nucleus' requires Acrosome. C: The answers have been interchanged: 'Transfer of an embryo with more than eight blastomeres into the uterus' requires IUT, and 'Cells nourishing developing germ cells' requires Sertoli cells. D: The answers have been interchanged: 'Cap-like structure over the sperm nucleus' requires Acrosome, and 'Transfer of an embryo with more than eight blastomeres into the uterus' requires IUT.

BIO-12-CH02-0056 | Medium | Menstrual cycle

Correct answer: D. FSH

Solution: The chapter links Pituitary hormone promoting follicular growth with FSH, so FSH completes the statement.

Distractor review: A: Tubectomy belongs with 'Surgical interruption of the fallopian tubes', whereas the tested description is 'Pituitary hormone promoting follicular growth'. B: Ovary belongs with 'Organ producing ova, estrogen and progesterone', whereas the tested description is 'Pituitary hormone promoting follicular growth'. C: Primary spermatocyte belongs with 'Cell completing meiosis I to form secondary spermatocytes', whereas the tested description is 'Pituitary hormone promoting follicular growth'.

BIO-12-CH02-0057 | Easy | Gametogenesis

Correct answer: A. Diploid male germ cells multiplying by mitosis

Solution: Spermatogonia represents 'Diploid male germ cells multiplying by mitosis'. Each alternative describes a different term or result.

Distractor review: B: 'Duct carrying sperm away from the epididymis' identifies Vas deferens, not Spermatogonia. C: 'Site of sperm formation' identifies Seminiferous tubules, not Spermatogonia. D: 'Uterine layer receiving the implanting blastocyst' identifies Endometrium, not Spermatogonia.

BIO-12-CH02-0058 | Medium | Asexual reproduction

Correct answer: A. Binary fission - Amoeba

Solution: The accurate entry is 'Binary fission - Amoeba'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Hormone chiefly maintaining the secretory endometrium' should be paired with Progesterone, not Endometrium. C: 'Bacterial sexually transmitted infection caused by Neisseria' should be paired with Gonorrhoea, not Vas deferens. D: 'Pituitary hormone promoting follicular growth' should be paired with FSH, not AIDS.

BIO-12-CH02-0059 | Hard | Gametogenesis

Correct answer: C. Diploid male germ cells multiplying by mitosis - Spermatogonia; Duct carrying sperm away from the epididymis - Vas deferens

Solution: Both parts of 'Diploid male germ cells multiplying by mitosis - Spermatogonia; Duct carrying sperm away from the epididymis - Vas deferens' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Uterine layer receiving the implanting blastocyst' requires Endometrium, and 'Budding' requires Hydra. B: The answers have been interchanged: 'Pituitary hormone promoting follicular growth' requires FSH, and 'Uterine layer receiving the implanting blastocyst' requires Endometrium. D: The answers have been interchanged: 'Site of sperm formation' requires Seminiferous tubules, and 'Pituitary hormone promoting follicular growth' requires FSH.

BIO-12-CH02-0060 | Easy | Menstrual cycle

Correct answer: D. Temporary ovarian body formed after ovulation

Solution: Corpus luteum represents 'Temporary ovarian body formed after ovulation'. Each alternative describes a different term or result.

Distractor review: A: 'Transfer of an embryo with more than eight blastomeres into the uterus' identifies IUT, not Corpus luteum. B: 'Duct carrying sperm away from the epididymis' identifies Vas deferens, not Corpus luteum. C: 'Immediate endocrine trigger for ovulation' identifies LH surge, not Corpus luteum.

BIO-12-CH02-0061 | Medium | Asexual reproduction**Correct answer: B. Planaria**

Solution: The observation 'Regeneration producing complete individuals' points to Planaria; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Leydig cells belongs with 'Cells secreting testicular androgens', whereas the tested description is 'Regeneration producing complete individuals'. C: Copper-T belongs with 'Copper-releasing device placed in the uterus', whereas the tested description is 'Regeneration producing complete individuals'. D: ZIFT belongs with 'Transfer of an early embryo with up to eight blastomeres into the oviduct', whereas the tested description is 'Regeneration producing complete individuals'.

BIO-12-CH02-0062 | Easy | Asexual reproduction**Correct answer: C. Freshwater sponge**

Solution: 'Gemmule formation' is the defining description for Freshwater sponge. The other choices belong to different chapter concepts.

Distractor review: A: Tubectomy belongs with 'Surgical interruption of the fallopian tubes', whereas the tested description is 'Gemmule formation'. B: Copper-T belongs with 'Copper-releasing device placed in the uterus', whereas the tested description is 'Gemmule formation'. D: Amoeba belongs with 'Binary fission', whereas the tested description is 'Gemmule formation'.

BIO-12-CH02-0063 | Challenging | Early development**Correct answer: D. I, III and IV only**

Solution: Statement I is correct because 'Process producing the three primary germ layers' corresponds to Gastrulation. Statement II is incorrect: 'Rapid mitotic divisions of the zygote without overall growth' corresponds to Cleavage, not Morula. Statement III is correct because 'Solid ball of about 8 to 16 blastomeres' corresponds to Morula. Statement IV is correct because 'Binary fission' corresponds to Amoeba. Therefore, the valid combination is I, III and IV only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) IV. B: This combination omits true statement(s) I, IV. C: This combination includes false statement(s) II and omits true statement(s) III, IV.

BIO-12-CH02-0064 | Hard | Early development**Correct answer: D. Solid ball of about 8 to 16 blastomeres - Morula; Diploid male germ cells multiplying by mitosis - Spermatogonia**

Solution: Both parts of 'Solid ball of about 8 to 16 blastomeres - Morula; Diploid male germ cells multiplying by mitosis - Spermatogonia' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Cells nourishing developing germ cells' requires Sertoli cells, and 'Organ producing ova, estrogen and progesterone' requires Ovary. B: The answers have been interchanged: 'Transfer of an early embryo with up to eight blastomeres into the oviduct' requires ZIFT, and 'Hormone chiefly maintaining the secretory endometrium' requires Progesterone. C: The answers have been interchanged: 'Hormone chiefly maintaining the secretory endometrium' requires Progesterone, and 'Site of sperm formation' requires Seminiferous tubules.

BIO-12-CH02-0065 | Easy | Assisted reproduction and infection**Correct answer: A. Transfer of an early embryo with up to eight blastomeres into the oviduct**

Solution: ZIFT represents 'Transfer of an early embryo with up to eight blastomeres into the oviduct'. Each alternative describes a different term or result.

Distractor review: B: 'Binary fission' identifies Amoeba, not ZIFT. C: 'Site of sperm formation' identifies Seminiferous tubules, not ZIFT. D: 'Viral syndrome caused by HIV' identifies AIDS, not ZIFT.

BIO-12-CH02-0066 | Easy | Female reproductive system**Correct answer: D. Usual site of fertilization**

Solution: Ampulla of the fallopian tube represents 'Usual site of fertilization'. Each alternative describes a different term or result.

Distractor review: A: 'Cells nourishing developing germ cells' identifies Sertoli cells, not Ampulla of the fallopian tube. B: 'Uterine layer receiving the implanting blastocyst' identifies Endometrium, not Ampulla of the fallopian tube. C: 'Diploid male germ cells multiplying by mitosis' identifies Spermatogonia, not Ampulla of the fallopian tube.

BIO-12-CH02-0067 | Hard | Female reproductive system

Correct answer: A. Usual site of fertilization - Ampulla of the fallopian tube; Cells nourishing developing germ cells - Sertoli cells

Solution: Both parts of 'Usual site of fertilization - Ampulla of the fallopian tube; Cells nourishing developing germ cells - Sertoli cells' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Diploid male germ cells multiplying by mitosis' requires Spermatogonia, and 'Gemmule formation' requires Freshwater sponge. C: The answers have been interchanged: 'Gemmule formation' requires Freshwater sponge, and 'Surgical interruption of the vas deferens' requires Vasectomy. D: The answers have been interchanged: 'Surgical interruption of the vas deferens' requires Vasectomy, and 'Site of sperm formation' requires Seminiferous tubules.

BIO-12-CH02-0068 | Hard | Early development

Correct answer: D. Embryonic stage with trophoblast and inner cell mass - Spermiogenesis

Solution: 'Embryonic stage with trophoblast and inner cell mass' should be paired with Blastocyst, not Spermiogenesis. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Duct carrying sperm away from the epididymis - Vas deferens. B: This is a valid association: Transfer of an embryo with more than eight blastomeres into the uterus - IUT. C: This is a valid association: Surgical interruption of the fallopian tubes - Tubectomy.

BIO-12-CH02-0069 | Hard | Menstrual cycle

Correct answer: B. Temporary ovarian body formed after ovulation - Corpus luteum; Surgical interruption of the vas deferens - Vasectomy

Solution: Both parts of 'Temporary ovarian body formed after ovulation - Corpus luteum; Surgical interruption of the vas deferens - Vasectomy' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Cell completing meiosis I to form secondary spermatocytes' requires Primary spermatocyte, and 'Copper-releasing device placed in the uterus' requires Copper-T. C: The answers have been interchanged: 'Duct carrying sperm away from the epididymis' requires Vas deferens, and 'Cells secreting testicular androgens' requires Leydig cells. D: The answers have been interchanged: 'Process producing the three primary germ layers' requires Gastrulation, and 'Duct carrying sperm away from the epididymis' requires Vas deferens.

BIO-12-CH02-0070 | Medium | Early development

Correct answer: C. Embryonic stage with trophoblast and inner cell mass - Blastocyst

Solution: The accurate entry is 'Embryonic stage with trophoblast and inner cell mass - Blastocyst'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Transfer of an embryo with more than eight blastomeres into the uterus' should be paired with IUT, not Tubectomy. B: 'Duct carrying sperm away from the epididymis' should be paired with Vas deferens, not Blastocyst. D: 'Transformation of spermatids into spermatozoa' should be paired with Spermiogenesis, not IUT.

BIO-12-CH02-0071 | Hard | Assisted reproduction and infection

Correct answer: D. Transfer of an early embryo with up to eight blastomeres into the oviduct - Infundibulum

Solution: 'Transfer of an early embryo with up to eight blastomeres into the oviduct' should be paired with ZIFT, not Infundibulum. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Site of sperm formation - Seminiferous tubules. B: This is a valid association: Binary fission - Amoeba. C: This is a valid association: Transformation of spermatids into spermatozoa - Spermiogenesis.

BIO-12-CH02-0072 | Medium | Assisted reproduction and infection

Correct answer: C. Viral syndrome caused by HIV - AIDS

Solution: The accurate entry is 'Viral syndrome caused by HIV - AIDS'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Solid ball of about 8 to 16 blastomeres' should be paired with Morula, not Planaria. B: 'Copper-releasing device placed in the uterus' should be paired with Copper-T, not AIDS. D: 'Funnel-like oviduct part bearing fimbriae' should be paired with Infundibulum, not Sertoli cells.

BIO-12-CH02-0073 | Medium | Asexual reproduction**Correct answer: B. Regeneration producing complete individuals - Planaria**

Solution: The accurate entry is 'Regeneration producing complete individuals - Planaria'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Copper-releasing device placed in the uterus' should be paired with Copper-T, not ZIFT. C: 'Cells secreting testicular androgens' should be paired with Leydig cells, not Blastocyst. D: 'Cap-like structure over the sperm nucleus' should be paired with Acrosome, not Sertoli cells.

BIO-12-CH02-0074 | Easy | Early development**Correct answer: D. Process producing the three primary germ layers**

Solution: Gastrulation represents 'Process producing the three primary germ layers'. Each alternative describes a different term or result.

Distractor review: A: 'Usual site of fertilization' identifies Ampulla of the fallopian tube, not Gastrulation. B: 'Rapid mitotic divisions of the zygote without overall growth' identifies Cleavage, not Gastrulation. C: 'Solid ball of about 8 to 16 blastomeres' identifies Morula, not Gastrulation.

BIO-12-CH02-0075 | Medium | Assisted reproduction and infection**Correct answer: C. ZIFT**

Solution: The chapter links Transfer of an early embryo with up to eight blastomeres into the oviduct with ZIFT, so ZIFT completes the statement.

Distractor review: A: AIDS belongs with 'Viral syndrome caused by HIV', whereas the tested description is 'Transfer of an early embryo with up to eight blastomeres into the oviduct'. B: Infundibulum belongs with 'Funnel-like oviduct part bearing fimbriae', whereas the tested description is 'Transfer of an early embryo with up to eight blastomeres into the oviduct'. D: Spermiogenesis belongs with 'Transformation of spermatids into spermatozoa', whereas the tested description is 'Transfer of an early embryo with up to eight blastomeres into the oviduct'.

BIO-12-CH02-0076 | Easy | Male reproductive system**Correct answer: C. Duct carrying sperm away from the epididymis**

Solution: Vas deferens represents 'Duct carrying sperm away from the epididymis'. Each alternative describes a different term or result.

Distractor review: A: 'Transfer of an embryo with more than eight blastomeres into the uterus' identifies IUT, not Vas deferens. B: 'Process producing the three primary germ layers' identifies Gastrulation, not Vas deferens. D: 'Organ producing ova, estrogen and progesterone' identifies Ovary, not Vas deferens.

BIO-12-CH02-0077 | Hard | Assisted reproduction and infection**Correct answer: A. Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea; Regeneration producing complete individuals - Planaria**

Solution: Both parts of 'Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea; Regeneration producing complete individuals - Planaria' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Uterine layer receiving the implanting blastocyst' requires Endometrium, and 'Barrier method that also lowers STI transmission' requires Condom. C: The answers have been interchanged: 'Funnel-like oviduct part bearing fimbriae' requires Infundibulum, and 'Transformation of spermatids into spermatozoa' requires Spermiogenesis. D: The answers have been interchanged: 'Diploid male germ cells multiplying by mitosis' requires Spermatogonia, and 'Copper-releasing device placed in the uterus' requires Copper-T.

BIO-12-CH02-0078 | Medium | Gametogenesis**Correct answer: C. Diploid male germ cells multiplying by mitosis - Spermatogonia**

Solution: The accurate entry is 'Diploid male germ cells multiplying by mitosis - Spermatogonia'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Duct carrying sperm away from the epididymis' should be paired with Vas deferens, not FSH. B: 'Uterine layer receiving the implanting blastocyst' should be paired with Endometrium, not AIDS. D: 'Pituitary hormone promoting follicular growth' should be paired with FSH, not Hydra.

BIO-12-CH02-0079 | Easy | Early development**Correct answer: B. Blastocyst**

Solution: 'Embryonic stage with trophoblast and inner cell mass' is the defining description for Blastocyst. The other choices belong to different chapter concepts.

Distractor review: A: Spermiogenesis belongs with 'Transformation of spermatids into spermatozoa', whereas the tested description is 'Embryonic stage with trophoblast and inner cell mass'. C: LH surge belongs with 'Immediate endocrine trigger for ovulation', whereas the tested description is 'Embryonic stage with trophoblast and inner cell mass'. D: Morula belongs with 'Solid ball of about 8 to 16 blastomeres', whereas the tested description is 'Embryonic stage with trophoblast and inner cell mass'.

BIO-12-CH02-0080 | Medium | Gametogenesis**Correct answer: D. Acrosome**

Solution: The observation 'Cap-like structure over the sperm nucleus' points to Acrosome; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Blastocyst belongs with 'Embryonic stage with trophoblast and inner cell mass', whereas the tested description is 'Cap-like structure over the sperm nucleus'. B: Spermiogenesis belongs with 'Transformation of spermatids into spermatozoa', whereas the tested description is 'Cap-like structure over the sperm nucleus'. C: Ampulla of the fallopian tube belongs with 'Usual site of fertilization', whereas the tested description is 'Cap-like structure over the sperm nucleus'.

BIO-12-CH02-0081 | Challenging | Early development**Correct answer: D. I and II only**

Solution: Statement I is correct because 'Embryonic stage with trophoblast and inner cell mass' corresponds to Blastocyst. Statement II is correct because 'Process producing the three primary germ layers' corresponds to Gastrulation. Statement III is incorrect: 'Rapid mitotic divisions of the zygote without overall growth' corresponds to Cleavage, not AIDS. Statement IV is incorrect: 'Viral syndrome caused by HIV' corresponds to AIDS, not Blastocyst. Therefore, the valid combination is I and II only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) I. B: This combination includes false statement(s) III. C: This combination includes false statement(s) III, IV.

BIO-12-CH02-0082 | Medium | Early development**Correct answer: C. Morula**

Solution: The chapter links Solid ball of about 8 to 16 blastomeres with Morula, so Morula completes the statement.

Distractor review: A: Cleavage belongs with 'Rapid mitotic divisions of the zygote without overall growth', whereas the tested description is 'Solid ball of about 8 to 16 blastomeres'. B: Spermatogonia belongs with 'Diploid male germ cells multiplying by mitosis', whereas the tested description is 'Solid ball of about 8 to 16 blastomeres'. D: Blastocyst belongs with 'Embryonic stage with trophoblast and inner cell mass', whereas the tested description is 'Solid ball of about 8 to 16 blastomeres'.

BIO-12-CH02-0083 | Easy | Male reproductive system**Correct answer: D. Vas deferens**

Solution: 'Duct carrying sperm away from the epididymis' is the defining description for Vas deferens. The other choices belong to different chapter concepts.

Distractor review: A: Leydig cells belongs with 'Cells secreting testicular androgens', whereas the tested description is 'Duct carrying sperm away from the epididymis'. B: IUT belongs with 'Transfer of an embryo with more than eight blastomeres into the uterus', whereas the tested description is 'Duct carrying sperm away from the epididymis'. C: Acrosome belongs with 'Cap-like structure over the sperm nucleus', whereas the tested description is 'Duct carrying sperm away from the epididymis'.

BIO-12-CH02-0084 | Medium | Menstrual cycle**Correct answer: B. Progesterone**

Solution: The chapter links Hormone chiefly maintaining the secretory endometrium with Progesterone, so Progesterone completes the statement.

Distractor review: A: Spermiogenesis belongs with 'Transformation of spermatids into spermatozoa', whereas the tested description is 'Hormone chiefly maintaining the secretory endometrium'. C: Freshwater sponge belongs with 'Gemmule formation', whereas the tested description is 'Hormone chiefly maintaining the secretory endometrium'. D: Blastocyst belongs with 'Embryonic stage with trophoblast and inner cell mass', whereas the tested description is 'Hormone chiefly maintaining the secretory endometrium'.

BIO-12-CH02-0085 | Medium | Male reproductive system**Correct answer: A. Sertoli cells**

Solution: The chapter links Cells nourishing developing germ cells with Sertoli cells, so Sertoli cells completes the statement.

Distractor review: B: Spermiogenesis belongs with 'Transformation of spermatids into spermatozoa', whereas the tested description is 'Cells nourishing developing germ cells'. C: ZIFT belongs with 'Transfer of an early embryo with up to eight blastomeres into the oviduct', whereas the tested description is 'Cells nourishing developing germ cells'. D: Spermatogonia belongs with 'Diploid male germ cells multiplying by mitosis', whereas the tested description is 'Cells nourishing developing germ cells'.

BIO-12-CH02-0086 | Medium | Assisted reproduction and infection**Correct answer: C. AIDS**

Solution: The observation 'Viral syndrome caused by HIV' points to AIDS; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Morula belongs with 'Solid ball of about 8 to 16 blastomeres', whereas the tested description is 'Viral syndrome caused by HIV'. B: Gonorrhoea belongs with 'Bacterial sexually transmitted infection caused by Neisseria', whereas the tested description is 'Viral syndrome caused by HIV'. D: IUT belongs with 'Transfer of an embryo with more than eight blastomeres into the uterus', whereas the tested description is 'Viral syndrome caused by HIV'.

BIO-12-CH02-0087 | Hard | Male reproductive system**Correct answer: D. Duct carrying sperm away from the epididymis - IUT**

Solution: 'Duct carrying sperm away from the epididymis' should be paired with Vas deferens, not IUT. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Cap-like structure over the sperm nucleus - Acrosome. B: This is a valid association: Process producing the three primary germ layers - Gastrulation. C: This is a valid association: Organ producing ova, estrogen and progesterone - Ovary.

BIO-12-CH02-0088 | Medium | Gametogenesis**Correct answer: B. Transformation of spermatids into spermatozoa - Spermiogenesis**

Solution: The accurate entry is 'Transformation of spermatids into spermatozoa - Spermiogenesis'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Temporary ovarian body formed after ovulation' should be paired with Corpus luteum, not Spermiogenesis. C: 'Cells nourishing developing germ cells' should be paired with Sertoli cells, not Planaria. D: 'Viral syndrome caused by HIV' should be paired with AIDS, not Vasectomy.

BIO-12-CH02-0089 | Easy | Assisted reproduction and infection**Correct answer: A. Bacterial sexually transmitted infection caused by Neisseria**

Solution: Gonorrhoea represents 'Bacterial sexually transmitted infection caused by Neisseria'. Each alternative describes a different term or result.

Distractor review: B: 'Uterine layer receiving the implanting blastocyst' identifies Endometrium, not Gonorrhoea. C: 'Transformation of spermatids into spermatozoa' identifies Spermiogenesis, not Gonorrhoea. D: 'Regeneration producing complete individuals' identifies Planaria, not Gonorrhoea.

BIO-12-CH02-0090 | Medium | Assisted reproduction and infection**Correct answer: C. Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea**

Solution: The accurate entry is 'Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Barrier method that also lowers STI transmission' should be paired with Condom, not Copper-T. B: 'Diploid male germ cells multiplying by mitosis' should be paired with Spermatogonia, not Gonorrhoea. D: 'Transformation of spermatids into spermatozoa' should be paired with Spermiogenesis, not Condom.

BIO-12-CH02-0091 | Medium | Female reproductive system**Correct answer: A. Endometrium**

Solution: The chapter links Uterine layer receiving the implanting blastocyst with Endometrium, so Endometrium completes the statement.

Distractor review: B: Planaria belongs with 'Regeneration producing complete individuals', whereas the tested description is 'Uterine layer receiving the implanting blastocyst'. C: LH surge belongs with 'Immediate endocrine trigger for ovulation', whereas the tested description is 'Uterine layer receiving the implanting blastocyst'. D: Ovary belongs with 'Organ producing ova, estrogen and progesterone', whereas the tested description is 'Uterine layer receiving the implanting blastocyst'.

BIO-12-CH02-0092 | Medium | Female reproductive system**Correct answer: D. Endometrium**

Solution: The observation 'Uterine layer receiving the implanting blastocyst' points to Endometrium; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Infundibulum belongs with 'Funnel-like oviduct part bearing fimbriae', whereas the tested description is 'Uterine layer receiving the implanting blastocyst'. B: LH surge belongs with 'Immediate endocrine trigger for ovulation', whereas the tested description is 'Uterine layer receiving the implanting blastocyst'. C: Corpus luteum belongs with 'Temporary ovarian body formed after ovulation', whereas the tested description is 'Uterine layer receiving the implanting blastocyst'.

BIO-12-CH02-0093 | Hard | Menstrual cycle**Correct answer: B. Hormone chiefly maintaining the secretory endometrium - Spermiogenesis**

Solution: 'Hormone chiefly maintaining the secretory endometrium' should be paired with Progesterone, not Spermiogenesis. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Surgical interruption of the vas deferens - Vasectomy. C: This is a valid association: Embryonic stage with trophoblast and inner cell mass - Blastocyst. D: This is a valid association: Organ producing ova, estrogen and progesterone - Ovary.

BIO-12-CH02-0094 | Hard | Birth control and reproductive health**Correct answer: C. Copper-releasing device placed in the uterus - Copper-T; Pituitary hormone promoting follicular growth - FSH**

Solution: Both parts of 'Copper-releasing device placed in the uterus - Copper-T; Pituitary hormone promoting follicular growth - FSH' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Cell completing meiosis I to form secondary spermatocytes' requires Primary spermatocyte, and 'Diploid male germ cells multiplying by mitosis' requires Spermatogonia. B: The answers have been interchanged: 'Diploid male germ cells multiplying by mitosis' requires Spermatogonia, and 'Uterine layer receiving the implanting blastocyst' requires Endometrium. D: The answers have been interchanged: 'Site of sperm formation' requires Seminiferous tubules, and 'Duct carrying sperm away from the epididymis' requires Vas deferens.

BIO-12-CH02-0095 | Challenging | Asexual reproduction**Correct answer: A. I, II and IV only**

Solution: Statement I is correct because 'Binary fission' corresponds to Amoeba. Statement II is correct because 'Gemmule formation' corresponds to Freshwater sponge. Statement III is incorrect: 'Budding' corresponds to Hydra, not Condom. Statement IV is correct because 'Barrier method that also lowers STI transmission' corresponds to Condom. Therefore, the valid combination is I, II and IV only.

Distractor review: B: This combination includes false statement(s) III and omits true statement(s) IV. C: This combination includes false statement(s) III and omits true statement(s) I, II. D: This combination omits true statement(s) I, II.

BIO-12-CH02-0096 | Hard | Early development**Correct answer: D. Process producing the three primary germ layers - Gastrulation; Usual site of fertilization - Ampulla of the fallopian tube**

Solution: Both parts of 'Process producing the three primary germ layers - Gastrulation; Usual site of fertilization - Ampulla of the fallopian tube' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Regeneration producing complete individuals' requires Planaria, and 'Pituitary hormone promoting follicular growth' requires FSH. B: The answers have been interchanged: 'Cells secreting testicular androgens' requires Leydig cells, and 'Duct carrying sperm away from the epididymis' requires Vas deferens. C: The answers have been interchanged: 'Pituitary hormone promoting follicular growth' requires FSH, and 'Cells secreting testicular androgens' requires Leydig cells.

BIO-12-CH02-0097 | Hard | Early development**Correct answer: C. Rapid mitotic divisions of the zygote without overall growth - Cleavage; Cap-like structure over the sperm nucleus - Acrosome**

Solution: Both parts of 'Rapid mitotic divisions of the zygote without overall growth - Cleavage; Cap-like structure over the sperm nucleus - Acrosome' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Duct carrying sperm away from the epididymis' requires Vas deferens, and 'Cell completing meiosis I to form secondary spermatocytes' requires Primary spermatocyte. B: The answers have been interchanged: 'Cell completing meiosis I to form secondary spermatocytes' requires Primary spermatocyte, and 'Gemmule formation' requires Freshwater sponge. D: The answers have been interchanged: 'Pituitary hormone promoting follicular growth' requires FSH, and 'Budding' requires Hydra.

BIO-12-CH02-0098 | Easy | Menstrual cycle**Correct answer: B. Progesterone**

Solution: 'Hormone chiefly maintaining the secretory endometrium' is the defining description for Progesterone. The other choices belong to different chapter concepts.

Distractor review: A: Freshwater sponge belongs with 'Gemmule formation', whereas the tested description is 'Hormone chiefly maintaining the secretory endometrium'. C: FSH belongs with 'Pituitary hormone promoting follicular growth', whereas the tested description is 'Hormone chiefly maintaining the secretory endometrium'. D: Blastocyst belongs with 'Embryonic stage with trophoblast and inner cell mass', whereas the tested description is 'Hormone chiefly maintaining the secretory endometrium'.

BIO-12-CH02-0099 | Hard | Asexual reproduction**Correct answer: B. Gemmule formation - Gonorrhoea**

Solution: 'Gemmule formation' should be paired with Freshwater sponge, not Gonorrhoea. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Rapid mitotic divisions of the zygote without overall growth - Cleavage. C: This is a valid association: Copper-releasing device placed in the uterus - Copper-T. D: This is a valid association: Usual site of fertilization - Ampulla of the fallopian tube.

BIO-12-CH02-0100 | Easy | Early development**Correct answer: D. Rapid mitotic divisions of the zygote without overall growth**

Solution: Cleavage represents 'Rapid mitotic divisions of the zygote without overall growth'. Each alternative describes a different term or result.

Distractor review: A: 'Duct carrying sperm away from the epididymis' identifies Vas deferens, not Cleavage. B: 'Solid ball of about 8 to 16 blastomeres' identifies Morula, not Cleavage. C: 'Pituitary hormone promoting follicular growth' identifies FSH, not Cleavage.

BIO-12-CH02-0101 | Hard | Birth control and reproductive health**Correct answer: A. Copper-releasing device placed in the uterus - FSH**

Solution: 'Copper-releasing device placed in the uterus' should be paired with Copper-T, not FSH. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Duct carrying sperm away from the epididymis - Vas deferens. C: This is a valid association: Diploid male germ cells multiplying by mitosis - Spermatogonia. D: This is a valid association: Site of sperm formation - Seminiferous tubules.

BIO-12-CH02-0102 | Easy | Asexual reproduction**Correct answer: A. Gemmule formation**

Solution: Freshwater sponge represents 'Gemmule formation'. Each alternative describes a different term or result.

Distractor review: B: 'Immediate endocrine trigger for ovulation' identifies LH surge, not Freshwater sponge. C: 'Surgical interruption of the fallopian tubes' identifies Tubectomy, not Freshwater sponge. D: 'Copper-releasing device placed in the uterus' identifies Copper-T, not Freshwater sponge.

BIO-12-CH02-0103 | Hard | Gametogenesis**Correct answer: C. Cell completing meiosis I to form secondary spermatocytes - Infundibulum**

Solution: 'Cell completing meiosis I to form secondary spermatocytes' should be paired with Primary spermatocyte, not Infundibulum. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Process producing the three primary germ layers - Gastrulation. B: This is a valid association: Budding - Hydra. D: This is a valid association: Site of sperm formation - Seminiferous tubules.

BIO-12-CH02-0104 | Medium | Birth control and reproductive health**Correct answer: A. Tubectomy**

Solution: The chapter links Surgical interruption of the fallopian tubes with Tubectomy, so Tubectomy completes the statement.

Distractor review: B: Freshwater sponge belongs with 'Gemmule formation', whereas the tested description is 'Surgical interruption of the fallopian tubes'. C: Vasectomy belongs with 'Surgical interruption of the vas deferens', whereas the tested description is 'Surgical interruption of the fallopian tubes'. D: Acrosome belongs with 'Cap-like structure over the sperm nucleus', whereas the tested description is 'Surgical interruption of the fallopian tubes'.

BIO-12-CH02-0105 | Easy | Assisted reproduction and infection**Correct answer: A. Viral syndrome caused by HIV**

Solution: AIDS represents 'Viral syndrome caused by HIV'. Each alternative describes a different term or result.

Distractor review: B: 'Transfer of an embryo with more than eight blastomeres into the uterus' identifies IUT, not AIDS. C: 'Regeneration producing complete individuals' identifies Planaria, not AIDS. D: 'Site of sperm formation' identifies Seminiferous tubules, not AIDS.

BIO-12-CH02-0106 | Medium | Female reproductive system**Correct answer: B. Ampulla of the fallopian tube**

Solution: The observation 'Usual site of fertilization' points to Ampulla of the fallopian tube; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Endometrium belongs with 'Uterine layer receiving the implanting blastocyst', whereas the tested description is 'Usual site of fertilization'. C: Freshwater sponge belongs with 'Gemmule formation', whereas the tested description is 'Usual site of fertilization'. D: Planaria belongs with 'Regeneration producing complete individuals', whereas the tested description is 'Usual site of fertilization'.

BIO-12-CH02-0107 | Medium | Assisted reproduction and infection**Correct answer: C. Transfer of an embryo with more than eight blastomeres into the uterus - IUT**

Solution: The accurate entry is 'Transfer of an embryo with more than eight blastomeres into the uterus - IUT'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Organ producing ova, estrogen and progesterone' should be paired with Ovary, not IUT. B: 'Transformation of spermatids into spermatozoa' should be paired with Spermiogenesis, not Amoeba. D: 'Temporary ovarian body formed after ovulation' should be paired with Corpus luteum, not Spermiogenesis.

BIO-12-CH02-0108 | Medium | Menstrual cycle**Correct answer: C. Corpus luteum**

Solution: The observation 'Temporary ovarian body formed after ovulation' points to Corpus luteum; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Leydig cells belongs with 'Cells secreting testicular androgens', whereas the tested description is 'Temporary ovarian body formed after ovulation'. B: Progesterone belongs with 'Hormone chiefly maintaining the secretory endometrium', whereas the tested description is 'Temporary ovarian body formed after ovulation'. D: LH surge belongs with 'Immediate endocrine trigger for ovulation', whereas the tested description is 'Temporary ovarian body formed after ovulation'.

BIO-12-CH02-0109 | Challenging | Male reproductive system**Correct answer: B. IV only**

Solution: Statement I is incorrect: 'Duct carrying sperm away from the epididymis' corresponds to Vas deferens, not Sertoli cells. Statement II is incorrect: 'Cells nourishing developing germ cells' corresponds to Sertoli cells, not Leydig cells. Statement III is incorrect: 'Cells secreting testicular androgens' corresponds to Leydig cells, not FSH. Statement IV is correct because 'Pituitary hormone promoting follicular growth' corresponds to FSH. Therefore, the valid combination is IV only.

Distractor review: A: This combination includes false statement(s) II, III. C: This combination includes false statement(s) II, III and omits true statement(s) IV. D: This combination includes false statement(s) I.

BIO-12-CH02-0110 | Challenging | Assisted reproduction and infection**Correct answer: C. I, II and III only**

Solution: Statement I is correct because 'Bacterial sexually transmitted infection caused by Neisseria' corresponds to Gonorrhoea. Statement II is correct because 'Transfer of an early embryo with up to eight blastomeres into the oviduct' corresponds to ZIFT. Statement III is correct because 'Viral syndrome caused by HIV' corresponds to AIDS. Statement IV is incorrect: 'Pituitary hormone promoting follicular growth' corresponds to FSH, not Gonorrhoea. Therefore, the valid combination is I, II and III only.

Distractor review: A: This combination omits true statement(s) I, III. B: This combination includes false statement(s) IV and omits true statement(s) II. D: This combination omits true statement(s) II.

BIO-12-CH02-0111 | Medium | Male reproductive system**Correct answer: D. Site of sperm formation - Seminiferous tubules**

Solution: The accurate entry is 'Site of sperm formation - Seminiferous tubules'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Uterine layer receiving the implanting blastocyst' should be paired with Endometrium, not Tubectomy. B: 'Organ producing ova, estrogen and progesterone' should be paired with Ovary, not LH surge. C: 'Hormone chiefly maintaining the secretory endometrium' should be paired with Progesterone, not Spermatogonia.

BIO-12-CH02-0112 | Medium | Menstrual cycle**Correct answer: A. Immediate endocrine trigger for ovulation - LH surge**

Solution: The accurate entry is 'Immediate endocrine trigger for ovulation - LH surge'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Transfer of an early embryo with up to eight blastomeres into the oviduct' should be paired with ZIFT, not AIDS. C: 'Uterine layer receiving the implanting blastocyst' should be paired with Endometrium, not Vasectomy. D: 'Organ producing ova, estrogen and progesterone' should be paired with Ovary, not ZIFT.

BIO-12-CH02-0113 | Medium | Gametogenesis**Correct answer: D. Spermiogenesis**

Solution: The observation 'Transformation of spermatids into spermatozoa' points to Spermiogenesis; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: AIDS belongs with 'Viral syndrome caused by HIV', whereas the tested description is 'Transformation of spermatids into spermatozoa'. B: Primary spermatocyte belongs with 'Cell completing meiosis I to form secondary spermatocytes', whereas the tested description is 'Transformation of spermatids into spermatozoa'. C: Vasectomy belongs with 'Surgical interruption of the vas deferens', whereas the tested description is 'Transformation of spermatids into spermatozoa'.

BIO-12-CH02-0114 | Easy | Menstrual cycle**Correct answer: C. FSH**

Solution: 'Pituitary hormone promoting follicular growth' is the defining description for FSH. The other choices belong to different chapter concepts.

Distractor review: A: Tubectomy belongs with 'Surgical interruption of the fallopian tubes', whereas the tested description is 'Pituitary hormone promoting follicular growth'. B: ZIFT belongs with 'Transfer of an early embryo with up to eight blastomeres into the oviduct', whereas the tested description is 'Pituitary hormone promoting follicular growth'. D: Blastocyst belongs with 'Embryonic stage with trophoblast and inner cell mass', whereas the tested description is 'Pituitary hormone promoting follicular growth'.

BIO-12-CH02-0115 | Challenging | Female reproductive system**Correct answer: B. III only**

Solution: Statement I is incorrect: 'Funnel-like oviduct part bearing fimbriae' corresponds to Infundibulum, not Ampulla of the fallopian tube. Statement II is incorrect: 'Usual site of fertilization' corresponds to Ampulla of the fallopian tube, not Endometrium. Statement III is correct because 'Uterine layer receiving the implanting blastocyst' corresponds to Endometrium. Statement IV is incorrect: 'Temporary ovarian body formed after ovulation' corresponds to Corpus luteum, not Infundibulum. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) I, II. C: This combination includes false statement(s) I. D: This combination includes false statement(s) IV.

BIO-12-CH02-0116 | Hard | Birth control and reproductive health**Correct answer: B. Surgical interruption of the vas deferens - Cleavage**

Solution: 'Surgical interruption of the vas deferens' should be paired with Vasectomy, not Cleavage. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Funnel-like oviduct part bearing fimbriae - Infundibulum. C: This is a valid association: Process producing the three primary germ layers - Gastrulation. D: This is a valid association: Usual site of fertilization - Ampulla of the fallopian tube.

BIO-12-CH02-0117 | Easy | Early development**Correct answer: C. Solid ball of about 8 to 16 blastomeres**

Solution: Morula represents 'Solid ball of about 8 to 16 blastomeres'. Each alternative describes a different term or result.

Distractor review: A: 'Organ producing ova, estrogen and progesterone' identifies Ovary, not Morula. B: 'Hormone chiefly maintaining the secretory endometrium' identifies Progesterone, not Morula. D: 'Rapid mitotic divisions of the zygote without overall growth' identifies Cleavage, not Morula.

BIO-12-CH02-0118 | Medium | Female reproductive system

Correct answer: A. Infundibulum

Solution: The chapter links Funnel-like oviduct part bearing fimbriae with Infundibulum, so Infundibulum completes the statement.

Distractor review: B: Amoeba belongs with 'Binary fission', whereas the tested description is 'Funnel-like oviduct part bearing fimbriae'. C: Ampulla of the fallopian tube belongs with 'Usual site of fertilization', whereas the tested description is 'Funnel-like oviduct part bearing fimbriae'. D: Gonorrhoea belongs with 'Bacterial sexually transmitted infection caused by Neisseria', whereas the tested description is 'Funnel-like oviduct part bearing fimbriae'.

BIO-12-CH02-0119 | Hard | Menstrual cycle

Correct answer: C. Immediate endocrine trigger for ovulation - Seminiferous tubules

Solution: 'Immediate endocrine trigger for ovulation' should be paired with LH surge, not Seminiferous tubules. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Organ producing ova, estrogen and progesterone - Ovary. B: This is a valid association: Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT. D: This is a valid association: Surgical interruption of the vas deferens - Vasectomy.

BIO-12-CH02-0120 | Easy | Gametogenesis

Correct answer: C. Acrosome

Solution: 'Cap-like structure over the sperm nucleus' is the defining description for Acrosome. The other choices belong to different chapter concepts.

Distractor review: A: LH surge belongs with 'Immediate endocrine trigger for ovulation', whereas the tested description is 'Cap-like structure over the sperm nucleus'. B: Spermiogenesis belongs with 'Transformation of spermatids into spermatozoa', whereas the tested description is 'Cap-like structure over the sperm nucleus'. D: Amoeba belongs with 'Binary fission', whereas the tested description is 'Cap-like structure over the sperm nucleus'.

Standard 12 | Chapter 03

Inheritance and Variation

BIO-12-CH03-0001 | Medium | X-linked recessive inheritance

Correct answer: A. One-half of the sons

Solution: Each son receives Y from the father and has a one-half chance of receiving the affected X allele from the carrier mother.

Distractor review: B: The alternative 'All sons' conflicts with the labelled relationship, scale or mechanism in the figure. Each son receives Y from the father and has a one-half chance of receiving the affected X allele from the carrier mother. C: The alternative 'One-quarter of the sons' conflicts with the labelled relationship, scale or mechanism in the figure. Each son receives Y from the father and has a one-half chance of receiving the affected X allele from the carrier mother. D: The alternative 'No sons' conflicts with the labelled relationship, scale or mechanism in the figure. Each son receives Y from the father and has a one-half chance of receiving the affected X allele from the carrier mother.

BIO-12-CH03-0002 | Easy | Monohybrid cross: Tt x Tt

Correct answer: B. 1 TT : 2 Tt : 1 tt

Solution: The four cells are TT, Tt, Tt and tt, giving 1:2:1.

Distractor review: A: The alternative '3 TT : 1 tt' conflicts with the labelled relationship, scale or mechanism in the figure. The four cells are TT, Tt, Tt and tt, giving 1:2:1. C: The alternative '1 Tt : 1 tt' conflicts with the labelled relationship, scale or mechanism in the figure. The four cells are TT, Tt, Tt and tt, giving 1:2:1. D: The alternative '9 : 3 : 3 : 1' conflicts with the labelled relationship, scale or mechanism in the figure. The four cells are TT, Tt, Tt and tt, giving 1:2:1.

BIO-12-CH03-0003 | Medium | Sex-linked inheritance

Correct answer: A. All daughters and no sons

Solution: The chapter links A male transmitting an X-linked allele to with All daughters and no sons, so All daughters and no sons completes the statement.

Distractor review: B: Law of segregation belongs with 'Separation of allele pairs during gamete formation', whereas the tested description is 'A male transmitting an X-linked allele to'. C: 3:1 belongs with 'Monohybrid F2 phenotypic ratio under complete dominance', whereas the tested description is 'A male transmitting an X-linked allele to'. D: Carrier belongs with 'A female heterozygous for a recessive X-linked allele', whereas the tested description is 'A male transmitting an X-linked allele to'.

BIO-12-CH03-0004 | Medium | Sex-linked inheritance

Correct answer: B. Carrier

Solution: The observation 'A female heterozygous for a recessive X-linked allele' points to Carrier; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Test cross belongs with 'Cross with a homozygous recessive individual', whereas the tested description is 'A female heterozygous for a recessive X-linked allele'. C: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'A female heterozygous for a recessive X-linked allele'. D: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'A female heterozygous for a recessive X-linked allele'.

BIO-12-CH03-0005 | Medium | Crosses

Correct answer: C. Expected test-cross ratio for a heterozygous monohybrid - 1:1

Solution: The accurate entry is 'Expected test-cross ratio for a heterozygous monohybrid - 1:1'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Genetic constitution of an organism' should be paired with Genotype, not Male. B: 'Separation of allele pairs during gamete formation' should be paired with Law of segregation, not 1:1. D: 'XXY constitution in a male' should be paired with Klinefelter syndrome, not Codominance.

BIO-12-CH03-0006 | Hard | Monohybrid cross: Tt x Tt

Correct answer: C. 1/4

Solution: Only genotype tt produces the recessive phenotype, and it occupies one of four equally likely cells.

Distractor review: A: The alternative '1/2' conflicts with the labelled relationship, scale or mechanism in the figure. Only genotype tt produces the recessive phenotype, and it occupies one of four equally likely cells. B: The alternative '3/4' conflicts with the labelled relationship, scale or mechanism in the figure. Only genotype tt produces the recessive phenotype, and it occupies one of four equally likely cells. D: The alternative '1' conflicts with the labelled relationship, scale or mechanism in the figure. Only genotype tt produces the recessive phenotype, and it occupies one of four equally likely cells.

BIO-12-CH03-0007 | Medium | Sex-linked inheritance**Correct answer: D. A female heterozygous for a recessive X-linked allele - Carrier**

Solution: The accurate entry is 'A female heterozygous for a recessive X-linked allele - Carrier'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Genes on the same chromosome tending to be inherited together' should be paired with Linkage, not Genotype. B: 'Cross of an F1 individual with either parent' should be paired with Back cross, not 1:1. C: 'Cross with a homozygous recessive individual' should be paired with Test cross, not 3:1.

BIO-12-CH03-0008 | Hard | Non-Mendelian patterns**Correct answer: A. Heterozygote showing an intermediate phenotype - Incomplete dominance; Genes on the same chromosome tending to be inherited together - Linkage**

Solution: Both parts of 'Heterozygote showing an intermediate phenotype - Incomplete dominance; Genes on the same chromosome tending to be inherited together - Linkage' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Trisomy of chromosome 21' requires Down syndrome, and 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1. C: The answers have been interchanged: 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1, and 'Number of linkage groups in a species corresponds to its' requires Haploid chromosome number. D: The answers have been interchanged: 'Observable expression of a character' requires Phenotype, and 'Trisomy of chromosome 21' requires Down syndrome.

BIO-12-CH03-0009 | Challenging | X-linked recessive inheritance**Correct answer: C. Affected sons can occur, but affected daughters are not expected**

Solution: The unaffected father supplies a normal X to every daughter, while a son can receive the affected X from the carrier mother.

Distractor review: A: The alternative 'Affected daughters occur but affected sons do not' conflicts with the labelled relationship, scale or mechanism in the figure. The unaffected father supplies a normal X to every daughter, while a son can receive the affected X from the carrier mother. B: The alternative 'All daughters and all sons are affected' conflicts with the labelled relationship, scale or mechanism in the figure. The unaffected father supplies a normal X to every daughter, while a son can receive the affected X from the carrier mother. D: The alternative 'No child can inherit the recessive allele' conflicts with the labelled relationship, scale or mechanism in the figure. The unaffected father supplies a normal X to every daughter, while a son can receive the affected X from the carrier mother.

BIO-12-CH03-0010 | Easy | Sex determination**Correct answer: D. Sex chromosome system in human females**

Solution: XX represents 'Sex chromosome system in human females'. Each alternative describes a different term or result.

Distractor review: A: 'Trisomy of chromosome 21' identifies Down syndrome, not XX. B: 'Human example of multiple alleles cited in the chapter' identifies ABO blood grouping, not XX. C: 'Heterogametic sex in humans' identifies Male, not XX.

BIO-12-CH03-0011 | Easy | Sex-linked inheritance**Correct answer: C. A male transmitting an X-linked allele to**

Solution: All daughters and no sons represents 'A male transmitting an X-linked allele to'. Each alternative describes a different term or result.

Distractor review: A: 'Monohybrid F2 phenotypic ratio under complete dominance' identifies 3:1, not All daughters and no sons. B: 'Trisomy of chromosome 21' identifies Down syndrome, not All daughters and no sons. D: 'A female heterozygous for a recessive X-linked allele' identifies Carrier, not All daughters and no sons.

BIO-12-CH03-0012 | Easy | Mendelian inheritance**Correct answer: C. 3:1**

Solution: 'Monohybrid F2 phenotypic ratio under complete dominance' is the defining description for 3:1. The other choices belong to different chapter concepts.

Distractor review: A: 9:3:3:1 belongs with 'Dihybrid F2 phenotypic ratio under independent assortment', whereas the tested description is 'Monohybrid F2 phenotypic ratio under complete dominance'. B: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'Monohybrid F2 phenotypic ratio under complete dominance'. D: Test cross belongs with 'Cross with a homozygous recessive individual', whereas the tested description is 'Monohybrid F2 phenotypic ratio under complete dominance'.

BIO-12-CH03-0013 | Hard | Chromosomal disorders

Correct answer: A. Monosomy X - Turner syndrome; Exchange between non-sister chromatids of homologous chromosomes - Crossing over

Solution: Both parts of 'Monosomy X - Turner syndrome; Exchange between non-sister chromatids of homologous chromosomes - Crossing over' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Genetic constitution of an organism' requires Genotype, and 'Heterozygote showing an intermediate phenotype' requires Incomplete dominance. C: The answers have been interchanged: 'Monohybrid F2 phenotypic ratio under complete dominance' requires 3:1, and 'Genetic constitution of an organism' requires Genotype. D: The answers have been interchanged: 'Stage at which crossing over occurs' requires Pachytene of prophase I, and 'Monohybrid F2 phenotypic ratio under complete dominance' requires 3:1.

BIO-12-CH03-0014 | Easy | Crosses

Correct answer: B. Expected dihybrid test-cross ratio with independent assortment

Solution: 1:1:1:1 represents 'Expected dihybrid test-cross ratio with independent assortment'. Each alternative describes a different term or result.

Distractor review: A: 'Observable expression of a character' identifies Phenotype, not 1:1:1:1. C: 'Cross with a homozygous recessive individual' identifies Test cross, not 1:1:1:1. D: 'Classic X-linked human trait involving red-green vision' identifies Colour blindness, not 1:1:1:1.

BIO-12-CH03-0015 | Hard | Crosses

Correct answer: D. Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1; Classic X-linked human trait involving red-green vision - Colour blindness

Solution: Both parts of 'Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1; Classic X-linked human trait involving red-green vision - Colour blindness' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Heterozygote showing an intermediate phenotype' requires Incomplete dominance, and 'Alternative forms of the same gene' requires Alleles. B: The answers have been interchanged: 'A female heterozygous for a recessive X-linked allele' requires Carrier, and 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1. C: The answers have been interchanged: 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1, and 'Observable expression of a character' requires Phenotype.

BIO-12-CH03-0016 | Easy | Mendelian inheritance

Correct answer: D. 9:3:3:1

Solution: 'Dihybrid F2 phenotypic ratio under independent assortment' is the defining description for 9:3:3:1. The other choices belong to different chapter concepts.

Distractor review: A: Law of independent assortment belongs with 'Independent behaviour of different allele pairs', whereas the tested description is 'Dihybrid F2 phenotypic ratio under independent assortment'. B: X chromosome belongs with 'Chromosome carrying the common haemophilia allele', whereas the tested description is 'Dihybrid F2 phenotypic ratio under independent assortment'. C: ZW female and ZZ male belongs with 'Sex chromosome system in birds', whereas the tested description is 'Dihybrid F2 phenotypic ratio under independent assortment'.

BIO-12-CH03-0017 | Hard | Chromosomes linkage and crossing over

Correct answer: B. Exchange between non-sister chromatids of homologous chromosomes - Crossing over; Classic X-linked human trait involving red-green vision - Colour blindness

Solution: Both parts of 'Exchange between non-sister chromatids of homologous chromosomes - Crossing over; Classic X-linked human trait involving red-green vision - Colour blindness' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Heterozygote showing an intermediate phenotype' requires Incomplete dominance, and 'Sex chromosome system in human females' requires XX. C: The answers have been interchanged: 'Loss or gain of one or a few chromosomes' requires Aneuploidy, and 'Heterozygote showing an intermediate phenotype' requires Incomplete dominance. D: The answers have been interchanged: 'One gene influencing several characters' requires Pleiotropy, and 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1.

BIO-12-CH03-0018 | Medium | Mendelian inheritance**Correct answer: D. Law of segregation**

Solution: The chapter links Separation of allele pairs during gamete formation with Law of segregation, so Law of segregation completes the statement.

Distractor review: A: 1:1:1:1 belongs with 'Expected dihybrid test-cross ratio with independent assortment', whereas the tested description is 'Separation of allele pairs during gamete formation'. B: X chromosome belongs with 'Chromosome carrying the common haemophilia allele', whereas the tested description is 'Separation of allele pairs during gamete formation'. C: 3:1 belongs with 'Monohybrid F2 phenotypic ratio under complete dominance', whereas the tested description is 'Separation of allele pairs during gamete formation'.

BIO-12-CH03-0019 | Medium | Sex determination**Correct answer: A. Sex determination in honey bees - Haplo-diploid type**

Solution: The accurate entry is 'Sex determination in honey bees - Haplo-diploid type'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Monosomy X' should be paired with Turner syndrome, not Aneuploidy. C: 'Expected test-cross ratio for a heterozygous monohybrid' should be paired with 1:1, not 1:1:1:1. D: 'Classic X-linked human trait involving red-green vision' should be paired with Colour blindness, not Turner syndrome.

BIO-12-CH03-0020 | Medium | Sex determination**Correct answer: B. XX**

Solution: The observation 'Sex chromosome system in human females' points to XX; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: X chromosome belongs with 'Chromosome carrying the common haemophilia allele', whereas the tested description is 'Sex chromosome system in human females'. C: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'Sex chromosome system in human females'. D: Pachytene of prophase I belongs with 'Stage at which crossing over occurs', whereas the tested description is 'Sex chromosome system in human females'.

BIO-12-CH03-0021 | Medium | Genetic terminology**Correct answer: A. Phenotype**

Solution: The chapter links Observable expression of a character with Phenotype, so Phenotype completes the statement.

Distractor review: B: Locus belongs with 'Position occupied by a gene on a chromosome', whereas the tested description is 'Observable expression of a character'. C: Crossing over belongs with 'Exchange between non-sister chromatids of homologous chromosomes', whereas the tested description is 'Observable expression of a character'. D: Pachytene of prophase I belongs with 'Stage at which crossing over occurs', whereas the tested description is 'Observable expression of a character'.

BIO-12-CH03-0022 | Easy | Genetic terminology**Correct answer: A. Alleles**

Solution: 'Alternative forms of the same gene' is the defining description for Alleles. The other choices belong to different chapter concepts.

Distractor review: B: Locus belongs with 'Position occupied by a gene on a chromosome', whereas the tested description is 'Alternative forms of the same gene'. C: Genotype belongs with 'Genetic constitution of an organism', whereas the tested description is 'Alternative forms of the same gene'. D: Crossing over belongs with 'Exchange between non-sister chromatids of homologous chromosomes', whereas the tested description is 'Alternative forms of the same gene'.

BIO-12-CH03-0023 | Hard | Sex-linked inheritance**Correct answer: B. Chromosome carrying the common haemophilia allele - X chromosome; Cross with a homozygous recessive individual - Test cross**

Solution: Both parts of 'Chromosome carrying the common haemophilia allele - X chromosome; Cross with a homozygous recessive individual - Test cross' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Genes on the same chromosome tending to be inherited together' requires Linkage, and 'One gene influencing several characters' requires Pleiotropy. C: The answers have been interchanged: 'Exchange between non-sister chromatids of homologous chromosomes' requires Crossing over, and 'Trisomy of chromosome 21' requires Down syndrome. D: The answers have been interchanged: 'Monosomy X' requires Turner syndrome, and 'Genes on the same chromosome tending to be inherited together' requires Linkage.

BIO-12-CH03-0024 | Hard | Non-Mendelian patterns

Correct answer: B. Human example of multiple alleles cited in the chapter - ABO blood grouping; Classic X-linked human trait involving red-green vision - Colour blindness

Solution: Both parts of 'Human example of multiple alleles cited in the chapter - ABO blood grouping; Classic X-linked human trait involving red-green vision - Colour blindness' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Chromosome carrying the common haemophilia allele' requires X chromosome, and 'Stage at which crossing over occurs' requires Pachytene of prophase I. C: The answers have been interchanged: 'Stage at which crossing over occurs' requires Pachytene of prophase I, and 'Expected test-cross ratio for a heterozygous monohybrid' requires 1:1. D: The answers have been interchanged: 'Monosomy X' requires Turner syndrome, and 'Observable expression of a character' requires Phenotype.

BIO-12-CH03-0025 | Challenging | Genetic terminology

Correct answer: A. I and II only

Solution: Statement I is correct because 'Position occupied by a gene on a chromosome' corresponds to Locus. Statement II is correct because 'Genetic constitution of an organism' corresponds to Genotype. Statement III is incorrect: 'Alternative forms of the same gene' corresponds to Alleles, not Law of independent assortment. Statement IV is incorrect: 'Independent behaviour of different allele pairs' corresponds to Law of independent assortment, not Locus. Therefore, the valid combination is I and II only.

Distractor review: B: This combination includes false statement(s) IV and omits true statement(s) I. C: This combination omits true statement(s) II. D: This combination includes false statement(s) III, IV and omits true statement(s) I, II.

BIO-12-CH03-0026 | Challenging | Sex determination

Correct answer: A. I, II and IV only

Solution: Statement I is correct because 'Sex chromosome system in human females' corresponds to XX. Statement II is correct because 'Sex determination in honey bees' corresponds to Haplo-diploid type. Statement III is incorrect: 'Heterogametic sex in humans' corresponds to Male, not Back cross. Statement IV is correct because 'Cross of an F1 individual with either parent' corresponds to Back cross. Therefore, the valid combination is I, II and IV only.

Distractor review: B: This combination omits true statement(s) I, IV. C: This combination omits true statement(s) II. D: This combination omits true statement(s) I, II.

BIO-12-CH03-0027 | Challenging | Chromosomes linkage and crossing over

Correct answer: D. IV only

Solution: Statement I is incorrect: 'Number of linkage groups in a species corresponds to its' corresponds to Haploid chromosome number, not Linkage. Statement II is incorrect: 'Genes on the same chromosome tending to be inherited together' corresponds to Linkage, not Crossing over. Statement III is incorrect: 'Exchange between non-sister chromatids of homologous chromosomes' corresponds to Crossing over, not Genotype. Statement IV is correct because 'Genetic constitution of an organism' corresponds to Genotype. Therefore, the valid combination is IV only.

Distractor review: A: This combination includes false statement(s) I, II, III and omits true statement(s) IV. B: This combination includes false statement(s) I, III and omits true statement(s) IV. C: This combination includes false statement(s) I, II, III.

BIO-12-CH03-0028 | Hard | Sex determination

Correct answer: D. Sex chromosome system in human females - XX; Trisomy of chromosome 21 - Down syndrome

Solution: Both parts of 'Sex chromosome system in human females - XX; Trisomy of chromosome 21 - Down syndrome' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1, and 'Stage at which crossing over occurs' requires Pachytene of prophase I. B: The answers have been interchanged: 'Chromosome carrying the common haemophilia allele' requires X chromosome, and 'Human example of multiple alleles cited in the chapter' requires ABO blood grouping. C: The answers have been interchanged: 'Human example of multiple alleles cited in the chapter' requires ABO blood grouping, and 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1.

BIO-12-CH03-0029 | Hard | Sex-linked inheritance**Correct answer: C. Chromosome carrying the common haemophilia allele - Test cross**

Solution: 'Chromosome carrying the common haemophilia allele' should be paired with X chromosome, not Test cross. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Genes on the same chromosome tending to be inherited together - Linkage. B: This is a valid association: Trisomy of chromosome 21 - Down syndrome. D: This is a valid association: One gene influencing several characters - Pleiotropy.

BIO-12-CH03-0030 | Easy | Chromosomal disorders**Correct answer: C. Monosomy X**

Solution: Turner syndrome represents 'Monosomy X'. Each alternative describes a different term or result.

Distractor review: A: 'XXY constitution in a male' identifies Klinefelter syndrome, not Turner syndrome. B: 'Monohybrid F2 phenotypic ratio under complete dominance' identifies 3:1, not Turner syndrome. D: 'Genetic constitution of an organism' identifies Genotype, not Turner syndrome.

BIO-12-CH03-0031 | Hard | Sex-linked inheritance**Correct answer: A. A female heterozygous for a recessive X-linked allele - Back cross**

Solution: 'A female heterozygous for a recessive X-linked allele' should be paired with Carrier, not Back cross. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Monohybrid F2 phenotypic ratio under complete dominance - 3:1. C: This is a valid association: Expected test-cross ratio for a heterozygous monohybrid - 1:1. D: This is a valid association: Both alleles expressed in a heterozygote - Codominance.

BIO-12-CH03-0032 | Easy | Non-Mendelian patterns**Correct answer: B. ABO blood grouping**

Solution: 'Human example of multiple alleles cited in the chapter' is the defining description for ABO blood grouping. The other choices belong to different chapter concepts.

Distractor review: A: Turner syndrome belongs with 'Monosomy X', whereas the tested description is 'Human example of multiple alleles cited in the chapter'. C: Pleiotropy belongs with 'One gene influencing several characters', whereas the tested description is 'Human example of multiple alleles cited in the chapter'. D: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'Human example of multiple alleles cited in the chapter'.

BIO-12-CH03-0033 | Hard | Mendelian inheritance**Correct answer: B. Dihybrid F2 phenotypic ratio under independent assortment - Back cross**

Solution: 'Dihybrid F2 phenotypic ratio under independent assortment' should be paired with 9:3:3:1, not Back cross. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Expected test-cross ratio for a heterozygous monohybrid - 1:1. C: This is a valid association: Chromosome carrying the common haemophilia allele - X chromosome. D: This is a valid association: Heterogametic sex in humans - Male.

BIO-12-CH03-0034 | Medium | Sex-linked inheritance**Correct answer: C. A male transmitting an X-linked allele to - All daughters and no sons**

Solution: The accurate entry is 'A male transmitting an X-linked allele to - All daughters and no sons'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Observable expression of a character' should be paired with Phenotype, not Down syndrome. B: 'Sex chromosome system in human females' should be paired with XX, not All daughters and no sons. D: 'Trisomy of chromosome 21' should be paired with Down syndrome, not 3:1.

BIO-12-CH03-0035 | Challenging | Sex-linked inheritance**Correct answer: D. I, II and III only**

Solution: Statement I is correct because 'A female heterozygous for a recessive X-linked allele' corresponds to Carrier. Statement II is correct because 'Chromosome carrying the common haemophilia allele' corresponds to X chromosome. Statement III is correct because 'A male transmitting an X-linked allele to' corresponds to All daughters and no sons. Statement IV is incorrect: 'Heterogametic sex in humans' corresponds to Male, not Carrier. Therefore, the valid combination is I, II and III only.

Distractor review: A: This combination omits true statement(s) III. B: This combination includes false statement(s) IV and omits true statement(s) II. C: This combination omits true statement(s) I, III.

BIO-12-CH03-0036 | Medium | Chromosomes linkage and crossing over**Correct answer: D. Crossing over**

Solution: The observation 'Exchange between non-sister chromatids of homologous chromosomes' points to Crossing over; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Incomplete dominance belongs with 'Heterozygote showing an intermediate phenotype', whereas the tested description is 'Exchange between non-sister chromatids of homologous chromosomes'. B: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'Exchange between non-sister chromatids of homologous chromosomes'. C: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'Exchange between non-sister chromatids of homologous chromosomes'.

BIO-12-CH03-0037 | Medium | Crosses**Correct answer: B. Test cross**

Solution: The chapter links Cross with a homozygous recessive individual with Test cross, so Test cross completes the statement.

Distractor review: A: Male belongs with 'Heterogametic sex in humans', whereas the tested description is 'Cross with a homozygous recessive individual'. C: Back cross belongs with 'Cross of an F1 individual with either parent', whereas the tested description is 'Cross with a homozygous recessive individual'. D: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'Cross with a homozygous recessive individual'.

BIO-12-CH03-0038 | Medium | Chromosomal disorders**Correct answer: A. Down syndrome**

Solution: The chapter links Trisomy of chromosome 21 with Down syndrome, so Down syndrome completes the statement.

Distractor review: B: Turner syndrome belongs with 'Monosomy X', whereas the tested description is 'Trisomy of chromosome 21'. C: Crossing over belongs with 'Exchange between non-sister chromatids of homologous chromosomes', whereas the tested description is 'Trisomy of chromosome 21'. D: Incomplete dominance belongs with 'Heterozygote showing an intermediate phenotype', whereas the tested description is 'Trisomy of chromosome 21'.

BIO-12-CH03-0039 | Easy | Non-Mendelian patterns**Correct answer: D. One gene influencing several characters**

Solution: Pleiotropy represents 'One gene influencing several characters'. Each alternative describes a different term or result.

Distractor review: A: 'Trisomy of chromosome 21' identifies Down syndrome, not Pleiotropy. B: 'Heterozygote showing an intermediate phenotype' identifies Incomplete dominance, not Pleiotropy. C: 'Heterogametic sex in humans' identifies Male, not Pleiotropy.

BIO-12-CH03-0040 | Medium | Sex-linked inheritance**Correct answer: C. Colour blindness**

Solution: The chapter links Classic X-linked human trait involving red-green vision with Colour blindness, so Colour blindness completes the statement.

Distractor review: A: 9:3:3:1 belongs with 'Dihybrid F2 phenotypic ratio under independent assortment', whereas the tested description is 'Classic X-linked human trait involving red-green vision'. B: Carrier belongs with 'A female heterozygous for a recessive X-linked allele', whereas the tested description is 'Classic X-linked human trait involving red-green vision'. D: X chromosome belongs with 'Chromosome carrying the common haemophilia allele', whereas the tested description is 'Classic X-linked human trait involving red-green vision'.

BIO-12-CH03-0041 | Easy | Sex determination**Correct answer: B. XX**

Solution: 'Sex chromosome system in human females' is the defining description for XX. The other choices belong to different chapter concepts.

Distractor review: A: 9:3:3:1 belongs with 'Dihybrid F2 phenotypic ratio under independent assortment', whereas the tested description is 'Sex chromosome system in human females'. C: Pachytene of prophase I belongs with 'Stage at which crossing over occurs', whereas the tested description is 'Sex chromosome system in human females'. D: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'Sex chromosome system in human females'.

BIO-12-CH03-0042 | Medium | Chromosomes linkage and crossing over**Correct answer: C. Linkage**

Solution: The chapter links Genes on the same chromosome tending to be inherited together with Linkage, so Linkage completes the statement.

Distractor review: A: 1:1 belongs with 'Expected test-cross ratio for a heterozygous monohybrid', whereas the tested description is 'Genes on the same chromosome tending to be inherited together'. B: Haploid chromosome number belongs with 'Number of linkage groups in a species corresponds to its', whereas the tested description is 'Genes on the same chromosome tending to be inherited together'. D: Locus belongs with 'Position occupied by a gene on a chromosome', whereas the tested description is 'Genes on the same chromosome tending to be inherited together'.

BIO-12-CH03-0043 | Medium | Non-Mendelian patterns**Correct answer: D. ABO blood grouping**

Solution: The chapter links Human example of multiple alleles cited in the chapter with ABO blood grouping, so ABO blood grouping completes the statement.

Distractor review: A: Pleiotropy belongs with 'One gene influencing several characters', whereas the tested description is 'Human example of multiple alleles cited in the chapter'. B: Pachytene of prophase I belongs with 'Stage at which crossing over occurs', whereas the tested description is 'Human example of multiple alleles cited in the chapter'. C: Codominance belongs with 'Both alleles expressed in a heterozygote', whereas the tested description is 'Human example of multiple alleles cited in the chapter'.

BIO-12-CH03-0044 | Easy | Chromosomes linkage and crossing over**Correct answer: B. Crossing over**

Solution: 'Exchange between non-sister chromatids of homologous chromosomes' is the defining description for Crossing over. The other choices belong to different chapter concepts.

Distractor review: A: Alleles belongs with 'Alternative forms of the same gene', whereas the tested description is 'Exchange between non-sister chromatids of homologous chromosomes'. C: Aneuploidy belongs with 'Loss or gain of one or a few chromosomes', whereas the tested description is 'Exchange between non-sister chromatids of homologous chromosomes'. D: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'Exchange between non-sister chromatids of homologous chromosomes'.

BIO-12-CH03-0045 | Medium | Chromosomes linkage and crossing over**Correct answer: C. Crossing over**

Solution: The chapter links Exchange between non-sister chromatids of homologous chromosomes with Crossing over, so Crossing over completes the statement.

Distractor review: A: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'Exchange between non-sister chromatids of homologous chromosomes'. B: XX belongs with 'Sex chromosome system in human females', whereas the tested description is 'Exchange between non-sister chromatids of homologous chromosomes'. D: Aneuploidy belongs with 'Loss or gain of one or a few chromosomes', whereas the tested description is 'Exchange between non-sister chromatids of homologous chromosomes'.

BIO-12-CH03-0046 | Medium | Mendelian inheritance**Correct answer: C. Law of independent assortment**

Solution: The chapter links Independent behaviour of different allele pairs with Law of independent assortment, so Law of independent assortment completes the statement.

Distractor review: A: 3:1 belongs with 'Monohybrid F2 phenotypic ratio under complete dominance', whereas the tested description is 'Independent behaviour of different allele pairs'. B: 1:1:1:1 belongs with 'Expected dihybrid test-cross ratio with independent assortment', whereas the tested description is 'Independent behaviour of different allele pairs'. D: Law of segregation belongs with 'Separation of allele pairs during gamete formation', whereas the tested description is 'Independent behaviour of different allele pairs'.

BIO-12-CH03-0047 | Hard | Sex determination**Correct answer: A. Sex determination in honey bees - 9:3:3:1**

Solution: 'Sex determination in honey bees' should be paired with Haplo-diploid type, not 9:3:3:1. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Monosomy X - Turner syndrome. C: This is a valid association: Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1. D: This is a valid association: Expected test-cross ratio for a heterozygous monohybrid - 1:1.

BIO-12-CH03-0048 | Medium | Sex determination**Correct answer: A. Male**

Solution: The chapter links Heterogametic sex in humans with Male, so Male completes the statement.

Distractor review: B: Law of segregation belongs with 'Separation of allele pairs during gamete formation', whereas the tested description is 'Heterogametic sex in humans'. C: 1:1:1:1 belongs with 'Expected dihybrid test-cross ratio with independent assortment', whereas the tested description is 'Heterogametic sex in humans'. D: Aneuploidy belongs with 'Loss or gain of one or a few chromosomes', whereas the tested description is 'Heterogametic sex in humans'.

BIO-12-CH03-0049 | Hard | Crosses**Correct answer: B. Cross with a homozygous recessive individual - Male**

Solution: 'Cross with a homozygous recessive individual' should be paired with Test cross, not Male. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Monosomy X - Turner syndrome. C: This is a valid association: Monohybrid F2 phenotypic ratio under complete dominance - 3:1. D: This is a valid association: Genes on the same chromosome tending to be inherited together - Linkage.

BIO-12-CH03-0050 | Medium | Crosses**Correct answer: D. Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1**

Solution: The accurate entry is 'Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Exchange between non-sister chromatids of homologous chromosomes' should be paired with Crossing over, not Alleles. B: 'Heterozygote showing an intermediate phenotype' should be paired with Incomplete dominance, not 1:1:1:1. C: 'A female heterozygous for a recessive X-linked allele' should be paired with Carrier, not Phenotype.

BIO-12-CH03-0051 | Hard | Mendelian inheritance**Correct answer: A. Independent behaviour of different allele pairs - Locus**

Solution: 'Independent behaviour of different allele pairs' should be paired with Law of independent assortment, not Locus. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: One gene influencing several characters - Pleiotropy. C: This is a valid association: Both alleles expressed in a heterozygote - Codominance. D: This is a valid association: A male transmitting an X-linked allele to - All daughters and no sons.

BIO-12-CH03-0052 | Hard | Chromosomal disorders**Correct answer: A. Trisomy of chromosome 21 - ABO blood grouping**

Solution: 'Trisomy of chromosome 21' should be paired with Down syndrome, not ABO blood grouping. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Separation of allele pairs during gamete formation - Law of segregation. C: This is a valid association: Heterozygote showing an intermediate phenotype - Incomplete dominance. D: This is a valid association: A female heterozygous for a recessive X-linked allele - Carrier.

BIO-12-CH03-0053 | Hard | Genetic terminology**Correct answer: D. Alternative forms of the same gene - Klinefelter syndrome**

Solution: 'Alternative forms of the same gene' should be paired with Alleles, not Klinefelter syndrome. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Number of linkage groups in a species corresponds to its - Haploid chromosome number. B: This is a valid association: Separation of allele pairs during gamete formation - Law of segregation. C: This is a valid association: Heterogametic sex in humans - Male.

BIO-12-CH03-0054 | Hard | Non-Mendelian patterns**Correct answer: B. Human example of multiple alleles cited in the chapter - Colour blindness**

Solution: 'Human example of multiple alleles cited in the chapter' should be paired with ABO blood grouping, not Colour blindness. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Chromosome carrying the common haemophilia allele - X chromosome. C: This is a valid association: Observable expression of a character - Phenotype. D: This is a valid association: Expected test-cross ratio for a heterozygous monohybrid - 1:1.

BIO-12-CH03-0055 | Medium | Genetic terminology**Correct answer: C. Locus**

Solution: The observation 'Position occupied by a gene on a chromosome' points to Locus; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: 9:3:3:1 belongs with 'Dihybrid F2 phenotypic ratio under independent assortment', whereas the tested description is 'Position occupied by a gene on a chromosome'. B: Crossing over belongs with 'Exchange between non-sister chromatids of homologous chromosomes', whereas the tested description is 'Position occupied by a gene on a chromosome'. D: Phenotype belongs with 'Observable expression of a character', whereas the tested description is 'Position occupied by a gene on a chromosome'.

BIO-12-CH03-0056 | Hard | Mendelian inheritance**Correct answer: D. Monohybrid F2 phenotypic ratio under complete dominance - ZW female and ZZ male**

Solution: 'Monohybrid F2 phenotypic ratio under complete dominance' should be paired with 3:1, not ZW female and ZZ male. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Classic X-linked human trait involving red-green vision - Colour blindness. B: This is a valid association: Stage at which crossing over occurs - Pachytene of prophase I. C: This is a valid association: Observable expression of a character - Phenotype.

BIO-12-CH03-0057 | Easy | Crosses**Correct answer: C. Test cross**

Solution: 'Cross with a homozygous recessive individual' is the defining description for Test cross. The other choices belong to different chapter concepts.

Distractor review: A: 1:1 belongs with 'Expected test-cross ratio for a heterozygous monohybrid', whereas the tested description is 'Cross with a homozygous recessive individual'. B: 3:1 belongs with 'Monohybrid F2 phenotypic ratio under complete dominance', whereas the tested description is 'Cross with a homozygous recessive individual'. D: Male belongs with 'Heterogametic sex in humans', whereas the tested description is 'Cross with a homozygous recessive individual'.

BIO-12-CH03-0058 | Hard | Crosses**Correct answer: C. Cross with a homozygous recessive individual - Test cross; Heterogametic sex in humans - Male**

Solution: Both parts of 'Cross with a homozygous recessive individual - Test cross; Heterogametic sex in humans - Male' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Exchange between non-sister chromatids of homologous chromosomes' requires Crossing over, and 'A male transmitting an X-linked allele to' requires All daughters and no sons. B: The answers have been interchanged: 'Monosomy X' requires Turner syndrome, and 'Genes on the same chromosome tending to be inherited together' requires Linkage. D: The answers have been interchanged: 'A male transmitting an X-linked allele to' requires All daughters and no sons, and 'Number of linkage groups in a species corresponds to its' requires Haploid chromosome number.

BIO-12-CH03-0059 | Easy | Chromosomal disorders**Correct answer: D. Klinefelter syndrome**

Solution: 'XXY constitution in a male' is the defining description for Klinefelter syndrome. The other choices belong to different chapter concepts.

Distractor review: A: Aneuploidy belongs with 'Loss or gain of one or a few chromosomes', whereas the tested description is 'XXY constitution in a male'. B: 3:1 belongs with 'Monohybrid F2 phenotypic ratio under complete dominance', whereas the tested description is 'XXY constitution in a male'. C: Law of segregation belongs with 'Separation of allele pairs during gamete formation', whereas the tested description is 'XXY constitution in a male'.

BIO-12-CH03-0060 | Medium | Chromosomal disorders**Correct answer: B. Down syndrome**

Solution: The observation 'Trisomy of chromosome 21' points to Down syndrome; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Crossing over belongs with 'Exchange between non-sister chromatids of homologous chromosomes', whereas the tested description is 'Trisomy of chromosome 21'. C: Incomplete dominance belongs with 'Heterozygote showing an intermediate phenotype', whereas the tested description is 'Trisomy of chromosome 21'. D: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'Trisomy of chromosome 21'.

BIO-12-CH03-0061 | Challenging | Chromosomal disorders**Correct answer: C. III only**

Solution: Statement I is incorrect: 'XXY constitution in a male' corresponds to Klinefelter syndrome, not Aneuploidy. Statement II is incorrect: 'Loss or gain of one or a few chromosomes' corresponds to Aneuploidy, not Turner syndrome. Statement III is correct because 'Monosomy X' corresponds to Turner syndrome. Statement IV is incorrect: 'Dihybrid F2 phenotypic ratio under independent assortment' corresponds to 9:3:3:1, not Klinefelter syndrome. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) II. B: This combination includes false statement(s) I, II, IV and omits true statement(s) III. D: This combination includes false statement(s) IV and omits true statement(s) III.

BIO-12-CH03-0062 | Easy | Genetic terminology**Correct answer: D. Genotype**

Solution: 'Genetic constitution of an organism' is the defining description for Genotype. The other choices belong to different chapter concepts.

Distractor review: A: Klinefelter syndrome belongs with 'XXY constitution in a male', whereas the tested description is 'Genetic constitution of an organism'. B: 1:1 belongs with 'Expected test-cross ratio for a heterozygous monohybrid', whereas the tested description is 'Genetic constitution of an organism'. C: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'Genetic constitution of an organism'.

BIO-12-CH03-0063 | Medium | Genetic terminology**Correct answer: A. Observable expression of a character - Phenotype**

Solution: The accurate entry is 'Observable expression of a character - Phenotype'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Stage at which crossing over occurs' should be paired with Pachytene of prophase I, not Incomplete dominance. C: 'Sex determination in honey bees' should be paired with Haplo-diploid type, not ABO blood grouping. D: 'Human example of multiple alleles cited in the chapter' should be paired with ABO blood grouping, not Phenotype.

BIO-12-CH03-0064 | Medium | Genetic terminology**Correct answer: D. Alleles**

Solution: The chapter links Alternative forms of the same gene with Alleles, so Alleles completes the statement.

Distractor review: A: Law of segregation belongs with 'Separation of allele pairs during gamete formation', whereas the tested description is 'Alternative forms of the same gene'. B: Klinefelter syndrome belongs with 'XXY constitution in a male', whereas the tested description is 'Alternative forms of the same gene'. C: Locus belongs with 'Position occupied by a gene on a chromosome', whereas the tested description is 'Alternative forms of the same gene'.

BIO-12-CH03-0065 | Medium | Sex determination**Correct answer: A. Sex chromosome system in human females - XX**

Solution: The accurate entry is 'Sex chromosome system in human females - XX'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Monosomy X' should be paired with Turner syndrome, not XX. C: 'Dihybrid F2 phenotypic ratio under independent assortment' should be paired with 9:3:3:1, not Turner syndrome. D: 'Stage at which crossing over occurs' should be paired with Pachytene of prophase I, not All daughters and no sons.

BIO-12-CH03-0066 | Medium | Chromosomes linkage and crossing over**Correct answer: A. Linkage**

Solution: The observation 'Genes on the same chromosome tending to be inherited together' points to Linkage; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: All daughters and no sons belongs with 'A male transmitting an X-linked allele to', whereas the tested description is 'Genes on the same chromosome tending to be inherited together'. C: Phenotype belongs with 'Observable expression of a character', whereas the tested description is 'Genes on the same chromosome tending to be inherited together'. D: Locus belongs with 'Position occupied by a gene on a chromosome', whereas the tested description is 'Genes on the same chromosome tending to be inherited together'.

BIO-12-CH03-0067 | Medium | Chromosomal disorders**Correct answer: D. Klinefelter syndrome**

Solution: The chapter links XXY constitution in a male with Klinefelter syndrome, so Klinefelter syndrome completes the statement.

Distractor review: A: ZW female and ZZ male belongs with 'Sex chromosome system in birds', whereas the tested description is 'XXY constitution in a male'. B: Law of segregation belongs with 'Separation of allele pairs during gamete formation', whereas the tested description is 'XXY constitution in a male'. C: All daughters and no sons belongs with 'A male transmitting an X-linked allele to', whereas the tested description is 'XXY constitution in a male'.

BIO-12-CH03-0068 | Easy | Non-Mendelian patterns**Correct answer: A. Codominance**

Solution: 'Both alleles expressed in a heterozygote' is the defining description for Codominance. The other choices belong to different chapter concepts.

Distractor review: B: 1:1:1:1 belongs with 'Expected dihybrid test-cross ratio with independent assortment', whereas the tested description is 'Both alleles expressed in a heterozygote'. C: Back cross belongs with 'Cross of an F1 individual with either parent', whereas the tested description is 'Both alleles expressed in a heterozygote'. D: 9:3:3:1 belongs with 'Dihybrid F2 phenotypic ratio under independent assortment', whereas the tested description is 'Both alleles expressed in a heterozygote'.

BIO-12-CH03-0069 | Medium | Non-Mendelian patterns**Correct answer: C. ABO blood grouping**

Solution: The observation 'Human example of multiple alleles cited in the chapter' points to ABO blood grouping; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Pachytene of prophase I belongs with 'Stage at which crossing over occurs', whereas the tested description is 'Human example of multiple alleles cited in the chapter'. B: Turner syndrome belongs with 'Monosomy X', whereas the tested description is 'Human example of multiple alleles cited in the chapter'. D: Pleiotropy belongs with 'One gene influencing several characters', whereas the tested description is 'Human example of multiple alleles cited in the chapter'.

BIO-12-CH03-0070 | Hard | Sex determination**Correct answer: A. Sex determination in honey bees - Haplo-diploid type; Dihybrid F2 phenotypic ratio under independent assortment - 9:3:3:1**

Solution: Both parts of 'Sex determination in honey bees - Haplo-diploid type; Dihybrid F2 phenotypic ratio under independent assortment - 9:3:3:1' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Expected dihybrid test-cross ratio with independent assortment' requires 1:1:1:1, and 'Loss or gain of one or a few chromosomes' requires Aneuploidy. C: The answers have been interchanged: 'Monosomy X' requires Turner syndrome, and 'Expected dihybrid test-cross ratio with independent assortment' requires 1:1:1:1. D: The answers have been interchanged: 'Classic X-linked human trait involving red-green vision' requires Colour blindness, and 'Expected test-cross ratio for a heterozygous monohybrid' requires 1:1.

BIO-12-CH03-0071 | Easy | Chromosomes linkage and crossing over**Correct answer: B. Number of linkage groups in a species corresponds to its**

Solution: Haploid chromosome number represents 'Number of linkage groups in a species corresponds to its'. Each alternative describes a different term or result.

Distractor review: A: 'Alternative forms of the same gene' identifies Alleles, not Haploid chromosome number. C: 'Human example of multiple alleles cited in the chapter' identifies ABO blood grouping, not Haploid chromosome number. D: 'Genetic constitution of an organism' identifies Genotype, not Haploid chromosome number.

BIO-12-CH03-0072 | Hard | Sex-linked inheritance**Correct answer: D. Classic X-linked human trait involving red-green vision - Colour blindness; Monosomy X - Turner syndrome**

Solution: Both parts of 'Classic X-linked human trait involving red-green vision - Colour blindness; Monosomy X - Turner syndrome' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Sex chromosome system in birds' requires ZW female and ZZ male, and 'Observable expression of a character' requires Phenotype. B: The answers have been interchanged: 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1, and 'Sex chromosome system in birds' requires ZW female and ZZ male. C: The answers have been interchanged: 'Observable expression of a character' requires Phenotype, and 'Sex chromosome system in human females' requires XX.

BIO-12-CH03-0073 | Easy | Mendelian inheritance**Correct answer: A. Separation of allele pairs during gamete formation**

Solution: Law of segregation represents 'Separation of allele pairs during gamete formation'. Each alternative describes a different term or result.

Distractor review: B: 'Expected dihybrid test-cross ratio with independent assortment' identifies 1:1:1:1, not Law of segregation. C: 'Position occupied by a gene on a chromosome' identifies Locus, not Law of segregation. D: 'Classic X-linked human trait involving red-green vision' identifies Colour blindness, not Law of segregation.

BIO-12-CH03-0074 | Medium | Mendelian inheritance**Correct answer: C. Dihybrid F2 phenotypic ratio under independent assortment - 9:3:3:1**

Solution: The accurate entry is 'Dihybrid F2 phenotypic ratio under independent assortment - 9:3:3:1'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'A female heterozygous for a recessive X-linked allele' should be paired with Carrier, not 9:3:3:1. B: 'Cross of an F1 individual with either parent' should be paired with Back cross, not Male. D: 'Expected test-cross ratio for a heterozygous monohybrid' should be paired with 1:1, not Linkage.

BIO-12-CH03-0075 | Challenging | Genetic terminology**Correct answer: A. III and IV only**

Solution: Statement I is incorrect: 'Alternative forms of the same gene' corresponds to Alleles, not Genotype. Statement II is incorrect: 'Genetic constitution of an organism' corresponds to Genotype, not Locus. Statement III is correct because 'Position occupied by a gene on a chromosome' corresponds to Locus. Statement IV is correct because 'Expected test-cross ratio for a heterozygous monohybrid' corresponds to 1:1. Therefore, the valid combination is III and IV only.

Distractor review: B: This combination includes false statement(s) I, II and omits true statement(s) III. C: This combination includes false statement(s) I, II and omits true statement(s) IV. D: This combination includes false statement(s) I, II.

BIO-12-CH03-0076 | Medium | Chromosomal disorders**Correct answer: B. Turner syndrome**

Solution: The chapter links Monosomy X with Turner syndrome, so Turner syndrome completes the statement.

Distractor review: A: Genotype belongs with 'Genetic constitution of an organism', whereas the tested description is 'Monosomy X'. C: Crossing over belongs with 'Exchange between non-sister chromatids of homologous chromosomes', whereas the tested description is 'Monosomy X'. D: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'Monosomy X'.

BIO-12-CH03-0077 | Easy | Mendelian inheritance**Correct answer: A. Independent behaviour of different allele pairs**

Solution: Law of independent assortment represents 'Independent behaviour of different allele pairs'. Each alternative describes a different term or result.

Distractor review: B: 'Expected dihybrid test-cross ratio with independent assortment' identifies 1:1:1:1, not Law of independent assortment. C: 'Separation of allele pairs during gamete formation' identifies Law of segregation, not Law of independent assortment. D: 'One gene influencing several characters' identifies Pleiotropy, not Law of independent assortment.

BIO-12-CH03-0078 | Medium | Sex-linked inheritance**Correct answer: C. Colour blindness**

Solution: The observation 'Classic X-linked human trait involving red-green vision' points to Colour blindness; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Turner syndrome belongs with 'Monosomy X', whereas the tested description is 'Classic X-linked human trait involving red-green vision'. B: Carrier belongs with 'A female heterozygous for a recessive X-linked allele', whereas the tested description is 'Classic X-linked human trait involving red-green vision'. D: X chromosome belongs with 'Chromosome carrying the common haemophilia allele', whereas the tested description is 'Classic X-linked human trait involving red-green vision'.

BIO-12-CH03-0079 | Easy | Non-Mendelian patterns**Correct answer: D. Heterozygote showing an intermediate phenotype**

Solution: Incomplete dominance represents 'Heterozygote showing an intermediate phenotype'. Each alternative describes a different term or result.

Distractor review: A: 'Observable expression of a character' identifies Phenotype, not Incomplete dominance. B: 'One gene influencing several characters' identifies Pleiotropy, not Incomplete dominance. C: 'Dihybrid F2 phenotypic ratio under independent assortment' identifies 9:3:3:1, not Incomplete dominance.

BIO-12-CH03-0080 | Easy | Crosses**Correct answer: D. Cross with a homozygous recessive individual**

Solution: Test cross represents 'Cross with a homozygous recessive individual'. Each alternative describes a different term or result.

Distractor review: A: 'Exchange between non-sister chromatids of homologous chromosomes' identifies Crossing over, not Test cross. B: 'Heterogametic sex in humans' identifies Male, not Test cross. C: 'Cross of an F1 individual with either parent' identifies Back cross, not Test cross.

BIO-12-CH03-0081 | Medium | Chromosomal disorders**Correct answer: C. Loss or gain of one or a few chromosomes - Aneuploidy**

Solution: The accurate entry is 'Loss or gain of one or a few chromosomes - Aneuploidy'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Exchange between non-sister chromatids of homologous chromosomes' should be paired with Crossing over, not Back cross. B: 'A female heterozygous for a recessive X-linked allele' should be paired with Carrier, not Law of segregation. D: 'Separation of allele pairs during gamete formation' should be paired with Law of segregation, not Aneuploidy.

BIO-12-CH03-0082 | Hard | Chromosomal disorders**Correct answer: D. XXY constitution in a male - Law of segregation**

Solution: 'XXY constitution in a male' should be paired with Klinefelter syndrome, not Law of segregation. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Sex chromosome system in human females - XX. B: This is a valid association: Monohybrid F2 phenotypic ratio under complete dominance - 3:1. C: This is a valid association: Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1.

BIO-12-CH03-0083 | Medium | Non-Mendelian patterns**Correct answer: A. Pleiotropy**

Solution: The chapter links One gene influencing several characters with Pleiotropy, so Pleiotropy completes the statement.

Distractor review: B: Male belongs with 'Heterogametic sex in humans', whereas the tested description is 'One gene influencing several characters'. C: Incomplete dominance belongs with 'Heterozygote showing an intermediate phenotype', whereas the tested description is 'One gene influencing several characters'. D: Test cross belongs with 'Cross with a homozygous recessive individual', whereas the tested description is 'One gene influencing several characters'.

BIO-12-CH03-0084 | Medium | Chromosomes linkage and crossing over**Correct answer: C. Number of linkage groups in a species corresponds to its - Haploid chromosome number**

Solution: The accurate entry is 'Number of linkage groups in a species corresponds to its - Haploid chromosome number'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Classic X-linked human trait involving red-green vision' should be paired with Colour blindness, not Haploid chromosome number. B: 'Alternative forms of the same gene' should be paired with Alleles, not Klinefelter syndrome. D: 'Sex determination in honey bees' should be paired with Haplo-diploid type, not Genotype.

BIO-12-CH03-0085 | Hard | Chromosomal disorders**Correct answer: A. XXY constitution in a male - Klinefelter syndrome; Separation of allele pairs during gamete formation - Law of segregation**

Solution: Both parts of 'XXY constitution in a male - Klinefelter syndrome; Separation of allele pairs during gamete formation - Law of segregation' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Expected dihybrid test-cross ratio with independent assortment' requires 1:1:1:1, and 'Cross with a homozygous recessive individual' requires Test cross. C: The answers have been interchanged: 'Sex chromosome system in birds' requires ZW female and ZZ male, and 'Monohybrid F2 phenotypic ratio under complete dominance' requires 3:1. D: The answers have been interchanged: 'A male transmitting an X-linked allele to' requires All daughters and no sons, and 'Expected dihybrid test-cross ratio with independent assortment' requires 1:1:1:1.

BIO-12-CH03-0086 | Easy | Chromosomes linkage and crossing over**Correct answer: D. Exchange between non-sister chromatids of homologous chromosomes**

Solution: Crossing over represents 'Exchange between non-sister chromatids of homologous chromosomes'. Each alternative describes a different term or result.

Distractor review: A: 'Alternative forms of the same gene' identifies Alleles, not Crossing over. B: 'Sex chromosome system in human females' identifies XX, not Crossing over. C: 'Stage at which crossing over occurs' identifies Pachytene of prophase I, not Crossing over.

BIO-12-CH03-0087 | Medium | Sex determination**Correct answer: C. XX**

Solution: The chapter links Sex chromosome system in human females with XX, so XX completes the statement.

Distractor review: A: Pachytene of prophase I belongs with 'Stage at which crossing over occurs', whereas the tested description is 'Sex chromosome system in human females'. B: Male belongs with 'Heterogametic sex in humans', whereas the tested description is 'Sex chromosome system in human females'. D: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'Sex chromosome system in human females'.

BIO-12-CH03-0088 | Medium | Chromosomes linkage and crossing over**Correct answer: A. Pachytene of prophase I**

Solution: The chapter links Stage at which crossing over occurs with Pachytene of prophase I, so Pachytene of prophase I completes the statement.

Distractor review: B: Haploid chromosome number belongs with 'Number of linkage groups in a species corresponds to its', whereas the tested description is 'Stage at which crossing over occurs'. C: Locus belongs with 'Position occupied by a gene on a chromosome', whereas the tested description is 'Stage at which crossing over occurs'. D: Crossing over belongs with 'Exchange between non-sister chromatids of homologous chromosomes', whereas the tested description is 'Stage at which crossing over occurs'.

BIO-12-CH03-0089 | Challenging | Mendelian inheritance**Correct answer: D. II only**

Solution: Statement I is incorrect: 'Dihybrid F2 phenotypic ratio under independent assortment' corresponds to 9:3:3:1, not 3:1. Statement II is correct because 'Monohybrid F2 phenotypic ratio under complete dominance' corresponds to 3:1. Statement III is incorrect: 'Independent behaviour of different allele pairs' corresponds to Law of independent assortment, not Pachytene of prophase I. Statement IV is incorrect: 'Stage at which crossing over occurs' corresponds to Pachytene of prophase I, not 9:3:3:1. Therefore, the valid combination is II only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) II. B: This combination includes false statement(s) I, III, IV and omits true statement(s) II. C: This combination includes false statement(s) I, III and omits true statement(s) II.

BIO-12-CH03-0090 | Hard | Chromosomes linkage and crossing over**Correct answer: C. Exchange between non-sister chromatids of homologous chromosomes - Colour blindness**

Solution: 'Exchange between non-sister chromatids of homologous chromosomes' should be paired with Crossing over, not Colour blindness. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: One gene influencing several characters - Pleiotropy. B: This is a valid association: Heterozygote showing an intermediate phenotype - Incomplete dominance. D: This is a valid association: Alternative forms of the same gene - Alleles.

BIO-12-CH03-0091 | Easy | Sex determination**Correct answer: B. Haplo-diploid type**

Solution: 'Sex determination in honey bees' is the defining description for Haplo-diploid type. The other choices belong to different chapter concepts.

Distractor review: A: 1:1:1:1 belongs with 'Expected dihybrid test-cross ratio with independent assortment', whereas the tested description is 'Sex determination in honey bees'. C: XX belongs with 'Sex chromosome system in human females', whereas the tested description is 'Sex determination in honey bees'. D: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'Sex determination in honey bees'.

BIO-12-CH03-0092 | Challenging | Mendelian inheritance**Correct answer: B. II and IV only**

Solution: Statement I is incorrect: 'Independent behaviour of different allele pairs' corresponds to Law of independent assortment, not 9:3:3:1. Statement II is correct because 'Dihybrid F2 phenotypic ratio under independent assortment' corresponds to 9:3:3:1. Statement III is incorrect: 'Separation of allele pairs during gamete formation' corresponds to Law of segregation, not Back cross. Statement IV is correct because 'Cross of an F1 individual with either parent' corresponds to Back cross. Therefore, the valid combination is II and IV only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) II. C: This combination includes false statement(s) III and omits true statement(s) IV. D: This combination includes false statement(s) III.

BIO-12-CH03-0093 | Medium | Non-Mendelian patterns**Correct answer: A. One gene influencing several characters - Pleiotropy**

Solution: The accurate entry is 'One gene influencing several characters - Pleiotropy'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Heterogametic sex in humans' should be paired with Male, not Aneuploidy. C: 'Cross with a homozygous recessive individual' should be paired with Test cross, not 3:1. D: 'Monohybrid F2 phenotypic ratio under complete dominance' should be paired with 3:1, not Pleiotropy.

BIO-12-CH03-0094 | Easy | Genetic terminology**Correct answer: C. Alternative forms of the same gene**

Solution: Alleles represents 'Alternative forms of the same gene'. Each alternative describes a different term or result.

Distractor review: A: 'Number of linkage groups in a species corresponds to its' identifies Haploid chromosome number, not Alleles. B: 'Position occupied by a gene on a chromosome' identifies Locus, not Alleles. D: 'XXY constitution in a male' identifies Klinefelter syndrome, not Alleles.

BIO-12-CH03-0095 | Easy | Sex-linked inheritance**Correct answer: D. X chromosome**

Solution: 'Chromosome carrying the common haemophilia allele' is the defining description for X chromosome. The other choices belong to different chapter concepts.

Distractor review: A: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'Chromosome carrying the common haemophilia allele'. B: Turner syndrome belongs with 'Monosomy X', whereas the tested description is 'Chromosome carrying the common haemophilia allele'. C: Test cross belongs with 'Cross with a homozygous recessive individual', whereas the tested description is 'Chromosome carrying the common haemophilia allele'.

BIO-12-CH03-0096 | Medium | Genetic terminology**Correct answer: B. Genetic constitution of an organism - Genotype**

Solution: The accurate entry is 'Genetic constitution of an organism - Genotype'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Sex chromosome system in birds' should be paired with ZW female and ZZ male, not Genotype. C: 'Classic X-linked human trait involving red-green vision' should be paired with Colour blindness, not Turner syndrome. D: 'XXY constitution in a male' should be paired with Klinefelter syndrome, not ABO blood grouping.

BIO-12-CH03-0097 | Challenging | Non-Mendelian patterns**Correct answer: B. I only**

Solution: Statement I is correct because 'One gene influencing several characters' corresponds to Pleiotropy. Statement II is incorrect: 'Human example of multiple alleles cited in the chapter' corresponds to ABO blood grouping, not Incomplete dominance. Statement III is incorrect: 'Heterozygote showing an intermediate phenotype' corresponds to Incomplete dominance, not Aneuploidy. Statement IV is incorrect: 'Loss or gain of one or a few chromosomes' corresponds to Aneuploidy, not Pleiotropy. Therefore, the valid combination is I only.

Distractor review: A: This combination includes false statement(s) II, IV and omits true statement(s) I. C: This combination includes false statement(s) II, III. D: This combination includes false statement(s) II and omits true statement(s) I.

BIO-12-CH03-0098 | Medium | Mendelian inheritance**Correct answer: D. Law of independent assortment**

Solution: The observation 'Independent behaviour of different allele pairs' points to Law of independent assortment; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: 3:1 belongs with 'Monohybrid F2 phenotypic ratio under complete dominance', whereas the tested description is 'Independent behaviour of different allele pairs'. B: ZW female and ZZ male belongs with 'Sex chromosome system in birds', whereas the tested description is 'Independent behaviour of different allele pairs'. C: All daughters and no sons belongs with 'A male transmitting an X-linked allele to', whereas the tested description is 'Independent behaviour of different allele pairs'.

BIO-12-CH03-0099 | Medium | Crosses**Correct answer: B. Cross with a homozygous recessive individual - Test cross**

Solution: The accurate entry is 'Cross with a homozygous recessive individual - Test cross'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Monohybrid F2 phenotypic ratio under complete dominance' should be paired with 3:1, not Linkage. C: 'Exchange between non-sister chromatids of homologous chromosomes' should be paired with Crossing over, not Haploid chromosome number. D: 'A male transmitting an X-linked allele to' should be paired with All daughters and no sons, not Test cross.

BIO-12-CH03-0100 | Challenging | Non-Mendelian patterns**Correct answer: D. II and III only**

Solution: Statement I is incorrect: 'Both alleles expressed in a heterozygote' corresponds to Codominance, not Incomplete dominance. Statement II is correct because 'Heterozygote showing an intermediate phenotype' corresponds to Incomplete dominance. Statement III is correct because 'One gene influencing several characters' corresponds to Pleiotropy. Statement IV is incorrect: 'Sex determination in honey bees' corresponds to Haplo-diploid type, not Codominance. Therefore, the valid combination is II and III only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) II, III. B: This combination omits true statement(s) III. C: This combination includes false statement(s) I and omits true statement(s) III.

BIO-12-CH03-0101 | Hard | Genetic terminology**Correct answer: C. Alternative forms of the same gene - Alleles; XXY constitution in a male - Klinefelter syndrome**

Solution: Both parts of 'Alternative forms of the same gene - Alleles; XXY constitution in a male - Klinefelter syndrome' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Exchange between non-sister chromatids of homologous chromosomes' requires Crossing over, and 'Number of linkage groups in a species corresponds to its' requires Haploid chromosome number. B: The answers have been interchanged: 'A male transmitting an X-linked allele to' requires All daughters and no sons, and 'Separation of allele pairs during gamete formation' requires Law of segregation. D: The answers have been interchanged: 'Heterogametic sex in humans' requires Male, and 'Cross with a homozygous recessive individual' requires Test cross.

BIO-12-CH03-0102 | Hard | Mendelian inheritance**Correct answer: A. Separation of allele pairs during gamete formation - Law of segregation; Position occupied by a gene on a chromosome - Locus**

Solution: Both parts of 'Separation of allele pairs during gamete formation - Law of segregation; Position occupied by a gene on a chromosome - Locus' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Chromosome carrying the common haemophilia allele' requires X chromosome, and 'Classic X-linked human trait involving red-green vision' requires Colour blindness. C: The answers have been interchanged: 'Monosomy X' requires Turner syndrome, and 'Number of linkage groups in a species corresponds to its' requires Haploid chromosome number. D: The answers have been interchanged: 'Expected dihybrid test-cross ratio with independent assortment' requires 1:1:1:1, and 'Monosomy X' requires Turner syndrome.

BIO-12-CH03-0103 | Medium | Genetic terminology**Correct answer: D. Alternative forms of the same gene - Alleles**

Solution: The accurate entry is 'Alternative forms of the same gene - Alleles'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Number of linkage groups in a species corresponds to its' should be paired with Haploid chromosome number, not Law of segregation. B: 'A male transmitting an X-linked allele to' should be paired with All daughters and no sons, not Male. C: 'XXY constitution in a male' should be paired with Klinefelter syndrome, not Haploid chromosome number.

BIO-12-CH03-0104 | Challenging | Sex determination**Correct answer: C. I, II, III and IV**

Solution: Statement I is correct because 'Sex determination in honey bees' corresponds to Haplo-diploid type. Statement II is correct because 'Heterogametic sex in humans' corresponds to Male. Statement III is correct because 'Sex chromosome system in human females' corresponds to XX. Statement IV is correct because 'Genes on the same chromosome tending to be inherited together' corresponds to Linkage. Therefore, the valid combination is I, II, III and IV.

Distractor review: A: This combination omits true statement(s) II, III, IV. B: This combination omits true statement(s) I, II. D: This combination omits true statement(s) I, II, IV.

BIO-12-CH03-0105 | Challenging | Crosses**Correct answer: A. I and IV only**

Solution: Statement I is correct because 'Expected dihybrid test-cross ratio with independent assortment' corresponds to 1:1:1:1. Statement II is incorrect: 'Cross of an F1 individual with either parent' corresponds to Back cross, not Test cross. Statement III is incorrect: 'Cross with a homozygous recessive individual' corresponds to Test cross, not Male. Statement IV is correct because 'Heterogametic sex in humans' corresponds to Male. Therefore, the valid combination is I and IV only.

Distractor review: B: This combination includes false statement(s) III and omits true statement(s) I, IV. C: This combination includes false statement(s) II. D: This combination includes false statement(s) III and omits true statement(s) IV.

BIO-12-CH03-0106 | Hard | Chromosomes linkage and crossing over

Correct answer: B. Number of linkage groups in a species corresponds to its - Haploid chromosome number; Human example of multiple alleles cited in the chapter - ABO blood grouping

Solution: Both parts of 'Number of linkage groups in a species corresponds to its - Haploid chromosome number; Human example of multiple alleles cited in the chapter - ABO blood grouping' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Sex determination in honey bees' requires Haplo-diploid type, and 'Cross of an F1 individual with either parent' requires Back cross. C: The answers have been interchanged: 'Cross of an F1 individual with either parent' requires Back cross, and 'Genetic constitution of an organism' requires Genotype. D: The answers have been interchanged: 'Alternative forms of the same gene' requires Alleles, and 'Classic X-linked human trait involving red-green vision' requires Colour blindness.

BIO-12-CH03-0107 | Medium | Sex determination

Correct answer: B. Male

Solution: The observation 'Heterogametic sex in humans' points to Male; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: XX belongs with 'Sex chromosome system in human females', whereas the tested description is 'Heterogametic sex in humans'. C: Incomplete dominance belongs with 'Heterozygote showing an intermediate phenotype', whereas the tested description is 'Heterogametic sex in humans'. D: Law of segregation belongs with 'Separation of allele pairs during gamete formation', whereas the tested description is 'Heterogametic sex in humans'.

BIO-12-CH03-0108 | Easy | Sex-linked inheritance

Correct answer: A. Chromosome carrying the common haemophilia allele

Solution: X chromosome represents 'Chromosome carrying the common haemophilia allele'. Each alternative describes a different term or result.

Distractor review: B: 'Classic X-linked human trait involving red-green vision' identifies Colour blindness, not X chromosome. C: 'Exchange between non-sister chromatids of homologous chromosomes' identifies Crossing over, not X chromosome. D: 'Cross with a homozygous recessive individual' identifies Test cross, not X chromosome.

BIO-12-CH03-0109 | Medium | Sex determination

Correct answer: A. Haplo-diploid type

Solution: The observation 'Sex determination in honey bees' points to Haplo-diploid type; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: ZW female and ZZ male belongs with 'Sex chromosome system in birds', whereas the tested description is 'Sex determination in honey bees'. C: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'Sex determination in honey bees'. D: 9:3:3:1 belongs with 'Dihybrid F2 phenotypic ratio under independent assortment', whereas the tested description is 'Sex determination in honey bees'.

BIO-12-CH03-0110 | Hard | Genetic terminology

Correct answer: B. Position occupied by a gene on a chromosome - Locus; Monosomy X - Turner syndrome

Solution: Both parts of 'Position occupied by a gene on a chromosome - Locus; Monosomy X - Turner syndrome' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Loss or gain of one or a few chromosomes' requires Aneuploidy, and 'Number of linkage groups in a species corresponds to its' requires Haploid chromosome number. C: The answers have been interchanged: 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1, and 'Loss or gain of one or a few chromosomes' requires Aneuploidy. D: The answers have been interchanged: 'Expected test-cross ratio for a heterozygous monohybrid' requires 1:1, and 'Dihybrid F2 phenotypic ratio under independent assortment' requires 9:3:3:1.

BIO-12-CH03-0111 | Medium | Crosses

Correct answer: B. Test cross

Solution: The observation 'Cross with a homozygous recessive individual' points to Test cross; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'Cross with a homozygous recessive individual'. C: 1:1 belongs with 'Expected test-cross ratio for a heterozygous monohybrid', whereas the tested description is 'Cross with a homozygous recessive individual'. D: Turner syndrome belongs with 'Monosomy X', whereas the tested description is 'Cross with a homozygous recessive individual'.

BIO-12-CH03-0112 | Medium | Mendelian inheritance**Correct answer: D. Separation of allele pairs during gamete formation - Law of segregation**

Solution: The accurate entry is 'Separation of allele pairs during gamete formation - Law of segregation'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Chromosome carrying the common haemophilia allele' should be paired with X chromosome, not Incomplete dominance. B: 'Classic X-linked human trait involving red-green vision' should be paired with Colour blindness, not 1:1:1:1. C: 'Monosomy X' should be paired with Turner syndrome, not Law of segregation.

BIO-12-CH03-0113 | Hard | Sex determination**Correct answer: C. Sex chromosome system in human females - Down syndrome**

Solution: 'Sex chromosome system in human females' should be paired with XX, not Down syndrome. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Human example of multiple alleles cited in the chapter - ABO blood grouping. B: This is a valid association: Chromosome carrying the common haemophilia allele - X chromosome. D: This is a valid association: Dihybrid F2 phenotypic ratio under independent assortment - 9:3:3:1.

BIO-12-CH03-0114 | Hard | Sex-linked inheritance**Correct answer: D. A male transmitting an X-linked allele to - All daughters and no sons; Observable expression of a character - Phenotype**

Solution: Both parts of 'A male transmitting an X-linked allele to - All daughters and no sons; Observable expression of a character - Phenotype' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Monohybrid F2 phenotypic ratio under complete dominance' requires 3:1, and 'Sex chromosome system in human females' requires XX. B: The answers have been interchanged: 'Trisomy of chromosome 21' requires Down syndrome, and 'Cross of an F1 individual with either parent' requires Back cross. C: The answers have been interchanged: 'Sex chromosome system in human females' requires XX, and 'Genes on the same chromosome tending to be inherited together' requires Linkage.

BIO-12-CH03-0115 | Easy | Chromosomes linkage and crossing over**Correct answer: C. Haploid chromosome number**

Solution: 'Number of linkage groups in a species corresponds to its' is the defining description for Haploid chromosome number. The other choices belong to different chapter concepts.

Distractor review: A: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'Number of linkage groups in a species corresponds to its'. B: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'Number of linkage groups in a species corresponds to its'. D: Haplo-diploid type belongs with 'Sex determination in honey bees', whereas the tested description is 'Number of linkage groups in a species corresponds to its'.

BIO-12-CH03-0116 | Medium | Mendelian inheritance**Correct answer: C. 3:1**

Solution: The observation 'Monohybrid F2 phenotypic ratio under complete dominance' points to 3:1; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Pachytene of prophase I belongs with 'Stage at which crossing over occurs', whereas the tested description is 'Monohybrid F2 phenotypic ratio under complete dominance'. B: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'Monohybrid F2 phenotypic ratio under complete dominance'. D: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'Monohybrid F2 phenotypic ratio under complete dominance'.

BIO-12-CH03-0117 | Medium | Mendelian inheritance**Correct answer: D. 9:3:3:1**

Solution: The chapter links Dihybrid F2 phenotypic ratio under independent assortment with 9:3:3:1, so 9:3:3:1 completes the statement.

Distractor review: A: X chromosome belongs with 'Chromosome carrying the common haemophilia allele', whereas the tested description is 'Dihybrid F2 phenotypic ratio under independent assortment'. B: 3:1 belongs with 'Monohybrid F2 phenotypic ratio under complete dominance', whereas the tested description is 'Dihybrid F2 phenotypic ratio under independent assortment'. C: Law of independent assortment belongs with 'Independent behaviour of different allele pairs', whereas the tested description is 'Dihybrid F2 phenotypic ratio under independent assortment'.

BIO-12-CH03-0118 | Easy | Sex-linked inheritance**Correct answer: B. Classic X-linked human trait involving red-green vision**

Solution: Colour blindness represents 'Classic X-linked human trait involving red-green vision'. Each alternative describes a different term or result.

Distractor review: A: 'A female heterozygous for a recessive X-linked allele' identifies Carrier, not Colour blindness. C: 'Dihybrid F2 phenotypic ratio under independent assortment' identifies 9:3:3:1, not Colour blindness. D: 'Observable expression of a character' identifies Phenotype, not Colour blindness.

BIO-12-CH03-0119 | Medium | Crosses**Correct answer: B. Cross of an F1 individual with either parent - Back cross**

Solution: The accurate entry is 'Cross of an F1 individual with either parent - Back cross'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Dihybrid F2 phenotypic ratio under independent assortment' should be paired with 9:3:3:1, not Male. C: 'Heterogametic sex in humans' should be paired with Male, not X chromosome. D: 'One gene influencing several characters' should be paired with Pleiotropy, not ABO blood grouping.

BIO-12-CH03-0120 | Medium | Sex-linked inheritance**Correct answer: C. Carrier**

Solution: The chapter links A female heterozygous for a recessive X-linked allele with Carrier, so Carrier completes the statement.

Distractor review: A: Back cross belongs with 'Cross of an F1 individual with either parent', whereas the tested description is 'A female heterozygous for a recessive X-linked allele'. B: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'A female heterozygous for a recessive X-linked allele'. D: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'A female heterozygous for a recessive X-linked allele'.

BIO-12-CH03-0121 | Hard | Genetic terminology**Correct answer: D. Genetic constitution of an organism - Klinefelter syndrome**

Solution: 'Genetic constitution of an organism' should be paired with Genotype, not Klinefelter syndrome. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Sex chromosome system in birds - ZW female and ZZ male. B: This is a valid association: Classic X-linked human trait involving red-green vision - Colour blindness. C: This is a valid association: Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1.

BIO-12-CH03-0122 | Easy | Mendelian inheritance**Correct answer: B. Law of independent assortment**

Solution: 'Independent behaviour of different allele pairs' is the defining description for Law of independent assortment. The other choices belong to different chapter concepts.

Distractor review: A: Locus belongs with 'Position occupied by a gene on a chromosome', whereas the tested description is 'Independent behaviour of different allele pairs'. C: All daughters and no sons belongs with 'A male transmitting an X-linked allele to', whereas the tested description is 'Independent behaviour of different allele pairs'. D: ZW female and ZZ male belongs with 'Sex chromosome system in birds', whereas the tested description is 'Independent behaviour of different allele pairs'.

BIO-12-CH03-0123 | Medium | Non-Mendelian patterns**Correct answer: A. Codominance**

Solution: The observation 'Both alleles expressed in a heterozygote' points to Codominance; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: 1:1:1:1 belongs with 'Expected dihybrid test-cross ratio with independent assortment', whereas the tested description is 'Both alleles expressed in a heterozygote'. C: Pleiotropy belongs with 'One gene influencing several characters', whereas the tested description is 'Both alleles expressed in a heterozygote'. D: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'Both alleles expressed in a heterozygote'.

BIO-12-CH03-0124 | Easy | Sex determination**Correct answer: C. Sex determination in honey bees**

Solution: Haplo-diploid type represents 'Sex determination in honey bees'. Each alternative describes a different term or result.

Distractor review: A: 'Monosomy X' identifies Turner syndrome, not Haplo-diploid type. B: 'Sex chromosome system in human females' identifies XX, not Haplo-diploid type. D: 'Expected dihybrid test-cross ratio with independent assortment' identifies 1:1:1:1, not Haplo-diploid type.

BIO-12-CH03-0125 | Medium | Non-Mendelian patterns**Correct answer: B. Pleiotropy**

Solution: The observation 'One gene influencing several characters' points to Pleiotropy; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: ABO blood grouping belongs with 'Human example of multiple alleles cited in the chapter', whereas the tested description is 'One gene influencing several characters'. C: Down syndrome belongs with 'Trisomy of chromosome 21', whereas the tested description is 'One gene influencing several characters'. D: Male belongs with 'Heterogametic sex in humans', whereas the tested description is 'One gene influencing several characters'.

BIO-12-CH03-0126 | Medium | Chromosomal disorders**Correct answer: D. Turner syndrome**

Solution: The observation 'Monosomy X' points to Turner syndrome; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'Monosomy X'. B: Down syndrome belongs with 'Trisomy of chromosome 21', whereas the tested description is 'Monosomy X'. C: 3:1 belongs with 'Monohybrid F2 phenotypic ratio under complete dominance', whereas the tested description is 'Monosomy X'.

BIO-12-CH03-0127 | Hard | Chromosomes linkage and crossing over**Correct answer: B. Number of linkage groups in a species corresponds to its - ABO blood grouping**

Solution: 'Number of linkage groups in a species corresponds to its' should be paired with Haploid chromosome number, not ABO blood grouping. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Classic X-linked human trait involving red-green vision - Colour blindness. C: This is a valid association: Genetic constitution of an organism - Genotype. D: This is a valid association: Sex determination in honey bees - Haplo-diploid type.

BIO-12-CH03-0128 | Medium | Chromosomal disorders**Correct answer: A. Klinefelter syndrome**

Solution: The observation 'XXY constitution in a male' points to Klinefelter syndrome; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: 3:1 belongs with 'Monohybrid F2 phenotypic ratio under complete dominance', whereas the tested description is 'XXY constitution in a male'. C: XX belongs with 'Sex chromosome system in human females', whereas the tested description is 'XXY constitution in a male'. D: Aneuploidy belongs with 'Loss or gain of one or a few chromosomes', whereas the tested description is 'XXY constitution in a male'.

BIO-12-CH03-0129 | Hard | Crosses**Correct answer: C. Expected test-cross ratio for a heterozygous monohybrid - 1:1; XXY constitution in a male - Klinefelter syndrome**

Solution: Both parts of 'Expected test-cross ratio for a heterozygous monohybrid - 1:1; XXY constitution in a male - Klinefelter syndrome' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Loss or gain of one or a few chromosomes' requires Aneuploidy, and 'Both alleles expressed in a heterozygote' requires Codominance. B: The answers have been interchanged: 'Both alleles expressed in a heterozygote' requires Codominance, and 'Stage at which crossing over occurs' requires Pachytene of prophase I. D: The answers have been interchanged: 'Genetic constitution of an organism' requires Genotype, and 'Separation of allele pairs during gamete formation' requires Law of segregation.

BIO-12-CH03-0130 | Hard | Non-Mendelian patterns**Correct answer: B. Both alleles expressed in a heterozygote - Haplo-diploid type**

Solution: 'Both alleles expressed in a heterozygote' should be paired with Codominance, not Haplo-diploid type. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Expected test-cross ratio for a heterozygous monohybrid - 1:1. C: This is a valid association: Chromosome carrying the common haemophilia allele - X chromosome. D: This is a valid association: Cross of an F1 individual with either parent - Back cross.

BIO-12-CH03-0131 | Easy | Genetic terminology**Correct answer: B. Position occupied by a gene on a chromosome**

Solution: Locus represents 'Position occupied by a gene on a chromosome'. Each alternative describes a different term or result.

Distractor review: A: 'Exchange between non-sister chromatids of homologous chromosomes' identifies Crossing over, not Locus. C: 'Expected test-cross ratio for a heterozygous monohybrid' identifies 1:1, not Locus. D: 'Genetic constitution of an organism' identifies Genotype, not Locus.

BIO-12-CH03-0132 | Easy | Non-Mendelian patterns**Correct answer: D. Human example of multiple alleles cited in the chapter**

Solution: ABO blood grouping represents 'Human example of multiple alleles cited in the chapter'. Each alternative describes a different term or result.

Distractor review: A: 'Stage at which crossing over occurs' identifies Pachytene of prophase I, not ABO blood grouping. B: 'Classic X-linked human trait involving red-green vision' identifies Colour blindness, not ABO blood grouping. C: 'Observable expression of a character' identifies Phenotype, not ABO blood grouping.

BIO-12-CH03-0133 | Hard | Mendelian inheritance**Correct answer: A. Independent behaviour of different allele pairs - Law of independent assortment; Position occupied by a gene on a chromosome - Locus**

Solution: Both parts of 'Independent behaviour of different allele pairs - Law of independent assortment; Position occupied by a gene on a chromosome - Locus' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Sex chromosome system in birds' requires ZW female and ZZ male, and 'Expected dihybrid test-cross ratio with independent assortment' requires 1:1:1:1. C: The answers have been interchanged: 'Both alleles expressed in a heterozygote' requires Codominance, and 'Heterogametic sex in humans' requires Male. D: The answers have been interchanged: 'A male transmitting an X-linked allele to' requires All daughters and no sons, and 'Sex chromosome system in birds' requires ZW female and ZZ male.

BIO-12-CH03-0134 | Easy | Crosses**Correct answer: C. Expected test-cross ratio for a heterozygous monohybrid**

Solution: 1:1 represents 'Expected test-cross ratio for a heterozygous monohybrid'. Each alternative describes a different term or result.

Distractor review: A: 'Genetic constitution of an organism' identifies Genotype, not 1:1. B: 'Cross with a homozygous recessive individual' identifies Test cross, not 1:1. D: 'Stage at which crossing over occurs' identifies Pachytene of prophase I, not 1:1.

BIO-12-CH03-0135 | Challenging | Sex-linked inheritance**Correct answer: B. II, III and IV only**

Solution: Statement I is incorrect: 'Chromosome carrying the common haemophilia allele' corresponds to X chromosome, not Carrier. Statement II is correct because 'A female heterozygous for a recessive X-linked allele' corresponds to Carrier. Statement III is correct because 'A male transmitting an X-linked allele to' corresponds to All daughters and no sons. Statement IV is correct because 'Sex chromosome system in birds' corresponds to ZW female and ZZ male. Therefore, the valid combination is II, III and IV only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) II, IV. C: This combination omits true statement(s) III. D: This combination omits true statement(s) III, IV.

BIO-12-CH03-0136 | Medium | Genetic terminology**Correct answer: C. Genotype**

Solution: The observation 'Genetic constitution of an organism' points to Genotype; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Locus belongs with 'Position occupied by a gene on a chromosome', whereas the tested description is 'Genetic constitution of an organism'. B: Klinefelter syndrome belongs with 'XXY constitution in a male', whereas the tested description is 'Genetic constitution of an organism'. D: 1:1 belongs with 'Expected test-cross ratio for a heterozygous monohybrid', whereas the tested description is 'Genetic constitution of an organism'.

BIO-12-CH03-0137 | Medium | Non-Mendelian patterns**Correct answer: A. Incomplete dominance**

Solution: The chapter links Heterozygote showing an intermediate phenotype with Incomplete dominance, so Incomplete dominance completes the statement.

Distractor review: B: 9:3:3:1 belongs with 'Dihybrid F2 phenotypic ratio under independent assortment', whereas the tested description is 'Heterozygote showing an intermediate phenotype'. C: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'Heterozygote showing an intermediate phenotype'. D: Down syndrome belongs with 'Trisomy of chromosome 21', whereas the tested description is 'Heterozygote showing an intermediate phenotype'.

BIO-12-CH03-0138 | Medium | Chromosomes linkage and crossing over**Correct answer: C. Haploid chromosome number**

Solution: The chapter links Number of linkage groups in a species corresponds to its with Haploid chromosome number, so Haploid chromosome number completes the statement.

Distractor review: A: Back cross belongs with 'Cross of an F1 individual with either parent', whereas the tested description is 'Number of linkage groups in a species corresponds to its'. B: Genotype belongs with 'Genetic constitution of an organism', whereas the tested description is 'Number of linkage groups in a species corresponds to its'. D: Linkage belongs with 'Genes on the same chromosome tending to be inherited together', whereas the tested description is 'Number of linkage groups in a species corresponds to its'.

BIO-12-CH03-0139 | Easy | Chromosomal disorders**Correct answer: B. XXY constitution in a male**

Solution: Klinefelter syndrome represents 'XXY constitution in a male'. Each alternative describes a different term or result.

Distractor review: A: 'Sex chromosome system in human females' identifies XX, not Klinefelter syndrome. C: 'Separation of allele pairs during gamete formation' identifies Law of segregation, not Klinefelter syndrome. D: 'Trisomy of chromosome 21' identifies Down syndrome, not Klinefelter syndrome.

BIO-12-CH03-0140 | Easy | Sex-linked inheritance**Correct answer: D. Carrier**

Solution: 'A female heterozygous for a recessive X-linked allele' is the defining description for Carrier. The other choices belong to different chapter concepts.

Distractor review: A: 1:1 belongs with 'Expected test-cross ratio for a heterozygous monohybrid', whereas the tested description is 'A female heterozygous for a recessive X-linked allele'. B: Test cross belongs with 'Cross with a homozygous recessive individual', whereas the tested description is 'A female heterozygous for a recessive X-linked allele'. C: Colour blindness belongs with 'Classic X-linked human trait involving red-green vision', whereas the tested description is 'A female heterozygous for a recessive X-linked allele'.

Standard 12 | Chapter 04

Molecular Basis of Inheritance

BIO-12-CH04-0001 | Medium | Regulation of the lac operon

Correct answer: A. The active repressor binds the operator

Solution: Without inducer, active repressor occupies the operator and obstructs transcription.

Distractor review: B: The alternative 'Allolactose binds RNA polymerase' conflicts with the labelled relationship, scale or mechanism in the figure. Without inducer, active repressor occupies the operator and obstructs transcription. C: The alternative 'The operator is removed from DNA' conflicts with the labelled relationship, scale or mechanism in the figure. Without inducer, active repressor occupies the operator and obstructs transcription. D: The alternative 'The structural genes become ribosomal RNA' conflicts with the labelled relationship, scale or mechanism in the figure. Without inducer, active repressor occupies the operator and obstructs transcription.

BIO-12-CH04-0002 | Hard | Events at a DNA replication fork

Correct answer: A. It joins adjacent Okazaki fragments

Solution: Discontinuous lagging-strand synthesis creates Okazaki fragments that DNA ligase joins.

Distractor review: B: The alternative 'It unwinds the parental helix' conflicts with the labelled relationship, scale or mechanism in the figure. Discontinuous lagging-strand synthesis creates Okazaki fragments that DNA ligase joins. C: The alternative 'It synthesizes RNA primers' conflicts with the labelled relationship, scale or mechanism in the figure. Discontinuous lagging-strand synthesis creates Okazaki fragments that DNA ligase joins. D: The alternative 'It reads mRNA codons' conflicts with the labelled relationship, scale or mechanism in the figure. Discontinuous lagging-strand synthesis creates Okazaki fragments that DNA ligase joins.

BIO-12-CH04-0003 | Medium | DNA structure

Correct answer: D. Three

Solution: The chapter links Number of hydrogen bonds in a G-C pair with Three, so Three completes the statement.

Distractor review: A: Genome belongs with 'Complete DNA content of an organism', whereas the tested description is 'Number of hydrogen bonds in a G-C pair'. B: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'Number of hydrogen bonds in a G-C pair'. C: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'Number of hydrogen bonds in a G-C pair'.

BIO-12-CH04-0004 | Challenging | Regulation of the lac operon

Correct answer: D. Lactose present and glucose scarce

Solution: Lactose-derived allolactose removes repression, while low glucose favours strong catabolite-activator action.

Distractor review: A: The alternative 'Lactose absent and glucose abundant' conflicts with the labelled relationship, scale or mechanism in the figure. Lactose-derived allolactose removes repression, while low glucose favours strong catabolite-activator action. B: The alternative 'Lactose absent and glucose scarce' conflicts with the labelled relationship, scale or mechanism in the figure. Lactose-derived allolactose removes repression, while low glucose favours strong catabolite-activator action. C: The alternative 'Lactose present with an active operator-bound repressor' conflicts with the labelled relationship, scale or mechanism in the figure. Lactose-derived allolactose removes repression, while low glucose favours strong catabolite-activator action.

BIO-12-CH04-0005 | Medium | Gene regulation and genomics

Correct answer: A. Genome

Solution: The observation 'Complete DNA content of an organism' points to Genome; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: DNA fingerprinting belongs with 'Technique identifying individuals from variable DNA regions', whereas the tested description is 'Complete DNA content of an organism'. C: Deoxyribose belongs with 'Sugar present in DNA', whereas the tested description is 'Complete DNA content of an organism'. D: Degeneracy belongs with 'Property in which several codons specify one amino acid', whereas the tested description is 'Complete DNA content of an organism'.

BIO-12-CH04-0006 | Medium | DNA packaging

Correct answer: C. Histone octamer

Solution: The observation 'Core proteins around which DNA is wrapped' points to Histone octamer; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Friedrich Miescher belongs with 'Scientist who first isolated nuclein', whereas the tested description is 'Core proteins around which DNA is wrapped'. B: Repressor belongs with 'Regulatory protein that blocks the operator', whereas the tested description is 'Core proteins around which DNA is wrapped'. D: Heterochromatin belongs with 'Highly condensed transcriptionally inactive chromatin', whereas the tested description is 'Core proteins around which DNA is wrapped'.

BIO-12-CH04-0007 | Medium | DNA replication**Correct answer: A. Okazaki fragments**

Solution: The observation 'Short DNA segments formed on the lagging strand' points to Okazaki fragments; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Helicase belongs with 'Enzyme unwinding the DNA double helix', whereas the tested description is 'Short DNA segments formed on the lagging strand'. C: Deoxyribose belongs with 'Sugar present in DNA', whereas the tested description is 'Short DNA segments formed on the lagging strand'. D: DNA ligase belongs with 'Enzyme joining Okazaki fragments', whereas the tested description is 'Short DNA segments formed on the lagging strand'.

BIO-12-CH04-0008 | Hard | Transcription and RNA**Correct answer: B. RNA carrying codons from DNA to ribosomes - A site**

Solution: 'RNA carrying codons from DNA to ribosomes' should be paired with mRNA, not A site. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Short DNA segments formed on the lagging strand - Okazaki fragments. C: This is a valid association: Hershey-Chase label used for phage DNA - Radioactive phosphorus-32. D: This is a valid association: Sugar present in DNA - Deoxyribose.

BIO-12-CH04-0009 | Hard | DNA structure**Correct answer: C. Sugar present in DNA - AUG**

Solution: 'Sugar present in DNA' should be paired with Deoxyribose, not AUG. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Complete DNA content of an organism - Genome. B: This is a valid association: Basic repeating unit of chromatin - Nucleosome. D: This is a valid association: Mode of DNA replication - Semiconservative.

BIO-12-CH04-0010 | Hard | Evidence for genetic material**Correct answer: B. Hershey-Chase label used for phage DNA - mRNA**

Solution: 'Hershey-Chase label used for phage DNA' should be paired with Radioactive phosphorus-32, not mRNA. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Core proteins around which DNA is wrapped - Histone octamer. C: This is a valid association: Highly condensed transcriptionally inactive chromatin - Heterochromatin. D: This is a valid association: Purine pairing with thymine - Adenine.

BIO-12-CH04-0011 | Medium | DNA packaging**Correct answer: C. Linker histone - H1**

Solution: The accurate entry is 'Linker histone - H1'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Approximate base pairs per turn of B-DNA' should be paired with Ten, not Degeneracy. B: 'Sugar present in DNA' should be paired with Deoxyribose, not Allolactose. D: 'Inducer of the lac operon' should be paired with Allolactose, not mRNA.

BIO-12-CH04-0012 | Medium | DNA packaging**Correct answer: D. Nucleosome**

Solution: The observation 'Basic repeating unit of chromatin' points to Nucleosome; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'Basic repeating unit of chromatin'. B: Helicase belongs with 'Enzyme unwinding the DNA double helix', whereas the tested description is 'Basic repeating unit of chromatin'. C: Degeneracy belongs with 'Property in which several codons specify one amino acid', whereas the tested description is 'Basic repeating unit of chromatin'.

BIO-12-CH04-0013 | Medium | Genetic code and translation**Correct answer: B. Degeneracy**

Solution: The observation 'Property in which several codons specify one amino acid' points to Degeneracy; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Repressor belongs with 'Regulatory protein that blocks the operator', whereas the tested description is 'Property in which several codons specify one amino acid'. C: Genome belongs with 'Complete DNA content of an organism', whereas the tested description is 'Property in which several codons specify one amino acid'. D: Histone octamer belongs with 'Core proteins around which DNA is wrapped', whereas the tested description is 'Property in which several codons specify one amino acid'.

BIO-12-CH04-0014 | Medium | DNA replication**Correct answer: C. Semiconservative**

Solution: The chapter links Mode of DNA replication with Semiconservative, so Semiconservative completes the statement.

Distractor review: A: A site belongs with 'Ribosomal site accepting an incoming aminoacyl-tRNA', whereas the tested description is 'Mode of DNA replication'. B: Radioactive phosphorus-32 belongs with 'Hershey-Chase label used for phage DNA', whereas the tested description is 'Mode of DNA replication'. D: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'Mode of DNA replication'.

BIO-12-CH04-0015 | Easy | Events at a DNA replication fork**Correct answer: A. Helicase**

Solution: Helicase unwinds and separates the DNA strands.

Distractor review: B: The alternative 'Ligase' conflicts with the labelled relationship, scale or mechanism in the figure. Helicase unwinds and separates the DNA strands. C: The alternative 'Primase' conflicts with the labelled relationship, scale or mechanism in the figure. Helicase unwinds and separates the DNA strands. D: The alternative 'RNA polymerase' conflicts with the labelled relationship, scale or mechanism in the figure. Helicase unwinds and separates the DNA strands.

BIO-12-CH04-0016 | Hard | DNA replication**Correct answer: B. Enzyme joining Okazaki fragments - A site**

Solution: 'Enzyme joining Okazaki fragments' should be paired with DNA ligase, not A site. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Transforming principle demonstrated in pneumococcus - DNA. C: This is a valid association: Approximate base pairs per turn of B-DNA - Ten. D: This is a valid association: Highly condensed transcriptionally inactive chromatin - Heterochromatin.

BIO-12-CH04-0017 | Challenging | DNA structure**Correct answer: D. I only**

Solution: Statement I is correct because 'Purine pairing with thymine' corresponds to Adenine. Statement II is incorrect: 'Sugar present in DNA' corresponds to Deoxyribose, not Three. Statement III is incorrect: 'Number of hydrogen bonds in a G-C pair' corresponds to Three, not Allolactose. Statement IV is incorrect: 'Inducer of the lac operon' corresponds to Allolactose, not Adenine. Therefore, the valid combination is I only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) I. B: This combination includes false statement(s) II, III, IV and omits true statement(s) I. C: This combination includes false statement(s) III.

BIO-12-CH04-0018 | Challenging | Genetic code and translation**Correct answer: A. II and III only**

Solution: Statement I is incorrect: 'Ribosomal site accepting an incoming aminoacyl-tRNA' corresponds to A site, not UAA. Statement II is correct because 'A termination codon' corresponds to UAA. Statement III is correct because 'Codon that usually initiates translation' corresponds to AUG. Statement IV is incorrect: 'Hershey-Chase label used for phage DNA' corresponds to Radioactive phosphorus-32, not A site. Therefore, the valid combination is II and III only.

Distractor review: B: This combination includes false statement(s) IV and omits true statement(s) II. C: This combination includes false statement(s) I and omits true statement(s) II, III. D: This combination includes false statement(s) IV and omits true statement(s) III.

BIO-12-CH04-0019 | Hard | Gene regulation and genomics**Correct answer: A. Regulatory protein that blocks the operator - Nucleosome**

Solution: 'Regulatory protein that blocks the operator' should be paired with Repressor, not Nucleosome. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: RNA carrying codons from DNA to ribosomes - mRNA. C: This is a valid association: Purine pairing with thymine - Adenine. D: This is a valid association: Core proteins around which DNA is wrapped - Histone octamer.

BIO-12-CH04-0020 | Medium | DNA structure**Correct answer: D. Approximate base pairs per turn of B-DNA - Ten**

Solution: The accurate entry is 'Approximate base pairs per turn of B-DNA - Ten'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Scientist who first isolated nuclein' should be paired with Friedrich Miescher, not Repressor. B: 'RNA forming a major structural and catalytic part of ribosomes' should be paired with rRNA, not Allolactose. C: 'RNA carrying codons from DNA to ribosomes' should be paired with mRNA, not rRNA.

BIO-12-CH04-0021 | Easy | DNA replication**Correct answer: C. Enzyme unwinding the DNA double helix**

Solution: Helicase represents 'Enzyme unwinding the DNA double helix'. Each alternative describes a different term or result.

Distractor review: A: 'Mode of DNA replication' identifies Semiconservative, not Helicase. B: 'Enzyme that abolished transformation in the Avery experiment' identifies DNase, not Helicase. D: 'Scientist who first isolated nuclein' identifies Friedrich Miescher, not Helicase.

BIO-12-CH04-0022 | Hard | Genetic code and translation**Correct answer: B. Ribosomal site accepting an incoming aminoacyl-tRNA - A site; Approximate base pairs per turn of B-DNA - Ten**

Solution: Both parts of 'Ribosomal site accepting an incoming aminoacyl-tRNA - A site; Approximate base pairs per turn of B-DNA - Ten' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Inducer of the lac operon' requires Allolactose, and 'RNA forming a major structural and catalytic part of ribosomes' requires rRNA. C: The answers have been interchanged: 'Enzyme that abolished transformation in the Avery experiment' requires DNase, and 'DNA strand with a sequence corresponding to RNA except T for U' requires Coding strand. D: The answers have been interchanged: 'RNA bearing an anticodon and amino acid' requires tRNA, and 'Enzyme joining Okazaki fragments' requires DNA ligase.

BIO-12-CH04-0023 | Hard | Transcription and RNA**Correct answer: B. RNA bearing an anticodon and amino acid - DNA ligase**

Solution: 'RNA bearing an anticodon and amino acid' should be paired with tRNA, not DNA ligase. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Highly condensed transcriptionally inactive chromatin - Heterochromatin. C: This is a valid association: Hershey-Chase label used for phage DNA - Radioactive phosphorus-32. D: This is a valid association: Core proteins around which DNA is wrapped - Histone octamer.

BIO-12-CH04-0024 | Hard | DNA packaging**Correct answer: D. Highly condensed transcriptionally inactive chromatin - Degeneracy**

Solution: 'Highly condensed transcriptionally inactive chromatin' should be paired with Heterochromatin, not Degeneracy. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Technique identifying individuals from variable DNA regions - DNA fingerprinting. B: This is a valid association: Scientist who first isolated nuclein - Friedrich Miescher. C: This is a valid association: RNA bearing an anticodon and amino acid - tRNA.

BIO-12-CH04-0025 | Easy | Transcription and RNA**Correct answer: A. rRNA**

Solution: 'RNA forming a major structural and catalytic part of ribosomes' is the defining description for rRNA. The other choices belong to different chapter concepts.

Distractor review: B: Heterochromatin belongs with 'Highly condensed transcriptionally inactive chromatin', whereas the tested description is 'RNA forming a major structural and catalytic part of ribosomes'. C: H1 belongs with 'Linker histone', whereas the tested description is 'RNA forming a major structural and catalytic part of ribosomes'. D: tRNA belongs with 'RNA bearing an anticodon and amino acid', whereas the tested description is 'RNA forming a major structural and catalytic part of ribosomes'.

BIO-12-CH04-0026 | Hard | DNA structure**Correct answer: C. Approximate base pairs per turn of B-DNA - Ten; Scientist who first isolated nuclein - Friedrich Miescher**

Solution: Both parts of 'Approximate base pairs per turn of B-DNA - Ten; Scientist who first isolated nuclein - Friedrich Miescher' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Ribosomal site accepting an incoming aminoacyl-tRNA' requires A site, and 'Inducer of the lac operon' requires Allolactose. B: The answers have been interchanged: 'RNA forming a major structural and catalytic part of ribosomes' requires rRNA, and 'Ribosomal site accepting an incoming aminoacyl-tRNA' requires A site. D: The answers have been interchanged: 'Regulatory protein that blocks the operator' requires Repressor, and 'RNA forming a major structural and catalytic part of ribosomes' requires rRNA.

BIO-12-CH04-0027 | Medium | Transcription and RNA**Correct answer: B. mRNA**

Solution: The chapter links RNA carrying codons from DNA to ribosomes with mRNA, so mRNA completes the statement.

Distractor review: A: DNA belongs with 'Transforming principle demonstrated in pneumococcus', whereas the tested description is 'RNA carrying codons from DNA to ribosomes'. C: rRNA belongs with 'RNA forming a major structural and catalytic part of ribosomes', whereas the tested description is 'RNA carrying codons from DNA to ribosomes'. D: tRNA belongs with 'RNA bearing an anticodon and amino acid', whereas the tested description is 'RNA carrying codons from DNA to ribosomes'.

BIO-12-CH04-0028 | Medium | DNA replication**Correct answer: C. Enzyme joining Okazaki fragments - DNA ligase**

Solution: The accurate entry is 'Enzyme joining Okazaki fragments - DNA ligase'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Transforming principle demonstrated in pneumococcus' should be paired with DNA, not Ten. B: 'Approximate base pairs per turn of B-DNA' should be paired with Ten, not DNA ligase. D: 'Highly condensed transcriptionally inactive chromatin' should be paired with Heterochromatin, not DNA.

BIO-12-CH04-0029 | Easy | Gene regulation and genomics**Correct answer: B. Allolactose**

Solution: 'Inducer of the lac operon' is the defining description for Allolactose. The other choices belong to different chapter concepts.

Distractor review: A: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'Inducer of the lac operon'. C: Genome belongs with 'Complete DNA content of an organism', whereas the tested description is 'Inducer of the lac operon'. D: H1 belongs with 'Linker histone', whereas the tested description is 'Inducer of the lac operon'.

BIO-12-CH04-0030 | Hard | Genetic code and translation**Correct answer: C. A termination codon - Allolactose**

Solution: 'A termination codon' should be paired with UAA, not Allolactose. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Sugar present in DNA - Deoxyribose. B: This is a valid association: Highly condensed transcriptionally inactive chromatin - Heterochromatin. D: This is a valid association: Technique identifying individuals from variable DNA regions - DNA fingerprinting.

BIO-12-CH04-0031 | Hard | Genetic code and translation**Correct answer: A. Codon that usually initiates translation - AUG; DNA strand with a sequence corresponding to RNA except T for U - Coding strand**

Solution: Both parts of 'Codon that usually initiates translation - AUG; DNA strand with a sequence corresponding to RNA except T for U - Coding strand' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Scientist who first isolated nuclein' requires Friedrich Miescher, and 'Enzyme unwinding the DNA double helix' requires Helicase. C: The answers have been interchanged: 'RNA carrying codons from DNA to ribosomes' requires mRNA, and 'Number of hydrogen bonds in a G-C pair' requires Three. D: The answers have been interchanged: 'Enzyme unwinding the DNA double helix' requires Helicase, and 'RNA carrying codons from DNA to ribosomes' requires mRNA.

BIO-12-CH04-0032 | Challenging | DNA replication**Correct answer: D. I, II, III and IV**

Solution: Statement I is correct because 'Enzyme joining Okazaki fragments' corresponds to DNA ligase. Statement II is correct because 'Enzyme unwinding the DNA double helix' corresponds to Helicase. Statement III is correct because 'Short DNA segments formed on the lagging strand' corresponds to Okazaki fragments. Statement IV is correct because 'Property in which several codons specify one amino acid' corresponds to Degeneracy. Therefore, the valid combination is I, II, III and IV.

Distractor review: A: This combination omits true statement(s) III, IV. B: This combination omits true statement(s) II, IV. C: This combination omits true statement(s) II, III, IV.

BIO-12-CH04-0033 | Easy | Transcription and RNA**Correct answer: D. mRNA**

Solution: 'RNA carrying codons from DNA to ribosomes' is the defining description for mRNA. The other choices belong to different chapter concepts.

Distractor review: A: rRNA belongs with 'RNA forming a major structural and catalytic part of ribosomes', whereas the tested description is 'RNA carrying codons from DNA to ribosomes'. B: Okazaki fragments belongs with 'Short DNA segments formed on the lagging strand', whereas the tested description is 'RNA carrying codons from DNA to ribosomes'. C: UAA belongs with 'A termination codon', whereas the tested description is 'RNA carrying codons from DNA to ribosomes'.

BIO-12-CH04-0034 | Easy | Evidence for genetic material**Correct answer: A. Radioactive phosphorus-32**

Solution: 'Hershey-Chase label used for phage DNA' is the defining description for Radioactive phosphorus-32. The other choices belong to different chapter concepts.

Distractor review: B: Histone octamer belongs with 'Core proteins around which DNA is wrapped', whereas the tested description is 'Hershey-Chase label used for phage DNA'. C: Friedrich Miescher belongs with 'Scientist who first isolated nuclein', whereas the tested description is 'Hershey-Chase label used for phage DNA'. D: Ten belongs with 'Approximate base pairs per turn of B-DNA', whereas the tested description is 'Hershey-Chase label used for phage DNA'.

BIO-12-CH04-0035 | Easy | DNA replication**Correct answer: C. Short DNA segments formed on the lagging strand**

Solution: Okazaki fragments represents 'Short DNA segments formed on the lagging strand'. Each alternative describes a different term or result.

Distractor review: A: 'Sugar present in DNA' identifies Deoxyribose, not Okazaki fragments. B: 'A termination codon' identifies UAA, not Okazaki fragments. D: 'Basic repeating unit of chromatin' identifies Nucleosome, not Okazaki fragments.

BIO-12-CH04-0036 | Medium | DNA replication**Correct answer: B. Helicase**

Solution: The chapter links Enzyme unwinding the DNA double helix with Helicase, so Helicase completes the statement.

Distractor review: A: Three belongs with 'Number of hydrogen bonds in a G-C pair', whereas the tested description is 'Enzyme unwinding the DNA double helix'. C: DNase belongs with 'Enzyme that abolished transformation in the Avery experiment', whereas the tested description is 'Enzyme unwinding the DNA double helix'. D: Okazaki fragments belongs with 'Short DNA segments formed on the lagging strand', whereas the tested description is 'Enzyme unwinding the DNA double helix'.

BIO-12-CH04-0037 | Medium | Genetic code and translation**Correct answer: C. A site**

Solution: The chapter links Ribosomal site accepting an incoming aminoacyl-tRNA with A site, so A site completes the statement.

Distractor review: A: Ten belongs with 'Approximate base pairs per turn of B-DNA', whereas the tested description is 'Ribosomal site accepting an incoming aminoacyl-tRNA'. B: rRNA belongs with 'RNA forming a major structural and catalytic part of ribosomes', whereas the tested description is 'Ribosomal site accepting an incoming aminoacyl-tRNA'. D: DNA ligase belongs with 'Enzyme joining Okazaki fragments', whereas the tested description is 'Ribosomal site accepting an incoming aminoacyl-tRNA'.

BIO-12-CH04-0038 | Medium | Gene regulation and genomics**Correct answer: C. Complete DNA content of an organism - Genome**

Solution: The accurate entry is 'Complete DNA content of an organism - Genome'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Property in which several codons specify one amino acid' should be paired with Degeneracy, not tRNA. B: 'A termination codon' should be paired with UAA, not DNA ligase. D: 'Sugar present in DNA' should be paired with Deoxyribose, not UAA.

BIO-12-CH04-0039 | Medium | Genetic code and translation**Correct answer: B. UAA**

Solution: The chapter links A termination codon with UAA, so UAA completes the statement.

Distractor review: A: Deoxyribose belongs with 'Sugar present in DNA', whereas the tested description is 'A termination codon'. C: DNA fingerprinting belongs with 'Technique identifying individuals from variable DNA regions', whereas the tested description is 'A termination codon'. D: Allolactose belongs with 'Inducer of the lac operon', whereas the tested description is 'A termination codon'.

BIO-12-CH04-0040 | Medium | Genetic code and translation**Correct answer: B. A termination codon - UAA**

Solution: The accurate entry is 'A termination codon - UAA'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Inducer of the lac operon' should be paired with Allolactose, not Deoxyribose. C: 'Highly condensed transcriptionally inactive chromatin' should be paired with Heterochromatin, not Friedrich Miescher. D: 'Scientist who first isolated nuclein' should be paired with Friedrich Miescher, not UAA.

BIO-12-CH04-0041 | Challenging | DNA replication**Correct answer: C. I and IV only**

Solution: Statement I is correct because 'Short DNA segments formed on the lagging strand' corresponds to Okazaki fragments. Statement II is incorrect: 'Enzyme unwinding the DNA double helix' corresponds to Helicase, not DNA ligase. Statement III is incorrect: 'Enzyme joining Okazaki fragments' corresponds to DNA ligase, not Degeneracy. Statement IV is correct because 'Property in which several codons specify one amino acid' corresponds to Degeneracy. Therefore, the valid combination is I and IV only.

Distractor review: A: This combination includes false statement(s) II. B: This combination includes false statement(s) II and omits true statement(s) I, IV. D: This combination includes false statement(s) III.

BIO-12-CH04-0042 | Easy | DNA packaging**Correct answer: A. Basic repeating unit of chromatin**

Solution: Nucleosome represents 'Basic repeating unit of chromatin'. Each alternative describes a different term or result.

Distractor review: B: 'RNA carrying codons from DNA to ribosomes' identifies mRNA, not Nucleosome. C: 'Linker histone' identifies H1, not Nucleosome. D: 'Enzyme unwinding the DNA double helix' identifies Helicase, not Nucleosome.

BIO-12-CH04-0043 | Hard | DNA replication**Correct answer: D. Mode of DNA replication - Radioactive phosphorus-32**

Solution: 'Mode of DNA replication' should be paired with Semiconservative, not Radioactive phosphorus-32. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: RNA carrying codons from DNA to ribosomes - mRNA. B: This is a valid association: Enzyme that abolished transformation in the Avery experiment - DNase. C: This is a valid association: Property in which several codons specify one amino acid - Degeneracy.

BIO-12-CH04-0044 | Easy | DNA replication**Correct answer: D. Enzyme joining Okazaki fragments**

Solution: DNA ligase represents 'Enzyme joining Okazaki fragments'. Each alternative describes a different term or result.

Distractor review: A: 'Enzyme unwinding the DNA double helix' identifies Helicase, not DNA ligase. B: 'A termination codon' identifies UAA, not DNA ligase. C: 'Mode of DNA replication' identifies Semiconservative, not DNA ligase.

BIO-12-CH04-0045 | Hard | DNA packaging**Correct answer: C. Basic repeating unit of chromatin - Nucleosome; Property in which several codons specify one amino acid - Degeneracy**

Solution: Both parts of 'Basic repeating unit of chromatin - Nucleosome; Property in which several codons specify one amino acid - Degeneracy' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Transforming principle demonstrated in pneumococcus' requires DNA, and 'RNA carrying codons from DNA to ribosomes' requires mRNA. B: The answers have been interchanged: 'Enzyme unwinding the DNA double helix' requires Helicase, and 'Transforming principle demonstrated in pneumococcus' requires DNA. D: The answers have been interchanged: 'Inducer of the lac operon' requires Allolactose, and 'Number of hydrogen bonds in a G-C pair' requires Three.

BIO-12-CH04-0046 | Medium | Evidence for genetic material**Correct answer: B. Friedrich Miescher**

Solution: The observation 'Scientist who first isolated nuclein' points to Friedrich Miescher; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: DNA belongs with 'Transforming principle demonstrated in pneumococcus', whereas the tested description is 'Scientist who first isolated nuclein'. C: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'Scientist who first isolated nuclein'. D: tRNA belongs with 'RNA bearing an anticodon and amino acid', whereas the tested description is 'Scientist who first isolated nuclein'.

BIO-12-CH04-0047 | Medium | Evidence for genetic material**Correct answer: D. DNA**

Solution: The chapter links Transforming principle demonstrated in pneumococcus with DNA, so DNA completes the statement.

Distractor review: A: DNase belongs with 'Enzyme that abolished transformation in the Avery experiment', whereas the tested description is 'Transforming principle demonstrated in pneumococcus'. B: Okazaki fragments belongs with 'Short DNA segments formed on the lagging strand', whereas the tested description is 'Transforming principle demonstrated in pneumococcus'. C: Semiconservative belongs with 'Mode of DNA replication', whereas the tested description is 'Transforming principle demonstrated in pneumococcus'.

BIO-12-CH04-0048 | Medium | DNA structure**Correct answer: A. Number of hydrogen bonds in a G-C pair - Three**

Solution: The accurate entry is 'Number of hydrogen bonds in a G-C pair - Three'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'RNA carrying codons from DNA to ribosomes' should be paired with mRNA, not DNA. C: 'Transforming principle demonstrated in pneumococcus' should be paired with DNA, not DNase. D: 'Complete DNA content of an organism' should be paired with Genome, not mRNA.

BIO-12-CH04-0049 | Medium | Gene regulation and genomics**Correct answer: D. Genome**

Solution: The chapter links Complete DNA content of an organism with Genome, so Genome completes the statement.

Distractor review: A: Repressor belongs with 'Regulatory protein that blocks the operator', whereas the tested description is 'Complete DNA content of an organism'. B: DNA fingerprinting belongs with 'Technique identifying individuals from variable DNA regions', whereas the tested description is 'Complete DNA content of an organism'. C: UAA belongs with 'A termination codon', whereas the tested description is 'Complete DNA content of an organism'.

BIO-12-CH04-0050 | Hard | Genetic code and translation**Correct answer: D. Ribosomal site accepting an incoming aminoacyl-tRNA - Ten**

Solution: 'Ribosomal site accepting an incoming aminoacyl-tRNA' should be paired with A site, not Ten. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Enzyme joining Okazaki fragments - DNA ligase. B: This is a valid association: RNA forming a major structural and catalytic part of ribosomes - rRNA. C: This is a valid association: RNA bearing an anticodon and amino acid - tRNA.

BIO-12-CH04-0051 | Hard | Gene regulation and genomics**Correct answer: C. Technique identifying individuals from variable DNA regions - Friedrich Miescher**

Solution: 'Technique identifying individuals from variable DNA regions' should be paired with DNA fingerprinting, not Friedrich Miescher. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Enzyme joining Okazaki fragments - DNA ligase. B: This is a valid association: Approximate base pairs per turn of B-DNA - Ten. D: This is a valid association: Transforming principle demonstrated in pneumococcus - DNA.

BIO-12-CH04-0052 | Medium | DNA structure**Correct answer: D. Deoxyribose**

Solution: The chapter links Sugar present in DNA with Deoxyribose, so Deoxyribose completes the statement.

Distractor review: A: Genome belongs with 'Complete DNA content of an organism', whereas the tested description is 'Sugar present in DNA'. B: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'Sugar present in DNA'. C: rRNA belongs with 'RNA forming a major structural and catalytic part of ribosomes', whereas the tested description is 'Sugar present in DNA'.

BIO-12-CH04-0053 | Hard | DNA structure**Correct answer: A. Sugar present in DNA - Deoxyribose; Codon that usually initiates translation - AUG**

Solution: Both parts of 'Sugar present in DNA - Deoxyribose; Codon that usually initiates translation - AUG' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Mode of DNA replication' requires Semiconservative, and 'Enzyme that abolished transformation in the Avery experiment' requires DNase. C: The answers have been interchanged: 'RNA forming a major structural and catalytic part of ribosomes' requires rRNA, and 'Complete DNA content of an organism' requires Genome. D: The answers have been interchanged: 'RNA carrying codons from DNA to ribosomes' requires mRNA, and 'RNA forming a major structural and catalytic part of ribosomes' requires rRNA.

BIO-12-CH04-0054 | Medium | DNA packaging**Correct answer: B. Heterochromatin**

Solution: The chapter links Highly condensed transcriptionally inactive chromatin with Heterochromatin, so Heterochromatin completes the statement.

Distractor review: A: Degeneracy belongs with 'Property in which several codons specify one amino acid', whereas the tested description is 'Highly condensed transcriptionally inactive chromatin'. C: Friedrich Miescher belongs with 'Scientist who first isolated nuclein', whereas the tested description is 'Highly condensed transcriptionally inactive chromatin'. D: DNA fingerprinting belongs with 'Technique identifying individuals from variable DNA regions', whereas the tested description is 'Highly condensed transcriptionally inactive chromatin'.

BIO-12-CH04-0055 | Hard | Genetic code and translation**Correct answer: A. Property in which several codons specify one amino acid - Degeneracy; Regulatory protein that blocks the operator - Repressor**

Solution: Both parts of 'Property in which several codons specify one amino acid - Degeneracy; Regulatory protein that blocks the operator - Repressor' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Core proteins around which DNA is wrapped' requires Histone octamer, and 'Purine pairing with thymine' requires Adenine. C: The answers have been interchanged: 'Technique identifying individuals from variable DNA regions' requires DNA fingerprinting, and 'Enzyme unwinding the DNA double helix' requires Helicase. D: The answers have been interchanged: 'RNA bearing an anticodon and amino acid' requires tRNA, and 'Complete DNA content of an organism' requires Genome.

BIO-12-CH04-0056 | Easy | Evidence for genetic material**Correct answer: B. Enzyme that abolished transformation in the Avery experiment**

Solution: DNase represents 'Enzyme that abolished transformation in the Avery experiment'. Each alternative describes a different term or result.

Distractor review: A: 'RNA bearing an anticodon and amino acid' identifies tRNA, not DNase. C: 'Scientist who first isolated nuclein' identifies Friedrich Miescher, not DNase. D: 'Codon that usually initiates translation' identifies AUG, not DNase.

BIO-12-CH04-0057 | Medium | DNA structure**Correct answer: B. Deoxyribose**

Solution: The observation 'Sugar present in DNA' points to Deoxyribose; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Genome belongs with 'Complete DNA content of an organism', whereas the tested description is 'Sugar present in DNA'. C: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'Sugar present in DNA'. D: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'Sugar present in DNA'.

BIO-12-CH04-0058 | Medium | DNA packaging**Correct answer: D. H1**

Solution: The observation 'Linker histone' points to H1; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Ten belongs with 'Approximate base pairs per turn of B-DNA', whereas the tested description is 'Linker histone'. B: Allolactose belongs with 'Inducer of the lac operon', whereas the tested description is 'Linker histone'. C: Histone octamer belongs with 'Core proteins around which DNA is wrapped', whereas the tested description is 'Linker histone'.

BIO-12-CH04-0059 | Easy | Genetic code and translation**Correct answer: C. Codon that usually initiates translation**

Solution: AUG represents 'Codon that usually initiates translation'. Each alternative describes a different term or result.

Distractor review: A: 'Enzyme unwinding the DNA double helix' identifies Helicase, not AUG. B: 'RNA carrying codons from DNA to ribosomes' identifies mRNA, not AUG. D: 'Number of hydrogen bonds in a G-C pair' identifies Three, not AUG.

BIO-12-CH04-0060 | Easy | Gene regulation and genomics**Correct answer: B. Complete DNA content of an organism**

Solution: Genome represents 'Complete DNA content of an organism'. Each alternative describes a different term or result.

Distractor review: A: 'A termination codon' identifies UAA, not Genome. C: 'Technique identifying individuals from variable DNA regions' identifies DNA fingerprinting, not Genome. D: 'RNA bearing an anticodon and amino acid' identifies tRNA, not Genome.

BIO-12-CH04-0061 | Medium | DNA structure**Correct answer: D. Purine pairing with thymine - Adenine**

Solution: The accurate entry is 'Purine pairing with thymine - Adenine'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Ribosomal site accepting an incoming aminoacyl-tRNA' should be paired with A site, not Radioactive phosphorus-32. B: 'Hershey-Chase label used for phage DNA' should be paired with Radioactive phosphorus-32, not Adenine. C: 'Enzyme unwinding the DNA double helix' should be paired with Helicase, not A site.

BIO-12-CH04-0062 | Challenging | Evidence for genetic material**Correct answer: A. II, III and IV only**

Solution: Statement I is incorrect: 'Hershey-Chase label used for phage DNA' corresponds to Radioactive phosphorus-32, not DNA. Statement II is correct because 'Transforming principle demonstrated in pneumococcus' corresponds to DNA. Statement III is correct because 'Enzyme that abolished transformation in the Avery experiment' corresponds to DNase. Statement IV is correct because 'Ribosomal site accepting an incoming aminoacyl-tRNA' corresponds to A site. Therefore, the valid combination is II, III and IV only.

Distractor review: B: This combination includes false statement(s) I and omits true statement(s) IV. C: This combination omits true statement(s) II, IV. D: This combination includes false statement(s) I and omits true statement(s) II, III, IV.

BIO-12-CH04-0063 | Easy | Gene regulation and genomics**Correct answer: D. DNA fingerprinting**

Solution: 'Technique identifying individuals from variable DNA regions' is the defining description for DNA fingerprinting. The other choices belong to different chapter concepts.

Distractor review: A: Friedrich Miescher belongs with 'Scientist who first isolated nuclein', whereas the tested description is 'Technique identifying individuals from variable DNA regions'. B: Helicase belongs with 'Enzyme unwinding the DNA double helix', whereas the tested description is 'Technique identifying individuals from variable DNA regions'. C: DNA belongs with 'Transforming principle demonstrated in pneumococcus', whereas the tested description is 'Technique identifying individuals from variable DNA regions'.

BIO-12-CH04-0064 | Hard | Evidence for genetic material**Correct answer: A. Hershey-Chase label used for phage DNA - Radioactive phosphorus-32; RNA carrying codons from DNA to ribosomes - mRNA**

Solution: Both parts of 'Hershey-Chase label used for phage DNA - Radioactive phosphorus-32; RNA carrying codons from DNA to ribosomes - mRNA' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Purine pairing with thymine' requires Adenine, and 'Approximate base pairs per turn of B-DNA' requires Ten. C: The answers have been interchanged: 'Approximate base pairs per turn of B-DNA' requires Ten, and 'Core proteins around which DNA is wrapped' requires Histone octamer. D: The answers have been interchanged: 'Core proteins around which DNA is wrapped' requires Histone octamer, and 'Ribosomal site accepting an incoming aminoacyl-tRNA' requires A site.

BIO-12-CH04-0065 | Medium | Transcription and RNA**Correct answer: A. tRNA**

Solution: The chapter links RNA bearing an anticodon and amino acid with tRNA, so tRNA completes the statement.

Distractor review: B: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'RNA bearing an anticodon and amino acid'. C: Radioactive phosphorus-32 belongs with 'Hershey-Chase label used for phage DNA', whereas the tested description is 'RNA bearing an anticodon and amino acid'. D: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'RNA bearing an anticodon and amino acid'.

BIO-12-CH04-0066 | Challenging | Evidence for genetic material**Correct answer: B. I and III only**

Solution: Statement I is correct because 'Scientist who first isolated nuclein' corresponds to Friedrich Miescher. Statement II is incorrect: 'Transforming principle demonstrated in pneumococcus' corresponds to DNA, not DNase. Statement III is correct because 'Enzyme that abolished transformation in the Avery experiment' corresponds to DNase. Statement IV is incorrect: 'RNA carrying codons from DNA to ribosomes' corresponds to mRNA, not Friedrich Miescher. Therefore, the valid combination is I and III only.

Distractor review: A: This combination includes false statement(s) II. C: This combination includes false statement(s) II, IV and omits true statement(s) I, III. D: This combination includes false statement(s) II and omits true statement(s) I, III.

BIO-12-CH04-0067 | Hard | Gene regulation and genomics**Correct answer: B. Complete DNA content of an organism - Genome; Codon that usually initiates translation - AUG**

Solution: Both parts of 'Complete DNA content of an organism - Genome; Codon that usually initiates translation - AUG' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Enzyme joining Okazaki fragments' requires DNA ligase, and 'Scientist who first isolated nuclein' requires Friedrich Miescher. C: The answers have been interchanged: 'Sugar present in DNA' requires Deoxyribose, and 'Property in which several codons specify one amino acid' requires Degeneracy. D: The answers have been interchanged: 'Property in which several codons specify one amino acid' requires Degeneracy, and 'A termination codon' requires UAA.

BIO-12-CH04-0068 | Hard | Evidence for genetic material**Correct answer: D. Enzyme that abolished transformation in the Avery experiment - DNase; RNA bearing an anticodon and amino acid - tRNA**

Solution: Both parts of 'Enzyme that abolished transformation in the Avery experiment - DNase; RNA bearing an anticodon and amino acid - tRNA' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Basic repeating unit of chromatin' requires Nucleosome, and 'Approximate base pairs per turn of B-DNA' requires Ten. B: The answers have been interchanged: 'Approximate base pairs per turn of B-DNA' requires Ten, and 'Codon that usually initiates translation' requires AUG. C: The answers have been interchanged: 'Ribosomal site accepting an incoming aminoacyl-tRNA' requires A site, and 'Sugar present in DNA' requires Deoxyribose.

BIO-12-CH04-0069 | Medium | DNA replication**Correct answer: D. DNA ligase**

Solution: The chapter links Enzyme joining Okazaki fragments with DNA ligase, so DNA ligase completes the statement.

Distractor review: A: A site belongs with 'Ribosomal site accepting an incoming aminoacyl-tRNA', whereas the tested description is 'Enzyme joining Okazaki fragments'. B: Semiconservative belongs with 'Mode of DNA replication', whereas the tested description is 'Enzyme joining Okazaki fragments'. C: Heterochromatin belongs with 'Highly condensed transcriptionally inactive chromatin', whereas the tested description is 'Enzyme joining Okazaki fragments'.

BIO-12-CH04-0070 | Medium | Evidence for genetic material**Correct answer: B. Radioactive phosphorus-32**

Solution: The observation 'Hershey-Chase label used for phage DNA' points to Radioactive phosphorus-32; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: DNase belongs with 'Enzyme that abolished transformation in the Avery experiment', whereas the tested description is 'Hershey-Chase label used for phage DNA'. C: A site belongs with 'Ribosomal site accepting an incoming aminoacyl-tRNA', whereas the tested description is 'Hershey-Chase label used for phage DNA'. D: Ten belongs with 'Approximate base pairs per turn of B-DNA', whereas the tested description is 'Hershey-Chase label used for phage DNA'.

BIO-12-CH04-0071 | Medium | Gene regulation and genomics**Correct answer: B. DNA fingerprinting**

Solution: The chapter links Technique identifying individuals from variable DNA regions with DNA fingerprinting, so DNA fingerprinting completes the statement.

Distractor review: A: Helicase belongs with 'Enzyme unwinding the DNA double helix', whereas the tested description is 'Technique identifying individuals from variable DNA regions'. C: Ten belongs with 'Approximate base pairs per turn of B-DNA', whereas the tested description is 'Technique identifying individuals from variable DNA regions'. D: Allolactose belongs with 'Inducer of the lac operon', whereas the tested description is 'Technique identifying individuals from variable DNA regions'.

BIO-12-CH04-0072 | Easy | Genetic code and translation**Correct answer: C. A site**

Solution: 'Ribosomal site accepting an incoming aminoacyl-tRNA' is the defining description for A site. The other choices belong to different chapter concepts.

Distractor review: A: Allolactose belongs with 'Inducer of the lac operon', whereas the tested description is 'Ribosomal site accepting an incoming aminoacyl-tRNA'. B: DNA ligase belongs with 'Enzyme joining Okazaki fragments', whereas the tested description is 'Ribosomal site accepting an incoming aminoacyl-tRNA'. D: AUG belongs with 'Codon that usually initiates translation', whereas the tested description is 'Ribosomal site accepting an incoming aminoacyl-tRNA'.

BIO-12-CH04-0073 | Easy | DNA structure**Correct answer: D. Approximate base pairs per turn of B-DNA**

Solution: Ten represents 'Approximate base pairs per turn of B-DNA'. Each alternative describes a different term or result.

Distractor review: A: 'Scientist who first isolated nuclein' identifies Friedrich Miescher, not Ten. B: 'RNA forming a major structural and catalytic part of ribosomes' identifies rRNA, not Ten. C: 'Regulatory protein that blocks the operator' identifies Repressor, not Ten.

BIO-12-CH04-0074 | Medium | DNA packaging**Correct answer: D. Basic repeating unit of chromatin - Nucleosome**

Solution: The accurate entry is 'Basic repeating unit of chromatin - Nucleosome'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Enzyme that abolished transformation in the Avery experiment' should be paired with DNase, not DNA. B: 'Inducer of the lac operon' should be paired with Allolactose, not Nucleosome. C: 'Transforming principle demonstrated in pneumococcus' should be paired with DNA, not Allolactose.

BIO-12-CH04-0075 | Easy | Transcription and RNA**Correct answer: B. tRNA**

Solution: 'RNA bearing an anticodon and amino acid' is the defining description for tRNA. The other choices belong to different chapter concepts.

Distractor review: A: Heterochromatin belongs with 'Highly condensed transcriptionally inactive chromatin', whereas the tested description is 'RNA bearing an anticodon and amino acid'. C: DNA ligase belongs with 'Enzyme joining Okazaki fragments', whereas the tested description is 'RNA bearing an anticodon and amino acid'. D: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'RNA bearing an anticodon and amino acid'.

BIO-12-CH04-0076 | Easy | Genetic code and translation**Correct answer: C. UAA**

Solution: 'A termination codon' is the defining description for UAA. The other choices belong to different chapter concepts.

Distractor review: A: Genome belongs with 'Complete DNA content of an organism', whereas the tested description is 'A termination codon'. B: Deoxyribose belongs with 'Sugar present in DNA', whereas the tested description is 'A termination codon'. D: DNA fingerprinting belongs with 'Technique identifying individuals from variable DNA regions', whereas the tested description is 'A termination codon'.

BIO-12-CH04-0077 | Hard | DNA replication**Correct answer: C. Short DNA segments formed on the lagging strand - Okazaki fragments; Highly condensed transcriptionally inactive chromatin - Heterochromatin**

Solution: Both parts of 'Short DNA segments formed on the lagging strand - Okazaki fragments; Highly condensed transcriptionally inactive chromatin - Heterochromatin' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'RNA forming a major structural and catalytic part of ribosomes' requires rRNA, and 'Technique identifying individuals from variable DNA regions' requires DNA fingerprinting. B: The answers have been interchanged: 'Sugar present in DNA' requires Deoxyribose, and 'A termination codon' requires UAA. D: The answers have been interchanged: 'Technique identifying individuals from variable DNA regions' requires DNA fingerprinting, and 'Inducer of the lac operon' requires Allolactose.

BIO-12-CH04-0078 | Easy | DNA structure**Correct answer: D. Deoxyribose**

Solution: 'Sugar present in DNA' is the defining description for Deoxyribose. The other choices belong to different chapter concepts.

Distractor review: A: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'Sugar present in DNA'. B: AUG belongs with 'Codon that usually initiates translation', whereas the tested description is 'Sugar present in DNA'. C: Ten belongs with 'Approximate base pairs per turn of B-DNA', whereas the tested description is 'Sugar present in DNA'.

BIO-12-CH04-0079 | Easy | Evidence for genetic material**Correct answer: B. DNA**

Solution: 'Transforming principle demonstrated in pneumococcus' is the defining description for DNA. The other choices belong to different chapter concepts.

Distractor review: A: DNA fingerprinting belongs with 'Technique identifying individuals from variable DNA regions', whereas the tested description is 'Transforming principle demonstrated in pneumococcus'. C: Okazaki fragments belongs with 'Short DNA segments formed on the lagging strand', whereas the tested description is 'Transforming principle demonstrated in pneumococcus'. D: Semiconservative belongs with 'Mode of DNA replication', whereas the tested description is 'Transforming principle demonstrated in pneumococcus'.

BIO-12-CH04-0080 | Medium | DNA replication**Correct answer: C. Helicase**

Solution: The observation 'Enzyme unwinding the DNA double helix' points to Helicase; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Semiconservative belongs with 'Mode of DNA replication', whereas the tested description is 'Enzyme unwinding the DNA double helix'. B: Three belongs with 'Number of hydrogen bonds in a G-C pair', whereas the tested description is 'Enzyme unwinding the DNA double helix'. D: Heterochromatin belongs with 'Highly condensed transcriptionally inactive chromatin', whereas the tested description is 'Enzyme unwinding the DNA double helix'.

BIO-12-CH04-0081 | Medium | Evidence for genetic material**Correct answer: A. Radioactive phosphorus-32**

Solution: The chapter links Hershey-Chase label used for phage DNA with Radioactive phosphorus-32, so Radioactive phosphorus-32 completes the statement.

Distractor review: B: Ten belongs with 'Approximate base pairs per turn of B-DNA', whereas the tested description is 'Hershey-Chase label used for phage DNA'. C: DNase belongs with 'Enzyme that abolished transformation in the Avery experiment', whereas the tested description is 'Hershey-Chase label used for phage DNA'. D: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'Hershey-Chase label used for phage DNA'.

BIO-12-CH04-0082 | Hard | DNA replication**Correct answer: D. Enzyme joining Okazaki fragments - DNA ligase; Ribosomal site accepting an incoming aminoacyl-tRNA - A site**

Solution: Both parts of 'Enzyme joining Okazaki fragments - DNA ligase; Ribosomal site accepting an incoming aminoacyl-tRNA - A site' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Approximate base pairs per turn of B-DNA' requires Ten, and 'Technique identifying individuals from variable DNA regions' requires DNA fingerprinting. B: The answers have been interchanged: 'Transforming principle demonstrated in pneumococcus' requires DNA, and 'Enzyme that abolished transformation in the Avery experiment' requires DNase. C: The answers have been interchanged: 'Highly condensed transcriptionally inactive chromatin' requires Heterochromatin, and 'A termination codon' requires UAA.

BIO-12-CH04-0083 | Medium | Evidence for genetic material**Correct answer: D. DNase**

Solution: The observation 'Enzyme that abolished transformation in the Avery experiment' points to DNase; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: AUG belongs with 'Codon that usually initiates translation', whereas the tested description is 'Enzyme that abolished transformation in the Avery experiment'. B: Nucleosome belongs with 'Basic repeating unit of chromatin', whereas the tested description is 'Enzyme that abolished transformation in the Avery experiment'. C: Radioactive phosphorus-32 belongs with 'Hershey-Chase label used for phage DNA', whereas the tested description is 'Enzyme that abolished transformation in the Avery experiment'.

BIO-12-CH04-0084 | Easy | DNA replication**Correct answer: A. DNA ligase**

Solution: 'Enzyme joining Okazaki fragments' is the defining description for DNA ligase. The other choices belong to different chapter concepts.

Distractor review: B: Helicase belongs with 'Enzyme unwinding the DNA double helix', whereas the tested description is 'Enzyme joining Okazaki fragments'. C: Semiconservative belongs with 'Mode of DNA replication', whereas the tested description is 'Enzyme joining Okazaki fragments'. D: DNase belongs with 'Enzyme that abolished transformation in the Avery experiment', whereas the tested description is 'Enzyme joining Okazaki fragments'.

BIO-12-CH04-0085 | Hard | Gene regulation and genomics**Correct answer: C. Inducer of the lac operon - A site**

Solution: 'Inducer of the lac operon' should be paired with Allolactose, not A site. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Highly condensed transcriptionally inactive chromatin - Heterochromatin. B: This is a valid association: Scientist who first isolated nuclein - Friedrich Miescher. D: This is a valid association: Purine pairing with thymine - Adenine.

BIO-12-CH04-0086 | Challenging | DNA structure**Correct answer: D. I, III and IV only**

Solution: Statement I is correct because 'Number of hydrogen bonds in a G-C pair' corresponds to Three. Statement II is incorrect: 'Sugar present in DNA' corresponds to Deoxyribose, not Ten. Statement III is correct because 'Approximate base pairs per turn of B-DNA' corresponds to Ten. Statement IV is correct because 'Ribosomal site accepting an incoming aminoacyl-tRNA' corresponds to A site. Therefore, the valid combination is I, III and IV only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) I, IV. B: This combination includes false statement(s) II and omits true statement(s) I. C: This combination includes false statement(s) II and omits true statement(s) III, IV.

BIO-12-CH04-0087 | Medium | Gene regulation and genomics**Correct answer: B. Repressor**

Solution: The observation 'Regulatory protein that blocks the operator' points to Repressor; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'Regulatory protein that blocks the operator'. C: Nucleosome belongs with 'Basic repeating unit of chromatin', whereas the tested description is 'Regulatory protein that blocks the operator'. D: DNA fingerprinting belongs with 'Technique identifying individuals from variable DNA regions', whereas the tested description is 'Regulatory protein that blocks the operator'.

BIO-12-CH04-0088 | Hard | Transcription and RNA**Correct answer: A. RNA forming a major structural and catalytic part of ribosomes - Ten**

Solution: 'RNA forming a major structural and catalytic part of ribosomes' should be paired with rRNA, not Ten. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Linker histone - H1. C: This is a valid association: Purine pairing with thymine - Adenine. D: This is a valid association: Basic repeating unit of chromatin - Nucleosome.

BIO-12-CH04-0089 | Easy | DNA replication**Correct answer: C. Semiconservative**

Solution: 'Mode of DNA replication' is the defining description for Semiconservative. The other choices belong to different chapter concepts.

Distractor review: A: Helicase belongs with 'Enzyme unwinding the DNA double helix', whereas the tested description is 'Mode of DNA replication'. B: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'Mode of DNA replication'. D: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'Mode of DNA replication'.

BIO-12-CH04-0090 | Challenging | Transcription and RNA**Correct answer: B. I and II only**

Solution: Statement I is correct because 'RNA forming a major structural and catalytic part of ribosomes' corresponds to rRNA. Statement II is correct because 'DNA strand with a sequence corresponding to RNA except T for U' corresponds to Coding strand. Statement III is incorrect: 'RNA bearing an anticodon and amino acid' corresponds to tRNA, not DNA. Statement IV is incorrect: 'Transforming principle demonstrated in pneumococcus' corresponds to DNA, not rRNA. Therefore, the valid combination is I and II only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) II. C: This combination includes false statement(s) IV. D: This combination includes false statement(s) III.

BIO-12-CH04-0091 | Medium | Transcription and RNA**Correct answer: A. rRNA**

Solution: The chapter links RNA forming a major structural and catalytic part of ribosomes with rRNA, so rRNA completes the statement.

Distractor review: B: Ten belongs with 'Approximate base pairs per turn of B-DNA', whereas the tested description is 'RNA forming a major structural and catalytic part of ribosomes'. C: tRNA belongs with 'RNA bearing an anticodon and amino acid', whereas the tested description is 'RNA forming a major structural and catalytic part of ribosomes'. D: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'RNA forming a major structural and catalytic part of ribosomes'.

BIO-12-CH04-0092 | Easy | DNA packaging**Correct answer: B. Heterochromatin**

Solution: 'Highly condensed transcriptionally inactive chromatin' is the defining description for Heterochromatin. The other choices belong to different chapter concepts.

Distractor review: A: Friedrich Miescher belongs with 'Scientist who first isolated nuclein', whereas the tested description is 'Highly condensed transcriptionally inactive chromatin'. C: Nucleosome belongs with 'Basic repeating unit of chromatin', whereas the tested description is 'Highly condensed transcriptionally inactive chromatin'. D: Three belongs with 'Number of hydrogen bonds in a G-C pair', whereas the tested description is 'Highly condensed transcriptionally inactive chromatin'.

BIO-12-CH04-0093 | Medium | Evidence for genetic material**Correct answer: A. Enzyme that abolished transformation in the Avery experiment - DNase**

Solution: The accurate entry is 'Enzyme that abolished transformation in the Avery experiment - DNase'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'RNA bearing an anticodon and amino acid' should be paired with tRNA, not Nucleosome. C: 'Ribosomal site accepting an incoming aminoacyl-tRNA' should be paired with A site, not DNase. D: 'Regulatory protein that blocks the operator' should be paired with Repressor, not Ten.

BIO-12-CH04-0094 | Easy | Transcription and RNA**Correct answer: C. RNA bearing an anticodon and amino acid**

Solution: tRNA represents 'RNA bearing an anticodon and amino acid'. Each alternative describes a different term or result.

Distractor review: A: 'DNA strand with a sequence corresponding to RNA except T for U' identifies Coding strand, not tRNA. B: 'Highly condensed transcriptionally inactive chromatin' identifies Heterochromatin, not tRNA. D: 'Hershey-Chase label used for phage DNA' identifies Radioactive phosphorus-32, not tRNA.

BIO-12-CH04-0095 | Medium | DNA replication**Correct answer: C. Semiconservative**

Solution: The observation 'Mode of DNA replication' points to Semiconservative; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Okazaki fragments belongs with 'Short DNA segments formed on the lagging strand', whereas the tested description is 'Mode of DNA replication'. B: Radioactive phosphorus-32 belongs with 'Hershey-Chase label used for phage DNA', whereas the tested description is 'Mode of DNA replication'. D: DNase belongs with 'Enzyme that abolished transformation in the Avery experiment', whereas the tested description is 'Mode of DNA replication'.

BIO-12-CH04-0096 | Medium | Genetic code and translation**Correct answer: A. Degeneracy**

Solution: The chapter links Property in which several codons specify one amino acid with Degeneracy, so Degeneracy completes the statement.

Distractor review: B: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'Property in which several codons specify one amino acid'. C: tRNA belongs with 'RNA bearing an anticodon and amino acid', whereas the tested description is 'Property in which several codons specify one amino acid'. D: Repressor belongs with 'Regulatory protein that blocks the operator', whereas the tested description is 'Property in which several codons specify one amino acid'.

BIO-12-CH04-0097 | Medium | Genetic code and translation**Correct answer: C. Codon that usually initiates translation - AUG**

Solution: The accurate entry is 'Codon that usually initiates translation - AUG'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'DNA strand with a sequence corresponding to RNA except T for U' should be paired with Coding strand, not Helicase. B: 'RNA carrying codons from DNA to ribosomes' should be paired with mRNA, not tRNA. D: 'Number of hydrogen bonds in a G-C pair' should be paired with Three, not Allolactose.

BIO-12-CH04-0098 | Easy | Evidence for genetic material**Correct answer: D. Transforming principle demonstrated in pneumococcus**

Solution: DNA represents 'Transforming principle demonstrated in pneumococcus'. Each alternative describes a different term or result.

Distractor review: A: 'Highly condensed transcriptionally inactive chromatin' identifies Heterochromatin, not DNA. B: 'Short DNA segments formed on the lagging strand' identifies Okazaki fragments, not DNA. C: 'Inducer of the lac operon' identifies Allolactose, not DNA.

BIO-12-CH04-0099 | Easy | Gene regulation and genomics**Correct answer: B. Repressor**

Solution: 'Regulatory protein that blocks the operator' is the defining description for Repressor. The other choices belong to different chapter concepts.

Distractor review: A: DNA fingerprinting belongs with 'Technique identifying individuals from variable DNA regions', whereas the tested description is 'Regulatory protein that blocks the operator'. C: Radioactive phosphorus-32 belongs with 'Hershey-Chase label used for phage DNA', whereas the tested description is 'Regulatory protein that blocks the operator'. D: Adenine belongs with 'Purine pairing with thymine', whereas the tested description is 'Regulatory protein that blocks the operator'.

BIO-12-CH04-0100 | Medium | Genetic code and translation**Correct answer: C. AUG**

Solution: The observation 'Codon that usually initiates translation' points to AUG; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Helicase belongs with 'Enzyme unwinding the DNA double helix', whereas the tested description is 'Codon that usually initiates translation'. B: Friedrich Miescher belongs with 'Scientist who first isolated nuclein', whereas the tested description is 'Codon that usually initiates translation'. D: UAA belongs with 'A termination codon', whereas the tested description is 'Codon that usually initiates translation'.

BIO-12-CH04-0101 | Hard | DNA replication**Correct answer: D. Enzyme unwinding the DNA double helix - Helicase; Enzyme that abolished transformation in the Avery experiment - DNase**

Solution: Both parts of 'Enzyme unwinding the DNA double helix - Helicase; Enzyme that abolished transformation in the Avery experiment - DNase' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Inducer of the lac operon' requires Allolactose, and 'Purine pairing with thymine' requires Adenine. B: The answers have been interchanged: 'Scientist who first isolated nuclein' requires Friedrich Miescher, and 'Inducer of the lac operon' requires Allolactose. C: The answers have been interchanged: 'Number of hydrogen bonds in a G-C pair' requires Three, and 'Scientist who first isolated nuclein' requires Friedrich Miescher.

BIO-12-CH04-0102 | Medium | DNA packaging**Correct answer: B. Highly condensed transcriptionally inactive chromatin - Heterochromatin**

Solution: The accurate entry is 'Highly condensed transcriptionally inactive chromatin - Heterochromatin'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Technique identifying individuals from variable DNA regions' should be paired with DNA fingerprinting, not tRNA. C: 'DNA strand with a sequence corresponding to RNA except T for U' should be paired with Coding strand, not Heterochromatin. D: 'Scientist who first isolated nuclein' should be paired with Friedrich Miescher, not Three.

BIO-12-CH04-0103 | Medium | Transcription and RNA**Correct answer: A. RNA forming a major structural and catalytic part of ribosomes - rRNA**

Solution: The accurate entry is 'RNA forming a major structural and catalytic part of ribosomes - rRNA'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Highly condensed transcriptionally inactive chromatin' should be paired with Heterochromatin, not Nucleosome. C: 'Linker histone' should be paired with H1, not Adenine. D: 'Short DNA segments formed on the lagging strand' should be paired with Okazaki fragments, not rRNA.

BIO-12-CH04-0104 | Easy | Genetic code and translation**Correct answer: C. Property in which several codons specify one amino acid**

Solution: Degeneracy represents 'Property in which several codons specify one amino acid'. Each alternative describes a different term or result.

Distractor review: A: 'RNA bearing an anticodon and amino acid' identifies tRNA, not Degeneracy. B: 'A termination codon' identifies UAA, not Degeneracy. D: 'Core proteins around which DNA is wrapped' identifies Histone octamer, not Degeneracy.

BIO-12-CH04-0105 | Hard | DNA structure**Correct answer: B. Number of hydrogen bonds in a G-C pair - Three; Complete DNA content of an organism - Genome**

Solution: Both parts of 'Number of hydrogen bonds in a G-C pair - Three; Complete DNA content of an organism - Genome' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Enzyme joining Okazaki fragments' requires DNA ligase, and 'RNA carrying codons from DNA to ribosomes' requires mRNA. C: The answers have been interchanged: 'RNA carrying codons from DNA to ribosomes' requires mRNA, and 'Technique identifying individuals from variable DNA regions' requires DNA fingerprinting. D: The answers have been interchanged: 'Transforming principle demonstrated in pneumococcus' requires DNA, and 'Inducer of the lac operon' requires Allolactose.

BIO-12-CH04-0106 | Medium | DNA replication**Correct answer: A. Enzyme unwinding the DNA double helix - Helicase**

Solution: The accurate entry is 'Enzyme unwinding the DNA double helix - Helicase'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Inducer of the lac operon' should be paired with Allolactose, not Degeneracy. C: 'Number of hydrogen bonds in a G-C pair' should be paired with Three, not Allolactose. D: 'Purine pairing with thymine' should be paired with Adenine, not Helicase.

BIO-12-CH04-0107 | Hard | DNA structure**Correct answer: B. Approximate base pairs per turn of B-DNA - Friedrich Miescher**

Solution: 'Approximate base pairs per turn of B-DNA' should be paired with Ten, not Friedrich Miescher. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: RNA forming a major structural and catalytic part of ribosomes - rRNA. C: This is a valid association: RNA carrying codons from DNA to ribosomes - mRNA. D: This is a valid association: Regulatory protein that blocks the operator - Repressor.

BIO-12-CH04-0108 | Easy | DNA structure**Correct answer: B. Number of hydrogen bonds in a G-C pair**

Solution: Three represents 'Number of hydrogen bonds in a G-C pair'. Each alternative describes a different term or result.

Distractor review: A: 'Purine pairing with thymine' identifies Adenine, not Three. C: 'RNA carrying codons from DNA to ribosomes' identifies mRNA, not Three. D: 'Technique identifying individuals from variable DNA regions' identifies DNA fingerprinting, not Three.

BIO-12-CH04-0109 | Hard | Evidence for genetic material**Correct answer: C. Transforming principle demonstrated in pneumococcus - Heterochromatin**

Solution: 'Transforming principle demonstrated in pneumococcus' should be paired with DNA, not Heterochromatin. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Mode of DNA replication - Semiconservative. B: This is a valid association: Short DNA segments formed on the lagging strand - Okazaki fragments. D: This is a valid association: Inducer of the lac operon - Allolactose.

BIO-12-CH04-0110 | Medium | Gene regulation and genomics**Correct answer: D. Allolactose**

Solution: The observation 'Inducer of the lac operon' points to Allolactose; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: H1 belongs with 'Linker histone', whereas the tested description is 'Inducer of the lac operon'. B: Genome belongs with 'Complete DNA content of an organism', whereas the tested description is 'Inducer of the lac operon'. C: Repressor belongs with 'Regulatory protein that blocks the operator', whereas the tested description is 'Inducer of the lac operon'.

BIO-12-CH04-0111 | Easy | DNA structure**Correct answer: A. Sugar present in DNA**

Solution: Deoxyribose represents 'Sugar present in DNA'. Each alternative describes a different term or result.

Distractor review: B: 'Codon that usually initiates translation' identifies AUG, not Deoxyribose. C: 'Complete DNA content of an organism' identifies Genome, not Deoxyribose. D: 'Purine pairing with thymine' identifies Adenine, not Deoxyribose.

BIO-12-CH04-0112 | Medium | Transcription and RNA**Correct answer: C. tRNA**

Solution: The observation 'RNA bearing an anticodon and amino acid' points to tRNA; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Radioactive phosphorus-32 belongs with 'Hershey-Chase label used for phage DNA', whereas the tested description is 'RNA bearing an anticodon and amino acid'. B: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'RNA bearing an anticodon and amino acid'. D: Coding strand belongs with 'DNA strand with a sequence corresponding to RNA except T for U', whereas the tested description is 'RNA bearing an anticodon and amino acid'.

BIO-12-CH04-0113 | Challenging | Transcription and RNA**Correct answer: D. I, II and IV only**

Solution: Statement I is correct because 'DNA strand with a sequence corresponding to RNA except T for U' corresponds to Coding strand. Statement II is correct because 'RNA carrying codons from DNA to ribosomes' corresponds to mRNA. Statement III is incorrect: 'RNA bearing an anticodon and amino acid' corresponds to tRNA, not UAA. Statement IV is correct because 'A termination codon' corresponds to UAA. Therefore, the valid combination is I, II and IV only.

Distractor review: A: This combination omits true statement(s) I, II. B: This combination omits true statement(s) IV. C: This combination omits true statement(s) II, IV.

BIO-12-CH04-0114 | Challenging | Gene regulation and genomics**Correct answer: D. IV only**

Solution: Statement I is incorrect: 'Regulatory protein that blocks the operator' corresponds to Repressor, not Genome. Statement II is incorrect: 'Complete DNA content of an organism' corresponds to Genome, not Allolactose. Statement III is incorrect: 'Inducer of the lac operon' corresponds to Allolactose, not UAA. Statement IV is correct because 'A termination codon' corresponds to UAA. Therefore, the valid combination is IV only.

Distractor review: A: This combination includes false statement(s) III. B: This combination includes false statement(s) I, III and omits true statement(s) IV. C: This combination includes false statement(s) I and omits true statement(s) IV.

BIO-12-CH04-0115 | Hard | DNA packaging**Correct answer: B. Core proteins around which DNA is wrapped - Friedrich Miescher**

Solution: 'Core proteins around which DNA is wrapped' should be paired with Histone octamer, not Friedrich Miescher. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Property in which several codons specify one amino acid - Degeneracy. C: This is a valid association: Enzyme unwinding the DNA double helix - Helicase. D: This is a valid association: RNA bearing an anticodon and amino acid - tRNA.

BIO-12-CH04-0116 | Medium | Evidence for genetic material**Correct answer: A. Friedrich Miescher**

Solution: The chapter links Scientist who first isolated nuclein with Friedrich Miescher, so Friedrich Miescher completes the statement.

Distractor review: B: DNase belongs with 'Enzyme that abolished transformation in the Avery experiment', whereas the tested description is 'Scientist who first isolated nuclein'. C: DNA belongs with 'Transforming principle demonstrated in pneumococcus', whereas the tested description is 'Scientist who first isolated nuclein'. D: Allolactose belongs with 'Inducer of the lac operon', whereas the tested description is 'Scientist who first isolated nuclein'.

BIO-12-CH04-0117 | Medium | Gene regulation and genomics**Correct answer: A. DNA fingerprinting**

Solution: The observation 'Technique identifying individuals from variable DNA regions' points to DNA fingerprinting; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Friedrich Miescher belongs with 'Scientist who first isolated nuclein', whereas the tested description is 'Technique identifying individuals from variable DNA regions'. C: mRNA belongs with 'RNA carrying codons from DNA to ribosomes', whereas the tested description is 'Technique identifying individuals from variable DNA regions'. D: Helicase belongs with 'Enzyme unwinding the DNA double helix', whereas the tested description is 'Technique identifying individuals from variable DNA regions'.

BIO-12-CH04-0118 | Medium | Evidence for genetic material**Correct answer: C. Transforming principle demonstrated in pneumococcus - DNA**

Solution: The accurate entry is 'Transforming principle demonstrated in pneumococcus - DNA'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'DNA strand with a sequence corresponding to RNA except T for U' should be paired with Coding strand, not DNA. B: 'Short DNA segments formed on the lagging strand' should be paired with Okazaki fragments, not DNA fingerprinting. D: 'Inducer of the lac operon' should be paired with Allolactose, not Semiconservative.

BIO-12-CH04-0119 | Challenging | Genetic code and translation**Correct answer: A. I, II and III only**

Solution: Statement I is correct because 'Property in which several codons specify one amino acid' corresponds to Degeneracy. Statement II is correct because 'Codon that usually initiates translation' corresponds to AUG. Statement III is correct because 'A termination codon' corresponds to UAA. Statement IV is incorrect: 'Technique identifying individuals from variable DNA regions' corresponds to DNA fingerprinting, not Degeneracy. Therefore, the valid combination is I, II and III only.

Distractor review: B: This combination includes false statement(s) IV and omits true statement(s) II. C: This combination omits true statement(s) II, III. D: This combination includes false statement(s) IV and omits true statement(s) I.

BIO-12-CH04-0120 | Medium | Gene regulation and genomics**Correct answer: B. Regulatory protein that blocks the operator - Repressor**

Solution: The accurate entry is 'Regulatory protein that blocks the operator - Repressor'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'A termination codon' should be paired with UAA, not mRNA. C: 'Core proteins around which DNA is wrapped' should be paired with Histone octamer, not Repressor. D: 'Basic repeating unit of chromatin' should be paired with Nucleosome, not UAA.

BIO-12-CH04-0121 | Easy | DNA packaging**Correct answer: A. Histone octamer**

Solution: 'Core proteins around which DNA is wrapped' is the defining description for Histone octamer. The other choices belong to different chapter concepts.

Distractor review: B: tRNA belongs with 'RNA bearing an anticodon and amino acid', whereas the tested description is 'Core proteins around which DNA is wrapped'. C: Helicase belongs with 'Enzyme unwinding the DNA double helix', whereas the tested description is 'Core proteins around which DNA is wrapped'. D: Repressor belongs with 'Regulatory protein that blocks the operator', whereas the tested description is 'Core proteins around which DNA is wrapped'.

BIO-12-CH04-0122 | Medium | DNA packaging**Correct answer: A. Core proteins around which DNA is wrapped - Histone octamer**

Solution: The accurate entry is 'Core proteins around which DNA is wrapped - Histone octamer'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'RNA bearing an anticodon and amino acid' should be paired with tRNA, not Helicase. C: 'Regulatory protein that blocks the operator' should be paired with Repressor, not Radioactive phosphorus-32. D: 'Hershey-Chase label used for phage DNA' should be paired with Radioactive phosphorus-32, not Degeneracy.

BIO-12-CH04-0123 | Hard | Transcription and RNA**Correct answer: C. RNA bearing an anticodon and amino acid - tRNA; Enzyme joining Okazaki fragments - DNA ligase**

Solution: Both parts of 'RNA bearing an anticodon and amino acid - tRNA; Enzyme joining Okazaki fragments - DNA ligase' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Hershey-Chase label used for phage DNA' requires Radioactive phosphorus-32, and 'Purine pairing with thymine' requires Adenine. B: The answers have been interchanged: 'Core proteins around which DNA is wrapped' requires Histone octamer, and 'Ribosomal site accepting an incoming aminoacyl-tRNA' requires A site. D: The answers have been interchanged: 'Purine pairing with thymine' requires Adenine, and 'Technique identifying individuals from variable DNA regions' requires DNA fingerprinting.

BIO-12-CH04-0124 | Medium | DNA structure**Correct answer: C. Adenine**

Solution: The observation 'Purine pairing with thymine' points to Adenine; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: A site belongs with 'Ribosomal site accepting an incoming aminoacyl-tRNA', whereas the tested description is 'Purine pairing with thymine'. B: Nucleosome belongs with 'Basic repeating unit of chromatin', whereas the tested description is 'Purine pairing with thymine'. D: Allolactose belongs with 'Inducer of the lac operon', whereas the tested description is 'Purine pairing with thymine'.

BIO-12-CH04-0125 | Easy | DNA packaging**Correct answer: D. Nucleosome**

Solution: 'Basic repeating unit of chromatin' is the defining description for Nucleosome. The other choices belong to different chapter concepts.

Distractor review: A: DNase belongs with 'Enzyme that abolished transformation in the Avery experiment', whereas the tested description is 'Basic repeating unit of chromatin'. B: H1 belongs with 'Linker histone', whereas the tested description is 'Basic repeating unit of chromatin'. C: Histone octamer belongs with 'Core proteins around which DNA is wrapped', whereas the tested description is 'Basic repeating unit of chromatin'.

BIO-12-CH04-0126 | Easy | Genetic code and translation**Correct answer: C. Ribosomal site accepting an incoming aminoacyl-tRNA**

Solution: A site represents 'Ribosomal site accepting an incoming aminoacyl-tRNA'. Each alternative describes a different term or result.

Distractor review: A: 'Property in which several codons specify one amino acid' identifies Degeneracy, not A site. B: 'Enzyme joining Okazaki fragments' identifies DNA ligase, not A site. D: 'Codon that usually initiates translation' identifies AUG, not A site.

BIO-12-CH04-0127 | Hard | DNA packaging**Correct answer: D. Basic repeating unit of chromatin - Degeneracy**

Solution: 'Basic repeating unit of chromatin' should be paired with Nucleosome, not Degeneracy. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: RNA carrying codons from DNA to ribosomes - mRNA. B: This is a valid association: Enzyme that abolished transformation in the Avery experiment - DNase. C: This is a valid association: Transforming principle demonstrated in pneumococcus - DNA.

BIO-12-CH04-0128 | Medium | Transcription and RNA**Correct answer: A. DNA strand with a sequence corresponding to RNA except T for U - Coding strand**

Solution: The accurate entry is 'DNA strand with a sequence corresponding to RNA except T for U - Coding strand'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Codon that usually initiates translation' should be paired with AUG, not UAA. C: 'A termination codon' should be paired with UAA, not Radioactive phosphorus-32. D: 'Hershey-Chase label used for phage DNA' should be paired with Radioactive phosphorus-32, not Allolactose.

BIO-12-CH04-0129 | Easy | Evidence for genetic material**Correct answer: C. Hershey-Chase label used for phage DNA**

Solution: Radioactive phosphorus-32 represents 'Hershey-Chase label used for phage DNA'. Each alternative describes a different term or result.

Distractor review: A: 'Core proteins around which DNA is wrapped' identifies Histone octamer, not Radioactive phosphorus-32. B: 'Purine pairing with thymine' identifies Adenine, not Radioactive phosphorus-32. D: 'Ribosomal site accepting an incoming aminoacyl-tRNA' identifies A site, not Radioactive phosphorus-32.

BIO-12-CH04-0130 | Hard | Gene regulation and genomics**Correct answer: B. Complete DNA content of an organism - AUG**

Solution: 'Complete DNA content of an organism' should be paired with Genome, not AUG. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Enzyme joining Okazaki fragments - DNA ligase. C: This is a valid association: Property in which several codons specify one amino acid - Degeneracy. D: This is a valid association: Sugar present in DNA - Deoxyribose.

BIO-12-CH04-0131 | Easy | DNA structure**Correct answer: D. Ten**

Solution: 'Approximate base pairs per turn of B-DNA' is the defining description for Ten. The other choices belong to different chapter concepts.

Distractor review: A: Three belongs with 'Number of hydrogen bonds in a G-C pair', whereas the tested description is 'Approximate base pairs per turn of B-DNA'. B: Repressor belongs with 'Regulatory protein that blocks the operator', whereas the tested description is 'Approximate base pairs per turn of B-DNA'. C: Friedrich Miescher belongs with 'Scientist who first isolated nuclein', whereas the tested description is 'Approximate base pairs per turn of B-DNA'.

BIO-12-CH04-0132 | Challenging | DNA packaging**Correct answer: A. III only**

Solution: Statement I is incorrect: 'Highly condensed transcriptionally inactive chromatin' corresponds to Heterochromatin, not Histone octamer. Statement II is incorrect: 'Core proteins around which DNA is wrapped' corresponds to Histone octamer, not Nucleosome. Statement III is correct because 'Basic repeating unit of chromatin' corresponds to Nucleosome. Statement IV is incorrect: 'Hershey-Chase label used for phage DNA' corresponds to Radioactive phosphorus-32, not Heterochromatin. Therefore, the valid combination is III only.

Distractor review: B: This combination includes false statement(s) IV. C: This combination includes false statement(s) II, IV and omits true statement(s) III. D: This combination includes false statement(s) II.

Standard 12 | Chapter 05

Origin and Evolution of Life

BIO-12-CH05-0001 | Challenging | Hardy-Weinberg equilibrium

Correct answer: B. Absence of genetic drift in a large population

Solution: A bottleneck intensifies genetic drift, contradicting the large-population/no-drift assumption.

Distractor review: A: The alternative 'Allele frequencies must sum to one' conflicts with the labelled relationship, scale or mechanism in the figure. A bottleneck intensifies genetic drift, contradicting the large-population/no-drift assumption. C: The alternative 'Diploid organisms possess two alleles' conflicts with the labelled relationship, scale or mechanism in the figure. A bottleneck intensifies genetic drift, contradicting the large-population/no-drift assumption. D: The alternative 'Dominant alleles are always common' conflicts with the labelled relationship, scale or mechanism in the figure. A bottleneck intensifies genetic drift, contradicting the large-population/no-drift assumption.

BIO-12-CH05-0002 | Medium | Hardy-Weinberg equilibrium

Correct answer: D. $2pq = 0.42$

Solution: The heterozygote frequency is $2pq = 2 \times 0.7 \times 0.3 = 0.42$.

Distractor review: A: The alternative 'p squared = 0.49' conflicts with the labelled relationship, scale or mechanism in the figure. The heterozygote frequency is $2pq = 2 \times 0.7 \times 0.3 = 0.42$. B: The alternative 'q squared = 0.09' conflicts with the labelled relationship, scale or mechanism in the figure. The heterozygote frequency is $2pq = 2 \times 0.7 \times 0.3 = 0.42$. C: The alternative 'p + q = 0.21' conflicts with the labelled relationship, scale or mechanism in the figure. The heterozygote frequency is $2pq = 2 \times 0.7 \times 0.3 = 0.42$.

BIO-12-CH05-0003 | Medium | Evidence of evolution

Correct answer: B. Embryological evidence

Solution: The chapter links Developmental similarities among embryos with Embryological evidence, so Embryological evidence completes the statement.

Distractor review: A: Gene flow belongs with 'Movement of alleles between populations', whereas the tested description is 'Developmental similarities among embryos'. C: Genetic drift belongs with 'Random change in allele frequency, strongest in small populations', whereas the tested description is 'Developmental similarities among embryos'. D: Holocene belongs with 'Current geological epoch', whereas the tested description is 'Developmental similarities among embryos'.

BIO-12-CH05-0004 | Challenging | Geological history

Correct answer: D. I and II only

Solution: Statement I is correct because 'Era known for dominance of reptiles' corresponds to Mesozoic. Statement II is correct because 'Current geological epoch' corresponds to Holocene. Statement III is incorrect: 'Era in which mammals diversified extensively' corresponds to Cenozoic, not Inheritance of acquired characters. Statement IV is incorrect: 'Lamarckian mechanism' corresponds to Inheritance of acquired characters, not Mesozoic. Therefore, the valid combination is I and II only.

Distractor review: A: This combination includes false statement(s) III, IV. B: This combination includes false statement(s) IV and omits true statement(s) I. C: This combination omits true statement(s) I.

BIO-12-CH05-0005 | Medium | Origin of life

Correct answer: D. Miller-Urey experiment produced - Organic molecules including amino acids

Solution: The accurate entry is 'Miller-Urey experiment produced - Organic molecules including amino acids'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Modern synthetic theory combines selection with' should be paired with Population genetics, not Homo sapiens. B: 'Same basic structure with different functions' should be paired with Homologous organs, not Reproductive isolation. C: 'Hardy-Weinberg sum of allele frequencies' should be paired with $p + q = 1$, not Population genetics.

BIO-12-CH05-0006 | Medium | Human evolution

Correct answer: A. Australopithecus

Solution: The observation 'Genus of the famous fossil Lucy' points to Australopithecus; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Mutation belongs with 'De Vries emphasized', whereas the tested description is 'Genus of the famous fossil Lucy'. C: Holocene belongs with 'Current geological epoch', whereas the tested description is 'Genus of the famous fossil Lucy'. D: Adaptive radiation belongs with 'Formation of different species from a common ancestor in one region', whereas the tested description is 'Genus of the famous fossil Lucy'.

BIO-12-CH05-0007 | Challenging | Evidence of evolution**Correct answer: B. I, II and IV only**

Solution: Statement I is correct because 'Reduced structures with little current function' corresponds to Vestigial organs. Statement II is correct because 'Same basic structure with different functions' corresponds to Homologous organs. Statement III is incorrect: 'Different origin but similar function' corresponds to Analogous organs, not Biogenesis. Statement IV is correct because 'Theory proposing life came from pre-existing life' corresponds to Biogenesis. Therefore, the valid combination is I, II and IV only.

Distractor review: A: This combination omits true statement(s) I, IV. C: This combination omits true statement(s) I, II. D: This combination omits true statement(s) II, IV.

BIO-12-CH05-0008 | Medium | Geological history**Correct answer: A. Jurassic**

Solution: The observation 'Period associated with the first birds' points to Jurassic; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Cenozoic belongs with 'Era in which mammals diversified extensively', whereas the tested description is 'Period associated with the first birds'. C: Chemical evolution belongs with 'Oparin-Haldane proposal for origin of life', whereas the tested description is 'Period associated with the first birds'. D: Mutation belongs with 'De Vries emphasized', whereas the tested description is 'Period associated with the first birds'.

BIO-12-CH05-0009 | Easy | Three modes of natural selection**Correct answer: B. Stabilizing selection**

Solution: Stabilizing selection concentrates the distribution around the mean.

Distractor review: A: The alternative 'Directional selection' conflicts with the labelled relationship, scale or mechanism in the figure. Stabilizing selection concentrates the distribution around the mean. C: The alternative 'Disruptive selection' conflicts with the labelled relationship, scale or mechanism in the figure. Stabilizing selection concentrates the distribution around the mean. D: The alternative 'Artificial hybridization' conflicts with the labelled relationship, scale or mechanism in the figure. Stabilizing selection concentrates the distribution around the mean.

BIO-12-CH05-0010 | Medium | Speciation and adaptation**Correct answer: B. Barrier preventing gene flow between populations - Reproductive isolation**

Solution: The accurate entry is 'Barrier preventing gene flow between populations - Reproductive isolation'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Early human associated with systematic tool use' should be paired with Homo habilis, not Reproductive isolation. C: 'Random change in allele frequency, strongest in small populations' should be paired with Genetic drift, not Inheritance of acquired characters. D: 'Current geological epoch' should be paired with Holocene, not Genetic drift.

BIO-12-CH05-0011 | Hard | Three modes of natural selection**Correct answer: D. Disruptive selection**

Solution: Disruptive selection favours both extremes over intermediate phenotypes.

Distractor review: A: The alternative 'Stabilizing selection' conflicts with the labelled relationship, scale or mechanism in the figure. Disruptive selection favours both extremes over intermediate phenotypes. B: The alternative 'Directional selection' conflicts with the labelled relationship, scale or mechanism in the figure. Disruptive selection favours both extremes over intermediate phenotypes. C: The alternative 'No selection' conflicts with the labelled relationship, scale or mechanism in the figure. Disruptive selection favours both extremes over intermediate phenotypes.

BIO-12-CH05-0012 | Easy | Population genetics**Correct answer: D. Movement of alleles between populations**

Solution: Gene flow represents 'Movement of alleles between populations'. Each alternative describes a different term or result.

Distractor review: A: 'Random change in allele frequency, strongest in small populations' identifies Genetic drift, not Gene flow. B: 'Current geological epoch' identifies Holocene, not Gene flow. C: 'Same basic structure with different functions' identifies Homologous organs, not Gene flow.

BIO-12-CH05-0013 | Medium | Geological history**Correct answer: A. Holocene**

Solution: The observation 'Current geological epoch' points to Holocene; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Embryological evidence belongs with 'Developmental similarities among embryos', whereas the tested description is 'Current geological epoch'. C: Population genetics belongs with 'Modern synthetic theory combines selection with', whereas the tested description is 'Current geological epoch'. D: Cenozoic belongs with 'Era in which mammals diversified extensively', whereas the tested description is 'Current geological epoch'.

BIO-12-CH05-0014 | Challenging | Population genetics**Correct answer: D. IV only**

Solution: Statement I is incorrect: 'Hardy-Weinberg genotype equation' corresponds to $p^2 + 2pq + q^2 = 1$, not $p + q = 1$. Statement II is incorrect: 'Hardy-Weinberg sum of allele frequencies' corresponds to $p + q = 1$, not Gene flow. Statement III is incorrect: 'Movement of alleles between populations' corresponds to Gene flow, not Stabilizing selection. Statement IV is correct because 'Selection favouring the average phenotype' corresponds to Stabilizing selection. Therefore, the valid combination is IV only.

Distractor review: A: This combination includes false statement(s) I, II, III and omits true statement(s) IV. B: This combination includes false statement(s) I, II. C: This combination includes false statement(s) III and omits true statement(s) IV.

BIO-12-CH05-0015 | Easy | Origin of life**Correct answer: B. Early aggregates of organic molecules in water**

Solution: Coacervates represents 'Early aggregates of organic molecules in water'. Each alternative describes a different term or result.

Distractor review: A: 'Genus of the famous fossil Lucy' identifies Australopithecus, not Coacervates. C: 'Hardy-Weinberg genotype equation' identifies $p^2 + 2pq + q^2 = 1$, not Coacervates. D: 'Era in which mammals diversified extensively' identifies Cenozoic, not Coacervates.

BIO-12-CH05-0016 | Medium | Population genetics**Correct answer: C. Genetic drift**

Solution: The observation 'Random change in allele frequency, strongest in small populations' points to Genetic drift; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Embryological evidence belongs with 'Developmental similarities among embryos', whereas the tested description is 'Random change in allele frequency, strongest in small populations'. B: $p^2 + 2pq + q^2 = 1$ belongs with 'Hardy-Weinberg genotype equation', whereas the tested description is 'Random change in allele frequency, strongest in small populations'. D: Stabilizing selection belongs with 'Selection favouring the average phenotype', whereas the tested description is 'Random change in allele frequency, strongest in small populations'.

BIO-12-CH05-0017 | Hard | Geological history**Correct answer: A. Current geological epoch - Stabilizing selection**

Solution: 'Current geological epoch' should be paired with Holocene, not Stabilizing selection. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Random change in allele frequency, strongest in small populations - Genetic drift. C: This is a valid association: Modern synthetic theory combines selection with - Population genetics. D: This is a valid association: Reduced structures with little current function - Vestigial organs.

BIO-12-CH05-0018 | Medium | Evolutionary theories**Correct answer: C. Natural selection**

Solution: The chapter links Darwinian mechanism with Natural selection, so Natural selection completes the statement.

Distractor review: A: Population genetics belongs with 'Modern synthetic theory combines selection with', whereas the tested description is 'Darwinian mechanism'. B: Coacervates belongs with 'Early aggregates of organic molecules in water', whereas the tested description is 'Darwinian mechanism'. D: Inheritance of acquired characters belongs with 'Lamarckian mechanism', whereas the tested description is 'Darwinian mechanism'.

BIO-12-CH05-0019 | Medium | Human evolution**Correct answer: A. Early human associated with systematic tool use - Homo habilis**

Solution: The accurate entry is 'Early human associated with systematic tool use - Homo habilis'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Selection favouring the average phenotype' should be paired with Stabilizing selection, not Cenozoic. C: 'Era in which mammals diversified extensively' should be paired with Cenozoic, not Biogenesis. D: 'De Vries emphasized' should be paired with Mutation, not $p^2 + 2pq + q^2 = 1$.

BIO-12-CH05-0020 | Easy | Human evolution**Correct answer: B. Scientific name of modern humans**

Solution: Homo sapiens represents 'Scientific name of modern humans'. Each alternative describes a different term or result.

Distractor review: A: 'Genus of the famous fossil Lucy' identifies Australopithecus, not Homo sapiens. C: 'Same basic structure with different functions' identifies Homologous organs, not Homo sapiens. D: 'De Vries emphasized' identifies Mutation, not Homo sapiens.

BIO-12-CH05-0021 | Hard | Evidence of evolution**Correct answer: B. Reduced structures with little current function - Coacervates**

Solution: 'Reduced structures with little current function' should be paired with Vestigial organs, not Coacervates. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$. C: This is a valid association: Era in which mammals diversified extensively - Cenozoic. D: This is a valid association: De Vries emphasized - Mutation.

BIO-12-CH05-0022 | Medium | Population genetics**Correct answer: C. $p + q = 1$**

Solution: The chapter links Hardy-Weinberg sum of allele frequencies with $p + q = 1$, so $p + q = 1$ completes the statement.

Distractor review: A: Holocene belongs with 'Current geological epoch', whereas the tested description is 'Hardy-Weinberg sum of allele frequencies'. B: $p^2 + 2pq + q^2 = 1$ belongs with 'Hardy-Weinberg genotype equation', whereas the tested description is 'Hardy-Weinberg sum of allele frequencies'. D: Embryological evidence belongs with 'Developmental similarities among embryos', whereas the tested description is 'Hardy-Weinberg sum of allele frequencies'.

BIO-12-CH05-0023 | Hard | Speciation and adaptation**Correct answer: B. Selection favouring both extreme phenotypes - Disruptive selection; De Vries emphasized - Mutation**

Solution: Both parts of 'Selection favouring both extreme phenotypes - Disruptive selection; De Vries emphasized - Mutation' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Different origin but similar function' requires Analogous organs, and 'Theory proposing life came from pre-existing life' requires Biogenesis. C: The answers have been interchanged: 'Theory proposing life came from pre-existing life' requires Biogenesis, and 'Modern synthetic theory combines selection with' requires Population genetics. D: The answers have been interchanged: 'Random change in allele frequency, strongest in small populations' requires Genetic drift, and 'Different origin but similar function' requires Analogous organs.

BIO-12-CH05-0024 | Hard | Speciation and adaptation**Correct answer: B. Formation of different species from a common ancestor in one region - Adaptive radiation; Genus of the famous fossil Lucy - Australopithecus**

Solution: Both parts of 'Formation of different species from a common ancestor in one region - Adaptive radiation; Genus of the famous fossil Lucy - Australopithecus' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Era in which mammals diversified extensively' requires Cenozoic, and 'Developmental similarities among embryos' requires Embryological evidence. C: The answers have been interchanged: 'Reduced structures with little current function' requires Vestigial organs, and 'Theory proposing life came from pre-existing life' requires Biogenesis. D: The answers have been interchanged: 'Developmental similarities among embryos' requires Embryological evidence, and 'Oparin-Haldane proposal for origin of life' requires Chemical evolution.

BIO-12-CH05-0025 | Hard | Human evolution**Correct answer: C. Early human associated with systematic tool use - Population genetics**

Solution: 'Early human associated with systematic tool use' should be paired with Homo habilis, not Population genetics. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Miller-Urey experiment produced - Organic molecules including amino acids. B: This is a valid association: De Vries emphasized - Mutation. D: This is a valid association: Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$.

BIO-12-CH05-0026 | Easy | Population genetics**Correct answer: B. Hardy-Weinberg sum of allele frequencies**

Solution: $p + q = 1$ represents 'Hardy-Weinberg sum of allele frequencies'. Each alternative describes a different term or result.

Distractor review: A: 'Hardy-Weinberg genotype equation' identifies $p^2 + 2pq + q^2 = 1$, not $p + q = 1$. C: 'Random change in allele frequency, strongest in small populations' identifies Genetic drift, not $p + q = 1$. D: 'Different origin but similar function' identifies Analogous organs, not $p + q = 1$.

BIO-12-CH05-0027 | Medium | Human evolution**Correct answer: A. Homo sapiens**

Solution: The observation 'Scientific name of modern humans' points to Homo sapiens; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Population genetics belongs with 'Modern synthetic theory combines selection with', whereas the tested description is 'Scientific name of modern humans'. C: Australopithecus belongs with 'Genus of the famous fossil Lucy', whereas the tested description is 'Scientific name of modern humans'. D: Chemical evolution belongs with 'Oparin-Haldane proposal for origin of life', whereas the tested description is 'Scientific name of modern humans'.

BIO-12-CH05-0028 | Medium | Evidence of evolution**Correct answer: A. Reduced structures with little current function - Vestigial organs**

Solution: The accurate entry is 'Reduced structures with little current function - Vestigial organs'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Human species associated with Java man' should be paired with Homo erectus, not Cenozoic. C: 'Oparin-Haldane proposal for origin of life' should be paired with Chemical evolution, not Homo erectus. D: 'Hardy-Weinberg genotype equation' should be paired with $p^2 + 2pq + q^2 = 1$, not Gene flow.

BIO-12-CH05-0029 | Easy | Evolutionary theories**Correct answer: B. Population genetics**

Solution: 'Modern synthetic theory combines selection with' is the defining description for Population genetics. The other choices belong to different chapter concepts.

Distractor review: A: Chemical evolution belongs with 'Oparin-Haldane proposal for origin of life', whereas the tested description is 'Modern synthetic theory combines selection with'. C: Homo habilis belongs with 'Early human associated with systematic tool use', whereas the tested description is 'Modern synthetic theory combines selection with'. D: Natural selection belongs with 'Darwinian mechanism', whereas the tested description is 'Modern synthetic theory combines selection with'.

BIO-12-CH05-0030 | Medium | Geological history**Correct answer: C. Cenozoic**

Solution: The chapter links Era in which mammals diversified extensively with Cenozoic, so Cenozoic completes the statement.

Distractor review: A: Coacervates belongs with 'Early aggregates of organic molecules in water', whereas the tested description is 'Era in which mammals diversified extensively'. B: Jurassic belongs with 'Period associated with the first birds', whereas the tested description is 'Era in which mammals diversified extensively'. D: Homologous organs belongs with 'Same basic structure with different functions', whereas the tested description is 'Era in which mammals diversified extensively'.

BIO-12-CH05-0031 | Medium | Evolutionary theories**Correct answer: C. Natural selection**

Solution: The observation 'Darwinian mechanism' points to Natural selection; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Inheritance of acquired characters belongs with 'Lamarckian mechanism', whereas the tested description is 'Darwinian mechanism'. B: Coacervates belongs with 'Early aggregates of organic molecules in water', whereas the tested description is 'Darwinian mechanism'. D: Homo erectus belongs with 'Human species associated with Java man', whereas the tested description is 'Darwinian mechanism'.

BIO-12-CH05-0032 | Hard | Origin of life**Correct answer: B. Early aggregates of organic molecules in water - Cenozoic**

Solution: 'Early aggregates of organic molecules in water' should be paired with Coacervates, not Cenozoic. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Different origin but similar function - Analogous organs. C: This is a valid association: Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$. D: This is a valid association: Genus of the famous fossil Lucy - Australopithecus.

BIO-12-CH05-0033 | Hard | Human evolution**Correct answer: C. Human species associated with Java man - Homo erectus; Darwinian mechanism - Natural selection**

Solution: Both parts of 'Human species associated with Java man - Homo erectus; Darwinian mechanism - Natural selection' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Theory proposing life came from pre-existing life' requires Biogenesis, and 'Same basic structure with different functions' requires Homologous organs. B: The answers have been interchanged: 'Reduced structures with little current function' requires Vestigial organs, and 'Hardy-Weinberg sum of allele frequencies' requires $p + q = 1$. D: The answers have been interchanged: 'Same basic structure with different functions' requires Homologous organs, and 'Period associated with the first birds' requires Jurassic.

BIO-12-CH05-0034 | Easy | Evidence of evolution**Correct answer: C. Same basic structure with different functions**

Solution: Homologous organs represents 'Same basic structure with different functions'. Each alternative describes a different term or result.

Distractor review: A: 'Developmental similarities among embryos' identifies Embryological evidence, not Homologous organs. B: 'Random change in allele frequency, strongest in small populations' identifies Genetic drift, not Homologous organs. D: 'Selection favouring both extreme phenotypes' identifies Disruptive selection, not Homologous organs.

BIO-12-CH05-0035 | Medium | Evidence of evolution**Correct answer: A. Embryological evidence**

Solution: The observation 'Developmental similarities among embryos' points to Embryological evidence; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Vestigial organs belongs with 'Reduced structures with little current function', whereas the tested description is 'Developmental similarities among embryos'. C: Holocene belongs with 'Current geological epoch', whereas the tested description is 'Developmental similarities among embryos'. D: Genetic drift belongs with 'Random change in allele frequency, strongest in small populations', whereas the tested description is 'Developmental similarities among embryos'.

BIO-12-CH05-0036 | Easy | Population genetics**Correct answer: B. Gene flow**

Solution: 'Movement of alleles between populations' is the defining description for Gene flow. The other choices belong to different chapter concepts.

Distractor review: A: Coacervates belongs with 'Early aggregates of organic molecules in water', whereas the tested description is 'Movement of alleles between populations'. C: Holocene belongs with 'Current geological epoch', whereas the tested description is 'Movement of alleles between populations'. D: Genetic drift belongs with 'Random change in allele frequency, strongest in small populations', whereas the tested description is 'Movement of alleles between populations'.

BIO-12-CH05-0037 | Easy | Speciation and adaptation**Correct answer: C. Disruptive selection**

Solution: 'Selection favouring both extreme phenotypes' is the defining description for Disruptive selection. The other choices belong to different chapter concepts.

Distractor review: A: Genetic drift belongs with 'Random change in allele frequency, strongest in small populations', whereas the tested description is 'Selection favouring both extreme phenotypes'. B: Mutation belongs with 'De Vries emphasized', whereas the tested description is 'Selection favouring both extreme phenotypes'. D: Biogenesis belongs with 'Theory proposing life came from pre-existing life', whereas the tested description is 'Selection favouring both extreme phenotypes'.

BIO-12-CH05-0038 | Hard | Population genetics**Correct answer: C. Movement of alleles between populations - Gene flow; Oparin-Haldane proposal for origin of life - Chemical evolution**

Solution: Both parts of 'Movement of alleles between populations - Gene flow; Oparin-Haldane proposal for origin of life - Chemical evolution' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Genus of the famous fossil Lucy' requires Australopithecus, and 'Period associated with the first birds' requires Jurassic. B: The answers have been interchanged: 'Same basic structure with different functions' requires Homologous organs, and 'Current geological epoch' requires Holocene. D: The answers have been interchanged: 'Human species associated with Java man' requires Homo erectus, and 'Genus of the famous fossil Lucy' requires Australopithecus.

BIO-12-CH05-0039 | Hard | Evolutionary theories**Correct answer: D. Lamarckian mechanism - Australopithecus**

Solution: 'Lamarckian mechanism' should be paired with Inheritance of acquired characters, not Australopithecus. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Reduced structures with little current function - Vestigial organs. B: This is a valid association: Same basic structure with different functions - Homologous organs. C: This is a valid association: Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$.

BIO-12-CH05-0040 | Medium | Geological history**Correct answer: B. Current geological epoch - Holocene**

Solution: The accurate entry is 'Current geological epoch - Holocene'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Modern synthetic theory combines selection with' should be paired with Population genetics, not Vestigial organs. C: 'Reduced structures with little current function' should be paired with Vestigial organs, not Coacervates. D: 'Genus of the famous fossil Lucy' should be paired with Australopithecus, not Holocene.

BIO-12-CH05-0041 | Medium | Origin of life**Correct answer: D. Biogenesis**

Solution: The observation 'Theory proposing life came from pre-existing life' points to Biogenesis; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Inheritance of acquired characters belongs with 'Lamarckian mechanism', whereas the tested description is 'Theory proposing life came from pre-existing life'. B: Mutation belongs with 'De Vries emphasized', whereas the tested description is 'Theory proposing life came from pre-existing life'. C: Genetic drift belongs with 'Random change in allele frequency, strongest in small populations', whereas the tested description is 'Theory proposing life came from pre-existing life'.

BIO-12-CH05-0042 | Easy | Speciation and adaptation**Correct answer: A. Selection favouring both extreme phenotypes**

Solution: Disruptive selection represents 'Selection favouring both extreme phenotypes'. Each alternative describes a different term or result.

Distractor review: B: 'Random change in allele frequency, strongest in small populations' identifies Genetic drift, not Disruptive selection. C: 'De Vries emphasized' identifies Mutation, not Disruptive selection. D: 'Theory proposing life came from pre-existing life' identifies Biogenesis, not Disruptive selection.

BIO-12-CH05-0043 | Challenging | Human evolution**Correct answer: A. I, II and III only**

Solution: Statement I is correct because 'Early human associated with systematic tool use' corresponds to Homo habilis. Statement II is correct because 'Human species associated with Java man' corresponds to Homo erectus. Statement III is correct because 'Scientific name of modern humans' corresponds to Homo sapiens. Statement IV is incorrect: 'Different origin but similar function' corresponds to Analogous organs, not Homo habilis. Therefore, the valid combination is I, II and III only.

Distractor review: B: This combination includes false statement(s) IV. C: This combination omits true statement(s) I, II. D: This combination includes false statement(s) IV and omits true statement(s) III.

BIO-12-CH05-0044 | Easy | Geological history**Correct answer: D. Mesozoic**

Solution: 'Era known for dominance of reptiles' is the defining description for Mesozoic. The other choices belong to different chapter concepts.

Distractor review: A: Genetic drift belongs with 'Random change in allele frequency, strongest in small populations', whereas the tested description is 'Era known for dominance of reptiles'. B: Holocene belongs with 'Current geological epoch', whereas the tested description is 'Era known for dominance of reptiles'. C: Homo habilis belongs with 'Early human associated with systematic tool use', whereas the tested description is 'Era known for dominance of reptiles'.

BIO-12-CH05-0045 | Easy | Speciation and adaptation**Correct answer: C. Reproductive isolation**

Solution: 'Barrier preventing gene flow between populations' is the defining description for Reproductive isolation. The other choices belong to different chapter concepts.

Distractor review: A: Homologous organs belongs with 'Same basic structure with different functions', whereas the tested description is 'Barrier preventing gene flow between populations'. B: Holocene belongs with 'Current geological epoch', whereas the tested description is 'Barrier preventing gene flow between populations'. D: Inheritance of acquired characters belongs with 'Lamarckian mechanism', whereas the tested description is 'Barrier preventing gene flow between populations'.

BIO-12-CH05-0046 | Hard | Speciation and adaptation**Correct answer: A. Selection favouring both extreme phenotypes - Mutation**

Solution: 'Selection favouring both extreme phenotypes' should be paired with Disruptive selection, not Mutation. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Modern synthetic theory combines selection with - Population genetics. C: This is a valid association: Theory proposing life came from pre-existing life - Biogenesis. D: This is a valid association: Era known for dominance of reptiles - Mesozoic.

BIO-12-CH05-0047 | Challenging | Origin of life**Correct answer: D. III only**

Solution: Statement I is incorrect: 'Early aggregates of organic molecules in water' corresponds to Coacervates, not Chemical evolution. Statement II is incorrect: 'Oparin-Haldane proposal for origin of life' corresponds to Chemical evolution, not Biogenesis. Statement III is correct because 'Theory proposing life came from pre-existing life' corresponds to Biogenesis. Statement IV is incorrect: 'Darwinian mechanism' corresponds to Natural selection, not Coacervates. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) I. B: This combination includes false statement(s) I, II. C: This combination includes false statement(s) II and omits true statement(s) III.

BIO-12-CH05-0048 | Hard | Origin of life**Correct answer: C. Oparin-Haldane proposal for origin of life - Stabilizing selection**

Solution: 'Oparin-Haldane proposal for origin of life' should be paired with Chemical evolution, not Stabilizing selection. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Movement of alleles between populations - Gene flow. B: This is a valid association: Random change in allele frequency, strongest in small populations - Genetic drift. D: This is a valid association: Selection favouring both extreme phenotypes - Disruptive selection.

BIO-12-CH05-0049 | Medium | Population genetics**Correct answer: B. Random change in allele frequency, strongest in small populations - Genetic drift**

Solution: The accurate entry is 'Random change in allele frequency, strongest in small populations - Genetic drift'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Developmental similarities among embryos' should be paired with Embryological evidence, not Cenozoic. C: 'Era in which mammals diversified extensively' should be paired with Cenozoic, not Genetic drift. D: 'Selection favouring the average phenotype' should be paired with Stabilizing selection, not Organic molecules including amino acids.

BIO-12-CH05-0050 | Hard | Human evolution**Correct answer: B. Scientific name of modern humans - Homo sapiens; De Vries emphasized - Mutation**

Solution: Both parts of 'Scientific name of modern humans - Homo sapiens; De Vries emphasized - Mutation' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Reduced structures with little current function' requires Vestigial organs, and 'Theory proposing life came from pre-existing life' requires Biogenesis. C: The answers have been interchanged: 'Modern synthetic theory combines selection with' requires Population genetics, and 'Same basic structure with different functions' requires Homologous organs. D: The answers have been interchanged: 'Same basic structure with different functions' requires Homologous organs, and 'Reduced structures with little current function' requires Vestigial organs.

BIO-12-CH05-0051 | Medium | Origin of life**Correct answer: C. Organic molecules including amino acids**

Solution: The chapter links Miller-Urey experiment produced with Organic molecules including amino acids, so Organic molecules including amino acids completes the statement.

Distractor review: A: Coacervates belongs with 'Early aggregates of organic molecules in water', whereas the tested description is 'Miller-Urey experiment produced'. B: Population genetics belongs with 'Modern synthetic theory combines selection with', whereas the tested description is 'Miller-Urey experiment produced'. D: Chemical evolution belongs with 'Oparin-Haldane proposal for origin of life', whereas the tested description is 'Miller-Urey experiment produced'.

BIO-12-CH05-0052 | Challenging | Geological history**Correct answer: A. III and IV only**

Solution: Statement I is incorrect: 'Era in which mammals diversified extensively' corresponds to Cenozoic, not Mesozoic. Statement II is incorrect: 'Era known for dominance of reptiles' corresponds to Mesozoic, not Holocene. Statement III is correct because 'Current geological epoch' corresponds to Holocene. Statement IV is correct because 'Miller-Urey experiment produced' corresponds to Organic molecules including amino acids. Therefore, the valid combination is III and IV only.

Distractor review: B: This combination includes false statement(s) I, II and omits true statement(s) IV. C: This combination includes false statement(s) I, II. D: This combination includes false statement(s) I.

BIO-12-CH05-0053 | Medium | Speciation and adaptation**Correct answer: A. Stabilizing selection**

Solution: The chapter links Selection favouring the average phenotype with Stabilizing selection, so Stabilizing selection completes the statement.

Distractor review: B: $p^2 + 2pq + q^2 = 1$ belongs with 'Hardy-Weinberg genotype equation', whereas the tested description is 'Selection favouring the average phenotype'. C: Reproductive isolation belongs with 'Barrier preventing gene flow between populations', whereas the tested description is 'Selection favouring the average phenotype'. D: Population genetics belongs with 'Modern synthetic theory combines selection with', whereas the tested description is 'Selection favouring the average phenotype'.

BIO-12-CH05-0054 | Hard | Human evolution**Correct answer: D. Scientific name of modern humans - Mutation**

Solution: 'Scientific name of modern humans' should be paired with Homo sapiens, not Mutation. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Same basic structure with different functions - Homologous organs. B: This is a valid association: Oparin-Haldane proposal for origin of life - Chemical evolution. C: This is a valid association: Miller-Urey experiment produced - Organic molecules including amino acids.

BIO-12-CH05-0055 | Easy | Evidence of evolution**Correct answer: B. Analogous organs**

Solution: 'Different origin but similar function' is the defining description for Analogous organs. The other choices belong to different chapter concepts.

Distractor review: A: Homologous organs belongs with 'Same basic structure with different functions', whereas the tested description is 'Different origin but similar function'. C: Australopithecus belongs with 'Genus of the famous fossil Lucy', whereas the tested description is 'Different origin but similar function'. D: Gene flow belongs with 'Movement of alleles between populations', whereas the tested description is 'Different origin but similar function'.

BIO-12-CH05-0056 | Medium | Geological history**Correct answer: C. Holocene**

Solution: The chapter links Current geological epoch with Holocene, so Holocene completes the statement.

Distractor review: A: Inheritance of acquired characters belongs with 'Lamarckian mechanism', whereas the tested description is 'Current geological epoch'. B: Mesozoic belongs with 'Era known for dominance of reptiles', whereas the tested description is 'Current geological epoch'. D: Homo erectus belongs with 'Human species associated with Java man', whereas the tested description is 'Current geological epoch'.

BIO-12-CH05-0057 | Medium | Population genetics**Correct answer: D. Movement of alleles between populations - Gene flow**

Solution: The accurate entry is 'Movement of alleles between populations - Gene flow'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Same basic structure with different functions' should be paired with Homologous organs, not Coacervates. B: 'Early aggregates of organic molecules in water' should be paired with Coacervates, not Australopithecus. C: 'Human species associated with Java man' should be paired with Homo erectus, not Jurassic.

BIO-12-CH05-0058 | Easy | Population genetics**Correct answer: A. Genetic drift**

Solution: 'Random change in allele frequency, strongest in small populations' is the defining description for Genetic drift. The other choices belong to different chapter concepts.

Distractor review: B: Embryological evidence belongs with 'Developmental similarities among embryos', whereas the tested description is 'Random change in allele frequency, strongest in small populations'. C: Australopithecus belongs with 'Genus of the famous fossil Lucy', whereas the tested description is 'Random change in allele frequency, strongest in small populations'. D: $p + q = 1$ belongs with 'Hardy-Weinberg sum of allele frequencies', whereas the tested description is 'Random change in allele frequency, strongest in small populations'.

BIO-12-CH05-0059 | Medium | Speciation and adaptation**Correct answer: D. Stabilizing selection**

Solution: The observation 'Selection favouring the average phenotype' points to Stabilizing selection; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Jurassic belongs with 'Period associated with the first birds', whereas the tested description is 'Selection favouring the average phenotype'. B: Vestigial organs belongs with 'Reduced structures with little current function', whereas the tested description is 'Selection favouring the average phenotype'. C: Mutation belongs with 'De Vries emphasized', whereas the tested description is 'Selection favouring the average phenotype'.

BIO-12-CH05-0060 | Medium | Origin of life**Correct answer: C. Organic molecules including amino acids**

Solution: The observation 'Miller-Urey experiment produced' points to Organic molecules including amino acids; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Coacervates belongs with 'Early aggregates of organic molecules in water', whereas the tested description is 'Miller-Urey experiment produced'. B: $p + q = 1$ belongs with 'Hardy-Weinberg sum of allele frequencies', whereas the tested description is 'Miller-Urey experiment produced'. D: Reproductive isolation belongs with 'Barrier preventing gene flow between populations', whereas the tested description is 'Miller-Urey experiment produced'.

BIO-12-CH05-0061 | Hard | Evidence of evolution**Correct answer: D. Different origin but similar function - Adaptive radiation**

Solution: 'Different origin but similar function' should be paired with Analogous organs, not Adaptive radiation. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Movement of alleles between populations - Gene flow. B: This is a valid association: Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$. C: This is a valid association: Period associated with the first birds - Jurassic.

BIO-12-CH05-0062 | Medium | Evolutionary theories**Correct answer: D. Darwinian mechanism - Natural selection**

Solution: The accurate entry is 'Darwinian mechanism - Natural selection'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Selection favouring the average phenotype' should be paired with Stabilizing selection, not Jurassic. B: 'Early human associated with systematic tool use' should be paired with Homo habilis, not Natural selection. C: 'Different origin but similar function' should be paired with Analogous organs, not Homo habilis.

BIO-12-CH05-0063 | Medium | Human evolution**Correct answer: A. Homo habilis**

Solution: The chapter links Early human associated with systematic tool use with Homo habilis, so Homo habilis completes the statement.

Distractor review: B: Mutation belongs with 'De Vries emphasized', whereas the tested description is 'Early human associated with systematic tool use'. C: Australopithecus belongs with 'Genus of the famous fossil Lucy', whereas the tested description is 'Early human associated with systematic tool use'. D: Organic molecules including amino acids belongs with 'Miller-Urey experiment produced', whereas the tested description is 'Early human associated with systematic tool use'.

BIO-12-CH05-0064 | Easy | Geological history**Correct answer: A. Holocene**

Solution: 'Current geological epoch' is the defining description for Holocene. The other choices belong to different chapter concepts.

Distractor review: B: Embryological evidence belongs with 'Developmental similarities among embryos', whereas the tested description is 'Current geological epoch'. C: Genetic drift belongs with 'Random change in allele frequency, strongest in small populations', whereas the tested description is 'Current geological epoch'. D: Stabilizing selection belongs with 'Selection favouring the average phenotype', whereas the tested description is 'Current geological epoch'.

BIO-12-CH05-0065 | Medium | Evolutionary theories**Correct answer: D. Lamarckian mechanism - Inheritance of acquired characters**

Solution: The accurate entry is 'Lamarckian mechanism - Inheritance of acquired characters'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Genus of the famous fossil Lucy' should be paired with Australopithecus, not Holocene. B: 'Reduced structures with little current function' should be paired with Vestigial organs, not $p^2 + 2pq + q^2 = 1$. C: 'Same basic structure with different functions' should be paired with Homologous organs, not Inheritance of acquired characters.

BIO-12-CH05-0066 | Hard | Population genetics**Correct answer: C. Random change in allele frequency, strongest in small populations - Australopithecus**

Solution: 'Random change in allele frequency, strongest in small populations' should be paired with Genetic drift, not Australopithecus. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Developmental similarities among embryos - Embryological evidence. B: This is a valid association: Era in which mammals diversified extensively - Cenozoic. D: This is a valid association: Early aggregates of organic molecules in water - Coacervates.

BIO-12-CH05-0067 | Hard | Geological history**Correct answer: D. Era known for dominance of reptiles - Australopithecus**

Solution: 'Era known for dominance of reptiles' should be paired with Mesozoic, not Australopithecus. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Modern synthetic theory combines selection with - Population genetics. B: This is a valid association: Movement of alleles between populations - Gene flow. C: This is a valid association: Random change in allele frequency, strongest in small populations - Genetic drift.

BIO-12-CH05-0068 | Medium | Evolutionary theories**Correct answer: B. Inheritance of acquired characters**

Solution: The observation 'Lamarckian mechanism' points to Inheritance of acquired characters; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Australopithecus belongs with 'Genus of the famous fossil Lucy', whereas the tested description is 'Lamarckian mechanism'. C: Natural selection belongs with 'Darwinian mechanism', whereas the tested description is 'Lamarckian mechanism'. D: Adaptive radiation belongs with 'Formation of different species from a common ancestor in one region', whereas the tested description is 'Lamarckian mechanism'.

BIO-12-CH05-0069 | Medium | Evidence of evolution**Correct answer: B. Analogous organs**

Solution: The chapter links Different origin but similar function with Analogous organs, so Analogous organs completes the statement.

Distractor review: A: Jurassic belongs with 'Period associated with the first birds', whereas the tested description is 'Different origin but similar function'. C: Adaptive radiation belongs with 'Formation of different species from a common ancestor in one region', whereas the tested description is 'Different origin but similar function'. D: Vestigial organs belongs with 'Reduced structures with little current function', whereas the tested description is 'Different origin but similar function'.

BIO-12-CH05-0070 | Easy | Human evolution**Correct answer: C. Human species associated with Java man**

Solution: Homo erectus represents 'Human species associated with Java man'. Each alternative describes a different term or result.

Distractor review: A: 'Same basic structure with different functions' identifies Homologous organs, not Homo erectus. B: 'Reduced structures with little current function' identifies Vestigial organs, not Homo erectus. D: 'Genus of the famous fossil Lucy' identifies Australopithecus, not Homo erectus.

BIO-12-CH05-0071 | Easy | Evolutionary theories**Correct answer: A. Darwinian mechanism**

Solution: Natural selection represents 'Darwinian mechanism'. Each alternative describes a different term or result.

Distractor review: B: 'Selection favouring the average phenotype' identifies Stabilizing selection, not Natural selection. C: 'Different origin but similar function' identifies Analogous organs, not Natural selection. D: 'Modern synthetic theory combines selection with' identifies Population genetics, not Natural selection.

BIO-12-CH05-0072 | Easy | Evolutionary theories**Correct answer: B. Inheritance of acquired characters**

Solution: 'Lamarckian mechanism' is the defining description for Inheritance of acquired characters. The other choices belong to different chapter concepts.

Distractor review: A: $p^2 + 2pq + q^2 = 1$ belongs with 'Hardy-Weinberg genotype equation', whereas the tested description is 'Lamarckian mechanism'. C: Adaptive radiation belongs with 'Formation of different species from a common ancestor in one region', whereas the tested description is 'Lamarckian mechanism'. D: Holocene belongs with 'Current geological epoch', whereas the tested description is 'Lamarckian mechanism'.

BIO-12-CH05-0073 | Medium | Speciation and adaptation**Correct answer: D. Selection favouring the average phenotype - Stabilizing selection**

Solution: The accurate entry is 'Selection favouring the average phenotype - Stabilizing selection'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Period associated with the first birds' should be paired with Jurassic, not Mutation. B: 'Reduced structures with little current function' should be paired with Vestigial organs, not Genetic drift. C: 'De Vries emphasized' should be paired with Mutation, not Cenozoic.

BIO-12-CH05-0074 | Medium | Speciation and adaptation**Correct answer: D. Adaptive radiation**

Solution: The chapter links Formation of different species from a common ancestor in one region with Adaptive radiation, so Adaptive radiation completes the statement.

Distractor review: A: Biogenesis belongs with 'Theory proposing life came from pre-existing life', whereas the tested description is 'Formation of different species from a common ancestor in one region'. B: Stabilizing selection belongs with 'Selection favouring the average phenotype', whereas the tested description is 'Formation of different species from a common ancestor in one region'. C: Vestigial organs belongs with 'Reduced structures with little current function', whereas the tested description is 'Formation of different species from a common ancestor in one region'.

BIO-12-CH05-0075 | Medium | Evidence of evolution**Correct answer: A. Vestigial organs**

Solution: The observation 'Reduced structures with little current function' points to Vestigial organs; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: $p^2 + 2pq + q^2 = 1$ belongs with 'Hardy-Weinberg genotype equation', whereas the tested description is 'Reduced structures with little current function'. C: Homologous organs belongs with 'Same basic structure with different functions', whereas the tested description is 'Reduced structures with little current function'. D: Homo erectus belongs with 'Human species associated with Java man', whereas the tested description is 'Reduced structures with little current function'.

BIO-12-CH05-0076 | Medium | Population genetics**Correct answer: C. Hardy-Weinberg sum of allele frequencies - $p + q = 1$**

Solution: The accurate entry is 'Hardy-Weinberg sum of allele frequencies - $p + q = 1$ '. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Current geological epoch' should be paired with Holocene, not Analogous organs. B: 'Era in which mammals diversified extensively' should be paired with Cenozoic, not Embryological evidence. D: 'Human species associated with Java man' should be paired with Homo erectus, not Mutation.

BIO-12-CH05-0077 | Challenging | Origin of life**Correct answer: C. II only**

Solution: Statement I is incorrect: 'Miller-Urey experiment produced' corresponds to Organic molecules including amino acids, not Chemical evolution. Statement II is correct because 'Oparin-Haldane proposal for origin of life' corresponds to Chemical evolution. Statement III is incorrect: 'Theory proposing life came from pre-existing life' corresponds to Biogenesis, not Stabilizing selection. Statement IV is incorrect: 'Selection favouring the average phenotype' corresponds to Stabilizing selection, not Organic molecules including amino acids. Therefore, the valid combination is II only.

Distractor review: A: This combination includes false statement(s) III, IV and omits true statement(s) II. B: This combination includes false statement(s) III and omits true statement(s) II. D: This combination includes false statement(s) IV and omits true statement(s) II.

BIO-12-CH05-0078 | Medium | Evolutionary theories**Correct answer: A. Population genetics**

Solution: The chapter links Modern synthetic theory combines selection with with Population genetics, so Population genetics completes the statement.

Distractor review: B: Genetic drift belongs with 'Random change in allele frequency, strongest in small populations', whereas the tested description is 'Modern synthetic theory combines selection with'. C: Chemical evolution belongs with 'Oparin-Haldane proposal for origin of life', whereas the tested description is 'Modern synthetic theory combines selection with'. D: Natural selection belongs with 'Darwinian mechanism', whereas the tested description is 'Modern synthetic theory combines selection with'.

BIO-12-CH05-0079 | Medium | Human evolution**Correct answer: C. Australopithecus**

Solution: The chapter links Genus of the famous fossil Lucy with Australopithecus, so Australopithecus completes the statement.

Distractor review: A: Homo sapiens belongs with 'Scientific name of modern humans', whereas the tested description is 'Genus of the famous fossil Lucy'. B: Embryological evidence belongs with 'Developmental similarities among embryos', whereas the tested description is 'Genus of the famous fossil Lucy'. D: Mutation belongs with 'De Vries emphasized', whereas the tested description is 'Genus of the famous fossil Lucy'.

BIO-12-CH05-0080 | Easy | Human evolution**Correct answer: B. Homo habilis**

Solution: 'Early human associated with systematic tool use' is the defining description for Homo habilis. The other choices belong to different chapter concepts.

Distractor review: A: Stabilizing selection belongs with 'Selection favouring the average phenotype', whereas the tested description is 'Early human associated with systematic tool use'. C: Cenozoic belongs with 'Era in which mammals diversified extensively', whereas the tested description is 'Early human associated with systematic tool use'. D: Organic molecules including amino acids belongs with 'Miller-Urey experiment produced', whereas the tested description is 'Early human associated with systematic tool use'.

BIO-12-CH05-0081 | Hard | Evolutionary theories**Correct answer: A. Darwinian mechanism - Natural selection; Hardy-Weinberg sum of allele frequencies - $p + q = 1$**

Solution: Both parts of 'Darwinian mechanism - Natural selection; Hardy-Weinberg sum of allele frequencies - $p + q = 1$ ' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Different origin but similar function' requires Analogous organs, and 'Selection favouring the average phenotype' requires Stabilizing selection. C: The answers have been interchanged: 'Early human associated with systematic tool use' requires Homo habilis, and 'Period associated with the first birds' requires Jurassic. D: The answers have been interchanged: 'Early aggregates of organic molecules in water' requires Coacervates, and 'Different origin but similar function' requires Analogous organs.

BIO-12-CH05-0082 | Easy | Geological history**Correct answer: D. Era known for dominance of reptiles**

Solution: Mesozoic represents 'Era known for dominance of reptiles'. Each alternative describes a different term or result.

Distractor review: A: 'Early human associated with systematic tool use' identifies Homo habilis, not Mesozoic. B: 'Modern synthetic theory combines selection with' identifies Population genetics, not Mesozoic. C: 'Genus of the famous fossil Lucy' identifies Australopithecus, not Mesozoic.

BIO-12-CH05-0083 | Medium | Geological history**Correct answer: B. Jurassic**

Solution: The chapter links Period associated with the first birds with Jurassic, so Jurassic completes the statement.

Distractor review: A: Vestigial organs belongs with 'Reduced structures with little current function', whereas the tested description is 'Period associated with the first birds'. C: Mesozoic belongs with 'Era known for dominance of reptiles', whereas the tested description is 'Period associated with the first birds'. D: Population genetics belongs with 'Modern synthetic theory combines selection with', whereas the tested description is 'Period associated with the first birds'.

BIO-12-CH05-0084 | Hard | Speciation and adaptation**Correct answer: D. Barrier preventing gene flow between populations - Holocene**

Solution: 'Barrier preventing gene flow between populations' should be paired with Reproductive isolation, not Holocene. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Same basic structure with different functions - Homologous organs. B: This is a valid association: Miller-Urey experiment produced - Organic molecules including amino acids. C: This is a valid association: Random change in allele frequency, strongest in small populations - Genetic drift.

BIO-12-CH05-0085 | Easy | Origin of life**Correct answer: D. Chemical evolution**

Solution: 'Oparin-Haldane proposal for origin of life' is the defining description for Chemical evolution. The other choices belong to different chapter concepts.

Distractor review: A: Organic molecules including amino acids belongs with 'Miller-Urey experiment produced', whereas the tested description is 'Oparin-Haldane proposal for origin of life'. B: Stabilizing selection belongs with 'Selection favouring the average phenotype', whereas the tested description is 'Oparin-Haldane proposal for origin of life'. C: Biogenesis belongs with 'Theory proposing life came from pre-existing life', whereas the tested description is 'Oparin-Haldane proposal for origin of life'.

BIO-12-CH05-0086 | Hard | Geological history

Correct answer: A. Era known for dominance of reptiles - Mesozoic; Genus of the famous fossil Lucy - Australopithecus

Solution: Both parts of 'Era known for dominance of reptiles - Mesozoic; Genus of the famous fossil Lucy - Australopithecus' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Early human associated with systematic tool use' requires Homo habilis, and 'Movement of alleles between populations' requires Gene flow. C: The answers have been interchanged: 'Modern synthetic theory combines selection with' requires Population genetics, and 'Random change in allele frequency, strongest in small populations' requires Genetic drift. D: The answers have been interchanged: 'Random change in allele frequency, strongest in small populations' requires Genetic drift, and 'Early human associated with systematic tool use' requires Homo habilis.

BIO-12-CH05-0087 | Hard | Evidence of evolution

Correct answer: C. Same basic structure with different functions - Homologous organs; Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$

Solution: Both parts of 'Same basic structure with different functions - Homologous organs; Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$ ' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Random change in allele frequency, strongest in small populations' requires Genetic drift, and 'Movement of alleles between populations' requires Gene flow. B: The answers have been interchanged: 'De Vries emphasized' requires Mutation, and 'Era known for dominance of reptiles' requires Mesozoic. D: The answers have been interchanged: 'Darwinian mechanism' requires Natural selection, and 'De Vries emphasized' requires Mutation.

BIO-12-CH05-0088 | Hard | Origin of life

Correct answer: D. Miller-Urey experiment produced - $p + q = 1$

Solution: 'Miller-Urey experiment produced' should be paired with Organic molecules including amino acids, not $p + q = 1$. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Random change in allele frequency, strongest in small populations - Genetic drift. B: This is a valid association: Barrier preventing gene flow between populations - Reproductive isolation. C: This is a valid association: Modern synthetic theory combines selection with - Population genetics.

BIO-12-CH05-0089 | Easy | Evidence of evolution

Correct answer: D. Vestigial organs

Solution: 'Reduced structures with little current function' is the defining description for Vestigial organs. The other choices belong to different chapter concepts.

Distractor review: A: Homo erectus belongs with 'Human species associated with Java man', whereas the tested description is 'Reduced structures with little current function'. B: Chemical evolution belongs with 'Oparin-Haldane proposal for origin of life', whereas the tested description is 'Reduced structures with little current function'. C: $p^2 + 2pq + q^2 = 1$ belongs with 'Hardy-Weinberg genotype equation', whereas the tested description is 'Reduced structures with little current function'.

BIO-12-CH05-0090 | Easy | Origin of life

Correct answer: C. Coacervates

Solution: 'Early aggregates of organic molecules in water' is the defining description for Coacervates. The other choices belong to different chapter concepts.

Distractor review: A: Cenozoic belongs with 'Era in which mammals diversified extensively', whereas the tested description is 'Early aggregates of organic molecules in water'. B: Analogous organs belongs with 'Different origin but similar function', whereas the tested description is 'Early aggregates of organic molecules in water'. D: Biogenesis belongs with 'Theory proposing life came from pre-existing life', whereas the tested description is 'Early aggregates of organic molecules in water'.

BIO-12-CH05-0091 | Hard | Population genetics

Correct answer: B. Movement of alleles between populations - Chemical evolution

Solution: 'Movement of alleles between populations' should be paired with Gene flow, not Chemical evolution. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Same basic structure with different functions - Homologous organs. C: This is a valid association: Current geological epoch - Holocene. D: This is a valid association: Early aggregates of organic molecules in water - Coacervates.

BIO-12-CH05-0092 | Medium | Evidence of evolution**Correct answer: D. Analogous organs**

Solution: The observation 'Different origin but similar function' points to Analogous organs; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Jurassic belongs with 'Period associated with the first birds', whereas the tested description is 'Different origin but similar function'. B: Australopithecus belongs with 'Genus of the famous fossil Lucy', whereas the tested description is 'Different origin but similar function'. C: Homologous organs belongs with 'Same basic structure with different functions', whereas the tested description is 'Different origin but similar function'.

BIO-12-CH05-0093 | Medium | Speciation and adaptation**Correct answer: C. Adaptive radiation**

Solution: The observation 'Formation of different species from a common ancestor in one region' points to Adaptive radiation; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Biogenesis belongs with 'Theory proposing life came from pre-existing life', whereas the tested description is 'Formation of different species from a common ancestor in one region'. B: Disruptive selection belongs with 'Selection favouring both extreme phenotypes', whereas the tested description is 'Formation of different species from a common ancestor in one region'. D: Homo erectus belongs with 'Human species associated with Java man', whereas the tested description is 'Formation of different species from a common ancestor in one region'.

BIO-12-CH05-0094 | Hard | Origin of life**Correct answer: A. Early aggregates of organic molecules in water - Coacervates; Era in which mammals diversified extensively - Cenozoic**

Solution: Both parts of 'Early aggregates of organic molecules in water - Coacervates; Era in which mammals diversified extensively - Cenozoic' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Same basic structure with different functions' requires Homologous organs, and 'Hardy-Weinberg sum of allele frequencies' requires $p + q = 1$. C: The answers have been interchanged: 'Genus of the famous fossil Lucy' requires Australopithecus, and 'Same basic structure with different functions' requires Homologous organs. D: The answers have been interchanged: 'Hardy-Weinberg genotype equation' requires $p^2 + 2pq + q^2 = 1$, and 'Different origin but similar function' requires Analogous organs.

BIO-12-CH05-0095 | Challenging | Evolutionary theories**Correct answer: A. II and IV only**

Solution: Statement I is incorrect: 'Modern synthetic theory combines selection with' corresponds to Population genetics, not Natural selection. Statement II is correct because 'Darwinian mechanism' corresponds to Natural selection. Statement III is incorrect: 'De Vries emphasized' corresponds to Mutation, not Australopithecus. Statement IV is correct because 'Genus of the famous fossil Lucy' corresponds to Australopithecus. Therefore, the valid combination is II and IV only.

Distractor review: B: This combination omits true statement(s) II. C: This combination includes false statement(s) I and omits true statement(s) IV. D: This combination includes false statement(s) I, III and omits true statement(s) IV.

BIO-12-CH05-0096 | Easy | Origin of life**Correct answer: C. Organic molecules including amino acids**

Solution: 'Miller-Urey experiment produced' is the defining description for Organic molecules including amino acids. The other choices belong to different chapter concepts.

Distractor review: A: Population genetics belongs with 'Modern synthetic theory combines selection with', whereas the tested description is 'Miller-Urey experiment produced'. B: Homologous organs belongs with 'Same basic structure with different functions', whereas the tested description is 'Miller-Urey experiment produced'. D: Homo sapiens belongs with 'Scientific name of modern humans', whereas the tested description is 'Miller-Urey experiment produced'.

BIO-12-CH05-0097 | Hard | Population genetics**Correct answer: C. Hardy-Weinberg sum of allele frequencies - $p + q = 1$; Era in which mammals diversified extensively - Cenozoic**

Solution: Both parts of 'Hardy-Weinberg sum of allele frequencies - $p + q = 1$; Era in which mammals diversified extensively - Cenozoic' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Developmental similarities among embryos' requires Embryological evidence, and 'Different origin but similar function' requires Analogous organs. B: The answers have been interchanged: 'Different origin but similar function' requires Analogous organs, and 'Human species associated with Java man' requires Homo erectus. D: The answers have been interchanged: 'Current geological epoch' requires Holocene, and 'Developmental similarities among embryos' requires Embryological evidence.

BIO-12-CH05-0098 | Medium | Origin of life**Correct answer: B. Coacervates**

Solution: The chapter links Early aggregates of organic molecules in water with Coacervates, so Coacervates completes the statement.

Distractor review: A: Cenozoic belongs with 'Era in which mammals diversified extensively', whereas the tested description is 'Early aggregates of organic molecules in water'. C: Biogenesis belongs with 'Theory proposing life came from pre-existing life', whereas the tested description is 'Early aggregates of organic molecules in water'. D: $p^2 + 2pq + q^2 = 1$ belongs with 'Hardy-Weinberg genotype equation', whereas the tested description is 'Early aggregates of organic molecules in water'.

BIO-12-CH05-0099 | Challenging | Speciation and adaptation**Correct answer: A. I only**

Solution: Statement I is correct because 'Barrier preventing gene flow between populations' corresponds to Reproductive isolation. Statement II is incorrect: 'Selection favouring the average phenotype' corresponds to Stabilizing selection, not Adaptive radiation. Statement III is incorrect: 'Formation of different species from a common ancestor in one region' corresponds to Adaptive radiation, not Biogenesis. Statement IV is incorrect: 'Theory proposing life came from pre-existing life' corresponds to Biogenesis, not Reproductive isolation. Therefore, the valid combination is I only.

Distractor review: B: This combination includes false statement(s) II, III and omits true statement(s) I. C: This combination includes false statement(s) II, IV. D: This combination includes false statement(s) III, IV and omits true statement(s) I.

BIO-12-CH05-0100 | Easy | Speciation and adaptation**Correct answer: A. Formation of different species from a common ancestor in one region**

Solution: Adaptive radiation represents 'Formation of different species from a common ancestor in one region'. Each alternative describes a different term or result.

Distractor review: B: 'Reduced structures with little current function' identifies Vestigial organs, not Adaptive radiation. C: 'Genus of the famous fossil Lucy' identifies Australopithecus, not Adaptive radiation. D: 'Theory proposing life came from pre-existing life' identifies Biogenesis, not Adaptive radiation.

Standard 12 | Chapter 06

Plant Water Relation

BIO-12-CH06-0001 | Hard | Water-potential gradient through a plant

Correct answer: B. Reduced transpiration pull and reduced carbon-dioxide entry

Solution: Closure conserves water by reducing transpiration, but it also restricts carbon-dioxide diffusion into the leaf.

Distractor review: A: The alternative 'Increased transpiration and increased carbon-dioxide entry' conflicts with the labelled relationship, scale or mechanism in the figure. Closure conserves water by reducing transpiration, but it also restricts carbon-dioxide diffusion into the leaf. C: The alternative 'Reversal of xylem flow toward soil' conflicts with the labelled relationship, scale or mechanism in the figure. Closure conserves water by reducing transpiration, but it also restricts carbon-dioxide diffusion into the leaf. D: The alternative 'Permanent conversion of xylem into phloem' conflicts with the labelled relationship, scale or mechanism in the figure. Closure conserves water by reducing transpiration, but it also restricts carbon-dioxide diffusion into the leaf.

BIO-12-CH06-0002 | Medium | Guard-cell control of a stoma

Correct answer: A. Turgid guard cells

Solution: Solute accumulation draws in water, makes guard cells turgid and opens the pore.

Distractor review: B: The alternative 'Flaccid guard cells' conflicts with the labelled relationship, scale or mechanism in the figure. Solute accumulation draws in water, makes guard cells turgid and opens the pore. C: The alternative 'Dead guard cells' conflicts with the labelled relationship, scale or mechanism in the figure. Solute accumulation draws in water, makes guard cells turgid and opens the pore. D: The alternative 'Guard cells lacking water and potassium' conflicts with the labelled relationship, scale or mechanism in the figure. Solute accumulation draws in water, makes guard cells turgid and opens the pore.

BIO-12-CH06-0003 | Challenging | Guard-cell control of a stoma

Correct answer: B. Water exits guard cells, turgor falls and the pore closes

Solution: Ion loss raises guard-cell water potential, water exits, turgor decreases and the stomatal pore closes.

Distractor review: A: The alternative 'Water enters guard cells, turgor rises and the pore closes' conflicts with the labelled relationship, scale or mechanism in the figure. Ion loss raises guard-cell water potential, water exits, turgor decreases and the stomatal pore closes. C: The alternative 'Chlorophyll leaves guard cells and the pore opens' conflicts with the labelled relationship, scale or mechanism in the figure. Ion loss raises guard-cell water potential, water exits, turgor decreases and the stomatal pore closes. D: The alternative 'Xylem pressure rises and permanently locks the pore open' conflicts with the labelled relationship, scale or mechanism in the figure. Ion loss raises guard-cell water potential, water exits, turgor decreases and the stomatal pore closes.

BIO-12-CH06-0004 | Medium | Ascent of sap

Correct answer: D. Theory chiefly explaining ascent of sap - Cohesion-tension theory

Solution: The accurate entry is 'Theory chiefly explaining ascent of sap - Cohesion-tension theory'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Movement through connected cytoplasm via plasmodesmata' should be paired with Symplast pathway, not Cuticular transpiration. B: 'Principal water-absorbing region of a young root' should be paired with Root-hair zone, not Lenticular transpiration. C: 'Water loss directly through the cuticle' should be paired with Cuticular transpiration, not Adhesion.

BIO-12-CH06-0005 | Easy | Water-potential gradient through a plant

Correct answer: A. higher to lower water potential

Solution: Water moves down its water-potential gradient, from the less negative soil value to more negative tissues.

Distractor review: B: The alternative 'lower to higher water potential' conflicts with the labelled relationship, scale or mechanism in the figure. Water moves down its water-potential gradient, from the less negative soil value to more negative tissues. C: The alternative 'higher to lower solute concentration only' conflicts with the labelled relationship, scale or mechanism in the figure. Water moves down its water-potential gradient, from the less negative soil value to more negative tissues. D: The alternative 'xylem to soil against the gradient' conflicts with the labelled relationship, scale or mechanism in the figure. Water moves down its water-potential gradient, from the less negative soil value to more negative tissues.

BIO-12-CH06-0006 | Hard | Water potential and osmosis

Correct answer: C. Shrinkage of a protoplast in a hypertonic solution - Plasmolysis; Energy-dependent uptake of mineral ions - Active absorption

Solution: Both parts of 'Shrinkage of a protoplast in a hypertonic solution - Plasmolysis; Energy-dependent uptake of mineral ions - Active absorption' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Typical sink tissue' requires Developing fruit or storage organ, and 'Major route of transpiration' requires Stomata. B: The answers have been interchanged: 'Main tissue conducting water upward' requires Xylem, and 'Principal water-absorbing region of a young root' requires Root-hair zone. D: The answers have been interchanged: 'Movement through connected cytoplasm via plasmodesmata' requires Symplast pathway, and 'Force generated by evaporation from leaves' requires Transpiration pull.

BIO-12-CH06-0007 | Easy | Root absorption

Correct answer: A. Movement through connected cytoplasm via plasmodesmata

Solution: Symplast pathway represents 'Movement through connected cytoplasm via plasmodesmata'. Each alternative describes a different term or result.

Distractor review: B: 'Pressure-driven movement through sieve tubes' identifies Mass flow, not Symplast pathway. C: 'Barrier forcing water through living cells at the endodermis' identifies Casparian strip, not Symplast pathway. D: 'Net movement of water through a selectively permeable membrane' identifies Osmosis, not Symplast pathway.

BIO-12-CH06-0008 | Hard | Mineral and food transport

Correct answer: D. Energy-dependent uptake of mineral ions - Capillary water

Solution: 'Energy-dependent uptake of mineral ions' should be paired with Active absorption, not Capillary water. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Attraction between water molecules - Cohesion. B: This is a valid association: Guard-cell shape typical of grasses - Dumb-bell shaped. C: This is a valid association: Ion whose accumulation promotes guard-cell opening - Potassium.

BIO-12-CH06-0009 | Hard | Mineral and food transport

Correct answer: D. Typical sink tissue - Developing fruit or storage organ; Cells controlling the stomatal pore - Guard cells

Solution: Both parts of 'Typical sink tissue - Developing fruit or storage organ; Cells controlling the stomatal pore - Guard cells' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Main tissue conducting water upward' requires Xylem, and 'Soil water most readily available to plants' requires Capillary water. B: The answers have been interchanged: 'Passive movement through cell walls and intercellular spaces' requires Apoplast pathway, and 'Component of water potential produced by dissolved solute' requires Solute potential. C: The answers have been interchanged: 'Water potential of pure water at standard conditions' requires Zero, and 'Passive movement through cell walls and intercellular spaces' requires Apoplast pathway.

BIO-12-CH06-0010 | Medium | Mineral and food transport

Correct answer: A. Mass flow

Solution: The chapter links Pressure-driven movement through sieve tubes with Mass flow, so Mass flow completes the statement.

Distractor review: B: Developing fruit or storage organ belongs with 'Typical sink tissue', whereas the tested description is 'Pressure-driven movement through sieve tubes'. C: Mature photosynthetic leaf belongs with 'Source tissue in phloem transport', whereas the tested description is 'Pressure-driven movement through sieve tubes'. D: Cohesion-tension theory belongs with 'Theory chiefly explaining ascent of sap', whereas the tested description is 'Pressure-driven movement through sieve tubes'.

BIO-12-CH06-0011 | Easy | Water potential and osmosis

Correct answer: C. Shrinkage of a protoplast in a hypertonic solution

Solution: Plasmolysis represents 'Shrinkage of a protoplast in a hypertonic solution'. Each alternative describes a different term or result.

Distractor review: A: 'Energy-dependent uptake of mineral ions' identifies Active absorption, not Plasmolysis. B: 'Force generated by evaporation from leaves' identifies Transpiration pull, not Plasmolysis. D: 'Water potential of pure water at standard conditions' identifies Zero, not Plasmolysis.

BIO-12-CH06-0012 | Easy | Water potential and osmosis**Correct answer: B. Zero**

Solution: 'Water potential of pure water at standard conditions' is the defining description for Zero. The other choices belong to different chapter concepts.

Distractor review: A: Osmosis belongs with 'Net movement of water through a selectively permeable membrane', whereas the tested description is 'Water potential of pure water at standard conditions'. C: Plasmolysis belongs with 'Shrinkage of a protoplast in a hypertonic solution', whereas the tested description is 'Water potential of pure water at standard conditions'. D: Stomata belongs with 'Major route of transpiration', whereas the tested description is 'Water potential of pure water at standard conditions'.

BIO-12-CH06-0013 | Hard | Water potential and osmosis**Correct answer: D. Water potential of pure water at standard conditions - Stomata**

Solution: 'Water potential of pure water at standard conditions' should be paired with Zero, not Stomata. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Theory chiefly explaining ascent of sap - Cohesion-tension theory. B: This is a valid association: Positive pressure developed in roots - Root pressure. C: This is a valid association: Loss of water as vapour from aerial plant parts - Transpiration.

BIO-12-CH06-0014 | Hard | Root absorption**Correct answer: C. Passive movement through cell walls and intercellular spaces - Apoplast pathway; Guard-cell shape typical of dicots - Kidney-shaped**

Solution: Both parts of 'Passive movement through cell walls and intercellular spaces - Apoplast pathway; Guard-cell shape typical of dicots - Kidney-shaped' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Cells controlling the stomatal pore' requires Guard cells, and 'Attraction between water molecules' requires Cohesion. B: The answers have been interchanged: 'Positive pressure developed in roots' requires Root pressure, and 'Loss of water as vapour from aerial plant parts' requires Transpiration. D: The answers have been interchanged: 'Component of water potential produced by dissolved solute' requires Solute potential, and 'Cells controlling the stomatal pore' requires Guard cells.

BIO-12-CH06-0015 | Hard | Stomatal apparatus**Correct answer: D. Guard-cell shape typical of grasses - Capillary water**

Solution: 'Guard-cell shape typical of grasses' should be paired with Dumb-bell shaped, not Capillary water. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Passive movement through cell walls and intercellular spaces - Apoplast pathway. B: This is a valid association: Source tissue in phloem transport - Mature photosynthetic leaf. C: This is a valid association: Soil water held tightly to particle surfaces - Hygroscopic water.

BIO-12-CH06-0016 | Easy | Root absorption**Correct answer: B. Root-hair zone**

Solution: 'Principal water-absorbing region of a young root' is the defining description for Root-hair zone. The other choices belong to different chapter concepts.

Distractor review: A: Plasmolysis belongs with 'Shrinkage of a protoplast in a hypertonic solution', whereas the tested description is 'Principal water-absorbing region of a young root'. C: Active absorption belongs with 'Energy-dependent uptake of mineral ions', whereas the tested description is 'Principal water-absorbing region of a young root'. D: Symplast pathway belongs with 'Movement through connected cytoplasm via plasmodesmata', whereas the tested description is 'Principal water-absorbing region of a young root'.

BIO-12-CH06-0017 | Challenging | Transpiration**Correct answer: B. III only**

Solution: Statement I is incorrect: 'Water loss directly through the cuticle' corresponds to Cuticular transpiration, not Lenticular transpiration. Statement II is incorrect: 'Water loss through lenticels' corresponds to Lenticular transpiration, not Transpiration. Statement III is correct because 'Loss of water as vapour from aerial plant parts' corresponds to Transpiration. Statement IV is incorrect: 'Component of water potential produced by dissolved solute' corresponds to Solute potential, not Cuticular transpiration. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) II. C: This combination includes false statement(s) IV. D: This combination includes false statement(s) I, IV and omits true statement(s) III.

BIO-12-CH06-0018 | Hard | Root absorption**Correct answer: C. Principal water-absorbing region of a young root - Plasmolysis**

Solution: 'Principal water-absorbing region of a young root' should be paired with Root-hair zone, not Plasmolysis. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Cells controlling the stomatal pore - Guard cells. B: This is a valid association: Water loss directly through the cuticle - Cuticular transpiration. D: This is a valid association: Energy-dependent uptake of mineral ions - Active absorption.

BIO-12-CH06-0019 | Medium | Water potential and osmosis**Correct answer: A. Solute potential**

Solution: The chapter links Component of water potential produced by dissolved solute with Solute potential, so Solute potential completes the statement.

Distractor review: B: Hygroscopic water belongs with 'Soil water held tightly to particle surfaces', whereas the tested description is 'Component of water potential produced by dissolved solute'. C: Plasmolysis belongs with 'Shrinkage of a protoplast in a hypertonic solution', whereas the tested description is 'Component of water potential produced by dissolved solute'. D: Zero belongs with 'Water potential of pure water at standard conditions', whereas the tested description is 'Component of water potential produced by dissolved solute'.

BIO-12-CH06-0020 | Medium | Root absorption**Correct answer: D. Passive movement through cell walls and intercellular spaces - Apoplast pathway**

Solution: The accurate entry is 'Passive movement through cell walls and intercellular spaces - Apoplast pathway'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Component of water potential produced by dissolved solute' should be paired with Solute potential, not Cohesion. B: 'Loss of water as vapour from aerial plant parts' should be paired with Transpiration, not Guard cells. C: 'Attraction between water molecules' should be paired with Cohesion, not Apoplast pathway.

BIO-12-CH06-0021 | Hard | Root absorption**Correct answer: D. Movement through connected cytoplasm via plasmodesmata - Symplast pathway; Net movement of water through a selectively permeable membrane - Osmosis**

Solution: Both parts of 'Movement through connected cytoplasm via plasmodesmata - Symplast pathway; Net movement of water through a selectively permeable membrane - Osmosis' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Guard-cell shape typical of grasses' requires Dumb-bell shaped, and 'Water loss directly through the cuticle' requires Cuticular transpiration. B: The answers have been interchanged: 'Pressure-driven movement through sieve tubes' requires Mass flow, and 'Guard-cell shape typical of grasses' requires Dumb-bell shaped. C: The answers have been interchanged: 'Shrinkage of a protoplast in a hypertonic solution' requires Plasmolysis, and 'Attraction of water to a different surface' requires Adhesion.

BIO-12-CH06-0022 | Medium | Root absorption**Correct answer: C. Movement through connected cytoplasm via plasmodesmata - Symplast pathway**

Solution: The accurate entry is 'Movement through connected cytoplasm via plasmodesmata - Symplast pathway'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Source tissue in phloem transport' should be paired with Mature photosynthetic leaf, not Adhesion. B: 'Guard-cell shape typical of grasses' should be paired with Dumb-bell shaped, not Symplast pathway. D: 'Attraction of water to a different surface' should be paired with Adhesion, not Dumb-bell shaped.

BIO-12-CH06-0023 | Medium | Transpiration**Correct answer: B. Transpiration**

Solution: The chapter links Loss of water as vapour from aerial plant parts with Transpiration, so Transpiration completes the statement.

Distractor review: A: Stomata belongs with 'Major route of transpiration', whereas the tested description is 'Loss of water as vapour from aerial plant parts'. C: Plasmolysis belongs with 'Shrinkage of a protoplast in a hypertonic solution', whereas the tested description is 'Loss of water as vapour from aerial plant parts'. D: Osmosis belongs with 'Net movement of water through a selectively permeable membrane', whereas the tested description is 'Loss of water as vapour from aerial plant parts'.

BIO-12-CH06-0024 | Medium | Stomatal apparatus**Correct answer: D. Guard cells**

Solution: The chapter links Cells controlling the stomatal pore with Guard cells, so Guard cells completes the statement.

Distractor review: A: Developing fruit or storage organ belongs with 'Typical sink tissue', whereas the tested description is 'Cells controlling the stomatal pore'. B: Mature photosynthetic leaf belongs with 'Source tissue in phloem transport', whereas the tested description is 'Cells controlling the stomatal pore'. C: Kidney-shaped belongs with 'Guard-cell shape typical of dicots', whereas the tested description is 'Cells controlling the stomatal pore'.

BIO-12-CH06-0025 | Medium | Root absorption**Correct answer: B. Root-hair zone**

Solution: The chapter links Principal water-absorbing region of a young root with Root-hair zone, so Root-hair zone completes the statement.

Distractor review: A: Apoplast pathway belongs with 'Passive movement through cell walls and intercellular spaces', whereas the tested description is 'Principal water-absorbing region of a young root'. C: Mature photosynthetic leaf belongs with 'Source tissue in phloem transport', whereas the tested description is 'Principal water-absorbing region of a young root'. D: Symplast pathway belongs with 'Movement through connected cytoplasm via plasmodesmata', whereas the tested description is 'Principal water-absorbing region of a young root'.

BIO-12-CH06-0026 | Medium | Properties and forms of water**Correct answer: A. Attraction between water molecules - Cohesion**

Solution: The accurate entry is 'Attraction between water molecules - Cohesion'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Barrier forcing water through living cells at the endodermis' should be paired with Casparian strip, not Lenticular transpiration. C: 'Passive movement through cell walls and intercellular spaces' should be paired with Apoplast pathway, not Cohesion. D: 'Component of water potential produced by dissolved solute' should be paired with Solute potential, not Apoplast pathway.

BIO-12-CH06-0027 | Medium | Root absorption**Correct answer: C. Principal water-absorbing region of a young root - Root-hair zone**

Solution: The accurate entry is 'Principal water-absorbing region of a young root - Root-hair zone'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Energy-dependent uptake of mineral ions' should be paired with Active absorption, not Guard cells. B: 'Cells controlling the stomatal pore' should be paired with Guard cells, not Root-hair zone. D: 'Typical sink tissue' should be paired with Developing fruit or storage organ, not Mature photosynthetic leaf.

BIO-12-CH06-0028 | Challenging | Root absorption**Correct answer: A. III and IV only**

Solution: Statement I is incorrect: 'Passive movement through cell walls and intercellular spaces' corresponds to Apoplast pathway, not Casparian strip. Statement II is incorrect: 'Barrier forcing water through living cells at the endodermis' corresponds to Casparian strip, not Root-hair zone. Statement III is correct because 'Principal water-absorbing region of a young root' corresponds to Root-hair zone. Statement IV is correct because 'Component of water potential produced by dissolved solute' corresponds to Solute potential. Therefore, the valid combination is III and IV only.

Distractor review: B: This combination includes false statement(s) I. C: This combination includes false statement(s) I and omits true statement(s) III, IV. D: This combination includes false statement(s) I, II and omits true statement(s) III.

BIO-12-CH06-0029 | Medium | Stomatal apparatus**Correct answer: C. Ion whose accumulation promotes guard-cell opening - Potassium**

Solution: The accurate entry is 'Ion whose accumulation promotes guard-cell opening - Potassium'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Barrier forcing water through living cells at the endodermis' should be paired with Casparian strip, not Apoplast pathway. B: 'Water potential of pure water at standard conditions' should be paired with Zero, not Potassium. D: 'Passive movement through cell walls and intercellular spaces' should be paired with Apoplast pathway, not Zero.

BIO-12-CH06-0030 | Easy | Properties and forms of water**Correct answer: B. Attraction between water molecules**

Solution: Cohesion represents 'Attraction between water molecules'. Each alternative describes a different term or result.

Distractor review: A: 'Barrier forcing water through living cells at the endodermis' identifies Casparian strip, not Cohesion. C: 'Soil water most readily available to plants' identifies Capillary water, not Cohesion. D: 'Water loss through lenticels' identifies Lenticular transpiration, not Cohesion.

BIO-12-CH06-0031 | Hard | Ascent of sap**Correct answer: C. Positive pressure developed in roots - Root pressure; Attraction between water molecules - Cohesion**

Solution: Both parts of 'Positive pressure developed in roots - Root pressure; Attraction between water molecules - Cohesion' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Source tissue in phloem transport' requires Mature photosynthetic leaf, and 'Pressure-driven movement through sieve tubes' requires Mass flow. B: The answers have been interchanged: 'Water potential of pure water at standard conditions' requires Zero, and 'Attraction of water to a different surface' requires Adhesion. D: The answers have been interchanged: 'Attraction of water to a different surface' requires Adhesion, and 'Source tissue in phloem transport' requires Mature photosynthetic leaf.

BIO-12-CH06-0032 | Easy | Water potential and osmosis**Correct answer: C. Osmosis**

Solution: 'Net movement of water through a selectively permeable membrane' is the defining description for Osmosis. The other choices belong to different chapter concepts.

Distractor review: A: Solute potential belongs with 'Component of water potential produced by dissolved solute', whereas the tested description is 'Net movement of water through a selectively permeable membrane'. B: Dumb-bell shaped belongs with 'Guard-cell shape typical of grasses', whereas the tested description is 'Net movement of water through a selectively permeable membrane'. D: Symplast pathway belongs with 'Movement through connected cytoplasm via plasmodesmata', whereas the tested description is 'Net movement of water through a selectively permeable membrane'.

BIO-12-CH06-0033 | Medium | Mineral and food transport**Correct answer: B. Active absorption**

Solution: The chapter links Energy-dependent uptake of mineral ions with Active absorption, so Active absorption completes the statement.

Distractor review: A: Kidney-shaped belongs with 'Guard-cell shape typical of dicots', whereas the tested description is 'Energy-dependent uptake of mineral ions'. C: Potassium belongs with 'Ion whose accumulation promotes guard-cell opening', whereas the tested description is 'Energy-dependent uptake of mineral ions'. D: Capillary water belongs with 'Soil water most readily available to plants', whereas the tested description is 'Energy-dependent uptake of mineral ions'.

BIO-12-CH06-0034 | Hard | Stomatal apparatus**Correct answer: D. Guard-cell shape typical of grasses - Dumb-bell shaped; Soil water most readily available to plants - Capillary water**

Solution: Both parts of 'Guard-cell shape typical of grasses - Dumb-bell shaped; Soil water most readily available to plants - Capillary water' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Source tissue in phloem transport' requires Mature photosynthetic leaf, and 'Passive movement through cell walls and intercellular spaces' requires Apoplast pathway. B: The answers have been interchanged: 'Passive movement through cell walls and intercellular spaces' requires Apoplast pathway, and 'Positive pressure developed in roots' requires Root pressure. C: The answers have been interchanged: 'Soil water held tightly to particle surfaces' requires Hygroscopic water, and 'Force generated by evaporation from leaves' requires Transpiration pull.

BIO-12-CH06-0035 | Easy | Transpiration**Correct answer: A. Transpiration**

Solution: 'Loss of water as vapour from aerial plant parts' is the defining description for Transpiration. The other choices belong to different chapter concepts.

Distractor review: B: Hygroscopic water belongs with 'Soil water held tightly to particle surfaces', whereas the tested description is 'Loss of water as vapour from aerial plant parts'. C: Plasmolysis belongs with 'Shrinkage of a protoplast in a hypertonic solution', whereas the tested description is 'Loss of water as vapour from aerial plant parts'. D: Osmosis belongs with 'Net movement of water through a selectively permeable membrane', whereas the tested description is 'Loss of water as vapour from aerial plant parts'.

BIO-12-CH06-0036 | Medium | Water potential and osmosis**Correct answer: A. Osmosis**

Solution: The observation 'Net movement of water through a selectively permeable membrane' points to Osmosis; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Plasmolysis belongs with 'Shrinkage of a protoplast in a hypertonic solution', whereas the tested description is 'Net movement of water through a selectively permeable membrane'. C: Solute potential belongs with 'Component of water potential produced by dissolved solute', whereas the tested description is 'Net movement of water through a selectively permeable membrane'. D: Active absorption belongs with 'Energy-dependent uptake of mineral ions', whereas the tested description is 'Net movement of water through a selectively permeable membrane'.

BIO-12-CH06-0037 | Medium | Transpiration**Correct answer: B. Water loss through lenticels - Lenticular transpiration**

Solution: The accurate entry is 'Water loss through lenticels - Lenticular transpiration'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Theory chiefly explaining ascent of sap' should be paired with Cohesion-tension theory, not Guard cells. C: 'Cells controlling the stomatal pore' should be paired with Guard cells, not Capillary water. D: 'Principal water-absorbing region of a young root' should be paired with Root-hair zone, not Apoplast pathway.

BIO-12-CH06-0038 | Hard | Mineral and food transport**Correct answer: C. Pressure-driven movement through sieve tubes - Xylem**

Solution: 'Pressure-driven movement through sieve tubes' should be paired with Mass flow, not Xylem. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Barrier forcing water through living cells at the endodermis - Casparian strip. B: This is a valid association: Theory chiefly explaining ascent of sap - Cohesion-tension theory. D: This is a valid association: Component of water potential produced by dissolved solute - Solute potential.

BIO-12-CH06-0039 | Easy | Stomatal apparatus**Correct answer: B. Potassium**

Solution: 'Ion whose accumulation promotes guard-cell opening' is the defining description for Potassium. The other choices belong to different chapter concepts.

Distractor review: A: Root pressure belongs with 'Positive pressure developed in roots', whereas the tested description is 'Ion whose accumulation promotes guard-cell opening'. C: Guard cells belongs with 'Cells controlling the stomatal pore', whereas the tested description is 'Ion whose accumulation promotes guard-cell opening'. D: Dumb-bell shaped belongs with 'Guard-cell shape typical of grasses', whereas the tested description is 'Ion whose accumulation promotes guard-cell opening'.

BIO-12-CH06-0040 | Medium | Mineral and food transport**Correct answer: C. Developing fruit or storage organ**

Solution: The observation 'Typical sink tissue' points to Developing fruit or storage organ; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Zero belongs with 'Water potential of pure water at standard conditions', whereas the tested description is 'Typical sink tissue'. B: Guard cells belongs with 'Cells controlling the stomatal pore', whereas the tested description is 'Typical sink tissue'. D: Apoplast pathway belongs with 'Passive movement through cell walls and intercellular spaces', whereas the tested description is 'Typical sink tissue'.

BIO-12-CH06-0041 | Easy | Stomatal apparatus**Correct answer: C. Guard-cell shape typical of grasses**

Solution: Dumb-bell shaped represents 'Guard-cell shape typical of grasses'. Each alternative describes a different term or result.

Distractor review: A: 'Guard-cell shape typical of dicots' identifies Kidney-shaped, not Dumb-bell shaped. B: 'Typical sink tissue' identifies Developing fruit or storage organ, not Dumb-bell shaped. D: 'Ion whose accumulation promotes guard-cell opening' identifies Potassium, not Dumb-bell shaped.

BIO-12-CH06-0042 | Easy | Transpiration**Correct answer: D. Loss of water as vapour from aerial plant parts**

Solution: Transpiration represents 'Loss of water as vapour from aerial plant parts'. Each alternative describes a different term or result.

Distractor review: A: 'Soil water held tightly to particle surfaces' identifies Hygroscopic water, not Transpiration. B: 'Water loss through lenticels' identifies Lenticular transpiration, not Transpiration. C: 'Major route of transpiration' identifies Stomata, not Transpiration.

BIO-12-CH06-0043 | Challenging | Properties and forms of water**Correct answer: A. II only**

Solution: Statement I is incorrect: 'Soil water held tightly to particle surfaces' corresponds to Hygroscopic water, not Cohesion. Statement II is correct because 'Attraction between water molecules' corresponds to Cohesion. Statement III is incorrect: 'Attraction of water to a different surface' corresponds to Adhesion, not Casparian strip. Statement IV is incorrect: 'Barrier forcing water through living cells at the endodermis' corresponds to Casparian strip, not Hygroscopic water. Therefore, the valid combination is II only.

Distractor review: B: This combination includes false statement(s) I, III and omits true statement(s) II. C: This combination includes false statement(s) I and omits true statement(s) II. D: This combination includes false statement(s) III.

BIO-12-CH06-0044 | Hard | Transpiration**Correct answer: B. Loss of water as vapour from aerial plant parts - Casparian strip**

Solution: 'Loss of water as vapour from aerial plant parts' should be paired with Transpiration, not Casparian strip. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Movement through connected cytoplasm via plasmodesmata - Symplast pathway. C: This is a valid association: Soil water held tightly to particle surfaces - Hygroscopic water. D: This is a valid association: Net movement of water through a selectively permeable membrane - Osmosis.

BIO-12-CH06-0045 | Medium | Water potential and osmosis**Correct answer: C. Shrinkage of a protoplast in a hypertonic solution - Plasmolysis**

Solution: The accurate entry is 'Shrinkage of a protoplast in a hypertonic solution - Plasmolysis'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Movement through connected cytoplasm via plasmodesmata' should be paired with Symplast pathway, not Developing fruit or storage organ. B: 'Force generated by evaporation from leaves' should be paired with Transpiration pull, not Stomata. D: 'Main tissue conducting water upward' should be paired with Xylem, not Plasmolysis.

BIO-12-CH06-0046 | Medium | Transpiration**Correct answer: C. Water loss directly through the cuticle - Cuticular transpiration**

Solution: The accurate entry is 'Water loss directly through the cuticle - Cuticular transpiration'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Soil water most readily available to plants' should be paired with Capillary water, not Kidney-shaped. B: 'Soil water held tightly to particle surfaces' should be paired with Hygroscopic water, not Cuticular transpiration. D: 'Movement through connected cytoplasm via plasmodesmata' should be paired with Symplast pathway, not Xylem.

BIO-12-CH06-0047 | Hard | Stomatal apparatus**Correct answer: B. Ion whose accumulation promotes guard-cell opening - Symplast pathway**

Solution: 'Ion whose accumulation promotes guard-cell opening' should be paired with Potassium, not Symplast pathway. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Passive movement through cell walls and intercellular spaces - Apoplast pathway. C: This is a valid association: Barrier forcing water through living cells at the endodermis - Casparian strip. D: This is a valid association: Attraction of water to a different surface - Adhesion.

BIO-12-CH06-0048 | Medium | Properties and forms of water**Correct answer: D. Cohesion**

Solution: The observation 'Attraction between water molecules' points to Cohesion; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Capillary water belongs with 'Soil water most readily available to plants', whereas the tested description is 'Attraction between water molecules'. B: Lenticular transpiration belongs with 'Water loss through lenticels', whereas the tested description is 'Attraction between water molecules'. C: Adhesion belongs with 'Attraction of water to a different surface', whereas the tested description is 'Attraction between water molecules'.

BIO-12-CH06-0049 | Easy | Stomatal apparatus**Correct answer: A. Dumb-bell shaped**

Solution: 'Guard-cell shape typical of grasses' is the defining description for Dumb-bell shaped. The other choices belong to different chapter concepts.

Distractor review: B: Kidney-shaped belongs with 'Guard-cell shape typical of dicots', whereas the tested description is 'Guard-cell shape typical of grasses'. C: Mature photosynthetic leaf belongs with 'Source tissue in phloem transport', whereas the tested description is 'Guard-cell shape typical of grasses'. D: Potassium belongs with 'Ion whose accumulation promotes guard-cell opening', whereas the tested description is 'Guard-cell shape typical of grasses'.

BIO-12-CH06-0050 | Easy | Mineral and food transport**Correct answer: A. Typical sink tissue**

Solution: Developing fruit or storage organ represents 'Typical sink tissue'. Each alternative describes a different term or result.

Distractor review: B: 'Water potential of pure water at standard conditions' identifies Zero, not Developing fruit or storage organ. C: 'Source tissue in phloem transport' identifies Mature photosynthetic leaf, not Developing fruit or storage organ. D: 'Passive movement through cell walls and intercellular spaces' identifies Apoplast pathway, not Developing fruit or storage organ.

BIO-12-CH06-0051 | Challenging | Ascent of sap**Correct answer: B. II and IV only**

Solution: Statement I is incorrect: 'Positive pressure developed in roots' corresponds to Root pressure, not Cohesion-tension theory. Statement II is correct because 'Theory chiefly explaining ascent of sap' corresponds to Cohesion-tension theory. Statement III is incorrect: 'Main tissue conducting water upward' corresponds to Xylem, not Root-hair zone. Statement IV is correct because 'Principal water-absorbing region of a young root' corresponds to Root-hair zone. Therefore, the valid combination is II and IV only.

Distractor review: A: This combination includes false statement(s) I, III and omits true statement(s) II. C: This combination includes false statement(s) I, III and omits true statement(s) IV. D: This combination includes false statement(s) I, III.

BIO-12-CH06-0052 | Medium | Transpiration**Correct answer: B. Cuticular transpiration**

Solution: The observation 'Water loss directly through the cuticle' points to Cuticular transpiration; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Osmosis belongs with 'Net movement of water through a selectively permeable membrane', whereas the tested description is 'Water loss directly through the cuticle'. C: Symplast pathway belongs with 'Movement through connected cytoplasm via plasmodesmata', whereas the tested description is 'Water loss directly through the cuticle'. D: Transpiration pull belongs with 'Force generated by evaporation from leaves', whereas the tested description is 'Water loss directly through the cuticle'.

BIO-12-CH06-0053 | Easy | Properties and forms of water**Correct answer: D. Cohesion**

Solution: 'Attraction between water molecules' is the defining description for Cohesion. The other choices belong to different chapter concepts.

Distractor review: A: Lenticular transpiration belongs with 'Water loss through lenticels', whereas the tested description is 'Attraction between water molecules'. B: Transpiration belongs with 'Loss of water as vapour from aerial plant parts', whereas the tested description is 'Attraction between water molecules'. C: Adhesion belongs with 'Attraction of water to a different surface', whereas the tested description is 'Attraction between water molecules'.

BIO-12-CH06-0054 | Medium | Properties and forms of water**Correct answer: A. Cohesion**

Solution: The chapter links Attraction between water molecules with Cohesion, so Cohesion completes the statement.

Distractor review: B: Transpiration belongs with 'Loss of water as vapour from aerial plant parts', whereas the tested description is 'Attraction between water molecules'. C: Casparian strip belongs with 'Barrier forcing water through living cells at the endodermis', whereas the tested description is 'Attraction between water molecules'. D: Capillary water belongs with 'Soil water most readily available to plants', whereas the tested description is 'Attraction between water molecules'.

BIO-12-CH06-0055 | Easy | Mineral and food transport**Correct answer: C. Mass flow**

Solution: 'Pressure-driven movement through sieve tubes' is the defining description for Mass flow. The other choices belong to different chapter concepts.

Distractor review: A: Stomata belongs with 'Major route of transpiration', whereas the tested description is 'Pressure-driven movement through sieve tubes'. B: Mature photosynthetic leaf belongs with 'Source tissue in phloem transport', whereas the tested description is 'Pressure-driven movement through sieve tubes'. D: Solute potential belongs with 'Component of water potential produced by dissolved solute', whereas the tested description is 'Pressure-driven movement through sieve tubes'.

BIO-12-CH06-0056 | Medium | Stomatal apparatus**Correct answer: A. Potassium**

Solution: The chapter links Ion whose accumulation promotes guard-cell opening with Potassium, so Potassium completes the statement.

Distractor review: B: Symplast pathway belongs with 'Movement through connected cytoplasm via plasmodesmata', whereas the tested description is 'Ion whose accumulation promotes guard-cell opening'. C: Apoplast pathway belongs with 'Passive movement through cell walls and intercellular spaces', whereas the tested description is 'Ion whose accumulation promotes guard-cell opening'. D: Casparian strip belongs with 'Barrier forcing water through living cells at the endodermis', whereas the tested description is 'Ion whose accumulation promotes guard-cell opening'.

BIO-12-CH06-0057 | Medium | Ascent of sap**Correct answer: D. Xylem**

Solution: The observation 'Main tissue conducting water upward' points to Xylem; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Hygroscopic water belongs with 'Soil water held tightly to particle surfaces', whereas the tested description is 'Main tissue conducting water upward'. B: Apoplast pathway belongs with 'Passive movement through cell walls and intercellular spaces', whereas the tested description is 'Main tissue conducting water upward'. C: Dumb-bell shaped belongs with 'Guard-cell shape typical of grasses', whereas the tested description is 'Main tissue conducting water upward'.

BIO-12-CH06-0058 | Hard | Properties and forms of water**Correct answer: C. Attraction between water molecules - Cohesion; Barrier forcing water through living cells at the endodermis - Casparian strip**

Solution: Both parts of 'Attraction between water molecules - Cohesion; Barrier forcing water through living cells at the endodermis - Casparian strip' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Net movement of water through a selectively permeable membrane' requires Osmosis, and 'Passive movement through cell walls and intercellular spaces' requires Apoplast pathway. B: The answers have been interchanged: 'Component of water potential produced by dissolved solute' requires Solute potential, and 'Net movement of water through a selectively permeable membrane' requires Osmosis. D: The answers have been interchanged: 'Water loss through lenticels' requires Lenticular transpiration, and 'Component of water potential produced by dissolved solute' requires Solute potential.

BIO-12-CH06-0059 | Medium | Ascent of sap**Correct answer: D. Force generated by evaporation from leaves - Transpiration pull**

Solution: The accurate entry is 'Force generated by evaporation from leaves - Transpiration pull'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Attraction of water to a different surface' should be paired with Adhesion, not Developing fruit or storage organ. B: 'Typical sink tissue' should be paired with Developing fruit or storage organ, not Guard cells. C: 'Source tissue in phloem transport' should be paired with Mature photosynthetic leaf, not Adhesion.

BIO-12-CH06-0060 | Medium | Water potential and osmosis**Correct answer: B. Solute potential**

Solution: The observation 'Component of water potential produced by dissolved solute' points to Solute potential; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Plasmolysis belongs with 'Shrinkage of a protoplast in a hypertonic solution', whereas the tested description is 'Component of water potential produced by dissolved solute'. C: Zero belongs with 'Water potential of pure water at standard conditions', whereas the tested description is 'Component of water potential produced by dissolved solute'. D: Apoplast pathway belongs with 'Passive movement through cell walls and intercellular spaces', whereas the tested description is 'Component of water potential produced by dissolved solute'.

BIO-12-CH06-0061 | Challenging | Transpiration**Correct answer: A. I only**

Solution: Statement I is correct because 'Loss of water as vapour from aerial plant parts' corresponds to Transpiration. Statement II is incorrect: 'Water loss through lenticels' corresponds to Lenticular transpiration, not Stomata. Statement III is incorrect: 'Major route of transpiration' corresponds to Stomata, not Capillary water. Statement IV is incorrect: 'Soil water most readily available to plants' corresponds to Capillary water, not Transpiration. Therefore, the valid combination is I only.

Distractor review: B: This combination includes false statement(s) II, III. C: This combination includes false statement(s) II, III, IV. D: This combination includes false statement(s) II, III and omits true statement(s) I.

BIO-12-CH06-0062 | Easy | Root absorption**Correct answer: B. Passive movement through cell walls and intercellular spaces**

Solution: Apoplast pathway represents 'Passive movement through cell walls and intercellular spaces'. Each alternative describes a different term or result.

Distractor review: A: 'Positive pressure developed in roots' identifies Root pressure, not Apoplast pathway. C: 'Component of water potential produced by dissolved solute' identifies Solute potential, not Apoplast pathway. D: 'Movement through connected cytoplasm via plasmodesmata' identifies Symplast pathway, not Apoplast pathway.

BIO-12-CH06-0063 | Challenging | Water potential and osmosis**Correct answer: D. II and III only**

Solution: Statement I is incorrect: 'Component of water potential produced by dissolved solute' corresponds to Solute potential, not Zero. Statement II is correct because 'Water potential of pure water at standard conditions' corresponds to Zero. Statement III is correct because 'Shrinkage of a protoplast in a hypertonic solution' corresponds to Plasmolysis. Statement IV is incorrect: 'Pressure-driven movement through sieve tubes' corresponds to Mass flow, not Solute potential. Therefore, the valid combination is II and III only.

Distractor review: A: This combination omits true statement(s) III. B: This combination includes false statement(s) I, IV. C: This combination includes false statement(s) I, IV and omits true statement(s) II.

BIO-12-CH06-0064 | Easy | Ascent of sap**Correct answer: A. Force generated by evaporation from leaves**

Solution: Transpiration pull represents 'Force generated by evaporation from leaves'. Each alternative describes a different term or result.

Distractor review: B: 'Source tissue in phloem transport' identifies Mature photosynthetic leaf, not Transpiration pull. C: 'Guard-cell shape typical of dicots' identifies Kidney-shaped, not Transpiration pull. D: 'Theory chiefly explaining ascent of sap' identifies Cohesion-tension theory, not Transpiration pull.

BIO-12-CH06-0065 | Easy | Mineral and food transport**Correct answer: A. Active absorption**

Solution: 'Energy-dependent uptake of mineral ions' is the defining description for Active absorption. The other choices belong to different chapter concepts.

Distractor review: B: Kidney-shaped belongs with 'Guard-cell shape typical of dicots', whereas the tested description is 'Energy-dependent uptake of mineral ions'. C: Developing fruit or storage organ belongs with 'Typical sink tissue', whereas the tested description is 'Energy-dependent uptake of mineral ions'. D: Mature photosynthetic leaf belongs with 'Source tissue in phloem transport', whereas the tested description is 'Energy-dependent uptake of mineral ions'.

BIO-12-CH06-0066 | Hard | Ascent of sap**Correct answer: D. Force generated by evaporation from leaves - Transpiration pull; Source tissue in phloem transport - Mature photosynthetic leaf**

Solution: Both parts of 'Force generated by evaporation from leaves - Transpiration pull; Source tissue in phloem transport - Mature photosynthetic leaf' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Soil water most readily available to plants' requires Capillary water, and 'Cells controlling the stomatal pore' requires Guard cells. B: The answers have been interchanged: 'Attraction of water to a different surface' requires Adhesion, and 'Guard-cell shape typical of dicots' requires Kidney-shaped. C: The answers have been interchanged: 'Typical sink tissue' requires Developing fruit or storage organ, and 'Soil water most readily available to plants' requires Capillary water.

BIO-12-CH06-0067 | Medium | Stomatal apparatus**Correct answer: A. Dumb-bell shaped**

Solution: The chapter links Guard-cell shape typical of grasses with Dumb-bell shaped, so Dumb-bell shaped completes the statement.

Distractor review: B: Capillary water belongs with 'Soil water most readily available to plants', whereas the tested description is 'Guard-cell shape typical of grasses'. C: Root pressure belongs with 'Positive pressure developed in roots', whereas the tested description is 'Guard-cell shape typical of grasses'. D: Kidney-shaped belongs with 'Guard-cell shape typical of dicots', whereas the tested description is 'Guard-cell shape typical of grasses'.

BIO-12-CH06-0068 | Medium | Ascent of sap**Correct answer: C. Positive pressure developed in roots - Root pressure**

Solution: The accurate entry is 'Positive pressure developed in roots - Root pressure'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Pressure-driven movement through sieve tubes' should be paired with Mass flow, not Lenticular transpiration. B: 'Net movement of water through a selectively permeable membrane' should be paired with Osmosis, not Root pressure. D: 'Attraction of water to a different surface' should be paired with Adhesion, not Mass flow.

BIO-12-CH06-0069 | Challenging | Root absorption**Correct answer: D. I, II, III and IV**

Solution: Statement I is correct because 'Movement through connected cytoplasm via plasmodesmata' corresponds to Symplast pathway. Statement II is correct because 'Principal water-absorbing region of a young root' corresponds to Root-hair zone. Statement III is correct because 'Barrier forcing water through living cells at the endodermis' corresponds to Casparian strip. Statement IV is correct because 'Guard-cell shape typical of grasses' corresponds to Dumb-bell shaped. Therefore, the valid combination is I, II, III and IV.

Distractor review: A: This combination omits true statement(s) III. B: This combination omits true statement(s) IV. C: This combination omits true statement(s) I, II.

BIO-12-CH06-0070 | Hard | Mineral and food transport**Correct answer: A. Source tissue in phloem transport - Apoplast pathway**

Solution: 'Source tissue in phloem transport' should be paired with Mature photosynthetic leaf, not Apoplast pathway. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Attraction between water molecules - Cohesion. C: This is a valid association: Movement through connected cytoplasm via plasmodesmata - Symplast pathway. D: This is a valid association: Guard-cell shape typical of grasses - Dumb-bell shaped.

BIO-12-CH06-0071 | Medium | Properties and forms of water**Correct answer: A. Hygroscopic water**

Solution: The chapter links Soil water held tightly to particle surfaces with Hygroscopic water, so Hygroscopic water completes the statement.

Distractor review: B: Cohesion-tension theory belongs with 'Theory chiefly explaining ascent of sap', whereas the tested description is 'Soil water held tightly to particle surfaces'. C: Transpiration belongs with 'Loss of water as vapour from aerial plant parts', whereas the tested description is 'Soil water held tightly to particle surfaces'. D: Cohesion belongs with 'Attraction between water molecules', whereas the tested description is 'Soil water held tightly to particle surfaces'.

BIO-12-CH06-0072 | Easy | Transpiration**Correct answer: C. Water loss through lenticels**

Solution: Lenticular transpiration represents 'Water loss through lenticels'. Each alternative describes a different term or result.

Distractor review: A: 'Loss of water as vapour from aerial plant parts' identifies Transpiration, not Lenticular transpiration. B: 'Soil water most readily available to plants' identifies Capillary water, not Lenticular transpiration. D: 'Theory chiefly explaining ascent of sap' identifies Cohesion-tension theory, not Lenticular transpiration.

BIO-12-CH06-0073 | Medium | Mineral and food transport**Correct answer: A. Developing fruit or storage organ**

Solution: The chapter links Typical sink tissue with Developing fruit or storage organ, so Developing fruit or storage organ completes the statement.

Distractor review: B: Zero belongs with 'Water potential of pure water at standard conditions', whereas the tested description is 'Typical sink tissue'. C: Mass flow belongs with 'Pressure-driven movement through sieve tubes', whereas the tested description is 'Typical sink tissue'. D: Mature photosynthetic leaf belongs with 'Source tissue in phloem transport', whereas the tested description is 'Typical sink tissue'.

BIO-12-CH06-0074 | Hard | Water potential and osmosis**Correct answer: C. Net movement of water through a selectively permeable membrane - Active absorption**

Solution: 'Net movement of water through a selectively permeable membrane' should be paired with Osmosis, not Active absorption. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Attraction between water molecules - Cohesion. B: This is a valid association: Source tissue in phloem transport - Mature photosynthetic leaf. D: This is a valid association: Movement through connected cytoplasm via plasmodesmata - Symplast pathway.

BIO-12-CH06-0075 | Hard | Ascent of sap**Correct answer: B. Theory chiefly explaining ascent of sap - Cohesion-tension theory; Principal water-absorbing region of a young root - Root-hair zone**

Solution: Both parts of 'Theory chiefly explaining ascent of sap - Cohesion-tension theory; Principal water-absorbing region of a young root - Root-hair zone' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Movement through connected cytoplasm via plasmodesmata' requires Symplast pathway, and 'Water loss through lenticels' requires Lenticular transpiration. C: The answers have been interchanged: 'Guard-cell shape typical of grasses' requires Dumb-bell shaped, and 'Attraction of water to a different surface' requires Adhesion. D: The answers have been interchanged: 'Attraction of water to a different surface' requires Adhesion, and 'Soil water most readily available to plants' requires Capillary water.

BIO-12-CH06-0076 | Easy | Mineral and food transport**Correct answer: B. Mature photosynthetic leaf**

Solution: 'Source tissue in phloem transport' is the defining description for Mature photosynthetic leaf. The other choices belong to different chapter concepts.

Distractor review: A: Cohesion belongs with 'Attraction between water molecules', whereas the tested description is 'Source tissue in phloem transport'. C: Symplast pathway belongs with 'Movement through connected cytoplasm via plasmodesmata', whereas the tested description is 'Source tissue in phloem transport'. D: Zero belongs with 'Water potential of pure water at standard conditions', whereas the tested description is 'Source tissue in phloem transport'.

BIO-12-CH06-0077 | Medium | Transpiration**Correct answer: D. Loss of water as vapour from aerial plant parts - Transpiration**

Solution: The accurate entry is 'Loss of water as vapour from aerial plant parts - Transpiration'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Main tissue conducting water upward' should be paired with Xylem, not Transpiration. B: 'Shrinkage of a protoplast in a hypertonic solution' should be paired with Plasmolysis, not Symplast pathway. C: 'Barrier forcing water through living cells at the endodermis' should be paired with Casparian strip, not Plasmolysis.

BIO-12-CH06-0078 | Hard | Properties and forms of water**Correct answer: A. Attraction between water molecules - Casparian strip**

Solution: 'Attraction between water molecules' should be paired with Cohesion, not Casparian strip. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Net movement of water through a selectively permeable membrane - Osmosis. C: This is a valid association: Component of water potential produced by dissolved solute - Solute potential. D: This is a valid association: Passive movement through cell walls and intercellular spaces - Apoplast pathway.

BIO-12-CH06-0079 | Easy | Ascent of sap**Correct answer: D. Theory chiefly explaining ascent of sap**

Solution: Cohesion-tension theory represents 'Theory chiefly explaining ascent of sap'. Each alternative describes a different term or result.

Distractor review: A: 'Guard-cell shape typical of grasses' identifies Dumb-bell shaped, not Cohesion-tension theory. B: 'Force generated by evaporation from leaves' identifies Transpiration pull, not Cohesion-tension theory. C: 'Water loss through lenticels' identifies Lenticular transpiration, not Cohesion-tension theory.

BIO-12-CH06-0080 | Medium | Water potential and osmosis**Correct answer: B. Net movement of water through a selectively permeable membrane - Osmosis**

Solution: The accurate entry is 'Net movement of water through a selectively permeable membrane - Osmosis'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Energy-dependent uptake of mineral ions' should be paired with Active absorption, not Dumb-bell shaped. C: 'Main tissue conducting water upward' should be paired with Xylem, not Mature photosynthetic leaf. D: 'Movement through connected cytoplasm via plasmodesmata' should be paired with Symplast pathway, not Xylem.

BIO-12-CH06-0081 | Medium | Root absorption**Correct answer: B. Apoplast pathway**

Solution: The observation 'Passive movement through cell walls and intercellular spaces' points to Apoplast pathway; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Symplast pathway belongs with 'Movement through connected cytoplasm via plasmodesmata', whereas the tested description is 'Passive movement through cell walls and intercellular spaces'. C: Root pressure belongs with 'Positive pressure developed in roots', whereas the tested description is 'Passive movement through cell walls and intercellular spaces'. D: Transpiration belongs with 'Loss of water as vapour from aerial plant parts', whereas the tested description is 'Passive movement through cell walls and intercellular spaces'.

BIO-12-CH06-0082 | Medium | Properties and forms of water**Correct answer: D. Attraction of water to a different surface - Adhesion**

Solution: The accurate entry is 'Attraction of water to a different surface - Adhesion'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Shrinkage of a protoplast in a hypertonic solution' should be paired with Plasmolysis, not Adhesion. B: 'Source tissue in phloem transport' should be paired with Mature photosynthetic leaf, not Zero. C: 'Water potential of pure water at standard conditions' should be paired with Zero, not Plasmolysis.

BIO-12-CH06-0083 | Hard | Transpiration**Correct answer: B. Loss of water as vapour from aerial plant parts - Transpiration; Barrier forcing water through living cells at the endodermis - Casparian strip**

Solution: Both parts of 'Loss of water as vapour from aerial plant parts - Transpiration; Barrier forcing water through living cells at the endodermis - Casparian strip' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Soil water held tightly to particle surfaces' requires Hygroscopic water, and 'Shrinkage of a protoplast in a hypertonic solution' requires Plasmolysis. C: The answers have been interchanged: 'Main tissue conducting water upward' requires Xylem, and 'Energy-dependent uptake of mineral ions' requires Active absorption. D: The answers have been interchanged: 'Net movement of water through a selectively permeable membrane' requires Osmosis, and 'Movement through connected cytoplasm via plasmodesmata' requires Symplast pathway.

BIO-12-CH06-0084 | Medium | Stomatal apparatus**Correct answer: C. Guard cells**

Solution: The observation 'Cells controlling the stomatal pore' points to Guard cells; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Mature photosynthetic leaf belongs with 'Source tissue in phloem transport', whereas the tested description is 'Cells controlling the stomatal pore'. B: Dumb-bell shaped belongs with 'Guard-cell shape typical of grasses', whereas the tested description is 'Cells controlling the stomatal pore'. D: Mass flow belongs with 'Pressure-driven movement through sieve tubes', whereas the tested description is 'Cells controlling the stomatal pore'.

BIO-12-CH06-0085 | Challenging | Stomatal apparatus**Correct answer: D. I and IV only**

Solution: Statement I is correct because 'Guard-cell shape typical of grasses' corresponds to Dumb-bell shaped. Statement II is incorrect: 'Guard-cell shape typical of dicots' corresponds to Kidney-shaped, not Potassium. Statement III is incorrect: 'Ion whose accumulation promotes guard-cell opening' corresponds to Potassium, not Root-hair zone. Statement IV is correct because 'Principal water-absorbing region of a young root' corresponds to Root-hair zone. Therefore, the valid combination is I and IV only.

Distractor review: A: This combination includes false statement(s) II, III. B: This combination includes false statement(s) II and omits true statement(s) I, IV. C: This combination includes false statement(s) III and omits true statement(s) I, IV.

BIO-12-CH06-0086 | Medium | Mineral and food transport**Correct answer: C. Mass flow**

Solution: The observation 'Pressure-driven movement through sieve tubes' points to Mass flow; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Mature photosynthetic leaf belongs with 'Source tissue in phloem transport', whereas the tested description is 'Pressure-driven movement through sieve tubes'. B: Xylem belongs with 'Main tissue conducting water upward', whereas the tested description is 'Pressure-driven movement through sieve tubes'. D: Solute potential belongs with 'Component of water potential produced by dissolved solute', whereas the tested description is 'Pressure-driven movement through sieve tubes'.

BIO-12-CH06-0087 | Easy | Properties and forms of water**Correct answer: C. Soil water most readily available to plants**

Solution: Capillary water represents 'Soil water most readily available to plants'. Each alternative describes a different term or result.

Distractor review: A: 'Source tissue in phloem transport' identifies Mature photosynthetic leaf, not Capillary water. B: 'Energy-dependent uptake of mineral ions' identifies Active absorption, not Capillary water. D: 'Theory chiefly explaining ascent of sap' identifies Cohesion-tension theory, not Capillary water.

BIO-12-CH06-0088 | Hard | Transpiration**Correct answer: D. Water loss through lenticels - Lenticular transpiration; Theory chiefly explaining ascent of sap - Cohesion-tension theory**

Solution: Both parts of 'Water loss through lenticels - Lenticular transpiration; Theory chiefly explaining ascent of sap - Cohesion-tension theory' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Soil water most readily available to plants' requires Capillary water, and 'Soil water held tightly to particle surfaces' requires Hygroscopic water. B: The answers have been interchanged: 'Principal water-absorbing region of a young root' requires Root-hair zone, and 'Cells controlling the stomatal pore' requires Guard cells. C: The answers have been interchanged: 'Soil water held tightly to particle surfaces' requires Hygroscopic water, and 'Attraction between water molecules' requires Cohesion.

Standard 12 | Chapter 07

Plant Growth and Mineral Nutrition

BIO-12-CH07-0001 | Hard | Sigmoid growth curve

Correct answer: C. Resources become limiting and growth rate falls

Solution: Limiting resources and other constraints reduce net growth as the system approaches its maximum.

Distractor review: A: The alternative 'All cells immediately stop metabolism' conflicts with the labelled relationship, scale or mechanism in the figure. Limiting resources and other constraints reduce net growth as the system approaches its maximum. B: The alternative 'Carrying capacity increases without limit' conflicts with the labelled relationship, scale or mechanism in the figure. Limiting resources and other constraints reduce net growth as the system approaches its maximum. D: The alternative 'The organism loses all genetic material' conflicts with the labelled relationship, scale or mechanism in the figure. Limiting resources and other constraints reduce net growth as the system approaches its maximum.

BIO-12-CH07-0002 | Challenging | Nitrogen assimilation in a plant

Correct answer: A. Impaired nitrate assimilation and reduced amino-acid synthesis

Solution: Reduced nitrate reduction limits ammonium supply for amino-acid and protein synthesis.

Distractor review: B: The alternative 'Faster conversion of nitrate to protein' conflicts with the labelled relationship, scale or mechanism in the figure. Reduced nitrate reduction limits ammonium supply for amino-acid and protein synthesis. C: The alternative 'Immediate conversion of nitrate into atmospheric nitrogen' conflicts with the labelled relationship, scale or mechanism in the figure. Reduced nitrate reduction limits ammonium supply for amino-acid and protein synthesis. D: The alternative 'Selective increase in chlorophyll without nitrogen uptake' conflicts with the labelled relationship, scale or mechanism in the figure. Reduced nitrate reduction limits ammonium supply for amino-acid and protein synthesis.

BIO-12-CH07-0003 | Easy | Sigmoid growth curve

Correct answer: B. Log or exponential phase

Solution: The steep log phase shows the greatest increase per unit time.

Distractor review: A: The alternative 'Initial lag only' conflicts with the labelled relationship, scale or mechanism in the figure. The steep log phase shows the greatest increase per unit time. C: The alternative 'Stationary phase' conflicts with the labelled relationship, scale or mechanism in the figure. The steep log phase shows the greatest increase per unit time. D: The alternative 'After all cells die' conflicts with the labelled relationship, scale or mechanism in the figure. The steep log phase shows the greatest increase per unit time.

BIO-12-CH07-0004 | Medium | Nitrogen assimilation in a plant

Correct answer: C. Nitrate reductase

Solution: Nitrate reductase converts nitrate to nitrite during nitrate assimilation.

Distractor review: A: The alternative 'Nitrogenase' conflicts with the labelled relationship, scale or mechanism in the figure. Nitrate reductase converts nitrate to nitrite during nitrate assimilation. B: The alternative 'Rubisco' conflicts with the labelled relationship, scale or mechanism in the figure. Nitrate reductase converts nitrate to nitrite during nitrate assimilation. D: The alternative 'Pepsin' conflicts with the labelled relationship, scale or mechanism in the figure. Nitrate reductase converts nitrate to nitrite during nitrate assimilation.

BIO-12-CH07-0005 | Medium | Photoperiodism and vernalization

Correct answer: C. Photoperiodism

Solution: The observation 'Response of flowering to relative day and night length' points to Photoperiodism; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Geometric growth belongs with 'Increase proportional to the existing size', whereas the tested description is 'Response of flowering to relative day and night length'. B: Vernalization belongs with 'Cold treatment promoting flowering', whereas the tested description is 'Response of flowering to relative day and night length'. D: Nitrogen fixation belongs with 'Conversion of atmospheric nitrogen to ammonia', whereas the tested description is 'Response of flowering to relative day and night length'.

BIO-12-CH07-0006 | Easy | Essential mineral elements**Correct answer: A. Phosphorus**

Solution: 'Element important in ATP and nucleic acids' is the defining description for Phosphorus. The other choices belong to different chapter concepts.

Distractor review: B: Phytochrome belongs with 'Pigment involved in photoperiod perception', whereas the tested description is 'Element important in ATP and nucleic acids'. C: Nitrogen belongs with 'Element required for synthesis of amino acids and nucleic acids', whereas the tested description is 'Element important in ATP and nucleic acids'. D: Absolute growth rate belongs with 'Total growth per unit time', whereas the tested description is 'Element important in ATP and nucleic acids'.

BIO-12-CH07-0007 | Hard | Plant hormones II**Correct answer: C. Hormone associated with stomatal closure and stress - Gibberellin**

Solution: 'Hormone associated with stomatal closure and stress' should be paired with Absciscic acid, not Gibberellin. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Central atom of chlorophyll - Magnesium. B: This is a valid association: Growth occurring throughout life in meristematic regions - Indeterminate growth. D: This is a valid association: Increase by a constant amount per unit time - Arithmetic growth.

BIO-12-CH07-0008 | Hard | Photoperiodism and vernalization**Correct answer: D. Cold treatment promoting flowering - Sigmoid curve**

Solution: 'Cold treatment promoting flowering' should be paired with Vernalization, not Sigmoid curve. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Element required for functioning of stomata and enzyme activation - Potassium. B: This is a valid association: Phase in which cells acquire specialized form and function - Maturation phase. C: This is a valid association: Element required for synthesis of amino acids and nucleic acids - Nitrogen.

BIO-12-CH07-0009 | Medium | Growth fundamentals**Correct answer: D. Growth measurable as increase per unit time - Growth rate**

Solution: The accurate entry is 'Growth measurable as increase per unit time - Growth rate'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Growth per unit initial size per unit time' should be paired with Relative growth rate, not Arithmetic growth. B: 'Increase proportional to the existing size' should be paired with Geometric growth, not Growth rate. C: 'Hormone delaying leaf senescence' should be paired with Cytokinin, not Relative growth rate.

BIO-12-CH07-0010 | Hard | Growth rates**Correct answer: B. Increase proportional to the existing size - Geometric growth; Permanent irreversible increase in size - Growth**

Solution: Both parts of 'Increase proportional to the existing size - Geometric growth; Permanent irreversible increase in size - Growth' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Hormone associated with stomatal closure and stress' requires Absciscic acid, and 'Gaseous hormone promoting fruit ripening' requires Ethylene. C: The answers have been interchanged: 'Conversion of atmospheric nitrogen to ammonia' requires Nitrogen fixation, and 'Hormone associated with stomatal closure and stress' requires Absciscic acid. D: The answers have been interchanged: 'Gaseous hormone promoting fruit ripening' requires Ethylene, and 'Overall sequence from germination to senescence' requires Development.

BIO-12-CH07-0011 | Medium | Plant hormones II**Correct answer: B. Absciscic acid**

Solution: The chapter links Hormone associated with stomatal closure and stress with Absciscic acid, so Absciscic acid completes the statement.

Distractor review: A: Magnesium belongs with 'Central atom of chlorophyll', whereas the tested description is 'Hormone associated with stomatal closure and stress'. C: Auxin belongs with 'Hormone used to promote rooting in stem cuttings', whereas the tested description is 'Hormone associated with stomatal closure and stress'. D: Gibberellin belongs with 'Hormone promoting stem elongation and bolting', whereas the tested description is 'Hormone associated with stomatal closure and stress'.

BIO-12-CH07-0012 | Medium | Essential mineral elements**Correct answer: C. Central atom of chlorophyll - Magnesium**

Solution: The accurate entry is 'Central atom of chlorophyll - Magnesium'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Overall sequence from germination to senescence' should be paired with Development, not Magnesium. B: 'Permanent irreversible increase in size' should be paired with Growth, not Maturation phase. D: 'Typical growth curve of a whole organ' should be paired with Sigmoid curve, not Denitrification.

BIO-12-CH07-0013 | Easy | Nitrogen metabolism**Correct answer: C. Conversion of ammonia to nitrite and nitrate**

Solution: Nitrification represents 'Conversion of ammonia to nitrite and nitrate'. Each alternative describes a different term or result.

Distractor review: A: 'Element required for synthesis of amino acids and nucleic acids' identifies Nitrogen, not Nitrification. B: 'Symbiotic nitrogen-fixing bacterium in legume nodules' identifies Rhizobium, not Nitrification. D: 'Pigment involved in photoperiod perception' identifies Phytochrome, not Nitrification.

BIO-12-CH07-0014 | Challenging | Plant hormones I**Correct answer: D. II and IV only**

Solution: Statement I is incorrect: 'Hormone promoting cell division' corresponds to Cytokinin, not Gibberellin. Statement II is correct because 'Hormone promoting stem elongation and bolting' corresponds to Gibberellin. Statement III is incorrect: 'Hormone promoting cell elongation and apical dominance' corresponds to Auxin, not Arithmetic growth. Statement IV is correct because 'Increase by a constant amount per unit time' corresponds to Arithmetic growth. Therefore, the valid combination is II and IV only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) II. B: This combination includes false statement(s) I and omits true statement(s) IV. C: This combination includes false statement(s) I, III and omits true statement(s) II.

BIO-12-CH07-0015 | Hard | Nitrogen metabolism**Correct answer: B. Reduction of nitrate to atmospheric nitrogen - Vernalization**

Solution: 'Reduction of nitrate to atmospheric nitrogen' should be paired with Denitrification, not Vernalization. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Permanent irreversible increase in size - Growth. C: This is a valid association: Element required for functioning of stomata and enzyme activation - Potassium. D: This is a valid association: Plant flowering when day length is shorter than a critical value - Short-day plant.

BIO-12-CH07-0016 | Medium | Phases and curves**Correct answer: D. Sigmoid curve**

Solution: The observation 'Typical growth curve of a whole organ' points to Sigmoid curve; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Ethylene belongs with 'Gaseous hormone promoting fruit ripening', whereas the tested description is 'Typical growth curve of a whole organ'. B: Phytochrome belongs with 'Pigment involved in photoperiod perception', whereas the tested description is 'Typical growth curve of a whole organ'. C: Vernalization belongs with 'Cold treatment promoting flowering', whereas the tested description is 'Typical growth curve of a whole organ'.

BIO-12-CH07-0017 | Hard | Growth fundamentals**Correct answer: A. Growth occurring throughout life in meristematic regions - 2,4-D**

Solution: 'Growth occurring throughout life in meristematic regions' should be paired with Indeterminate growth, not 2,4-D. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Conversion of atmospheric nitrogen to ammonia - Nitrogen fixation. C: This is a valid association: Element required for synthesis of amino acids and nucleic acids - Nitrogen. D: This is a valid association: Hormone promoting cell elongation and apical dominance - Auxin.

BIO-12-CH07-0018 | Hard | Plant hormones I

Correct answer: B. Gaseous hormone promoting fruit ripening - Ethylene; Hormone associated with stomatal closure and stress - Absciscic acid

Solution: Both parts of 'Gaseous hormone promoting fruit ripening - Ethylene; Hormone associated with stomatal closure and stress - Absciscic acid' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Overall sequence from germination to senescence' requires Development, and 'Reduction of nitrate to atmospheric nitrogen' requires Denitrification. C: The answers have been interchanged: 'Element important in ATP and nucleic acids' requires Phosphorus, and 'Increase proportional to the existing size' requires Geometric growth. D: The answers have been interchanged: 'Reduction of nitrate to atmospheric nitrogen' requires Denitrification, and 'Hormone delaying leaf senescence' requires Cytokinin.

BIO-12-CH07-0019 | Easy | Phases and curves

Correct answer: B. Phase dominated by cell enlargement

Solution: Elongation phase represents 'Phase dominated by cell enlargement'. Each alternative describes a different term or result.

Distractor review: A: 'Overall sequence from germination to senescence' identifies Development, not Elongation phase. C: 'Element required for synthesis of amino acids and nucleic acids' identifies Nitrogen, not Elongation phase. D: 'Growth phase dominated by active cell division' identifies Meristematic phase, not Elongation phase.

BIO-12-CH07-0020 | Medium | Growth rates

Correct answer: A. Arithmetic growth

Solution: The chapter links Increase by a constant amount per unit time with Arithmetic growth, so Arithmetic growth completes the statement.

Distractor review: B: Denitrification belongs with 'Reduction of nitrate to atmospheric nitrogen', whereas the tested description is 'Increase by a constant amount per unit time'. C: Elongation phase belongs with 'Phase dominated by cell enlargement', whereas the tested description is 'Increase by a constant amount per unit time'. D: Magnesium belongs with 'Central atom of chlorophyll', whereas the tested description is 'Increase by a constant amount per unit time'.

BIO-12-CH07-0021 | Easy | Photoperiodism and vernalization

Correct answer: C. Phytochrome

Solution: 'Pigment involved in photoperiod perception' is the defining description for Phytochrome. The other choices belong to different chapter concepts.

Distractor review: A: Short-day plant belongs with 'Plant flowering when day length is shorter than a critical value', whereas the tested description is 'Pigment involved in photoperiod perception'. B: Photoperiodism belongs with 'Response of flowering to relative day and night length', whereas the tested description is 'Pigment involved in photoperiod perception'. D: Growth belongs with 'Permanent irreversible increase in size', whereas the tested description is 'Pigment involved in photoperiod perception'.

BIO-12-CH07-0022 | Hard | Essential mineral elements

Correct answer: B. Element important in ATP and nucleic acids - Phosphorus; Gaseous hormone promoting fruit ripening - Ethylene

Solution: Both parts of 'Element important in ATP and nucleic acids - Phosphorus; Gaseous hormone promoting fruit ripening - Ethylene' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Conversion of atmospheric nitrogen to ammonia' requires Nitrogen fixation, and 'Response of flowering to relative day and night length' requires Photoperiodism. C: The answers have been interchanged: 'Total growth per unit time' requires Absolute growth rate, and 'Growth per unit initial size per unit time' requires Relative growth rate. D: The answers have been interchanged: 'Response of flowering to relative day and night length' requires Photoperiodism, and 'Total growth per unit time' requires Absolute growth rate.

BIO-12-CH07-0023 | Easy | Plant hormones II

Correct answer: B. Cytokinin

Solution: 'Hormone delaying leaf senescence' is the defining description for Cytokinin. The other choices belong to different chapter concepts.

Distractor review: A: 2,4-D belongs with 'Synthetic auxin widely used as a selective weedicide', whereas the tested description is 'Hormone delaying leaf senescence'. C: Geometric growth belongs with 'Increase proportional to the existing size', whereas the tested description is 'Hormone delaying leaf senescence'. D: Absciscic acid belongs with 'Hormone associated with stomatal closure and stress', whereas the tested description is 'Hormone delaying leaf senescence'.

BIO-12-CH07-0024 | Hard | Growth rates

Correct answer: A. Growth per unit initial size per unit time - Relative growth rate; Hormone promoting cell division - Cytokinin

Solution: Both parts of 'Growth per unit initial size per unit time - Relative growth rate; Hormone promoting cell division - Cytokinin' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Hormone delaying leaf senescence' requires Cytokinin, and 'Hormone used to promote rooting in stem cuttings' requires Auxin. C: The answers have been interchanged: 'Growth phase dominated by active cell division' requires Meristematic phase, and 'Phase in which cells acquire specialized form and function' requires Maturation phase. D: The answers have been interchanged: 'Hormone used to promote rooting in stem cuttings' requires Auxin, and 'Growth measurable as increase per unit time' requires Growth rate.

BIO-12-CH07-0025 | Medium | Essential mineral elements

Correct answer: A. Potassium

Solution: The observation 'Element required for functioning of stomata and enzyme activation' points to Potassium; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Geometric growth belongs with 'Increase proportional to the existing size', whereas the tested description is 'Element required for functioning of stomata and enzyme activation'. C: Magnesium belongs with 'Central atom of chlorophyll', whereas the tested description is 'Element required for functioning of stomata and enzyme activation'. D: Cytokinin belongs with 'Hormone promoting cell division', whereas the tested description is 'Element required for functioning of stomata and enzyme activation'.

BIO-12-CH07-0026 | Easy | Growth rates

Correct answer: D. Increase proportional to the existing size

Solution: Geometric growth represents 'Increase proportional to the existing size'. Each alternative describes a different term or result.

Distractor review: A: 'Overall sequence from germination to senescence' identifies Development, not Geometric growth. B: 'Gaseous hormone promoting fruit ripening' identifies Ethylene, not Geometric growth. C: 'Hormone associated with stomatal closure and stress' identifies Absciscic acid, not Geometric growth.

BIO-12-CH07-0027 | Medium | Nitrogen metabolism

Correct answer: B. Conversion of atmospheric nitrogen to ammonia - Nitrogen fixation

Solution: The accurate entry is 'Conversion of atmospheric nitrogen to ammonia - Nitrogen fixation'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Response of flowering to relative day and night length' should be paired with Photoperiodism, not Absciscic acid. C: 'Growth phase dominated by active cell division' should be paired with Meristematic phase, not 2,4-D. D: 'Hormone used to promote rooting in stem cuttings' should be paired with Auxin, not Nitrogen fixation.

BIO-12-CH07-0028 | Challenging | Nitrogen metabolism

Correct answer: A. I only

Solution: Statement I is correct because 'Conversion of atmospheric nitrogen to ammonia' corresponds to Nitrogen fixation. Statement II is incorrect: 'Symbiotic nitrogen-fixing bacterium in legume nodules' corresponds to Rhizobium, not Denitrification. Statement III is incorrect: 'Reduction of nitrate to atmospheric nitrogen' corresponds to Denitrification, not Phosphorus. Statement IV is incorrect: 'Element important in ATP and nucleic acids' corresponds to Phosphorus, not Nitrogen fixation. Therefore, the valid combination is I only.

Distractor review: B: This combination includes false statement(s) II. C: This combination includes false statement(s) III and omits true statement(s) I. D: This combination includes false statement(s) II, III, IV and omits true statement(s) I.

BIO-12-CH07-0029 | Hard | Plant hormones II

Correct answer: D. Hormone delaying leaf senescence - Denitrification

Solution: 'Hormone delaying leaf senescence' should be paired with Cytokinin, not Denitrification. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Conversion of atmospheric nitrogen to ammonia - Nitrogen fixation. B: This is a valid association: Central atom of chlorophyll - Magnesium. C: This is a valid association: Growth occurring throughout life in meristematic regions - Indeterminate growth.

BIO-12-CH07-0030 | Medium | Plant hormones II**Correct answer: B. Hormone associated with stomatal closure and stress - Absciscic acid**

Solution: The accurate entry is 'Hormone associated with stomatal closure and stress - Absciscic acid'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Hormone promoting stem elongation and bolting' should be paired with Gibberellin, not Magnesium. C: 'Increase by a constant amount per unit time' should be paired with Arithmetic growth, not Absciscic acid. D: 'Symbiotic nitrogen-fixing bacterium in legume nodules' should be paired with Rhizobium, not Absolute growth rate.

BIO-12-CH07-0031 | Easy | Growth fundamentals**Correct answer: C. Overall sequence from germination to senescence**

Solution: Development represents 'Overall sequence from germination to senescence'. Each alternative describes a different term or result.

Distractor review: A: 'Growth per unit initial size per unit time' identifies Relative growth rate, not Development. B: 'Permanent irreversible increase in size' identifies Growth, not Development. D: 'Reduction of nitrate to atmospheric nitrogen' identifies Denitrification, not Development.

BIO-12-CH07-0032 | Hard | Plant hormones II**Correct answer: C. Synthetic auxin widely used as a selective weedicide - 2,4-D; Typical growth curve of a whole organ - Sigmoid curve**

Solution: Both parts of 'Synthetic auxin widely used as a selective weedicide - 2,4-D; Typical growth curve of a whole organ - Sigmoid curve' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Increase proportional to the existing size' requires Geometric growth, and 'Growth per unit initial size per unit time' requires Relative growth rate. B: The answers have been interchanged: 'Total growth per unit time' requires Absolute growth rate, and 'Element required for synthesis of amino acids and nucleic acids' requires Nitrogen. D: The answers have been interchanged: 'Element required for synthesis of amino acids and nucleic acids' requires Nitrogen, and 'Increase proportional to the existing size' requires Geometric growth.

BIO-12-CH07-0033 | Medium | Plant hormones II**Correct answer: D. Synthetic auxin widely used as a selective weedicide - 2,4-D**

Solution: The accurate entry is 'Synthetic auxin widely used as a selective weedicide - 2,4-D'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Total growth per unit time' should be paired with Absolute growth rate, not Geometric growth. B: 'Hormone promoting cell elongation and apical dominance' should be paired with Auxin, not Nitrogen. C: 'Element required for synthesis of amino acids and nucleic acids' should be paired with Nitrogen, not Relative growth rate.

BIO-12-CH07-0034 | Easy | Growth fundamentals**Correct answer: D. Growth occurring throughout life in meristematic regions**

Solution: Indeterminate growth represents 'Growth occurring throughout life in meristematic regions'. Each alternative describes a different term or result.

Distractor review: A: 'Overall sequence from germination to senescence' identifies Development, not Indeterminate growth. B: 'Permanent irreversible increase in size' identifies Growth, not Indeterminate growth. C: 'Conversion of atmospheric nitrogen to ammonia' identifies Nitrogen fixation, not Indeterminate growth.

BIO-12-CH07-0035 | Medium | Phases and curves**Correct answer: C. Sigmoid curve**

Solution: The chapter links Typical growth curve of a whole organ with Sigmoid curve, so Sigmoid curve completes the statement.

Distractor review: A: Maturation phase belongs with 'Phase in which cells acquire specialized form and function', whereas the tested description is 'Typical growth curve of a whole organ'. B: Vernalization belongs with 'Cold treatment promoting flowering', whereas the tested description is 'Typical growth curve of a whole organ'. D: Phytochrome belongs with 'Pigment involved in photoperiod perception', whereas the tested description is 'Typical growth curve of a whole organ'.

BIO-12-CH07-0036 | Easy | Plant hormones I**Correct answer: B. Gaseous hormone promoting fruit ripening**

Solution: Ethylene represents 'Gaseous hormone promoting fruit ripening'. Each alternative describes a different term or result.

Distractor review: A: 'Hormone associated with stomatal closure and stress' identifies Absciscic acid, not Ethylene. C: 'Hormone promoting cell elongation and apical dominance' identifies Auxin, not Ethylene. D: 'Element important in ATP and nucleic acids' identifies Phosphorus, not Ethylene.

BIO-12-CH07-0037 | Medium | Growth rates**Correct answer: C. Arithmetic growth**

Solution: The observation 'Increase by a constant amount per unit time' points to Arithmetic growth; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Denitrification belongs with 'Reduction of nitrate to atmospheric nitrogen', whereas the tested description is 'Increase by a constant amount per unit time'. B: Elongation phase belongs with 'Phase dominated by cell enlargement', whereas the tested description is 'Increase by a constant amount per unit time'. D: 2,4-D belongs with 'Synthetic auxin widely used as a selective weedicide', whereas the tested description is 'Increase by a constant amount per unit time'.

BIO-12-CH07-0038 | Medium | Photoperiodism and vernalization**Correct answer: C. Photoperiodism**

Solution: The chapter links Response of flowering to relative day and night length with Photoperiodism, so Photoperiodism completes the statement.

Distractor review: A: Cytokinin belongs with 'Hormone delaying leaf senescence', whereas the tested description is 'Response of flowering to relative day and night length'. B: Nitrogen fixation belongs with 'Conversion of atmospheric nitrogen to ammonia', whereas the tested description is 'Response of flowering to relative day and night length'. D: Vernalization belongs with 'Cold treatment promoting flowering', whereas the tested description is 'Response of flowering to relative day and night length'.

BIO-12-CH07-0039 | Easy | Phases and curves**Correct answer: B. Growth phase dominated by active cell division**

Solution: Meristematic phase represents 'Growth phase dominated by active cell division'. Each alternative describes a different term or result.

Distractor review: A: 'Phase dominated by cell enlargement' identifies Elongation phase, not Meristematic phase. C: 'Typical growth curve of a whole organ' identifies Sigmoid curve, not Meristematic phase. D: 'Cold treatment promoting flowering' identifies Vernalization, not Meristematic phase.

BIO-12-CH07-0040 | Medium | Phases and curves**Correct answer: A. Elongation phase**

Solution: The observation 'Phase dominated by cell enlargement' points to Elongation phase; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Development belongs with 'Overall sequence from germination to senescence', whereas the tested description is 'Phase dominated by cell enlargement'. C: Sigmoid curve belongs with 'Typical growth curve of a whole organ', whereas the tested description is 'Phase dominated by cell enlargement'. D: Auxin belongs with 'Hormone used to promote rooting in stem cuttings', whereas the tested description is 'Phase dominated by cell enlargement'.

BIO-12-CH07-0041 | Easy | Growth fundamentals**Correct answer: C. Indeterminate growth**

Solution: 'Growth occurring throughout life in meristematic regions' is the defining description for Indeterminate growth. The other choices belong to different chapter concepts.

Distractor review: A: Growth belongs with 'Permanent irreversible increase in size', whereas the tested description is 'Growth occurring throughout life in meristematic regions'. B: Development belongs with 'Overall sequence from germination to senescence', whereas the tested description is 'Growth occurring throughout life in meristematic regions'. D: Nitrogen fixation belongs with 'Conversion of atmospheric nitrogen to ammonia', whereas the tested description is 'Growth occurring throughout life in meristematic regions'.

BIO-12-CH07-0042 | Hard | Phases and curves**Correct answer: B. Growth phase dominated by active cell division - Vernalization**

Solution: 'Growth phase dominated by active cell division' should be paired with Meristematic phase, not Vernalization. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Hormone promoting cell division - Cytokinin. C: This is a valid association: Hormone promoting stem elongation and bolting - Gibberellin. D: This is a valid association: Response of flowering to relative day and night length - Photoperiodism.

BIO-12-CH07-0043 | Medium | Plant hormones I**Correct answer: A. Auxin**

Solution: The observation 'Hormone promoting cell elongation and apical dominance' points to Auxin; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Growth rate belongs with 'Growth measurable as increase per unit time', whereas the tested description is 'Hormone promoting cell elongation and apical dominance'. C: Development belongs with 'Overall sequence from germination to senescence', whereas the tested description is 'Hormone promoting cell elongation and apical dominance'. D: Nitrogen belongs with 'Element required for synthesis of amino acids and nucleic acids', whereas the tested description is 'Hormone promoting cell elongation and apical dominance'.

BIO-12-CH07-0044 | Easy | Plant hormones II**Correct answer: B. Auxin**

Solution: 'Hormone used to promote rooting in stem cuttings' is the defining description for Auxin. The other choices belong to different chapter concepts.

Distractor review: A: 2,4-D belongs with 'Synthetic auxin widely used as a selective weedicide', whereas the tested description is 'Hormone used to promote rooting in stem cuttings'. C: Indeterminate growth belongs with 'Growth occurring throughout life in meristematic regions', whereas the tested description is 'Hormone used to promote rooting in stem cuttings'. D: Rhizobium belongs with 'Symbiotic nitrogen-fixing bacterium in legume nodules', whereas the tested description is 'Hormone used to promote rooting in stem cuttings'.

BIO-12-CH07-0045 | Easy | Plant hormones II**Correct answer: B. Absciscic acid**

Solution: 'Hormone associated with stomatal closure and stress' is the defining description for Absciscic acid. The other choices belong to different chapter concepts.

Distractor review: A: Denitrification belongs with 'Reduction of nitrate to atmospheric nitrogen', whereas the tested description is 'Hormone associated with stomatal closure and stress'. C: Indeterminate growth belongs with 'Growth occurring throughout life in meristematic regions', whereas the tested description is 'Hormone associated with stomatal closure and stress'. D: Gibberellin belongs with 'Hormone promoting stem elongation and bolting', whereas the tested description is 'Hormone associated with stomatal closure and stress'.

BIO-12-CH07-0046 | Medium | Growth rates**Correct answer: C. Geometric growth**

Solution: The chapter links Increase proportional to the existing size with Geometric growth, so Geometric growth completes the statement.

Distractor review: A: Growth belongs with 'Permanent irreversible increase in size', whereas the tested description is 'Increase proportional to the existing size'. B: Development belongs with 'Overall sequence from germination to senescence', whereas the tested description is 'Increase proportional to the existing size'. D: Absciscic acid belongs with 'Hormone associated with stomatal closure and stress', whereas the tested description is 'Increase proportional to the existing size'.

BIO-12-CH07-0047 | Hard | Phases and curves**Correct answer: A. Growth phase dominated by active cell division - Meristematic phase; Cold treatment promoting flowering - Vernalization**

Solution: Both parts of 'Growth phase dominated by active cell division - Meristematic phase; Cold treatment promoting flowering - Vernalization' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Reduction of nitrate to atmospheric nitrogen' requires Denitrification, and 'Hormone promoting stem elongation and bolting' requires Gibberellin. C: The answers have been interchanged: 'Hormone promoting stem elongation and bolting' requires Gibberellin, and 'Hormone promoting cell division' requires Cytokinin. D: The answers have been interchanged: 'Response of flowering to relative day and night length' requires Photoperiodism, and 'Gaseous hormone promoting fruit ripening' requires Ethylene.

BIO-12-CH07-0048 | Easy | Nitrogen metabolism**Correct answer: D. Denitrification**

Solution: 'Reduction of nitrate to atmospheric nitrogen' is the defining description for Denitrification. The other choices belong to different chapter concepts.

Distractor review: A: Vernalization belongs with 'Cold treatment promoting flowering', whereas the tested description is 'Reduction of nitrate to atmospheric nitrogen'. B: Short-day plant belongs with 'Plant flowering when day length is shorter than a critical value', whereas the tested description is 'Reduction of nitrate to atmospheric nitrogen'. C: Ethylene belongs with 'Gaseous hormone promoting fruit ripening', whereas the tested description is 'Reduction of nitrate to atmospheric nitrogen'.

BIO-12-CH07-0049 | Easy | Nitrogen metabolism**Correct answer: A. Nitrogen fixation**

Solution: 'Conversion of atmospheric nitrogen to ammonia' is the defining description for Nitrogen fixation. The other choices belong to different chapter concepts.

Distractor review: B: Denitrification belongs with 'Reduction of nitrate to atmospheric nitrogen', whereas the tested description is 'Conversion of atmospheric nitrogen to ammonia'. C: Photoperiodism belongs with 'Response of flowering to relative day and night length', whereas the tested description is 'Conversion of atmospheric nitrogen to ammonia'. D: Nitrification belongs with 'Conversion of ammonia to nitrite and nitrate', whereas the tested description is 'Conversion of atmospheric nitrogen to ammonia'.

BIO-12-CH07-0050 | Easy | Essential mineral elements**Correct answer: D. Potassium**

Solution: 'Element required for functioning of stomata and enzyme activation' is the defining description for Potassium. The other choices belong to different chapter concepts.

Distractor review: A: Cytokinin belongs with 'Hormone promoting cell division', whereas the tested description is 'Element required for functioning of stomata and enzyme activation'. B: Nitrogen belongs with 'Element required for synthesis of amino acids and nucleic acids', whereas the tested description is 'Element required for functioning of stomata and enzyme activation'. C: Absciscic acid belongs with 'Hormone associated with stomatal closure and stress', whereas the tested description is 'Element required for functioning of stomata and enzyme activation'.

BIO-12-CH07-0051 | Hard | Essential mineral elements**Correct answer: C. Element required for functioning of stomata and enzyme activation - Geometric growth**

Solution: 'Element required for functioning of stomata and enzyme activation' should be paired with Potassium, not Geometric growth. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Response of flowering to relative day and night length - Photoperiodism. B: This is a valid association: Hormone promoting cell division - Cytokinin. D: This is a valid association: Synthetic auxin widely used as a selective weedicide - 2,4-D.

BIO-12-CH07-0052 | Medium | Nitrogen metabolism**Correct answer: D. Rhizobium**

Solution: The chapter links Symbiotic nitrogen-fixing bacterium in legume nodules with Rhizobium, so Rhizobium completes the statement.

Distractor review: A: Indeterminate growth belongs with 'Growth occurring throughout life in meristematic regions', whereas the tested description is 'Symbiotic nitrogen-fixing bacterium in legume nodules'. B: 2,4-D belongs with 'Synthetic auxin widely used as a selective weedicide', whereas the tested description is 'Symbiotic nitrogen-fixing bacterium in legume nodules'. C: Gibberellin belongs with 'Hormone promoting stem elongation and bolting', whereas the tested description is 'Symbiotic nitrogen-fixing bacterium in legume nodules'.

BIO-12-CH07-0053 | Hard | Nitrogen metabolism**Correct answer: D. Conversion of atmospheric nitrogen to ammonia - Photoperiodism**

Solution: 'Conversion of atmospheric nitrogen to ammonia' should be paired with Nitrogen fixation, not Photoperiodism. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Hormone delaying leaf senescence - Cytokinin. B: This is a valid association: Overall sequence from germination to senescence - Development. C: This is a valid association: Hormone associated with stomatal closure and stress - Absciscic acid.

BIO-12-CH07-0054 | Medium | Phases and curves**Correct answer: A. Phase dominated by cell enlargement - Elongation phase**

Solution: The accurate entry is 'Phase dominated by cell enlargement - Elongation phase'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Hormone delaying leaf senescence' should be paired with Cytokinin, not Phosphorus. C: 'Symbiotic nitrogen-fixing bacterium in legume nodules' should be paired with Rhizobium, not Elongation phase. D: 'Element required for synthesis of amino acids and nucleic acids' should be paired with Nitrogen, not Rhizobium.

BIO-12-CH07-0055 | Medium | Growth fundamentals**Correct answer: D. Growth rate**

Solution: The observation 'Growth measurable as increase per unit time' points to Growth rate; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Maturation phase belongs with 'Phase in which cells acquire specialized form and function', whereas the tested description is 'Growth measurable as increase per unit time'. B: Cytokinin belongs with 'Hormone delaying leaf senescence', whereas the tested description is 'Growth measurable as increase per unit time'. C: Auxin belongs with 'Hormone used to promote rooting in stem cuttings', whereas the tested description is 'Growth measurable as increase per unit time'.

BIO-12-CH07-0056 | Challenging | Plant hormones II**Correct answer: A. II and III only**

Solution: Statement I is incorrect: 'Synthetic auxin widely used as a selective weedicide' corresponds to 2,4-D, not Cytokinin. Statement II is correct because 'Hormone delaying leaf senescence' corresponds to Cytokinin. Statement III is correct because 'Hormone associated with stomatal closure and stress' corresponds to Absciscic acid. Statement IV is incorrect: 'Typical growth curve of a whole organ' corresponds to Sigmoid curve, not 2,4-D. Therefore, the valid combination is II and III only.

Distractor review: B: This combination includes false statement(s) I, IV. C: This combination includes false statement(s) I and omits true statement(s) II. D: This combination includes false statement(s) I.

BIO-12-CH07-0057 | Medium | Plant hormones I**Correct answer: A. Hormone promoting cell elongation and apical dominance - Auxin**

Solution: The accurate entry is 'Hormone promoting cell elongation and apical dominance - Auxin'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Conversion of atmospheric nitrogen to ammonia' should be paired with Nitrogen fixation, not Auxin. C: 'Element required for synthesis of amino acids and nucleic acids' should be paired with Nitrogen, not Vernalization. D: 'Growth measurable as increase per unit time' should be paired with Growth rate, not Nitrogen fixation.

BIO-12-CH07-0058 | Easy | Essential mineral elements**Correct answer: D. Element important in ATP and nucleic acids**

Solution: Phosphorus represents 'Element important in ATP and nucleic acids'. Each alternative describes a different term or result.

Distractor review: A: 'Conversion of atmospheric nitrogen to ammonia' identifies Nitrogen fixation, not Phosphorus. B: 'Pigment involved in photoperiod perception' identifies Phytochrome, not Phosphorus. C: 'Gaseous hormone promoting fruit ripening' identifies Ethylene, not Phosphorus.

BIO-12-CH07-0059 | Hard | Growth fundamentals**Correct answer: A. Growth occurring throughout life in meristematic regions - Indeterminate growth; Synthetic auxin widely used as a selective weedicide - 2,4-D**

Solution: Both parts of 'Growth occurring throughout life in meristematic regions - Indeterminate growth; Synthetic auxin widely used as a selective weedicide - 2,4-D' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Conversion of atmospheric nitrogen to ammonia' requires Nitrogen fixation, and 'Hormone promoting cell elongation and apical dominance' requires Auxin. C: The answers have been interchanged: 'Element required for synthesis of amino acids and nucleic acids' requires Nitrogen, and 'Increase proportional to the existing size' requires Geometric growth. D: The answers have been interchanged: 'Increase proportional to the existing size' requires Geometric growth, and 'Typical growth curve of a whole organ' requires Sigmoid curve.

BIO-12-CH07-0060 | Medium | Photoperiodism and vernalization**Correct answer: C. Pigment involved in photoperiod perception - Phytochrome**

Solution: The accurate entry is 'Pigment involved in photoperiod perception - Phytochrome'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Permanent irreversible increase in size' should be paired with Growth, not Meristematic phase. B: 'Hormone promoting stem elongation and bolting' should be paired with Gibberellin, not Magnesium. D: 'Growth phase dominated by active cell division' should be paired with Meristematic phase, not Phytochrome.

BIO-12-CH07-0061 | Challenging | Growth rates**Correct answer: B. I, II, III and IV**

Solution: Statement I is correct because 'Growth per unit initial size per unit time' corresponds to Relative growth rate. Statement II is correct because 'Increase proportional to the existing size' corresponds to Geometric growth. Statement III is correct because 'Total growth per unit time' corresponds to Absolute growth rate. Statement IV is correct because 'Conversion of ammonia to nitrite and nitrate' corresponds to Nitrification. Therefore, the valid combination is I, II, III and IV.

Distractor review: A: This combination omits true statement(s) II, III, IV. C: This combination omits true statement(s) II, IV. D: This combination omits true statement(s) III, IV.

BIO-12-CH07-0062 | Challenging | Phases and curves**Correct answer: D. I and IV only**

Solution: Statement I is correct because 'Typical growth curve of a whole organ' corresponds to Sigmoid curve. Statement II is incorrect: 'Phase dominated by cell enlargement' corresponds to Elongation phase, not Meristematic phase. Statement III is incorrect: 'Growth phase dominated by active cell division' corresponds to Meristematic phase, not Cytokinin. Statement IV is correct because 'Hormone promoting cell division' corresponds to Cytokinin. Therefore, the valid combination is I and IV only.

Distractor review: A: This combination includes false statement(s) II, III and omits true statement(s) I. B: This combination includes false statement(s) III and omits true statement(s) I. C: This combination omits true statement(s) I.

BIO-12-CH07-0063 | Medium | Growth fundamentals**Correct answer: B. Indeterminate growth**

Solution: The observation 'Growth occurring throughout life in meristematic regions' points to Indeterminate growth; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Nitrogen belongs with 'Element required for synthesis of amino acids and nucleic acids', whereas the tested description is 'Growth occurring throughout life in meristematic regions'. C: Development belongs with 'Overall sequence from germination to senescence', whereas the tested description is 'Growth occurring throughout life in meristematic regions'. D: 2,4-D belongs with 'Synthetic auxin widely used as a selective weedicide', whereas the tested description is 'Growth occurring throughout life in meristematic regions'.

BIO-12-CH07-0064 | Medium | Essential mineral elements**Correct answer: A. Phosphorus**

Solution: The chapter links Element important in ATP and nucleic acids with Phosphorus, so Phosphorus completes the statement.

Distractor review: B: Magnesium belongs with 'Central atom of chlorophyll', whereas the tested description is 'Element important in ATP and nucleic acids'. C: Nitrogen belongs with 'Element required for synthesis of amino acids and nucleic acids', whereas the tested description is 'Element important in ATP and nucleic acids'. D: Absolute growth rate belongs with 'Total growth per unit time', whereas the tested description is 'Element important in ATP and nucleic acids'.

BIO-12-CH07-0065 | Medium | Photoperiodism and vernalization**Correct answer: A. Phytochrome**

Solution: The observation 'Pigment involved in photoperiod perception' points to Phytochrome; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Short-day plant belongs with 'Plant flowering when day length is shorter than a critical value', whereas the tested description is 'Pigment involved in photoperiod perception'. C: Photoperiodism belongs with 'Response of flowering to relative day and night length', whereas the tested description is 'Pigment involved in photoperiod perception'. D: Ethylene belongs with 'Gaseous hormone promoting fruit ripening', whereas the tested description is 'Pigment involved in photoperiod perception'.

BIO-12-CH07-0066 | Medium | Plant hormones II**Correct answer: D. Cytokinin**

Solution: The chapter links Hormone delaying leaf senescence with Cytokinin, so Cytokinin completes the statement.

Distractor review: A: Denitrification belongs with 'Reduction of nitrate to atmospheric nitrogen', whereas the tested description is 'Hormone delaying leaf senescence'. B: 2,4-D belongs with 'Synthetic auxin widely used as a selective weedicide', whereas the tested description is 'Hormone delaying leaf senescence'. C: Geometric growth belongs with 'Increase proportional to the existing size', whereas the tested description is 'Hormone delaying leaf senescence'.

BIO-12-CH07-0067 | Hard | Phases and curves**Correct answer: D. Phase dominated by cell enlargement - Elongation phase; Hormone used to promote rooting in stem cuttings - Auxin**

Solution: Both parts of 'Phase dominated by cell enlargement - Elongation phase; Hormone used to promote rooting in stem cuttings - Auxin' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Cold treatment promoting flowering' requires Vernalization, and 'Element required for synthesis of amino acids and nucleic acids' requires Nitrogen. B: The answers have been interchanged: 'Symbiotic nitrogen-fixing bacterium in legume nodules' requires Rhizobium, and 'Element important in ATP and nucleic acids' requires Phosphorus. C: The answers have been interchanged: 'Overall sequence from germination to senescence' requires Development, and 'Cold treatment promoting flowering' requires Vernalization.

BIO-12-CH07-0068 | Hard | Essential mineral elements**Correct answer: B. Element important in ATP and nucleic acids - Ethylene**

Solution: 'Element important in ATP and nucleic acids' should be paired with Phosphorus, not Ethylene. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Growth per unit initial size per unit time - Relative growth rate. C: This is a valid association: Response of flowering to relative day and night length - Photoperiodism. D: This is a valid association: Total growth per unit time - Absolute growth rate.

BIO-12-CH07-0069 | Medium | Growth fundamentals**Correct answer: A. Growth**

Solution: The chapter links Permanent irreversible increase in size with Growth, so Growth completes the statement.

Distractor review: B: Auxin belongs with 'Hormone used to promote rooting in stem cuttings', whereas the tested description is 'Permanent irreversible increase in size'. C: Growth rate belongs with 'Growth measurable as increase per unit time', whereas the tested description is 'Permanent irreversible increase in size'. D: Potassium belongs with 'Element required for functioning of stomata and enzyme activation', whereas the tested description is 'Permanent irreversible increase in size'.

BIO-12-CH07-0070 | Medium | Nitrogen metabolism**Correct answer: D. Symbiotic nitrogen-fixing bacterium in legume nodules - Rhizobium**

Solution: The accurate entry is 'Symbiotic nitrogen-fixing bacterium in legume nodules - Rhizobium'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Pigment involved in photoperiod perception' should be paired with Phytochrome, not Rhizobium. B: 'Total growth per unit time' should be paired with Absolute growth rate, not Sigmoid curve. C: 'Typical growth curve of a whole organ' should be paired with Sigmoid curve, not Phytochrome.

BIO-12-CH07-0071 | Medium | Phases and curves**Correct answer: C. Meristematic phase**

Solution: The chapter links Growth phase dominated by active cell division with Meristematic phase, so Meristematic phase completes the statement.

Distractor review: A: Gibberellin belongs with 'Hormone promoting stem elongation and bolting', whereas the tested description is 'Growth phase dominated by active cell division'. B: Sigmoid curve belongs with 'Typical growth curve of a whole organ', whereas the tested description is 'Growth phase dominated by active cell division'. D: Elongation phase belongs with 'Phase dominated by cell enlargement', whereas the tested description is 'Growth phase dominated by active cell division'.

BIO-12-CH07-0072 | Medium | Essential mineral elements**Correct answer: D. Magnesium**

Solution: The observation 'Central atom of chlorophyll' points to Magnesium; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Abscissic acid belongs with 'Hormone associated with stomatal closure and stress', whereas the tested description is 'Central atom of chlorophyll'. B: Nitrogen fixation belongs with 'Conversion of atmospheric nitrogen to ammonia', whereas the tested description is 'Central atom of chlorophyll'. C: Nitrogen belongs with 'Element required for synthesis of amino acids and nucleic acids', whereas the tested description is 'Central atom of chlorophyll'.

BIO-12-CH07-0073 | Medium | Nitrogen metabolism**Correct answer: C. Denitrification**

Solution: The chapter links Reduction of nitrate to atmospheric nitrogen with Denitrification, so Denitrification completes the statement.

Distractor review: A: Potassium belongs with 'Element required for functioning of stomata and enzyme activation', whereas the tested description is 'Reduction of nitrate to atmospheric nitrogen'. B: Vernalization belongs with 'Cold treatment promoting flowering', whereas the tested description is 'Reduction of nitrate to atmospheric nitrogen'. D: Ethylene belongs with 'Gaseous hormone promoting fruit ripening', whereas the tested description is 'Reduction of nitrate to atmospheric nitrogen'.

BIO-12-CH07-0074 | Hard | Photoperiodism and vernalization**Correct answer: C. Cold treatment promoting flowering - Vernalization; Typical growth curve of a whole organ - Sigmoid curve**

Solution: Both parts of 'Cold treatment promoting flowering - Vernalization; Typical growth curve of a whole organ - Sigmoid curve' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Symbiotic nitrogen-fixing bacterium in legume nodules' requires Rhizobium, and 'Element required for synthesis of amino acids and nucleic acids' requires Nitrogen. B: The answers have been interchanged: 'Phase in which cells acquire specialized form and function' requires Maturation phase, and 'Increase by a constant amount per unit time' requires Arithmetic growth. D: The answers have been interchanged: 'Element required for functioning of stomata and enzyme activation' requires Potassium, and 'Symbiotic nitrogen-fixing bacterium in legume nodules' requires Rhizobium.

BIO-12-CH07-0075 | Medium | Plant hormones I**Correct answer: B. Auxin**

Solution: The chapter links Hormone promoting cell elongation and apical dominance with Auxin, so Auxin completes the statement.

Distractor review: A: Gibberellin belongs with 'Hormone promoting stem elongation and bolting', whereas the tested description is 'Hormone promoting cell elongation and apical dominance'. C: Magnesium belongs with 'Central atom of chlorophyll', whereas the tested description is 'Hormone promoting cell elongation and apical dominance'. D: Development belongs with 'Overall sequence from germination to senescence', whereas the tested description is 'Hormone promoting cell elongation and apical dominance'.

BIO-12-CH07-0076 | Easy | Plant hormones II**Correct answer: B. Synthetic auxin widely used as a selective weedicide**

Solution: 2,4-D represents 'Synthetic auxin widely used as a selective weedicide'. Each alternative describes a different term or result.

Distractor review: A: 'Hormone delaying leaf senescence' identifies Cytokinin, not 2,4-D. C: 'Element required for synthesis of amino acids and nucleic acids' identifies Nitrogen, not 2,4-D. D: 'Increase proportional to the existing size' identifies Geometric growth, not 2,4-D.

BIO-12-CH07-0077 | Medium | Growth rates**Correct answer: D. Absolute growth rate**

Solution: The observation 'Total growth per unit time' points to Absolute growth rate; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Development belongs with 'Overall sequence from germination to senescence', whereas the tested description is 'Total growth per unit time'. B: Auxin belongs with 'Hormone promoting cell elongation and apical dominance', whereas the tested description is 'Total growth per unit time'. C: Elongation phase belongs with 'Phase dominated by cell enlargement', whereas the tested description is 'Total growth per unit time'.

BIO-12-CH07-0078 | Medium | Plant hormones II**Correct answer: D. Hormone used to promote rooting in stem cuttings - Auxin**

Solution: The accurate entry is 'Hormone used to promote rooting in stem cuttings - Auxin'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Cold treatment promoting flowering' should be paired with Vernalization, not Indeterminate growth. B: 'Element important in ATP and nucleic acids' should be paired with Phosphorus, not Auxin. C: 'Symbiotic nitrogen-fixing bacterium in legume nodules' should be paired with Rhizobium, not Phytochrome.

BIO-12-CH07-0079 | Medium | Growth fundamentals**Correct answer: C. Development**

Solution: The chapter links Overall sequence from germination to senescence with Development, so Development completes the statement.

Distractor review: A: Magnesium belongs with 'Central atom of chlorophyll', whereas the tested description is 'Overall sequence from germination to senescence'. B: Cytokinin belongs with 'Hormone delaying leaf senescence', whereas the tested description is 'Overall sequence from germination to senescence'. D: Denitrification belongs with 'Reduction of nitrate to atmospheric nitrogen', whereas the tested description is 'Overall sequence from germination to senescence'.

BIO-12-CH07-0080 | Easy | Photoperiodism and vernalization**Correct answer: D. Vernalization**

Solution: 'Cold treatment promoting flowering' is the defining description for Vernalization. The other choices belong to different chapter concepts.

Distractor review: A: Potassium belongs with 'Element required for functioning of stomata and enzyme activation', whereas the tested description is 'Cold treatment promoting flowering'. B: Maturation phase belongs with 'Phase in which cells acquire specialized form and function', whereas the tested description is 'Cold treatment promoting flowering'. C: Sigmoid curve belongs with 'Typical growth curve of a whole organ', whereas the tested description is 'Cold treatment promoting flowering'.

BIO-12-CH07-0081 | Medium | Photoperiodism and vernalization**Correct answer: B. Vernalization**

Solution: The chapter links Cold treatment promoting flowering with Vernalization, so Vernalization completes the statement.

Distractor review: A: Potassium belongs with 'Element required for functioning of stomata and enzyme activation', whereas the tested description is 'Cold treatment promoting flowering'. C: Short-day plant belongs with 'Plant flowering when day length is shorter than a critical value', whereas the tested description is 'Cold treatment promoting flowering'. D: Phytochrome belongs with 'Pigment involved in photoperiod perception', whereas the tested description is 'Cold treatment promoting flowering'.

BIO-12-CH07-0082 | Hard | Plant hormones I**Correct answer: B. Hormone promoting cell elongation and apical dominance - Auxin; Central atom of chlorophyll - Magnesium**

Solution: Both parts of 'Hormone promoting cell elongation and apical dominance - Auxin; Central atom of chlorophyll - Magnesium' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Element required for synthesis of amino acids and nucleic acids' requires Nitrogen, and 'Growth measurable as increase per unit time' requires Growth rate. C: The answers have been interchanged: 'Overall sequence from germination to senescence' requires Development, and 'Element required for synthesis of amino acids and nucleic acids' requires Nitrogen. D: The answers have been interchanged: 'Cold treatment promoting flowering' requires Vernalization, and 'Conversion of atmospheric nitrogen to ammonia' requires Nitrogen fixation.

BIO-12-CH07-0083 | Hard | Plant hormones I**Correct answer: C. Gaseous hormone promoting fruit ripening - Absciscic acid**

Solution: 'Gaseous hormone promoting fruit ripening' should be paired with Ethylene, not Absciscic acid. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Overall sequence from germination to senescence - Development. B: This is a valid association: Increase proportional to the existing size - Geometric growth. D: This is a valid association: Element required for synthesis of amino acids and nucleic acids - Nitrogen.

BIO-12-CH07-0084 | Challenging | Essential mineral elements**Correct answer: C. II, III and IV only**

Solution: Statement I is incorrect: 'Element important in ATP and nucleic acids' corresponds to Phosphorus, not Nitrogen. Statement II is correct because 'Element required for synthesis of amino acids and nucleic acids' corresponds to Nitrogen. Statement III is correct because 'Central atom of chlorophyll' corresponds to Magnesium. Statement IV is correct because 'Total growth per unit time' corresponds to Absolute growth rate. Therefore, the valid combination is II, III and IV only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) II, III. B: This combination includes false statement(s) I and omits true statement(s) IV. D: This combination includes false statement(s) I and omits true statement(s) III, IV.

BIO-12-CH07-0085 | Challenging | Photoperiodism and vernalization**Correct answer: A. I and III only**

Solution: Statement I is correct because 'Pigment involved in photoperiod perception' corresponds to Phytochrome. Statement II is incorrect: 'Plant flowering when day length is shorter than a critical value' corresponds to Short-day plant, not Vernalization. Statement III is correct because 'Cold treatment promoting flowering' corresponds to Vernalization. Statement IV is incorrect: 'Gaseous hormone promoting fruit ripening' corresponds to Ethylene, not Phytochrome. Therefore, the valid combination is I and III only.

Distractor review: B: This combination includes false statement(s) IV and omits true statement(s) I, III. C: This combination includes false statement(s) II, IV. D: This combination omits true statement(s) I.

BIO-12-CH07-0086 | Medium | Essential mineral elements**Correct answer: A. Element required for synthesis of amino acids and nucleic acids - Nitrogen**

Solution: The accurate entry is 'Element required for synthesis of amino acids and nucleic acids - Nitrogen'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Hormone used to promote rooting in stem cuttings' should be paired with Auxin, not Cytokinin. C: 'Growth occurring throughout life in meristematic regions' should be paired with Indeterminate growth, not Growth rate. D: 'Growth phase dominated by active cell division' should be paired with Meristematic phase, not Nitrogen.

BIO-12-CH07-0087 | Hard | Growth rates**Correct answer: D. Total growth per unit time - Nitrification**

Solution: 'Total growth per unit time' should be paired with Absolute growth rate, not Nitrification. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Hormone promoting cell elongation and apical dominance - Auxin. B: This is a valid association: Hormone associated with stomatal closure and stress - Absciscic acid. C: This is a valid association: Element required for synthesis of amino acids and nucleic acids - Nitrogen.

BIO-12-CH07-0088 | Hard | Growth fundamentals**Correct answer: A. Overall sequence from germination to senescence - Development; Conversion of ammonia to nitrite and nitrate - Nitrification**

Solution: Both parts of 'Overall sequence from germination to senescence - Development; Conversion of ammonia to nitrite and nitrate - Nitrification' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Reduction of nitrate to atmospheric nitrogen' requires Denitrification, and 'Growth per unit initial size per unit time' requires Relative growth rate. C: The answers have been interchanged: 'Response of flowering to relative day and night length' requires Photoperiodism, and 'Cold treatment promoting flowering' requires Vernalization. D: The answers have been interchanged: 'Growth per unit initial size per unit time' requires Relative growth rate, and 'Response of flowering to relative day and night length' requires Photoperiodism.

BIO-12-CH07-0089 | Challenging | Plant hormones I**Correct answer: C. I, III and IV only**

Solution: Statement I is correct because 'Gaseous hormone promoting fruit ripening' corresponds to Ethylene. Statement II is incorrect: 'Hormone promoting cell elongation and apical dominance' corresponds to Auxin, not Gibberellin. Statement III is correct because 'Hormone promoting stem elongation and bolting' corresponds to Gibberellin. Statement IV is correct because 'Reduction of nitrate to atmospheric nitrogen' corresponds to Denitrification. Therefore, the valid combination is I, III and IV only.

Distractor review: A: This combination omits true statement(s) I, III. B: This combination includes false statement(s) II and omits true statement(s) I, III. D: This combination includes false statement(s) II.

BIO-12-CH07-0090 | Easy | Plant hormones I**Correct answer: A. Ethylene**

Solution: 'Gaseous hormone promoting fruit ripening' is the defining description for Ethylene. The other choices belong to different chapter concepts.

Distractor review: B: Nitrogen belongs with 'Element required for synthesis of amino acids and nucleic acids', whereas the tested description is 'Gaseous hormone promoting fruit ripening'. C: Development belongs with 'Overall sequence from germination to senescence', whereas the tested description is 'Gaseous hormone promoting fruit ripening'. D: Absciscic acid belongs with 'Hormone associated with stomatal closure and stress', whereas the tested description is 'Gaseous hormone promoting fruit ripening'.

BIO-12-CH07-0091 | Easy | Growth rates**Correct answer: C. Absolute growth rate**

Solution: 'Total growth per unit time' is the defining description for Absolute growth rate. The other choices belong to different chapter concepts.

Distractor review: A: Absciscic acid belongs with 'Hormone associated with stomatal closure and stress', whereas the tested description is 'Total growth per unit time'. B: Nitrification belongs with 'Conversion of ammonia to nitrite and nitrate', whereas the tested description is 'Total growth per unit time'. D: Auxin belongs with 'Hormone promoting cell elongation and apical dominance', whereas the tested description is 'Total growth per unit time'.

BIO-12-CH07-0092 | Medium | Nitrogen metabolism**Correct answer: B. Conversion of ammonia to nitrite and nitrate - Nitrification**

Solution: The accurate entry is 'Conversion of ammonia to nitrite and nitrate - Nitrification'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Element required for synthesis of amino acids and nucleic acids' should be paired with Nitrogen, not Auxin. C: 'Increase proportional to the existing size' should be paired with Geometric growth, not Absolute growth rate. D: 'Permanent irreversible increase in size' should be paired with Growth, not Growth rate.

BIO-12-CH07-0093 | Medium | Growth rates**Correct answer: B. Increase by a constant amount per unit time - Arithmetic growth**

Solution: The accurate entry is 'Increase by a constant amount per unit time - Arithmetic growth'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Reduction of nitrate to atmospheric nitrogen' should be paired with Denitrification, not Ethylene. C: 'Central atom of chlorophyll' should be paired with Magnesium, not Elongation phase. D: 'Phase dominated by cell enlargement' should be paired with Elongation phase, not Photoperiodism.

BIO-12-CH07-0094 | Medium | Plant hormones I**Correct answer: A. Hormone promoting stem elongation and bolting - Gibberellin**

Solution: The accurate entry is 'Hormone promoting stem elongation and bolting - Gibberellin'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Phase dominated by cell enlargement' should be paired with Elongation phase, not 2,4-D. C: 'Total growth per unit time' should be paired with Absolute growth rate, not Nitrogen. D: 'Synthetic auxin widely used as a selective weedicide' should be paired with 2,4-D, not Short-day plant.

BIO-12-CH07-0095 | Hard | Photoperiodism and vernalization**Correct answer: C. Pigment involved in photoperiod perception - Cytokinin**

Solution: 'Pigment involved in photoperiod perception' should be paired with Phytochrome, not Cytokinin. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Gaseous hormone promoting fruit ripening - Ethylene. B: This is a valid association: Hormone promoting stem elongation and bolting - Gibberellin. D: This is a valid association: Growth phase dominated by active cell division - Meristematic phase.

BIO-12-CH07-0096 | Challenging | Growth fundamentals**Correct answer: D. I and II only**

Solution: Statement I is correct because 'Permanent irreversible increase in size' corresponds to Growth. Statement II is correct because 'Overall sequence from germination to senescence' corresponds to Development. Statement III is incorrect: 'Growth occurring throughout life in meristematic regions' corresponds to Indeterminate growth, not Magnesium. Statement IV is incorrect: 'Central atom of chlorophyll' corresponds to Magnesium, not Growth. Therefore, the valid combination is I and II only.

Distractor review: A: This combination omits true statement(s) I. B: This combination includes false statement(s) III and omits true statement(s) I. C: This combination includes false statement(s) IV and omits true statement(s) II.

BIO-12-CH07-0097 | Easy | Growth rates**Correct answer: A. Growth per unit initial size per unit time**

Solution: Relative growth rate represents 'Growth per unit initial size per unit time'. Each alternative describes a different term or result.

Distractor review: B: 'Growth measurable as increase per unit time' identifies Growth rate, not Relative growth rate. C: 'Increase proportional to the existing size' identifies Geometric growth, not Relative growth rate. D: 'Hormone promoting cell division' identifies Cytokinin, not Relative growth rate.

BIO-12-CH07-0098 | Hard | Plant hormones II**Correct answer: D. Hormone used to promote rooting in stem cuttings - Auxin**

Solution: 'Hormone used to promote rooting in stem cuttings' should be paired with Auxin, not Auxin. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Element important in ATP and nucleic acids - Phosphorus. B: This is a valid association: Symbiotic nitrogen-fixing bacterium in legume nodules - Rhizobium. C: This is a valid association: Growth occurring throughout life in meristematic regions - Indeterminate growth.

BIO-12-CH07-0099 | Easy | Phases and curves**Correct answer: A. Meristematic phase**

Solution: 'Growth phase dominated by active cell division' is the defining description for Meristematic phase. The other choices belong to different chapter concepts.

Distractor review: B: Denitrification belongs with 'Reduction of nitrate to atmospheric nitrogen', whereas the tested description is 'Growth phase dominated by active cell division'. C: Photoperiodism belongs with 'Response of flowering to relative day and night length', whereas the tested description is 'Growth phase dominated by active cell division'. D: Elongation phase belongs with 'Phase dominated by cell enlargement', whereas the tested description is 'Growth phase dominated by active cell division'.

BIO-12-CH07-0100 | Easy | Plant hormones I**Correct answer: B. Hormone promoting cell elongation and apical dominance**

Solution: Auxin represents 'Hormone promoting cell elongation and apical dominance'. Each alternative describes a different term or result.

Distractor review: A: 'Central atom of chlorophyll' identifies Magnesium, not Auxin. C: 'Cold treatment promoting flowering' identifies Vernalization, not Auxin. D: 'Growth measurable as increase per unit time' identifies Growth rate, not Auxin.

Standard 12 | Chapter 08

Respiration and Circulation

BIO-12-CH08-0001 | Medium | Electrical events in one cardiac cycle

Correct answer: A. QRS complex

Solution: The QRS complex represents ventricular depolarization.

Distractor review: B: The alternative 'P wave' conflicts with the labelled relationship, scale or mechanism in the figure. The QRS complex represents ventricular depolarization. C: The alternative 'T wave' conflicts with the labelled relationship, scale or mechanism in the figure. The QRS complex represents ventricular depolarization. D: The alternative 'Interval after T only' conflicts with the labelled relationship, scale or mechanism in the figure. The QRS complex represents ventricular depolarization.

BIO-12-CH08-0002 | Hard | Oxyhaemoglobin dissociation curves

Correct answer: B. More oxygen is unloaded to tissues

Solution: A right shift lowers affinity at a given oxygen pressure and promotes tissue unloading.

Distractor review: A: The alternative 'Less oxygen is unloaded to tissues' conflicts with the labelled relationship, scale or mechanism in the figure. A right shift lowers affinity at a given oxygen pressure and promotes tissue unloading. C: The alternative 'Haemoglobin affinity becomes permanently infinite' conflicts with the labelled relationship, scale or mechanism in the figure. A right shift lowers affinity at a given oxygen pressure and promotes tissue unloading. D: The alternative 'Oxygen cannot bind in the lungs' conflicts with the labelled relationship, scale or mechanism in the figure. A right shift lowers affinity at a given oxygen pressure and promotes tissue unloading.

BIO-12-CH08-0003 | Medium | Respiratory organs

Correct answer: B. Respiratory surface in earthworm - Moist skin

Solution: The accurate entry is 'Respiratory surface in earthworm - Moist skin'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'ECG wave representing atrial depolarization' should be paired with P wave, not 0.8 second. C: 'Anucleate biconcave cells transporting respiratory gases' should be paired with Erythrocytes, not Leucocytes. D: 'Vessels containing valves to prevent backflow' should be paired with Veins, not P wave.

BIO-12-CH08-0004 | Easy | Oxyhaemoglobin dissociation curves

Correct answer: D. Cooperative binding of oxygen to haemoglobin

Solution: Binding of one oxygen molecule increases haemoglobin affinity for subsequent oxygen molecules.

Distractor review: A: The alternative 'Oxygen is insoluble in plasma' conflicts with the labelled relationship, scale or mechanism in the figure. Binding of one oxygen molecule increases haemoglobin affinity for subsequent oxygen molecules. B: The alternative 'Each haemoglobin binds only one oxygen molecule' conflicts with the labelled relationship, scale or mechanism in the figure. Binding of one oxygen molecule increases haemoglobin affinity for subsequent oxygen molecules. C: The alternative 'Carbon dioxide irreversibly destroys haemoglobin' conflicts with the labelled relationship, scale or mechanism in the figure. Binding of one oxygen molecule increases haemoglobin affinity for subsequent oxygen molecules.

BIO-12-CH08-0005 | Easy | Human respiratory tract

Correct answer: A. Microscopic sites of gaseous exchange

Solution: Alveoli represents 'Microscopic sites of gaseous exchange'. Each alternative describes a different term or result.

Distractor review: B: 'Voice box' identifies Larynx, not Alveoli. C: 'Enzyme accelerating carbonic acid formation in red cells' identifies Carbonic anhydrase, not Alveoli. D: 'Change in thoracic volume during inspiration' identifies Increase, not Alveoli.

BIO-12-CH08-0006 | Hard | Heart anatomy

Correct answer: C. Vessel carrying blood from right ventricle to lungs - Lungs

Solution: 'Vessel carrying blood from right ventricle to lungs' should be paired with Pulmonary artery, not Lungs. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: ECG complex representing ventricular depolarization - QRS complex. B: This is a valid association: Fluid returning tissue proteins and absorbed fats to blood - Lymph. D: This is a valid association: ECG wave representing atrial depolarization - P wave.

BIO-12-CH08-0007 | Medium | Blood cells**Correct answer: B. Leucocytes**

Solution: The chapter links Cells chiefly responsible for defence with Leucocytes, so Leucocytes completes the statement.

Distractor review: A: Sinoatrial node belongs with 'Natural pacemaker of the heart', whereas the tested description is 'Cells chiefly responsible for defence'. C: Platelets belongs with 'Cell fragments important in clotting', whereas the tested description is 'Cells chiefly responsible for defence'. D: Veins belongs with 'Vessels containing valves to prevent backflow', whereas the tested description is 'Cells chiefly responsible for defence'.

BIO-12-CH08-0008 | Hard | Gas transport**Correct answer: B. Major form of carbon dioxide transport - QRS complex**

Solution: 'Major form of carbon dioxide transport' should be paired with Bicarbonate ions, not QRS complex. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Cell fragments important in clotting - Platelets. C: This is a valid association: Change in thoracic volume during inspiration - Increase. D: This is a valid association: ECG wave representing atrial depolarization - P wave.

BIO-12-CH08-0009 | Easy | Respiratory organs**Correct answer: C. Tracheal system**

Solution: 'Respiratory organs in insects' is the defining description for Tracheal system. The other choices belong to different chapter concepts.

Distractor review: A: 0.8 second belongs with 'Approximate duration of one cardiac cycle at 72 beats per minute', whereas the tested description is 'Respiratory organs in insects'. B: Platelets belongs with 'Cell fragments important in clotting', whereas the tested description is 'Respiratory organs in insects'. D: Bicuspid or mitral valve belongs with 'Valve between left atrium and left ventricle', whereas the tested description is 'Respiratory organs in insects'.

BIO-12-CH08-0010 | Hard | Mechanics of breathing**Correct answer: B. Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure; Shift promoting oxygen unloading at high carbon dioxide - Bohr effect**

Solution: Both parts of 'Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure; Shift promoting oxygen unloading at high carbon dioxide - Bohr effect' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Valve between left atrium and left ventricle' requires Bicuspid or mitral valve, and 'Common passage for food and air' requires Pharynx. C: The answers have been interchanged: 'ECG wave representing atrial depolarization' requires P wave, and 'ECG complex representing ventricular depolarization' requires QRS complex. D: The answers have been interchanged: 'Vessels carrying blood away from the heart' requires Arteries, and 'Voice box' requires Larynx.

BIO-12-CH08-0011 | Medium | Mechanics of breathing**Correct answer: D. Main muscle of quiet inspiration - Diaphragm**

Solution: The accurate entry is 'Main muscle of quiet inspiration - Diaphragm'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'ECG complex representing ventricular depolarization' should be paired with QRS complex, not Diaphragm. B: 'Vessel carrying blood from right ventricle to lungs' should be paired with Pulmonary artery, not Trachea. C: 'Windpipe supported by cartilaginous rings' should be paired with Trachea, not Bicarbonate ions.

BIO-12-CH08-0012 | Medium | Heart anatomy**Correct answer: A. Pulmonary artery**

Solution: The observation 'Vessel carrying blood from right ventricle to lungs' points to Pulmonary artery; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Lungs belongs with 'Primary respiratory organs in adult humans', whereas the tested description is 'Vessel carrying blood from right ventricle to lungs'. C: Sinoatrial node belongs with 'Natural pacemaker of the heart', whereas the tested description is 'Vessel carrying blood from right ventricle to lungs'. D: Bicuspid or mitral valve belongs with 'Valve between left atrium and left ventricle', whereas the tested description is 'Vessel carrying blood from right ventricle to lungs'.

BIO-12-CH08-0013 | Medium | Mechanics of breathing**Correct answer: C. Diaphragm**

Solution: The chapter links Main muscle of quiet inspiration with Diaphragm, so Diaphragm completes the statement.

Distractor review: A: Decrease below atmospheric pressure belongs with 'Change in intrapulmonary pressure during inspiration', whereas the tested description is 'Main muscle of quiet inspiration'. B: Increase belongs with 'Change in thoracic volume during inspiration', whereas the tested description is 'Main muscle of quiet inspiration'. D: Systolic pressure belongs with 'Pressure recorded during ventricular systole', whereas the tested description is 'Main muscle of quiet inspiration'.

BIO-12-CH08-0014 | Easy | Heart anatomy**Correct answer: D. Natural pacemaker of the heart**

Solution: Sinoatrial node represents 'Natural pacemaker of the heart'. Each alternative describes a different term or result.

Distractor review: A: 'Approximate duration of one cardiac cycle at 72 beats per minute' identifies 0.8 second, not Sinoatrial node. B: 'Valve between left atrium and left ventricle' identifies Bicuspid or mitral valve, not Sinoatrial node. C: 'Respiratory surface in earthworm' identifies Moist skin, not Sinoatrial node.

BIO-12-CH08-0015 | Medium | Human respiratory tract**Correct answer: B. Trachea**

Solution: The chapter links Windpipe supported by cartilaginous rings with Trachea, so Trachea completes the statement.

Distractor review: A: Decrease below atmospheric pressure belongs with 'Change in intrapulmonary pressure during inspiration', whereas the tested description is 'Windpipe supported by cartilaginous rings'. C: Leucocytes belongs with 'Cells chiefly responsible for defence', whereas the tested description is 'Windpipe supported by cartilaginous rings'. D: Mostly passive belongs with 'Quiet expiration at rest', whereas the tested description is 'Windpipe supported by cartilaginous rings'.

BIO-12-CH08-0016 | Medium | Gas transport**Correct answer: A. Major form of oxygen transport in blood - Oxyhaemoglobin**

Solution: The accurate entry is 'Major form of oxygen transport in blood - Oxyhaemoglobin'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Pressure recorded during ventricular systole' should be paired with Systolic pressure, not Tracheal system. C: 'Main muscle of quiet inspiration' should be paired with Diaphragm, not Decrease below atmospheric pressure. D: 'Voice box' should be paired with Larynx, not P wave.

BIO-12-CH08-0017 | Medium | Respiratory organs**Correct answer: A. Gills**

Solution: The observation 'Respiratory organs in most fishes' points to Gills; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Lungs belongs with 'Primary respiratory organs in adult humans', whereas the tested description is 'Respiratory organs in most fishes'. C: Mostly passive belongs with 'Quiet expiration at rest', whereas the tested description is 'Respiratory organs in most fishes'. D: Carbonic anhydrase belongs with 'Enzyme accelerating carbonic acid formation in red cells', whereas the tested description is 'Respiratory organs in most fishes'.

BIO-12-CH08-0018 | Hard | Heart anatomy**Correct answer: B. Natural pacemaker of the heart - Sinoatrial node; Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second**

Solution: Both parts of 'Natural pacemaker of the heart - Sinoatrial node; Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Cells chiefly responsible for defence' requires Leucocytes, and 'Respiratory surface in earthworm' requires Moist skin. C: The answers have been interchanged: 'ECG complex representing ventricular depolarization' requires QRS complex, and 'Primary respiratory organs in adult humans' requires Lungs. D: The answers have been interchanged: 'Windpipe supported by cartilaginous rings' requires Trachea, and 'Cells chiefly responsible for defence' requires Leucocytes.

BIO-12-CH08-0019 | Easy | Cardiac cycle and ECG**Correct answer: A. Approximate duration of one cardiac cycle at 72 beats per minute**

Solution: 0.8 second represents 'Approximate duration of one cardiac cycle at 72 beats per minute'. Each alternative describes a different term or result.

Distractor review: B: 'Vessels containing valves to prevent backflow' identifies Veins, not 0.8 second. C: 'Respiratory surface in earthworm' identifies Moist skin, not 0.8 second. D: 'Valve between right atrium and right ventricle' identifies Tricuspid valve, not 0.8 second.

BIO-12-CH08-0020 | Hard | Respiratory organs

Correct answer: A. Respiratory organs in insects - Tracheal system; Vessel carrying blood from right ventricle to lungs - Pulmonary artery

Solution: Both parts of 'Respiratory organs in insects - Tracheal system; Vessel carrying blood from right ventricle to lungs - Pulmonary artery' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Approximate duration of one cardiac cycle at 72 beats per minute' requires 0.8 second, and 'Quiet expiration at rest' requires Mostly passive. C: The answers have been interchanged: 'Cell fragments important in clotting' requires Platelets, and 'Approximate duration of one cardiac cycle at 72 beats per minute' requires 0.8 second. D: The answers have been interchanged: 'Valve between left atrium and left ventricle' requires Bicuspid or mitral valve, and 'Cell fragments important in clotting' requires Platelets.

BIO-12-CH08-0021 | Hard | Human respiratory tract

Correct answer: D. Microscopic sites of gaseous exchange - Alveoli; Vessels carrying blood away from the heart - Arteries

Solution: Both parts of 'Microscopic sites of gaseous exchange - Alveoli; Vessels carrying blood away from the heart - Arteries' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Change in intrapulmonary pressure during inspiration' requires Decrease below atmospheric pressure, and 'Main muscle of quiet inspiration' requires Diaphragm. B: The answers have been interchanged: 'Major form of oxygen transport in blood' requires Oxyhaemoglobin, and 'Change in intrapulmonary pressure during inspiration' requires Decrease below atmospheric pressure. C: The answers have been interchanged: 'Valve between left atrium and left ventricle' requires Bicuspid or mitral valve, and 'Change in thoracic volume during inspiration' requires Increase.

BIO-12-CH08-0022 | Challenging | Electrical events in one cardiac cycle

Correct answer: D. It allows atrial emptying and ventricular filling before ventricular contraction

Solution: The AV delay gives the atria time to complete ventricular filling before ventricular depolarization and contraction.

Distractor review: A: The alternative 'It prevents atrial depolarization' conflicts with the labelled relationship, scale or mechanism in the figure. The AV delay gives the atria time to complete ventricular filling before ventricular depolarization and contraction. B: The alternative 'It causes semilunar valves to remain permanently open' conflicts with the labelled relationship, scale or mechanism in the figure. The AV delay gives the atria time to complete ventricular filling before ventricular depolarization and contraction. C: The alternative 'It initiates oxygen binding by haemoglobin' conflicts with the labelled relationship, scale or mechanism in the figure. The AV delay gives the atria time to complete ventricular filling before ventricular depolarization and contraction.

BIO-12-CH08-0023 | Easy | Human respiratory tract

Correct answer: C. Trachea

Solution: 'Windpipe supported by cartilaginous rings' is the defining description for Trachea. The other choices belong to different chapter concepts.

Distractor review: A: Alveoli belongs with 'Microscopic sites of gaseous exchange', whereas the tested description is 'Windpipe supported by cartilaginous rings'. B: P wave belongs with 'ECG wave representing atrial depolarization', whereas the tested description is 'Windpipe supported by cartilaginous rings'. D: Pharynx belongs with 'Common passage for food and air', whereas the tested description is 'Windpipe supported by cartilaginous rings'.

BIO-12-CH08-0024 | Hard | Human respiratory tract

Correct answer: B. Common passage for food and air - Moist skin

Solution: 'Common passage for food and air' should be paired with Pharynx, not Moist skin. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Anucleate biconcave cells transporting respiratory gases - Erythrocytes. C: This is a valid association: Major form of oxygen transport in blood - Oxyhaemoglobin. D: This is a valid association: Major form of carbon dioxide transport - Bicarbonate ions.

BIO-12-CH08-0025 | Easy | Cardiac cycle and ECG

Correct answer: D. ECG wave representing atrial depolarization

Solution: P wave represents 'ECG wave representing atrial depolarization'. Each alternative describes a different term or result.

Distractor review: A: 'Approximate duration of one cardiac cycle at 72 beats per minute' identifies 0.8 second, not P wave. B: 'Major form of oxygen transport in blood' identifies Oxyhaemoglobin, not P wave. C: 'Fluid returning tissue proteins and absorbed fats to blood' identifies Lymph, not P wave.

BIO-12-CH08-0026 | Hard | Respiratory organs**Correct answer: A. Respiratory surface in earthworm - Decrease below atmospheric pressure**

Solution: 'Respiratory surface in earthworm' should be paired with Moist skin, not Decrease below atmospheric pressure. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second. C: This is a valid association: Most abundant granulocyte and active phagocyte - Neutrophil. D: This is a valid association: ECG wave representing atrial depolarization - P wave.

BIO-12-CH08-0027 | Medium | Blood cells**Correct answer: C. Platelets**

Solution: The chapter links Cell fragments important in clotting with Platelets, so Platelets completes the statement.

Distractor review: A: Larynx belongs with 'Voice box', whereas the tested description is 'Cell fragments important in clotting'. B: Trachea belongs with 'Windpipe supported by cartilaginous rings', whereas the tested description is 'Cell fragments important in clotting'. D: Oxyhaemoglobin belongs with 'Major form of oxygen transport in blood', whereas the tested description is 'Cell fragments important in clotting'.

BIO-12-CH08-0028 | Medium | Cardiac cycle and ECG**Correct answer: C. QRS complex**

Solution: The observation 'ECG complex representing ventricular depolarization' points to QRS complex; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Gills belongs with 'Respiratory organs in most fishes', whereas the tested description is 'ECG complex representing ventricular depolarization'. B: Increase belongs with 'Change in thoracic volume during inspiration', whereas the tested description is 'ECG complex representing ventricular depolarization'. D: T wave belongs with 'ECG wave representing ventricular repolarization', whereas the tested description is 'ECG complex representing ventricular depolarization'.

BIO-12-CH08-0029 | Medium | Heart anatomy**Correct answer: A. Vessel carrying blood from right ventricle to lungs - Pulmonary artery**

Solution: The accurate entry is 'Vessel carrying blood from right ventricle to lungs - Pulmonary artery'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Fluid returning tissue proteins and absorbed fats to blood' should be paired with Lymph, not P wave. C: 'Primary respiratory organs in adult humans' should be paired with Lungs, not QRS complex. D: 'Microscopic sites of gaseous exchange' should be paired with Alveoli, not Lymph.

BIO-12-CH08-0030 | Hard | Mechanics of breathing**Correct answer: A. Main muscle of quiet inspiration - Moist skin**

Solution: 'Main muscle of quiet inspiration' should be paired with Diaphragm, not Moist skin. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Pressure recorded during ventricular systole - Systolic pressure. C: This is a valid association: Respiratory organs in insects - Tracheal system. D: This is a valid association: Windpipe supported by cartilaginous rings - Trachea.

BIO-12-CH08-0031 | Medium | Gas transport**Correct answer: C. Bohr effect**

Solution: The observation 'Shift promoting oxygen unloading at high carbon dioxide' points to Bohr effect; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Carbonic anhydrase belongs with 'Enzyme accelerating carbonic acid formation in red cells', whereas the tested description is 'Shift promoting oxygen unloading at high carbon dioxide'. B: Larynx belongs with 'Voice box', whereas the tested description is 'Shift promoting oxygen unloading at high carbon dioxide'. D: Bicarbonate ions belongs with 'Major form of carbon dioxide transport', whereas the tested description is 'Shift promoting oxygen unloading at high carbon dioxide'.

BIO-12-CH08-0032 | Medium | Vessels pressure and lymph**Correct answer: D. Lymph**

Solution: The observation 'Fluid returning tissue proteins and absorbed fats to blood' points to Lymph; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Pulmonary artery belongs with 'Vessel carrying blood from right ventricle to lungs', whereas the tested description is 'Fluid returning tissue proteins and absorbed fats to blood'. B: Arteries belongs with 'Vessels carrying blood away from the heart', whereas the tested description is 'Fluid returning tissue proteins and absorbed fats to blood'. C: Veins belongs with 'Vessels containing valves to prevent backflow', whereas the tested description is 'Fluid returning tissue proteins and absorbed fats to blood'.

BIO-12-CH08-0033 | Medium | Human respiratory tract**Correct answer: A. Common passage for food and air - Pharynx**

Solution: The accurate entry is 'Common passage for food and air - Pharynx'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Major form of carbon dioxide transport' should be paired with Bicarbonate ions, not Oxyhaemoglobin. C: 'Major form of oxygen transport in blood' should be paired with Oxyhaemoglobin, not Pharynx. D: 'Respiratory surface in earthworm' should be paired with Moist skin, not Erythrocytes.

BIO-12-CH08-0034 | Hard | Cardiac cycle and ECG**Correct answer: A. Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second; Vessels containing valves to prevent backflow - Veins**

Solution: Both parts of 'Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second; Vessels containing valves to prevent backflow - Veins' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Valve between left atrium and left ventricle' requires Bicuspid or mitral valve, and 'Cells chiefly responsible for defence' requires Leucocytes. C: The answers have been interchanged: 'Primary respiratory organs in adult humans' requires Lungs, and 'Valve between left atrium and left ventricle' requires Bicuspid or mitral valve. D: The answers have been interchanged: 'Respiratory surface in earthworm' requires Moist skin, and 'Valve between right atrium and right ventricle' requires Tricuspid valve.

BIO-12-CH08-0035 | Challenging | Vessels pressure and lymph**Correct answer: D. I and IV only**

Solution: Statement I is correct because 'Vessels carrying blood away from the heart' corresponds to Arteries. Statement II is incorrect: 'Fluid returning tissue proteins and absorbed fats to blood' corresponds to Lymph, not Veins. Statement III is incorrect: 'Vessels containing valves to prevent backflow' corresponds to Veins, not Pulmonary artery. Statement IV is correct because 'Vessel carrying blood from right ventricle to lungs' corresponds to Pulmonary artery. Therefore, the valid combination is I and IV only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) I. B: This combination omits true statement(s) IV. C: This combination includes false statement(s) III and omits true statement(s) I, IV.

BIO-12-CH08-0036 | Medium | Gas transport**Correct answer: C. Carbonic anhydrase**

Solution: The chapter links Enzyme accelerating carbonic acid formation in red cells with Carbonic anhydrase, so Carbonic anhydrase completes the statement.

Distractor review: A: Oxyhaemoglobin belongs with 'Major form of oxygen transport in blood', whereas the tested description is 'Enzyme accelerating carbonic acid formation in red cells'. B: P wave belongs with 'ECG wave representing atrial depolarization', whereas the tested description is 'Enzyme accelerating carbonic acid formation in red cells'. D: Tricuspid valve belongs with 'Valve between right atrium and right ventricle', whereas the tested description is 'Enzyme accelerating carbonic acid formation in red cells'.

BIO-12-CH08-0037 | Medium | Vessels pressure and lymph**Correct answer: C. Veins**

Solution: The observation 'Vessels containing valves to prevent backflow' points to Veins; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Erythrocytes belongs with 'Anucleate biconcave cells transporting respiratory gases', whereas the tested description is 'Vessels containing valves to prevent backflow'. B: Larynx belongs with 'Voice box', whereas the tested description is 'Vessels containing valves to prevent backflow'. D: Neutrophil belongs with 'Most abundant granulocyte and active phagocyte', whereas the tested description is 'Vessels containing valves to prevent backflow'.

BIO-12-CH08-0038 | Challenging | Human respiratory tract**Correct answer: D. II, III and IV only**

Solution: Statement I is incorrect: 'Windpipe supported by cartilaginous rings' corresponds to Trachea, not Larynx. Statement II is correct because 'Voice box' corresponds to Larynx. Statement III is correct because 'Common passage for food and air' corresponds to Pharynx. Statement IV is correct because 'ECG complex representing ventricular depolarization' corresponds to QRS complex. Therefore, the valid combination is II, III and IV only.

Distractor review: A: This combination includes false statement(s) I. B: This combination omits true statement(s) IV. C: This combination omits true statement(s) II.

BIO-12-CH08-0039 | Medium | Respiratory organs**Correct answer: D. Tracheal system**

Solution: The chapter links Respiratory organs in insects with Tracheal system, so Tracheal system completes the statement.

Distractor review: A: Mostly passive belongs with 'Quiet expiration at rest', whereas the tested description is 'Respiratory organs in insects'. B: Bicuspid or mitral valve belongs with 'Valve between left atrium and left ventricle', whereas the tested description is 'Respiratory organs in insects'. C: 0.8 second belongs with 'Approximate duration of one cardiac cycle at 72 beats per minute', whereas the tested description is 'Respiratory organs in insects'.

BIO-12-CH08-0040 | Medium | Human respiratory tract**Correct answer: B. Voice box - Larynx**

Solution: The accurate entry is 'Voice box - Larynx'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Quiet expiration at rest' should be paired with Mostly passive, not Larynx. C: 'Vessels containing valves to prevent backflow' should be paired with Veins, not Moist skin. D: 'Respiratory surface in earthworm' should be paired with Moist skin, not Systolic pressure.

BIO-12-CH08-0041 | Hard | Heart anatomy**Correct answer: D. Valve between left atrium and left ventricle - Bicuspid or mitral valve; Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure**

Solution: Both parts of 'Valve between left atrium and left ventricle - Bicuspid or mitral valve; Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Fluid returning tissue proteins and absorbed fats to blood' requires Lymph, and 'ECG complex representing ventricular depolarization' requires QRS complex. B: The answers have been interchanged: 'Enzyme accelerating carbonic acid formation in red cells' requires Carbonic anhydrase, and 'Fluid returning tissue proteins and absorbed fats to blood' requires Lymph. C: The answers have been interchanged: 'ECG complex representing ventricular depolarization' requires QRS complex, and 'Most abundant granulocyte and active phagocyte' requires Neutrophil.

BIO-12-CH08-0042 | Challenging | Mechanics of breathing**Correct answer: C. I and III only**

Solution: Statement I is correct because 'Change in intrapulmonary pressure during inspiration' corresponds to Decrease below atmospheric pressure. Statement II is incorrect: 'Change in thoracic volume during inspiration' corresponds to Increase, not Mostly passive. Statement III is correct because 'Quiet expiration at rest' corresponds to Mostly passive. Statement IV is incorrect: 'ECG complex representing ventricular depolarization' corresponds to QRS complex, not Decrease below atmospheric pressure. Therefore, the valid combination is I and III only.

Distractor review: A: This combination omits true statement(s) III. B: This combination omits true statement(s) I. D: This combination includes false statement(s) II and omits true statement(s) III.

BIO-12-CH08-0043 | Medium | Cardiac cycle and ECG**Correct answer: C. 0.8 second**

Solution: The chapter links Approximate duration of one cardiac cycle at 72 beats per minute with 0.8 second, so 0.8 second completes the statement.

Distractor review: A: Tricuspid valve belongs with 'Valve between right atrium and right ventricle', whereas the tested description is 'Approximate duration of one cardiac cycle at 72 beats per minute'. B: Moist skin belongs with 'Respiratory surface in earthworm', whereas the tested description is 'Approximate duration of one cardiac cycle at 72 beats per minute'. D: Veins belongs with 'Vessels containing valves to prevent backflow', whereas the tested description is 'Approximate duration of one cardiac cycle at 72 beats per minute'.

BIO-12-CH08-0044 | Easy | Human respiratory tract**Correct answer: D. Pharynx**

Solution: 'Common passage for food and air' is the defining description for Pharynx. The other choices belong to different chapter concepts.

Distractor review: A: Decrease below atmospheric pressure belongs with 'Change in intrapulmonary pressure during inspiration', whereas the tested description is 'Common passage for food and air'. B: Bicarbonate ions belongs with 'Major form of carbon dioxide transport', whereas the tested description is 'Common passage for food and air'. C: Erythrocytes belongs with 'Anucleate biconcave cells transporting respiratory gases', whereas the tested description is 'Common passage for food and air'.

BIO-12-CH08-0045 | Easy | Mechanics of breathing**Correct answer: C. Change in intrapulmonary pressure during inspiration**

Solution: Decrease below atmospheric pressure represents 'Change in intrapulmonary pressure during inspiration'. Each alternative describes a different term or result.

Distractor review: A: 'Valve between left atrium and left ventricle' identifies Bicuspid or mitral valve, not Decrease below atmospheric pressure. B: 'Shift promoting oxygen unloading at high carbon dioxide' identifies Bohr effect, not Decrease below atmospheric pressure. D: 'Vessels carrying blood away from the heart' identifies Arteries, not Decrease below atmospheric pressure.

BIO-12-CH08-0046 | Hard | Gas transport**Correct answer: B. Enzyme accelerating carbonic acid formation in red cells - Carbonic anhydrase; Voice box - Larynx**

Solution: Both parts of 'Enzyme accelerating carbonic acid formation in red cells - Carbonic anhydrase; Voice box - Larynx' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'ECG wave representing ventricular repolarization' requires T wave, and 'Cells chiefly responsible for defence' requires Leucocytes. C: The answers have been interchanged: 'Most abundant granulocyte and active phagocyte' requires Neutrophil, and 'Cell fragments important in clotting' requires Platelets. D: The answers have been interchanged: 'ECG wave representing atrial depolarization' requires P wave, and 'Valve between right atrium and right ventricle' requires Tricuspid valve.

BIO-12-CH08-0047 | Easy | Heart anatomy**Correct answer: C. Pulmonary artery**

Solution: 'Vessel carrying blood from right ventricle to lungs' is the defining description for Pulmonary artery. The other choices belong to different chapter concepts.

Distractor review: A: Lymph belongs with 'Fluid returning tissue proteins and absorbed fats to blood', whereas the tested description is 'Vessel carrying blood from right ventricle to lungs'. B: Sinoatrial node belongs with 'Natural pacemaker of the heart', whereas the tested description is 'Vessel carrying blood from right ventricle to lungs'. D: QRS complex belongs with 'ECG complex representing ventricular depolarization', whereas the tested description is 'Vessel carrying blood from right ventricle to lungs'.

BIO-12-CH08-0048 | Medium | Human respiratory tract**Correct answer: C. Larynx**

Solution: The observation 'Voice box' points to Larynx; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Trachea belongs with 'Windpipe supported by cartilaginous rings', whereas the tested description is 'Voice box'. B: Moist skin belongs with 'Respiratory surface in earthworm', whereas the tested description is 'Voice box'. D: QRS complex belongs with 'ECG complex representing ventricular depolarization', whereas the tested description is 'Voice box'.

BIO-12-CH08-0049 | Challenging | Human respiratory tract**Correct answer: B. I, III and IV only**

Solution: Statement I is correct because 'Voice box' corresponds to Larynx. Statement II is incorrect: 'Windpipe supported by cartilaginous rings' corresponds to Trachea, not Alveoli. Statement III is correct because 'Microscopic sites of gaseous exchange' corresponds to Alveoli. Statement IV is correct because 'Pressure recorded during ventricular systole' corresponds to Systolic pressure. Therefore, the valid combination is I, III and IV only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) III. C: This combination omits true statement(s) III. D: This combination includes false statement(s) II.

BIO-12-CH08-0050 | Easy | Heart anatomy**Correct answer: C. Tricuspid valve**

Solution: 'Valve between right atrium and right ventricle' is the defining description for Tricuspid valve. The other choices belong to different chapter concepts.

Distractor review: A: Lymph belongs with 'Fluid returning tissue proteins and absorbed fats to blood', whereas the tested description is 'Valve between right atrium and right ventricle'. B: Bohr effect belongs with 'Shift promoting oxygen unloading at high carbon dioxide', whereas the tested description is 'Valve between right atrium and right ventricle'. D: Platelets belongs with 'Cell fragments important in clotting', whereas the tested description is 'Valve between right atrium and right ventricle'.

BIO-12-CH08-0051 | Hard | Respiratory organs**Correct answer: C. Respiratory organs in most fishes - Neutrophil**

Solution: 'Respiratory organs in most fishes' should be paired with Gills, not Neutrophil. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Shift promoting oxygen unloading at high carbon dioxide - Bohr effect. B: This is a valid association: Enzyme accelerating carbonic acid formation in red cells - Carbonic anhydrase. D: This is a valid association: Natural pacemaker of the heart - Sinoatrial node.

BIO-12-CH08-0052 | Medium | Gas transport**Correct answer: B. Oxyhaemoglobin**

Solution: The observation 'Major form of oxygen transport in blood' points to Oxyhaemoglobin; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Diaphragm belongs with 'Main muscle of quiet inspiration', whereas the tested description is 'Major form of oxygen transport in blood'. C: Larynx belongs with 'Voice box', whereas the tested description is 'Major form of oxygen transport in blood'. D: Systolic pressure belongs with 'Pressure recorded during ventricular systole', whereas the tested description is 'Major form of oxygen transport in blood'.

BIO-12-CH08-0053 | Medium | Respiratory organs**Correct answer: C. Respiratory organs in insects - Tracheal system**

Solution: The accurate entry is 'Respiratory organs in insects - Tracheal system'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Cell fragments important in clotting' should be paired with Platelets, not Mostly passive. B: 'Natural pacemaker of the heart' should be paired with Sinoatrial node, not Tracheal system. D: 'Quiet expiration at rest' should be paired with Mostly passive, not QRS complex.

BIO-12-CH08-0054 | Hard | Blood cells**Correct answer: A. Anucleate biconcave cells transporting respiratory gases - Erythrocytes; Respiratory organs in most fishes - Gills**

Solution: Both parts of 'Anucleate biconcave cells transporting respiratory gases - Erythrocytes; Respiratory organs in most fishes - Gills' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Valve between left atrium and left ventricle' requires Bicuspid or mitral valve, and 'Fluid returning tissue proteins and absorbed fats to blood' requires Lymph. C: The answers have been interchanged: 'Vessels containing valves to prevent backflow' requires Veins, and 'Valve between right atrium and right ventricle' requires Tricuspid valve. D: The answers have been interchanged: 'Change in intrapulmonary pressure during inspiration' requires Decrease below atmospheric pressure, and 'Vessels containing valves to prevent backflow' requires Veins.

BIO-12-CH08-0055 | Easy | Human respiratory tract**Correct answer: B. Common passage for food and air**

Solution: Pharynx represents 'Common passage for food and air'. Each alternative describes a different term or result.

Distractor review: A: 'Change in intrapulmonary pressure during inspiration' identifies Decrease below atmospheric pressure, not Pharynx. C: 'Microscopic sites of gaseous exchange' identifies Alveoli, not Pharynx. D: 'Major form of carbon dioxide transport' identifies Bicarbonate ions, not Pharynx.

BIO-12-CH08-0056 | Hard | Human respiratory tract**Correct answer: D. Common passage for food and air - Pharynx; Respiratory surface in earthworm - Moist skin**

Solution: Both parts of 'Common passage for food and air - Pharynx; Respiratory surface in earthworm - Moist skin' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Major form of carbon dioxide transport' requires Bicarbonate ions, and 'Change in intrapulmonary pressure during inspiration' requires Decrease below atmospheric pressure. B: The answers have been interchanged: 'Anucleate biconcave cells transporting respiratory gases' requires Erythrocytes, and 'Major form of carbon dioxide transport' requires Bicarbonate ions. C: The answers have been interchanged: 'Fluid returning tissue proteins and absorbed fats to blood' requires Lymph, and 'Anucleate biconcave cells transporting respiratory gases' requires Erythrocytes.

BIO-12-CH08-0057 | Challenging | Vessels pressure and lymph**Correct answer: B. I and II only**

Solution: Statement I is correct because 'Vessels containing valves to prevent backflow' corresponds to Veins. Statement II is correct because 'Pressure recorded during ventricular systole' corresponds to Systolic pressure. Statement III is incorrect: 'Vessels carrying blood away from the heart' corresponds to Arteries, not 0.8 second. Statement IV is incorrect: 'Approximate duration of one cardiac cycle at 72 beats per minute' corresponds to 0.8 second, not Veins. Therefore, the valid combination is I and II only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) I. C: This combination includes false statement(s) IV and omits true statement(s) I, II. D: This combination includes false statement(s) III, IV and omits true statement(s) I.

BIO-12-CH08-0058 | Easy | Blood cells**Correct answer: A. Anucleate biconcave cells transporting respiratory gases**

Solution: Erythrocytes represents 'Anucleate biconcave cells transporting respiratory gases'. Each alternative describes a different term or result.

Distractor review: B: 'Vessels containing valves to prevent backflow' identifies Veins, not Erythrocytes. C: 'Valve between right atrium and right ventricle' identifies Tricuspid valve, not Erythrocytes. D: 'Change in intrapulmonary pressure during inspiration' identifies Decrease below atmospheric pressure, not Erythrocytes.

BIO-12-CH08-0059 | Medium | Mechanics of breathing**Correct answer: A. Increase**

Solution: The observation 'Change in thoracic volume during inspiration' points to Increase; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Lymph belongs with 'Fluid returning tissue proteins and absorbed fats to blood', whereas the tested description is 'Change in thoracic volume during inspiration'. C: Trachea belongs with 'Windpipe supported by cartilaginous rings', whereas the tested description is 'Change in thoracic volume during inspiration'. D: Carbonic anhydrase belongs with 'Enzyme accelerating carbonic acid formation in red cells', whereas the tested description is 'Change in thoracic volume during inspiration'.

BIO-12-CH08-0060 | Medium | Human respiratory tract**Correct answer: D. Pharynx**

Solution: The observation 'Common passage for food and air' points to Pharynx; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Lymph belongs with 'Fluid returning tissue proteins and absorbed fats to blood', whereas the tested description is 'Common passage for food and air'. B: Larynx belongs with 'Voice box', whereas the tested description is 'Common passage for food and air'. C: Moist skin belongs with 'Respiratory surface in earthworm', whereas the tested description is 'Common passage for food and air'.

BIO-12-CH08-0061 | Easy | Cardiac cycle and ECG**Correct answer: B. T wave**

Solution: 'ECG wave representing ventricular repolarization' is the defining description for T wave. The other choices belong to different chapter concepts.

Distractor review: A: Systolic pressure belongs with 'Pressure recorded during ventricular systole', whereas the tested description is 'ECG wave representing ventricular repolarization'. C: Erythrocytes belongs with 'Anucleate biconcave cells transporting respiratory gases', whereas the tested description is 'ECG wave representing ventricular repolarization'. D: Neutrophil belongs with 'Most abundant granulocyte and active phagocyte', whereas the tested description is 'ECG wave representing ventricular repolarization'.

BIO-12-CH08-0062 | Easy | Respiratory organs**Correct answer: D. Respiratory organs in insects**

Solution: Tracheal system represents 'Respiratory organs in insects'. Each alternative describes a different term or result.

Distractor review: A: 'Respiratory organs in most fishes' identifies Gills, not Tracheal system. B: 'Approximate duration of one cardiac cycle at 72 beats per minute' identifies 0.8 second, not Tracheal system. C: 'Quiet expiration at rest' identifies Mostly passive, not Tracheal system.

BIO-12-CH08-0063 | Hard | Mechanics of breathing**Correct answer: D. Quiet expiration at rest - Mostly passive; Valve between right atrium and right ventricle - Tricuspid valve**

Solution: Both parts of 'Quiet expiration at rest - Mostly passive; Valve between right atrium and right ventricle - Tricuspid valve' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Respiratory organs in most fishes' requires Gills, and 'Approximate duration of one cardiac cycle at 72 beats per minute' requires 0.8 second. B: The answers have been interchanged: 'Microscopic sites of gaseous exchange' requires Alveoli, and 'Common passage for food and air' requires Pharynx. C: The answers have been interchanged: 'Most abundant granulocyte and active phagocyte' requires Neutrophil, and 'Respiratory organs in most fishes' requires Gills.

BIO-12-CH08-0064 | Hard | Gas transport**Correct answer: C. Shift promoting oxygen unloading at high carbon dioxide - Veins**

Solution: 'Shift promoting oxygen unloading at high carbon dioxide' should be paired with Bohr effect, not Veins. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Common passage for food and air - Pharynx. B: This is a valid association: Quiet expiration at rest - Mostly passive. D: This is a valid association: Voice box - Larynx.

BIO-12-CH08-0065 | Easy | Blood cells**Correct answer: A. Leucocytes**

Solution: 'Cells chiefly responsible for defence' is the defining description for Leucocytes. The other choices belong to different chapter concepts.

Distractor review: B: Veins belongs with 'Vessels containing valves to prevent backflow', whereas the tested description is 'Cells chiefly responsible for defence'. C: Sinoatrial node belongs with 'Natural pacemaker of the heart', whereas the tested description is 'Cells chiefly responsible for defence'. D: Decrease below atmospheric pressure belongs with 'Change in intrapulmonary pressure during inspiration', whereas the tested description is 'Cells chiefly responsible for defence'.

BIO-12-CH08-0066 | Hard | Vessels pressure and lymph**Correct answer: A. Fluid returning tissue proteins and absorbed fats to blood - Lymph; Vessel carrying blood from right ventricle to lungs - Pulmonary artery**

Solution: Both parts of 'Fluid returning tissue proteins and absorbed fats to blood - Lymph; Vessel carrying blood from right ventricle to lungs - Pulmonary artery' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Microscopic sites of gaseous exchange' requires Alveoli, and 'Cells chiefly responsible for defence' requires Leucocytes. C: The answers have been interchanged: 'Cell fragments important in clotting' requires Platelets, and 'Approximate duration of one cardiac cycle at 72 beats per minute' requires 0.8 second. D: The answers have been interchanged: 'Approximate duration of one cardiac cycle at 72 beats per minute' requires 0.8 second, and 'Major form of carbon dioxide transport' requires Bicarbonate ions.

BIO-12-CH08-0067 | Medium | Vessels pressure and lymph**Correct answer: D. Pressure recorded during ventricular systole - Systolic pressure**

Solution: The accurate entry is 'Pressure recorded during ventricular systole - Systolic pressure'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Change in thoracic volume during inspiration' should be paired with Increase, not T wave. B: 'Vessel carrying blood from right ventricle to lungs' should be paired with Pulmonary artery, not Pharynx. C: 'Primary respiratory organs in adult humans' should be paired with Lungs, not Systolic pressure.

BIO-12-CH08-0068 | Medium | Cardiac cycle and ECG**Correct answer: B. ECG wave representing ventricular repolarization - T wave**

Solution: The accurate entry is 'ECG wave representing ventricular repolarization - T wave'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Vessels containing valves to prevent backflow' should be paired with Veins, not Arteries. C: 'Quiet expiration at rest' should be paired with Mostly passive, not Tricuspid valve. D: 'Vessels carrying blood away from the heart' should be paired with Arteries, not T wave.

BIO-12-CH08-0069 | Easy | Blood cells**Correct answer: A. Neutrophil**

Solution: 'Most abundant granulocyte and active phagocyte' is the defining description for Neutrophil. The other choices belong to different chapter concepts.

Distractor review: B: Carbonic anhydrase belongs with 'Enzyme accelerating carbonic acid formation in red cells', whereas the tested description is 'Most abundant granulocyte and active phagocyte'. C: Larynx belongs with 'Voice box', whereas the tested description is 'Most abundant granulocyte and active phagocyte'. D: Leucocytes belongs with 'Cells chiefly responsible for defence', whereas the tested description is 'Most abundant granulocyte and active phagocyte'.

BIO-12-CH08-0070 | Medium | Mechanics of breathing**Correct answer: A. Diaphragm**

Solution: The observation 'Main muscle of quiet inspiration' points to Diaphragm; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Pulmonary artery belongs with 'Vessel carrying blood from right ventricle to lungs', whereas the tested description is 'Main muscle of quiet inspiration'. C: Tracheal system belongs with 'Respiratory organs in insects', whereas the tested description is 'Main muscle of quiet inspiration'. D: Increase belongs with 'Change in thoracic volume during inspiration', whereas the tested description is 'Main muscle of quiet inspiration'.

BIO-12-CH08-0071 | Medium | Mechanics of breathing**Correct answer: B. Decrease below atmospheric pressure**

Solution: The chapter links Change in intrapulmonary pressure during inspiration with Decrease below atmospheric pressure, so Decrease below atmospheric pressure completes the statement.

Distractor review: A: Increase belongs with 'Change in thoracic volume during inspiration', whereas the tested description is 'Change in intrapulmonary pressure during inspiration'. C: Larynx belongs with 'Voice box', whereas the tested description is 'Change in intrapulmonary pressure during inspiration'. D: Arteries belongs with 'Vessels carrying blood away from the heart', whereas the tested description is 'Change in intrapulmonary pressure during inspiration'.

BIO-12-CH08-0072 | Hard | Vessels pressure and lymph**Correct answer: B. Vessels carrying blood away from the heart - P wave**

Solution: 'Vessels carrying blood away from the heart' should be paired with Arteries, not P wave. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Valve between right atrium and right ventricle - Tricuspid valve. C: This is a valid association: Quiet expiration at rest - Mostly passive. D: This is a valid association: Main muscle of quiet inspiration - Diaphragm.

BIO-12-CH08-0073 | Challenging | Blood cells**Correct answer: D. I, II and IV only**

Solution: Statement I is correct because 'Most abundant granulocyte and active phagocyte' corresponds to Neutrophil. Statement II is correct because 'Anucleate biconcave cells transporting respiratory gases' corresponds to Erythrocytes. Statement III is incorrect: 'Cells chiefly responsible for defence' corresponds to Leucocytes, not Lymph. Statement IV is correct because 'Fluid returning tissue proteins and absorbed fats to blood' corresponds to Lymph. Therefore, the valid combination is I, II and IV only.

Distractor review: A: This combination includes false statement(s) III and omits true statement(s) II. B: This combination includes false statement(s) III and omits true statement(s) I, II. C: This combination omits true statement(s) I, IV.

BIO-12-CH08-0074 | Medium | Heart anatomy**Correct answer: D. Valve between right atrium and right ventricle - Tricuspid valve**

Solution: The accurate entry is 'Valve between right atrium and right ventricle - Tricuspid valve'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Fluid returning tissue proteins and absorbed fats to blood' should be paired with Lymph, not Platelets. B: 'Major form of oxygen transport in blood' should be paired with Oxyhaemoglobin, not Trachea. C: 'Shift promoting oxygen unloading at high carbon dioxide' should be paired with Bohr effect, not Oxyhaemoglobin.

BIO-12-CH08-0075 | Medium | Blood cells**Correct answer: B. Erythrocytes**

Solution: The chapter links Anucleate biconcave cells transporting respiratory gases with Erythrocytes, so Erythrocytes completes the statement.

Distractor review: A: Decrease below atmospheric pressure belongs with 'Change in intrapulmonary pressure during inspiration', whereas the tested description is 'Anucleate biconcave cells transporting respiratory gases'. C: Leucocytes belongs with 'Cells chiefly responsible for defence', whereas the tested description is 'Anucleate biconcave cells transporting respiratory gases'. D: QRS complex belongs with 'ECG complex representing ventricular depolarization', whereas the tested description is 'Anucleate biconcave cells transporting respiratory gases'.

BIO-12-CH08-0076 | Easy | Gas transport**Correct answer: C. Enzyme accelerating carbonic acid formation in red cells**

Solution: Carbonic anhydrase represents 'Enzyme accelerating carbonic acid formation in red cells'. Each alternative describes a different term or result.

Distractor review: A: 'Major form of oxygen transport in blood' identifies Oxyhaemoglobin, not Carbonic anhydrase. B: 'ECG wave representing atrial depolarization' identifies P wave, not Carbonic anhydrase. D: 'Most abundant granulocyte and active phagocyte' identifies Neutrophil, not Carbonic anhydrase.

BIO-12-CH08-0077 | Easy | Mechanics of breathing**Correct answer: C. Diaphragm**

Solution: 'Main muscle of quiet inspiration' is the defining description for Diaphragm. The other choices belong to different chapter concepts.

Distractor review: A: Decrease below atmospheric pressure belongs with 'Change in intrapulmonary pressure during inspiration', whereas the tested description is 'Main muscle of quiet inspiration'. B: Moist skin belongs with 'Respiratory surface in earthworm', whereas the tested description is 'Main muscle of quiet inspiration'. D: Increase belongs with 'Change in thoracic volume during inspiration', whereas the tested description is 'Main muscle of quiet inspiration'.

BIO-12-CH08-0078 | Easy | Gas transport**Correct answer: B. Carbonic anhydrase**

Solution: 'Enzyme accelerating carbonic acid formation in red cells' is the defining description for Carbonic anhydrase. The other choices belong to different chapter concepts.

Distractor review: A: Neutrophil belongs with 'Most abundant granulocyte and active phagocyte', whereas the tested description is 'Enzyme accelerating carbonic acid formation in red cells'. C: Oxyhaemoglobin belongs with 'Major form of oxygen transport in blood', whereas the tested description is 'Enzyme accelerating carbonic acid formation in red cells'. D: Larynx belongs with 'Voice box', whereas the tested description is 'Enzyme accelerating carbonic acid formation in red cells'.

BIO-12-CH08-0079 | Medium | Gas transport**Correct answer: A. Oxyhaemoglobin**

Solution: The chapter links Major form of oxygen transport in blood with Oxyhaemoglobin, so Oxyhaemoglobin completes the statement.

Distractor review: B: Systolic pressure belongs with 'Pressure recorded during ventricular systole', whereas the tested description is 'Major form of oxygen transport in blood'. C: P wave belongs with 'ECG wave representing atrial depolarization', whereas the tested description is 'Major form of oxygen transport in blood'. D: Carbonic anhydrase belongs with 'Enzyme accelerating carbonic acid formation in red cells', whereas the tested description is 'Major form of oxygen transport in blood'.

BIO-12-CH08-0080 | Hard | Human respiratory tract**Correct answer: D. Windpipe supported by cartilaginous rings - Pulmonary artery**

Solution: 'Windpipe supported by cartilaginous rings' should be paired with Trachea, not Pulmonary artery. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: ECG wave representing atrial depolarization - P wave. B: This is a valid association: Quiet expiration at rest - Mostly passive. C: This is a valid association: Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure.

BIO-12-CH08-0081 | Easy | Respiratory organs**Correct answer: C. Gills**

Solution: 'Respiratory organs in most fishes' is the defining description for Gills. The other choices belong to different chapter concepts.

Distractor review: A: Bohr effect belongs with 'Shift promoting oxygen unloading at high carbon dioxide', whereas the tested description is 'Respiratory organs in most fishes'. B: Lungs belongs with 'Primary respiratory organs in adult humans', whereas the tested description is 'Respiratory organs in most fishes'. D: Carbonic anhydrase belongs with 'Enzyme accelerating carbonic acid formation in red cells', whereas the tested description is 'Respiratory organs in most fishes'.

BIO-12-CH08-0082 | Hard | Cardiac cycle and ECG**Correct answer: A. ECG wave representing atrial depolarization - P wave; Windpipe supported by cartilaginous rings - Trachea**

Solution: Both parts of 'ECG wave representing atrial depolarization - P wave; Windpipe supported by cartilaginous rings - Trachea' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Fluid returning tissue proteins and absorbed fats to blood' requires Lymph, and 'Microscopic sites of gaseous exchange' requires Alveoli. C: The answers have been interchanged: 'Microscopic sites of gaseous exchange' requires Alveoli, and 'Natural pacemaker of the heart' requires Sinoatrial node. D: The answers have been interchanged: 'Major form of oxygen transport in blood' requires Oxyhaemoglobin, and 'Respiratory organs in insects' requires Tracheal system.

BIO-12-CH08-0083 | Easy | Gas transport**Correct answer: C. Oxyhaemoglobin**

Solution: 'Major form of oxygen transport in blood' is the defining description for Oxyhaemoglobin. The other choices belong to different chapter concepts.

Distractor review: A: Bicarbonate ions belongs with 'Major form of carbon dioxide transport', whereas the tested description is 'Major form of oxygen transport in blood'. B: Tracheal system belongs with 'Respiratory organs in insects', whereas the tested description is 'Major form of oxygen transport in blood'. D: Larynx belongs with 'Voice box', whereas the tested description is 'Major form of oxygen transport in blood'.

BIO-12-CH08-0084 | Hard | Cardiac cycle and ECG**Correct answer: D. ECG wave representing ventricular repolarization - Neutrophil**

Solution: 'ECG wave representing ventricular repolarization' should be paired with T wave, not Neutrophil. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Pressure recorded during ventricular systole - Systolic pressure. B: This is a valid association: Anucleate biconcave cells transporting respiratory gases - Erythrocytes. C: This is a valid association: Vessels containing valves to prevent backflow - Veins.

BIO-12-CH08-0085 | Easy | Vessels pressure and lymph**Correct answer: A. Arteries**

Solution: 'Vessels carrying blood away from the heart' is the defining description for Arteries. The other choices belong to different chapter concepts.

Distractor review: B: Mostly passive belongs with 'Quiet expiration at rest', whereas the tested description is 'Vessels carrying blood away from the heart'. C: Diaphragm belongs with 'Main muscle of quiet inspiration', whereas the tested description is 'Vessels carrying blood away from the heart'. D: P wave belongs with 'ECG wave representing atrial depolarization', whereas the tested description is 'Vessels carrying blood away from the heart'.

BIO-12-CH08-0086 | Medium | Vessels pressure and lymph**Correct answer: B. Arteries**

Solution: The observation 'Vessels carrying blood away from the heart' points to Arteries; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: P wave belongs with 'ECG wave representing atrial depolarization', whereas the tested description is 'Vessels carrying blood away from the heart'. C: Mostly passive belongs with 'Quiet expiration at rest', whereas the tested description is 'Vessels carrying blood away from the heart'. D: Veins belongs with 'Vessels containing valves to prevent backflow', whereas the tested description is 'Vessels carrying blood away from the heart'.

BIO-12-CH08-0087 | Hard | Vessels pressure and lymph**Correct answer: B. Vessels containing valves to prevent backflow - Erythrocytes**

Solution: 'Vessels containing valves to prevent backflow' should be paired with Veins, not Erythrocytes. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Voice box - Larynx. C: This is a valid association: Windpipe supported by cartilaginous rings - Trachea. D: This is a valid association: Common passage for food and air - Pharynx.

BIO-12-CH08-0088 | Easy | Vessels pressure and lymph**Correct answer: D. Fluid returning tissue proteins and absorbed fats to blood**

Solution: Lymph represents 'Fluid returning tissue proteins and absorbed fats to blood'. Each alternative describes a different term or result.

Distractor review: A: 'Cell fragments important in clotting' identifies Platelets, not Lymph. B: 'Vessel carrying blood from right ventricle to lungs' identifies Pulmonary artery, not Lymph. C: 'Vessels containing valves to prevent backflow' identifies Veins, not Lymph.

BIO-12-CH08-0089 | Easy | Heart anatomy**Correct answer: B. Valve between left atrium and left ventricle**

Solution: Bicuspid or mitral valve represents 'Valve between left atrium and left ventricle'. Each alternative describes a different term or result.

Distractor review: A: 'Change in intrapulmonary pressure during inspiration' identifies Decrease below atmospheric pressure, not Bicuspid or mitral valve. C: 'Cell fragments important in clotting' identifies Platelets, not Bicuspid or mitral valve. D: 'Shift promoting oxygen unloading at high carbon dioxide' identifies Bohr effect, not Bicuspid or mitral valve.

BIO-12-CH08-0090 | Hard | Vessels pressure and lymph**Correct answer: A. Pressure recorded during ventricular systole - Systolic pressure; Major form of oxygen transport in blood - Oxyhaemoglobin**

Solution: Both parts of 'Pressure recorded during ventricular systole - Systolic pressure; Major form of oxygen transport in blood - Oxyhaemoglobin' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'ECG wave representing ventricular repolarization' requires T wave, and 'Common passage for food and air' requires Pharynx. C: The answers have been interchanged: 'Primary respiratory organs in adult humans' requires Lungs, and 'Anucleate biconcave cells transporting respiratory gases' requires Erythrocytes. D: The answers have been interchanged: 'Vessel carrying blood from right ventricle to lungs' requires Pulmonary artery, and 'ECG wave representing ventricular repolarization' requires T wave.

BIO-12-CH08-0091 | Easy | Vessels pressure and lymph**Correct answer: D. Pressure recorded during ventricular systole**

Solution: Systolic pressure represents 'Pressure recorded during ventricular systole'. Each alternative describes a different term or result.

Distractor review: A: 'Fluid returning tissue proteins and absorbed fats to blood' identifies Lymph, not Systolic pressure. B: 'Common passage for food and air' identifies Pharynx, not Systolic pressure. C: 'ECG wave representing ventricular repolarization' identifies T wave, not Systolic pressure.

BIO-12-CH08-0092 | Medium | Gas transport**Correct answer: B. Bohr effect**

Solution: The chapter links Shift promoting oxygen unloading at high carbon dioxide with Bohr effect, so Bohr effect completes the statement.

Distractor review: A: Diaphragm belongs with 'Main muscle of quiet inspiration', whereas the tested description is 'Shift promoting oxygen unloading at high carbon dioxide'. C: Pharynx belongs with 'Common passage for food and air', whereas the tested description is 'Shift promoting oxygen unloading at high carbon dioxide'. D: Veins belongs with 'Vessels containing valves to prevent backflow', whereas the tested description is 'Shift promoting oxygen unloading at high carbon dioxide'.

BIO-12-CH08-0093 | Hard | Heart anatomy**Correct answer: C. Valve between right atrium and right ventricle - Lymph**

Solution: 'Valve between right atrium and right ventricle' should be paired with Tricuspid valve, not Lymph. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Vessels carrying blood away from the heart - Arteries. B: This is a valid association: Windpipe supported by cartilaginous rings - Trachea. D: This is a valid association: Major form of oxygen transport in blood - Oxyhaemoglobin.

BIO-12-CH08-0094 | Hard | Gas transport**Correct answer: A. Enzyme accelerating carbonic acid formation in red cells - Larynx**

Solution: 'Enzyme accelerating carbonic acid formation in red cells' should be paired with Carbonic anhydrase, not Larynx. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Cell fragments important in clotting - Platelets. C: This is a valid association: Valve between right atrium and right ventricle - Tricuspid valve. D: This is a valid association: ECG wave representing ventricular repolarization - T wave.

BIO-12-CH08-0095 | Medium | Cardiac cycle and ECG**Correct answer: D. ECG complex representing ventricular depolarization - QRS complex**

Solution: The accurate entry is 'ECG complex representing ventricular depolarization - QRS complex'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Change in thoracic volume during inspiration' should be paired with Increase, not Erythrocytes. B: 'Anucleate biconcave cells transporting respiratory gases' should be paired with Erythrocytes, not Pulmonary artery. C: 'Cell fragments important in clotting' should be paired with Platelets, not Gills.

BIO-12-CH08-0096 | Challenging | Cardiac cycle and ECG**Correct answer: C. IV only**

Solution: Statement I is incorrect: 'ECG complex representing ventricular depolarization' corresponds to QRS complex, not 0.8 second. Statement II is incorrect: 'Approximate duration of one cardiac cycle at 72 beats per minute' corresponds to 0.8 second, not P wave. Statement III is incorrect: 'ECG wave representing atrial depolarization' corresponds to P wave, not Tracheal system. Statement IV is correct because 'Respiratory organs in insects' corresponds to Tracheal system. Therefore, the valid combination is IV only.

Distractor review: A: This combination includes false statement(s) I, II and omits true statement(s) IV. B: This combination includes false statement(s) III and omits true statement(s) IV. D: This combination includes false statement(s) I, II, III and omits true statement(s) IV.

BIO-12-CH08-0097 | Hard | Cardiac cycle and ECG**Correct answer: C. Approximate duration of one cardiac cycle at 72 beats per minute - Veins**

Solution: 'Approximate duration of one cardiac cycle at 72 beats per minute' should be paired with 0.8 second, not Veins. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Valve between right atrium and right ventricle - Tricuspid valve. B: This is a valid association: Major form of oxygen transport in blood - Oxyhaemoglobin. D: This is a valid association: Respiratory surface in earthworm - Moist skin.

BIO-12-CH08-0098 | Hard | Blood cells**Correct answer: A. Cells chiefly responsible for defence - Veins**

Solution: 'Cells chiefly responsible for defence' should be paired with Leucocytes, not Veins. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Natural pacemaker of the heart - Sinoatrial node. C: This is a valid association: Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure. D: This is a valid association: Pressure recorded during ventricular systole - Systolic pressure.

BIO-12-CH08-0099 | Easy | Vessels pressure and lymph**Correct answer: D. Veins**

Solution: 'Vessels containing valves to prevent backflow' is the defining description for Veins. The other choices belong to different chapter concepts.

Distractor review: A: Neutrophil belongs with 'Most abundant granulocyte and active phagocyte', whereas the tested description is 'Vessels containing valves to prevent backflow'. B: Trachea belongs with 'Windpipe supported by cartilaginous rings', whereas the tested description is 'Vessels containing valves to prevent backflow'. C: Erythrocytes belongs with 'Anucleate biconcave cells transporting respiratory gases', whereas the tested description is 'Vessels containing valves to prevent backflow'.

BIO-12-CH08-0100 | Challenging | Mechanics of breathing**Correct answer: B. I, II and III only**

Solution: Statement I is correct because 'Main muscle of quiet inspiration' corresponds to Diaphragm. Statement II is correct because 'Change in thoracic volume during inspiration' corresponds to Increase. Statement III is correct because 'Change in intrapulmonary pressure during inspiration' corresponds to Decrease below atmospheric pressure. Statement IV is incorrect: 'Most abundant granulocyte and active phagocyte' corresponds to Neutrophil, not Diaphragm. Therefore, the valid combination is I, II and III only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) I, II, III. C: This combination omits true statement(s) I. D: This combination omits true statement(s) I, III.

BIO-12-CH08-0101 | Hard | Respiratory organs**Correct answer: A. Respiratory organs in insects - Pulmonary artery**

Solution: 'Respiratory organs in insects' should be paired with Tracheal system, not Pulmonary artery. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second. C: This is a valid association: Valve between left atrium and left ventricle - Bicuspid or mitral valve. D: This is a valid association: Natural pacemaker of the heart - Sinoatrial node.

BIO-12-CH08-0102 | Easy | Gas transport**Correct answer: A. Bicarbonate ions**

Solution: 'Major form of carbon dioxide transport' is the defining description for Bicarbonate ions. The other choices belong to different chapter concepts.

Distractor review: B: Carbonic anhydrase belongs with 'Enzyme accelerating carbonic acid formation in red cells', whereas the tested description is 'Major form of carbon dioxide transport'. C: QRS complex belongs with 'ECG complex representing ventricular depolarization', whereas the tested description is 'Major form of carbon dioxide transport'. D: Bicuspid or mitral valve belongs with 'Valve between left atrium and left ventricle', whereas the tested description is 'Major form of carbon dioxide transport'.

BIO-12-CH08-0103 | Medium | Human respiratory tract**Correct answer: D. Trachea**

Solution: The observation 'Windpipe supported by cartilaginous rings' points to Trachea; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Leucocytes belongs with 'Cells chiefly responsible for defence', whereas the tested description is 'Windpipe supported by cartilaginous rings'. B: Mostly passive belongs with 'Quiet expiration at rest', whereas the tested description is 'Windpipe supported by cartilaginous rings'. C: Pharynx belongs with 'Common passage for food and air', whereas the tested description is 'Windpipe supported by cartilaginous rings'.

BIO-12-CH08-0104 | Easy | Blood cells**Correct answer: B. Most abundant granulocyte and active phagocyte**

Solution: Neutrophil represents 'Most abundant granulocyte and active phagocyte'. Each alternative describes a different term or result.

Distractor review: A: 'Vessel carrying blood from right ventricle to lungs' identifies Pulmonary artery, not Neutrophil. C: 'ECG complex representing ventricular depolarization' identifies QRS complex, not Neutrophil. D: 'Voice box' identifies Larynx, not Neutrophil.

BIO-12-CH08-0105 | Challenging | Heart anatomy**Correct answer: B. III only**

Solution: Statement I is incorrect: 'Vessel carrying blood from right ventricle to lungs' corresponds to Pulmonary artery, not Bicuspid or mitral valve. Statement II is incorrect: 'Valve between left atrium and left ventricle' corresponds to Bicuspid or mitral valve, not Tricuspid valve. Statement III is correct because 'Valve between right atrium and right ventricle' corresponds to Tricuspid valve. Statement IV is incorrect: 'Cell fragments important in clotting' corresponds to Platelets, not Pulmonary artery. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) I, IV and omits true statement(s) III. C: This combination includes false statement(s) I, II, IV and omits true statement(s) III. D: This combination includes false statement(s) II, IV.

BIO-12-CH08-0106 | Easy | Mechanics of breathing**Correct answer: D. Increase**

Solution: 'Change in thoracic volume during inspiration' is the defining description for Increase. The other choices belong to different chapter concepts.

Distractor review: A: Decrease below atmospheric pressure belongs with 'Change in intrapulmonary pressure during inspiration', whereas the tested description is 'Change in thoracic volume during inspiration'. B: Trachea belongs with 'Windpipe supported by cartilaginous rings', whereas the tested description is 'Change in thoracic volume during inspiration'. C: Lymph belongs with 'Fluid returning tissue proteins and absorbed fats to blood', whereas the tested description is 'Change in thoracic volume during inspiration'.

BIO-12-CH08-0107 | Challenging | Heart anatomy**Correct answer: B. III and IV only**

Solution: Statement I is incorrect: 'Natural pacemaker of the heart' corresponds to Sinoatrial node, not Pulmonary artery. Statement II is incorrect: 'Vessel carrying blood from right ventricle to lungs' corresponds to Pulmonary artery, not Tricuspid valve. Statement III is correct because 'Valve between right atrium and right ventricle' corresponds to Tricuspid valve. Statement IV is correct because 'Anucleate biconcave cells transporting respiratory gases' corresponds to Erythrocytes. Therefore, the valid combination is III and IV only.

Distractor review: A: This combination includes false statement(s) II. C: This combination includes false statement(s) I, II. D: This combination omits true statement(s) IV.

BIO-12-CH08-0108 | Medium | Respiratory organs**Correct answer: D. Lungs**

Solution: The chapter links Primary respiratory organs in adult humans with Lungs, so Lungs completes the statement.

Distractor review: A: QRS complex belongs with 'ECG complex representing ventricular depolarization', whereas the tested description is 'Primary respiratory organs in adult humans'. B: Decrease below atmospheric pressure belongs with 'Change in intrapulmonary pressure during inspiration', whereas the tested description is 'Primary respiratory organs in adult humans'. C: Gills belongs with 'Respiratory organs in most fishes', whereas the tested description is 'Primary respiratory organs in adult humans'.

BIO-12-CH08-0109 | Medium | Blood cells**Correct answer: C. Cell fragments important in clotting - Platelets**

Solution: The accurate entry is 'Cell fragments important in clotting - Platelets'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Major form of oxygen transport in blood' should be paired with Oxyhaemoglobin, not Mostly passive. B: 'Vessels carrying blood away from the heart' should be paired with Arteries, not Bohr effect. D: 'Common passage for food and air' should be paired with Pharynx, not Platelets.

BIO-12-CH08-0110 | Medium | Cardiac cycle and ECG**Correct answer: A. P wave**

Solution: The chapter links ECG wave representing atrial depolarization with P wave, so P wave completes the statement.

Distractor review: B: Trachea belongs with 'Windpipe supported by cartilaginous rings', whereas the tested description is 'ECG wave representing atrial depolarization'. C: Sinoatrial node belongs with 'Natural pacemaker of the heart', whereas the tested description is 'ECG wave representing atrial depolarization'. D: QRS complex belongs with 'ECG complex representing ventricular depolarization', whereas the tested description is 'ECG wave representing atrial depolarization'.

BIO-12-CH08-0111 | Medium | Vessels pressure and lymph**Correct answer: B. Vessels carrying blood away from the heart - Arteries**

Solution: The accurate entry is 'Vessels carrying blood away from the heart - Arteries'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Voice box' should be paired with Larynx, not QRS complex. C: 'Quiet expiration at rest' should be paired with Mostly passive, not Diaphragm. D: 'ECG wave representing atrial depolarization' should be paired with P wave, not Larynx.

BIO-12-CH08-0112 | Medium | Cardiac cycle and ECG**Correct answer: C. T wave**

Solution: The chapter links ECG wave representing ventricular repolarization with T wave, so T wave completes the statement.

Distractor review: A: Veins belongs with 'Vessels containing valves to prevent backflow', whereas the tested description is 'ECG wave representing ventricular repolarization'. B: QRS complex belongs with 'ECG complex representing ventricular depolarization', whereas the tested description is 'ECG wave representing ventricular repolarization'. D: Mostly passive belongs with 'Quiet expiration at rest', whereas the tested description is 'ECG wave representing ventricular repolarization'.

BIO-12-CH08-0113 | Hard | Blood cells**Correct answer: A. Most abundant granulocyte and active phagocyte - Neutrophil; Voice box - Larynx**

Solution: Both parts of 'Most abundant granulocyte and active phagocyte - Neutrophil; Voice box - Larynx' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Enzyme accelerating carbonic acid formation in red cells' requires Carbonic anhydrase, and 'ECG wave representing atrial depolarization' requires P wave. C: The answers have been interchanged: 'Vessel carrying blood from right ventricle to lungs' requires Pulmonary artery, and 'Pressure recorded during ventricular systole' requires Systolic pressure. D: The answers have been interchanged: 'ECG wave representing atrial depolarization' requires P wave, and 'Respiratory organs in insects' requires Tracheal system.

BIO-12-CH08-0114 | Medium | Heart anatomy**Correct answer: A. Pulmonary artery**

Solution: The chapter links Vessel carrying blood from right ventricle to lungs with Pulmonary artery, so Pulmonary artery completes the statement.

Distractor review: B: Bicuspid or mitral valve belongs with 'Valve between left atrium and left ventricle', whereas the tested description is 'Vessel carrying blood from right ventricle to lungs'. C: Sinoatrial node belongs with 'Natural pacemaker of the heart', whereas the tested description is 'Vessel carrying blood from right ventricle to lungs'. D: Lymph belongs with 'Fluid returning tissue proteins and absorbed fats to blood', whereas the tested description is 'Vessel carrying blood from right ventricle to lungs'.

BIO-12-CH08-0115 | Medium | Heart anatomy**Correct answer: B. Sinoatrial node**

Solution: The chapter links Natural pacemaker of the heart with Sinoatrial node, so Sinoatrial node completes the statement.

Distractor review: A: Bicuspid or mitral valve belongs with 'Valve between left atrium and left ventricle', whereas the tested description is 'Natural pacemaker of the heart'. C: Trachea belongs with 'Windpipe supported by cartilaginous rings', whereas the tested description is 'Natural pacemaker of the heart'. D: Moist skin belongs with 'Respiratory surface in earthworm', whereas the tested description is 'Natural pacemaker of the heart'.

BIO-12-CH08-0116 | Medium | Respiratory organs**Correct answer: D. Gills**

Solution: The chapter links Respiratory organs in most fishes with Gills, so Gills completes the statement.

Distractor review: A: Mostly passive belongs with 'Quiet expiration at rest', whereas the tested description is 'Respiratory organs in most fishes'. B: Lungs belongs with 'Primary respiratory organs in adult humans', whereas the tested description is 'Respiratory organs in most fishes'. C: Tracheal system belongs with 'Respiratory organs in insects', whereas the tested description is 'Respiratory organs in most fishes'.

BIO-12-CH08-0117 | Medium | Vessels pressure and lymph**Correct answer: A. Systolic pressure**

Solution: The chapter links Pressure recorded during ventricular systole with Systolic pressure, so Systolic pressure completes the statement.

Distractor review: B: Pulmonary artery belongs with 'Vessel carrying blood from right ventricle to lungs', whereas the tested description is 'Pressure recorded during ventricular systole'. C: T wave belongs with 'ECG wave representing ventricular repolarization', whereas the tested description is 'Pressure recorded during ventricular systole'. D: Pharynx belongs with 'Common passage for food and air', whereas the tested description is 'Pressure recorded during ventricular systole'.

BIO-12-CH08-0118 | Medium | Blood cells**Correct answer: A. Anucleate biconcave cells transporting respiratory gases - Erythrocytes**

Solution: The accurate entry is 'Anucleate biconcave cells transporting respiratory gases - Erythrocytes'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Vessels containing valves to prevent backflow' should be paired with Veins, not QRS complex. C: 'Change in intrapulmonary pressure during inspiration' should be paired with Decrease below atmospheric pressure, not Tricuspid valve. D: 'Valve between left atrium and left ventricle' should be paired with Bicuspid or mitral valve, not Erythrocytes.

BIO-12-CH08-0119 | Hard | Gas transport**Correct answer: C. Major form of oxygen transport in blood - Systolic pressure**

Solution: 'Major form of oxygen transport in blood' should be paired with Oxyhaemoglobin, not Systolic pressure. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Cells chiefly responsible for defence - Leucocytes. B: This is a valid association: ECG wave representing atrial depolarization - P wave. D: This is a valid association: Respiratory organs in insects - Tracheal system.

BIO-12-CH08-0120 | Challenging | Gas transport**Correct answer: B. II only**

Solution: Statement I is incorrect: 'Shift promoting oxygen unloading at high carbon dioxide' corresponds to Bohr effect, not Carbonic anhydrase. Statement II is correct because 'Enzyme accelerating carbonic acid formation in red cells' corresponds to Carbonic anhydrase. Statement III is incorrect: 'Major form of carbon dioxide transport' corresponds to Bicarbonate ions, not Alveoli. Statement IV is incorrect: 'Microscopic sites of gaseous exchange' corresponds to Alveoli, not Bohr effect. Therefore, the valid combination is II only.

Distractor review: A: This combination includes false statement(s) III and omits true statement(s) II. C: This combination includes false statement(s) I and omits true statement(s) II. D: This combination includes false statement(s) IV.

BIO-12-CH08-0121 | Medium | Vessels pressure and lymph**Correct answer: B. Veins**

Solution: The chapter links Vessels containing valves to prevent backflow with Veins, so Veins completes the statement.

Distractor review: A: Trachea belongs with 'Windpipe supported by cartilaginous rings', whereas the tested description is 'Vessels containing valves to prevent backflow'. C: Arteries belongs with 'Vessels carrying blood away from the heart', whereas the tested description is 'Vessels containing valves to prevent backflow'. D: Neutrophil belongs with 'Most abundant granulocyte and active phagocyte', whereas the tested description is 'Vessels containing valves to prevent backflow'.

BIO-12-CH08-0122 | Medium | Blood cells**Correct answer: C. Neutrophil**

Solution: The chapter links Most abundant granulocyte and active phagocyte with Neutrophil, so Neutrophil completes the statement.

Distractor review: A: Leucocytes belongs with 'Cells chiefly responsible for defence', whereas the tested description is 'Most abundant granulocyte and active phagocyte'. B: Larynx belongs with 'Voice box', whereas the tested description is 'Most abundant granulocyte and active phagocyte'. D: Systolic pressure belongs with 'Pressure recorded during ventricular systole', whereas the tested description is 'Most abundant granulocyte and active phagocyte'.

BIO-12-CH08-0123 | Medium | Gas transport**Correct answer: D. Shift promoting oxygen unloading at high carbon dioxide - Bohr effect**

Solution: The accurate entry is 'Shift promoting oxygen unloading at high carbon dioxide - Bohr effect'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Quiet expiration at rest' should be paired with Mostly passive, not Larynx. B: 'Primary respiratory organs in adult humans' should be paired with Lungs, not Bohr effect. C: 'Vessels containing valves to prevent backflow' should be paired with Veins, not Diaphragm.

BIO-12-CH08-0124 | Easy | Respiratory organs**Correct answer: D. Moist skin**

Solution: 'Respiratory surface in earthworm' is the defining description for Moist skin. The other choices belong to different chapter concepts.

Distractor review: A: Decrease below atmospheric pressure belongs with 'Change in intrapulmonary pressure during inspiration', whereas the tested description is 'Respiratory surface in earthworm'. B: Veins belongs with 'Vessels containing valves to prevent backflow', whereas the tested description is 'Respiratory surface in earthworm'. C: Tracheal system belongs with 'Respiratory organs in insects', whereas the tested description is 'Respiratory surface in earthworm'.

BIO-12-CH08-0125 | Medium | Heart anatomy**Correct answer: C. Bicuspid or mitral valve**

Solution: The chapter links Valve between left atrium and left ventricle with Bicuspid or mitral valve, so Bicuspid or mitral valve completes the statement.

Distractor review: A: Carbonic anhydrase belongs with 'Enzyme accelerating carbonic acid formation in red cells', whereas the tested description is 'Valve between left atrium and left ventricle'. B: Sinoatrial node belongs with 'Natural pacemaker of the heart', whereas the tested description is 'Valve between left atrium and left ventricle'. D: Lymph belongs with 'Fluid returning tissue proteins and absorbed fats to blood', whereas the tested description is 'Valve between left atrium and left ventricle'.

BIO-12-CH08-0126 | Medium | Mechanics of breathing**Correct answer: D. Mostly passive**

Solution: The observation 'Quiet expiration at rest' points to Mostly passive; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Pharynx belongs with 'Common passage for food and air', whereas the tested description is 'Quiet expiration at rest'. B: Alveoli belongs with 'Microscopic sites of gaseous exchange', whereas the tested description is 'Quiet expiration at rest'. C: Diaphragm belongs with 'Main muscle of quiet inspiration', whereas the tested description is 'Quiet expiration at rest'.

BIO-12-CH08-0127 | Easy | Mechanics of breathing**Correct answer: C. Quiet expiration at rest**

Solution: Mostly passive represents 'Quiet expiration at rest'. Each alternative describes a different term or result.

Distractor review: A: 'Change in intrapulmonary pressure during inspiration' identifies Decrease below atmospheric pressure, not Mostly passive. B: 'Valve between right atrium and right ventricle' identifies Tricuspid valve, not Mostly passive. D: 'Vessel carrying blood from right ventricle to lungs' identifies Pulmonary artery, not Mostly passive.

BIO-12-CH08-0128 | Medium | Heart anatomy**Correct answer: A. Tricuspid valve**

Solution: The observation 'Valve between right atrium and right ventricle' points to Tricuspid valve; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Pulmonary artery belongs with 'Vessel carrying blood from right ventricle to lungs', whereas the tested description is 'Valve between right atrium and right ventricle'. C: Bicuspid or mitral valve belongs with 'Valve between left atrium and left ventricle', whereas the tested description is 'Valve between right atrium and right ventricle'. D: Lymph belongs with 'Fluid returning tissue proteins and absorbed fats to blood', whereas the tested description is 'Valve between right atrium and right ventricle'.

BIO-12-CH08-0129 | Hard | Mechanics of breathing**Correct answer: C. Change in thoracic volume during inspiration - Leucocytes**

Solution: 'Change in thoracic volume during inspiration' should be paired with Increase, not Leucocytes. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Natural pacemaker of the heart - Sinoatrial node. B: This is a valid association: Fluid returning tissue proteins and absorbed fats to blood - Lymph. D: This is a valid association: Major form of oxygen transport in blood - Oxyhaemoglobin.

BIO-12-CH08-0130 | Easy | Gas transport**Correct answer: D. Bohr effect**

Solution: 'Shift promoting oxygen unloading at high carbon dioxide' is the defining description for Bohr effect. The other choices belong to different chapter concepts.

Distractor review: A: Diaphragm belongs with 'Main muscle of quiet inspiration', whereas the tested description is 'Shift promoting oxygen unloading at high carbon dioxide'. B: Pharynx belongs with 'Common passage for food and air', whereas the tested description is 'Shift promoting oxygen unloading at high carbon dioxide'. C: Mostly passive belongs with 'Quiet expiration at rest', whereas the tested description is 'Shift promoting oxygen unloading at high carbon dioxide'.

BIO-12-CH08-0131 | Medium | Blood cells

Correct answer: B. Erythrocytes

Solution: The observation 'Anucleate biconcave cells transporting respiratory gases' points to Erythrocytes; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Platelets belongs with 'Cell fragments important in clotting', whereas the tested description is 'Anucleate biconcave cells transporting respiratory gases'. C: QRS complex belongs with 'ECG complex representing ventricular depolarization', whereas the tested description is 'Anucleate biconcave cells transporting respiratory gases'. D: Gills belongs with 'Respiratory organs in most fishes', whereas the tested description is 'Anucleate biconcave cells transporting respiratory gases'.

BIO-12-CH08-0132 | Challenging | Gas transport

Correct answer: C. II and IV only

Solution: Statement I is incorrect: 'Enzyme accelerating carbonic acid formation in red cells' corresponds to Carbonic anhydrase, not Oxyhaemoglobin. Statement II is correct because 'Major form of oxygen transport in blood' corresponds to Oxyhaemoglobin. Statement III is incorrect: 'Shift promoting oxygen unloading at high carbon dioxide' corresponds to Bohr effect, not Systolic pressure. Statement IV is correct because 'Pressure recorded during ventricular systole' corresponds to Systolic pressure. Therefore, the valid combination is II and IV only.

Distractor review: A: This combination includes false statement(s) III and omits true statement(s) II. B: This combination includes false statement(s) I and omits true statement(s) IV. D: This combination includes false statement(s) I.

Standard 12 | Chapter 09

Control and Co-ordination

BIO-12-CH09-0001 | Hard | Reflex arc

Correct answer: A. loss of afferent input from the receptor

Solution: The sensory root carries afferent impulses into the spinal cord, so its damage interrupts receptor input.

Distractor review: B: The alternative 'continuous motor output with normal sensation' conflicts with the labelled relationship, scale or mechanism in the figure. The sensory root carries afferent impulses into the spinal cord, so its damage interrupts receptor input. C: The alternative 'faster neurotransmitter removal' conflicts with the labelled relationship, scale or mechanism in the figure. The sensory root carries afferent impulses into the spinal cord, so its damage interrupts receptor input. D: The alternative 'increased hormone secretion by the pituitary' conflicts with the labelled relationship, scale or mechanism in the figure. The sensory root carries afferent impulses into the spinal cord, so its damage interrupts receptor input.

BIO-12-CH09-0002 | Challenging | Chemical synaptic transmission

Correct answer: A. Prolonged activation of postsynaptic receptors

Solution: Slower acetylcholine breakdown prolongs its presence in the cleft and extends postsynaptic stimulation.

Distractor review: B: The alternative 'Failure to release any neurotransmitter' conflicts with the labelled relationship, scale or mechanism in the figure. Slower acetylcholine breakdown prolongs its presence in the cleft and extends postsynaptic stimulation. C: The alternative 'Instant destruction of all receptors' conflicts with the labelled relationship, scale or mechanism in the figure. Slower acetylcholine breakdown prolongs its presence in the cleft and extends postsynaptic stimulation. D: The alternative 'Conversion of acetylcholine into DNA' conflicts with the labelled relationship, scale or mechanism in the figure. Slower acetylcholine breakdown prolongs its presence in the cleft and extends postsynaptic stimulation.

BIO-12-CH09-0003 | Easy | Reflex arc

Correct answer: D. Sensory neuron

Solution: The sensory or afferent neuron transmits information from the receptor to the central nervous system.

Distractor review: A: The alternative 'Motor neuron' conflicts with the labelled relationship, scale or mechanism in the figure. The sensory or afferent neuron transmits information from the receptor to the central nervous system. B: The alternative 'Postganglionic autonomic neuron only' conflicts with the labelled relationship, scale or mechanism in the figure. The sensory or afferent neuron transmits information from the receptor to the central nervous system. C: The alternative 'Neurosecretory cell' conflicts with the labelled relationship, scale or mechanism in the figure. The sensory or afferent neuron transmits information from the receptor to the central nervous system.

BIO-12-CH09-0004 | Easy | Central nervous system

Correct answer: C. Meninges

Solution: 'Protective membranes around brain and spinal cord' is the defining description for Meninges. The other choices belong to different chapter concepts.

Distractor review: A: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Protective membranes around brain and spinal cord'. B: Axon belongs with 'Long process conducting impulses away from the cell body', whereas the tested description is 'Protective membranes around brain and spinal cord'. D: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Protective membranes around brain and spinal cord'.

BIO-12-CH09-0005 | Medium | Chemical synaptic transmission

Correct answer: B. Calcium-ion entry

Solution: Arrival of the action potential opens voltage-gated calcium channels; calcium entry triggers vesicle fusion.

Distractor review: A: The alternative 'Sodium removal from the synaptic cleft' conflicts with the labelled relationship, scale or mechanism in the figure. Arrival of the action potential opens voltage-gated calcium channels; calcium entry triggers vesicle fusion. C: The alternative 'DNA replication' conflicts with the labelled relationship, scale or mechanism in the figure. Arrival of the action potential opens voltage-gated calcium channels; calcium entry triggers vesicle fusion. D: The alternative 'Myelin breakdown' conflicts with the labelled relationship, scale or mechanism in the figure. Arrival of the action potential opens voltage-gated calcium channels; calcium entry triggers vesicle fusion.

BIO-12-CH09-0006 | Medium | Neuron structure**Correct answer: B. Neuron processes usually receiving impulses - Dendrites**

Solution: The accurate entry is 'Neuron processes usually receiving impulses - Dendrites'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Junction between two neurons' should be paired with Synapse, not Neurotransmitter. C: 'Thyroid hormone containing iodine' should be paired with Thyroxine, not Synapse. D: 'Receptors detecting light' should be paired with Photoreceptors, not Adrenaline.

BIO-12-CH09-0007 | Easy | Neuron structure**Correct answer: A. Dendrites**

Solution: 'Neuron processes usually receiving impulses' is the defining description for Dendrites. The other choices belong to different chapter concepts.

Distractor review: B: Axon belongs with 'Long process conducting impulses away from the cell body', whereas the tested description is 'Neuron processes usually receiving impulses'. C: Thyroxine belongs with 'Thyroid hormone containing iodine', whereas the tested description is 'Neuron processes usually receiving impulses'. D: Medulla oblongata belongs with 'Region controlling breathing and cardiovascular reflexes', whereas the tested description is 'Neuron processes usually receiving impulses'.

BIO-12-CH09-0008 | Easy | Impulse and synapse**Correct answer: A. Membrane state with inside negative relative to outside**

Solution: Resting potential represents 'Membrane state with inside negative relative to outside'. Each alternative describes a different term or result.

Distractor review: B: 'Protective membranes around brain and spinal cord' identifies Meninges, not Resting potential. C: 'Neural pathway mediating a reflex' identifies Reflex arc, not Resting potential. D: 'Iodine-deficiency enlargement of thyroid' identifies Simple goitre, not Resting potential.

BIO-12-CH09-0009 | Medium | Nervous organization**Correct answer: D. Hydra**

Solution: The observation 'Diffuse nerve network without a central brain' points to Hydra; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Simple goitre belongs with 'Iodine-deficiency enlargement of thyroid', whereas the tested description is 'Diffuse nerve network without a central brain'. B: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Diffuse nerve network without a central brain'. C: Planaria belongs with 'Ladder-like nervous system', whereas the tested description is 'Diffuse nerve network without a central brain'.

BIO-12-CH09-0010 | Easy | Endocrine disorders**Correct answer: D. Acromegaly**

Solution: 'Excess growth hormone in adults' is the defining description for Acromegaly. The other choices belong to different chapter concepts.

Distractor review: A: Dwarfism belongs with 'Childhood deficiency of growth hormone', whereas the tested description is 'Excess growth hormone in adults'. B: Neuron belongs with 'Structural and functional unit of nervous tissue', whereas the tested description is 'Excess growth hormone in adults'. C: Depolarization belongs with 'Rapid reversal of membrane polarity', whereas the tested description is 'Excess growth hormone in adults'.

BIO-12-CH09-0011 | Easy | Pancreas adrenal and gonads**Correct answer: A. Insulin**

Solution: 'Pancreatic hormone lowering blood glucose' is the defining description for Insulin. The other choices belong to different chapter concepts.

Distractor review: B: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Pancreatic hormone lowering blood glucose'. C: Simple goitre belongs with 'Iodine-deficiency enlargement of thyroid', whereas the tested description is 'Pancreatic hormone lowering blood glucose'. D: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Pancreatic hormone lowering blood glucose'.

BIO-12-CH09-0012 | Medium | Receptors and reflexes**Correct answer: C. Reflex action**

Solution: The observation 'Rapid automatic response through a neural pathway' points to Reflex action; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Calcitonin belongs with 'Hormone lowering blood calcium', whereas the tested description is 'Rapid automatic response through a neural pathway'. B: Medulla oblongata belongs with 'Region controlling breathing and cardiovascular reflexes', whereas the tested description is 'Rapid automatic response through a neural pathway'. D: Dendrites belongs with 'Neuron processes usually receiving impulses', whereas the tested description is 'Rapid automatic response through a neural pathway'.

BIO-12-CH09-0013 | Medium | Pituitary and thyroid**Correct answer: A. Calcitonin**

Solution: The chapter links Hormone lowering blood calcium with Calcitonin, so Calcitonin completes the statement.

Distractor review: B: Testosterone belongs with 'Principal androgen secreted by testes', whereas the tested description is 'Hormone lowering blood calcium'. C: Growth hormone belongs with 'Pituitary hormone promoting body growth', whereas the tested description is 'Hormone lowering blood calcium'. D: Dendrites belongs with 'Neuron processes usually receiving impulses', whereas the tested description is 'Hormone lowering blood calcium'.

BIO-12-CH09-0014 | Easy | Pancreas adrenal and gonads**Correct answer: C. Adrenaline**

Solution: 'Adrenal medullary fight-or-flight hormone' is the defining description for Adrenaline. The other choices belong to different chapter concepts.

Distractor review: A: Depolarization belongs with 'Rapid reversal of membrane polarity', whereas the tested description is 'Adrenal medullary fight-or-flight hormone'. B: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Adrenal medullary fight-or-flight hormone'. D: Planaria belongs with 'Ladder-like nervous system', whereas the tested description is 'Adrenal medullary fight-or-flight hormone'.

BIO-12-CH09-0015 | Medium | Central nervous system**Correct answer: A. Meninges**

Solution: The observation 'Protective membranes around brain and spinal cord' points to Meninges; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Protective membranes around brain and spinal cord'. C: TSH belongs with 'Pituitary hormone stimulating the thyroid', whereas the tested description is 'Protective membranes around brain and spinal cord'. D: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Protective membranes around brain and spinal cord'.

BIO-12-CH09-0016 | Challenging | Receptors and reflexes**Correct answer: C. I, II and III only**

Solution: Statement I is correct because 'Rapid automatic response through a neural pathway' corresponds to Reflex action. Statement II is correct because 'Receptors detecting sound and pressure' corresponds to Mechanoreceptors. Statement III is correct because 'Neural pathway mediating a reflex' corresponds to Reflex arc. Statement IV is incorrect: 'Iodine-deficiency enlargement of thyroid' corresponds to Simple goitre, not Reflex action. Therefore, the valid combination is I, II and III only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) I. B: This combination omits true statement(s) II, III. D: This combination includes false statement(s) IV and omits true statement(s) III.

BIO-12-CH09-0017 | Medium | Endocrine disorders**Correct answer: A. Iodine-deficiency enlargement of thyroid - Simple goitre**

Solution: The accurate entry is 'Iodine-deficiency enlargement of thyroid - Simple goitre'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Region controlling breathing and cardiovascular reflexes' should be paired with Medulla oblongata, not Synapse. C: 'Insulating sheath around many axons' should be paired with Myelin sheath, not Medulla oblongata. D: 'Membrane state with inside negative relative to outside' should be paired with Resting potential, not Cerebrum.

BIO-12-CH09-0018 | Hard | Neuron structure**Correct answer: D. Neuron processes usually receiving impulses - Medulla oblongata**

Solution: 'Neuron processes usually receiving impulses' should be paired with Dendrites, not Medulla oblongata. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Receptors detecting light - Photoreceptors. B: This is a valid association: Adrenal medullary fight-or-flight hormone - Adrenaline. C: This is a valid association: Junction between two neurons - Synapse.

BIO-12-CH09-0019 | Medium | Neuron structure**Correct answer: A. Dendrites**

Solution: The chapter links Neuron processes usually receiving impulses with Dendrites, so Dendrites completes the statement.

Distractor review: B: Axon belongs with 'Long process conducting impulses away from the cell body', whereas the tested description is 'Neuron processes usually receiving impulses'. C: Adrenaline belongs with 'Adrenal medullary fight-or-flight hormone', whereas the tested description is 'Neuron processes usually receiving impulses'. D: Synapse belongs with 'Junction between two neurons', whereas the tested description is 'Neuron processes usually receiving impulses'.

BIO-12-CH09-0020 | Medium | Neuron structure**Correct answer: C. Nodes of Ranvier**

Solution: The observation 'Gaps between adjacent myelin segments' points to Nodes of Ranvier; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Glucagon belongs with 'Pancreatic hormone raising blood glucose', whereas the tested description is 'Gaps between adjacent myelin segments'. B: Axon belongs with 'Long process conducting impulses away from the cell body', whereas the tested description is 'Gaps between adjacent myelin segments'. D: Cerebellum belongs with 'Brain region coordinating posture and balance', whereas the tested description is 'Gaps between adjacent myelin segments'.

BIO-12-CH09-0021 | Hard | Impulse and synapse**Correct answer: B. Membrane state with inside negative relative to outside - Resting potential; Brain region coordinating posture and balance - Cerebellum**

Solution: Both parts of 'Membrane state with inside negative relative to outside - Resting potential; Brain region coordinating posture and balance - Cerebellum' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Hormone lowering blood calcium' requires Calcitonin, and 'Neural pathway mediating a reflex' requires Reflex arc. C: The answers have been interchanged: 'Protective membranes around brain and spinal cord' requires Meninges, and 'Receptors detecting light' requires Photoreceptors. D: The answers have been interchanged: 'Iodine-deficiency enlargement of thyroid' requires Simple goitre, and 'Protective membranes around brain and spinal cord' requires Meninges.

BIO-12-CH09-0022 | Hard | Pancreas adrenal and gonads**Correct answer: C. Principal androgen secreted by testes - Testosterone; Pituitary hormone promoting body growth - Growth hormone**

Solution: Both parts of 'Principal androgen secreted by testes - Testosterone; Pituitary hormone promoting body growth - Growth hormone' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Receptors detecting light' requires Photoreceptors, and 'Receptors detecting sound and pressure' requires Mechanoreceptors. B: The answers have been interchanged: 'Neural pathway mediating a reflex' requires Reflex arc, and 'Ladder-like nervous system' requires Planaria. D: The answers have been interchanged: 'Thyroid hormone containing iodine' requires Thyroxine, and 'Rapid automatic response through a neural pathway' requires Reflex action.

BIO-12-CH09-0023 | Hard | Nervous organization**Correct answer: A. Structural and functional unit of nervous tissue - Synapse**

Solution: 'Structural and functional unit of nervous tissue' should be paired with Neuron, not Synapse. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Thyroid hormone containing iodine - Thyroxine. C: This is a valid association: Protective membranes around brain and spinal cord - Meninges. D: This is a valid association: Pancreatic hormone raising blood glucose - Glucagon.

BIO-12-CH09-0024 | Medium | Neuron structure**Correct answer: C. Gaps between adjacent myelin segments - Nodes of Ranvier**

Solution: The accurate entry is 'Gaps between adjacent myelin segments - Nodes of Ranvier'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Pancreatic hormone raising blood glucose' should be paired with Glucagon, not Resting potential. B: 'Insulin deficiency or ineffective action' should be paired with Diabetes mellitus, not Hydra. D: 'Brain region coordinating posture and balance' should be paired with Cerebellum, not Mechanoreceptors.

BIO-12-CH09-0025 | Medium | Pancreas adrenal and gonads**Correct answer: A. Glucagon**

Solution: The chapter links Pancreatic hormone raising blood glucose with Glucagon, so Glucagon completes the statement.

Distractor review: B: Hydra belongs with 'Diffuse nerve network without a central brain', whereas the tested description is 'Pancreatic hormone raising blood glucose'. C: Reflex arc belongs with 'Neural pathway mediating a reflex', whereas the tested description is 'Pancreatic hormone raising blood glucose'. D: Photoreceptors belongs with 'Receptors detecting light', whereas the tested description is 'Pancreatic hormone raising blood glucose'.

BIO-12-CH09-0026 | Hard | Neuron structure**Correct answer: D. Gaps between adjacent myelin segments - Glucagon**

Solution: 'Gaps between adjacent myelin segments' should be paired with Nodes of Ranvier, not Glucagon. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Insulin deficiency or ineffective action - Diabetes mellitus. B: This is a valid association: Diffuse nerve network without a central brain - Hydra. C: This is a valid association: Membrane state with inside negative relative to outside - Resting potential.

BIO-12-CH09-0027 | Medium | Receptors and reflexes**Correct answer: B. Receptors detecting sound and pressure - Mechanoreceptors**

Solution: The accurate entry is 'Receptors detecting sound and pressure - Mechanoreceptors'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Adrenal medullary fight-or-flight hormone' should be paired with Adrenaline, not Mechanoreceptors. C: 'Protective membranes around brain and spinal cord' should be paired with Meninges, not Depolarization. D: 'Childhood deficiency of growth hormone' should be paired with Dwarfism, not Medulla oblongata.

BIO-12-CH09-0028 | Hard | Central nervous system**Correct answer: D. Protective membranes around brain and spinal cord - Simple goitre**

Solution: 'Protective membranes around brain and spinal cord' should be paired with Meninges, not Simple goitre. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Pituitary hormone stimulating the thyroid - TSH. B: This is a valid association: Rapid automatic response through a neural pathway - Reflex action. C: This is a valid association: Long process conducting impulses away from the cell body - Axon.

BIO-12-CH09-0029 | Hard | Pancreas adrenal and gonads**Correct answer: C. Pancreatic hormone lowering blood glucose - Insulin; Rapid automatic response through a neural pathway - Reflex action**

Solution: Both parts of 'Pancreatic hormone lowering blood glucose - Insulin; Rapid automatic response through a neural pathway - Reflex action' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Iodine-deficiency enlargement of thyroid' requires Simple goitre, and 'Supporting cells of nervous tissue' requires Neuroglia. B: The answers have been interchanged: 'Ladder-like nervous system' requires Planaria, and 'Membrane state with inside negative relative to outside' requires Resting potential. D: The answers have been interchanged: 'Insulin deficiency or ineffective action' requires Diabetes mellitus, and 'Ladder-like nervous system' requires Planaria.

BIO-12-CH09-0030 | Medium | Receptors and reflexes**Correct answer: D. Neural pathway mediating a reflex - Reflex arc**

Solution: The accurate entry is 'Neural pathway mediating a reflex - Reflex arc'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Insulating sheath around many axons' should be paired with Myelin sheath, not Glucagon. B: 'Pancreatic hormone raising blood glucose' should be paired with Glucagon, not Nodes of Ranvier. C: 'Gaps between adjacent myelin segments' should be paired with Nodes of Ranvier, not Diabetes mellitus.

BIO-12-CH09-0031 | Medium | Pancreas adrenal and gonads**Correct answer: C. Insulin**

Solution: The chapter links Pancreatic hormone lowering blood glucose with Insulin, so Insulin completes the statement.

Distractor review: A: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Pancreatic hormone lowering blood glucose'. B: Neuroglia belongs with 'Supporting cells of nervous tissue', whereas the tested description is 'Pancreatic hormone lowering blood glucose'. D: Cerebrum belongs with 'Largest part of the human brain', whereas the tested description is 'Pancreatic hormone lowering blood glucose'.

BIO-12-CH09-0032 | Challenging | Central nervous system**Correct answer: A. III only**

Solution: Statement I is incorrect: 'Region controlling breathing and cardiovascular reflexes' corresponds to Medulla oblongata, not Cerebrum. Statement II is incorrect: 'Largest part of the human brain' corresponds to Cerebrum, not Meninges. Statement III is correct because 'Protective membranes around brain and spinal cord' corresponds to Meninges. Statement IV is incorrect: 'Hormone lowering blood calcium' corresponds to Calcitonin, not Medulla oblongata. Therefore, the valid combination is III only.

Distractor review: B: This combination includes false statement(s) IV. C: This combination includes false statement(s) I, II and omits true statement(s) III. D: This combination includes false statement(s) IV and omits true statement(s) III.

BIO-12-CH09-0033 | Medium | Nervous organization**Correct answer: B. Neuroglia**

Solution: The observation 'Supporting cells of nervous tissue' points to Neuroglia; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Cerebrum belongs with 'Largest part of the human brain', whereas the tested description is 'Supporting cells of nervous tissue'. C: Resting potential belongs with 'Membrane state with inside negative relative to outside', whereas the tested description is 'Supporting cells of nervous tissue'. D: Synapse belongs with 'Junction between two neurons', whereas the tested description is 'Supporting cells of nervous tissue'.

BIO-12-CH09-0034 | Hard | Pituitary and thyroid**Correct answer: B. Pituitary hormone stimulating the thyroid - Reflex arc**

Solution: 'Pituitary hormone stimulating the thyroid' should be paired with TSH, not Reflex arc. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Insulating sheath around many axons - Myelin sheath. C: This is a valid association: Receptors detecting sound and pressure - Mechanoreceptors. D: This is a valid association: Region controlling breathing and cardiovascular reflexes - Medulla oblongata.

BIO-12-CH09-0035 | Medium | Nervous organization**Correct answer: A. Neuroglia**

Solution: The chapter links Supporting cells of nervous tissue with Neuroglia, so Neuroglia completes the statement.

Distractor review: B: Adrenaline belongs with 'Adrenal medullary fight-or-flight hormone', whereas the tested description is 'Supporting cells of nervous tissue'. C: Planaria belongs with 'Ladder-like nervous system', whereas the tested description is 'Supporting cells of nervous tissue'. D: Resting potential belongs with 'Membrane state with inside negative relative to outside', whereas the tested description is 'Supporting cells of nervous tissue'.

BIO-12-CH09-0036 | Hard | Pancreas adrenal and gonads**Correct answer: B. Adrenal medullary fight-or-flight hormone - Depolarization**

Solution: 'Adrenal medullary fight-or-flight hormone' should be paired with Adrenaline, not Depolarization. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Insulating sheath around many axons - Myelin sheath. C: This is a valid association: Ladder-like nervous system - Planaria. D: This is a valid association: Structural and functional unit of nervous tissue - Neuron.

BIO-12-CH09-0037 | Easy | Endocrine disorders**Correct answer: A. Diabetes mellitus**

Solution: 'Insulin deficiency or ineffective action' is the defining description for Diabetes mellitus. The other choices belong to different chapter concepts.

Distractor review: B: Growth hormone belongs with 'Pituitary hormone promoting body growth', whereas the tested description is 'Insulin deficiency or ineffective action'. C: Simple goitre belongs with 'Iodine-deficiency enlargement of thyroid', whereas the tested description is 'Insulin deficiency or ineffective action'. D: Meninges belongs with 'Protective membranes around brain and spinal cord', whereas the tested description is 'Insulin deficiency or ineffective action'.

BIO-12-CH09-0038 | Medium | Impulse and synapse**Correct answer: B. Depolarization**

Solution: The chapter links Rapid reversal of membrane polarity with Depolarization, so Depolarization completes the statement.

Distractor review: A: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Rapid reversal of membrane polarity'. C: Synapse belongs with 'Junction between two neurons', whereas the tested description is 'Rapid reversal of membrane polarity'. D: Resting potential belongs with 'Membrane state with inside negative relative to outside', whereas the tested description is 'Rapid reversal of membrane polarity'.

BIO-12-CH09-0039 | Medium | Neuron structure**Correct answer: A. Dendrites**

Solution: The observation 'Neuron processes usually receiving impulses' points to Dendrites; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Nodes of Ranvier belongs with 'Gaps between adjacent myelin segments', whereas the tested description is 'Neuron processes usually receiving impulses'. C: Photoreceptors belongs with 'Receptors detecting light', whereas the tested description is 'Neuron processes usually receiving impulses'. D: Medulla oblongata belongs with 'Region controlling breathing and cardiovascular reflexes', whereas the tested description is 'Neuron processes usually receiving impulses'.

BIO-12-CH09-0040 | Hard | Receptors and reflexes**Correct answer: A. Receptors detecting light - Photoreceptors; Pituitary hormone promoting body growth - Growth hormone**

Solution: Both parts of 'Receptors detecting light - Photoreceptors; Pituitary hormone promoting body growth - Growth hormone' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Largest part of the human brain' requires Cerebrum, and 'Pancreatic hormone lowering blood glucose' requires Insulin. C: The answers have been interchanged: 'Pituitary hormone stimulating the thyroid' requires TSH, and 'Largest part of the human brain' requires Cerebrum. D: The answers have been interchanged: 'Thyroid hormone containing iodine' requires Thyroxine, and 'Pituitary hormone stimulating the thyroid' requires TSH.

BIO-12-CH09-0041 | Hard | Endocrine disorders**Correct answer: C. Insulin deficiency or ineffective action - Planaria**

Solution: 'Insulin deficiency or ineffective action' should be paired with Diabetes mellitus, not Planaria. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Chemical released from a presynaptic terminal - Neurotransmitter. B: This is a valid association: Protective membranes around brain and spinal cord - Meninges. D: This is a valid association: Largest part of the human brain - Cerebrum.

BIO-12-CH09-0042 | Hard | Impulse and synapse**Correct answer: B. Junction between two neurons - Synapse; Rapid automatic response through a neural pathway - Reflex action**

Solution: Both parts of 'Junction between two neurons - Synapse; Rapid automatic response through a neural pathway - Reflex action' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Adrenal medullary fight-or-flight hormone' requires Adrenaline, and 'Diffuse nerve network without a central brain' requires Hydra. C: The answers have been interchanged: 'Thyroid hormone containing iodine' requires Thyroxine, and 'Protective membranes around brain and spinal cord' requires Meninges. D: The answers have been interchanged: 'Insulating sheath around many axons' requires Myelin sheath, and 'Thyroid hormone containing iodine' requires Thyroxine.

BIO-12-CH09-0043 | Easy | Impulse and synapse**Correct answer: B. Chemical released from a presynaptic terminal**

Solution: Neurotransmitter represents 'Chemical released from a presynaptic terminal'. Each alternative describes a different term or result.

Distractor review: A: 'Neuron processes usually receiving impulses' identifies Dendrites, not Neurotransmitter. C: 'Junction between two neurons' identifies Synapse, not Neurotransmitter. D: 'Excess growth hormone in adults' identifies Acromegaly, not Neurotransmitter.

BIO-12-CH09-0044 | Hard | Nervous organization**Correct answer: C. Supporting cells of nervous tissue - Neuroglia; Pancreatic hormone lowering blood glucose - Insulin**

Solution: Both parts of 'Supporting cells of nervous tissue - Neuroglia; Pancreatic hormone lowering blood glucose - Insulin' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Adrenal medullary fight-or-flight hormone' requires Adrenaline, and 'Largest part of the human brain' requires Cerebrum. B: The answers have been interchanged: 'Childhood deficiency of growth hormone' requires Dwarfism, and 'Receptors detecting light' requires Photoreceptors. D: The answers have been interchanged: 'Membrane state with inside negative relative to outside' requires Resting potential, and 'Childhood deficiency of growth hormone' requires Dwarfism.

BIO-12-CH09-0045 | Hard | Pituitary and thyroid**Correct answer: A. Pituitary hormone promoting body growth - Myelin sheath**

Solution: 'Pituitary hormone promoting body growth' should be paired with Growth hormone, not Myelin sheath. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Rapid automatic response through a neural pathway - Reflex action. C: This is a valid association: Long process conducting impulses away from the cell body - Axon. D: This is a valid association: Neuron processes usually receiving impulses - Dendrites.

BIO-12-CH09-0046 | Medium | Pancreas adrenal and gonads**Correct answer: B. Glucagon**

Solution: The observation 'Pancreatic hormone raising blood glucose' points to Glucagon; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Reflex arc belongs with 'Neural pathway mediating a reflex', whereas the tested description is 'Pancreatic hormone raising blood glucose'. C: Hydra belongs with 'Diffuse nerve network without a central brain', whereas the tested description is 'Pancreatic hormone raising blood glucose'. D: Testosterone belongs with 'Principal androgen secreted by testes', whereas the tested description is 'Pancreatic hormone raising blood glucose'.

BIO-12-CH09-0047 | Medium | Impulse and synapse**Correct answer: C. Neurotransmitter**

Solution: The chapter links Chemical released from a presynaptic terminal with Neurotransmitter, so Neurotransmitter completes the statement.

Distractor review: A: Meninges belongs with 'Protective membranes around brain and spinal cord', whereas the tested description is 'Chemical released from a presynaptic terminal'. B: Photoreceptors belongs with 'Receptors detecting light', whereas the tested description is 'Chemical released from a presynaptic terminal'. D: Acromegaly belongs with 'Excess growth hormone in adults', whereas the tested description is 'Chemical released from a presynaptic terminal'.

BIO-12-CH09-0048 | Hard | Impulse and synapse**Correct answer: C. Junction between two neurons - Reflex action**

Solution: 'Junction between two neurons' should be paired with Synapse, not Reflex action. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Thyroid hormone containing iodine - Thyroxine. B: This is a valid association: Adrenal medullary fight-or-flight hormone - Adrenaline. D: This is a valid association: Insulin deficiency or ineffective action - Diabetes mellitus.

BIO-12-CH09-0049 | Easy | Endocrine disorders**Correct answer: D. Dwarfism**

Solution: 'Childhood deficiency of growth hormone' is the defining description for Dwarfism. The other choices belong to different chapter concepts.

Distractor review: A: Simple goitre belongs with 'Iodine-deficiency enlargement of thyroid', whereas the tested description is 'Childhood deficiency of growth hormone'. B: Acromegaly belongs with 'Excess growth hormone in adults', whereas the tested description is 'Childhood deficiency of growth hormone'. C: TSH belongs with 'Pituitary hormone stimulating the thyroid', whereas the tested description is 'Childhood deficiency of growth hormone'.

BIO-12-CH09-0050 | Easy | Pancreas adrenal and gonads**Correct answer: C. Principal androgen secreted by testes**

Solution: Testosterone represents 'Principal androgen secreted by testes'. Each alternative describes a different term or result.

Distractor review: A: 'Rapid automatic response through a neural pathway' identifies Reflex action, not Testosterone. B: 'Thyroid hormone containing iodine' identifies Thyroxine, not Testosterone. D: 'Ladder-like nervous system' identifies Planaria, not Testosterone.

BIO-12-CH09-0051 | Medium | Endocrine disorders**Correct answer: A. Diabetes mellitus**

Solution: The chapter links Insulin deficiency or ineffective action with Diabetes mellitus, so Diabetes mellitus completes the statement.

Distractor review: B: Acromegaly belongs with 'Excess growth hormone in adults', whereas the tested description is 'Insulin deficiency or ineffective action'. C: Growth hormone belongs with 'Pituitary hormone promoting body growth', whereas the tested description is 'Insulin deficiency or ineffective action'. D: Planaria belongs with 'Ladder-like nervous system', whereas the tested description is 'Insulin deficiency or ineffective action'.

BIO-12-CH09-0052 | Medium | Nervous organization**Correct answer: D. Ladder-like nervous system - Planaria**

Solution: The accurate entry is 'Ladder-like nervous system - Planaria'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Thyroid hormone containing iodine' should be paired with Thyroxine, not TSH. B: 'Excess growth hormone in adults' should be paired with Acromegaly, not Mechanoreceptors. C: 'Hormone lowering blood calcium' should be paired with Calcitonin, not Thyroxine.

BIO-12-CH09-0053 | Medium | Pancreas adrenal and gonads**Correct answer: B. Adrenaline**

Solution: The observation 'Adrenal medullary fight-or-flight hormone' points to Adrenaline; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Glucagon belongs with 'Pancreatic hormone raising blood glucose', whereas the tested description is 'Adrenal medullary fight-or-flight hormone'. C: Neuron belongs with 'Structural and functional unit of nervous tissue', whereas the tested description is 'Adrenal medullary fight-or-flight hormone'. D: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Adrenal medullary fight-or-flight hormone'.

BIO-12-CH09-0054 | Medium | Pituitary and thyroid**Correct answer: D. Pituitary hormone promoting body growth - Growth hormone**

Solution: The accurate entry is 'Pituitary hormone promoting body growth - Growth hormone'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Adrenal medullary fight-or-flight hormone' should be paired with Adrenaline, not Reflex action. B: 'Insulating sheath around many axons' should be paired with Myelin sheath, not Dendrites. C: 'Diffuse nerve network without a central brain' should be paired with Hydra, not Growth hormone.

BIO-12-CH09-0055 | Medium | Pancreas adrenal and gonads**Correct answer: A. Principal androgen secreted by testes - Testosterone**

Solution: The accurate entry is 'Principal androgen secreted by testes - Testosterone'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Thyroid hormone containing iodine' should be paired with Thyroxine, not Photoreceptors. C: 'Neural pathway mediating a reflex' should be paired with Reflex arc, not Thyroxine. D: 'Ladder-like nervous system' should be paired with Planaria, not Reflex action.

BIO-12-CH09-0056 | Challenging | Pancreas adrenal and gonads**Correct answer: B. III and IV only**

Solution: Statement I is incorrect: 'Principal androgen secreted by testes' corresponds to Testosterone, not Glucagon. Statement II is incorrect: 'Pancreatic hormone raising blood glucose' corresponds to Glucagon, not Insulin. Statement III is correct because 'Pancreatic hormone lowering blood glucose' corresponds to Insulin. Statement IV is correct because 'Thyroid hormone containing iodine' corresponds to Thyroxine. Therefore, the valid combination is III and IV only.

Distractor review: A: This combination includes false statement(s) I, II and omits true statement(s) IV. C: This combination includes false statement(s) I, II. D: This combination includes false statement(s) II and omits true statement(s) IV.

BIO-12-CH09-0057 | Medium | Impulse and synapse**Correct answer: A. Resting potential**

Solution: The chapter links Membrane state with inside negative relative to outside with Resting potential, so Resting potential completes the statement.

Distractor review: B: Neurotransmitter belongs with 'Chemical released from a presynaptic terminal', whereas the tested description is 'Membrane state with inside negative relative to outside'. C: Simple goitre belongs with 'Iodine-deficiency enlargement of thyroid', whereas the tested description is 'Membrane state with inside negative relative to outside'. D: Depolarization belongs with 'Rapid reversal of membrane polarity', whereas the tested description is 'Membrane state with inside negative relative to outside'.

BIO-12-CH09-0058 | Easy | Nervous organization**Correct answer: D. Neuron**

Solution: 'Structural and functional unit of nervous tissue' is the defining description for Neuron. The other choices belong to different chapter concepts.

Distractor review: A: Testosterone belongs with 'Principal androgen secreted by testes', whereas the tested description is 'Structural and functional unit of nervous tissue'. B: Neuroglia belongs with 'Supporting cells of nervous tissue', whereas the tested description is 'Structural and functional unit of nervous tissue'. C: Glucagon belongs with 'Pancreatic hormone raising blood glucose', whereas the tested description is 'Structural and functional unit of nervous tissue'.

BIO-12-CH09-0059 | Challenging | Receptors and reflexes**Correct answer: C. II only**

Solution: Statement I is incorrect: 'Neural pathway mediating a reflex' corresponds to Reflex arc, not Mechanoreceptors. Statement II is correct because 'Receptors detecting sound and pressure' corresponds to Mechanoreceptors. Statement III is incorrect: 'Rapid automatic response through a neural pathway' corresponds to Reflex action, not Cerebrum. Statement IV is incorrect: 'Largest part of the human brain' corresponds to Cerebrum, not Reflex arc. Therefore, the valid combination is II only.

Distractor review: A: This combination includes false statement(s) I, IV and omits true statement(s) II. B: This combination includes false statement(s) IV and omits true statement(s) II. D: This combination includes false statement(s) III, IV and omits true statement(s) II.

BIO-12-CH09-0060 | Medium | Receptors and reflexes**Correct answer: C. Mechanoreceptors**

Solution: The observation 'Receptors detecting sound and pressure' points to Mechanoreceptors; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Neurotransmitter belongs with 'Chemical released from a presynaptic terminal', whereas the tested description is 'Receptors detecting sound and pressure'. B: Depolarization belongs with 'Rapid reversal of membrane polarity', whereas the tested description is 'Receptors detecting sound and pressure'. D: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Receptors detecting sound and pressure'.

BIO-12-CH09-0061 | Medium | Neuron structure**Correct answer: D. Long process conducting impulses away from the cell body - Axon**

Solution: The accurate entry is 'Long process conducting impulses away from the cell body - Axon'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Insulin deficiency or ineffective action' should be paired with Diabetes mellitus, not Hydra. B: 'Diffuse nerve network without a central brain' should be paired with Hydra, not Axon. C: 'Hormone lowering blood calcium' should be paired with Calcitonin, not Adrenaline.

BIO-12-CH09-0062 | Hard | Nervous organization**Correct answer: C. Structural and functional unit of nervous tissue - Neuron; Junction between two neurons - Synapse**

Solution: Both parts of 'Structural and functional unit of nervous tissue - Neuron; Junction between two neurons - Synapse' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Protective membranes around brain and spinal cord' requires Meninges, and 'Principal androgen secreted by testes' requires Testosterone. B: The answers have been interchanged: 'Insulin deficiency or ineffective action' requires Diabetes mellitus, and 'Protective membranes around brain and spinal cord' requires Meninges. D: The answers have been interchanged: 'Principal androgen secreted by testes' requires Testosterone, and 'Pancreatic hormone raising blood glucose' requires Glucagon.

BIO-12-CH09-0063 | Medium | Central nervous system**Correct answer: B. Cerebellum**

Solution: The observation 'Brain region coordinating posture and balance' points to Cerebellum; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Testosterone belongs with 'Principal androgen secreted by testes', whereas the tested description is 'Brain region coordinating posture and balance'. C: Neurotransmitter belongs with 'Chemical released from a presynaptic terminal', whereas the tested description is 'Brain region coordinating posture and balance'. D: Dendrites belongs with 'Neuron processes usually receiving impulses', whereas the tested description is 'Brain region coordinating posture and balance'.

BIO-12-CH09-0064 | Hard | Pancreas adrenal and gonads**Correct answer: D. Pancreatic hormone lowering blood glucose - Reflex action**

Solution: 'Pancreatic hormone lowering blood glucose' should be paired with Insulin, not Reflex action. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Supporting cells of nervous tissue - Neuroglia. B: This is a valid association: Ladder-like nervous system - Planaria. C: This is a valid association: Largest part of the human brain - Cerebrum.

BIO-12-CH09-0065 | Medium | Pituitary and thyroid**Correct answer: B. Growth hormone**

Solution: The chapter links Pituitary hormone promoting body growth with Growth hormone, so Growth hormone completes the statement.

Distractor review: A: Adrenaline belongs with 'Adrenal medullary fight-or-flight hormone', whereas the tested description is 'Pituitary hormone promoting body growth'. C: Myelin sheath belongs with 'Insulating sheath around many axons', whereas the tested description is 'Pituitary hormone promoting body growth'. D: Calcitonin belongs with 'Hormone lowering blood calcium', whereas the tested description is 'Pituitary hormone promoting body growth'.

BIO-12-CH09-0066 | Easy | Pituitary and thyroid**Correct answer: C. Pituitary hormone stimulating the thyroid**

Solution: TSH represents 'Pituitary hormone stimulating the thyroid'. Each alternative describes a different term or result.

Distractor review: A: 'Region controlling breathing and cardiovascular reflexes' identifies Medulla oblongata, not TSH. B: 'Pituitary hormone promoting body growth' identifies Growth hormone, not TSH. D: 'Neural pathway mediating a reflex' identifies Reflex arc, not TSH.

BIO-12-CH09-0067 | Easy | Impulse and synapse**Correct answer: D. Resting potential**

Solution: 'Membrane state with inside negative relative to outside' is the defining description for Resting potential. The other choices belong to different chapter concepts.

Distractor review: A: Neurotransmitter belongs with 'Chemical released from a presynaptic terminal', whereas the tested description is 'Membrane state with inside negative relative to outside'. B: Calcitonin belongs with 'Hormone lowering blood calcium', whereas the tested description is 'Membrane state with inside negative relative to outside'. C: Cerebellum belongs with 'Brain region coordinating posture and balance', whereas the tested description is 'Membrane state with inside negative relative to outside'.

BIO-12-CH09-0068 | Challenging | Impulse and synapse**Correct answer: B. II and IV only**

Solution: Statement I is incorrect: 'Rapid reversal of membrane polarity' corresponds to Depolarization, not Neurotransmitter. Statement II is correct because 'Chemical released from a presynaptic terminal' corresponds to Neurotransmitter. Statement III is incorrect: 'Junction between two neurons' corresponds to Synapse, not Cerebrum. Statement IV is correct because 'Largest part of the human brain' corresponds to Cerebrum. Therefore, the valid combination is II and IV only.

Distractor review: A: This combination includes false statement(s) III and omits true statement(s) IV. C: This combination omits true statement(s) IV. D: This combination includes false statement(s) III and omits true statement(s) II.

BIO-12-CH09-0069 | Medium | Impulse and synapse**Correct answer: C. Membrane state with inside negative relative to outside - Resting potential**

Solution: The accurate entry is 'Membrane state with inside negative relative to outside - Resting potential'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Brain region coordinating posture and balance' should be paired with Cerebellum, not Reflex arc. B: 'Iodine-deficiency enlargement of thyroid' should be paired with Simple goitre, not Photoreceptors. D: 'Receptors detecting light' should be paired with Photoreceptors, not Resting potential.

BIO-12-CH09-0070 | Easy | Pancreas adrenal and gonads**Correct answer: D. Pancreatic hormone lowering blood glucose**

Solution: Insulin represents 'Pancreatic hormone lowering blood glucose'. Each alternative describes a different term or result.

Distractor review: A: 'Largest part of the human brain' identifies Cerebrum, not Insulin. B: 'Principal androgen secreted by testes' identifies Testosterone, not Insulin. C: 'Insulin deficiency or ineffective action' identifies Diabetes mellitus, not Insulin.

BIO-12-CH09-0071 | Medium | Pituitary and thyroid**Correct answer: B. Pituitary hormone stimulating the thyroid - TSH**

Solution: The accurate entry is 'Pituitary hormone stimulating the thyroid - TSH'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Neural pathway mediating a reflex' should be paired with Reflex arc, not Testosterone. C: 'Insulin deficiency or ineffective action' should be paired with Diabetes mellitus, not Mechanoreceptors. D: 'Receptors detecting sound and pressure' should be paired with Mechanoreceptors, not Myelin sheath.

BIO-12-CH09-0072 | Medium | Impulse and synapse**Correct answer: B. Chemical released from a presynaptic terminal - Neurotransmitter**

Solution: The accurate entry is 'Chemical released from a presynaptic terminal - Neurotransmitter'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Receptors detecting light' should be paired with Photoreceptors, not Neuron. C: 'Excess growth hormone in adults' should be paired with Acromegaly, not Photoreceptors. D: 'Pituitary hormone promoting body growth' should be paired with Growth hormone, not Meninges.

BIO-12-CH09-0073 | Medium | Endocrine disorders**Correct answer: C. Insulin deficiency or ineffective action - Diabetes mellitus**

Solution: The accurate entry is 'Insulin deficiency or ineffective action - Diabetes mellitus'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Pituitary hormone promoting body growth' should be paired with Growth hormone, not Adrenaline. B: 'Adrenal medullary fight-or-flight hormone' should be paired with Adrenaline, not Reflex action. D: 'Ladder-like nervous system' should be paired with Planaria, not Growth hormone.

BIO-12-CH09-0074 | Medium | Receptors and reflexes**Correct answer: C. Reflex arc**

Solution: The observation 'Neural pathway mediating a reflex' points to Reflex arc; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Glucagon belongs with 'Pancreatic hormone raising blood glucose', whereas the tested description is 'Neural pathway mediating a reflex'. B: Mechanoreceptors belongs with 'Receptors detecting sound and pressure', whereas the tested description is 'Neural pathway mediating a reflex'. D: Insulin belongs with 'Pancreatic hormone lowering blood glucose', whereas the tested description is 'Neural pathway mediating a reflex'.

BIO-12-CH09-0075 | Medium | Neuron structure**Correct answer: D. Insulating sheath around many axons - Myelin sheath**

Solution: The accurate entry is 'Insulating sheath around many axons - Myelin sheath'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Receptors detecting light' should be paired with Photoreceptors, not Thyroxine. B: 'Pituitary hormone promoting body growth' should be paired with Growth hormone, not Depolarization. C: 'Rapid reversal of membrane polarity' should be paired with Depolarization, not Photoreceptors.

BIO-12-CH09-0076 | Hard | Receptors and reflexes**Correct answer: C. Rapid automatic response through a neural pathway - Medulla oblongata**

Solution: 'Rapid automatic response through a neural pathway' should be paired with Reflex action, not Medulla oblongata. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Hormone lowering blood calcium - Calcitonin. B: This is a valid association: Structural and functional unit of nervous tissue - Neuron. D: This is a valid association: Neuron processes usually receiving impulses - Dendrites.

BIO-12-CH09-0077 | Hard | Neuron structure

Correct answer: B. Long process conducting impulses away from the cell body - Calcitonin

Solution: 'Long process conducting impulses away from the cell body' should be paired with Axon, not Calcitonin. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Insulin deficiency or ineffective action - Diabetes mellitus. C: This is a valid association: Diffuse nerve network without a central brain - Hydra. D: This is a valid association: Structural and functional unit of nervous tissue - Neuron.

BIO-12-CH09-0078 | Easy | Pituitary and thyroid

Correct answer: C. Growth hormone

Solution: 'Pituitary hormone promoting body growth' is the defining description for Growth hormone. The other choices belong to different chapter concepts.

Distractor review: A: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Pituitary hormone promoting body growth'. B: Myelin sheath belongs with 'Insulating sheath around many axons', whereas the tested description is 'Pituitary hormone promoting body growth'. D: Dendrites belongs with 'Neuron processes usually receiving impulses', whereas the tested description is 'Pituitary hormone promoting body growth'.

BIO-12-CH09-0079 | Medium | Central nervous system

Correct answer: B. Medulla oblongata

Solution: The chapter links Region controlling breathing and cardiovascular reflexes with Medulla oblongata, so Medulla oblongata completes the statement.

Distractor review: A: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Region controlling breathing and cardiovascular reflexes'. C: Meninges belongs with 'Protective membranes around brain and spinal cord', whereas the tested description is 'Region controlling breathing and cardiovascular reflexes'. D: Resting potential belongs with 'Membrane state with inside negative relative to outside', whereas the tested description is 'Region controlling breathing and cardiovascular reflexes'.

BIO-12-CH09-0080 | Easy | Endocrine disorders

Correct answer: D. Insulin deficiency or ineffective action

Solution: Diabetes mellitus represents 'Insulin deficiency or ineffective action'. Each alternative describes a different term or result.

Distractor review: A: 'Adrenal medullary fight-or-flight hormone' identifies Adrenaline, not Diabetes mellitus. B: 'Protective membranes around brain and spinal cord' identifies Meninges, not Diabetes mellitus. C: 'Largest part of the human brain' identifies Cerebrum, not Diabetes mellitus.

BIO-12-CH09-0081 | Easy | Nervous organization

Correct answer: B. Hydra

Solution: 'Diffuse nerve network without a central brain' is the defining description for Hydra. The other choices belong to different chapter concepts.

Distractor review: A: Nodes of Ranvier belongs with 'Gaps between adjacent myelin segments', whereas the tested description is 'Diffuse nerve network without a central brain'. C: Planaria belongs with 'Ladder-like nervous system', whereas the tested description is 'Diffuse nerve network without a central brain'. D: Neuroglia belongs with 'Supporting cells of nervous tissue', whereas the tested description is 'Diffuse nerve network without a central brain'.

BIO-12-CH09-0082 | Hard | Pituitary and thyroid

Correct answer: C. Pituitary hormone promoting body growth - Growth hormone; Insulating sheath around many axons - Myelin sheath

Solution: Both parts of 'Pituitary hormone promoting body growth - Growth hormone; Insulating sheath around many axons - Myelin sheath' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Rapid automatic response through a neural pathway' requires Reflex action, and 'Long process conducting impulses away from the cell body' requires Axon. B: The answers have been interchanged: 'Long process conducting impulses away from the cell body' requires Axon, and 'Diffuse nerve network without a central brain' requires Hydra. D: The answers have been interchanged: 'Neuron processes usually receiving impulses' requires Dendrites, and 'Rapid automatic response through a neural pathway' requires Reflex action.

BIO-12-CH09-0083 | Easy | Receptors and reflexes**Correct answer: D. Neural pathway mediating a reflex**

Solution: Reflex arc represents 'Neural pathway mediating a reflex'. Each alternative describes a different term or result.

Distractor review: A: 'Pancreatic hormone lowering blood glucose' identifies Insulin, not Reflex arc. B: 'Gaps between adjacent myelin segments' identifies Nodes of Ranvier, not Reflex arc. C: 'Insulating sheath around many axons' identifies Myelin sheath, not Reflex arc.

BIO-12-CH09-0084 | Easy | Central nervous system**Correct answer: B. Medulla oblongata**

Solution: 'Region controlling breathing and cardiovascular reflexes' is the defining description for Medulla oblongata. The other choices belong to different chapter concepts.

Distractor review: A: Dendrites belongs with 'Neuron processes usually receiving impulses', whereas the tested description is 'Region controlling breathing and cardiovascular reflexes'. C: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Region controlling breathing and cardiovascular reflexes'. D: Cerebrum belongs with 'Largest part of the human brain', whereas the tested description is 'Region controlling breathing and cardiovascular reflexes'.

BIO-12-CH09-0085 | Medium | Pituitary and thyroid**Correct answer: D. Thyroxine**

Solution: The chapter links Thyroid hormone containing iodine with Thyroxine, so Thyroxine completes the statement.

Distractor review: A: Growth hormone belongs with 'Pituitary hormone promoting body growth', whereas the tested description is 'Thyroid hormone containing iodine'. B: Nodes of Ranvier belongs with 'Gaps between adjacent myelin segments', whereas the tested description is 'Thyroid hormone containing iodine'. C: Axon belongs with 'Long process conducting impulses away from the cell body', whereas the tested description is 'Thyroid hormone containing iodine'.

BIO-12-CH09-0086 | Medium | Central nervous system**Correct answer: B. Cerebellum**

Solution: The chapter links Brain region coordinating posture and balance with Cerebellum, so Cerebellum completes the statement.

Distractor review: A: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Brain region coordinating posture and balance'. C: Cerebrum belongs with 'Largest part of the human brain', whereas the tested description is 'Brain region coordinating posture and balance'. D: Meninges belongs with 'Protective membranes around brain and spinal cord', whereas the tested description is 'Brain region coordinating posture and balance'.

BIO-12-CH09-0087 | Easy | Neuron structure**Correct answer: D. Gaps between adjacent myelin segments**

Solution: Nodes of Ranvier represents 'Gaps between adjacent myelin segments'. Each alternative describes a different term or result.

Distractor review: A: 'Insulating sheath around many axons' identifies Myelin sheath, not Nodes of Ranvier. B: 'Membrane state with inside negative relative to outside' identifies Resting potential, not Nodes of Ranvier. C: 'Long process conducting impulses away from the cell body' identifies Axon, not Nodes of Ranvier.

BIO-12-CH09-0088 | Easy | Central nervous system**Correct answer: A. Region controlling breathing and cardiovascular reflexes**

Solution: Medulla oblongata represents 'Region controlling breathing and cardiovascular reflexes'. Each alternative describes a different term or result.

Distractor review: B: 'Largest part of the human brain' identifies Cerebrum, not Medulla oblongata. C: 'Long process conducting impulses away from the cell body' identifies Axon, not Medulla oblongata. D: 'Diffuse nerve network without a central brain' identifies Hydra, not Medulla oblongata.

BIO-12-CH09-0089 | Easy | Nervous organization**Correct answer: B. Structural and functional unit of nervous tissue**

Solution: Neuron represents 'Structural and functional unit of nervous tissue'. Each alternative describes a different term or result.

Distractor review: A: 'Principal androgen secreted by testes' identifies Testosterone, not Neuron. C: 'Pancreatic hormone raising blood glucose' identifies Glucagon, not Neuron. D: 'Supporting cells of nervous tissue' identifies Neuroglia, not Neuron.

BIO-12-CH09-0090 | Medium | Receptors and reflexes**Correct answer: A. Rapid automatic response through a neural pathway - Reflex action**

Solution: The accurate entry is 'Rapid automatic response through a neural pathway - Reflex action'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Pancreatic hormone raising blood glucose' should be paired with Glucagon, not Reflex action. C: 'Hormone lowering blood calcium' should be paired with Calcitonin, not Glucagon. D: 'Gaps between adjacent myelin segments' should be paired with Nodes of Ranvier, not Calcitonin.

BIO-12-CH09-0091 | Hard | Impulse and synapse**Correct answer: C. Membrane state with inside negative relative to outside - Cerebellum**

Solution: 'Membrane state with inside negative relative to outside' should be paired with Resting potential, not Cerebellum. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Iodine-deficiency enlargement of thyroid - Simple goitre. B: This is a valid association: Hormone lowering blood calcium - Calcitonin. D: This is a valid association: Neural pathway mediating a reflex - Reflex arc.

BIO-12-CH09-0092 | Challenging | Neuron structure**Correct answer: D. I only**

Solution: Statement I is correct because 'Insulating sheath around many axons' corresponds to Myelin sheath. Statement II is incorrect: 'Gaps between adjacent myelin segments' corresponds to Nodes of Ranvier, not Dendrites. Statement III is incorrect: 'Neuron processes usually receiving impulses' corresponds to Dendrites, not Dwarfism. Statement IV is incorrect: 'Childhood deficiency of growth hormone' corresponds to Dwarfism, not Myelin sheath. Therefore, the valid combination is I only.

Distractor review: A: This combination includes false statement(s) II, III, IV. B: This combination includes false statement(s) III and omits true statement(s) I. C: This combination includes false statement(s) II, III, IV and omits true statement(s) I.

BIO-12-CH09-0093 | Medium | Nervous organization**Correct answer: C. Structural and functional unit of nervous tissue - Neuron**

Solution: The accurate entry is 'Structural and functional unit of nervous tissue - Neuron'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Insulin deficiency or ineffective action' should be paired with Diabetes mellitus, not Testosterone. B: 'Thyroid hormone containing iodine' should be paired with Thyroxine, not Neuron. D: 'Principal androgen secreted by testes' should be paired with Testosterone, not Thyroxine.

BIO-12-CH09-0094 | Easy | Neuron structure**Correct answer: D. Nodes of Ranvier**

Solution: 'Gaps between adjacent myelin segments' is the defining description for Nodes of Ranvier. The other choices belong to different chapter concepts.

Distractor review: A: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Gaps between adjacent myelin segments'. B: Axon belongs with 'Long process conducting impulses away from the cell body', whereas the tested description is 'Gaps between adjacent myelin segments'. C: Glucagon belongs with 'Pancreatic hormone raising blood glucose', whereas the tested description is 'Gaps between adjacent myelin segments'.

BIO-12-CH09-0095 | Hard | Central nervous system**Correct answer: A. Protective membranes around brain and spinal cord - Meninges; Iodine-deficiency enlargement of thyroid - Simple goitre**

Solution: Both parts of 'Protective membranes around brain and spinal cord - Meninges; Iodine-deficiency enlargement of thyroid - Simple goitre' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Long process conducting impulses away from the cell body' requires Axon, and 'Pituitary hormone stimulating the thyroid' requires TSH. C: The answers have been interchanged: 'Pituitary hormone stimulating the thyroid' requires TSH, and 'Structural and functional unit of nervous tissue' requires Neuron. D: The answers have been interchanged: 'Insulin deficiency or ineffective action' requires Diabetes mellitus, and 'Rapid automatic response through a neural pathway' requires Reflex action.

BIO-12-CH09-0096 | Challenging | Pituitary and thyroid**Correct answer: D. II and III only**

Solution: Statement I is incorrect: 'Hormone lowering blood calcium' corresponds to Calcitonin, not TSH. Statement II is correct because 'Pituitary hormone stimulating the thyroid' corresponds to TSH. Statement III is correct because 'Pituitary hormone promoting body growth' corresponds to Growth hormone. Statement IV is incorrect: 'Structural and functional unit of nervous tissue' corresponds to Neuron, not Calcitonin. Therefore, the valid combination is II and III only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) II. B: This combination omits true statement(s) III. C: This combination includes false statement(s) I and omits true statement(s) III.

BIO-12-CH09-0097 | Challenging | Endocrine disorders**Correct answer: A. I, II, III and IV**

Solution: Statement I is correct because 'Iodine-deficiency enlargement of thyroid' corresponds to Simple goitre. Statement II is correct because 'Excess growth hormone in adults' corresponds to Acromegaly. Statement III is correct because 'Insulin deficiency or ineffective action' corresponds to Diabetes mellitus. Statement IV is correct because 'Rapid automatic response through a neural pathway' corresponds to Reflex action. Therefore, the valid combination is I, II, III and IV.

Distractor review: B: This combination omits true statement(s) I, II. C: This combination omits true statement(s) II. D: This combination omits true statement(s) I.

BIO-12-CH09-0098 | Medium | Pancreas adrenal and gonads**Correct answer: A. Testosterone**

Solution: The chapter links Principal androgen secreted by testes with Testosterone, so Testosterone completes the statement.

Distractor review: B: Glucagon belongs with 'Pancreatic hormone raising blood glucose', whereas the tested description is 'Principal androgen secreted by testes'. C: Reflex arc belongs with 'Neural pathway mediating a reflex', whereas the tested description is 'Principal androgen secreted by testes'. D: Growth hormone belongs with 'Pituitary hormone promoting body growth', whereas the tested description is 'Principal androgen secreted by testes'.

BIO-12-CH09-0099 | Medium | Impulse and synapse**Correct answer: C. Synapse**

Solution: The chapter links Junction between two neurons with Synapse, so Synapse completes the statement.

Distractor review: A: Neurotransmitter belongs with 'Chemical released from a presynaptic terminal', whereas the tested description is 'Junction between two neurons'. B: Reflex action belongs with 'Rapid automatic response through a neural pathway', whereas the tested description is 'Junction between two neurons'. D: Depolarization belongs with 'Rapid reversal of membrane polarity', whereas the tested description is 'Junction between two neurons'.

BIO-12-CH09-0100 | Easy | Impulse and synapse**Correct answer: D. Junction between two neurons**

Solution: Synapse represents 'Junction between two neurons'. Each alternative describes a different term or result.

Distractor review: A: 'Rapid reversal of membrane polarity' identifies Depolarization, not Synapse. B: 'Insulating sheath around many axons' identifies Myelin sheath, not Synapse. C: 'Insulin deficiency or ineffective action' identifies Diabetes mellitus, not Synapse.

BIO-12-CH09-0101 | Medium | Pituitary and thyroid**Correct answer: D. Calcitonin**

Solution: The observation 'Hormone lowering blood calcium' points to Calcitonin; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: TSH belongs with 'Pituitary hormone stimulating the thyroid', whereas the tested description is 'Hormone lowering blood calcium'. B: Neuron belongs with 'Structural and functional unit of nervous tissue', whereas the tested description is 'Hormone lowering blood calcium'. C: Growth hormone belongs with 'Pituitary hormone promoting body growth', whereas the tested description is 'Hormone lowering blood calcium'.

BIO-12-CH09-0102 | Medium | Pancreas adrenal and gonads**Correct answer: B. Adrenaline**

Solution: The chapter links Adrenal medullary fight-or-flight hormone with Adrenaline, so Adrenaline completes the statement.

Distractor review: A: Depolarization belongs with 'Rapid reversal of membrane polarity', whereas the tested description is 'Adrenal medullary fight-or-flight hormone'. C: Reflex arc belongs with 'Neural pathway mediating a reflex', whereas the tested description is 'Adrenal medullary fight-or-flight hormone'. D: Neuron belongs with 'Structural and functional unit of nervous tissue', whereas the tested description is 'Adrenal medullary fight-or-flight hormone'.

BIO-12-CH09-0103 | Medium | Central nervous system**Correct answer: B. Largest part of the human brain - Cerebrum**

Solution: The accurate entry is 'Largest part of the human brain - Cerebrum'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Junction between two neurons' should be paired with Synapse, not Simple goitre. C: 'Neuron processes usually receiving impulses' should be paired with Dendrites, not Synapse. D: 'Principal androgen secreted by testes' should be paired with Testosterone, not Cerebrum.

BIO-12-CH09-0104 | Challenging | Nervous organization**Correct answer: D. I and IV only**

Solution: Statement I is correct because 'Structural and functional unit of nervous tissue' corresponds to Neuron. Statement II is incorrect: 'Diffuse nerve network without a central brain' corresponds to Hydra, not Planaria. Statement III is incorrect: 'Ladder-like nervous system' corresponds to Planaria, not Adrenaline. Statement IV is correct because 'Adrenal medullary fight-or-flight hormone' corresponds to Adrenaline. Therefore, the valid combination is I and IV only.

Distractor review: A: This combination includes false statement(s) II, III and omits true statement(s) I. B: This combination includes false statement(s) III and omits true statement(s) IV. C: This combination includes false statement(s) II, III and omits true statement(s) I, IV.

BIO-12-CH09-0105 | Medium | Receptors and reflexes**Correct answer: A. Reflex action**

Solution: The chapter links Rapid automatic response through a neural pathway with Reflex action, so Reflex action completes the statement.

Distractor review: B: Calcitonin belongs with 'Hormone lowering blood calcium', whereas the tested description is 'Rapid automatic response through a neural pathway'. C: Medulla oblongata belongs with 'Region controlling breathing and cardiovascular reflexes', whereas the tested description is 'Rapid automatic response through a neural pathway'. D: Neuron belongs with 'Structural and functional unit of nervous tissue', whereas the tested description is 'Rapid automatic response through a neural pathway'.

BIO-12-CH09-0106 | Easy | Pituitary and thyroid**Correct answer: C. Pituitary hormone promoting body growth**

Solution: Growth hormone represents 'Pituitary hormone promoting body growth'. Each alternative describes a different term or result.

Distractor review: A: 'Adrenal medullary fight-or-flight hormone' identifies Adrenaline, not Growth hormone. B: 'Neuron processes usually receiving impulses' identifies Dendrites, not Growth hormone. D: 'Long process conducting impulses away from the cell body' identifies Axon, not Growth hormone.

BIO-12-CH09-0107 | Medium | Central nervous system**Correct answer: C. Protective membranes around brain and spinal cord - Meninges**

Solution: The accurate entry is 'Protective membranes around brain and spinal cord - Meninges'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Pituitary hormone stimulating the thyroid' should be paired with TSH, not Testosterone. B: 'Structural and functional unit of nervous tissue' should be paired with Neuron, not Meninges. D: 'Insulin deficiency or ineffective action' should be paired with Diabetes mellitus, not Axon.

BIO-12-CH09-0108 | Hard | Central nervous system**Correct answer: B. Region controlling breathing and cardiovascular reflexes - Resting potential**

Solution: 'Region controlling breathing and cardiovascular reflexes' should be paired with Medulla oblongata, not Resting potential. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Pituitary hormone stimulating the thyroid - TSH. C: This is a valid association: Insulin deficiency or ineffective action - Diabetes mellitus. D: This is a valid association: Long process conducting impulses away from the cell body - Axon.

BIO-12-CH09-0109 | Medium | Endocrine disorders**Correct answer: C. Diabetes mellitus**

Solution: The observation 'Insulin deficiency or ineffective action' points to Diabetes mellitus; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Cerebrum belongs with 'Largest part of the human brain', whereas the tested description is 'Insulin deficiency or ineffective action'. B: Acromegaly belongs with 'Excess growth hormone in adults', whereas the tested description is 'Insulin deficiency or ineffective action'. D: Planaria belongs with 'Ladder-like nervous system', whereas the tested description is 'Insulin deficiency or ineffective action'.

BIO-12-CH09-0110 | Challenging | Pituitary and thyroid**Correct answer: A. II, III and IV only**

Solution: Statement I is incorrect: 'Pituitary hormone promoting body growth' corresponds to Growth hormone, not TSH. Statement II is correct because 'Pituitary hormone stimulating the thyroid' corresponds to TSH. Statement III is correct because 'Hormone lowering blood calcium' corresponds to Calcitonin. Statement IV is correct because 'Long process conducting impulses away from the cell body' corresponds to Axon. Therefore, the valid combination is II, III and IV only.

Distractor review: B: This combination includes false statement(s) I and omits true statement(s) III. C: This combination includes false statement(s) I and omits true statement(s) IV. D: This combination omits true statement(s) II, III.

BIO-12-CH09-0111 | Hard | Impulse and synapse**Correct answer: B. Chemical released from a presynaptic terminal - Neurotransmitter; Neuron processes usually receiving impulses - Dendrites**

Solution: Both parts of 'Chemical released from a presynaptic terminal - Neurotransmitter; Neuron processes usually receiving impulses - Dendrites' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Excess growth hormone in adults' requires Acromegaly, and 'Protective membranes around brain and spinal cord' requires Meninges. C: The answers have been interchanged: 'Receptors detecting light' requires Photoreceptors, and 'Long process conducting impulses away from the cell body' requires Axon. D: The answers have been interchanged: 'Protective membranes around brain and spinal cord' requires Meninges, and 'Receptors detecting light' requires Photoreceptors.

BIO-12-CH09-0112 | Easy | Receptors and reflexes**Correct answer: D. Receptors detecting light**

Solution: Photoreceptors represents 'Receptors detecting light'. Each alternative describes a different term or result.

Distractor review: A: 'Pituitary hormone stimulating the thyroid' identifies TSH, not Photoreceptors. B: 'Neural pathway mediating a reflex' identifies Reflex arc, not Photoreceptors. C: 'Pituitary hormone promoting body growth' identifies Growth hormone, not Photoreceptors.

BIO-12-CH09-0113 | Medium | Nervous organization**Correct answer: C. Planaria**

Solution: The observation 'Ladder-like nervous system' points to Planaria; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Thyroxine belongs with 'Thyroid hormone containing iodine', whereas the tested description is 'Ladder-like nervous system'. B: Neuroglia belongs with 'Supporting cells of nervous tissue', whereas the tested description is 'Ladder-like nervous system'. D: Calcitonin belongs with 'Hormone lowering blood calcium', whereas the tested description is 'Ladder-like nervous system'.

BIO-12-CH09-0114 | Easy | Central nervous system**Correct answer: C. Protective membranes around brain and spinal cord**

Solution: Meninges represents 'Protective membranes around brain and spinal cord'. Each alternative describes a different term or result.

Distractor review: A: 'Largest part of the human brain' identifies Cerebrum, not Meninges. B: 'Rapid automatic response through a neural pathway' identifies Reflex action, not Meninges. D: 'Long process conducting impulses away from the cell body' identifies Axon, not Meninges.

BIO-12-CH09-0115 | Easy | Receptors and reflexes**Correct answer: A. Reflex action**

Solution: 'Rapid automatic response through a neural pathway' is the defining description for Reflex action. The other choices belong to different chapter concepts.

Distractor review: B: Neuron belongs with 'Structural and functional unit of nervous tissue', whereas the tested description is 'Rapid automatic response through a neural pathway'. C: Dendrites belongs with 'Neuron processes usually receiving impulses', whereas the tested description is 'Rapid automatic response through a neural pathway'. D: Mechanoreceptors belongs with 'Receptors detecting sound and pressure', whereas the tested description is 'Rapid automatic response through a neural pathway'.

BIO-12-CH09-0116 | Hard | Neuron structure**Correct answer: D. Gaps between adjacent myelin segments - Nodes of Ranvier; Pancreatic hormone raising blood glucose - Glucagon**

Solution: Both parts of 'Gaps between adjacent myelin segments - Nodes of Ranvier; Pancreatic hormone raising blood glucose - Glucagon' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Membrane state with inside negative relative to outside' requires Resting potential, and 'Diffuse nerve network without a central brain' requires Hydra. B: The answers have been interchanged: 'Diffuse nerve network without a central brain' requires Hydra, and 'Brain region coordinating posture and balance' requires Cerebellum. C: The answers have been interchanged: 'Brain region coordinating posture and balance' requires Cerebellum, and 'Structural and functional unit of nervous tissue' requires Neuron.

BIO-12-CH09-0117 | Easy | Pituitary and thyroid**Correct answer: A. TSH**

Solution: 'Pituitary hormone stimulating the thyroid' is the defining description for TSH. The other choices belong to different chapter concepts.

Distractor review: B: Reflex arc belongs with 'Neural pathway mediating a reflex', whereas the tested description is 'Pituitary hormone stimulating the thyroid'. C: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Pituitary hormone stimulating the thyroid'. D: Testosterone belongs with 'Principal androgen secreted by testes', whereas the tested description is 'Pituitary hormone stimulating the thyroid'.

BIO-12-CH09-0118 | Challenging | Nervous organization**Correct answer: D. I and III only**

Solution: Statement I is correct because 'Diffuse nerve network without a central brain' corresponds to Hydra. Statement II is incorrect: 'Supporting cells of nervous tissue' corresponds to Neuroglia, not Neuron. Statement III is correct because 'Structural and functional unit of nervous tissue' corresponds to Neuron. Statement IV is incorrect: 'Insulin deficiency or ineffective action' corresponds to Diabetes mellitus, not Hydra. Therefore, the valid combination is I and III only.

Distractor review: A: This combination includes false statement(s) II, IV. B: This combination includes false statement(s) IV and omits true statement(s) I, III. C: This combination includes false statement(s) II and omits true statement(s) I.

BIO-12-CH09-0119 | Hard | Receptors and reflexes**Correct answer: B. Neural pathway mediating a reflex - Reflex arc; Insulating sheath around many axons - Myelin sheath**

Solution: Both parts of 'Neural pathway mediating a reflex - Reflex arc; Insulating sheath around many axons - Myelin sheath' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Pancreatic hormone raising blood glucose' requires Glucagon, and 'Region controlling breathing and cardiovascular reflexes' requires Medulla oblongata. C: The answers have been interchanged: 'Junction between two neurons' requires Synapse, and 'Insulin deficiency or ineffective action' requires Diabetes mellitus. D: The answers have been interchanged: 'Region controlling breathing and cardiovascular reflexes' requires Medulla oblongata, and 'Gaps between adjacent myelin segments' requires Nodes of Ranvier.

BIO-12-CH09-0120 | Easy | Impulse and synapse**Correct answer: A. Synapse**

Solution: 'Junction between two neurons' is the defining description for Synapse. The other choices belong to different chapter concepts.

Distractor review: B: Hydra belongs with 'Diffuse nerve network without a central brain', whereas the tested description is 'Junction between two neurons'. C: Depolarization belongs with 'Rapid reversal of membrane polarity', whereas the tested description is 'Junction between two neurons'. D: Neurotransmitter belongs with 'Chemical released from a presynaptic terminal', whereas the tested description is 'Junction between two neurons'.

BIO-12-CH09-0121 | Challenging | Endocrine disorders**Correct answer: B. I, III and IV only**

Solution: Statement I is correct because 'Childhood deficiency of growth hormone' corresponds to Dwarfism. Statement II is incorrect: 'Iodine-deficiency enlargement of thyroid' corresponds to Simple goitre, not Diabetes mellitus. Statement III is correct because 'Insulin deficiency or ineffective action' corresponds to Diabetes mellitus. Statement IV is correct because 'Insulating sheath around many axons' corresponds to Myelin sheath. Therefore, the valid combination is I, III and IV only.

Distractor review: A: This combination omits true statement(s) III. C: This combination includes false statement(s) II and omits true statement(s) I, III. D: This combination omits true statement(s) I, III.

BIO-12-CH09-0122 | Hard | Nervous organization**Correct answer: D. Diffuse nerve network without a central brain - Dendrites**

Solution: 'Diffuse nerve network without a central brain' should be paired with Hydra, not Dendrites. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Excess growth hormone in adults - Acromegaly. B: This is a valid association: Insulin deficiency or ineffective action - Diabetes mellitus. C: This is a valid association: Brain region coordinating posture and balance - Cerebellum.

BIO-12-CH09-0123 | Easy | Nervous organization**Correct answer: B. Supporting cells of nervous tissue**

Solution: Neuroglia represents 'Supporting cells of nervous tissue'. Each alternative describes a different term or result.

Distractor review: A: 'Adrenal medullary fight-or-flight hormone' identifies Adrenaline, not Neuroglia. C: 'Pancreatic hormone lowering blood glucose' identifies Insulin, not Neuroglia. D: 'Ladder-like nervous system' identifies Planaria, not Neuroglia.

BIO-12-CH09-0124 | Hard | Central nervous system**Correct answer: D. Region controlling breathing and cardiovascular reflexes - Medulla oblongata; Membrane state with inside negative relative to outside - Resting potential**

Solution: Both parts of 'Region controlling breathing and cardiovascular reflexes - Medulla oblongata; Membrane state with inside negative relative to outside - Resting potential' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Diffuse nerve network without a central brain' requires Hydra, and 'Neuron processes usually receiving impulses' requires Dendrites. B: The answers have been interchanged: 'Insulin deficiency or ineffective action' requires Diabetes mellitus, and 'Pituitary hormone stimulating the thyroid' requires TSH. C: The answers have been interchanged: 'Long process conducting impulses away from the cell body' requires Axon, and 'Insulin deficiency or ineffective action' requires Diabetes mellitus.

BIO-12-CH09-0125 | Easy | Neuron structure**Correct answer: D. Axon**

Solution: 'Long process conducting impulses away from the cell body' is the defining description for Axon. The other choices belong to different chapter concepts.

Distractor review: A: Adrenaline belongs with 'Adrenal medullary fight-or-flight hormone', whereas the tested description is 'Long process conducting impulses away from the cell body'. B: Neuroglia belongs with 'Supporting cells of nervous tissue', whereas the tested description is 'Long process conducting impulses away from the cell body'. C: Diabetes mellitus belongs with 'Insulin deficiency or ineffective action', whereas the tested description is 'Long process conducting impulses away from the cell body'.

BIO-12-CH09-0126 | Hard | Receptors and reflexes**Correct answer: A. Receptors detecting light - Growth hormone**

Solution: 'Receptors detecting light' should be paired with Photoreceptors, not Growth hormone. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Excess growth hormone in adults - Acromegaly. C: This is a valid association: Iodine-deficiency enlargement of thyroid - Simple goitre. D: This is a valid association: Pituitary hormone stimulating the thyroid - TSH.

BIO-12-CH09-0127 | Hard | Endocrine disorders**Correct answer: D. Insulin deficiency or ineffective action - Diabetes mellitus; Ladder-like nervous system - Planaria**

Solution: Both parts of 'Insulin deficiency or ineffective action - Diabetes mellitus; Ladder-like nervous system - Planaria' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Protective membranes around brain and spinal cord' requires Meninges, and 'Pituitary hormone promoting body growth' requires Growth hormone. B: The answers have been interchanged: 'Chemical released from a presynaptic terminal' requires Neurotransmitter, and 'Rapid automatic response through a neural pathway' requires Reflex action. C: The answers have been interchanged: 'Largest part of the human brain' requires Cerebrum, and 'Adrenal medullary fight-or-flight hormone' requires Adrenaline.

BIO-12-CH09-0128 | Medium | Impulse and synapse**Correct answer: B. Neurotransmitter**

Solution: The observation 'Chemical released from a presynaptic terminal' points to Neurotransmitter; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Depolarization belongs with 'Rapid reversal of membrane polarity', whereas the tested description is 'Chemical released from a presynaptic terminal'. C: Acromegaly belongs with 'Excess growth hormone in adults', whereas the tested description is 'Chemical released from a presynaptic terminal'. D: Growth hormone belongs with 'Pituitary hormone promoting body growth', whereas the tested description is 'Chemical released from a presynaptic terminal'.

BIO-12-CH09-0129 | Medium | Endocrine disorders**Correct answer: A. Excess growth hormone in adults - Acromegaly**

Solution: The accurate entry is 'Excess growth hormone in adults - Acromegaly'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Ladder-like nervous system' should be paired with Planaria, not Testosterone. C: 'Principal androgen secreted by testes' should be paired with Testosterone, not Cerebellum. D: 'Structural and functional unit of nervous tissue' should be paired with Neuron, not Mechanoreceptors.

BIO-12-CH09-0130 | Hard | Pituitary and thyroid**Correct answer: C. Pituitary hormone stimulating the thyroid - TSH; Neural pathway mediating a reflex - Reflex arc**

Solution: Both parts of 'Pituitary hormone stimulating the thyroid - TSH; Neural pathway mediating a reflex - Reflex arc' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Principal androgen secreted by testes' requires Testosterone, and 'Receptors detecting sound and pressure' requires Mechanoreceptors. B: The answers have been interchanged: 'Insulin deficiency or ineffective action' requires Diabetes mellitus, and 'Principal androgen secreted by testes' requires Testosterone. D: The answers have been interchanged: 'Insulating sheath around many axons' requires Myelin sheath, and 'Iodine-deficiency enlargement of thyroid' requires Simple goitre.

BIO-12-CH09-0131 | Hard | Endocrine disorders**Correct answer: B. Childhood deficiency of growth hormone - Growth hormone**

Solution: 'Childhood deficiency of growth hormone' should be paired with Dwarfism, not Growth hormone. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Pancreatic hormone lowering blood glucose - Insulin. C: This is a valid association: Largest part of the human brain - Cerebrum. D: This is a valid association: Pituitary hormone stimulating the thyroid - TSH.

BIO-12-CH09-0132 | Medium | Nervous organization**Correct answer: A. Neuron**

Solution: The chapter links Structural and functional unit of nervous tissue with Neuron, so Neuron completes the statement.

Distractor review: B: Planaria belongs with 'Ladder-like nervous system', whereas the tested description is 'Structural and functional unit of nervous tissue'. C: Neuroglia belongs with 'Supporting cells of nervous tissue', whereas the tested description is 'Structural and functional unit of nervous tissue'. D: Synapse belongs with 'Junction between two neurons', whereas the tested description is 'Structural and functional unit of nervous tissue'.

Standard 12 | Chapter 10

Human Health and Diseases

BIO-12-CH10-0001 | Challenging | Key transfers in the malaria life cycle

Correct answer: A. Transmission from mosquito to human is greatly reduced

Solution: Sporozoites must reach the salivary glands to be inoculated into a human during feeding.

Distractor review: B: The alternative 'Human red cells release more merozoites' conflicts with the labelled relationship, scale or mechanism in the figure. Sporozoites must reach the salivary glands to be inoculated into a human during feeding. C: The alternative 'All human gametocytes become sporozoites' conflicts with the labelled relationship, scale or mechanism in the figure. Sporozoites must reach the salivary glands to be inoculated into a human during feeding. D: The alternative 'Mosquitoes can transmit malaria without biting' conflicts with the labelled relationship, scale or mechanism in the figure. Sporozoites must reach the salivary glands to be inoculated into a human during feeding.

BIO-12-CH10-0002 | Medium | Key transfers in the malaria life cycle

Correct answer: B. Sporozoites

Solution: Sporozoites in mosquito saliva enter the human host during a bite.

Distractor review: A: The alternative 'Merozoites' conflicts with the labelled relationship, scale or mechanism in the figure. Sporozoites in mosquito saliva enter the human host during a bite. C: The alternative 'Gametocytes' conflicts with the labelled relationship, scale or mechanism in the figure. Sporozoites in mosquito saliva enter the human host during a bite. D: The alternative 'Trophozoites inside red cells' conflicts with the labelled relationship, scale or mechanism in the figure. Sporozoites in mosquito saliva enter the human host during a bite.

BIO-12-CH10-0003 | Hard | Primary and secondary antibody responses

Correct answer: B. Rapid clonal expansion of antigen-specific memory cells

Solution: Memory B cells formed after first exposure rapidly generate many antibody-secreting cells after a booster.

Distractor review: A: The alternative 'Immediate loss of all plasma cells' conflicts with the labelled relationship, scale or mechanism in the figure. Memory B cells formed after first exposure rapidly generate many antibody-secreting cells after a booster. C: The alternative 'Nonspecific phagocytosis alone' conflicts with the labelled relationship, scale or mechanism in the figure. Memory B cells formed after first exposure rapidly generate many antibody-secreting cells after a booster. D: The alternative 'A permanent fall in antibody affinity' conflicts with the labelled relationship, scale or mechanism in the figure. Memory B cells formed after first exposure rapidly generate many antibody-secreting cells after a booster.

BIO-12-CH10-0004 | Medium | Immunity

Correct answer: A. Acquired immunity

Solution: The observation 'Specific defence developing after exposure' points to Acquired immunity; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Vaccination belongs with 'Administration of antigenic material to induce immune memory', whereas the tested description is 'Specific defence developing after exposure'. C: Innate immunity belongs with 'Non-specific defence present from birth', whereas the tested description is 'Specific defence developing after exposure'. D: Carcinogens belongs with 'Agents capable of causing cancer', whereas the tested description is 'Specific defence developing after exposure'.

BIO-12-CH10-0005 | Medium | Cancer and immunity disorders

Correct answer: D. Spread of malignant cells to distant sites - Metastasis

Solution: The accurate entry is 'Spread of malignant cells to distant sites - Metastasis'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Administration of antigenic material to induce immune memory' should be paired with Vaccination, not Active immunity. B: 'Stimulant derived from coca plant' should be paired with Cocaine, not Vaccination. C: 'Control approach that interrupts a pathogen's vector' should be paired with Vector control, not Salmonella typhi.

BIO-12-CH10-0006 | Challenging | Antibody structure**Correct answer: B. I, II, III and IV**

Solution: Statement I is correct because 'Basic shape of an antibody molecule' corresponds to Y-shaped. Statement II is correct because 'Stem region interacting with immune components' corresponds to Constant Fc region. Statement III is correct because 'Chains present in each antibody molecule' corresponds to Two heavy and two light chains. Statement IV is correct because 'Control approach that interrupts a pathogen's vector' corresponds to Vector control. Therefore, the valid combination is I, II, III and IV.

Distractor review: A: This combination omits true statement(s) III. C: This combination omits true statement(s) I, II. D: This combination omits true statement(s) I, III.

BIO-12-CH10-0007 | Easy | Primary and secondary antibody responses**Correct answer: D. Memory lymphocytes respond rapidly**

Solution: Memory cells generated during the primary response expand rapidly on re-exposure.

Distractor review: A: The alternative 'All antibodies become hormones' conflicts with the labelled relationship, scale or mechanism in the figure. Memory cells generated during the primary response expand rapidly on re-exposure. B: The alternative 'The antigen no longer has epitopes' conflicts with the labelled relationship, scale or mechanism in the figure. Memory cells generated during the primary response expand rapidly on re-exposure. C: The alternative 'Innate immunity permanently replaces adaptive immunity' conflicts with the labelled relationship, scale or mechanism in the figure. Memory cells generated during the primary response expand rapidly on re-exposure.

BIO-12-CH10-0008 | Medium | Viral and helminth diseases**Correct answer: D. Wuchereria**

Solution: The chapter links Filarial worm transmitted by mosquitoes with Wuchereria, so Wuchereria completes the statement.

Distractor review: A: HIV belongs with 'Virus that targets helper T lymphocytes', whereas the tested description is 'Filarial worm transmitted by mosquitoes'. B: Vector control belongs with 'Control approach that interrupts a pathogen's vector', whereas the tested description is 'Filarial worm transmitted by mosquitoes'. C: Cannabis sativa belongs with 'Cannabinoid source plant', whereas the tested description is 'Filarial worm transmitted by mosquitoes'.

BIO-12-CH10-0009 | Medium | Bacterial and protozoan diseases**Correct answer: C. Plasmodium**

Solution: The chapter links Protozoan causing malaria with Plasmodium, so Plasmodium completes the statement.

Distractor review: A: Female Anopheles belongs with 'Mosquito transmitting malaria', whereas the tested description is 'Protozoan causing malaria'. B: ELISA belongs with 'Diagnostic test commonly used to detect HIV antibodies', whereas the tested description is 'Protozoan causing malaria'. D: Streptococcus pneumoniae belongs with 'Causative bacterium of pneumonia commonly cited in the chapter', whereas the tested description is 'Protozoan causing malaria'.

BIO-12-CH10-0010 | Challenging | Bacterial and protozoan diseases**Correct answer: C. I and IV only**

Solution: Statement I is correct because 'Causative bacterium of pneumonia commonly cited in the chapter' corresponds to Streptococcus pneumoniae. Statement II is incorrect: 'Protozoan causing malaria' corresponds to Plasmodium, not Female Anopheles. Statement III is incorrect: 'Mosquito transmitting malaria' corresponds to Female Anopheles, not Wuchereria. Statement IV is correct because 'Filarial worm transmitted by mosquitoes' corresponds to Wuchereria. Therefore, the valid combination is I and IV only.

Distractor review: A: This combination includes false statement(s) II. B: This combination includes false statement(s) II, III and omits true statement(s) I, IV. D: This combination includes false statement(s) II, III and omits true statement(s) IV.

BIO-12-CH10-0011 | Challenging | Immunity**Correct answer: D. II, III and IV only**

Solution: Statement I is incorrect: 'Non-specific defence present from birth' corresponds to Innate immunity, not Acquired immunity. Statement II is correct because 'Specific defence developing after exposure' corresponds to Acquired immunity. Statement III is correct because 'Immediate immunity produced by transferred antibodies' corresponds to Passive immunity. Statement IV is correct because 'Virus that targets helper T lymphocytes' corresponds to HIV. Therefore, the valid combination is II, III and IV only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) II, III, IV. B: This combination omits true statement(s) II. C: This combination includes false statement(s) I and omits true statement(s) II, III.

BIO-12-CH10-0012 | Easy | Drugs and addiction**Correct answer: D. Opioid obtained from poppy latex**

Solution: Morphine represents 'Opioid obtained from poppy latex'. Each alternative describes a different term or result.

Distractor review: A: 'Responsible adolescent practice that prevents substance misuse' identifies Avoiding peer pressure and seeking guidance, not Morphine. B: 'Repeated compulsive substance use despite harm' identifies Addiction, not Morphine. C: 'Non-specific defence present from birth' identifies Innate immunity, not Morphine.

BIO-12-CH10-0013 | Medium | Drugs and addiction**Correct answer: C. Addiction**

Solution: The chapter links Repeated compulsive substance use despite harm with Addiction, so Addiction completes the statement.

Distractor review: A: Passive immunity belongs with 'Immediate immunity produced by transferred antibodies', whereas the tested description is 'Repeated compulsive substance use despite harm'. B: Morphine belongs with 'Opioid obtained from poppy latex', whereas the tested description is 'Repeated compulsive substance use despite harm'. D: Cannabis sativa belongs with 'Cannabinoid source plant', whereas the tested description is 'Repeated compulsive substance use despite harm'.

BIO-12-CH10-0014 | Medium | Antibody structure**Correct answer: A. Variable regions**

Solution: The chapter links Antigen-binding regions with Variable regions, so Variable regions completes the statement.

Distractor review: B: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Antigen-binding regions'. C: Morphine belongs with 'Opioid obtained from poppy latex', whereas the tested description is 'Antigen-binding regions'. D: Two heavy and two light chains belongs with 'Chains present in each antibody molecule', whereas the tested description is 'Antigen-binding regions'.

BIO-12-CH10-0015 | Easy | Immunity**Correct answer: A. Innate immunity**

Solution: 'Non-specific defence present from birth' is the defining description for Innate immunity. The other choices belong to different chapter concepts.

Distractor review: B: Variable regions belongs with 'Antigen-binding regions', whereas the tested description is 'Non-specific defence present from birth'. C: Metastasis belongs with 'Spread of malignant cells to distant sites', whereas the tested description is 'Non-specific defence present from birth'. D: Acquired immunity belongs with 'Specific defence developing after exposure', whereas the tested description is 'Non-specific defence present from birth'.

BIO-12-CH10-0016 | Challenging | Cancer and immunity disorders**Correct answer: C. I and III only**

Solution: Statement I is correct because 'Spread of malignant cells to distant sites' corresponds to Metastasis. Statement II is incorrect: 'Uncontrolled abnormal cell division' corresponds to Cancer, not Carcinogens. Statement III is correct because 'Agents capable of causing cancer' corresponds to Carcinogens. Statement IV is incorrect: 'Control approach that interrupts a pathogen's vector' corresponds to Vector control, not Metastasis. Therefore, the valid combination is I and III only.

Distractor review: A: This combination includes false statement(s) II, IV and omits true statement(s) I, III. B: This combination omits true statement(s) I. D: This combination includes false statement(s) IV and omits true statement(s) I, III.

BIO-12-CH10-0017 | Hard | Viral and helminth diseases**Correct answer: B. Roundworm causing ascariasis - Ascaris; Opioid obtained from poppy latex - Morphine**

Solution: Both parts of 'Roundworm causing ascariasis - Ascaris; Opioid obtained from poppy latex - Morphine' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Stem region interacting with immune components' requires Constant Fc region, and 'Immediate immunity produced by transferred antibodies' requires Passive immunity. C: The answers have been interchanged: 'Causative bacterium of typhoid' requires Salmonella typhi, and 'Mosquito transmitting malaria' requires Female Anopheles. D: The answers have been interchanged: 'Cannabinoid source plant' requires Cannabis sativa, and 'Stem region interacting with immune components' requires Constant Fc region.

BIO-12-CH10-0018 | Hard | Viral and helminth diseases**Correct answer: D. Virus that targets helper T lymphocytes - Metastasis**

Solution: 'Virus that targets helper T lymphocytes' should be paired with HIV, not Metastasis. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Administration of antigenic material to induce immune memory - Vaccination. B: This is a valid association: Control approach that interrupts a pathogen's vector - Vector control. C: This is a valid association: Uncontrolled abnormal cell division - Cancer.

BIO-12-CH10-0019 | Medium | Antibody structure**Correct answer: A. Constant Fc region**

Solution: The observation 'Stem region interacting with immune components' points to Constant Fc region; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Carcinogens belongs with 'Agents capable of causing cancer', whereas the tested description is 'Stem region interacting with immune components'. C: Two heavy and two light chains belongs with 'Chains present in each antibody molecule', whereas the tested description is 'Stem region interacting with immune components'. D: Addiction belongs with 'Repeated compulsive substance use despite harm', whereas the tested description is 'Stem region interacting with immune components'.

BIO-12-CH10-0020 | Easy | Bacterial and protozoan diseases**Correct answer: B. Protozoan causing malaria**

Solution: Plasmodium represents 'Protozoan causing malaria'. Each alternative describes a different term or result.

Distractor review: A: 'Roundworm causing ascariasis' identifies Ascaris, not Plasmodium. C: 'Agents capable of causing cancer' identifies Carcinogens, not Plasmodium. D: 'Diagnostic test commonly used to detect HIV antibodies' identifies ELISA, not Plasmodium.

BIO-12-CH10-0021 | Medium | Cancer and immunity disorders**Correct answer: A. Carcinogens**

Solution: The chapter links Agents capable of causing cancer with Carcinogens, so Carcinogens completes the statement.

Distractor review: B: Metastasis belongs with 'Spread of malignant cells to distant sites', whereas the tested description is 'Agents capable of causing cancer'. C: Variable regions belongs with 'Antigen-binding regions', whereas the tested description is 'Agents capable of causing cancer'. D: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Agents capable of causing cancer'.

BIO-12-CH10-0022 | Easy | Immunity**Correct answer: C. Specific defence developing after exposure**

Solution: Acquired immunity represents 'Specific defence developing after exposure'. Each alternative describes a different term or result.

Distractor review: A: 'Causative bacterium of typhoid' identifies Salmonella typhi, not Acquired immunity. B: 'Cannabinoid source plant' identifies Cannabis sativa, not Acquired immunity. D: 'Non-specific defence present from birth' identifies Innate immunity, not Acquired immunity.

BIO-12-CH10-0023 | Challenging | Cancer and immunity disorders**Correct answer: C. I, III and IV only**

Solution: Statement I is correct because 'Agents capable of causing cancer' corresponds to Carcinogens. Statement II is incorrect: 'Spread of malignant cells to distant sites' corresponds to Metastasis, not Cancer. Statement III is correct because 'Uncontrolled abnormal cell division' corresponds to Cancer. Statement IV is correct because 'Specific defence developing after exposure' corresponds to Acquired immunity. Therefore, the valid combination is I, III and IV only.

Distractor review: A: This combination omits true statement(s) I, III. B: This combination includes false statement(s) II. D: This combination includes false statement(s) II and omits true statement(s) I.

BIO-12-CH10-0024 | Easy | Viral and helminth diseases**Correct answer: D. HIV**

Solution: 'Virus that targets helper T lymphocytes' is the defining description for HIV. The other choices belong to different chapter concepts.

Distractor review: A: Vaccination belongs with 'Administration of antigenic material to induce immune memory', whereas the tested description is 'Virus that targets helper T lymphocytes'. B: Innate immunity belongs with 'Non-specific defence present from birth', whereas the tested description is 'Virus that targets helper T lymphocytes'. C: Metastasis belongs with 'Spread of malignant cells to distant sites', whereas the tested description is 'Virus that targets helper T lymphocytes'.

BIO-12-CH10-0025 | Medium | Cancer and immunity disorders**Correct answer: D. Carcinogens**

Solution: The observation 'Agents capable of causing cancer' points to Carcinogens; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Metastasis belongs with 'Spread of malignant cells to distant sites', whereas the tested description is 'Agents capable of causing cancer'. B: ELISA belongs with 'Diagnostic test commonly used to detect HIV antibodies', whereas the tested description is 'Agents capable of causing cancer'. C: Passive immunity belongs with 'Immediate immunity produced by transferred antibodies', whereas the tested description is 'Agents capable of causing cancer'.

BIO-12-CH10-0026 | Easy | Viral and helminth diseases**Correct answer: C. Roundworm causing ascariasis**

Solution: Ascaris represents 'Roundworm causing ascariasis'. Each alternative describes a different term or result.

Distractor review: A: 'Cannabinoid source plant' identifies Cannabis sativa, not Ascaris. B: 'Immediate immunity produced by transferred antibodies' identifies Passive immunity, not Ascaris. D: 'Opioid obtained from poppy latex' identifies Morphine, not Ascaris.

BIO-12-CH10-0027 | Medium | Drugs and addiction**Correct answer: D. Cocaine**

Solution: The chapter links Stimulant derived from coca plant with Cocaine, so Cocaine completes the statement.

Distractor review: A: Acquired immunity belongs with 'Specific defence developing after exposure', whereas the tested description is 'Stimulant derived from coca plant'. B: Vector control belongs with 'Control approach that interrupts a pathogen's vector', whereas the tested description is 'Stimulant derived from coca plant'. C: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Stimulant derived from coca plant'.

BIO-12-CH10-0028 | Hard | Viral and helminth diseases**Correct answer: B. Filarial worm transmitted by mosquitoes - Addiction**

Solution: 'Filarial worm transmitted by mosquitoes' should be paired with Wuchereria, not Addiction. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Cannabinoid source plant - Cannabis sativa. C: This is a valid association: Opioid obtained from poppy latex - Morphine. D: This is a valid association: Agents capable of causing cancer - Carcinogens.

BIO-12-CH10-0029 | Medium | Immunity**Correct answer: B. Acquired immunity**

Solution: The chapter links Specific defence developing after exposure with Acquired immunity, so Acquired immunity completes the statement.

Distractor review: A: Active immunity belongs with 'Immunity produced by a person's own immune response', whereas the tested description is 'Specific defence developing after exposure'. C: Salmonella typhi belongs with 'Causative bacterium of typhoid', whereas the tested description is 'Specific defence developing after exposure'. D: Vaccination belongs with 'Administration of antigenic material to induce immune memory', whereas the tested description is 'Specific defence developing after exposure'.

BIO-12-CH10-0030 | Hard | Bacterial and protozoan diseases**Correct answer: C. Protozoan causing malaria - Plasmodium; Agents capable of causing cancer - Carcinogens**

Solution: Both parts of 'Protozoan causing malaria - Plasmodium; Agents capable of causing cancer - Carcinogens' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Spread of malignant cells to distant sites' requires Metastasis, and 'Specific defence developing after exposure' requires Acquired immunity. B: The answers have been interchanged: 'Cannabinoid source plant' requires Cannabis sativa, and 'Diagnostic test commonly used to detect HIV antibodies' requires ELISA. D: The answers have been interchanged: 'Non-specific defence present from birth' requires Innate immunity, and 'Cannabinoid source plant' requires Cannabis sativa.

BIO-12-CH10-0031 | Hard | Cancer and immunity disorders**Correct answer: A. Specialists who study diagnosis and treatment of cancer - Wuchereria**

Solution: 'Specialists who study diagnosis and treatment of cancer' should be paired with Oncologists, not Wuchereria. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Immunity produced by a person's own immune response - Active immunity. C: This is a valid association: Mosquito vector of dengue - Aedes. D: This is a valid association: Administration of antigenic material to induce immune memory - Vaccination.

BIO-12-CH10-0032 | Medium | Prevention and public health**Correct answer: A. Vaccination**

Solution: The chapter links Administration of antigenic material to induce immune memory with Vaccination, so Vaccination completes the statement.

Distractor review: B: Innate immunity belongs with 'Non-specific defence present from birth', whereas the tested description is 'Administration of antigenic material to induce immune memory'. C: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Administration of antigenic material to induce immune memory'. D: ELISA belongs with 'Diagnostic test commonly used to detect HIV antibodies', whereas the tested description is 'Administration of antigenic material to induce immune memory'.

BIO-12-CH10-0033 | Hard | Immunity**Correct answer: B. Non-specific defence present from birth - Y-shaped**

Solution: 'Non-specific defence present from birth' should be paired with Innate immunity, not Y-shaped. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Spread of malignant cells to distant sites - Metastasis. C: This is a valid association: Stem region interacting with immune components - Constant Fc region. D: This is a valid association: Uncontrolled abnormal cell division - Cancer.

BIO-12-CH10-0034 | Hard | Drugs and addiction**Correct answer: B. Cannabinoid source plant - Cannabis sativa; Basic shape of an antibody molecule - Y-shaped**

Solution: Both parts of 'Cannabinoid source plant - Cannabis sativa; Basic shape of an antibody molecule - Y-shaped' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Causative bacterium of pneumonia commonly cited in the chapter' requires Streptococcus pneumoniae, and 'Immunity produced by a person's own immune response' requires Active immunity. C: The answers have been interchanged: 'Uncontrolled abnormal cell division' requires Cancer, and 'Mosquito transmitting malaria' requires Female Anopheles. D: The answers have been interchanged: 'Protozoan causing malaria' requires Plasmodium, and 'Causative bacterium of pneumonia commonly cited in the chapter' requires Streptococcus pneumoniae.

BIO-12-CH10-0035 | Hard | Prevention and public health**Correct answer: D. Control approach that interrupts a pathogen's vector - Vector control; Uncontrolled abnormal cell division - Cancer**

Solution: Both parts of 'Control approach that interrupts a pathogen's vector - Vector control; Uncontrolled abnormal cell division - Cancer' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Cannabinoid source plant' requires Cannabis sativa, and 'Causative bacterium of pneumonia commonly cited in the chapter' requires Streptococcus pneumoniae. B: The answers have been interchanged: 'Causative bacterium of pneumonia commonly cited in the chapter' requires Streptococcus pneumoniae, and 'Stimulant derived from coca plant' requires Cocaine. C: The answers have been interchanged: 'Protozoan causing malaria' requires Plasmodium, and 'Mosquito transmitting malaria' requires Female Anopheles.

BIO-12-CH10-0036 | Medium | Immunity**Correct answer: B. Immediate immunity produced by transferred antibodies - Passive immunity**

Solution: The accurate entry is 'Immediate immunity produced by transferred antibodies - Passive immunity'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Administration of antigenic material to induce immune memory' should be paired with Vaccination, not Salmonella typhi. C: 'Causative bacterium of typhoid' should be paired with Salmonella typhi, not Cocaine. D: 'Stem region interacting with immune components' should be paired with Constant Fc region, not Cannabis sativa.

BIO-12-CH10-0037 | Easy | Bacterial and protozoan diseases**Correct answer: D. Plasmodium**

Solution: 'Protozoan causing malaria' is the defining description for Plasmodium. The other choices belong to different chapter concepts.

Distractor review: A: Cannabis sativa belongs with 'Cannabinoid source plant', whereas the tested description is 'Protozoan causing malaria'. B: Carcinogens belongs with 'Agents capable of causing cancer', whereas the tested description is 'Protozoan causing malaria'. C: Innate immunity belongs with 'Non-specific defence present from birth', whereas the tested description is 'Protozoan causing malaria'.

BIO-12-CH10-0038 | Medium | Antibody structure**Correct answer: D. Antigen-binding regions - Variable regions**

Solution: The accurate entry is 'Antigen-binding regions - Variable regions'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Opioid obtained from poppy latex' should be paired with Morphine, not Vector control. B: 'Mosquito vector of dengue' should be paired with Aedes, not Active immunity. C: 'Filarial worm transmitted by mosquitoes' should be paired with Wuchereria, not Variable regions.

BIO-12-CH10-0039 | Medium | Bacterial and protozoan diseases**Correct answer: C. Female Anopheles**

Solution: The chapter links Mosquito transmitting malaria with Female Anopheles, so Female Anopheles completes the statement.

Distractor review: A: Wuchereria belongs with 'Filarial worm transmitted by mosquitoes', whereas the tested description is 'Mosquito transmitting malaria'. B: Ascaris belongs with 'Roundworm causing ascariasis', whereas the tested description is 'Mosquito transmitting malaria'. D: Passive immunity belongs with 'Immediate immunity produced by transferred antibodies', whereas the tested description is 'Mosquito transmitting malaria'.

BIO-12-CH10-0040 | Medium | Viral and helminth diseases**Correct answer: A. Wuchereria**

Solution: The observation 'Filarial worm transmitted by mosquitoes' points to Wuchereria; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: ELISA belongs with 'Diagnostic test commonly used to detect HIV antibodies', whereas the tested description is 'Filarial worm transmitted by mosquitoes'. C: Vector control belongs with 'Control approach that interrupts a pathogen's vector', whereas the tested description is 'Filarial worm transmitted by mosquitoes'. D: Carcinogens belongs with 'Agents capable of causing cancer', whereas the tested description is 'Filarial worm transmitted by mosquitoes'.

BIO-12-CH10-0041 | Easy | Immunity**Correct answer: D. Passive immunity**

Solution: 'Immediate immunity produced by transferred antibodies' is the defining description for Passive immunity. The other choices belong to different chapter concepts.

Distractor review: A: Vaccination belongs with 'Administration of antigenic material to induce immune memory', whereas the tested description is 'Immediate immunity produced by transferred antibodies'. B: Acquired immunity belongs with 'Specific defence developing after exposure', whereas the tested description is 'Immediate immunity produced by transferred antibodies'. C: Salmonella typhi belongs with 'Causative bacterium of typhoid', whereas the tested description is 'Immediate immunity produced by transferred antibodies'.

BIO-12-CH10-0042 | Easy | Prevention and public health**Correct answer: A. Control approach that interrupts a pathogen's vector**

Solution: Vector control represents 'Control approach that interrupts a pathogen's vector'. Each alternative describes a different term or result.

Distractor review: B: 'Uncontrolled abnormal cell division' identifies Cancer, not Vector control. C: 'Specific defence developing after exposure' identifies Acquired immunity, not Vector control. D: 'Stimulant derived from coca plant' identifies Cocaine, not Vector control.

BIO-12-CH10-0043 | Hard | Drugs and addiction**Correct answer: C. Stimulant derived from coca plant - Aedes**

Solution: 'Stimulant derived from coca plant' should be paired with Cocaine, not Aedes. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Uncontrolled abnormal cell division - Cancer. B: This is a valid association: Immediate immunity produced by transferred antibodies - Passive immunity. D: This is a valid association: Specific defence developing after exposure - Acquired immunity.

BIO-12-CH10-0044 | Challenging | Antibody structure**Correct answer: C. I and II only**

Solution: Statement I is correct because 'Antigen-binding regions' corresponds to Variable regions. Statement II is correct because 'Stem region interacting with immune components' corresponds to Constant Fc region. Statement III is incorrect: 'Chains present in each antibody molecule' corresponds to Two heavy and two light chains, not ELISA. Statement IV is incorrect: 'Diagnostic test commonly used to detect HIV antibodies' corresponds to ELISA, not Variable regions. Therefore, the valid combination is I and II only.

Distractor review: A: This combination includes false statement(s) III, IV and omits true statement(s) I. B: This combination omits true statement(s) I. D: This combination includes false statement(s) III and omits true statement(s) I.

BIO-12-CH10-0045 | Easy | Prevention and public health**Correct answer: B. Avoiding peer pressure and seeking guidance**

Solution: 'Responsible adolescent practice that prevents substance misuse' is the defining description for Avoiding peer pressure and seeking guidance. The other choices belong to different chapter concepts.

Distractor review: A: Acquired immunity belongs with 'Specific defence developing after exposure', whereas the tested description is 'Responsible adolescent practice that prevents substance misuse'. C: ELISA belongs with 'Diagnostic test commonly used to detect HIV antibodies', whereas the tested description is 'Responsible adolescent practice that prevents substance misuse'. D: Carcinogens belongs with 'Agents capable of causing cancer', whereas the tested description is 'Responsible adolescent practice that prevents substance misuse'.

BIO-12-CH10-0046 | Easy | Immunity**Correct answer: A. Non-specific defence present from birth**

Solution: Innate immunity represents 'Non-specific defence present from birth'. Each alternative describes a different term or result.

Distractor review: B: 'Control approach that interrupts a pathogen's vector' identifies Vector control, not Innate immunity. C: 'Basic shape of an antibody molecule' identifies Y-shaped, not Innate immunity. D: 'Antigen-binding regions' identifies Variable regions, not Innate immunity.

BIO-12-CH10-0047 | Hard | Cancer and immunity disorders**Correct answer: D. Specialists who study diagnosis and treatment of cancer - Oncologists; Filarial worm transmitted by mosquitoes - Wuchereria**

Solution: Both parts of 'Specialists who study diagnosis and treatment of cancer - Oncologists; Filarial worm transmitted by mosquitoes - Wuchereria' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Immediate immunity produced by transferred antibodies' requires Passive immunity, and 'Virus that targets helper T lymphocytes' requires HIV. B: The answers have been interchanged: 'Mosquito vector of dengue' requires Aedes, and 'Immediate immunity produced by transferred antibodies' requires Passive immunity. C: The answers have been interchanged: 'Administration of antigenic material to induce immune memory' requires Vaccination, and 'Immunity produced by a person's own immune response' requires Active immunity.

BIO-12-CH10-0048 | Challenging | Bacterial and protozoan diseases**Correct answer: A. I, II and IV only**

Solution: Statement I is correct because 'Protozoan causing malaria' corresponds to Plasmodium. Statement II is correct because 'Causative bacterium of typhoid' corresponds to Salmonella typhi. Statement III is incorrect: 'Causative bacterium of pneumonia commonly cited in the chapter' corresponds to Streptococcus pneumoniae, not Constant Fc region. Statement IV is correct because 'Stem region interacting with immune components' corresponds to Constant Fc region. Therefore, the valid combination is I, II and IV only.

Distractor review: B: This combination includes false statement(s) III and omits true statement(s) I, IV. C: This combination includes false statement(s) III. D: This combination omits true statement(s) II, IV.

BIO-12-CH10-0049 | Medium | Bacterial and protozoan diseases**Correct answer: D. Causative bacterium of pneumonia commonly cited in the chapter - Streptococcus pneumoniae**

Solution: The accurate entry is 'Causative bacterium of pneumonia commonly cited in the chapter - Streptococcus pneumoniae'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Antigen-binding regions' should be paired with Variable regions, not Streptococcus pneumoniae. B: 'Basic shape of an antibody molecule' should be paired with Y-shaped, not Variable regions. C: 'Cannabinoid source plant' should be paired with Cannabis sativa, not Vector control.

BIO-12-CH10-0050 | Hard | Bacterial and protozoan diseases

Correct answer: B. Causative bacterium of pneumonia commonly cited in the chapter - Oncologists

Solution: 'Causative bacterium of pneumonia commonly cited in the chapter' should be paired with Streptococcus pneumoniae, not Oncologists. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Control approach that interrupts a pathogen's vector - Vector control. C: This is a valid association: Roundworm causing ascariasis - Ascaris. D: This is a valid association: Cannabinoid source plant - Cannabis sativa.

BIO-12-CH10-0051 | Medium | Drugs and addiction

Correct answer: B. Cannabinoid source plant - Cannabis sativa

Solution: The accurate entry is 'Cannabinoid source plant - Cannabis sativa'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Causative bacterium of pneumonia commonly cited in the chapter' should be paired with Streptococcus pneumoniae, not Cannabis sativa. C: 'Mosquito transmitting malaria' should be paired with Female Anopheles, not Plasmodium. D: 'Protozoan causing malaria' should be paired with Plasmodium, not Active immunity.

BIO-12-CH10-0052 | Easy | Antibody structure

Correct answer: C. Stem region interacting with immune components

Solution: Constant Fc region represents 'Stem region interacting with immune components'. Each alternative describes a different term or result.

Distractor review: A: 'Chains present in each antibody molecule' identifies Two heavy and two light chains, not Constant Fc region. B: 'Repeated compulsive substance use despite harm' identifies Addiction, not Constant Fc region. D: 'Antigen-binding regions' identifies Variable regions, not Constant Fc region.

BIO-12-CH10-0053 | Easy | Bacterial and protozoan diseases

Correct answer: A. Streptococcus pneumoniae

Solution: 'Causative bacterium of pneumonia commonly cited in the chapter' is the defining description for Streptococcus pneumoniae. The other choices belong to different chapter concepts.

Distractor review: B: Ascaris belongs with 'Roundworm causing ascariasis', whereas the tested description is 'Causative bacterium of pneumonia commonly cited in the chapter'. C: Oncologists belongs with 'Specialists who study diagnosis and treatment of cancer', whereas the tested description is 'Causative bacterium of pneumonia commonly cited in the chapter'. D: Plasmodium belongs with 'Protozoan causing malaria', whereas the tested description is 'Causative bacterium of pneumonia commonly cited in the chapter'.

BIO-12-CH10-0054 | Easy | Viral and helminth diseases

Correct answer: D. Wuchereria

Solution: 'Filarial worm transmitted by mosquitoes' is the defining description for Wuchereria. The other choices belong to different chapter concepts.

Distractor review: A: ELISA belongs with 'Diagnostic test commonly used to detect HIV antibodies', whereas the tested description is 'Filarial worm transmitted by mosquitoes'. B: Cannabis sativa belongs with 'Cannabinoid source plant', whereas the tested description is 'Filarial worm transmitted by mosquitoes'. C: Addiction belongs with 'Repeated compulsive substance use despite harm', whereas the tested description is 'Filarial worm transmitted by mosquitoes'.

BIO-12-CH10-0055 | Easy | Bacterial and protozoan diseases

Correct answer: A. Causative bacterium of typhoid

Solution: Salmonella typhi represents 'Causative bacterium of typhoid'. Each alternative describes a different term or result.

Distractor review: B: 'Antigen-binding regions' identifies Variable regions, not Salmonella typhi. C: 'Control approach that interrupts a pathogen's vector' identifies Vector control, not Salmonella typhi. D: 'Causative bacterium of pneumonia commonly cited in the chapter' identifies Streptococcus pneumoniae, not Salmonella typhi.

BIO-12-CH10-0056 | Easy | Prevention and public health

Correct answer: C. Administration of antigenic material to induce immune memory

Solution: Vaccination represents 'Administration of antigenic material to induce immune memory'. Each alternative describes a different term or result.

Distractor review: A: 'Non-specific defence present from birth' identifies Innate immunity, not Vaccination. B: 'Repeated compulsive substance use despite harm' identifies Addiction, not Vaccination. D: 'Diagnostic test commonly used to detect HIV antibodies' identifies ELISA, not Vaccination.

BIO-12-CH10-0057 | Medium | Drugs and addiction**Correct answer: D. Cannabis sativa**

Solution: The observation 'Cannabinoid source plant' points to Cannabis sativa; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Morphine belongs with 'Opioid obtained from poppy latex', whereas the tested description is 'Cannabinoid source plant'. B: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Cannabinoid source plant'. C: Variable regions belongs with 'Antigen-binding regions', whereas the tested description is 'Cannabinoid source plant'.

BIO-12-CH10-0058 | Medium | Bacterial and protozoan diseases**Correct answer: C. Mosquito transmitting malaria - Female Anopheles**

Solution: The accurate entry is 'Mosquito transmitting malaria - Female Anopheles'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Basic shape of an antibody molecule' should be paired with Y-shaped, not Ascaris. B: 'Roundworm causing ascariasis' should be paired with Ascaris, not Wuchereria. D: 'Administration of antigenic material to induce immune memory' should be paired with Vaccination, not Cannabis sativa.

BIO-12-CH10-0059 | Easy | Viral and helminth diseases**Correct answer: D. Filarial worm transmitted by mosquitoes**

Solution: Wuchereria represents 'Filarial worm transmitted by mosquitoes'. Each alternative describes a different term or result.

Distractor review: A: 'Repeated compulsive substance use despite harm' identifies Addiction, not Wuchereria. B: 'Virus that targets helper T lymphocytes' identifies HIV, not Wuchereria. C: 'Diagnostic test commonly used to detect HIV antibodies' identifies ELISA, not Wuchereria.

BIO-12-CH10-0060 | Hard | Antibody structure**Correct answer: B. Basic shape of an antibody molecule - Female Anopheles**

Solution: 'Basic shape of an antibody molecule' should be paired with Y-shaped, not Female Anopheles. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Causative bacterium of pneumonia commonly cited in the chapter - Streptococcus pneumoniae. C: This is a valid association: Immediate immunity produced by transferred antibodies - Passive immunity. D: This is a valid association: Repeated compulsive substance use despite harm - Addiction.

BIO-12-CH10-0061 | Medium | Immunity**Correct answer: D. Innate immunity**

Solution: The observation 'Non-specific defence present from birth' points to Innate immunity; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Vector control belongs with 'Control approach that interrupts a pathogen's vector', whereas the tested description is 'Non-specific defence present from birth'. B: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Non-specific defence present from birth'. C: Metastasis belongs with 'Spread of malignant cells to distant sites', whereas the tested description is 'Non-specific defence present from birth'.

BIO-12-CH10-0062 | Easy | Drugs and addiction**Correct answer: D. Repeated compulsive substance use despite harm**

Solution: Addiction represents 'Repeated compulsive substance use despite harm'. Each alternative describes a different term or result.

Distractor review: A: 'Immediate immunity produced by transferred antibodies' identifies Passive immunity, not Addiction. B: 'Mosquito transmitting malaria' identifies Female Anopheles, not Addiction. C: 'Causative bacterium of typhoid' identifies Salmonella typhi, not Addiction.

BIO-12-CH10-0063 | Hard | Antibody structure**Correct answer: A. Antigen-binding regions - Aedes**

Solution: 'Antigen-binding regions' should be paired with Variable regions, not Aedes. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Non-specific defence present from birth - Innate immunity. C: This is a valid association: Control approach that interrupts a pathogen's vector - Vector control. D: This is a valid association: Opioid obtained from poppy latex - Morphine.

BIO-12-CH10-0064 | Challenging | Prevention and public health**Correct answer: A. IV only**

Solution: Statement I is incorrect: 'Control approach that interrupts a pathogen's vector' corresponds to Vector control, not ELISA. Statement II is incorrect: 'Diagnostic test commonly used to detect HIV antibodies' corresponds to ELISA, not Vaccination. Statement III is incorrect: 'Administration of antigenic material to induce immune memory' corresponds to Vaccination, not Streptococcus pneumoniae. Statement IV is correct because 'Causative bacterium of pneumonia commonly cited in the chapter' corresponds to Streptococcus pneumoniae. Therefore, the valid combination is IV only.

Distractor review: B: This combination includes false statement(s) I, II, III. C: This combination includes false statement(s) II. D: This combination includes false statement(s) I, II and omits true statement(s) IV.

BIO-12-CH10-0065 | Hard | Cancer and immunity disorders**Correct answer: B. Spread of malignant cells to distant sites - Cocaine**

Solution: 'Spread of malignant cells to distant sites' should be paired with Metastasis, not Cocaine. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Causative bacterium of typhoid - Salmonella typhi. C: This is a valid association: Immediate immunity produced by transferred antibodies - Passive immunity. D: This is a valid association: Immunity produced by a person's own immune response - Active immunity.

BIO-12-CH10-0066 | Medium | Cancer and immunity disorders**Correct answer: A. Cancer**

Solution: The chapter links Uncontrolled abnormal cell division with Cancer, so Cancer completes the statement.

Distractor review: B: Female Anopheles belongs with 'Mosquito transmitting malaria', whereas the tested description is 'Uncontrolled abnormal cell division'. C: Ascaris belongs with 'Roundworm causing ascariasis', whereas the tested description is 'Uncontrolled abnormal cell division'. D: Constant Fc region belongs with 'Stem region interacting with immune components', whereas the tested description is 'Uncontrolled abnormal cell division'.

BIO-12-CH10-0067 | Hard | Drugs and addiction**Correct answer: C. Opioid obtained from poppy latex - Morphine; Non-specific defence present from birth - Innate immunity**

Solution: Both parts of 'Opioid obtained from poppy latex - Morphine; Non-specific defence present from birth - Innate immunity' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Specialists who study diagnosis and treatment of cancer' requires Oncologists, and 'Specific defence developing after exposure' requires Acquired immunity. B: The answers have been interchanged: 'Mosquito vector of dengue' requires Aedes, and 'Responsible adolescent practice that prevents substance misuse' requires Avoiding peer pressure and seeking guidance. D: The answers have been interchanged: 'Immunity produced by a person's own immune response' requires Active immunity, and 'Agents capable of causing cancer' requires Carcinogens.

BIO-12-CH10-0068 | Easy | Cancer and immunity disorders**Correct answer: B. Metastasis**

Solution: 'Spread of malignant cells to distant sites' is the defining description for Metastasis. The other choices belong to different chapter concepts.

Distractor review: A: Vector control belongs with 'Control approach that interrupts a pathogen's vector', whereas the tested description is 'Spread of malignant cells to distant sites'. C: Passive immunity belongs with 'Immediate immunity produced by transferred antibodies', whereas the tested description is 'Spread of malignant cells to distant sites'. D: Carcinogens belongs with 'Agents capable of causing cancer', whereas the tested description is 'Spread of malignant cells to distant sites'.

BIO-12-CH10-0069 | Easy | Antibody structure**Correct answer: A. Variable regions**

Solution: 'Antigen-binding regions' is the defining description for Variable regions. The other choices belong to different chapter concepts.

Distractor review: B: Aedes belongs with 'Mosquito vector of dengue', whereas the tested description is 'Antigen-binding regions'. C: Morphine belongs with 'Opioid obtained from poppy latex', whereas the tested description is 'Antigen-binding regions'. D: Innate immunity belongs with 'Non-specific defence present from birth', whereas the tested description is 'Antigen-binding regions'.

BIO-12-CH10-0070 | Easy | Cancer and immunity disorders**Correct answer: D. Agents capable of causing cancer**

Solution: Carcinogens represents 'Agents capable of causing cancer'. Each alternative describes a different term or result.

Distractor review: A: 'Diagnostic test commonly used to detect HIV antibodies' identifies ELISA, not Carcinogens. B: 'Protozoan causing malaria' identifies Plasmodium, not Carcinogens. C: 'Basic shape of an antibody molecule' identifies Y-shaped, not Carcinogens.

BIO-12-CH10-0071 | Medium | Cancer and immunity disorders**Correct answer: B. Oncologists**

Solution: The chapter links Specialists who study diagnosis and treatment of cancer with Oncologists, so Oncologists completes the statement.

Distractor review: A: Carcinogens belongs with 'Agents capable of causing cancer', whereas the tested description is 'Specialists who study diagnosis and treatment of cancer'. C: Active immunity belongs with 'Immunity produced by a person's own immune response', whereas the tested description is 'Specialists who study diagnosis and treatment of cancer'. D: Wuchereria belongs with 'Filarial worm transmitted by mosquitoes', whereas the tested description is 'Specialists who study diagnosis and treatment of cancer'.

BIO-12-CH10-0072 | Medium | Drugs and addiction**Correct answer: A. Addiction**

Solution: The observation 'Repeated compulsive substance use despite harm' points to Addiction; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Salmonella typhi belongs with 'Causative bacterium of typhoid', whereas the tested description is 'Repeated compulsive substance use despite harm'. C: Cannabis sativa belongs with 'Cannabinoid source plant', whereas the tested description is 'Repeated compulsive substance use despite harm'. D: Morphine belongs with 'Opioid obtained from poppy latex', whereas the tested description is 'Repeated compulsive substance use despite harm'.

BIO-12-CH10-0073 | Easy | Antibody structure**Correct answer: B. Chains present in each antibody molecule**

Solution: Two heavy and two light chains represents 'Chains present in each antibody molecule'. Each alternative describes a different term or result.

Distractor review: A: 'Repeated compulsive substance use despite harm' identifies Addiction, not Two heavy and two light chains. C: 'Responsible adolescent practice that prevents substance misuse' identifies Avoiding peer pressure and seeking guidance, not Two heavy and two light chains. D: 'Specific defence developing after exposure' identifies Acquired immunity, not Two heavy and two light chains.

BIO-12-CH10-0074 | Medium | Drugs and addiction**Correct answer: D. Cannabis sativa**

Solution: The chapter links Cannabinoid source plant with Cannabis sativa, so Cannabis sativa completes the statement.

Distractor review: A: Variable regions belongs with 'Antigen-binding regions', whereas the tested description is 'Cannabinoid source plant'. B: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Cannabinoid source plant'. C: Female Anopheles belongs with 'Mosquito transmitting malaria', whereas the tested description is 'Cannabinoid source plant'.

BIO-12-CH10-0075 | Hard | Immunity**Correct answer: D. Non-specific defence present from birth - Innate immunity; Basic shape of an antibody molecule - Y-shaped**

Solution: Both parts of 'Non-specific defence present from birth - Innate immunity; Basic shape of an antibody molecule - Y-shaped' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Spread of malignant cells to distant sites' requires Metastasis, and 'Uncontrolled abnormal cell division' requires Cancer. B: The answers have been interchanged: 'Uncontrolled abnormal cell division' requires Cancer, and 'Opioid obtained from poppy latex' requires Morphine. C: The answers have been interchanged: 'Control approach that interrupts a pathogen's vector' requires Vector control, and 'Antigen-binding regions' requires Variable regions.

BIO-12-CH10-0076 | Hard | Viral and helminth diseases

Correct answer: A. Filarial worm transmitted by mosquitoes - Wuchereria; Repeated compulsive substance use despite harm - Addiction

Solution: Both parts of 'Filarial worm transmitted by mosquitoes - Wuchereria; Repeated compulsive substance use despite harm - Addiction' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Opioid obtained from poppy latex' requires Morphine, and 'Uncontrolled abnormal cell division' requires Cancer. C: The answers have been interchanged: 'Agents capable of causing cancer' requires Carcinogens, and 'Control approach that interrupts a pathogen's vector' requires Vector control. D: The answers have been interchanged: 'Control approach that interrupts a pathogen's vector' requires Vector control, and 'Opioid obtained from poppy latex' requires Morphine.

BIO-12-CH10-0077 | Medium | Viral and helminth diseases

Correct answer: B. Virus that targets helper T lymphocytes - HIV

Solution: The accurate entry is 'Virus that targets helper T lymphocytes - HIV'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Administration of antigenic material to induce immune memory' should be paired with Vaccination, not Cannabis sativa. C: 'Causative bacterium of typhoid' should be paired with Salmonella typhi, not HIV. D: 'Spread of malignant cells to distant sites' should be paired with Metastasis, not Vector control.

BIO-12-CH10-0078 | Medium | Immunity

Correct answer: D. Specific defence developing after exposure - Acquired immunity

Solution: The accurate entry is 'Specific defence developing after exposure - Acquired immunity'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Agents capable of causing cancer' should be paired with Carcinogens, not Addiction. B: 'Mosquito transmitting malaria' should be paired with Female Anopheles, not Acquired immunity. C: 'Administration of antigenic material to induce immune memory' should be paired with Vaccination, not Salmonella typhi.

BIO-12-CH10-0079 | Medium | Bacterial and protozoan diseases

Correct answer: B. Streptococcus pneumoniae

Solution: The chapter links Causative bacterium of pneumonia commonly cited in the chapter with Streptococcus pneumoniae, so Streptococcus pneumoniae completes the statement.

Distractor review: A: Vector control belongs with 'Control approach that interrupts a pathogen's vector', whereas the tested description is 'Causative bacterium of pneumonia commonly cited in the chapter'. C: Oncologists belongs with 'Specialists who study diagnosis and treatment of cancer', whereas the tested description is 'Causative bacterium of pneumonia commonly cited in the chapter'. D: Plasmodium belongs with 'Protozoan causing malaria', whereas the tested description is 'Causative bacterium of pneumonia commonly cited in the chapter'.

BIO-12-CH10-0080 | Medium | Cancer and immunity disorders

Correct answer: C. Oncologists

Solution: The observation 'Specialists who study diagnosis and treatment of cancer' points to Oncologists; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Vaccination belongs with 'Administration of antigenic material to induce immune memory', whereas the tested description is 'Specialists who study diagnosis and treatment of cancer'. B: Cocaine belongs with 'Stimulant derived from coca plant', whereas the tested description is 'Specialists who study diagnosis and treatment of cancer'. D: Cancer belongs with 'Uncontrolled abnormal cell division', whereas the tested description is 'Specialists who study diagnosis and treatment of cancer'.

BIO-12-CH10-0081 | Hard | Antibody structure

Correct answer: B. Chains present in each antibody molecule - Two heavy and two light chains; Stimulant derived from coca plant - Cocaine

Solution: Both parts of 'Chains present in each antibody molecule - Two heavy and two light chains; Stimulant derived from coca plant - Cocaine' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Responsible adolescent practice that prevents substance misuse' requires Avoiding peer pressure and seeking guidance, and 'Specific defence developing after exposure' requires Acquired immunity. C: The answers have been interchanged: 'Spread of malignant cells to distant sites' requires Metastasis, and 'Diagnostic test commonly used to detect HIV antibodies' requires ELISA. D: The answers have been interchanged: 'Specific defence developing after exposure' requires Acquired immunity, and 'Repeated compulsive substance use despite harm' requires Addiction.

BIO-12-CH10-0082 | Medium | Drugs and addiction**Correct answer: D. Stimulant derived from coca plant - Cocaine**

Solution: The accurate entry is 'Stimulant derived from coca plant - Cocaine'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Uncontrolled abnormal cell division' should be paired with Cancer, not Cocaine. B: 'Control approach that interrupts a pathogen's vector' should be paired with Vector control, not Passive immunity. C: 'Immediate immunity produced by transferred antibodies' should be paired with Passive immunity, not Cancer.

BIO-12-CH10-0083 | Hard | Cancer and immunity disorders**Correct answer: B. Uncontrolled abnormal cell division - Cancer; Virus that targets helper T lymphocytes - HIV**

Solution: Both parts of 'Uncontrolled abnormal cell division - Cancer; Virus that targets helper T lymphocytes - HIV' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Mosquito transmitting malaria' requires Female Anopheles, and 'Opioid obtained from poppy latex' requires Morphine. C: The answers have been interchanged: 'Stem region interacting with immune components' requires Constant Fc region, and 'Mosquito transmitting malaria' requires Female Anopheles. D: The answers have been interchanged: 'Administration of antigenic material to induce immune memory' requires Vaccination, and 'Stem region interacting with immune components' requires Constant Fc region.

BIO-12-CH10-0084 | Easy | Cancer and immunity disorders**Correct answer: C. Oncologists**

Solution: 'Specialists who study diagnosis and treatment of cancer' is the defining description for Oncologists. The other choices belong to different chapter concepts.

Distractor review: A: Carcinogens belongs with 'Agents capable of causing cancer', whereas the tested description is 'Specialists who study diagnosis and treatment of cancer'. B: Cocaine belongs with 'Stimulant derived from coca plant', whereas the tested description is 'Specialists who study diagnosis and treatment of cancer'. D: Wuchereria belongs with 'Filarial worm transmitted by mosquitoes', whereas the tested description is 'Specialists who study diagnosis and treatment of cancer'.

BIO-12-CH10-0085 | Challenging | Drugs and addiction**Correct answer: C. I, II and III only**

Solution: Statement I is correct because 'Stimulant derived from coca plant' corresponds to Cocaine. Statement II is correct because 'Opioid obtained from poppy latex' corresponds to Morphine. Statement III is correct because 'Repeated compulsive substance use despite harm' corresponds to Addiction. Statement IV is incorrect: 'Specific defence developing after exposure' corresponds to Acquired immunity, not Cocaine. Therefore, the valid combination is I, II and III only.

Distractor review: A: This combination includes false statement(s) IV. B: This combination includes false statement(s) IV and omits true statement(s) III. D: This combination omits true statement(s) I, II.

BIO-12-CH10-0086 | Medium | Immunity**Correct answer: B. Innate immunity**

Solution: The chapter links Non-specific defence present from birth with Innate immunity, so Innate immunity completes the statement.

Distractor review: A: Constant Fc region belongs with 'Stem region interacting with immune components', whereas the tested description is 'Non-specific defence present from birth'. C: Acquired immunity belongs with 'Specific defence developing after exposure', whereas the tested description is 'Non-specific defence present from birth'. D: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Non-specific defence present from birth'.

BIO-12-CH10-0087 | Medium | Prevention and public health**Correct answer: C. Vaccination**

Solution: The observation 'Administration of antigenic material to induce immune memory' points to Vaccination; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Avoiding peer pressure and seeking guidance belongs with 'Responsible adolescent practice that prevents substance misuse', whereas the tested description is 'Administration of antigenic material to induce immune memory'. B: Oncologists belongs with 'Specialists who study diagnosis and treatment of cancer', whereas the tested description is 'Administration of antigenic material to induce immune memory'. D: Addiction belongs with 'Repeated compulsive substance use despite harm', whereas the tested description is 'Administration of antigenic material to induce immune memory'.

BIO-12-CH10-0088 | Challenging | Drugs and addiction**Correct answer: C. III only**

Solution: Statement I is incorrect: 'Repeated compulsive substance use despite harm' corresponds to Addiction, not Morphine. Statement II is incorrect: 'Opioid obtained from poppy latex' corresponds to Morphine, not Cannabis sativa. Statement III is correct because 'Cannabinoid source plant' corresponds to Cannabis sativa. Statement IV is incorrect: 'Immediate immunity produced by transferred antibodies' corresponds to Passive immunity, not Addiction. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) I, II and omits true statement(s) III. B: This combination includes false statement(s) II, IV and omits true statement(s) III. D: This combination includes false statement(s) IV.

BIO-12-CH10-0089 | Medium | Viral and helminth diseases**Correct answer: D. Roundworm causing ascariasis - Ascaris**

Solution: The accurate entry is 'Roundworm causing ascariasis - Ascaris'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Cannabinoid source plant' should be paired with Cannabis sativa, not Passive immunity. B: 'Stem region interacting with immune components' should be paired with Constant Fc region, not Plasmodium. C: 'Protozoan causing malaria' should be paired with Plasmodium, not Female Anopheles.

BIO-12-CH10-0090 | Challenging | Viral and helminth diseases**Correct answer: A. III and IV only**

Solution: Statement I is incorrect: 'Roundworm causing ascariasis' corresponds to Ascaris, not Aedes. Statement II is incorrect: 'Mosquito vector of dengue' corresponds to Aedes, not HIV. Statement III is correct because 'Virus that targets helper T lymphocytes' corresponds to HIV. Statement IV is correct because 'Cannabinoid source plant' corresponds to Cannabis sativa. Therefore, the valid combination is III and IV only.

Distractor review: B: This combination includes false statement(s) I, II and omits true statement(s) IV. C: This combination includes false statement(s) II. D: This combination includes false statement(s) II and omits true statement(s) III.

BIO-12-CH10-0091 | Easy | Viral and helminth diseases**Correct answer: A. Mosquito vector of dengue**

Solution: Aedes represents 'Mosquito vector of dengue'. Each alternative describes a different term or result.

Distractor review: B: 'Immunity produced by a person's own immune response' identifies Active immunity, not Aedes. C: 'Responsible adolescent practice that prevents substance misuse' identifies Avoiding peer pressure and seeking guidance, not Aedes. D: 'Filarial worm transmitted by mosquitoes' identifies Wuchereria, not Aedes.

BIO-12-CH10-0092 | Hard | Bacterial and protozoan diseases**Correct answer: D. Protozoan causing malaria - Carcinogens**

Solution: 'Protozoan causing malaria' should be paired with Plasmodium, not Carcinogens. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Non-specific defence present from birth - Innate immunity. B: This is a valid association: Roundworm causing ascariasis - Ascaris. C: This is a valid association: Cannabinoid source plant - Cannabis sativa.

BIO-12-CH10-0093 | Medium | Prevention and public health**Correct answer: A. Avoiding peer pressure and seeking guidance**

Solution: The observation 'Responsible adolescent practice that prevents substance misuse' points to Avoiding peer pressure and seeking guidance; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Acquired immunity belongs with 'Specific defence developing after exposure', whereas the tested description is 'Responsible adolescent practice that prevents substance misuse'. C: Metastasis belongs with 'Spread of malignant cells to distant sites', whereas the tested description is 'Responsible adolescent practice that prevents substance misuse'. D: ELISA belongs with 'Diagnostic test commonly used to detect HIV antibodies', whereas the tested description is 'Responsible adolescent practice that prevents substance misuse'.

BIO-12-CH10-0094 | Medium | Prevention and public health

Correct answer: C. Responsible adolescent practice that prevents substance misuse - Avoiding peer pressure and seeking guidance

Solution: The accurate entry is 'Responsible adolescent practice that prevents substance misuse - Avoiding peer pressure and seeking guidance'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Chains present in each antibody molecule' should be paired with Two heavy and two light chains, not Avoiding peer pressure and seeking guidance. B: 'Agents capable of causing cancer' should be paired with Carcinogens, not Metastasis. D: 'Non-specific defence present from birth' should be paired with Innate immunity, not Two heavy and two light chains.

BIO-12-CH10-0095 | Easy | Cancer and immunity disorders

Correct answer: C. Uncontrolled abnormal cell division

Solution: Cancer represents 'Uncontrolled abnormal cell division'. Each alternative describes a different term or result.

Distractor review: A: 'Roundworm causing ascariasis' identifies Ascaris, not Cancer. B: 'Spread of malignant cells to distant sites' identifies Metastasis, not Cancer. D: 'Administration of antigenic material to induce immune memory' identifies Vaccination, not Cancer.

BIO-12-CH10-0096 | Medium | Viral and helminth diseases

Correct answer: A. Mosquito vector of dengue - Aedes

Solution: The accurate entry is 'Mosquito vector of dengue - Aedes'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Immediate immunity produced by transferred antibodies' should be paired with Passive immunity, not Vaccination. C: 'Causative bacterium of pneumonia commonly cited in the chapter' should be paired with Streptococcus pneumoniae, not Aedes. D: 'Stem region interacting with immune components' should be paired with Constant Fc region, not Active immunity.

BIO-12-CH10-0097 | Hard | Prevention and public health

Correct answer: C. Administration of antigenic material to induce immune memory - Vaccination; Repeated compulsive substance use despite harm - Addiction

Solution: Both parts of 'Administration of antigenic material to induce immune memory - Vaccination; Repeated compulsive substance use despite harm - Addiction' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Basic shape of an antibody molecule' requires Y-shaped, and 'Specialists who study diagnosis and treatment of cancer' requires Oncologists. B: The answers have been interchanged: 'Non-specific defence present from birth' requires Innate immunity, and 'Cannabinoid source plant' requires Cannabis sativa. D: The answers have been interchanged: 'Causative bacterium of typhoid' requires Salmonella typhi, and 'Agents capable of causing cancer' requires Carcinogens.

BIO-12-CH10-0098 | Medium | Prevention and public health

Correct answer: B. Vector control

Solution: The observation 'Control approach that interrupts a pathogen's vector' points to Vector control; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Streptococcus pneumoniae belongs with 'Causative bacterium of pneumonia commonly cited in the chapter', whereas the tested description is 'Control approach that interrupts a pathogen's vector'. C: ELISA belongs with 'Diagnostic test commonly used to detect HIV antibodies', whereas the tested description is 'Control approach that interrupts a pathogen's vector'. D: Acquired immunity belongs with 'Specific defence developing after exposure', whereas the tested description is 'Control approach that interrupts a pathogen's vector'.

BIO-12-CH10-0099 | Easy | Cancer and immunity disorders

Correct answer: A. Specialists who study diagnosis and treatment of cancer

Solution: Oncologists represents 'Specialists who study diagnosis and treatment of cancer'. Each alternative describes a different term or result.

Distractor review: B: 'Immunity produced by a person's own immune response' identifies Active immunity, not Oncologists. C: 'Filarial worm transmitted by mosquitoes' identifies Wuchereria, not Oncologists. D: 'Agents capable of causing cancer' identifies Carcinogens, not Oncologists.

BIO-12-CH10-0100 | Hard | Antibody structure

Correct answer: D. Stem region interacting with immune components - Constant Fc region; Agents capable of causing cancer - Carcinogens

Solution: Both parts of 'Stem region interacting with immune components - Constant Fc region; Agents capable of causing cancer - Carcinogens' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Protozoan causing malaria' requires Plasmodium, and 'Immunity produced by a person's own immune response' requires Active immunity. B: The answers have been interchanged: 'Immunity produced by a person's own immune response' requires Active immunity, and 'Mosquito transmitting malaria' requires Female Anopheles. C: The answers have been interchanged: 'Repeated compulsive substance use despite harm' requires Addiction, and 'Specialists who study diagnosis and treatment of cancer' requires Oncologists.

BIO-12-CH10-0101 | Hard | Bacterial and protozoan diseases

Correct answer: A. Causative bacterium of typhoid - Salmonella typhi; Diagnostic test commonly used to detect HIV antibodies - ELISA

Solution: Both parts of 'Causative bacterium of typhoid - Salmonella typhi; Diagnostic test commonly used to detect HIV antibodies - ELISA' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Responsible adolescent practice that prevents substance misuse' requires Avoiding peer pressure and seeking guidance, and 'Antigen-binding regions' requires Variable regions. C: The answers have been interchanged: 'Control approach that interrupts a pathogen's vector' requires Vector control, and 'Immediate immunity produced by transferred antibodies' requires Passive immunity. D: The answers have been interchanged: 'Antigen-binding regions' requires Variable regions, and 'Virus that targets helper T lymphocytes' requires HIV.

BIO-12-CH10-0102 | Hard | Prevention and public health

Correct answer: B. Responsible adolescent practice that prevents substance misuse - Acquired immunity

Solution: 'Responsible adolescent practice that prevents substance misuse' should be paired with Avoiding peer pressure and seeking guidance, not Acquired immunity. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Virus that targets helper T lymphocytes - HIV. C: This is a valid association: Chains present in each antibody molecule - Two heavy and two light chains. D: This is a valid association: Non-specific defence present from birth - Innate immunity.

BIO-12-CH10-0103 | Hard | Immunity

Correct answer: B. Immediate immunity produced by transferred antibodies - Carcinogens

Solution: 'Immediate immunity produced by transferred antibodies' should be paired with Passive immunity, not Carcinogens. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Administration of antigenic material to induce immune memory - Vaccination. C: This is a valid association: Specialists who study diagnosis and treatment of cancer - Oncologists. D: This is a valid association: Causative bacterium of typhoid - Salmonella typhi.

BIO-12-CH10-0104 | Easy | Drugs and addiction

Correct answer: A. Cocaine

Solution: 'Stimulant derived from coca plant' is the defining description for Cocaine. The other choices belong to different chapter concepts.

Distractor review: B: Morphine belongs with 'Opioid obtained from poppy latex', whereas the tested description is 'Stimulant derived from coca plant'. C: Acquired immunity belongs with 'Specific defence developing after exposure', whereas the tested description is 'Stimulant derived from coca plant'. D: Cannabis sativa belongs with 'Cannabinoid source plant', whereas the tested description is 'Stimulant derived from coca plant'.

BIO-12-CH10-0105 | Hard | Prevention and public health

Correct answer: C. Administration of antigenic material to induce immune memory - Addiction

Solution: 'Administration of antigenic material to induce immune memory' should be paired with Vaccination, not Addiction. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Specialists who study diagnosis and treatment of cancer - Oncologists. B: This is a valid association: Cannabinoid source plant - Cannabis sativa. D: This is a valid association: Causative bacterium of typhoid - Salmonella typhi.

BIO-12-CH10-0106 | Medium | Cancer and immunity disorders**Correct answer: D. Cancer**

Solution: The observation 'Uncontrolled abnormal cell division' points to Cancer; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Oncologists belongs with 'Specialists who study diagnosis and treatment of cancer', whereas the tested description is 'Uncontrolled abnormal cell division'. B: Metastasis belongs with 'Spread of malignant cells to distant sites', whereas the tested description is 'Uncontrolled abnormal cell division'. C: Vaccination belongs with 'Administration of antigenic material to induce immune memory', whereas the tested description is 'Uncontrolled abnormal cell division'.

BIO-12-CH10-0107 | Medium | Antibody structure**Correct answer: B. Stem region interacting with immune components - Constant Fc region**

Solution: The accurate entry is 'Stem region interacting with immune components - Constant Fc region'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Mosquito transmitting malaria' should be paired with Female Anopheles, not Constant Fc region. C: 'Protozoan causing malaria' should be paired with Plasmodium, not Female Anopheles. D: 'Agents capable of causing cancer' should be paired with Carcinogens, not Oncologists.

BIO-12-CH10-0108 | Hard | Immunity**Correct answer: C. Specific defence developing after exposure - Acquired immunity; Agents capable of causing cancer - Carcinogens**

Solution: Both parts of 'Specific defence developing after exposure - Acquired immunity; Agents capable of causing cancer - Carcinogens' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Causative bacterium of typhoid' requires Salmonella typhi, and 'Cannabinoid source plant' requires Cannabis sativa. B: The answers have been interchanged: 'Repeated compulsive substance use despite harm' requires Addiction, and 'Causative bacterium of typhoid' requires Salmonella typhi. D: The answers have been interchanged: 'Administration of antigenic material to induce immune memory' requires Vaccination, and 'Repeated compulsive substance use despite harm' requires Addiction.

BIO-12-CH10-0109 | Easy | Prevention and public health**Correct answer: B. Vaccination**

Solution: 'Administration of antigenic material to induce immune memory' is the defining description for Vaccination. The other choices belong to different chapter concepts.

Distractor review: A: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Administration of antigenic material to induce immune memory'. C: Innate immunity belongs with 'Non-specific defence present from birth', whereas the tested description is 'Administration of antigenic material to induce immune memory'. D: Avoiding peer pressure and seeking guidance belongs with 'Responsible adolescent practice that prevents substance misuse', whereas the tested description is 'Administration of antigenic material to induce immune memory'.

BIO-12-CH10-0110 | Medium | Prevention and public health**Correct answer: C. ELISA**

Solution: The observation 'Diagnostic test commonly used to detect HIV antibodies' points to ELISA; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Avoiding peer pressure and seeking guidance belongs with 'Responsible adolescent practice that prevents substance misuse', whereas the tested description is 'Diagnostic test commonly used to detect HIV antibodies'. B: Plasmodium belongs with 'Protozoan causing malaria', whereas the tested description is 'Diagnostic test commonly used to detect HIV antibodies'. D: Carcinogens belongs with 'Agents capable of causing cancer', whereas the tested description is 'Diagnostic test commonly used to detect HIV antibodies'.

BIO-12-CH10-0111 | Medium | Antibody structure**Correct answer: C. Chains present in each antibody molecule - Two heavy and two light chains**

Solution: The accurate entry is 'Chains present in each antibody molecule - Two heavy and two light chains'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Repeated compulsive substance use despite harm' should be paired with Addiction, not ELISA. B: 'Mosquito transmitting malaria' should be paired with Female Anopheles, not Acquired immunity. D: 'Responsible adolescent practice that prevents substance misuse' should be paired with Avoiding peer pressure and seeking guidance, not Addiction.

BIO-12-CH10-0112 | Medium | Antibody structure**Correct answer: B. Variable regions**

Solution: The observation 'Antigen-binding regions' points to Variable regions; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Aedes belongs with 'Mosquito vector of dengue', whereas the tested description is 'Antigen-binding regions'. C: Innate immunity belongs with 'Non-specific defence present from birth', whereas the tested description is 'Antigen-binding regions'. D: Active immunity belongs with 'Immunity produced by a person's own immune response', whereas the tested description is 'Antigen-binding regions'.

BIO-12-CH10-0113 | Easy | Antibody structure**Correct answer: C. Y-shaped**

Solution: 'Basic shape of an antibody molecule' is the defining description for Y-shaped. The other choices belong to different chapter concepts.

Distractor review: A: Streptococcus pneumoniae belongs with 'Causative bacterium of pneumonia commonly cited in the chapter', whereas the tested description is 'Basic shape of an antibody molecule'. B: Addiction belongs with 'Repeated compulsive substance use despite harm', whereas the tested description is 'Basic shape of an antibody molecule'. D: Constant Fc region belongs with 'Stem region interacting with immune components', whereas the tested description is 'Basic shape of an antibody molecule'.

BIO-12-CH10-0114 | Hard | Drugs and addiction**Correct answer: A. Repeated compulsive substance use despite harm - Addiction; Causative bacterium of typhoid - Salmonella typhi**

Solution: Both parts of 'Repeated compulsive substance use despite harm - Addiction; Causative bacterium of typhoid - Salmonella typhi' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Specialists who study diagnosis and treatment of cancer' requires Oncologists, and 'Immediate immunity produced by transferred antibodies' requires Passive immunity. C: The answers have been interchanged: 'Immediate immunity produced by transferred antibodies' requires Passive immunity, and 'Mosquito transmitting malaria' requires Female Anopheles. D: The answers have been interchanged: 'Agents capable of causing cancer' requires Carcinogens, and 'Diagnostic test commonly used to detect HIV antibodies' requires ELISA.

BIO-12-CH10-0115 | Medium | Viral and helminth diseases**Correct answer: D. HIV**

Solution: The chapter links Virus that targets helper T lymphocytes with HIV, so HIV completes the statement.

Distractor review: A: Vaccination belongs with 'Administration of antigenic material to induce immune memory', whereas the tested description is 'Virus that targets helper T lymphocytes'. B: Aedes belongs with 'Mosquito vector of dengue', whereas the tested description is 'Virus that targets helper T lymphocytes'. C: Innate immunity belongs with 'Non-specific defence present from birth', whereas the tested description is 'Virus that targets helper T lymphocytes'.

BIO-12-CH10-0116 | Medium | Antibody structure**Correct answer: A. Two heavy and two light chains**

Solution: The chapter links Chains present in each antibody molecule with Two heavy and two light chains, so Two heavy and two light chains completes the statement.

Distractor review: B: Avoiding peer pressure and seeking guidance belongs with 'Responsible adolescent practice that prevents substance misuse', whereas the tested description is 'Chains present in each antibody molecule'. C: Constant Fc region belongs with 'Stem region interacting with immune components', whereas the tested description is 'Chains present in each antibody molecule'. D: Acquired immunity belongs with 'Specific defence developing after exposure', whereas the tested description is 'Chains present in each antibody molecule'.

BIO-12-CH10-0117 | Medium | Bacterial and protozoan diseases**Correct answer: C. Streptococcus pneumoniae**

Solution: The observation 'Causative bacterium of pneumonia commonly cited in the chapter' points to Streptococcus pneumoniae; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Y-shaped belongs with 'Basic shape of an antibody molecule', whereas the tested description is 'Causative bacterium of pneumonia commonly cited in the chapter'. B: Vector control belongs with 'Control approach that interrupts a pathogen's vector', whereas the tested description is 'Causative bacterium of pneumonia commonly cited in the chapter'. D: Female Anopheles belongs with 'Mosquito transmitting malaria', whereas the tested description is 'Causative bacterium of pneumonia commonly cited in the chapter'.

BIO-12-CH10-0118 | Medium | Viral and helminth diseases**Correct answer: A. Aedes**

Solution: The observation 'Mosquito vector of dengue' points to Aedes; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Ascaris belongs with 'Roundworm causing ascariasis', whereas the tested description is 'Mosquito vector of dengue'. C: Constant Fc region belongs with 'Stem region interacting with immune components', whereas the tested description is 'Mosquito vector of dengue'. D: Passive immunity belongs with 'Immediate immunity produced by transferred antibodies', whereas the tested description is 'Mosquito vector of dengue'.

BIO-12-CH10-0119 | Medium | Immunity**Correct answer: C. Non-specific defence present from birth - Innate immunity**

Solution: The accurate entry is 'Non-specific defence present from birth - Innate immunity'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Basic shape of an antibody molecule' should be paired with Y-shaped, not Variable regions. B: 'Antigen-binding regions' should be paired with Variable regions, not Metastasis. D: 'Uncontrolled abnormal cell division' should be paired with Cancer, not Innate immunity.

BIO-12-CH10-0120 | Hard | Cancer and immunity disorders**Correct answer: B. Agents capable of causing cancer - Carcinogens; Immediate immunity produced by transferred antibodies - Passive immunity**

Solution: Both parts of 'Agents capable of causing cancer - Carcinogens; Immediate immunity produced by transferred antibodies - Passive immunity' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Virus that targets helper T lymphocytes' requires HIV, and 'Opioid obtained from poppy latex' requires Morphine. C: The answers have been interchanged: 'Antigen-binding regions' requires Variable regions, and 'Protozoan causing malaria' requires Plasmodium. D: The answers have been interchanged: 'Diagnostic test commonly used to detect HIV antibodies' requires ELISA, and 'Virus that targets helper T lymphocytes' requires HIV.

Standard 12 | Chapter 11

Enhancement of Food Production

BIO-12-CH11-0001 | Medium | Single-cell protein production

Correct answer: A. Spirulina

Solution: Spirulina biomass is used as a protein-rich food supplement and SCP source.

Distractor review: B: The alternative 'Plasmodium' conflicts with the labelled relationship, scale or mechanism in the figure. Spirulina biomass is used as a protein-rich food supplement and SCP source. C: The alternative 'Taenia' conflicts with the labelled relationship, scale or mechanism in the figure. Spirulina biomass is used as a protein-rich food supplement and SCP source. D: The alternative 'Rhizopus causing bread spoilage only' conflicts with the labelled relationship, scale or mechanism in the figure. Spirulina biomass is used as a protein-rich food supplement and SCP source.

BIO-12-CH11-0002 | Hard | Plant micropropagation

Correct answer: D. Auxin-to-cytokinin ratio

Solution: Relative auxin and cytokinin concentrations strongly influence root and shoot induction in tissue culture.

Distractor review: A: The alternative 'Blood glucose concentration' conflicts with the labelled relationship, scale or mechanism in the figure. Relative auxin and cytokinin concentrations strongly influence root and shoot induction in tissue culture. B: The alternative 'Atmospheric noble-gas ratio' conflicts with the labelled relationship, scale or mechanism in the figure. Relative auxin and cytokinin concentrations strongly influence root and shoot induction in tissue culture. C: The alternative 'Antibody titre' conflicts with the labelled relationship, scale or mechanism in the figure. Relative auxin and cytokinin concentrations strongly influence root and shoot induction in tissue culture.

BIO-12-CH11-0003 | Hard | Plant breeding objectives

Correct answer: A. Improvement of protein, oil, vitamin or mineral content - Quality improvement; Fungal antagonist used against plant pathogens - Trichoderma

Solution: Both parts of 'Improvement of protein, oil, vitamin or mineral content - Quality improvement; Fungal antagonist used against plant pathogens - Trichoderma' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Microbe producing citric acid' requires *Aspergillus niger*, and 'Bacterial biocontrol agent producing insecticidal crystal proteins' requires *Bacillus thuringiensis*. C: The answers have been interchanged: 'Bacterial biocontrol agent producing insecticidal crystal proteins' requires *Bacillus thuringiensis*, and 'Microbial aggregates in aeration tanks' requires Flocs. D: The answers have been interchanged: 'Microbial aggregates in aeration tanks' requires Flocs, and 'Fungus associated with penicillin production' requires *Penicillium*.

BIO-12-CH11-0004 | Hard | Tissue culture

Correct answer: D. Small tissue piece used to start a culture - Explant; Development of crop varieties with higher output - Increased yield

Solution: Both parts of 'Small tissue piece used to start a culture - Explant; Development of crop varieties with higher output - Increased yield' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Methanogen commonly found in cattle rumen' requires *Methanobacterium*, and 'Testing a new line at several places before release' requires Multi-location evaluation. B: The answers have been interchanged: 'Testing parents for desirable characters' requires Evaluation and selection, and 'Improvement of protein, oil, vitamin or mineral content' requires Quality improvement. C: The answers have been interchanged: 'Development of crops tolerating drought or salinity' requires Abiotic stress resistance, and 'Testing parents for desirable characters' requires Evaluation and selection.

BIO-12-CH11-0005 | Challenging | Single-cell protein production

Correct answer: B. Contamination rises and useful biomass yield falls

Solution: Contaminants compete for nutrients, while inadequate aeration restricts aerobic growth and biomass productivity.

Distractor review: A: The alternative 'Pure-culture yield necessarily doubles' conflicts with the labelled relationship, scale or mechanism in the figure. Contaminants compete for nutrients, while inadequate aeration restricts aerobic growth and biomass productivity. C: The alternative 'The culture becomes genetically identical to a plantlet' conflicts with the labelled relationship, scale or mechanism in the figure. Contaminants compete for nutrients, while inadequate aeration restricts aerobic growth and biomass productivity. D: The alternative 'Protein production becomes independent of nutrients' conflicts with the labelled relationship, scale or mechanism in the figure. Contaminants compete for nutrients, while inadequate aeration restricts aerobic growth and biomass productivity.

BIO-12-CH11-0006 | Easy | Plant micropropagation**Correct answer: B. Explant**

Solution: A small sterile plant piece called an explant initiates tissue culture.

Distractor review: A: The alternative 'Mature pollen only' conflicts with the labelled relationship, scale or mechanism in the figure. A small sterile plant piece called an explant initiates tissue culture. C: The alternative 'Whole field soil' conflicts with the labelled relationship, scale or mechanism in the figure. A small sterile plant piece called an explant initiates tissue culture. D: The alternative 'Animal plasma' conflicts with the labelled relationship, scale or mechanism in the figure. A small sterile plant piece called an explant initiates tissue culture.

BIO-12-CH11-0007 | Easy | Industrial microbes**Correct answer: D. Yeast used in ethanol production**

Solution: *Saccharomyces cerevisiae* represents 'Yeast used in ethanol production'. Each alternative describes a different term or result.

Distractor review: A: 'Fungal antagonist used against plant pathogens' identifies *Trichoderma*, not *Saccharomyces cerevisiae*. B: 'Microbial biomass used as protein-rich food or feed' identifies Single-cell protein, not *Saccharomyces cerevisiae*. C: 'Scientific management of farm animals' identifies Animal husbandry, not *Saccharomyces cerevisiae*.

BIO-12-CH11-0008 | Hard | Biofortification and SCP**Correct answer: B. Breeding crops for improved nutritional quality - Biofortification; Breeding for resistance to pathogens - Disease resistance**

Solution: Both parts of 'Breeding crops for improved nutritional quality - Biofortification; Breeding for resistance to pathogens - Disease resistance' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Microbial aggregates in aeration tanks' requires Flocs, and 'Management of milk-producing animals' requires Dairy farming. C: The answers have been interchanged: 'Management of milk-producing animals' requires Dairy farming, and 'Bacterium producing lactic acid' requires *Lactobacillus*. D: The answers have been interchanged: 'Symbiotic nitrogen fixer in legume roots' requires *Rhizobium*, and 'Microbial aggregates in aeration tanks' requires Flocs.

BIO-12-CH11-0009 | Medium | Biofortification and SCP**Correct answer: C. Spirulina**

Solution: The chapter links Alga widely used as single-cell protein with *Spirulina*, so *Spirulina* completes the statement.

Distractor review: A: Biofortification belongs with 'Breeding crops for improved nutritional quality', whereas the tested description is 'Alga widely used as single-cell protein'. B: Fortified maize belongs with 'Biofortified crop with twice the amount of lysine and tryptophan', whereas the tested description is 'Alga widely used as single-cell protein'. D: Apiculture belongs with 'Rearing of honey bees', whereas the tested description is 'Alga widely used as single-cell protein'.

BIO-12-CH11-0010 | Easy | Plant breeding process**Correct answer: D. Hybridization**

Solution: 'Crossing genetically distinct selected parents' is the defining description for Hybridization. The other choices belong to different chapter concepts.

Distractor review: A: Callus belongs with 'Unorganized mass of cultured cells', whereas the tested description is 'Crossing genetically distinct selected parents'. B: Increased yield belongs with 'Development of crop varieties with higher output', whereas the tested description is 'Crossing genetically distinct selected parents'. C: Fortified maize belongs with 'Biofortified crop with twice the amount of lysine and tryptophan', whereas the tested description is 'Crossing genetically distinct selected parents'.

BIO-12-CH11-0011 | Medium | Plant breeding process**Correct answer: A. Evaluation and selection**

Solution: The observation 'Testing parents for desirable characters' points to Evaluation and selection; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: *Methanobacterium* belongs with 'Methanogen commonly found in cattle rumen', whereas the tested description is 'Testing parents for desirable characters'. C: Biofortification belongs with 'Breeding crops for improved nutritional quality', whereas the tested description is 'Testing parents for desirable characters'. D: Explant belongs with 'Small tissue piece used to start a culture', whereas the tested description is 'Testing parents for desirable characters'.

BIO-12-CH11-0012 | Easy | Plant breeding objectives**Correct answer: C. Abiotic stress resistance**

Solution: 'Development of crops tolerating drought or salinity' is the defining description for Abiotic stress resistance. The other choices belong to different chapter concepts.

Distractor review: A: Quality improvement belongs with 'Improvement of protein, oil, vitamin or mineral content', whereas the tested description is 'Development of crops tolerating drought or salinity'. B: Totipotency belongs with 'Capacity of a plant cell to regenerate a whole plant', whereas the tested description is 'Development of crops tolerating drought or salinity'. D: Flocs belongs with 'Microbial aggregates in aeration tanks', whereas the tested description is 'Development of crops tolerating drought or salinity'.

BIO-12-CH11-0013 | Challenging | Tissue culture**Correct answer: A. I, II and IV only**

Solution: Statement I is correct because 'Unorganized mass of cultured cells' corresponds to Callus. Statement II is correct because 'Small tissue piece used to start a culture' corresponds to Explant. Statement III is incorrect: 'Rapid clonal production through tissue culture' corresponds to Micropropagation, not Biogas. Statement IV is correct because 'Anaerobic digesters produce methane-rich' corresponds to Biogas. Therefore, the valid combination is I, II and IV only.

Distractor review: B: This combination includes false statement(s) III and omits true statement(s) I, IV. C: This combination omits true statement(s) I, II. D: This combination includes false statement(s) III and omits true statement(s) II.

BIO-12-CH11-0014 | Easy | Plant breeding objectives**Correct answer: B. Improvement of protein, oil, vitamin or mineral content**

Solution: Quality improvement represents 'Improvement of protein, oil, vitamin or mineral content'. Each alternative describes a different term or result.

Distractor review: A: 'Unorganized mass of cultured cells' identifies Callus, not Quality improvement. C: 'Breeding crops for improved nutritional quality' identifies Biofortification, not Quality improvement. D: 'Fungal antagonist used against plant pathogens' identifies Trichoderma, not Quality improvement.

BIO-12-CH11-0015 | Easy | Biocontrol and biofertilizers**Correct answer: C. Bacillus thuringiensis**

Solution: 'Bacterial biocontrol agent producing insecticidal crystal proteins' is the defining description for Bacillus thuringiensis. The other choices belong to different chapter concepts.

Distractor review: A: Micropropagation belongs with 'Rapid clonal production through tissue culture', whereas the tested description is 'Bacterial biocontrol agent producing insecticidal crystal proteins'. B: Trichoderma belongs with 'Fungal antagonist used against plant pathogens', whereas the tested description is 'Bacterial biocontrol agent producing insecticidal crystal proteins'. D: Mycorrhiza belongs with 'Association of fungi with plant roots improving mineral uptake', whereas the tested description is 'Bacterial biocontrol agent producing insecticidal crystal proteins'.

BIO-12-CH11-0016 | Medium | Biocontrol and biofertilizers**Correct answer: B. Mycorrhiza**

Solution: The chapter links Association of fungi with plant roots improving mineral uptake with Mycorrhiza, so Mycorrhiza completes the statement.

Distractor review: A: Germplasm collection belongs with 'Collection of diverse cultivated and wild forms', whereas the tested description is 'Association of fungi with plant roots improving mineral uptake'. C: Rhizobium belongs with 'Symbiotic nitrogen fixer in legume roots', whereas the tested description is 'Association of fungi with plant roots improving mineral uptake'. D: Fortified maize belongs with 'Biofortified crop with twice the amount of lysine and tryptophan', whereas the tested description is 'Association of fungi with plant roots improving mineral uptake'.

BIO-12-CH11-0017 | Hard | Animal husbandry**Correct answer: B. Rearing of honey bees - Hybridization**

Solution: 'Rearing of honey bees' should be paired with Apiculture, not Hybridization. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Development of crops tolerating drought or salinity - Abiotic stress resistance. C: This is a valid association: Improvement of protein, oil, vitamin or mineral content - Quality improvement. D: This is a valid association: Indicator of organic pollution reduced during secondary treatment - Biochemical oxygen demand.

BIO-12-CH11-0018 | Easy | Tissue culture**Correct answer: A. Callus**

Solution: 'Unorganized mass of cultured cells' is the defining description for Callus. The other choices belong to different chapter concepts.

Distractor review: B: Rhizobium belongs with 'Symbiotic nitrogen fixer in legume roots', whereas the tested description is 'Unorganized mass of cultured cells'. C: Totipotency belongs with 'Capacity of a plant cell to regenerate a whole plant', whereas the tested description is 'Unorganized mass of cultured cells'. D: Micropropagation belongs with 'Rapid clonal production through tissue culture', whereas the tested description is 'Unorganized mass of cultured cells'.

BIO-12-CH11-0019 | Medium | Biofortification and SCP**Correct answer: D. Alga widely used as single-cell protein - Spirulina**

Solution: The accurate entry is 'Alga widely used as single-cell protein - Spirulina'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Management of milk-producing animals' should be paired with Dairy farming, not Rhizobium. B: 'Symbiotic nitrogen fixer in legume roots' should be paired with Rhizobium, not Apiculture. C: 'Indicator of organic pollution reduced during secondary treatment' should be paired with Biochemical oxygen demand, not Poultry farming.

BIO-12-CH11-0020 | Challenging | Plant breeding process**Correct answer: C. IV only**

Solution: Statement I is incorrect: 'Testing parents for desirable characters' corresponds to Evaluation and selection, not Multi-location evaluation. Statement II is incorrect: 'Testing a new line at several places before release' corresponds to Multi-location evaluation, not Hybridization. Statement III is incorrect: 'Crossing genetically distinct selected parents' corresponds to Hybridization, not Abiotic stress resistance. Statement IV is correct because 'Development of crops tolerating drought or salinity' corresponds to Abiotic stress resistance. Therefore, the valid combination is IV only.

Distractor review: A: This combination includes false statement(s) I, III. B: This combination includes false statement(s) II, III and omits true statement(s) IV. D: This combination includes false statement(s) I and omits true statement(s) IV.

BIO-12-CH11-0021 | Medium | Biocontrol and biofertilizers**Correct answer: D. Rhizobium**

Solution: The chapter links Symbiotic nitrogen fixer in legume roots with Rhizobium, so Rhizobium completes the statement.

Distractor review: A: Multi-location evaluation belongs with 'Testing a new line at several places before release', whereas the tested description is 'Symbiotic nitrogen fixer in legume roots'. B: Explant belongs with 'Small tissue piece used to start a culture', whereas the tested description is 'Symbiotic nitrogen fixer in legume roots'. C: Mycorrhiza belongs with 'Association of fungi with plant roots improving mineral uptake', whereas the tested description is 'Symbiotic nitrogen fixer in legume roots'.

BIO-12-CH11-0022 | Easy | Sewage and biogas**Correct answer: D. Microbial aggregates in aeration tanks**

Solution: Flocs represents 'Microbial aggregates in aeration tanks'. Each alternative describes a different term or result.

Distractor review: A: 'Crossing genetically distinct selected parents' identifies Hybridization, not Flocs. B: 'Indicator of organic pollution reduced during secondary treatment' identifies Biochemical oxygen demand, not Flocs. C: 'Breeding for resistance to pathogens' identifies Disease resistance, not Flocs.

BIO-12-CH11-0023 | Medium | Industrial microbes**Correct answer: C. Fungus associated with penicillin production - Penicillium**

Solution: The accurate entry is 'Fungus associated with penicillin production - Penicillium'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Management of milk-producing animals' should be paired with Dairy farming, not Penicillium. B: 'Microbial aggregates in aeration tanks' should be paired with Flocs, not Germplasm collection. D: 'Rearing of honey bees' should be paired with Apiculture, not Explant.

BIO-12-CH11-0024 | Hard | Tissue culture**Correct answer: B. Rapid clonal production through tissue culture - Disease resistance**

Solution: 'Rapid clonal production through tissue culture' should be paired with Micropropagation, not Disease resistance. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Methanogen commonly found in cattle rumen - Methanobacterium. C: This is a valid association: Microbial biomass used as protein-rich food or feed - Single-cell protein. D: This is a valid association: Bacterial biocontrol agent producing insecticidal crystal proteins - Bacillus thuringiensis.

BIO-12-CH11-0025 | Medium | Sewage and biogas**Correct answer: B. Flocs**

Solution: The chapter links Microbial aggregates in aeration tanks with Flocs, so Flocs completes the statement.

Distractor review: A: Rhizobium belongs with 'Symbiotic nitrogen fixer in legume roots', whereas the tested description is 'Microbial aggregates in aeration tanks'. C: Quality improvement belongs with 'Improvement of protein, oil, vitamin or mineral content', whereas the tested description is 'Microbial aggregates in aeration tanks'. D: Methanobacterium belongs with 'Methanogen commonly found in cattle rumen', whereas the tested description is 'Microbial aggregates in aeration tanks'.

BIO-12-CH11-0026 | Medium | Plant breeding process**Correct answer: D. Multi-location evaluation**

Solution: The observation 'Testing a new line at several places before release' points to Multi-location evaluation; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Germplasm collection belongs with 'Collection of diverse cultivated and wild forms', whereas the tested description is 'Testing a new line at several places before release'. B: Biofortification belongs with 'Breeding crops for improved nutritional quality', whereas the tested description is 'Testing a new line at several places before release'. C: Hybridization belongs with 'Crossing genetically distinct selected parents', whereas the tested description is 'Testing a new line at several places before release'.

BIO-12-CH11-0027 | Medium | Plant breeding objectives**Correct answer: B. Increased yield**

Solution: The observation 'Development of crop varieties with higher output' points to Increased yield; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Penicillium belongs with 'Fungus associated with penicillin production', whereas the tested description is 'Development of crop varieties with higher output'. C: Aspergillus niger belongs with 'Microbe producing citric acid', whereas the tested description is 'Development of crop varieties with higher output'. D: Single-cell protein belongs with 'Microbial biomass used as protein-rich food or feed', whereas the tested description is 'Development of crop varieties with higher output'.

BIO-12-CH11-0028 | Hard | Biocontrol and biofertilizers**Correct answer: D. Bacterial biocontrol agent producing insecticidal crystal proteins - Poultry farming**

Solution: 'Bacterial biocontrol agent producing insecticidal crystal proteins' should be paired with Bacillus thuringiensis, not Poultry farming. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Microbial biomass used as protein-rich food or feed - Single-cell protein. B: This is a valid association: Rapid clonal production through tissue culture - Micropropagation. C: This is a valid association: Small tissue piece used to start a culture - Explant.

BIO-12-CH11-0029 | Medium | Industrial microbes**Correct answer: D. Aspergillus niger**

Solution: The observation 'Microbe producing citric acid' points to Aspergillus niger; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Dairy farming belongs with 'Management of milk-producing animals', whereas the tested description is 'Microbe producing citric acid'. B: Lactobacillus belongs with 'Bacterium producing lactic acid', whereas the tested description is 'Microbe producing citric acid'. C: Animal husbandry belongs with 'Scientific management of farm animals', whereas the tested description is 'Microbe producing citric acid'.

BIO-12-CH11-0030 | Medium | Plant breeding process**Correct answer: C. Germplasm collection**

Solution: The observation 'Collection of diverse cultivated and wild forms' points to Germplasm collection; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Hybridization belongs with 'Crossing genetically distinct selected parents', whereas the tested description is 'Collection of diverse cultivated and wild forms'. B: Biochemical oxygen demand belongs with 'Indicator of organic pollution reduced during secondary treatment', whereas the tested description is 'Collection of diverse cultivated and wild forms'. D: Methanobacterium belongs with 'Methanogen commonly found in cattle rumen', whereas the tested description is 'Collection of diverse cultivated and wild forms'.

BIO-12-CH11-0031 | Medium | Sewage and biogas

Correct answer: A. Indicator of organic pollution reduced during secondary treatment - Biochemical oxygen demand

Solution: The accurate entry is 'Indicator of organic pollution reduced during secondary treatment - Biochemical oxygen demand'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Fungus associated with penicillin production' should be paired with Penicillium, not Biochemical oxygen demand. C: 'Testing parents for desirable characters' should be paired with Evaluation and selection, not Animal husbandry. D: 'Scientific management of farm animals' should be paired with Animal husbandry, not Hybridization.

BIO-12-CH11-0032 | Medium | Sewage and biogas

Correct answer: C. Biogas

Solution: The observation 'Anaerobic digesters produce methane-rich' points to Biogas; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Biofortification belongs with 'Breeding crops for improved nutritional quality', whereas the tested description is 'Anaerobic digesters produce methane-rich'. B: Evaluation and selection belongs with 'Testing parents for desirable characters', whereas the tested description is 'Anaerobic digesters produce methane-rich'. D: Hybridization belongs with 'Crossing genetically distinct selected parents', whereas the tested description is 'Anaerobic digesters produce methane-rich'.

BIO-12-CH11-0033 | Medium | Animal husbandry

Correct answer: A. Poultry farming

Solution: The chapter links Rearing of birds for eggs and meat with Poultry farming, so Poultry farming completes the statement.

Distractor review: B: Flocs belongs with 'Microbial aggregates in aeration tanks', whereas the tested description is 'Rearing of birds for eggs and meat'. C: Animal husbandry belongs with 'Scientific management of farm animals', whereas the tested description is 'Rearing of birds for eggs and meat'. D: Mycorrhiza belongs with 'Association of fungi with plant roots improving mineral uptake', whereas the tested description is 'Rearing of birds for eggs and meat'.

BIO-12-CH11-0034 | Medium | Plant breeding process

Correct answer: D. Hybridization

Solution: The observation 'Crossing genetically distinct selected parents' points to Hybridization; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Biogas belongs with 'Anaerobic digesters produce methane-rich', whereas the tested description is 'Crossing genetically distinct selected parents'. B: Fortified maize belongs with 'Biofortified crop with twice the amount of lysine and tryptophan', whereas the tested description is 'Crossing genetically distinct selected parents'. C: Germplasm collection belongs with 'Collection of diverse cultivated and wild forms', whereas the tested description is 'Crossing genetically distinct selected parents'.

BIO-12-CH11-0035 | Medium | Animal husbandry

Correct answer: C. Animal husbandry

Solution: The chapter links Scientific management of farm animals with Animal husbandry, so Animal husbandry completes the statement.

Distractor review: A: Mycorrhiza belongs with 'Association of fungi with plant roots improving mineral uptake', whereas the tested description is 'Scientific management of farm animals'. B: Biochemical oxygen demand belongs with 'Indicator of organic pollution reduced during secondary treatment', whereas the tested description is 'Scientific management of farm animals'. D: Apiculture belongs with 'Rearing of honey bees', whereas the tested description is 'Scientific management of farm animals'.

BIO-12-CH11-0036 | Easy | Biofortification and SCP

Correct answer: B. Single-cell protein

Solution: 'Microbial biomass used as protein-rich food or feed' is the defining description for Single-cell protein. The other choices belong to different chapter concepts.

Distractor review: A: Hybridization belongs with 'Crossing genetically distinct selected parents', whereas the tested description is 'Microbial biomass used as protein-rich food or feed'. C: Fortified maize belongs with 'Biofortified crop with twice the amount of lysine and tryptophan', whereas the tested description is 'Microbial biomass used as protein-rich food or feed'. D: Saccharomyces cerevisiae belongs with 'Yeast used in ethanol production', whereas the tested description is 'Microbial biomass used as protein-rich food or feed'.

BIO-12-CH11-0037 | Easy | Tissue culture**Correct answer: D. Small tissue piece used to start a culture**

Solution: Explant represents 'Small tissue piece used to start a culture'. Each alternative describes a different term or result.

Distractor review: A: 'Methanogen commonly found in cattle rumen' identifies Methanobacterium, not Explant. B: 'Unorganized mass of cultured cells' identifies Callus, not Explant. C: 'Improvement of protein, oil, vitamin or mineral content' identifies Quality improvement, not Explant.

BIO-12-CH11-0038 | Medium | Tissue culture**Correct answer: B. Callus**

Solution: The observation 'Unorganized mass of cultured cells' points to Callus; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Micropropagation belongs with 'Rapid clonal production through tissue culture', whereas the tested description is 'Unorganized mass of cultured cells'. C: Saccharomyces cerevisiae belongs with 'Yeast used in ethanol production', whereas the tested description is 'Unorganized mass of cultured cells'. D: Totipotency belongs with 'Capacity of a plant cell to regenerate a whole plant', whereas the tested description is 'Unorganized mass of cultured cells'.

BIO-12-CH11-0039 | Hard | Plant breeding objectives**Correct answer: C. Development of crops tolerating drought or salinity - Totipotency**

Solution: 'Development of crops tolerating drought or salinity' should be paired with Abiotic stress resistance, not Totipotency. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Association of fungi with plant roots improving mineral uptake - Mycorrhiza. B: This is a valid association: Alga widely used as single-cell protein - Spirulina. D: This is a valid association: Symbiotic nitrogen fixer in legume roots - Rhizobium.

BIO-12-CH11-0040 | Hard | Sewage and biogas**Correct answer: A. Microbial aggregates in aeration tanks - Flocs; Bacterium producing lactic acid - Lactobacillus**

Solution: Both parts of 'Microbial aggregates in aeration tanks - Flocs; Bacterium producing lactic acid - Lactobacillus' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Breeding for resistance to pathogens' requires Disease resistance, and 'Improvement of protein, oil, vitamin or mineral content' requires Quality improvement. C: The answers have been interchanged: 'Symbiotic nitrogen fixer in legume roots' requires Rhizobium, and 'Breeding for resistance to pathogens' requires Disease resistance. D: The answers have been interchanged: 'Crossing genetically distinct selected parents' requires Hybridization, and 'Symbiotic nitrogen fixer in legume roots' requires Rhizobium.

BIO-12-CH11-0041 | Easy | Animal husbandry**Correct answer: C. Dairy farming**

Solution: 'Management of milk-producing animals' is the defining description for Dairy farming. The other choices belong to different chapter concepts.

Distractor review: A: Germplasm collection belongs with 'Collection of diverse cultivated and wild forms', whereas the tested description is 'Management of milk-producing animals'. B: Apiculture belongs with 'Rearing of honey bees', whereas the tested description is 'Management of milk-producing animals'. D: Quality improvement belongs with 'Improvement of protein, oil, vitamin or mineral content', whereas the tested description is 'Management of milk-producing animals'.

BIO-12-CH11-0042 | Easy | Industrial microbes**Correct answer: A. Bacterium producing lactic acid**

Solution: Lactobacillus represents 'Bacterium producing lactic acid'. Each alternative describes a different term or result.

Distractor review: B: 'Yeast used in ethanol production' identifies Saccharomyces cerevisiae, not Lactobacillus. C: 'Management of milk-producing animals' identifies Dairy farming, not Lactobacillus. D: 'Fungal antagonist used against plant pathogens' identifies Trichoderma, not Lactobacillus.

BIO-12-CH11-0043 | Challenging | Sewage and biogas**Correct answer: D. I, II and III only**

Solution: Statement I is correct because 'Microbial aggregates in aeration tanks' corresponds to Flocs. Statement II is correct because 'Anaerobic digesters produce methane-rich' corresponds to Biogas. Statement III is correct because 'Methanogen commonly found in cattle rumen' corresponds to Methanobacterium. Statement IV is incorrect: 'Scientific management of farm animals' corresponds to Animal husbandry, not Flocs. Therefore, the valid combination is I, II and III only.

Distractor review: A: This combination omits true statement(s) I. B: This combination includes false statement(s) IV and omits true statement(s) II, III. C: This combination omits true statement(s) II, III.

BIO-12-CH11-0044 | Medium | Plant breeding objectives**Correct answer: A. Development of crop varieties with higher output - Increased yield**

Solution: The accurate entry is 'Development of crop varieties with higher output - Increased yield'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Microbe producing citric acid' should be paired with *Aspergillus niger*, not Multi-location evaluation. C: 'Fungus associated with penicillin production' should be paired with *Penicillium*, not Explant. D: 'Symbiotic nitrogen fixer in legume roots' should be paired with *Rhizobium*, not Mycorrhiza.

BIO-12-CH11-0045 | Medium | Tissue culture**Correct answer: A. Micropropagation**

Solution: The observation 'Rapid clonal production through tissue culture' points to Micropropagation; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Disease resistance belongs with 'Breeding for resistance to pathogens', whereas the tested description is 'Rapid clonal production through tissue culture'. C: Methanobacterium belongs with 'Methanogen commonly found in cattle rumen', whereas the tested description is 'Rapid clonal production through tissue culture'. D: Flocs belongs with 'Microbial aggregates in aeration tanks', whereas the tested description is 'Rapid clonal production through tissue culture'.

BIO-12-CH11-0046 | Hard | Industrial microbes**Correct answer: C. Fungus associated with penicillin production - *Penicillium*; Testing parents for desirable characters - Evaluation and selection**

Solution: Both parts of 'Fungus associated with penicillin production - *Penicillium*; Testing parents for desirable characters - Evaluation and selection' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Small tissue piece used to start a culture' requires Explant, and 'Collection of diverse cultivated and wild forms' requires Germplasm collection. B: The answers have been interchanged: 'Rearing of honey bees' requires Apiculture, and 'Microbial aggregates in aeration tanks' requires Flocs. D: The answers have been interchanged: 'Management of milk-producing animals' requires Dairy farming, and 'Symbiotic nitrogen fixer in legume roots' requires *Rhizobium*.

BIO-12-CH11-0047 | Easy | Biocontrol and biofertilizers**Correct answer: B. Symbiotic nitrogen fixer in legume roots**

Solution: *Rhizobium* represents 'Symbiotic nitrogen fixer in legume roots'. Each alternative describes a different term or result.

Distractor review: A: 'Testing a new line at several places before release' identifies Multi-location evaluation, not *Rhizobium*. C: 'Fungal antagonist used against plant pathogens' identifies *Trichoderma*, not *Rhizobium*. D: 'Association of fungi with plant roots improving mineral uptake' identifies Mycorrhiza, not *Rhizobium*.

BIO-12-CH11-0048 | Hard | Biofortification and SCP**Correct answer: D. Alga widely used as single-cell protein - *Spirulina*; Management of milk-producing animals - Dairy farming**

Solution: Both parts of 'Alga widely used as single-cell protein - *Spirulina*; Management of milk-producing animals - Dairy farming' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Rearing of birds for eggs and meat' requires Poultry farming, and 'Testing a new line at several places before release' requires Multi-location evaluation. B: The answers have been interchanged: 'Rearing of honey bees' requires Apiculture, and 'Rearing of birds for eggs and meat' requires Poultry farming. C: The answers have been interchanged: 'Symbiotic nitrogen fixer in legume roots' requires *Rhizobium*, and 'Indicator of organic pollution reduced during secondary treatment' requires Biochemical oxygen demand.

BIO-12-CH11-0049 | Easy | Plant breeding objectives**Correct answer: A. Development of crop varieties with higher output**

Solution: Increased yield represents 'Development of crop varieties with higher output'. Each alternative describes a different term or result.

Distractor review: B: 'Microbe producing citric acid' identifies *Aspergillus niger*, not Increased yield. C: 'Symbiotic nitrogen fixer in legume roots' identifies *Rhizobium*, not Increased yield. D: 'Fungus associated with penicillin production' identifies *Penicillium*, not Increased yield.

BIO-12-CH11-0050 | Hard | Animal husbandry**Correct answer: B. Management of milk-producing animals - Quality improvement**

Solution: 'Management of milk-producing animals' should be paired with Dairy farming, not Quality improvement. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Collection of diverse cultivated and wild forms - Germplasm collection. C: This is a valid association: Rapid clonal production through tissue culture - Micropropagation. D: This is a valid association: Association of fungi with plant roots improving mineral uptake - Mycorrhiza.

BIO-12-CH11-0051 | Easy | Plant breeding process**Correct answer: B. Evaluation and selection**

Solution: 'Testing parents for desirable characters' is the defining description for Evaluation and selection. The other choices belong to different chapter concepts.

Distractor review: A: *Rhizobium* belongs with 'Symbiotic nitrogen fixer in legume roots', whereas the tested description is 'Testing parents for desirable characters'. C: *Aspergillus niger* belongs with 'Microbe producing citric acid', whereas the tested description is 'Testing parents for desirable characters'. D: Explant belongs with 'Small tissue piece used to start a culture', whereas the tested description is 'Testing parents for desirable characters'.

BIO-12-CH11-0052 | Medium | Plant breeding process**Correct answer: A. Collection of diverse cultivated and wild forms - Germplasm collection**

Solution: The accurate entry is 'Collection of diverse cultivated and wild forms - Germplasm collection'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Management of milk-producing animals' should be paired with Dairy farming, not Totipotency. C: 'Microbial aggregates in aeration tanks' should be paired with Flocs, not Germplasm collection. D: 'Scientific management of farm animals' should be paired with Animal husbandry, not Biochemical oxygen demand.

BIO-12-CH11-0053 | Easy | Animal husbandry**Correct answer: C. Apiculture**

Solution: 'Rearing of honey bees' is the defining description for Apiculture. The other choices belong to different chapter concepts.

Distractor review: A: Dairy farming belongs with 'Management of milk-producing animals', whereas the tested description is 'Rearing of honey bees'. B: *Bacillus thuringiensis* belongs with 'Bacterial biocontrol agent producing insecticidal crystal proteins', whereas the tested description is 'Rearing of honey bees'. D: Biochemical oxygen demand belongs with 'Indicator of organic pollution reduced during secondary treatment', whereas the tested description is 'Rearing of honey bees'.

BIO-12-CH11-0054 | Medium | Biocontrol and biofertilizers**Correct answer: B. *Bacillus thuringiensis***

Solution: The observation 'Bacterial biocontrol agent producing insecticidal crystal proteins' points to *Bacillus thuringiensis*; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Mycorrhiza belongs with 'Association of fungi with plant roots improving mineral uptake', whereas the tested description is 'Bacterial biocontrol agent producing insecticidal crystal proteins'. C: Explant belongs with 'Small tissue piece used to start a culture', whereas the tested description is 'Bacterial biocontrol agent producing insecticidal crystal proteins'. D: Micropropagation belongs with 'Rapid clonal production through tissue culture', whereas the tested description is 'Bacterial biocontrol agent producing insecticidal crystal proteins'.

BIO-12-CH11-0055 | Challenging | Biocontrol and biofertilizers**Correct answer: C. III only**

Solution: Statement I is incorrect: 'Fungal antagonist used against plant pathogens' corresponds to Trichoderma, not Bacillus thuringiensis. Statement II is incorrect: 'Bacterial biocontrol agent producing insecticidal crystal proteins' corresponds to Bacillus thuringiensis, not Rhizobium. Statement III is correct because 'Symbiotic nitrogen fixer in legume roots' corresponds to Rhizobium. Statement IV is incorrect: 'Breeding for resistance to pathogens' corresponds to Disease resistance, not Trichoderma. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) IV. B: This combination includes false statement(s) I, IV and omits true statement(s) III. D: This combination includes false statement(s) I, II and omits true statement(s) III.

BIO-12-CH11-0056 | Hard | Plant breeding objectives**Correct answer: D. Development of crop varieties with higher output - Increased yield; Symbiotic nitrogen fixer in legume roots - Rhizobium**

Solution: Both parts of 'Development of crop varieties with higher output - Increased yield; Symbiotic nitrogen fixer in legume roots - Rhizobium' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Microbial biomass used as protein-rich food or feed' requires Single-cell protein, and 'Association of fungi with plant roots improving mineral uptake' requires Mycorrhiza. B: The answers have been interchanged: 'Fungus associated with penicillin production' requires Penicillium, and 'Microbe producing citric acid' requires Aspergillus niger. C: The answers have been interchanged: 'Microbe producing citric acid' requires Aspergillus niger, and 'Small tissue piece used to start a culture' requires Explant.

BIO-12-CH11-0057 | Easy | Tissue culture**Correct answer: C. Micropropagation**

Solution: 'Rapid clonal production through tissue culture' is the defining description for Micropropagation. The other choices belong to different chapter concepts.

Distractor review: A: Single-cell protein belongs with 'Microbial biomass used as protein-rich food or feed', whereas the tested description is 'Rapid clonal production through tissue culture'. B: Hybridization belongs with 'Crossing genetically distinct selected parents', whereas the tested description is 'Rapid clonal production through tissue culture'. D: Callus belongs with 'Unorganized mass of cultured cells', whereas the tested description is 'Rapid clonal production through tissue culture'.

BIO-12-CH11-0058 | Hard | Sewage and biogas**Correct answer: B. Methanogen commonly found in cattle rumen - Methanobacterium; Microbe producing citric acid - Aspergillus niger**

Solution: Both parts of 'Methanogen commonly found in cattle rumen - Methanobacterium; Microbe producing citric acid - Aspergillus niger' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Rearing of honey bees' requires Apiculture, and 'Development of crops tolerating drought or salinity' requires Abiotic stress resistance. C: The answers have been interchanged: 'Development of crops tolerating drought or salinity' requires Abiotic stress resistance, and 'Microbial biomass used as protein-rich food or feed' requires Single-cell protein. D: The answers have been interchanged: 'Microbial biomass used as protein-rich food or feed' requires Single-cell protein, and 'Crossing genetically distinct selected parents' requires Hybridization.

BIO-12-CH11-0059 | Easy | Biocontrol and biofertilizers**Correct answer: C. Mycorrhiza**

Solution: 'Association of fungi with plant roots improving mineral uptake' is the defining description for Mycorrhiza. The other choices belong to different chapter concepts.

Distractor review: A: Rhizobium belongs with 'Symbiotic nitrogen fixer in legume roots', whereas the tested description is 'Association of fungi with plant roots improving mineral uptake'. B: Fortified maize belongs with 'Biofortified crop with twice the amount of lysine and tryptophan', whereas the tested description is 'Association of fungi with plant roots improving mineral uptake'. D: Bacillus thuringiensis belongs with 'Bacterial biocontrol agent producing insecticidal crystal proteins', whereas the tested description is 'Association of fungi with plant roots improving mineral uptake'.

BIO-12-CH11-0060 | Medium | Industrial microbes**Correct answer: D. Microbe producing citric acid - Aspergillus niger**

Solution: The accurate entry is 'Microbe producing citric acid - Aspergillus niger'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Microbial aggregates in aeration tanks' should be paired with Flocs, not Micropropagation. B: 'Biofortified crop with twice the amount of lysine and tryptophan' should be paired with Fortified maize, not Evaluation and selection. C: 'Indicator of organic pollution reduced during secondary treatment' should be paired with Biochemical oxygen demand, not Dairy farming.

BIO-12-CH11-0061 | Medium | Biocontrol and biofertilizers**Correct answer: C. Trichoderma**

Solution: The observation 'Fungal antagonist used against plant pathogens' points to Trichoderma; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Spirulina belongs with 'Alga widely used as single-cell protein', whereas the tested description is 'Fungal antagonist used against plant pathogens'. B: Bacillus thuringiensis belongs with 'Bacterial biocontrol agent producing insecticidal crystal proteins', whereas the tested description is 'Fungal antagonist used against plant pathogens'. D: Lactobacillus belongs with 'Bacterium producing lactic acid', whereas the tested description is 'Fungal antagonist used against plant pathogens'.

BIO-12-CH11-0062 | Medium | Tissue culture**Correct answer: A. Totipotency**

Solution: The observation 'Capacity of a plant cell to regenerate a whole plant' points to Totipotency; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Disease resistance belongs with 'Breeding for resistance to pathogens', whereas the tested description is 'Capacity of a plant cell to regenerate a whole plant'. C: Animal husbandry belongs with 'Scientific management of farm animals', whereas the tested description is 'Capacity of a plant cell to regenerate a whole plant'. D: Flocs belongs with 'Microbial aggregates in aeration tanks', whereas the tested description is 'Capacity of a plant cell to regenerate a whole plant'.

BIO-12-CH11-0063 | Medium | Sewage and biogas**Correct answer: C. Methanobacterium**

Solution: The chapter links Methanogen commonly found in cattle rumen with Methanobacterium, so Methanobacterium completes the statement.

Distractor review: A: Biofortification belongs with 'Breeding crops for improved nutritional quality', whereas the tested description is 'Methanogen commonly found in cattle rumen'. B: Apiculture belongs with 'Rearing of honey bees', whereas the tested description is 'Methanogen commonly found in cattle rumen'. D: Evaluation and selection belongs with 'Testing parents for desirable characters', whereas the tested description is 'Methanogen commonly found in cattle rumen'.

BIO-12-CH11-0064 | Medium | Tissue culture**Correct answer: A. Explant**

Solution: The observation 'Small tissue piece used to start a culture' points to Explant; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Quality improvement belongs with 'Improvement of protein, oil, vitamin or mineral content', whereas the tested description is 'Small tissue piece used to start a culture'. C: Methanobacterium belongs with 'Methanogen commonly found in cattle rumen', whereas the tested description is 'Small tissue piece used to start a culture'. D: Abiotic stress resistance belongs with 'Development of crops tolerating drought or salinity', whereas the tested description is 'Small tissue piece used to start a culture'.

BIO-12-CH11-0065 | Hard | Plant breeding process**Correct answer: A. Testing parents for desirable characters - Biofortification**

Solution: 'Testing parents for desirable characters' should be paired with Evaluation and selection, not Biofortification. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Symbiotic nitrogen fixer in legume roots - Rhizobium. C: This is a valid association: Methanogen commonly found in cattle rumen - Methanobacterium. D: This is a valid association: Microbe producing citric acid - Aspergillus niger.

BIO-12-CH11-0066 | Easy | Plant breeding process**Correct answer: D. Testing parents for desirable characters**

Solution: Evaluation and selection represents 'Testing parents for desirable characters'. Each alternative describes a different term or result.

Distractor review: A: 'Methanogen commonly found in cattle rumen' identifies Methanobacterium, not Evaluation and selection. B: 'Crossing genetically distinct selected parents' identifies Hybridization, not Evaluation and selection. C: 'Microbe producing citric acid' identifies Aspergillus niger, not Evaluation and selection.

BIO-12-CH11-0067 | Hard | Plant breeding process

Correct answer: A. Testing parents for desirable characters - Evaluation and selection; Breeding crops for improved nutritional quality - Biofortification

Solution: Both parts of 'Testing parents for desirable characters - Evaluation and selection; Breeding crops for improved nutritional quality - Biofortification' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Fungal antagonist used against plant pathogens' requires Trichoderma, and 'Improvement of protein, oil, vitamin or mineral content' requires Quality improvement. C: The answers have been interchanged: 'Microbe producing citric acid' requires Aspergillus niger, and 'Symbiotic nitrogen fixer in legume roots' requires Rhizobium. D: The answers have been interchanged: 'Methanogen commonly found in cattle rumen' requires Methanobacterium, and 'Small tissue piece used to start a culture' requires Explant.

BIO-12-CH11-0068 | Hard | Animal husbandry

Correct answer: D. Management of milk-producing animals - Dairy farming; Improvement of protein, oil, vitamin or mineral content - Quality improvement

Solution: Both parts of 'Management of milk-producing animals - Dairy farming; Improvement of protein, oil, vitamin or mineral content - Quality improvement' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Fungal antagonist used against plant pathogens' requires Trichoderma, and 'Symbiotic nitrogen fixer in legume roots' requires Rhizobium. B: The answers have been interchanged: 'Association of fungi with plant roots improving mineral uptake' requires Mycorrhiza, and 'Rapid clonal production through tissue culture' requires Micropropagation. C: The answers have been interchanged: 'Rapid clonal production through tissue culture' requires Micropropagation, and 'Bacterium producing lactic acid' requires Lactobacillus.

BIO-12-CH11-0069 | Easy | Biofortification and SCP

Correct answer: A. Alga widely used as single-cell protein

Solution: Spirulina represents 'Alga widely used as single-cell protein'. Each alternative describes a different term or result.

Distractor review: B: 'Indicator of organic pollution reduced during secondary treatment' identifies Biochemical oxygen demand, not Spirulina. C: 'Rearing of honey bees' identifies Apiculture, not Spirulina. D: 'Management of milk-producing animals' identifies Dairy farming, not Spirulina.

BIO-12-CH11-0070 | Hard | Tissue culture

Correct answer: C. Unorganized mass of cultured cells - Single-cell protein

Solution: 'Unorganized mass of cultured cells' should be paired with Callus, not Single-cell protein. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Symbiotic nitrogen fixer in legume roots - Rhizobium. B: This is a valid association: Collection of diverse cultivated and wild forms - Germplasm collection. D: This is a valid association: Testing a new line at several places before release - Multi-location evaluation.

BIO-12-CH11-0071 | Hard | Sewage and biogas

Correct answer: B. Indicator of organic pollution reduced during secondary treatment - Evaluation and selection

Solution: 'Indicator of organic pollution reduced during secondary treatment' should be paired with Biochemical oxygen demand, not Evaluation and selection. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Fungus associated with penicillin production - Penicillium. C: This is a valid association: Scientific management of farm animals - Animal husbandry. D: This is a valid association: Crossing genetically distinct selected parents - Hybridization.

BIO-12-CH11-0072 | Medium | Biofortification and SCP

Correct answer: D. Single-cell protein

Solution: The observation 'Microbial biomass used as protein-rich food or feed' points to Single-cell protein; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Disease resistance belongs with 'Breeding for resistance to pathogens', whereas the tested description is 'Microbial biomass used as protein-rich food or feed'. B: Saccharomyces cerevisiae belongs with 'Yeast used in ethanol production', whereas the tested description is 'Microbial biomass used as protein-rich food or feed'. C: Bacillus thuringiensis belongs with 'Bacterial biocontrol agent producing insecticidal crystal proteins', whereas the tested description is 'Microbial biomass used as protein-rich food or feed'.

BIO-12-CH11-0073 | Easy | Biofortification and SCP**Correct answer: A. Breeding crops for improved nutritional quality**

Solution: Biofortification represents 'Breeding crops for improved nutritional quality'. Each alternative describes a different term or result.

Distractor review: B: 'Breeding for resistance to pathogens' identifies Disease resistance, not Biofortification. C: 'Symbiotic nitrogen fixer in legume roots' identifies Rhizobium, not Biofortification. D: 'Alga widely used as single-cell protein' identifies Spirulina, not Biofortification.

BIO-12-CH11-0074 | Challenging | Plant breeding objectives**Correct answer: B. III and IV only**

Solution: Statement I is incorrect: 'Development of crop varieties with higher output' corresponds to Increased yield, not Quality improvement. Statement II is incorrect: 'Improvement of protein, oil, vitamin or mineral content' corresponds to Quality improvement, not Disease resistance. Statement III is correct because 'Breeding for resistance to pathogens' corresponds to Disease resistance. Statement IV is correct because 'Rapid clonal production through tissue culture' corresponds to Micropropagation. Therefore, the valid combination is III and IV only.

Distractor review: A: This combination includes false statement(s) I, II and omits true statement(s) III. C: This combination includes false statement(s) I, II and omits true statement(s) IV. D: This combination includes false statement(s) I and omits true statement(s) III, IV.

BIO-12-CH11-0075 | Hard | Biofortification and SCP**Correct answer: B. Microbial biomass used as protein-rich food or feed - Saccharomyces cerevisiae**

Solution: 'Microbial biomass used as protein-rich food or feed' should be paired with Single-cell protein, not Saccharomyces cerevisiae. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Methanogen commonly found in cattle rumen - Methanobacterium. C: This is a valid association: Crossing genetically distinct selected parents - Hybridization. D: This is a valid association: Rapid clonal production through tissue culture - Micropropagation.

BIO-12-CH11-0076 | Medium | Biocontrol and biofertilizers**Correct answer: D. Trichoderma**

Solution: The chapter links Fungal antagonist used against plant pathogens with Trichoderma, so Trichoderma completes the statement.

Distractor review: A: Explant belongs with 'Small tissue piece used to start a culture', whereas the tested description is 'Fungal antagonist used against plant pathogens'. B: Spirulina belongs with 'Alga widely used as single-cell protein', whereas the tested description is 'Fungal antagonist used against plant pathogens'. C: Mycorrhiza belongs with 'Association of fungi with plant roots improving mineral uptake', whereas the tested description is 'Fungal antagonist used against plant pathogens'.

BIO-12-CH11-0077 | Medium | Tissue culture**Correct answer: A. Callus**

Solution: The chapter links Unorganized mass of cultured cells with Callus, so Callus completes the statement.

Distractor review: B: Totipotency belongs with 'Capacity of a plant cell to regenerate a whole plant', whereas the tested description is 'Unorganized mass of cultured cells'. C: Single-cell protein belongs with 'Microbial biomass used as protein-rich food or feed', whereas the tested description is 'Unorganized mass of cultured cells'. D: Micropropagation belongs with 'Rapid clonal production through tissue culture', whereas the tested description is 'Unorganized mass of cultured cells'.

BIO-12-CH11-0078 | Easy | Industrial microbes**Correct answer: C. Fungus associated with penicillin production**

Solution: Penicillium represents 'Fungus associated with penicillin production'. Each alternative describes a different term or result.

Distractor review: A: 'Microbial aggregates in aeration tanks' identifies Flocs, not Penicillium. B: 'Yeast used in ethanol production' identifies Saccharomyces cerevisiae, not Penicillium. D: 'Microbe producing citric acid' identifies Aspergillus niger, not Penicillium.

BIO-12-CH11-0079 | Hard | Tissue culture**Correct answer: B. Capacity of a plant cell to regenerate a whole plant - Penicillium**

Solution: 'Capacity of a plant cell to regenerate a whole plant' should be paired with Totipotency, not Penicillium. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Breeding for resistance to pathogens - Disease resistance. C: This is a valid association: Collection of diverse cultivated and wild forms - Germplasm collection. D: This is a valid association: Fungal antagonist used against plant pathogens - Trichoderma.

BIO-12-CH11-0080 | Medium | Animal husbandry**Correct answer: C. Poultry farming**

Solution: The observation 'Rearing of birds for eggs and meat' points to Poultry farming; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Dairy farming belongs with 'Management of milk-producing animals', whereas the tested description is 'Rearing of birds for eggs and meat'. B: Animal husbandry belongs with 'Scientific management of farm animals', whereas the tested description is 'Rearing of birds for eggs and meat'. D: Evaluation and selection belongs with 'Testing parents for desirable characters', whereas the tested description is 'Rearing of birds for eggs and meat'.

BIO-12-CH11-0081 | Medium | Industrial microbes**Correct answer: D. Yeast used in ethanol production - *Saccharomyces cerevisiae***

Solution: The accurate entry is 'Yeast used in ethanol production - *Saccharomyces cerevisiae*'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Development of crops tolerating drought or salinity' should be paired with Abiotic stress resistance, not Biogas. B: 'Biofortified crop with twice the amount of lysine and tryptophan' should be paired with Fortified maize, not Abiotic stress resistance. C: 'Microbial biomass used as protein-rich food or feed' should be paired with Single-cell protein, not Animal husbandry.

BIO-12-CH11-0082 | Challenging | Industrial microbes**Correct answer: A. II only**

Solution: Statement I is incorrect: 'Fungus associated with penicillin production' corresponds to *Penicillium*, not *Aspergillus niger*. Statement II is correct because 'Microbe producing citric acid' corresponds to *Aspergillus niger*. Statement III is incorrect: 'Yeast used in ethanol production' corresponds to *Saccharomyces cerevisiae*, not *Methanobacterium*. Statement IV is incorrect: 'Methanogen commonly found in cattle rumen' corresponds to *Methanobacterium*, not *Penicillium*. Therefore, the valid combination is II only.

Distractor review: B: This combination includes false statement(s) III and omits true statement(s) II. C: This combination includes false statement(s) III, IV. D: This combination includes false statement(s) III, IV and omits true statement(s) II.

BIO-12-CH11-0083 | Medium | Plant breeding objectives**Correct answer: C. Disease resistance**

Solution: The chapter links Breeding for resistance to pathogens with Disease resistance, so Disease resistance completes the statement.

Distractor review: A: *Methanobacterium* belongs with 'Methanogen commonly found in cattle rumen', whereas the tested description is 'Breeding for resistance to pathogens'. B: *Bacillus thuringiensis* belongs with 'Bacterial biocontrol agent producing insecticidal crystal proteins', whereas the tested description is 'Breeding for resistance to pathogens'. D: Increased yield belongs with 'Development of crop varieties with higher output', whereas the tested description is 'Breeding for resistance to pathogens'.

BIO-12-CH11-0084 | Hard | Biocontrol and biofertilizers**Correct answer: B. Association of fungi with plant roots improving mineral uptake - Apiculture**

Solution: 'Association of fungi with plant roots improving mineral uptake' should be paired with Mycorrhiza, not Apiculture. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Biofortified crop with twice the amount of lysine and tryptophan - Fortified maize. C: This is a valid association: Collection of diverse cultivated and wild forms - Germplasm collection. D: This is a valid association: Rearing of birds for eggs and meat - Poultry farming.

BIO-12-CH11-0085 | Challenging | Sewage and biogas**Correct answer: C. II and IV only**

Solution: Statement I is incorrect: 'Indicator of organic pollution reduced during secondary treatment' corresponds to Biochemical oxygen demand, not Flocs. Statement II is correct because 'Microbial aggregates in aeration tanks' corresponds to Flocs. Statement III is incorrect: 'Anaerobic digesters produce methane-rich' corresponds to Biogas, not Totipotency. Statement IV is correct because 'Capacity of a plant cell to regenerate a whole plant' corresponds to Totipotency. Therefore, the valid combination is II and IV only.

Distractor review: A: This combination includes false statement(s) I, III and omits true statement(s) II, IV. B: This combination includes false statement(s) III and omits true statement(s) II. D: This combination includes false statement(s) I and omits true statement(s) II.

BIO-12-CH11-0086 | Medium | Sewage and biogas**Correct answer: B. Microbial aggregates in aeration tanks - Flocs**

Solution: The accurate entry is 'Microbial aggregates in aeration tanks - Flocs'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Crossing genetically distinct selected parents' should be paired with Hybridization, not Disease resistance. C: 'Fungus associated with penicillin production' should be paired with Penicillium, not Flocs. D: 'Bacterium producing lactic acid' should be paired with Lactobacillus, not Rhizobium.

BIO-12-CH11-0087 | Hard | Plant breeding process**Correct answer: C. Crossing genetically distinct selected parents - Fortified maize**

Solution: 'Crossing genetically distinct selected parents' should be paired with Hybridization, not Fortified maize. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Microbial biomass used as protein-rich food or feed - Single-cell protein. B: This is a valid association: Development of crop varieties with higher output - Increased yield. D: This is a valid association: Rearing of birds for eggs and meat - Poultry farming.

BIO-12-CH11-0088 | Medium | Industrial microbes**Correct answer: A. Penicillium**

Solution: The chapter links Fungus associated with penicillin production with Penicillium, so Penicillium completes the statement.

Distractor review: B: Apiculture belongs with 'Rearing of honey bees', whereas the tested description is 'Fungus associated with penicillin production'. C: Flocs belongs with 'Microbial aggregates in aeration tanks', whereas the tested description is 'Fungus associated with penicillin production'. D: Germplasm collection belongs with 'Collection of diverse cultivated and wild forms', whereas the tested description is 'Fungus associated with penicillin production'.

BIO-12-CH11-0089 | Medium | Plant breeding objectives**Correct answer: B. Improvement of protein, oil, vitamin or mineral content - Quality improvement**

Solution: The accurate entry is 'Improvement of protein, oil, vitamin or mineral content - Quality improvement'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Fungal antagonist used against plant pathogens' should be paired with Trichoderma, not Callus. C: 'Breeding crops for improved nutritional quality' should be paired with Biofortification, not Aspergillus niger. D: 'Unorganized mass of cultured cells' should be paired with Callus, not Bacillus thuringiensis.

BIO-12-CH11-0090 | Hard | Industrial microbes**Correct answer: A. Bacterium producing lactic acid - Lactobacillus; Rearing of honey bees - Apiculture**

Solution: Both parts of 'Bacterium producing lactic acid - Lactobacillus; Rearing of honey bees - Apiculture' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Development of crop varieties with higher output' requires Increased yield, and 'Association of fungi with plant roots improving mineral uptake' requires Mycorrhiza. C: The answers have been interchanged: 'Alga widely used as single-cell protein' requires Spirulina, and 'Development of crop varieties with higher output' requires Increased yield. D: The answers have been interchanged: 'Fungal antagonist used against plant pathogens' requires Trichoderma, and 'Management of milk-producing animals' requires Dairy farming.

BIO-12-CH11-0091 | Easy | Tissue culture**Correct answer: B. Totipotency**

Solution: 'Capacity of a plant cell to regenerate a whole plant' is the defining description for Totipotency. The other choices belong to different chapter concepts.

Distractor review: A: Explant belongs with 'Small tissue piece used to start a culture', whereas the tested description is 'Capacity of a plant cell to regenerate a whole plant'. C: Animal husbandry belongs with 'Scientific management of farm animals', whereas the tested description is 'Capacity of a plant cell to regenerate a whole plant'. D: Trichoderma belongs with 'Fungal antagonist used against plant pathogens', whereas the tested description is 'Capacity of a plant cell to regenerate a whole plant'.

BIO-12-CH11-0092 | Medium | Biofortification and SCP**Correct answer: D. Spirulina**

Solution: The observation 'Alga widely used as single-cell protein' points to Spirulina; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Apiculture belongs with 'Rearing of honey bees', whereas the tested description is 'Alga widely used as single-cell protein'. B: Dairy farming belongs with 'Management of milk-producing animals', whereas the tested description is 'Alga widely used as single-cell protein'. C: Fortified maize belongs with 'Biofortified crop with twice the amount of lysine and tryptophan', whereas the tested description is 'Alga widely used as single-cell protein'.

BIO-12-CH11-0093 | Hard | Biocontrol and biofertilizers**Correct answer: C. Symbiotic nitrogen fixer in legume roots - Rhizobium; Bacterium producing lactic acid - Lactobacillus**

Solution: Both parts of 'Symbiotic nitrogen fixer in legume roots - Rhizobium; Bacterium producing lactic acid - Lactobacillus' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Small tissue piece used to start a culture' requires Explant, and 'Rapid clonal production through tissue culture' requires Micropropagation. B: The answers have been interchanged: 'Development of crop varieties with higher output' requires Increased yield, and 'Crossing genetically distinct selected parents' requires Hybridization. D: The answers have been interchanged: 'Testing a new line at several places before release' requires Multi-location evaluation, and 'Microbe producing citric acid' requires Aspergillus niger.

BIO-12-CH11-0094 | Medium | Plant breeding objectives**Correct answer: A. Increased yield**

Solution: The chapter links Development of crop varieties with higher output with Increased yield, so Increased yield completes the statement.

Distractor review: B: Rhizobium belongs with 'Symbiotic nitrogen fixer in legume roots', whereas the tested description is 'Development of crop varieties with higher output'. C: Disease resistance belongs with 'Breeding for resistance to pathogens', whereas the tested description is 'Development of crop varieties with higher output'. D: Mycorrhiza belongs with 'Association of fungi with plant roots improving mineral uptake', whereas the tested description is 'Development of crop varieties with higher output'.

BIO-12-CH11-0095 | Challenging | Biofortification and SCP**Correct answer: D. I only**

Solution: Statement I is correct because 'Breeding crops for improved nutritional quality' corresponds to Biofortification. Statement II is incorrect: 'Biofortified crop with twice the amount of lysine and tryptophan' corresponds to Fortified maize, not Spirulina. Statement III is incorrect: 'Alga widely used as single-cell protein' corresponds to Spirulina, not Totipotency. Statement IV is incorrect: 'Capacity of a plant cell to regenerate a whole plant' corresponds to Totipotency, not Biofortification. Therefore, the valid combination is I only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) I. B: This combination includes false statement(s) II and omits true statement(s) I. C: This combination includes false statement(s) II, IV and omits true statement(s) I.

BIO-12-CH11-0096 | Challenging | Animal husbandry**Correct answer: D. II and III only**

Solution: Statement I is incorrect: 'Rearing of birds for eggs and meat' corresponds to Poultry farming, not Dairy farming. Statement II is correct because 'Management of milk-producing animals' corresponds to Dairy farming. Statement III is correct because 'Rearing of honey bees' corresponds to Apiculture. Statement IV is incorrect: 'Development of crop varieties with higher output' corresponds to Increased yield, not Poultry farming. Therefore, the valid combination is II and III only.

Distractor review: A: This combination includes false statement(s) I, IV and omits true statement(s) III. B: This combination includes false statement(s) I and omits true statement(s) II. C: This combination includes false statement(s) I, IV and omits true statement(s) II.

BIO-12-CH11-0097 | Medium | Animal husbandry**Correct answer: A. Dairy farming**

Solution: The chapter links Management of milk-producing animals with Dairy farming, so Dairy farming completes the statement.

Distractor review: B: Lactobacillus belongs with 'Bacterium producing lactic acid', whereas the tested description is 'Management of milk-producing animals'. C: Micropropagation belongs with 'Rapid clonal production through tissue culture', whereas the tested description is 'Management of milk-producing animals'. D: Apiculture belongs with 'Rearing of honey bees', whereas the tested description is 'Management of milk-producing animals'.

BIO-12-CH11-0098 | Easy | Animal husbandry

Correct answer: B. Management of milk-producing animals

Solution: Dairy farming represents 'Management of milk-producing animals'. Each alternative describes a different term or result.

Distractor review: A: 'Rearing of honey bees' identifies Apiculture, not Dairy farming. C: 'Improvement of protein, oil, vitamin or mineral content' identifies Quality improvement, not Dairy farming. D: 'Bacterium producing lactic acid' identifies Lactobacillus, not Dairy farming.

BIO-12-CH11-0099 | Easy | Sewage and biogas

Correct answer: D. Methanogen commonly found in cattle rumen

Solution: Methanobacterium represents 'Methanogen commonly found in cattle rumen'. Each alternative describes a different term or result.

Distractor review: A: 'Anaerobic digesters produce methane-rich' identifies Biogas, not Methanobacterium. B: 'Development of crops tolerating drought or salinity' identifies Abiotic stress resistance, not Methanobacterium. C: 'Rearing of honey bees' identifies Apiculture, not Methanobacterium.

BIO-12-CH11-0100 | Medium | Animal husbandry

Correct answer: C. Apiculture

Solution: The observation 'Rearing of honey bees' points to Apiculture; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Animal husbandry belongs with 'Scientific management of farm animals', whereas the tested description is 'Rearing of honey bees'. B: Quality improvement belongs with 'Improvement of protein, oil, vitamin or mineral content', whereas the tested description is 'Rearing of honey bees'. D: Single-cell protein belongs with 'Microbial biomass used as protein-rich food or feed', whereas the tested description is 'Rearing of honey bees'.

Standard 12 | Chapter 12

Biotechnology

BIO-12-CH12-0001 | Easy | Polymerase chain reaction

Correct answer: C. Denaturation

Solution: High-temperature denaturation disrupts hydrogen bonding and separates template strands.

Distractor review: A: The alternative 'Annealing' conflicts with the labelled relationship, scale or mechanism in the figure. High-temperature denaturation disrupts hydrogen bonding and separates template strands. B: The alternative 'Extension' conflicts with the labelled relationship, scale or mechanism in the figure. High-temperature denaturation disrupts hydrogen bonding and separates template strands. D: The alternative 'Ligation into a chromosome' conflicts with the labelled relationship, scale or mechanism in the figure. High-temperature denaturation disrupts hydrogen bonding and separates template strands.

BIO-12-CH12-0002 | Easy | rDNA methodology

Correct answer: C. Production of many identical copies from one transformed cell

Solution: Cloning represents 'Production of many identical copies from one transformed cell'. Each alternative describes a different term or result.

Distractor review: A: 'Enzymes cutting DNA at specific recognition sequences' identifies Restriction endonucleases, not Cloning. B: 'Bt crop protection targets' identifies Specific insect pests, not Cloning. D: 'Separation of DNA fragments according to size' identifies Gel electrophoresis, not Cloning.

BIO-12-CH12-0003 | Hard | Polymerase chain reaction

Correct answer: B. Approximately 2 to the power n

Solution: Under ideal 100% efficiency, each cycle doubles target copy number, giving about 2^n molecules.

Distractor review: A: The alternative 'Exactly n' conflicts with the labelled relationship, scale or mechanism in the figure. Under ideal 100% efficiency, each cycle doubles target copy number, giving about 2^n molecules. C: The alternative 'Approximately n squared' conflicts with the labelled relationship, scale or mechanism in the figure. Under ideal 100% efficiency, each cycle doubles target copy number, giving about 2^n molecules. D: The alternative 'Always two molecules' conflicts with the labelled relationship, scale or mechanism in the figure. Under ideal 100% efficiency, each cycle doubles target copy number, giving about 2^n molecules.

BIO-12-CH12-0004 | Medium | Simplified plasmid cloning vector

Correct answer: C. Origin of replication

Solution: The origin of replication permits autonomous plasmid replication in a suitable host.

Distractor review: A: The alternative 'Inserted gene alone' conflicts with the labelled relationship, scale or mechanism in the figure. The origin of replication permits autonomous plasmid replication in a suitable host. B: The alternative 'Restriction enzyme protein' conflicts with the labelled relationship, scale or mechanism in the figure. The origin of replication permits autonomous plasmid replication in a suitable host. D: The alternative 'Antibody-binding site' conflicts with the labelled relationship, scale or mechanism in the figure. The origin of replication permits autonomous plasmid replication in a suitable host.

BIO-12-CH12-0005 | Easy | Bioethics and ownership

Correct answer: D. Unfair commercial use of biological resources or traditional knowledge

Solution: Biopiracy represents 'Unfair commercial use of biological resources or traditional knowledge'. Each alternative describes a different term or result.

Distractor review: A: 'Ethical concern about altered organisms entering nature' identifies Environmental release of GMOs, not Biopiracy. B: 'PCR step in which polymerase synthesizes DNA' identifies Extension, not Biopiracy. C: 'Small circular bacterial DNA commonly used as a vector' identifies Plasmid, not Biopiracy.

BIO-12-CH12-0006 | Medium | Medical applications

Correct answer: A. Human insulin

Solution: The observation 'First therapeutic product made using recombinant DNA technology' points to Human insulin; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: PCR belongs with 'Amplification of a chosen DNA segment in vitro', whereas the tested description is 'First therapeutic product made using recombinant DNA technology'. C: PCR-based diagnosis belongs with 'Early molecular diagnosis using amplification', whereas the tested description is 'First therapeutic product made using recombinant DNA technology'. D: Gene therapy belongs with 'Treatment by introducing a functional gene', whereas the tested description is 'First therapeutic product made using recombinant DNA technology'.

BIO-12-CH12-0007 | Medium | Bioethics and ownership**Correct answer: A. Environmental release of GMOs**

Solution: The chapter links Ethical concern about altered organisms entering nature with Environmental release of GMOs, so Environmental release of GMOs completes the statement.

Distractor review: B: DNA ligase belongs with 'Enzyme sealing phosphodiester bonds between DNA fragments', whereas the tested description is 'Ethical concern about altered organisms entering nature'. C: Biopatent belongs with 'Legal protection for a novel biotechnological invention', whereas the tested description is 'Ethical concern about altered organisms entering nature'. D: Extension belongs with 'PCR step in which polymerase synthesizes DNA', whereas the tested description is 'Ethical concern about altered organisms entering nature'.

BIO-12-CH12-0008 | Hard | Restriction and joining enzymes**Correct answer: B. Enzymes cutting DNA at specific recognition sequences - Restriction endonucleases; Amplification of a chosen DNA segment in vitro - PCR**

Solution: Both parts of 'Enzymes cutting DNA at specific recognition sequences - Restriction endonucleases; Amplification of a chosen DNA segment in vitro - PCR' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Gene allowing identification of transformed cells' requires Selectable marker, and 'Antibody produced from a single clone of cells' requires Monoclonal antibody. C: The answers have been interchanged: 'Site permitting vector replication in a host' requires Origin of replication, and 'Bt crop protection targets' requires Specific insect pests. D: The answers have been interchanged: 'Antibody produced from a single clone of cells' requires Monoclonal antibody, and 'Site permitting vector replication in a host' requires Origin of replication.

BIO-12-CH12-0009 | Easy | Core principles**Correct answer: A. Maintenance of sterile conditions for industrial microbial growth**

Solution: Bioprocess engineering represents 'Maintenance of sterile conditions for industrial microbial growth'. Each alternative describes a different term or result.

Distractor review: B: 'DNA formed by joining sequences from different sources' identifies Recombinant DNA, not Bioprocess engineering. C: 'Organism carrying and expressing an introduced gene' identifies Transgenic organism, not Bioprocess engineering. D: 'Site permitting vector replication in a host' identifies Origin of replication, not Bioprocess engineering.

BIO-12-CH12-0010 | Medium | Restriction and joining enzymes**Correct answer: D. Restriction endonucleases**

Solution: The observation 'Enzymes cutting DNA at specific recognition sequences' points to Restriction endonucleases; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Selectable marker belongs with 'Gene allowing identification of transformed cells', whereas the tested description is 'Enzymes cutting DNA at specific recognition sequences'. B: PCR belongs with 'Amplification of a chosen DNA segment in vitro', whereas the tested description is 'Enzymes cutting DNA at specific recognition sequences'. C: Transformation belongs with 'Introduction of recombinant DNA into a host cell', whereas the tested description is 'Enzymes cutting DNA at specific recognition sequences'.

BIO-12-CH12-0011 | Medium | Core principles**Correct answer: B. DNA formed by joining sequences from different sources - Recombinant DNA**

Solution: The accurate entry is 'DNA formed by joining sequences from different sources - Recombinant DNA'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Treatment by introducing a functional gene' should be paired with Gene therapy, not Selectable marker. C: 'Early molecular diagnosis using amplification' should be paired with PCR-based diagnosis, not Recombinant DNA. D: 'Vector region containing several unique restriction sites' should be paired with Multiple cloning site, not Biopatent.

BIO-12-CH12-0012 | Challenging | Simplified plasmid cloning vector**Correct answer: D. Insertional inactivation changes the marker phenotype of recombinant colonies**

Solution: Disruption of a marker produces a detectable phenotype that distinguishes many recombinant from nonrecombinant colonies.

Distractor review: A: The alternative 'The insert makes every colony identical without testing' conflicts with the labelled relationship, scale or mechanism in the figure. Disruption of a marker produces a detectable phenotype that distinguishes many recombinant from nonrecombinant colonies. B: The alternative 'The origin of replication is no longer needed' conflicts with the labelled relationship, scale or mechanism in the figure. Disruption of a marker produces a detectable phenotype that distinguishes many recombinant from nonrecombinant colonies. C: The alternative 'Nonrecombinant plasmids lose all DNA' conflicts with the labelled relationship, scale or mechanism in the figure. Disruption of a marker produces a detectable phenotype that distinguishes many recombinant from nonrecombinant colonies.

BIO-12-CH12-0013 | Medium | Cloning vectors**Correct answer: D. Vector region containing several unique restriction sites - Multiple cloning site**

Solution: The accurate entry is 'Vector region containing several unique restriction sites - Multiple cloning site'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Separation of DNA fragments according to size' should be paired with Gel electrophoresis, not *Bacillus thuringiensis*. B: 'Legal protection for a novel biotechnological invention' should be paired with Biopatent, not Gel electrophoresis. C: 'Restriction sequence reading the same in opposite directions' should be paired with Palindromic sequence, not Multiple cloning site.

BIO-12-CH12-0014 | Easy | PCR and analysis**Correct answer: B. PCR step separating DNA strands**

Solution: Denaturation represents 'PCR step separating DNA strands'. Each alternative describes a different term or result.

Distractor review: A: 'PCR step in which polymerase synthesizes DNA' identifies Extension, not Denaturation. C: 'Nutritionally enhanced transgenic crop example' identifies Golden rice, not Denaturation. D: 'Enzymes cutting DNA at specific recognition sequences' identifies Restriction endonucleases, not Denaturation.

BIO-12-CH12-0015 | Hard | Bioethics and ownership**Correct answer: C. Unfair commercial use of biological resources or traditional knowledge - Biopiracy; Small circular bacterial DNA commonly used as a vector - Plasmid**

Solution: Both parts of 'Unfair commercial use of biological resources or traditional knowledge - Biopiracy; Small circular bacterial DNA commonly used as a vector - Plasmid' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Enzymes cutting DNA at specific recognition sequences' requires Restriction endonucleases, and 'DNA formed by joining sequences from different sources' requires Recombinant DNA. B: The answers have been interchanged: 'PCR step in which polymerase synthesizes DNA' requires Extension, and 'Transgenic plant known for insect resistance' requires Bt cotton. D: The answers have been interchanged: 'DNA formed by joining sequences from different sources' requires Recombinant DNA, and 'Enzyme sealing phosphodiester bonds between DNA fragments' requires DNA ligase.

BIO-12-CH12-0016 | Easy | PCR and analysis**Correct answer: B. PCR step in which polymerase synthesizes DNA**

Solution: Extension represents 'PCR step in which polymerase synthesizes DNA'. Each alternative describes a different term or result.

Distractor review: A: 'Single-stranded ends produced by staggered restriction cuts' identifies Sticky ends, not Extension. C: 'Separation of DNA fragments according to size' identifies Gel electrophoresis, not Extension. D: 'Vector region containing several unique restriction sites' identifies Multiple cloning site, not Extension.

BIO-12-CH12-0017 | Challenging | PCR and analysis**Correct answer: C. III and IV only**

Solution: Statement I is incorrect: 'PCR step allowing primers to bind' corresponds to Annealing, not Extension. Statement II is incorrect: 'PCR step in which polymerase synthesizes DNA' corresponds to Extension, not Denaturation. Statement III is correct because 'PCR step separating DNA strands' corresponds to Denaturation. Statement IV is correct because 'Early molecular diagnosis using amplification' corresponds to PCR-based diagnosis. Therefore, the valid combination is III and IV only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) III, IV. B: This combination omits true statement(s) III. D: This combination includes false statement(s) II and omits true statement(s) IV.

BIO-12-CH12-0018 | Medium | Medical applications**Correct answer: D. Early molecular diagnosis using amplification - PCR-based diagnosis**

Solution: The accurate entry is 'Early molecular diagnosis using amplification - PCR-based diagnosis'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Bt toxin gene source' should be paired with *Bacillus thuringiensis*, not Bt cotton. B: 'DNA formed by joining sequences from different sources' should be paired with Recombinant DNA, not PCR-based diagnosis. C: 'Small circular bacterial DNA commonly used as a vector' should be paired with Plasmid, not Recombinant DNA.

BIO-12-CH12-0019 | Medium | Cloning vectors**Correct answer: C. Plasmid**

Solution: The observation 'Small circular bacterial DNA commonly used as a vector' points to Plasmid; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Origin of replication belongs with 'Site permitting vector replication in a host', whereas the tested description is 'Small circular bacterial DNA commonly used as a vector'. B: Transgenic organism belongs with 'Organism carrying and expressing an introduced gene', whereas the tested description is 'Small circular bacterial DNA commonly used as a vector'. D: Bt cotton belongs with 'Transgenic plant known for insect resistance', whereas the tested description is 'Small circular bacterial DNA commonly used as a vector'.

BIO-12-CH12-0020 | Hard | PCR and analysis**Correct answer: B. PCR step allowing primers to bind - Annealing; Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase**

Solution: Both parts of 'PCR step allowing primers to bind - Annealing; Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Bt toxin gene source' requires *Bacillus thuringiensis*, and 'Organism carrying and expressing an introduced gene' requires Transgenic organism. C: The answers have been interchanged: 'Indian committee that decides validity and safety of GMO research' requires GEAC, and 'Site permitting vector replication in a host' requires Origin of replication. D: The answers have been interchanged: 'Site permitting vector replication in a host' requires Origin of replication, and 'Antibody produced from a single clone of cells' requires Monoclonal antibody.

BIO-12-CH12-0021 | Medium | Agricultural applications**Correct answer: D. *Bacillus thuringiensis***

Solution: The observation 'Bt toxin gene source' points to *Bacillus thuringiensis*; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Gene therapy belongs with 'Treatment by introducing a functional gene', whereas the tested description is 'Bt toxin gene source'. B: Sticky ends belongs with 'Single-stranded ends produced by staggered restriction cuts', whereas the tested description is 'Bt toxin gene source'. C: Golden rice belongs with 'Nutritionally enhanced transgenic crop example', whereas the tested description is 'Bt toxin gene source'.

BIO-12-CH12-0022 | Easy | Restriction and joining enzymes**Correct answer: D. Palindromic sequence**

Solution: 'Restriction sequence reading the same in opposite directions' is the defining description for Palindromic sequence. The other choices belong to different chapter concepts.

Distractor review: A: Transgenic organism belongs with 'Organism carrying and expressing an introduced gene', whereas the tested description is 'Restriction sequence reading the same in opposite directions'. B: Golden rice belongs with 'Nutritionally enhanced transgenic crop example', whereas the tested description is 'Restriction sequence reading the same in opposite directions'. C: Monoclonal antibody belongs with 'Antibody produced from a single clone of cells', whereas the tested description is 'Restriction sequence reading the same in opposite directions'.

BIO-12-CH12-0023 | Challenging | Agricultural applications**Correct answer: C. II only**

Solution: Statement I is incorrect: 'Bt toxin gene source' corresponds to *Bacillus thuringiensis*, not Golden rice. Statement II is correct because 'Nutritionally enhanced transgenic crop example' corresponds to Golden rice. Statement III is incorrect: 'Transgenic plant known for insect resistance' corresponds to Bt cotton, not Bioprocess engineering. Statement IV is incorrect: 'Maintenance of sterile conditions for industrial microbial growth' corresponds to Bioprocess engineering, not *Bacillus thuringiensis*. Therefore, the valid combination is II only.

Distractor review: A: This combination includes false statement(s) I, III and omits true statement(s) II. B: This combination includes false statement(s) IV. D: This combination includes false statement(s) I, III.

BIO-12-CH12-0024 | Challenging | Medical applications**Correct answer: A. II and IV only**

Solution: Statement I is incorrect: 'Early molecular diagnosis using amplification' corresponds to PCR-based diagnosis, not Human insulin. Statement II is correct because 'First therapeutic product made using recombinant DNA technology' corresponds to Human insulin. Statement III is incorrect: 'Treatment by introducing a functional gene' corresponds to Gene therapy, not Golden rice. Statement IV is correct because 'Nutritionally enhanced transgenic crop example' corresponds to Golden rice. Therefore, the valid combination is II and IV only.

Distractor review: B: This combination includes false statement(s) III and omits true statement(s) II. C: This combination includes false statement(s) I, III and omits true statement(s) II. D: This combination omits true statement(s) II.

BIO-12-CH12-0025 | Medium | Bioethics and ownership**Correct answer: A. Biopatent**

Solution: The observation 'Legal protection for a novel biotechnological invention' points to Biopatent; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Environmental release of GMOs belongs with 'Ethical concern about altered organisms entering nature', whereas the tested description is 'Legal protection for a novel biotechnological invention'. C: Biopiracy belongs with 'Unfair commercial use of biological resources or traditional knowledge', whereas the tested description is 'Legal protection for a novel biotechnological invention'. D: Bt cotton belongs with 'Transgenic plant known for insect resistance', whereas the tested description is 'Legal protection for a novel biotechnological invention'.

BIO-12-CH12-0026 | Easy | Medical applications**Correct answer: D. Antibody produced from a single clone of cells**

Solution: Monoclonal antibody represents 'Antibody produced from a single clone of cells'. Each alternative describes a different term or result.

Distractor review: A: 'Early molecular diagnosis using amplification' identifies PCR-based diagnosis, not Monoclonal antibody. B: 'Nutritionally enhanced transgenic crop example' identifies Golden rice, not Monoclonal antibody. C: 'Treatment by introducing a functional gene' identifies Gene therapy, not Monoclonal antibody.

BIO-12-CH12-0027 | Hard | Restriction and joining enzymes**Correct answer: B. Restriction sequence reading the same in opposite directions - Monoclonal antibody**

Solution: 'Restriction sequence reading the same in opposite directions' should be paired with Palindromic sequence, not Monoclonal antibody. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Organism carrying and expressing an introduced gene - Transgenic organism. C: This is a valid association: Bt crop protection targets - Specific insect pests. D: This is a valid association: PCR step in which polymerase synthesizes DNA - Extension.

BIO-12-CH12-0028 | Hard | PCR and analysis**Correct answer: C. PCR step in which polymerase synthesizes DNA - Extension; Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase**

Solution: Both parts of 'PCR step in which polymerase synthesizes DNA - Extension; Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Bt toxin gene source' requires *Bacillus thuringiensis*, and 'Legal protection for a novel biotechnological invention' requires Biopatent. B: The answers have been interchanged: 'Separation of DNA fragments according to size' requires Gel electrophoresis, and 'Single-stranded ends produced by staggered restriction cuts' requires Sticky ends. D: The answers have been interchanged: 'Amplification of a chosen DNA segment in vitro' requires PCR, and 'Vector region containing several unique restriction sites' requires Multiple cloning site.

BIO-12-CH12-0029 | Challenging | Medical applications**Correct answer: B. I only**

Solution: Statement I is correct because 'First therapeutic product made using recombinant DNA technology' corresponds to Human insulin. Statement II is incorrect: 'Treatment by introducing a functional gene' corresponds to Gene therapy, not Monoclonal antibody. Statement III is incorrect: 'Antibody produced from a single clone of cells' corresponds to Monoclonal antibody, not Plasmid. Statement IV is incorrect: 'Small circular bacterial DNA commonly used as a vector' corresponds to Plasmid, not Human insulin. Therefore, the valid combination is I only.

Distractor review: A: This combination includes false statement(s) II, III and omits true statement(s) I. C: This combination includes false statement(s) II, III. D: This combination includes false statement(s) III and omits true statement(s) I.

BIO-12-CH12-0030 | Medium | rDNA methodology**Correct answer: B. Cloning**

Solution: The observation 'Production of many identical copies from one transformed cell' points to Cloning; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Restriction endonucleases belongs with 'Enzymes cutting DNA at specific recognition sequences', whereas the tested description is 'Production of many identical copies from one transformed cell'. C: Transformation belongs with 'Introduction of recombinant DNA into a host cell', whereas the tested description is 'Production of many identical copies from one transformed cell'. D: Gel electrophoresis belongs with 'Separation of DNA fragments according to size', whereas the tested description is 'Production of many identical copies from one transformed cell'.

BIO-12-CH12-0031 | Hard | Core principles

Correct answer: A. Maintenance of sterile conditions for industrial microbial growth - Bioprocess engineering; Site permitting vector replication in a host - Origin of replication

Solution: Both parts of 'Maintenance of sterile conditions for industrial microbial growth - Bioprocess engineering; Site permitting vector replication in a host - Origin of replication' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Enzymes cutting DNA at specific recognition sequences' requires Restriction endonucleases, and 'Vector region containing several unique restriction sites' requires Multiple cloning site. C: The answers have been interchanged: 'Early molecular diagnosis using amplification' requires PCR-based diagnosis, and 'Gene allowing identification of transformed cells' requires Selectable marker. D: The answers have been interchanged: 'Ethical concern about altered organisms entering nature' requires Environmental release of GMOs, and 'Enzyme sealing phosphodiester bonds between DNA fragments' requires DNA ligase.

BIO-12-CH12-0032 | Medium | rDNA methodology

Correct answer: D. Gel electrophoresis

Solution: The chapter links Separation of DNA fragments according to size with Gel electrophoresis, so Gel electrophoresis completes the statement.

Distractor review: A: Human insulin belongs with 'First therapeutic product made using recombinant DNA technology', whereas the tested description is 'Separation of DNA fragments according to size'. B: PCR-based diagnosis belongs with 'Early molecular diagnosis using amplification', whereas the tested description is 'Separation of DNA fragments according to size'. C: PCR belongs with 'Amplification of a chosen DNA segment in vitro', whereas the tested description is 'Separation of DNA fragments according to size'.

BIO-12-CH12-0033 | Medium | Core principles

Correct answer: C. Recombinant DNA

Solution: The chapter links DNA formed by joining sequences from different sources with Recombinant DNA, so Recombinant DNA completes the statement.

Distractor review: A: Multiple cloning site belongs with 'Vector region containing several unique restriction sites', whereas the tested description is 'DNA formed by joining sequences from different sources'. B: Genetic engineering belongs with 'Deliberate alteration of genetic material', whereas the tested description is 'DNA formed by joining sequences from different sources'. D: Human insulin belongs with 'First therapeutic product made using recombinant DNA technology', whereas the tested description is 'DNA formed by joining sequences from different sources'.

BIO-12-CH12-0034 | Hard | Agricultural applications

Correct answer: A. Bt toxin gene source - Human insulin

Solution: 'Bt toxin gene source' should be paired with *Bacillus thuringiensis*, not Human insulin. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Production of many identical copies from one transformed cell - Cloning. C: This is a valid association: Small circular bacterial DNA commonly used as a vector - Plasmid. D: This is a valid association: Single-stranded ends produced by staggered restriction cuts - Sticky ends.

BIO-12-CH12-0035 | Medium | Restriction and joining enzymes

Correct answer: A. Sticky ends

Solution: The chapter links Single-stranded ends produced by staggered restriction cuts with Sticky ends, so Sticky ends completes the statement.

Distractor review: B: DNA ligase belongs with 'Enzyme sealing phosphodiester bonds between DNA fragments', whereas the tested description is 'Single-stranded ends produced by staggered restriction cuts'. C: PCR belongs with 'Amplification of a chosen DNA segment in vitro', whereas the tested description is 'Single-stranded ends produced by staggered restriction cuts'. D: Palindromic sequence belongs with 'Restriction sequence reading the same in opposite directions', whereas the tested description is 'Single-stranded ends produced by staggered restriction cuts'.

BIO-12-CH12-0036 | Easy | Medical applications**Correct answer: C. Monoclonal antibody**

Solution: 'Antibody produced from a single clone of cells' is the defining description for Monoclonal antibody. The other choices belong to different chapter concepts.

Distractor review: A: PCR-based diagnosis belongs with 'Early molecular diagnosis using amplification', whereas the tested description is 'Antibody produced from a single clone of cells'. B: Gene therapy belongs with 'Treatment by introducing a functional gene', whereas the tested description is 'Antibody produced from a single clone of cells'. D: Taq polymerase belongs with 'Thermostable polymerase commonly used in PCR', whereas the tested description is 'Antibody produced from a single clone of cells'.

BIO-12-CH12-0037 | Hard | rDNA methodology**Correct answer: D. Production of many identical copies from one transformed cell - Specific insect pests**

Solution: 'Production of many identical copies from one transformed cell' should be paired with Cloning, not Specific insect pests. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: PCR step allowing primers to bind - Annealing. B: This is a valid association: Legal protection for a novel biotechnological invention - Biopatent. C: This is a valid association: Single-stranded ends produced by staggered restriction cuts - Sticky ends.

BIO-12-CH12-0038 | Easy | Bioethics and ownership**Correct answer: C. GEAC**

Solution: 'Indian committee that decides validity and safety of GMO research' is the defining description for GEAC. The other choices belong to different chapter concepts.

Distractor review: A: Environmental release of GMOs belongs with 'Ethical concern about altered organisms entering nature', whereas the tested description is 'Indian committee that decides validity and safety of GMO research'. B: Gene therapy belongs with 'Treatment by introducing a functional gene', whereas the tested description is 'Indian committee that decides validity and safety of GMO research'. D: Specific insect pests belongs with 'Bt crop protection targets', whereas the tested description is 'Indian committee that decides validity and safety of GMO research'.

BIO-12-CH12-0039 | Hard | Medical applications**Correct answer: A. Antibody produced from a single clone of cells - Cloning**

Solution: 'Antibody produced from a single clone of cells' should be paired with Monoclonal antibody, not Cloning. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: DNA formed by joining sequences from different sources - Recombinant DNA. C: This is a valid association: Thermostable polymerase commonly used in PCR - Taq polymerase. D: This is a valid association: Nutritionally enhanced transgenic crop example - Golden rice.

BIO-12-CH12-0040 | Hard | Core principles**Correct answer: D. Deliberate alteration of genetic material - Bacillus thuringiensis**

Solution: 'Deliberate alteration of genetic material' should be paired with Genetic engineering, not Bacillus thuringiensis. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Ethical concern about altered organisms entering nature - Environmental release of GMOs. B: This is a valid association: Thermostable polymerase commonly used in PCR - Taq polymerase. C: This is a valid association: Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase.

BIO-12-CH12-0041 | Challenging | Agricultural applications**Correct answer: C. II and III only**

Solution: Statement I is incorrect: 'Nutritionally enhanced transgenic crop example' corresponds to Golden rice, not Bacillus thuringiensis. Statement II is correct because 'Bt toxin gene source' corresponds to Bacillus thuringiensis. Statement III is correct because 'Transgenic plant known for insect resistance' corresponds to Bt cotton. Statement IV is incorrect: 'Organism carrying and expressing an introduced gene' corresponds to Transgenic organism, not Golden rice. Therefore, the valid combination is II and III only.

Distractor review: A: This combination omits true statement(s) II. B: This combination includes false statement(s) IV. D: This combination includes false statement(s) I.

BIO-12-CH12-0042 | Medium | Bioethics and ownership**Correct answer: C. Biopatent**

Solution: The chapter links Legal protection for a novel biotechnological invention with Biopatent, so Biopatent completes the statement.

Distractor review: A: Sticky ends belongs with 'Single-stranded ends produced by staggered restriction cuts', whereas the tested description is 'Legal protection for a novel biotechnological invention'. B: Environmental release of GMOs belongs with 'Ethical concern about altered organisms entering nature', whereas the tested description is 'Legal protection for a novel biotechnological invention'. D: Transgenic organism belongs with 'Organism carrying and expressing an introduced gene', whereas the tested description is 'Legal protection for a novel biotechnological invention'.

BIO-12-CH12-0043 | Easy | rDNA methodology**Correct answer: B. Separation of DNA fragments according to size**

Solution: Gel electrophoresis represents 'Separation of DNA fragments according to size'. Each alternative describes a different term or result.

Distractor review: A: 'Treatment by introducing a functional gene' identifies Gene therapy, not Gel electrophoresis. C: 'Amplification of a chosen DNA segment in vitro' identifies PCR, not Gel electrophoresis. D: 'PCR step separating DNA strands' identifies Denaturation, not Gel electrophoresis.

BIO-12-CH12-0044 | Medium | Core principles**Correct answer: A. Organism carrying and expressing an introduced gene - Transgenic organism**

Solution: The accurate entry is 'Organism carrying and expressing an introduced gene - Transgenic organism'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Unfair commercial use of biological resources or traditional knowledge' should be paired with Biopiracy, not GEAC. C: 'Introduction of recombinant DNA into a host cell' should be paired with Transformation, not Biopatent. D: 'Indian committee that decides validity and safety of GMO research' should be paired with GEAC, not Selectable marker.

BIO-12-CH12-0045 | Easy | rDNA methodology**Correct answer: D. Cloning**

Solution: 'Production of many identical copies from one transformed cell' is the defining description for Cloning. The other choices belong to different chapter concepts.

Distractor review: A: Restriction endonucleases belongs with 'Enzymes cutting DNA at specific recognition sequences', whereas the tested description is 'Production of many identical copies from one transformed cell'. B: Biopatent belongs with 'Legal protection for a novel biotechnological invention', whereas the tested description is 'Production of many identical copies from one transformed cell'. C: Sticky ends belongs with 'Single-stranded ends produced by staggered restriction cuts', whereas the tested description is 'Production of many identical copies from one transformed cell'.

BIO-12-CH12-0046 | Easy | Medical applications**Correct answer: A. Early molecular diagnosis using amplification**

Solution: PCR-based diagnosis represents 'Early molecular diagnosis using amplification'. Each alternative describes a different term or result.

Distractor review: B: 'Transgenic plant known for insect resistance' identifies Bt cotton, not PCR-based diagnosis. C: 'Thermostable polymerase commonly used in PCR' identifies Taq polymerase, not PCR-based diagnosis. D: 'Bt toxin gene source' identifies Bacillus thuringiensis, not PCR-based diagnosis.

BIO-12-CH12-0047 | Medium | Restriction and joining enzymes**Correct answer: C. Single-stranded ends produced by staggered restriction cuts - Sticky ends**

Solution: The accurate entry is 'Single-stranded ends produced by staggered restriction cuts - Sticky ends'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'First therapeutic product made using recombinant DNA technology' should be paired with Human insulin, not Bioprocess engineering. B: 'Amplification of a chosen DNA segment in vitro' should be paired with PCR, not Recombinant DNA. D: 'Organism carrying and expressing an introduced gene' should be paired with Transgenic organism, not Human insulin.

BIO-12-CH12-0048 | Hard | Cloning vectors**Correct answer: C. Vector region containing several unique restriction sites - Biopiracy**

Solution: 'Vector region containing several unique restriction sites' should be paired with Multiple cloning site, not Biopiracy. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Legal protection for a novel biotechnological invention - Biopatent. B: This is a valid association: Antibody produced from a single clone of cells - Monoclonal antibody. D: This is a valid association: Separation of DNA fragments according to size - Gel electrophoresis.

BIO-12-CH12-0049 | Hard | Agricultural applications**Correct answer: B. Bt crop protection targets - Specific insect pests; Indian committee that decides validity and safety of GMO research - GEAC**

Solution: Both parts of 'Bt crop protection targets - Specific insect pests; Indian committee that decides validity and safety of GMO research - GEAC' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Ethical concern about altered organisms entering nature' requires Environmental release of GMOs, and 'DNA formed by joining sequences from different sources' requires Recombinant DNA. C: The answers have been interchanged: 'Organism carrying and expressing an introduced gene' requires Transgenic organism, and 'Ethical concern about altered organisms entering nature' requires Environmental release of GMOs. D: The answers have been interchanged: 'DNA formed by joining sequences from different sources' requires Recombinant DNA, and 'Separation of DNA fragments according to size' requires Gel electrophoresis.

BIO-12-CH12-0050 | Easy | Agricultural applications**Correct answer: D. Bt crop protection targets**

Solution: Specific insect pests represents 'Bt crop protection targets'. Each alternative describes a different term or result.

Distractor review: A: 'Organism carrying and expressing an introduced gene' identifies Transgenic organism, not Specific insect pests. B: 'Nutritionally enhanced transgenic crop example' identifies Golden rice, not Specific insect pests. C: 'Ethical concern about altered organisms entering nature' identifies Environmental release of GMOs, not Specific insect pests.

BIO-12-CH12-0051 | Hard | PCR and analysis**Correct answer: C. Thermostable polymerase commonly used in PCR - Taq polymerase; Unfair commercial use of biological resources or traditional knowledge - Biopiracy**

Solution: Both parts of 'Thermostable polymerase commonly used in PCR - Taq polymerase; Unfair commercial use of biological resources or traditional knowledge - Biopiracy' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Organism carrying and expressing an introduced gene' requires Transgenic organism, and 'Ethical concern about altered organisms entering nature' requires Environmental release of GMOs. B: The answers have been interchanged: 'Site permitting vector replication in a host' requires Origin of replication, and 'Small circular bacterial DNA commonly used as a vector' requires Plasmid. D: The answers have been interchanged: 'Single-stranded ends produced by staggered restriction cuts' requires Sticky ends, and 'Organism carrying and expressing an introduced gene' requires Transgenic organism.

BIO-12-CH12-0052 | Medium | PCR and analysis**Correct answer: B. Taq polymerase**

Solution: The observation 'Thermostable polymerase commonly used in PCR' points to Taq polymerase; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Extension belongs with 'PCR step in which polymerase synthesizes DNA', whereas the tested description is 'Thermostable polymerase commonly used in PCR'. C: Sticky ends belongs with 'Single-stranded ends produced by staggered restriction cuts', whereas the tested description is 'Thermostable polymerase commonly used in PCR'. D: Environmental release of GMOs belongs with 'Ethical concern about altered organisms entering nature', whereas the tested description is 'Thermostable polymerase commonly used in PCR'.

BIO-12-CH12-0053 | Medium | PCR and analysis**Correct answer: C. Denaturation**

Solution: The chapter links PCR step separating DNA strands with Denaturation, so Denaturation completes the statement.

Distractor review: A: Restriction endonucleases belongs with 'Enzymes cutting DNA at specific recognition sequences', whereas the tested description is 'PCR step separating DNA strands'. B: Golden rice belongs with 'Nutritionally enhanced transgenic crop example', whereas the tested description is 'PCR step separating DNA strands'. D: Human insulin belongs with 'First therapeutic product made using recombinant DNA technology', whereas the tested description is 'PCR step separating DNA strands'.

BIO-12-CH12-0054 | Medium | Restriction and joining enzymes**Correct answer: D. Enzymes cutting DNA at specific recognition sequences - Restriction endonucleases**

Solution: The accurate entry is 'Enzymes cutting DNA at specific recognition sequences - Restriction endonucleases'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Bt crop protection targets' should be paired with Specific insect pests, not Restriction endonucleases. B: 'Amplification of a chosen DNA segment in vitro' should be paired with PCR, not Selectable marker. C: 'Introduction of recombinant DNA into a host cell' should be paired with Transformation, not Monoclonal antibody.

BIO-12-CH12-0055 | Medium | Restriction and joining enzymes**Correct answer: C. Restriction sequence reading the same in opposite directions - Palindromic sequence**

Solution: The accurate entry is 'Restriction sequence reading the same in opposite directions - Palindromic sequence'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'PCR step in which polymerase synthesizes DNA' should be paired with Extension, not Palindromic sequence. B: 'Antibody produced from a single clone of cells' should be paired with Monoclonal antibody, not Specific insect pests. D: 'Deliberate alteration of genetic material' should be paired with Genetic engineering, not Bt cotton.

BIO-12-CH12-0056 | Medium | Bioethics and ownership**Correct answer: D. GEAC**

Solution: The observation 'Indian committee that decides validity and safety of GMO research' points to GEAC; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Specific insect pests belongs with 'Bt crop protection targets', whereas the tested description is 'Indian committee that decides validity and safety of GMO research'. B: Bacillus thuringiensis belongs with 'Bt toxin gene source', whereas the tested description is 'Indian committee that decides validity and safety of GMO research'. C: Selectable marker belongs with 'Gene allowing identification of transformed cells', whereas the tested description is 'Indian committee that decides validity and safety of GMO research'.

BIO-12-CH12-0057 | Hard | PCR and analysis**Correct answer: D. PCR step separating DNA strands - Denaturation; Maintenance of sterile conditions for industrial microbial growth - Bioprocess engineering**

Solution: Both parts of 'PCR step separating DNA strands - Denaturation; Maintenance of sterile conditions for industrial microbial growth - Bioprocess engineering' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Site permitting vector replication in a host' requires Origin of replication, and 'First therapeutic product made using recombinant DNA technology' requires Human insulin. B: The answers have been interchanged: 'Production of many identical copies from one transformed cell' requires Cloning, and 'Legal protection for a novel biotechnological invention' requires Biopatent. C: The answers have been interchanged: 'Nutritionally enhanced transgenic crop example' requires Golden rice, and 'Production of many identical copies from one transformed cell' requires Cloning.

BIO-12-CH12-0058 | Medium | Medical applications**Correct answer: B. First therapeutic product made using recombinant DNA technology - Human insulin**

Solution: The accurate entry is 'First therapeutic product made using recombinant DNA technology - Human insulin'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Amplification of a chosen DNA segment in vitro' should be paired with PCR, not DNA ligase. C: 'Enzyme sealing phosphodiester bonds between DNA fragments' should be paired with DNA ligase, not Transformation. D: 'Thermostable polymerase commonly used in PCR' should be paired with Taq polymerase, not Extension.

BIO-12-CH12-0059 | Medium | PCR and analysis**Correct answer: C. Annealing**

Solution: The chapter links PCR step allowing primers to bind with Annealing, so Annealing completes the statement.

Distractor review: A: Golden rice belongs with 'Nutritionally enhanced transgenic crop example', whereas the tested description is 'PCR step allowing primers to bind'. B: Extension belongs with 'PCR step in which polymerase synthesizes DNA', whereas the tested description is 'PCR step allowing primers to bind'. D: Denaturation belongs with 'PCR step separating DNA strands', whereas the tested description is 'PCR step allowing primers to bind'.

BIO-12-CH12-0060 | Hard | Medical applications

Correct answer: B. Antibody produced from a single clone of cells - Monoclonal antibody; Production of many identical copies from one transformed cell - Cloning

Solution: Both parts of 'Antibody produced from a single clone of cells - Monoclonal antibody; Production of many identical copies from one transformed cell - Cloning' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'PCR step allowing primers to bind' requires Annealing, and 'Indian committee that decides validity and safety of GMO research' requires GEAC. C: The answers have been interchanged: 'DNA formed by joining sequences from different sources' requires Recombinant DNA, and 'Gene allowing identification of transformed cells' requires Selectable marker. D: The answers have been interchanged: 'Thermostable polymerase commonly used in PCR' requires Taq polymerase, and 'Nutritionally enhanced transgenic crop example' requires Golden rice.

BIO-12-CH12-0061 | Easy | PCR and analysis

Correct answer: C. Thermostable polymerase commonly used in PCR

Solution: Taq polymerase represents 'Thermostable polymerase commonly used in PCR'. Each alternative describes a different term or result.

Distractor review: A: 'PCR step in which polymerase synthesizes DNA' identifies Extension, not Taq polymerase. B: 'Organism carrying and expressing an introduced gene' identifies Transgenic organism, not Taq polymerase. D: 'Indian committee that decides validity and safety of GMO research' identifies GEAC, not Taq polymerase.

BIO-12-CH12-0062 | Easy | Agricultural applications

Correct answer: C. Golden rice

Solution: 'Nutritionally enhanced transgenic crop example' is the defining description for Golden rice. The other choices belong to different chapter concepts.

Distractor review: A: Bt cotton belongs with 'Transgenic plant known for insect resistance', whereas the tested description is 'Nutritionally enhanced transgenic crop example'. B: Sticky ends belongs with 'Single-stranded ends produced by staggered restriction cuts', whereas the tested description is 'Nutritionally enhanced transgenic crop example'. D: Restriction endonucleases belongs with 'Enzymes cutting DNA at specific recognition sequences', whereas the tested description is 'Nutritionally enhanced transgenic crop example'.

BIO-12-CH12-0063 | Hard | rDNA methodology

Correct answer: A. Separation of DNA fragments according to size - Gel electrophoresis; PCR step separating DNA strands - Denaturation

Solution: Both parts of 'Separation of DNA fragments according to size - Gel electrophoresis; PCR step separating DNA strands - Denaturation' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'First therapeutic product made using recombinant DNA technology' requires Human insulin, and 'Small circular bacterial DNA commonly used as a vector' requires Plasmid. C: The answers have been interchanged: 'Treatment by introducing a functional gene' requires Gene therapy, and 'First therapeutic product made using recombinant DNA technology' requires Human insulin. D: The answers have been interchanged: 'Early molecular diagnosis using amplification' requires PCR-based diagnosis, and 'Site permitting vector replication in a host' requires Origin of replication.

BIO-12-CH12-0064 | Easy | Core principles

Correct answer: C. Genetic engineering

Solution: 'Deliberate alteration of genetic material' is the defining description for Genetic engineering. The other choices belong to different chapter concepts.

Distractor review: A: Taq polymerase belongs with 'Thermostable polymerase commonly used in PCR', whereas the tested description is 'Deliberate alteration of genetic material'. B: Bacillus thuringiensis belongs with 'Bt toxin gene source', whereas the tested description is 'Deliberate alteration of genetic material'. D: DNA ligase belongs with 'Enzyme sealing phosphodiester bonds between DNA fragments', whereas the tested description is 'Deliberate alteration of genetic material'.

BIO-12-CH12-0065 | Medium | PCR and analysis

Correct answer: B. Taq polymerase

Solution: The chapter links Thermostable polymerase commonly used in PCR with Taq polymerase, so Taq polymerase completes the statement.

Distractor review: A: Sticky ends belongs with 'Single-stranded ends produced by staggered restriction cuts', whereas the tested description is 'Thermostable polymerase commonly used in PCR'. C: Biopiracy belongs with 'Unfair commercial use of biological resources or traditional knowledge', whereas the tested description is 'Thermostable polymerase commonly used in PCR'. D: GEAC belongs with 'Indian committee that decides validity and safety of GMO research', whereas the tested description is 'Thermostable polymerase commonly used in PCR'.

BIO-12-CH12-0066 | Medium | Cloning vectors**Correct answer: D. Multiple cloning site**

Solution: The chapter links Vector region containing several unique restriction sites with Multiple cloning site, so Multiple cloning site completes the statement.

Distractor review: A: Biopiracy belongs with 'Unfair commercial use of biological resources or traditional knowledge', whereas the tested description is 'Vector region containing several unique restriction sites'. B: Biopatent belongs with 'Legal protection for a novel biotechnological invention', whereas the tested description is 'Vector region containing several unique restriction sites'. C: Origin of replication belongs with 'Site permitting vector replication in a host', whereas the tested description is 'Vector region containing several unique restriction sites'.

BIO-12-CH12-0067 | Hard | Agricultural applications**Correct answer: B. Nutritionally enhanced transgenic crop example - Recombinant DNA**

Solution: 'Nutritionally enhanced transgenic crop example' should be paired with Golden rice, not Recombinant DNA. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Gene allowing identification of transformed cells - Selectable marker. C: This is a valid association: Single-stranded ends produced by staggered restriction cuts - Sticky ends. D: This is a valid association: Enzymes cutting DNA at specific recognition sequences - Restriction endonucleases.

BIO-12-CH12-0068 | Easy | Bioethics and ownership**Correct answer: A. Environmental release of GMOs**

Solution: 'Ethical concern about altered organisms entering nature' is the defining description for Environmental release of GMOs. The other choices belong to different chapter concepts.

Distractor review: B: Golden rice belongs with 'Nutritionally enhanced transgenic crop example', whereas the tested description is 'Ethical concern about altered organisms entering nature'. C: Extension belongs with 'PCR step in which polymerase synthesizes DNA', whereas the tested description is 'Ethical concern about altered organisms entering nature'. D: PCR belongs with 'Amplification of a chosen DNA segment in vitro', whereas the tested description is 'Ethical concern about altered organisms entering nature'.

BIO-12-CH12-0069 | Easy | Cloning vectors**Correct answer: B. Site permitting vector replication in a host**

Solution: Origin of replication represents 'Site permitting vector replication in a host'. Each alternative describes a different term or result.

Distractor review: A: 'Transgenic plant known for insect resistance' identifies Bt cotton, not Origin of replication. C: 'Gene allowing identification of transformed cells' identifies Selectable marker, not Origin of replication. D: 'Enzymes cutting DNA at specific recognition sequences' identifies Restriction endonucleases, not Origin of replication.

BIO-12-CH12-0070 | Medium | Agricultural applications**Correct answer: C. Bt toxin gene source - Bacillus thuringiensis**

Solution: The accurate entry is 'Bt toxin gene source - Bacillus thuringiensis'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Treatment by introducing a functional gene' should be paired with Gene therapy, not Sticky ends. B: 'Small circular bacterial DNA commonly used as a vector' should be paired with Plasmid, not Monoclonal antibody. D: 'First therapeutic product made using recombinant DNA technology' should be paired with Human insulin, not Gene therapy.

BIO-12-CH12-0071 | Easy | Agricultural applications**Correct answer: A. Transgenic plant known for insect resistance**

Solution: Bt cotton represents 'Transgenic plant known for insect resistance'. Each alternative describes a different term or result.

Distractor review: B: 'Nutritionally enhanced transgenic crop example' identifies Golden rice, not Bt cotton. C: 'Antibody produced from a single clone of cells' identifies Monoclonal antibody, not Bt cotton. D: 'Bt toxin gene source' identifies Bacillus thuringiensis, not Bt cotton.

BIO-12-CH12-0072 | Hard | Cloning vectors

Correct answer: D. Small circular bacterial DNA commonly used as a vector - Plasmid; Amplification of a chosen DNA segment in vitro - PCR

Solution: Both parts of 'Small circular bacterial DNA commonly used as a vector - Plasmid; Amplification of a chosen DNA segment in vitro - PCR' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Transgenic plant known for insect resistance' requires Bt cotton, and 'Restriction sequence reading the same in opposite directions' requires Palindromic sequence. B: The answers have been interchanged: 'Nutritionally enhanced transgenic crop example' requires Golden rice, and 'Transgenic plant known for insect resistance' requires Bt cotton. C: The answers have been interchanged: 'Enzymes cutting DNA at specific recognition sequences' requires Restriction endonucleases, and 'Organism carrying and expressing an introduced gene' requires Transgenic organism.

BIO-12-CH12-0073 | Hard | Agricultural applications

Correct answer: C. Transgenic plant known for insect resistance - Bt cotton; Antibody produced from a single clone of cells - Monoclonal antibody

Solution: Both parts of 'Transgenic plant known for insect resistance - Bt cotton; Antibody produced from a single clone of cells - Monoclonal antibody' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Amplification of a chosen DNA segment in vitro' requires PCR, and 'PCR step separating DNA strands' requires Denaturation. B: The answers have been interchanged: 'Single-stranded ends produced by staggered restriction cuts' requires Sticky ends, and 'Amplification of a chosen DNA segment in vitro' requires PCR. D: The answers have been interchanged: 'Organism carrying and expressing an introduced gene' requires Transgenic organism, and 'Enzyme sealing phosphodiester bonds between DNA fragments' requires DNA ligase.

BIO-12-CH12-0074 | Medium | Cloning vectors

Correct answer: D. Origin of replication

Solution: The observation 'Site permitting vector replication in a host' points to Origin of replication; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Restriction endonucleases belongs with 'Enzymes cutting DNA at specific recognition sequences', whereas the tested description is 'Site permitting vector replication in a host'. B: Gel electrophoresis belongs with 'Separation of DNA fragments according to size', whereas the tested description is 'Site permitting vector replication in a host'. C: Selectable marker belongs with 'Gene allowing identification of transformed cells', whereas the tested description is 'Site permitting vector replication in a host'.

BIO-12-CH12-0075 | Easy | Cloning vectors

Correct answer: B. Selectable marker

Solution: 'Gene allowing identification of transformed cells' is the defining description for Selectable marker. The other choices belong to different chapter concepts.

Distractor review: A: DNA ligase belongs with 'Enzyme sealing phosphodiester bonds between DNA fragments', whereas the tested description is 'Gene allowing identification of transformed cells'. C: Multiple cloning site belongs with 'Vector region containing several unique restriction sites', whereas the tested description is 'Gene allowing identification of transformed cells'. D: GEAC belongs with 'Indian committee that decides validity and safety of GMO research', whereas the tested description is 'Gene allowing identification of transformed cells'.

BIO-12-CH12-0076 | Easy | Core principles

Correct answer: A. Bioprocess engineering

Solution: 'Maintenance of sterile conditions for industrial microbial growth' is the defining description for Bioprocess engineering. The other choices belong to different chapter concepts.

Distractor review: B: Recombinant DNA belongs with 'DNA formed by joining sequences from different sources', whereas the tested description is 'Maintenance of sterile conditions for industrial microbial growth'. C: PCR-based diagnosis belongs with 'Early molecular diagnosis using amplification', whereas the tested description is 'Maintenance of sterile conditions for industrial microbial growth'. D: Selectable marker belongs with 'Gene allowing identification of transformed cells', whereas the tested description is 'Maintenance of sterile conditions for industrial microbial growth'.

BIO-12-CH12-0077 | Medium | rDNA methodology

Correct answer: A. Introduction of recombinant DNA into a host cell - Transformation

Solution: The accurate entry is 'Introduction of recombinant DNA into a host cell - Transformation'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'PCR step separating DNA strands' should be paired with Denaturation, not Bioprocess engineering. C: 'First therapeutic product made using recombinant DNA technology' should be paired with Human insulin, not Transformation. D: 'Maintenance of sterile conditions for industrial microbial growth' should be paired with Bioprocess engineering, not Genetic engineering.

BIO-12-CH12-0078 | Easy | PCR and analysis**Correct answer: D. PCR step allowing primers to bind**

Solution: Annealing represents 'PCR step allowing primers to bind'. Each alternative describes a different term or result.

Distractor review: A: 'Site permitting vector replication in a host' identifies Origin of replication, not Annealing. B: 'Antibody produced from a single clone of cells' identifies Monoclonal antibody, not Annealing. C: 'Enzyme sealing phosphodiester bonds between DNA fragments' identifies DNA ligase, not Annealing.

BIO-12-CH12-0079 | Challenging | rDNA methodology**Correct answer: C. I, II, III and IV**

Solution: Statement I is correct because 'Production of many identical copies from one transformed cell' corresponds to Cloning. Statement II is correct because 'Separation of DNA fragments according to size' corresponds to Gel electrophoresis. Statement III is correct because 'Amplification of a chosen DNA segment in vitro' corresponds to PCR. Statement IV is correct because 'Legal protection for a novel biotechnological invention' corresponds to Biopatent. Therefore, the valid combination is I, II, III and IV.

Distractor review: A: This combination omits true statement(s) I, IV. B: This combination omits true statement(s) I, II. D: This combination omits true statement(s) III.

BIO-12-CH12-0080 | Hard | Cloning vectors**Correct answer: D. Site permitting vector replication in a host - Origin of replication; Restriction sequence reading the same in opposite directions - Palindromic sequence**

Solution: Both parts of 'Site permitting vector replication in a host - Origin of replication; Restriction sequence reading the same in opposite directions - Palindromic sequence' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Enzymes cutting DNA at specific recognition sequences' requires Restriction endonucleases, and 'Transgenic plant known for insect resistance' requires Bt cotton. B: The answers have been interchanged: 'Single-stranded ends produced by staggered restriction cuts' requires Sticky ends, and 'Enzymes cutting DNA at specific recognition sequences' requires Restriction endonucleases. C: The answers have been interchanged: 'Bt crop protection targets' requires Specific insect pests, and 'Nutritionally enhanced transgenic crop example' requires Golden rice.

BIO-12-CH12-0081 | Medium | rDNA methodology**Correct answer: B. PCR**

Solution: The chapter links Amplification of a chosen DNA segment in vitro with PCR, so PCR completes the statement.

Distractor review: A: GEAC belongs with 'Indian committee that decides validity and safety of GMO research', whereas the tested description is 'Amplification of a chosen DNA segment in vitro'. C: Restriction endonucleases belongs with 'Enzymes cutting DNA at specific recognition sequences', whereas the tested description is 'Amplification of a chosen DNA segment in vitro'. D: Transformation belongs with 'Introduction of recombinant DNA into a host cell', whereas the tested description is 'Amplification of a chosen DNA segment in vitro'.

BIO-12-CH12-0082 | Medium | Medical applications**Correct answer: B. Gene therapy**

Solution: The observation 'Treatment by introducing a functional gene' points to Gene therapy; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Origin of replication belongs with 'Site permitting vector replication in a host', whereas the tested description is 'Treatment by introducing a functional gene'. C: Recombinant DNA belongs with 'DNA formed by joining sequences from different sources', whereas the tested description is 'Treatment by introducing a functional gene'. D: Human insulin belongs with 'First therapeutic product made using recombinant DNA technology', whereas the tested description is 'Treatment by introducing a functional gene'.

BIO-12-CH12-0083 | Challenging | Cloning vectors**Correct answer: A. I and IV only**

Solution: Statement I is correct because 'Gene allowing identification of transformed cells' corresponds to Selectable marker. Statement II is incorrect: 'Vector region containing several unique restriction sites' corresponds to Multiple cloning site, not Origin of replication. Statement III is incorrect: 'Site permitting vector replication in a host' corresponds to Origin of replication, not Transgenic organism. Statement IV is correct because 'Organism carrying and expressing an introduced gene' corresponds to Transgenic organism. Therefore, the valid combination is I and IV only.

Distractor review: B: This combination includes false statement(s) II, III and omits true statement(s) IV. C: This combination includes false statement(s) II, III and omits true statement(s) I. D: This combination includes false statement(s) III.

BIO-12-CH12-0084 | Hard | Bioethics and ownership**Correct answer: B. Ethical concern about altered organisms entering nature - Extension**

Solution: 'Ethical concern about altered organisms entering nature' should be paired with Environmental release of GMOs, not Extension. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Restriction sequence reading the same in opposite directions - Palindromic sequence. C: This is a valid association: Nutritionally enhanced transgenic crop example - Golden rice. D: This is a valid association: Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase.

BIO-12-CH12-0085 | Challenging | Restriction and joining enzymes**Correct answer: C. II, III and IV only**

Solution: Statement I is incorrect: 'Single-stranded ends produced by staggered restriction cuts' corresponds to Sticky ends, not Restriction endonucleases. Statement II is correct because 'Enzymes cutting DNA at specific recognition sequences' corresponds to Restriction endonucleases. Statement III is correct because 'Enzyme sealing phosphodiester bonds between DNA fragments' corresponds to DNA ligase. Statement IV is correct because 'Site permitting vector replication in a host' corresponds to Origin of replication. Therefore, the valid combination is II, III and IV only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) III. B: This combination omits true statement(s) II, IV. D: This combination includes false statement(s) I and omits true statement(s) III, IV.

BIO-12-CH12-0086 | Medium | Agricultural applications**Correct answer: A. Bt cotton**

Solution: The observation 'Transgenic plant known for insect resistance' points to Bt cotton; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: PCR-based diagnosis belongs with 'Early molecular diagnosis using amplification', whereas the tested description is 'Transgenic plant known for insect resistance'. C: Specific insect pests belongs with 'Bt crop protection targets', whereas the tested description is 'Transgenic plant known for insect resistance'. D: Monoclonal antibody belongs with 'Antibody produced from a single clone of cells', whereas the tested description is 'Transgenic plant known for insect resistance'.

BIO-12-CH12-0087 | Medium | rDNA methodology**Correct answer: B. Gel electrophoresis**

Solution: The observation 'Separation of DNA fragments according to size' points to Gel electrophoresis; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Plasmid belongs with 'Small circular bacterial DNA commonly used as a vector', whereas the tested description is 'Separation of DNA fragments according to size'. C: PCR-based diagnosis belongs with 'Early molecular diagnosis using amplification', whereas the tested description is 'Separation of DNA fragments according to size'. D: Gene therapy belongs with 'Treatment by introducing a functional gene', whereas the tested description is 'Separation of DNA fragments according to size'.

BIO-12-CH12-0088 | Medium | PCR and analysis**Correct answer: A. Extension**

Solution: The observation 'PCR step in which polymerase synthesizes DNA' points to Extension; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Gel electrophoresis belongs with 'Separation of DNA fragments according to size', whereas the tested description is 'PCR step in which polymerase synthesizes DNA'. C: Sticky ends belongs with 'Single-stranded ends produced by staggered restriction cuts', whereas the tested description is 'PCR step in which polymerase synthesizes DNA'. D: PCR belongs with 'Amplification of a chosen DNA segment in vitro', whereas the tested description is 'PCR step in which polymerase synthesizes DNA'.

BIO-12-CH12-0089 | Challenging | Bioethics and ownership**Correct answer: A. I and III only**

Solution: Statement I is correct because 'Ethical concern about altered organisms entering nature' corresponds to Environmental release of GMOs. Statement II is incorrect: 'Legal protection for a novel biotechnological invention' corresponds to Biopatent, not GEAC. Statement III is correct because 'Indian committee that decides validity and safety of GMO research' corresponds to GEAC. Statement IV is incorrect: 'Nutritionally enhanced transgenic crop example' corresponds to Golden rice, not Environmental release of GMOs. Therefore, the valid combination is I and III only.

Distractor review: B: This combination includes false statement(s) II and omits true statement(s) III. C: This combination omits true statement(s) I. D: This combination includes false statement(s) IV and omits true statement(s) I.

BIO-12-CH12-0090 | Hard | rDNA methodology

Correct answer: D. Introduction of recombinant DNA into a host cell - Transformation; PCR step separating DNA strands - Denaturation

Solution: Both parts of 'Introduction of recombinant DNA into a host cell - Transformation; PCR step separating DNA strands - Denaturation' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Maintenance of sterile conditions for industrial microbial growth' requires Bioprocess engineering, and 'Transgenic plant known for insect resistance' requires Bt cotton. B: The answers have been interchanged: 'Transgenic plant known for insect resistance' requires Bt cotton, and 'Deliberate alteration of genetic material' requires Genetic engineering. C: The answers have been interchanged: 'Antibody produced from a single clone of cells' requires Monoclonal antibody, and 'Maintenance of sterile conditions for industrial microbial growth' requires Bioprocess engineering.

BIO-12-CH12-0091 | Medium | Core principles

Correct answer: D. Maintenance of sterile conditions for industrial microbial growth - Bioprocess engineering

Solution: The accurate entry is 'Maintenance of sterile conditions for industrial microbial growth - Bioprocess engineering'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Vector region containing several unique restriction sites' should be paired with Multiple cloning site, not Selectable marker. B: 'Enzymes cutting DNA at specific recognition sequences' should be paired with Restriction endonucleases, not PCR-based diagnosis. C: 'Early molecular diagnosis using amplification' should be paired with PCR-based diagnosis, not Environmental release of GMOs.

BIO-12-CH12-0092 | Medium | Restriction and joining enzymes

Correct answer: A. Restriction endonucleases

Solution: The chapter links Enzymes cutting DNA at specific recognition sequences with Restriction endonucleases, so Restriction endonucleases completes the statement.

Distractor review: B: DNA ligase belongs with 'Enzyme sealing phosphodiester bonds between DNA fragments', whereas the tested description is 'Enzymes cutting DNA at specific recognition sequences'. C: Origin of replication belongs with 'Site permitting vector replication in a host', whereas the tested description is 'Enzymes cutting DNA at specific recognition sequences'. D: Monoclonal antibody belongs with 'Antibody produced from a single clone of cells', whereas the tested description is 'Enzymes cutting DNA at specific recognition sequences'.

BIO-12-CH12-0093 | Medium | rDNA methodology

Correct answer: D. Transformation

Solution: The chapter links Introduction of recombinant DNA into a host cell with Transformation, so Transformation completes the statement.

Distractor review: A: PCR belongs with 'Amplification of a chosen DNA segment in vitro', whereas the tested description is 'Introduction of recombinant DNA into a host cell'. B: Genetic engineering belongs with 'Deliberate alteration of genetic material', whereas the tested description is 'Introduction of recombinant DNA into a host cell'. C: Denaturation belongs with 'PCR step separating DNA strands', whereas the tested description is 'Introduction of recombinant DNA into a host cell'.

BIO-12-CH12-0094 | Medium | Core principles

Correct answer: A. Bioprocess engineering

Solution: The chapter links Maintenance of sterile conditions for industrial microbial growth with Bioprocess engineering, so Bioprocess engineering completes the statement.

Distractor review: B: Multiple cloning site belongs with 'Vector region containing several unique restriction sites', whereas the tested description is 'Maintenance of sterile conditions for industrial microbial growth'. C: Restriction endonucleases belongs with 'Enzymes cutting DNA at specific recognition sequences', whereas the tested description is 'Maintenance of sterile conditions for industrial microbial growth'. D: Origin of replication belongs with 'Site permitting vector replication in a host', whereas the tested description is 'Maintenance of sterile conditions for industrial microbial growth'.

BIO-12-CH12-0095 | Medium | Cloning vectors

Correct answer: D. Site permitting vector replication in a host - Origin of replication

Solution: The accurate entry is 'Site permitting vector replication in a host - Origin of replication'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Bt crop protection targets' should be paired with Specific insect pests, not Origin of replication. B: 'Single-stranded ends produced by staggered restriction cuts' should be paired with Sticky ends, not Bt cotton. C: 'Transgenic plant known for insect resistance' should be paired with Bt cotton, not Specific insect pests.

BIO-12-CH12-0096 | Hard | Core principles**Correct answer: B. Maintenance of sterile conditions for industrial microbial growth - Origin of replication**

Solution: 'Maintenance of sterile conditions for industrial microbial growth' should be paired with Bioprocess engineering, not Origin of replication. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Early molecular diagnosis using amplification - PCR-based diagnosis. C: This is a valid association: Ethical concern about altered organisms entering nature - Environmental release of GMOs. D: This is a valid association: Vector region containing several unique restriction sites - Multiple cloning site.

BIO-12-CH12-0097 | Medium | Agricultural applications**Correct answer: A. Nutritionally enhanced transgenic crop example - Golden rice**

Solution: The accurate entry is 'Nutritionally enhanced transgenic crop example - Golden rice'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Single-stranded ends produced by staggered restriction cuts' should be paired with Sticky ends, not GEAC. C: 'Gene allowing identification of transformed cells' should be paired with Selectable marker, not Golden rice. D: 'Enzymes cutting DNA at specific recognition sequences' should be paired with Restriction endonucleases, not Cloning.

BIO-12-CH12-0098 | Easy | Cloning vectors**Correct answer: B. Multiple cloning site**

Solution: 'Vector region containing several unique restriction sites' is the defining description for Multiple cloning site. The other choices belong to different chapter concepts.

Distractor review: A: Selectable marker belongs with 'Gene allowing identification of transformed cells', whereas the tested description is 'Vector region containing several unique restriction sites'. C: Gel electrophoresis belongs with 'Separation of DNA fragments according to size', whereas the tested description is 'Vector region containing several unique restriction sites'. D: Biopiracy belongs with 'Unfair commercial use of biological resources or traditional knowledge', whereas the tested description is 'Vector region containing several unique restriction sites'.

BIO-12-CH12-0099 | Easy | Agricultural applications**Correct answer: D. Bacillus thuringiensis**

Solution: 'Bt toxin gene source' is the defining description for Bacillus thuringiensis. The other choices belong to different chapter concepts.

Distractor review: A: Plasmid belongs with 'Small circular bacterial DNA commonly used as a vector', whereas the tested description is 'Bt toxin gene source'. B: Cloning belongs with 'Production of many identical copies from one transformed cell', whereas the tested description is 'Bt toxin gene source'. C: Bt cotton belongs with 'Transgenic plant known for insect resistance', whereas the tested description is 'Bt toxin gene source'.

BIO-12-CH12-0100 | Easy | rDNA methodology**Correct answer: D. Introduction of recombinant DNA into a host cell**

Solution: Transformation represents 'Introduction of recombinant DNA into a host cell'. Each alternative describes a different term or result.

Distractor review: A: 'PCR step separating DNA strands' identifies Denaturation, not Transformation. B: 'Deliberate alteration of genetic material' identifies Genetic engineering, not Transformation. C: 'Maintenance of sterile conditions for industrial microbial growth' identifies Bioprocess engineering, not Transformation.

BIO-12-CH12-0101 | Challenging | Core principles**Correct answer: B. I, III and IV only**

Solution: Statement I is correct because 'Deliberate alteration of genetic material' corresponds to Genetic engineering. Statement II is incorrect: 'Organism carrying and expressing an introduced gene' corresponds to Transgenic organism, not Bioprocess engineering. Statement III is correct because 'Maintenance of sterile conditions for industrial microbial growth' corresponds to Bioprocess engineering. Statement IV is correct because 'Transgenic plant known for insect resistance' corresponds to Bt cotton. Therefore, the valid combination is I, III and IV only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) I, III. C: This combination omits true statement(s) I, IV. D: This combination omits true statement(s) III, IV.

BIO-12-CH12-0102 | Hard | Bioethics and ownership**Correct answer: B. Indian committee that decides validity and safety of GMO research - Gene therapy**

Solution: 'Indian committee that decides validity and safety of GMO research' should be paired with GEAC, not Gene therapy. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Restriction sequence reading the same in opposite directions - Palindromic sequence. C: This is a valid association: Bt toxin gene source - Bacillus thuringiensis. D: This is a valid association: Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase.

BIO-12-CH12-0103 | Medium | Core principles**Correct answer: A. Transgenic organism**

Solution: The observation 'Organism carrying and expressing an introduced gene' points to Transgenic organism; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Selectable marker belongs with 'Gene allowing identification of transformed cells', whereas the tested description is 'Organism carrying and expressing an introduced gene'. C: Genetic engineering belongs with 'Deliberate alteration of genetic material', whereas the tested description is 'Organism carrying and expressing an introduced gene'. D: Recombinant DNA belongs with 'DNA formed by joining sequences from different sources', whereas the tested description is 'Organism carrying and expressing an introduced gene'.

BIO-12-CH12-0104 | Hard | Cloning vectors**Correct answer: A. Gene allowing identification of transformed cells - Specific insect pests**

Solution: 'Gene allowing identification of transformed cells' should be paired with Selectable marker, not Specific insect pests. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase. C: This is a valid association: Bt toxin gene source - Bacillus thuringiensis. D: This is a valid association: Amplification of a chosen DNA segment in vitro - PCR.

BIO-12-CH12-0105 | Medium | Agricultural applications**Correct answer: B. Bt cotton**

Solution: The chapter links Transgenic plant known for insect resistance with Bt cotton, so Bt cotton completes the statement.

Distractor review: A: Bacillus thuringiensis belongs with 'Bt toxin gene source', whereas the tested description is 'Transgenic plant known for insect resistance'. C: Sticky ends belongs with 'Single-stranded ends produced by staggered restriction cuts', whereas the tested description is 'Transgenic plant known for insect resistance'. D: Monoclonal antibody belongs with 'Antibody produced from a single clone of cells', whereas the tested description is 'Transgenic plant known for insect resistance'.

BIO-12-CH12-0106 | Hard | rDNA methodology**Correct answer: A. Production of many identical copies from one transformed cell - Cloning; Bt crop protection targets - Specific insect pests**

Solution: Both parts of 'Production of many identical copies from one transformed cell - Cloning; Bt crop protection targets - Specific insect pests' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'PCR step allowing primers to bind' requires Annealing, and 'PCR step separating DNA strands' requires Denaturation. C: The answers have been interchanged: 'Enzymes cutting DNA at specific recognition sequences' requires Restriction endonucleases, and 'Legal protection for a novel biotechnological invention' requires Biopatent. D: The answers have been interchanged: 'Single-stranded ends produced by staggered restriction cuts' requires Sticky ends, and 'PCR step allowing primers to bind' requires Annealing.

BIO-12-CH12-0107 | Hard | Medical applications**Correct answer: C. Early molecular diagnosis using amplification - PCR-based diagnosis; Bt toxin gene source - Bacillus thuringiensis**

Solution: Both parts of 'Early molecular diagnosis using amplification - PCR-based diagnosis; Bt toxin gene source - Bacillus thuringiensis' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'DNA formed by joining sequences from different sources' requires Recombinant DNA, and 'Restriction sequence reading the same in opposite directions' requires Palindromic sequence. B: The answers have been interchanged: 'Transgenic plant known for insect resistance' requires Bt cotton, and 'Small circular bacterial DNA commonly used as a vector' requires Plasmid. D: The answers have been interchanged: 'Small circular bacterial DNA commonly used as a vector' requires Plasmid, and 'Thermostable polymerase commonly used in PCR' requires Taq polymerase.

BIO-12-CH12-0108 | Easy | Restriction and joining enzymes**Correct answer: C. Enzymes cutting DNA at specific recognition sequences**

Solution: Restriction endonucleases represents 'Enzymes cutting DNA at specific recognition sequences'. Each alternative describes a different term or result.

Distractor review: A: 'Antibody produced from a single clone of cells' identifies Monoclonal antibody, not Restriction endonucleases. B: 'Single-stranded ends produced by staggered restriction cuts' identifies Sticky ends, not Restriction endonucleases. D: 'Site permitting vector replication in a host' identifies Origin of replication, not Restriction endonucleases.

BIO-12-CH12-0109 | Easy | Restriction and joining enzymes**Correct answer: B. Restriction endonucleases**

Solution: 'Enzymes cutting DNA at specific recognition sequences' is the defining description for Restriction endonucleases. The other choices belong to different chapter concepts.

Distractor review: A: PCR belongs with 'Amplification of a chosen DNA segment in vitro', whereas the tested description is 'Enzymes cutting DNA at specific recognition sequences'. C: Palindromic sequence belongs with 'Restriction sequence reading the same in opposite directions', whereas the tested description is 'Enzymes cutting DNA at specific recognition sequences'. D: Bt cotton belongs with 'Transgenic plant known for insect resistance', whereas the tested description is 'Enzymes cutting DNA at specific recognition sequences'.

BIO-12-CH12-0110 | Medium | Bioethics and ownership**Correct answer: A. Biopiracy**

Solution: The chapter links Unfair commercial use of biological resources or traditional knowledge with Biopiracy, so Biopiracy completes the statement.

Distractor review: B: Recombinant DNA belongs with 'DNA formed by joining sequences from different sources', whereas the tested description is 'Unfair commercial use of biological resources or traditional knowledge'. C: Environmental release of GMOs belongs with 'Ethical concern about altered organisms entering nature', whereas the tested description is 'Unfair commercial use of biological resources or traditional knowledge'. D: Restriction endonucleases belongs with 'Enzymes cutting DNA at specific recognition sequences', whereas the tested description is 'Unfair commercial use of biological resources or traditional knowledge'.

BIO-12-CH12-0111 | Medium | Agricultural applications**Correct answer: D. Specific insect pests**

Solution: The observation 'Bt crop protection targets' points to Specific insect pests; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Denaturation belongs with 'PCR step separating DNA strands', whereas the tested description is 'Bt crop protection targets'. B: Environmental release of GMOs belongs with 'Ethical concern about altered organisms entering nature', whereas the tested description is 'Bt crop protection targets'. C: Biopiracy belongs with 'Unfair commercial use of biological resources or traditional knowledge', whereas the tested description is 'Bt crop protection targets'.

BIO-12-CH12-0112 | Medium | Medical applications**Correct answer: A. Gene therapy**

Solution: The chapter links Treatment by introducing a functional gene with Gene therapy, so Gene therapy completes the statement.

Distractor review: B: Origin of replication belongs with 'Site permitting vector replication in a host', whereas the tested description is 'Treatment by introducing a functional gene'. C: Plasmid belongs with 'Small circular bacterial DNA commonly used as a vector', whereas the tested description is 'Treatment by introducing a functional gene'. D: PCR-based diagnosis belongs with 'Early molecular diagnosis using amplification', whereas the tested description is 'Treatment by introducing a functional gene'.

Standard 12 | Chapter 13

Organisms and Populations

BIO-12-CH13-0001 | Medium | Age structure of populations

Correct answer: D. The pyramid with a broad pre-reproductive base

Solution: A broad base shows many young individuals who will enter reproductive ages.

Distractor review: A: The alternative 'The top-heavy declining pyramid' conflicts with the labelled relationship, scale or mechanism in the figure. A broad base shows many young individuals who will enter reproductive ages. B: The alternative 'A pyramid with no pre-reproductive group' conflicts with the labelled relationship, scale or mechanism in the figure. A broad base shows many young individuals who will enter reproductive ages. C: The alternative 'All shapes necessarily show zero growth' conflicts with the labelled relationship, scale or mechanism in the figure. A broad base shows many young individuals who will enter reproductive ages.

BIO-12-CH13-0002 | Hard | Population growth models

Correct answer: A. Environmental resistance increases as resources become limiting

Solution: Density-dependent limits reduce per-capita growth as population size approaches carrying capacity.

Distractor review: B: The alternative 'Birth rate must become infinite' conflicts with the labelled relationship, scale or mechanism in the figure. Density-dependent limits reduce per-capita growth as population size approaches carrying capacity. C: The alternative 'Every individual changes species' conflicts with the labelled relationship, scale or mechanism in the figure. Density-dependent limits reduce per-capita growth as population size approaches carrying capacity. D: The alternative 'Competition disappears completely' conflicts with the labelled relationship, scale or mechanism in the figure. Density-dependent limits reduce per-capita growth as population size approaches carrying capacity.

BIO-12-CH13-0003 | Medium | Age pyramids

Correct answer: B. Broad-base age pyramid - Expanding population

Solution: The accurate entry is 'Broad-base age pyramid - Expanding population'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'One benefits and the other is unaffected' should be paired with Commensalism, not Exponential growth. C: 'Intrinsic rate of natural increase' should be paired with Biotic potential, not Expanding population. D: 'Proportion of individuals in different age groups' should be paired with Age structure, not Dormancy.

BIO-12-CH13-0004 | Easy | Population growth models

Correct answer: B. Carrying capacity

Solution: K is the maximum population size the environment can sustain under the stated conditions.

Distractor review: A: The alternative 'Intrinsic mutation rate' conflicts with the labelled relationship, scale or mechanism in the figure. K is the maximum population size the environment can sustain under the stated conditions. C: The alternative 'Number of chromosomes' conflicts with the labelled relationship, scale or mechanism in the figure. K is the maximum population size the environment can sustain under the stated conditions. D: The alternative 'Minimum viable age' conflicts with the labelled relationship, scale or mechanism in the figure. K is the maximum population size the environment can sustain under the stated conditions.

BIO-12-CH13-0005 | Medium | Negative interactions

Correct answer: A. One benefits while harming a host - Parasitism

Solution: The accurate entry is 'One benefits while harming a host - Parasitism'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Individuals before reproductive maturity' should be paired with Pre-reproductive group, not Stable population. C: 'Organisms maintaining a nearly constant internal temperature' should be paired with Regulators, not Exponential growth. D: 'Number of deaths during a given period' should be paired with Mortality, not Parasitism.

BIO-12-CH13-0006 | Hard | Positive interactions

Correct answer: C. Lichen partnership - Logistic growth

Solution: 'Lichen partnership' should be paired with Fungus and alga or cyanobacterium, not Logistic growth. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Narrow-base age pyramid - Declining population. B: This is a valid association: Number of individuals per unit area or volume - Population density. D: This is a valid association: One organism kills and consumes another - Predation.

BIO-12-CH13-0007 | Hard | Positive interactions

Correct answer: C. Both species benefit - Mutualism; Proportion of individuals in different age groups - Age structure

Solution: Both parts of 'Both species benefit - Mutualism; Proportion of individuals in different age groups - Age structure' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Organisms maintaining a nearly constant internal temperature' requires Regulators, and 'Production of antifreeze compounds in cold-water organisms' requires Biochemical cold adaptation. B: The answers have been interchanged: 'Number of births during a given period' requires Natality, and 'Number of deaths during a given period' requires Mortality. D: The answers have been interchanged: 'Number of deaths during a given period' requires Mortality, and 'Organisms maintaining a nearly constant internal temperature' requires Regulators.

BIO-12-CH13-0008 | Medium | Age pyramids

Correct answer: B. Nearly bell-shaped age pyramid - Stable population

Solution: The accurate entry is 'Nearly bell-shaped age pyramid - Stable population'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'J-shaped population growth under unlimited resources' should be paired with Exponential growth, not Amensalism. C: 'One is inhibited while the other is unaffected' should be paired with Amensalism, not Commensalism. D: 'One benefits while harming a host' should be paired with Parasitism, not Biotic potential.

BIO-12-CH13-0009 | Easy | Negative interactions

Correct answer: D. Competition

Solution: 'Both species are adversely affected while sharing a limiting resource' is the defining description for Competition. The other choices belong to different chapter concepts.

Distractor review: A: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'Both species are adversely affected while sharing a limiting resource'. B: Exponential growth belongs with 'J-shaped population growth under unlimited resources', whereas the tested description is 'Both species are adversely affected while sharing a limiting resource'. C: Natality belongs with 'Number of births during a given period', whereas the tested description is 'Both species are adversely affected while sharing a limiting resource'.

BIO-12-CH13-0010 | Medium | Population attributes

Correct answer: C. Natality

Solution: The observation 'Number of births during a given period' points to Natality; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Opuntia belongs with 'Desert plant whose leaves are reduced to spines', whereas the tested description is 'Number of births during a given period'. B: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'Number of births during a given period'. D: Population density belongs with 'Number of individuals per unit area or volume', whereas the tested description is 'Number of births during a given period'.

BIO-12-CH13-0011 | Hard | Age pyramids

Correct answer: A. Nearly bell-shaped age pyramid - Exponential growth

Solution: 'Nearly bell-shaped age pyramid' should be paired with Stable population, not Exponential growth. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: One benefits while harming a host - Parasitism. C: This is a valid association: One organism kills and consumes another - Predation. D: This is a valid association: One benefits and the other is unaffected - Commensalism.

BIO-12-CH13-0012 | Medium | Abiotic factors

Correct answer: C. Conformers

Solution: The observation 'Organisms whose body temperature changes with surroundings' points to Conformers; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Fungus and alga or cyanobacterium belongs with 'Lichen partnership', whereas the tested description is 'Organisms whose body temperature changes with surroundings'. B: Dormancy belongs with 'State of reduced activity during an unfavourable period', whereas the tested description is 'Organisms whose body temperature changes with surroundings'. D: Opuntia belongs with 'Desert plant whose leaves are reduced to spines', whereas the tested description is 'Organisms whose body temperature changes with surroundings'.

BIO-12-CH13-0013 | Medium | Population growth**Correct answer: D. Carrying capacity**

Solution: The chapter links Maximum population an environment can sustain with Carrying capacity, so Carrying capacity completes the statement.

Distractor review: A: Logistic growth belongs with 'S-shaped growth under resource limitation', whereas the tested description is 'Maximum population an environment can sustain'. B: Pre-reproductive group belongs with 'Individuals before reproductive maturity', whereas the tested description is 'Maximum population an environment can sustain'. C: Mutualism belongs with 'Both species benefit', whereas the tested description is 'Maximum population an environment can sustain'.

BIO-12-CH13-0014 | Medium | Positive interactions**Correct answer: A. Mutualism**

Solution: The chapter links Both species benefit with Mutualism, so Mutualism completes the statement.

Distractor review: B: Biotic potential belongs with 'Intrinsic rate of natural increase', whereas the tested description is 'Both species benefit'. C: Commensalism belongs with 'One benefits and the other is unaffected', whereas the tested description is 'Both species benefit'. D: Fungus and alga or cyanobacterium belongs with 'Lichen partnership', whereas the tested description is 'Both species benefit'.

BIO-12-CH13-0015 | Medium | Abiotic factors**Correct answer: B. Regulators**

Solution: The chapter links Organisms maintaining a nearly constant internal temperature with Regulators, so Regulators completes the statement.

Distractor review: A: Parasitism belongs with 'One benefits while harming a host', whereas the tested description is 'Organisms maintaining a nearly constant internal temperature'. C: Conformers belongs with 'Organisms whose body temperature changes with surroundings', whereas the tested description is 'Organisms maintaining a nearly constant internal temperature'. D: Exponential growth belongs with 'J-shaped population growth under unlimited resources', whereas the tested description is 'Organisms maintaining a nearly constant internal temperature'.

BIO-12-CH13-0016 | Easy | Population growth**Correct answer: B. Biotic potential**

Solution: 'Intrinsic rate of natural increase' is the defining description for Biotic potential. The other choices belong to different chapter concepts.

Distractor review: A: Regulators belongs with 'Organisms maintaining a nearly constant internal temperature', whereas the tested description is 'Intrinsic rate of natural increase'. C: Exponential growth belongs with 'J-shaped population growth under unlimited resources', whereas the tested description is 'Intrinsic rate of natural increase'. D: Symbiosis belongs with 'Close association often benefiting both partners', whereas the tested description is 'Intrinsic rate of natural increase'.

BIO-12-CH13-0017 | Hard | Age pyramids**Correct answer: A. Individuals before reproductive maturity - Commensalism**

Solution: 'Individuals before reproductive maturity' should be paired with Pre-reproductive group, not Commensalism. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: One benefits while harming a host - Parasitism. C: This is a valid association: Number of births during a given period - Natality. D: This is a valid association: State of reduced activity during an unfavourable period - Dormancy.

BIO-12-CH13-0018 | Medium | Population growth**Correct answer: C. Biotic potential**

Solution: The observation 'Intrinsic rate of natural increase' points to Biotic potential; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Regulators belongs with 'Organisms maintaining a nearly constant internal temperature', whereas the tested description is 'Intrinsic rate of natural increase'. B: Parasitism belongs with 'One benefits while harming a host', whereas the tested description is 'Intrinsic rate of natural increase'. D: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'Intrinsic rate of natural increase'.

BIO-12-CH13-0019 | Easy | Adaptations**Correct answer: A. Opuntia**

Solution: 'Desert plant whose leaves are reduced to spines' is the defining description for Opuntia. The other choices belong to different chapter concepts.

Distractor review: B: Commensalism belongs with 'One benefits and the other is unaffected', whereas the tested description is 'Desert plant whose leaves are reduced to spines'. C: Regulators belongs with 'Organisms maintaining a nearly constant internal temperature', whereas the tested description is 'Desert plant whose leaves are reduced to spines'. D: CAM belongs with 'Photosynthetic pathway that lets desert plants keep stomata closed by day', whereas the tested description is 'Desert plant whose leaves are reduced to spines'.

BIO-12-CH13-0020 | Medium | Adaptations**Correct answer: D. Photosynthetic pathway that lets desert plants keep stomata closed by day - CAM**

Solution: The accurate entry is 'Photosynthetic pathway that lets desert plants keep stomata closed by day - CAM'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Lichen partnership' should be paired with Fungus and alga or cyanobacterium, not Expanding population. B: 'Broad-base age pyramid' should be paired with Expanding population, not CAM. C: 'One organism kills and consumes another' should be paired with Predation, not Fungus and alga or cyanobacterium.

BIO-12-CH13-0021 | Challenging | Age structure of populations**Correct answer: A. Population size will tend to decline**

Solution: A narrow young cohort cannot fully replace the larger reproductive cohort as it ages, favouring decline.

Distractor review: B: The alternative 'Population size must grow exponentially' conflicts with the labelled relationship, scale or mechanism in the figure. A narrow young cohort cannot fully replace the larger reproductive cohort as it ages, favouring decline. C: The alternative 'Population instantly reaches infinite carrying capacity' conflicts with the labelled relationship, scale or mechanism in the figure. A narrow young cohort cannot fully replace the larger reproductive cohort as it ages, favouring decline. D: The alternative 'Age structure has no demographic implication' conflicts with the labelled relationship, scale or mechanism in the figure. A narrow young cohort cannot fully replace the larger reproductive cohort as it ages, favouring decline.

BIO-12-CH13-0022 | Medium | Positive interactions**Correct answer: A. Lichen partnership - Fungus and alga or cyanobacterium**

Solution: The accurate entry is 'Lichen partnership - Fungus and alga or cyanobacterium'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'One is inhibited while the other is unaffected' should be paired with Amensalism, not Population density. C: 'Number of individuals per unit area or volume' should be paired with Population density, not Fungus and alga or cyanobacterium. D: 'One organism kills and consumes another' should be paired with Predation, not Declining population.

BIO-12-CH13-0023 | Easy | Age pyramids**Correct answer: D. Stable population**

Solution: 'Nearly bell-shaped age pyramid' is the defining description for Stable population. The other choices belong to different chapter concepts.

Distractor review: A: Commensalism belongs with 'One benefits and the other is unaffected', whereas the tested description is 'Nearly bell-shaped age pyramid'. B: Predation belongs with 'One organism kills and consumes another', whereas the tested description is 'Nearly bell-shaped age pyramid'. C: Expanding population belongs with 'Broad-base age pyramid', whereas the tested description is 'Nearly bell-shaped age pyramid'.

BIO-12-CH13-0024 | Challenging | Population attributes**Correct answer: A. I, II, III and IV**

Solution: Statement I is correct because 'Proportion of individuals in different age groups' corresponds to Age structure. Statement II is correct because 'Number of individuals per unit area or volume' corresponds to Population density. Statement III is correct because 'Number of births during a given period' corresponds to Natality. Statement IV is correct because 'Narrow-base age pyramid' corresponds to Declining population. Therefore, the valid combination is I, II, III and IV.

Distractor review: B: This combination omits true statement(s) I, II. C: This combination omits true statement(s) III. D: This combination omits true statement(s) I.

BIO-12-CH13-0025 | Challenging | Positive interactions**Correct answer: A. I and IV only**

Solution: Statement I is correct because 'Both species benefit' corresponds to Mutualism. Statement II is incorrect: 'Close association often benefiting both partners' corresponds to Symbiosis, not Commensalism. Statement III is incorrect: 'One benefits and the other is unaffected' corresponds to Commensalism, not Age structure. Statement IV is correct because 'Proportion of individuals in different age groups' corresponds to Age structure. Therefore, the valid combination is I and IV only.

Distractor review: B: This combination includes false statement(s) II, III and omits true statement(s) I. C: This combination includes false statement(s) II, III and omits true statement(s) I, IV. D: This combination includes false statement(s) III and omits true statement(s) I, IV.

BIO-12-CH13-0026 | Medium | Positive interactions**Correct answer: C. One benefits and the other is unaffected - Commensalism**

Solution: The accurate entry is 'One benefits and the other is unaffected - Commensalism'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'State of reduced activity during an unfavourable period' should be paired with Dormancy, not Commensalism. B: 'Both species are adversely affected while sharing a limiting resource' should be paired with Competition, not Population density. D: 'Organisms whose body temperature changes with surroundings' should be paired with Conformers, not Dormancy.

BIO-12-CH13-0027 | Hard | Age pyramids**Correct answer: D. Narrow-base age pyramid - Conformers**

Solution: 'Narrow-base age pyramid' should be paired with Declining population, not Conformers. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: State of reduced activity during an unfavourable period - Dormancy. B: This is a valid association: Both species benefit - Mutualism. C: This is a valid association: Close association often benefiting both partners - Symbiosis.

BIO-12-CH13-0028 | Medium | Population growth**Correct answer: D. J-shaped population growth under unlimited resources - Exponential growth**

Solution: The accurate entry is 'J-shaped population growth under unlimited resources - Exponential growth'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Photosynthetic pathway that lets desert plants keep stomata closed by day' should be paired with CAM, not Dormancy. B: 'Number of individuals per unit area or volume' should be paired with Population density, not Pre-reproductive group. C: 'One organism kills and consumes another' should be paired with Predation, not CAM.

BIO-12-CH13-0029 | Medium | Age pyramids**Correct answer: A. Expanding population**

Solution: The observation 'Broad-base age pyramid' points to Expanding population; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Opuntia belongs with 'Desert plant whose leaves are reduced to spines', whereas the tested description is 'Broad-base age pyramid'. C: Stable population belongs with 'Nearly bell-shaped age pyramid', whereas the tested description is 'Broad-base age pyramid'. D: Age structure belongs with 'Proportion of individuals in different age groups', whereas the tested description is 'Broad-base age pyramid'.

BIO-12-CH13-0030 | Medium | Negative interactions**Correct answer: C. One is inhibited while the other is unaffected - Amensalism**

Solution: The accurate entry is 'One is inhibited while the other is unaffected - Amensalism'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'State of reduced activity during an unfavourable period' should be paired with Dormancy, not Biotic potential. B: 'Intrinsic rate of natural increase' should be paired with Biotic potential, not Water-conservation adaptations. D: 'Concentrated urine and reduced sweating in desert mammals' should be paired with Water-conservation adaptations, not Expanding population.

BIO-12-CH13-0031 | Hard | Negative interactions

Correct answer: D. One organism kills and consumes another - Predation; Individuals before reproductive maturity - Pre-reproductive group

Solution: Both parts of 'One organism kills and consumes another - Predation; Individuals before reproductive maturity - Pre-reproductive group' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Close association often benefiting both partners' requires Symbiosis, and 'Production of antifreeze compounds in cold-water organisms' requires Biochemical cold adaptation. B: The answers have been interchanged: 'State of reduced activity during an unfavourable period' requires Dormancy, and 'Close association often benefiting both partners' requires Symbiosis. C: The answers have been interchanged: 'Narrow-base age pyramid' requires Declining population, and 'State of reduced activity during an unfavourable period' requires Dormancy.

BIO-12-CH13-0032 | Medium | Positive interactions

Correct answer: C. Commensalism

Solution: The chapter links One benefits and the other is unaffected with Commensalism, so Commensalism completes the statement.

Distractor review: A: Conformers belongs with 'Organisms whose body temperature changes with surroundings', whereas the tested description is 'One benefits and the other is unaffected'. B: Parasitism belongs with 'One benefits while harming a host', whereas the tested description is 'One benefits and the other is unaffected'. D: Population density belongs with 'Number of individuals per unit area or volume', whereas the tested description is 'One benefits and the other is unaffected'.

BIO-12-CH13-0033 | Challenging | Population growth

Correct answer: D. II, III and IV only

Solution: Statement I is incorrect: 'S-shaped growth under resource limitation' corresponds to Logistic growth, not Carrying capacity. Statement II is correct because 'Maximum population an environment can sustain' corresponds to Carrying capacity. Statement III is correct because 'J-shaped population growth under unlimited resources' corresponds to Exponential growth. Statement IV is correct because 'One benefits while harming a host' corresponds to Parasitism. Therefore, the valid combination is II, III and IV only.

Distractor review: A: This combination omits true statement(s) III, IV. B: This combination includes false statement(s) I and omits true statement(s) II, III, IV. C: This combination includes false statement(s) I and omits true statement(s) III.

BIO-12-CH13-0034 | Challenging | Population growth

Correct answer: D. I and III only

Solution: Statement I is correct because 'Intrinsic rate of natural increase' corresponds to Biotic potential. Statement II is incorrect: 'J-shaped population growth under unlimited resources' corresponds to Exponential growth, not Carrying capacity. Statement III is correct because 'Maximum population an environment can sustain' corresponds to Carrying capacity. Statement IV is incorrect: 'Number of individuals per unit area or volume' corresponds to Population density, not Biotic potential. Therefore, the valid combination is I and III only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) I, III. B: This combination omits true statement(s) III. C: This combination includes false statement(s) II, IV and omits true statement(s) I, III.

BIO-12-CH13-0035 | Hard | Positive interactions

Correct answer: A. Both species benefit - Age structure

Solution: 'Both species benefit' should be paired with Mutualism, not Age structure. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Intrinsic rate of natural increase - Biotic potential. C: This is a valid association: Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation. D: This is a valid association: Number of births during a given period - Natality.

BIO-12-CH13-0036 | Easy | Population growth

Correct answer: D. Exponential growth

Solution: 'J-shaped population growth under unlimited resources' is the defining description for Exponential growth. The other choices belong to different chapter concepts.

Distractor review: A: Biotic potential belongs with 'Intrinsic rate of natural increase', whereas the tested description is 'J-shaped population growth under unlimited resources'. B: Carrying capacity belongs with 'Maximum population an environment can sustain', whereas the tested description is 'J-shaped population growth under unlimited resources'. C: Water-conservation adaptations belongs with 'Concentrated urine and reduced sweating in desert mammals', whereas the tested description is 'J-shaped population growth under unlimited resources'.

BIO-12-CH13-0037 | Medium | Negative interactions**Correct answer: D. Predation**

Solution: The observation 'One organism kills and consumes another' points to Predation; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Dormancy belongs with 'State of reduced activity during an unfavourable period', whereas the tested description is 'One organism kills and consumes another'. B: Declining population belongs with 'Narrow-base age pyramid', whereas the tested description is 'One organism kills and consumes another'. C: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'One organism kills and consumes another'.

BIO-12-CH13-0038 | Medium | Population attributes**Correct answer: C. Mortality**

Solution: The chapter links Number of deaths during a given period with Mortality, so Mortality completes the statement.

Distractor review: A: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'Number of deaths during a given period'. B: Commensalism belongs with 'One benefits and the other is unaffected', whereas the tested description is 'Number of deaths during a given period'. D: Conformers belongs with 'Organisms whose body temperature changes with surroundings', whereas the tested description is 'Number of deaths during a given period'.

BIO-12-CH13-0039 | Medium | Population growth**Correct answer: A. S-shaped growth under resource limitation - Logistic growth**

Solution: The accurate entry is 'S-shaped growth under resource limitation - Logistic growth'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Number of deaths during a given period' should be paired with Mortality, not Natality. C: 'Close association often benefiting both partners' should be paired with Symbiosis, not Logistic growth. D: 'Number of individuals per unit area or volume' should be paired with Population density, not Mortality.

BIO-12-CH13-0040 | Easy | Population growth**Correct answer: D. Carrying capacity**

Solution: 'Maximum population an environment can sustain' is the defining description for Carrying capacity. The other choices belong to different chapter concepts.

Distractor review: A: Mutualism belongs with 'Both species benefit', whereas the tested description is 'Maximum population an environment can sustain'. B: Exponential growth belongs with 'J-shaped population growth under unlimited resources', whereas the tested description is 'Maximum population an environment can sustain'. C: Expanding population belongs with 'Broad-base age pyramid', whereas the tested description is 'Maximum population an environment can sustain'.

BIO-12-CH13-0041 | Easy | Abiotic factors**Correct answer: A. State of reduced activity during an unfavourable period**

Solution: Dormancy represents 'State of reduced activity during an unfavourable period'. Each alternative describes a different term or result.

Distractor review: B: 'Seasonal movement to a favourable habitat' identifies Migration, not Dormancy. C: 'One benefits and the other is unaffected' identifies Commensalism, not Dormancy. D: 'Organisms maintaining a nearly constant internal temperature' identifies Regulators, not Dormancy.

BIO-12-CH13-0042 | Hard | Population growth**Correct answer: A. J-shaped population growth under unlimited resources - Population density**

Solution: 'J-shaped population growth under unlimited resources' should be paired with Exponential growth, not Population density. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: One organism kills and consumes another - Predation. C: This is a valid association: Individuals before reproductive maturity - Pre-reproductive group. D: This is a valid association: State of reduced activity during an unfavourable period - Dormancy.

BIO-12-CH13-0043 | Medium | Adaptations

Correct answer: B. Concentrated urine and reduced sweating in desert mammals - Water-conservation adaptations

Solution: The accurate entry is 'Concentrated urine and reduced sweating in desert mammals - Water-conservation adaptations'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Organisms whose body temperature changes with surroundings' should be paired with Conformers, not Expanding population. C: 'One benefits and the other is unaffected' should be paired with Commensalism, not Predation. D: 'Organisms maintaining a nearly constant internal temperature' should be paired with Regulators, not Commensalism.

BIO-12-CH13-0044 | Challenging | Abiotic factors

Correct answer: C. I, III and IV only

Solution: Statement I is correct because 'Organisms whose body temperature changes with surroundings' corresponds to Conformers. Statement II is incorrect: 'State of reduced activity during an unfavourable period' corresponds to Dormancy, not Regulators. Statement III is correct because 'Organisms maintaining a nearly constant internal temperature' corresponds to Regulators. Statement IV is correct because 'One benefits and the other is unaffected' corresponds to Commensalism. Therefore, the valid combination is I, III and IV only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) I, IV. B: This combination omits true statement(s) III, IV. D: This combination omits true statement(s) I, IV.

BIO-12-CH13-0045 | Hard | Population growth

Correct answer: A. Intrinsic rate of natural increase - Symbiosis

Solution: 'Intrinsic rate of natural increase' should be paired with Biotic potential, not Symbiosis. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Organisms maintaining a nearly constant internal temperature - Regulators. C: This is a valid association: One benefits while harming a host - Parasitism. D: This is a valid association: Photosynthetic pathway that lets desert plants keep stomata closed by day - CAM.

BIO-12-CH13-0046 | Easy | Positive interactions

Correct answer: B. Symbiosis

Solution: 'Close association often benefiting both partners' is the defining description for Symbiosis. The other choices belong to different chapter concepts.

Distractor review: A: Fungus and alga or cyanobacterium belongs with 'Lichen partnership', whereas the tested description is 'Close association often benefiting both partners'. C: Mutualism belongs with 'Both species benefit', whereas the tested description is 'Close association often benefiting both partners'. D: Migration belongs with 'Seasonal movement to a favourable habitat', whereas the tested description is 'Close association often benefiting both partners'.

BIO-12-CH13-0047 | Easy | Negative interactions

Correct answer: D. One is inhibited while the other is unaffected

Solution: Amensalism represents 'One is inhibited while the other is unaffected'. Each alternative describes a different term or result.

Distractor review: A: 'One benefits while harming a host' identifies Parasitism, not Amensalism. B: 'State of reduced activity during an unfavourable period' identifies Dormancy, not Amensalism. C: 'Intrinsic rate of natural increase' identifies Biotic potential, not Amensalism.

BIO-12-CH13-0048 | Medium | Age pyramids

Correct answer: D. Pre-reproductive group

Solution: The observation 'Individuals before reproductive maturity' points to Pre-reproductive group; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Commensalism belongs with 'One benefits and the other is unaffected', whereas the tested description is 'Individuals before reproductive maturity'. B: Parasitism belongs with 'One benefits while harming a host', whereas the tested description is 'Individuals before reproductive maturity'. C: Symbiosis belongs with 'Close association often benefiting both partners', whereas the tested description is 'Individuals before reproductive maturity'.

BIO-12-CH13-0049 | Medium | Positive interactions**Correct answer: B. Both species benefit - Mutualism**

Solution: The accurate entry is 'Both species benefit - Mutualism'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Number of deaths during a given period' should be paired with Mortality, not Biochemical cold adaptation. C: 'Production of antifreeze compounds in cold-water organisms' should be paired with Biochemical cold adaptation, not Mutualism. D: 'Number of births during a given period' should be paired with Natality, not Regulators.

BIO-12-CH13-0050 | Hard | Population attributes**Correct answer: C. Proportion of individuals in different age groups - Carrying capacity**

Solution: 'Proportion of individuals in different age groups' should be paired with Age structure, not Carrying capacity. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Photosynthetic pathway that lets desert plants keep stomata closed by day - CAM. B: This is a valid association: Close association often benefiting both partners - Symbiosis. D: This is a valid association: Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation.

BIO-12-CH13-0051 | Medium | Adaptations**Correct answer: D. CAM**

Solution: The observation 'Photosynthetic pathway that lets desert plants keep stomata closed by day' points to CAM; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Migration belongs with 'Seasonal movement to a favourable habitat', whereas the tested description is 'Photosynthetic pathway that lets desert plants keep stomata closed by day'. B: Fungus and alga or cyanobacterium belongs with 'Lichen partnership', whereas the tested description is 'Photosynthetic pathway that lets desert plants keep stomata closed by day'. C: Water-conservation adaptations belongs with 'Concentrated urine and reduced sweating in desert mammals', whereas the tested description is 'Photosynthetic pathway that lets desert plants keep stomata closed by day'.

BIO-12-CH13-0052 | Easy | Positive interactions**Correct answer: B. Mutualism**

Solution: 'Both species benefit' is the defining description for Mutualism. The other choices belong to different chapter concepts.

Distractor review: A: Mortality belongs with 'Number of deaths during a given period', whereas the tested description is 'Both species benefit'. C: Biotic potential belongs with 'Intrinsic rate of natural increase', whereas the tested description is 'Both species benefit'. D: Natality belongs with 'Number of births during a given period', whereas the tested description is 'Both species benefit'.

BIO-12-CH13-0053 | Medium | Negative interactions**Correct answer: B. Parasitism**

Solution: The chapter links One benefits while harming a host with Parasitism, so Parasitism completes the statement.

Distractor review: A: Pre-reproductive group belongs with 'Individuals before reproductive maturity', whereas the tested description is 'One benefits while harming a host'. C: Amensalism belongs with 'One is inhibited while the other is unaffected', whereas the tested description is 'One benefits while harming a host'. D: Stable population belongs with 'Nearly bell-shaped age pyramid', whereas the tested description is 'One benefits while harming a host'.

BIO-12-CH13-0054 | Easy | Adaptations**Correct answer: D. CAM**

Solution: 'Photosynthetic pathway that lets desert plants keep stomata closed by day' is the defining description for CAM. The other choices belong to different chapter concepts.

Distractor review: A: Mortality belongs with 'Number of deaths during a given period', whereas the tested description is 'Photosynthetic pathway that lets desert plants keep stomata closed by day'. B: Predation belongs with 'One organism kills and consumes another', whereas the tested description is 'Photosynthetic pathway that lets desert plants keep stomata closed by day'. C: Migration belongs with 'Seasonal movement to a favourable habitat', whereas the tested description is 'Photosynthetic pathway that lets desert plants keep stomata closed by day'.

BIO-12-CH13-0055 | Medium | Population attributes**Correct answer: C. Population density**

Solution: The chapter links Number of individuals per unit area or volume with Population density, so Population density completes the statement.

Distractor review: A: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'Number of individuals per unit area or volume'. B: Age structure belongs with 'Proportion of individuals in different age groups', whereas the tested description is 'Number of individuals per unit area or volume'. D: Water-conservation adaptations belongs with 'Concentrated urine and reduced sweating in desert mammals', whereas the tested description is 'Number of individuals per unit area or volume'.

BIO-12-CH13-0056 | Easy | Negative interactions**Correct answer: B. One organism kills and consumes another**

Solution: Predation represents 'One organism kills and consumes another'. Each alternative describes a different term or result.

Distractor review: A: 'Production of antifreeze compounds in cold-water organisms' identifies Biochemical cold adaptation, not Predation. C: 'State of reduced activity during an unfavourable period' identifies Dormancy, not Predation. D: 'Close association often benefiting both partners' identifies Symbiosis, not Predation.

BIO-12-CH13-0057 | Medium | Abiotic factors**Correct answer: D. Migration**

Solution: The chapter links Seasonal movement to a favourable habitat with Migration, so Migration completes the statement.

Distractor review: A: Regulators belongs with 'Organisms maintaining a nearly constant internal temperature', whereas the tested description is 'Seasonal movement to a favourable habitat'. B: Competition belongs with 'Both species are adversely affected while sharing a limiting resource', whereas the tested description is 'Seasonal movement to a favourable habitat'. C: Conformers belongs with 'Organisms whose body temperature changes with surroundings', whereas the tested description is 'Seasonal movement to a favourable habitat'.

BIO-12-CH13-0058 | Hard | Abiotic factors**Correct answer: A. Seasonal movement to a favourable habitat - Pre-reproductive group**

Solution: 'Seasonal movement to a favourable habitat' should be paired with Migration, not Pre-reproductive group. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Proportion of individuals in different age groups - Age structure. C: This is a valid association: Both species are adversely affected while sharing a limiting resource - Competition. D: This is a valid association: One organism kills and consumes another - Predation.

BIO-12-CH13-0059 | Medium | Population attributes**Correct answer: B. Age structure**

Solution: The observation 'Proportion of individuals in different age groups' points to Age structure; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Population density belongs with 'Number of individuals per unit area or volume', whereas the tested description is 'Proportion of individuals in different age groups'. C: Logistic growth belongs with 'S-shaped growth under resource limitation', whereas the tested description is 'Proportion of individuals in different age groups'. D: Carrying capacity belongs with 'Maximum population an environment can sustain', whereas the tested description is 'Proportion of individuals in different age groups'.

BIO-12-CH13-0060 | Medium | Positive interactions**Correct answer: B. Fungus and alga or cyanobacterium**

Solution: The observation 'Lichen partnership' points to Fungus and alga or cyanobacterium; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Symbiosis belongs with 'Close association often benefiting both partners', whereas the tested description is 'Lichen partnership'. C: Mortality belongs with 'Number of deaths during a given period', whereas the tested description is 'Lichen partnership'. D: Predation belongs with 'One organism kills and consumes another', whereas the tested description is 'Lichen partnership'.

BIO-12-CH13-0061 | Hard | Negative interactions

Correct answer: C. Both species are adversely affected while sharing a limiting resource - Exponential growth

Solution: 'Both species are adversely affected while sharing a limiting resource' should be paired with Competition, not Exponential growth. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation. B: This is a valid association: Broad-base age pyramid - Expanding population. D: This is a valid association: Number of births during a given period - Natality.

BIO-12-CH13-0062 | Hard | Adaptations

Correct answer: C. Desert plant whose leaves are reduced to spines - Regulators

Solution: 'Desert plant whose leaves are reduced to spines' should be paired with Opuntia, not Regulators. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Individuals before reproductive maturity - Pre-reproductive group. B: This is a valid association: One benefits and the other is unaffected - Commensalism. D: This is a valid association: Intrinsic rate of natural increase - Biotic potential.

BIO-12-CH13-0063 | Hard | Population attributes

Correct answer: A. Number of births during a given period - Dormancy

Solution: 'Number of births during a given period' should be paired with Natality, not Dormancy. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation. C: This is a valid association: Organisms maintaining a nearly constant internal temperature - Regulators. D: This is a valid association: Desert plant whose leaves are reduced to spines - Opuntia.

BIO-12-CH13-0064 | Hard | Abiotic factors

Correct answer: A. Organisms whose body temperature changes with surroundings - Conformers; S-shaped growth under resource limitation - Logistic growth

Solution: Both parts of 'Organisms whose body temperature changes with surroundings - Conformers; S-shaped growth under resource limitation - Logistic growth' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Proportion of individuals in different age groups' requires Age structure, and 'Both species are adversely affected while sharing a limiting resource' requires Competition. C: The answers have been interchanged: 'Narrow-base age pyramid' requires Declining population, and 'One is inhibited while the other is unaffected' requires Amensalism. D: The answers have been interchanged: 'Both species are adversely affected while sharing a limiting resource' requires Competition, and 'Narrow-base age pyramid' requires Declining population.

BIO-12-CH13-0065 | Easy | Negative interactions

Correct answer: D. Predation

Solution: 'One organism kills and consumes another' is the defining description for Predation. The other choices belong to different chapter concepts.

Distractor review: A: Amensalism belongs with 'One is inhibited while the other is unaffected', whereas the tested description is 'One organism kills and consumes another'. B: Declining population belongs with 'Narrow-base age pyramid', whereas the tested description is 'One organism kills and consumes another'. C: Symbiosis belongs with 'Close association often benefiting both partners', whereas the tested description is 'One organism kills and consumes another'.

BIO-12-CH13-0066 | Hard | Age pyramids

Correct answer: D. Broad-base age pyramid - Expanding population; Proportion of individuals in different age groups - Age structure

Solution: Both parts of 'Broad-base age pyramid - Expanding population; Proportion of individuals in different age groups - Age structure' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'J-shaped population growth under unlimited resources' requires Exponential growth, and 'Desert plant whose leaves are reduced to spines' requires Opuntia. B: The answers have been interchanged: 'Desert plant whose leaves are reduced to spines' requires Opuntia, and 'Intrinsic rate of natural increase' requires Biotic potential. C: The answers have been interchanged: 'Intrinsic rate of natural increase' requires Biotic potential, and 'Concentrated urine and reduced sweating in desert mammals' requires Water-conservation adaptations.

BIO-12-CH13-0067 | Challenging | Abiotic factors**Correct answer: B. I and II only**

Solution: Statement I is correct because 'State of reduced activity during an unfavourable period' corresponds to Dormancy. Statement II is correct because 'Organisms whose body temperature changes with surroundings' corresponds to Conformers. Statement III is incorrect: 'Organisms maintaining a nearly constant internal temperature' corresponds to Regulators, not Carrying capacity. Statement IV is incorrect: 'Maximum population an environment can sustain' corresponds to Carrying capacity, not Dormancy. Therefore, the valid combination is I and II only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) I. C: This combination includes false statement(s) IV and omits true statement(s) II. D: This combination includes false statement(s) III, IV and omits true statement(s) II.

BIO-12-CH13-0068 | Medium | Abiotic factors**Correct answer: C. Organisms whose body temperature changes with surroundings - Conformers**

Solution: The accurate entry is 'Organisms whose body temperature changes with surroundings - Conformers'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Proportion of individuals in different age groups' should be paired with Age structure, not Declining population. B: 'Desert plant whose leaves are reduced to spines' should be paired with Opuntia, not Competition. D: 'S-shaped growth under resource limitation' should be paired with Logistic growth, not Opuntia.

BIO-12-CH13-0069 | Hard | Abiotic factors**Correct answer: C. State of reduced activity during an unfavourable period - Dormancy; S-shaped growth under resource limitation - Logistic growth**

Solution: Both parts of 'State of reduced activity during an unfavourable period - Dormancy; S-shaped growth under resource limitation - Logistic growth' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Production of antifreeze compounds in cold-water organisms' requires Biochemical cold adaptation, and 'Both species benefit' requires Mutualism. B: The answers have been interchanged: 'Individuals before reproductive maturity' requires Pre-reproductive group, and 'Production of antifreeze compounds in cold-water organisms' requires Biochemical cold adaptation. D: The answers have been interchanged: 'One benefits and the other is unaffected' requires Commensalism, and 'Individuals before reproductive maturity' requires Pre-reproductive group.

BIO-12-CH13-0070 | Medium | Abiotic factors**Correct answer: B. Dormancy**

Solution: The chapter links State of reduced activity during an unfavourable period with Dormancy, so Dormancy completes the statement.

Distractor review: A: Commensalism belongs with 'One benefits and the other is unaffected', whereas the tested description is 'State of reduced activity during an unfavourable period'. C: Pre-reproductive group belongs with 'Individuals before reproductive maturity', whereas the tested description is 'State of reduced activity during an unfavourable period'. D: Mutualism belongs with 'Both species benefit', whereas the tested description is 'State of reduced activity during an unfavourable period'.

BIO-12-CH13-0071 | Easy | Population attributes**Correct answer: A. Age structure**

Solution: 'Proportion of individuals in different age groups' is the defining description for Age structure. The other choices belong to different chapter concepts.

Distractor review: B: Logistic growth belongs with 'S-shaped growth under resource limitation', whereas the tested description is 'Proportion of individuals in different age groups'. C: Population density belongs with 'Number of individuals per unit area or volume', whereas the tested description is 'Proportion of individuals in different age groups'. D: Mortality belongs with 'Number of deaths during a given period', whereas the tested description is 'Proportion of individuals in different age groups'.

BIO-12-CH13-0072 | Medium | Population attributes**Correct answer: B. Proportion of individuals in different age groups - Age structure**

Solution: The accurate entry is 'Proportion of individuals in different age groups - Age structure'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Maximum population an environment can sustain' should be paired with Carrying capacity, not Symbiosis. C: 'Photosynthetic pathway that lets desert plants keep stomata closed by day' should be paired with CAM, not Biochemical cold adaptation. D: 'Nearly bell-shaped age pyramid' should be paired with Stable population, not Age structure.

BIO-12-CH13-0073 | Hard | Population growth**Correct answer: B. Maximum population an environment can sustain - Mutualism**

Solution: 'Maximum population an environment can sustain' should be paired with Carrying capacity, not Mutualism. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Individuals before reproductive maturity - Pre-reproductive group. C: This is a valid association: Proportion of individuals in different age groups - Age structure. D: This is a valid association: Broad-base age pyramid - Expanding population.

BIO-12-CH13-0074 | Hard | Adaptations**Correct answer: C. Photosynthetic pathway that lets desert plants keep stomata closed by day - Symbiosis**

Solution: 'Photosynthetic pathway that lets desert plants keep stomata closed by day' should be paired with CAM, not Symbiosis. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Seasonal movement to a favourable habitat - Migration. B: This is a valid association: Number of deaths during a given period - Mortality. D: This is a valid association: Lichen partnership - Fungus and alga or cyanobacterium.

BIO-12-CH13-0075 | Easy | Population attributes**Correct answer: D. Natality**

Solution: 'Number of births during a given period' is the defining description for Natality. The other choices belong to different chapter concepts.

Distractor review: A: Regulators belongs with 'Organisms maintaining a nearly constant internal temperature', whereas the tested description is 'Number of births during a given period'. B: Opuntia belongs with 'Desert plant whose leaves are reduced to spines', whereas the tested description is 'Number of births during a given period'. C: Mortality belongs with 'Number of deaths during a given period', whereas the tested description is 'Number of births during a given period'.

BIO-12-CH13-0076 | Easy | Age pyramids**Correct answer: B. Broad-base age pyramid**

Solution: Expanding population represents 'Broad-base age pyramid'. Each alternative describes a different term or result.

Distractor review: A: 'State of reduced activity during an unfavourable period' identifies Dormancy, not Expanding population. C: 'J-shaped population growth under unlimited resources' identifies Exponential growth, not Expanding population. D: 'Individuals before reproductive maturity' identifies Pre-reproductive group, not Expanding population.

BIO-12-CH13-0077 | Medium | Negative interactions**Correct answer: C. Amensalism**

Solution: The chapter links One is inhibited while the other is unaffected with Amensalism, so Amensalism completes the statement.

Distractor review: A: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'One is inhibited while the other is unaffected'. B: Population density belongs with 'Number of individuals per unit area or volume', whereas the tested description is 'One is inhibited while the other is unaffected'. D: Water-conservation adaptations belongs with 'Concentrated urine and reduced sweating in desert mammals', whereas the tested description is 'One is inhibited while the other is unaffected'.

BIO-12-CH13-0078 | Easy | Adaptations**Correct answer: B. Water-conservation adaptations**

Solution: 'Concentrated urine and reduced sweating in desert mammals' is the defining description for Water-conservation adaptations. The other choices belong to different chapter concepts.

Distractor review: A: CAM belongs with 'Photosynthetic pathway that lets desert plants keep stomata closed by day', whereas the tested description is 'Concentrated urine and reduced sweating in desert mammals'. C: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'Concentrated urine and reduced sweating in desert mammals'. D: Regulators belongs with 'Organisms maintaining a nearly constant internal temperature', whereas the tested description is 'Concentrated urine and reduced sweating in desert mammals'.

BIO-12-CH13-0079 | Hard | Positive interactions**Correct answer: A. Close association often benefiting both partners - Amensalism**

Solution: 'Close association often benefiting both partners' should be paired with Symbiosis, not Amensalism. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Organisms whose body temperature changes with surroundings - Conformers. C: This is a valid association: Broad-base age pyramid - Expanding population. D: This is a valid association: Proportion of individuals in different age groups - Age structure.

BIO-12-CH13-0080 | Hard | Negative interactions**Correct answer: B. One organism kills and consumes another - Pre-reproductive group**

Solution: 'One organism kills and consumes another' should be paired with Predation, not Pre-reproductive group. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation. C: This is a valid association: Close association often benefiting both partners - Symbiosis. D: This is a valid association: State of reduced activity during an unfavourable period - Dormancy.

BIO-12-CH13-0081 | Hard | Population attributes**Correct answer: C. Number of individuals per unit area or volume - Population density; Narrow-base age pyramid - Declining population**

Solution: Both parts of 'Number of individuals per unit area or volume - Population density; Narrow-base age pyramid - Declining population' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Close association often benefiting both partners' requires Symbiosis, and 'Production of antifreeze compounds in cold-water organisms' requires Biochemical cold adaptation. B: The answers have been interchanged: 'One benefits while harming a host' requires Parasitism, and 'Close association often benefiting both partners' requires Symbiosis. D: The answers have been interchanged: 'Concentrated urine and reduced sweating in desert mammals' requires Water-conservation adaptations, and 'One benefits while harming a host' requires Parasitism.

BIO-12-CH13-0082 | Easy | Abiotic factors**Correct answer: B. Migration**

Solution: 'Seasonal movement to a favourable habitat' is the defining description for Migration. The other choices belong to different chapter concepts.

Distractor review: A: Exponential growth belongs with 'J-shaped population growth under unlimited resources', whereas the tested description is 'Seasonal movement to a favourable habitat'. C: Population density belongs with 'Number of individuals per unit area or volume', whereas the tested description is 'Seasonal movement to a favourable habitat'. D: Competition belongs with 'Both species are adversely affected while sharing a limiting resource', whereas the tested description is 'Seasonal movement to a favourable habitat'.

BIO-12-CH13-0083 | Easy | Positive interactions**Correct answer: B. Fungus and alga or cyanobacterium**

Solution: 'Lichen partnership' is the defining description for Fungus and alga or cyanobacterium. The other choices belong to different chapter concepts.

Distractor review: A: Logistic growth belongs with 'S-shaped growth under resource limitation', whereas the tested description is 'Lichen partnership'. C: Symbiosis belongs with 'Close association often benefiting both partners', whereas the tested description is 'Lichen partnership'. D: Mortality belongs with 'Number of deaths during a given period', whereas the tested description is 'Lichen partnership'.

BIO-12-CH13-0084 | Easy | Age pyramids**Correct answer: C. Pre-reproductive group**

Solution: 'Individuals before reproductive maturity' is the defining description for Pre-reproductive group. The other choices belong to different chapter concepts.

Distractor review: A: Natality belongs with 'Number of births during a given period', whereas the tested description is 'Individuals before reproductive maturity'. B: Mutualism belongs with 'Both species benefit', whereas the tested description is 'Individuals before reproductive maturity'. D: Declining population belongs with 'Narrow-base age pyramid', whereas the tested description is 'Individuals before reproductive maturity'.

BIO-12-CH13-0085 | Easy | Abiotic factors**Correct answer: A. Organisms whose body temperature changes with surroundings**

Solution: Conformers represents 'Organisms whose body temperature changes with surroundings'. Each alternative describes a different term or result.

Distractor review: B: 'Both species are adversely affected while sharing a limiting resource' identifies Competition, not Conformers. C: 'Lichen partnership' identifies Fungus and alga or cyanobacterium, not Conformers. D: 'Desert plant whose leaves are reduced to spines' identifies Opuntia, not Conformers.

BIO-12-CH13-0086 | Easy | Positive interactions**Correct answer: D. Both species benefit**

Solution: Mutualism represents 'Both species benefit'. Each alternative describes a different term or result.

Distractor review: A: 'Number of births during a given period' identifies Natality, not Mutualism. B: 'Number of deaths during a given period' identifies Mortality, not Mutualism. C: 'Lichen partnership' identifies Fungus and alga or cyanobacterium, not Mutualism.

BIO-12-CH13-0087 | Challenging | Negative interactions**Correct answer: C. I, II and IV only**

Solution: Statement I is correct because 'One organism kills and consumes another' corresponds to Predation. Statement II is correct because 'Both species are adversely affected while sharing a limiting resource' corresponds to Competition. Statement III is incorrect: 'One is inhibited while the other is unaffected' corresponds to Amensalism, not Mortality. Statement IV is correct because 'Number of deaths during a given period' corresponds to Mortality. Therefore, the valid combination is I, II and IV only.

Distractor review: A: This combination includes false statement(s) III and omits true statement(s) II. B: This combination omits true statement(s) I. D: This combination omits true statement(s) I, IV.

BIO-12-CH13-0088 | Medium | Population growth**Correct answer: A. Biotic potential**

Solution: The chapter links Intrinsic rate of natural increase with Biotic potential, so Biotic potential completes the statement.

Distractor review: B: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'Intrinsic rate of natural increase'. C: Logistic growth belongs with 'S-shaped growth under resource limitation', whereas the tested description is 'Intrinsic rate of natural increase'. D: Symbiosis belongs with 'Close association often benefiting both partners', whereas the tested description is 'Intrinsic rate of natural increase'.

BIO-12-CH13-0089 | Medium | Population growth**Correct answer: C. Logistic growth**

Solution: The chapter links S-shaped growth under resource limitation with Logistic growth, so Logistic growth completes the statement.

Distractor review: A: Regulators belongs with 'Organisms maintaining a nearly constant internal temperature', whereas the tested description is 'S-shaped growth under resource limitation'. B: Exponential growth belongs with 'J-shaped population growth under unlimited resources', whereas the tested description is 'S-shaped growth under resource limitation'. D: Commensalism belongs with 'One benefits and the other is unaffected', whereas the tested description is 'S-shaped growth under resource limitation'.

BIO-12-CH13-0090 | Medium | Age pyramids**Correct answer: D. Expanding population**

Solution: The chapter links Broad-base age pyramid with Expanding population, so Expanding population completes the statement.

Distractor review: A: Stable population belongs with 'Nearly bell-shaped age pyramid', whereas the tested description is 'Broad-base age pyramid'. B: Exponential growth belongs with 'J-shaped population growth under unlimited resources', whereas the tested description is 'Broad-base age pyramid'. C: Pre-reproductive group belongs with 'Individuals before reproductive maturity', whereas the tested description is 'Broad-base age pyramid'.

BIO-12-CH13-0091 | Easy | Population attributes**Correct answer: C. Population density**

Solution: 'Number of individuals per unit area or volume' is the defining description for Population density. The other choices belong to different chapter concepts.

Distractor review: A: Age structure belongs with 'Proportion of individuals in different age groups', whereas the tested description is 'Number of individuals per unit area or volume'. B: Parasitism belongs with 'One benefits while harming a host', whereas the tested description is 'Number of individuals per unit area or volume'. D: Symbiosis belongs with 'Close association often benefiting both partners', whereas the tested description is 'Number of individuals per unit area or volume'.

BIO-12-CH13-0092 | Easy | Age pyramids**Correct answer: B. Declining population**

Solution: 'Narrow-base age pyramid' is the defining description for Declining population. The other choices belong to different chapter concepts.

Distractor review: A: Biochemical cold adaptation belongs with 'Production of antifreeze compounds in cold-water organisms', whereas the tested description is 'Narrow-base age pyramid'. C: Age structure belongs with 'Proportion of individuals in different age groups', whereas the tested description is 'Narrow-base age pyramid'. D: Pre-reproductive group belongs with 'Individuals before reproductive maturity', whereas the tested description is 'Narrow-base age pyramid'.

BIO-12-CH13-0093 | Challenging | Adaptations**Correct answer: B. IV only**

Solution: Statement I is incorrect: 'Production of antifreeze compounds in cold-water organisms' corresponds to Biochemical cold adaptation, not Opuntia. Statement II is incorrect: 'Desert plant whose leaves are reduced to spines' corresponds to Opuntia, not Water-conservation adaptations. Statement III is incorrect: 'Concentrated urine and reduced sweating in desert mammals' corresponds to Water-conservation adaptations, not Conformers. Statement IV is correct because 'Organisms whose body temperature changes with surroundings' corresponds to Conformers. Therefore, the valid combination is IV only.

Distractor review: A: This combination includes false statement(s) I, III. C: This combination includes false statement(s) I and omits true statement(s) IV. D: This combination includes false statement(s) I, II and omits true statement(s) IV.

BIO-12-CH13-0094 | Medium | Adaptations**Correct answer: C. Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation**

Solution: The accurate entry is 'Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Broad-base age pyramid' should be paired with Expanding population, not Amensalism. B: 'Number of individuals per unit area or volume' should be paired with Population density, not Declining population. D: 'State of reduced activity during an unfavourable period' should be paired with Dormancy, not Expanding population.

BIO-12-CH13-0095 | Hard | Negative interactions**Correct answer: A. One is inhibited while the other is unaffected - Amensalism; State of reduced activity during an unfavourable period - Dormancy**

Solution: Both parts of 'One is inhibited while the other is unaffected - Amensalism; State of reduced activity during an unfavourable period - Dormancy' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Number of individuals per unit area or volume' requires Population density, and 'Concentrated urine and reduced sweating in desert mammals' requires Water-conservation adaptations. C: The answers have been interchanged: 'Production of antifreeze compounds in cold-water organisms' requires Biochemical cold adaptation, and 'Intrinsic rate of natural increase' requires Biotic potential. D: The answers have been interchanged: 'Organisms maintaining a nearly constant internal temperature' requires Regulators, and 'Broad-base age pyramid' requires Expanding population.

BIO-12-CH13-0096 | Medium | Population attributes**Correct answer: C. Mortality**

Solution: The observation 'Number of deaths during a given period' points to Mortality; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Population density belongs with 'Number of individuals per unit area or volume', whereas the tested description is 'Number of deaths during a given period'. B: Biotic potential belongs with 'Intrinsic rate of natural increase', whereas the tested description is 'Number of deaths during a given period'. D: Age structure belongs with 'Proportion of individuals in different age groups', whereas the tested description is 'Number of deaths during a given period'.

BIO-12-CH13-0097 | Hard | Population attributes**Correct answer: B. Number of individuals per unit area or volume - Declining population**

Solution: 'Number of individuals per unit area or volume' should be paired with Population density, not Declining population. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Close association often benefiting both partners - Symbiosis. C: This is a valid association: Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation. D: This is a valid association: Seasonal movement to a favourable habitat - Migration.

BIO-12-CH13-0098 | Challenging | Negative interactions**Correct answer: D. I, II and III only**

Solution: Statement I is correct because 'One is inhibited while the other is unaffected' corresponds to Amensalism. Statement II is correct because 'One organism kills and consumes another' corresponds to Predation. Statement III is correct because 'Both species are adversely affected while sharing a limiting resource' corresponds to Competition. Statement IV is incorrect: 'Both species benefit' corresponds to Mutualism, not Amensalism. Therefore, the valid combination is I, II and III only.

Distractor review: A: This combination omits true statement(s) I, II. B: This combination includes false statement(s) IV and omits true statement(s) II, III. C: This combination includes false statement(s) IV and omits true statement(s) I.

BIO-12-CH13-0099 | Medium | Adaptations**Correct answer: A. Water-conservation adaptations**

Solution: The chapter links Concentrated urine and reduced sweating in desert mammals with Water-conservation adaptations, so Water-conservation adaptations completes the statement.

Distractor review: B: Conformers belongs with 'Organisms whose body temperature changes with surroundings', whereas the tested description is 'Concentrated urine and reduced sweating in desert mammals'. C: Dormancy belongs with 'State of reduced activity during an unfavourable period', whereas the tested description is 'Concentrated urine and reduced sweating in desert mammals'. D: Regulators belongs with 'Organisms maintaining a nearly constant internal temperature', whereas the tested description is 'Concentrated urine and reduced sweating in desert mammals'.

BIO-12-CH13-0100 | Medium | Abiotic factors**Correct answer: D. State of reduced activity during an unfavourable period - Dormancy**

Solution: The accurate entry is 'State of reduced activity during an unfavourable period - Dormancy'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Maximum population an environment can sustain' should be paired with Carrying capacity, not Dormancy. B: 'Both species benefit' should be paired with Mutualism, not Expanding population. C: 'Production of antifreeze compounds in cold-water organisms' should be paired with Biochemical cold adaptation, not Carrying capacity.

Standard 12 | Chapter 14

Ecosystems and Energy Flow

BIO-12-CH14-0001 | Medium | Matter flow through an ecosystem

Correct answer: B. Decomposers

Solution: Decomposers mineralize organic matter and return inorganic nutrients to the abiotic pool.

Distractor review: A: The alternative 'Producers only' conflicts with the labelled relationship, scale or mechanism in the figure. Decomposers mineralize organic matter and return inorganic nutrients to the abiotic pool. C: The alternative 'Primary consumers only' conflicts with the labelled relationship, scale or mechanism in the figure. Decomposers mineralize organic matter and return inorganic nutrients to the abiotic pool. D: The alternative 'Higher consumers only' conflicts with the labelled relationship, scale or mechanism in the figure. Decomposers mineralize organic matter and return inorganic nutrients to the abiotic pool.

BIO-12-CH14-0002 | Hard | Pyramid of energy

Correct answer: A. Usable energy is lost at each trophic transfer

Solution: Metabolism and heat loss mean that less usable energy is available at each successive trophic level.

Distractor review: B: The alternative 'Energy is fully recycled by decomposers' conflicts with the labelled relationship, scale or mechanism in the figure. Metabolism and heat loss mean that less usable energy is available at each successive trophic level. C: The alternative 'Consumers manufacture more solar energy' conflicts with the labelled relationship, scale or mechanism in the figure. Metabolism and heat loss mean that less usable energy is available at each successive trophic level. D: The alternative 'Tertiary consumers always have the greatest biomass' conflicts with the labelled relationship, scale or mechanism in the figure. Metabolism and heat loss mean that less usable energy is available at each successive trophic level.

BIO-12-CH14-0003 | Hard | Succession and services

Correct answer: C. Succession beginning on a previously lifeless surface - Pyramid of numbers

Solution: 'Succession beginning on a previously lifeless surface' should be paired with Primary succession, not Pyramid of numbers. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Conversion of organic nutrients to inorganic forms - Mineralization. B: This is a valid association: Pyramid showing standing crop - Pyramid of biomass. D: This is a valid association: Gross primary productivity minus plant respiration - Net primary productivity.

BIO-12-CH14-0004 | Medium | Nutrient cycles

Correct answer: A. Sedimentary rocks

Solution: The chapter links Major reservoir of phosphorus with Sedimentary rocks, so Sedimentary rocks completes the statement.

Distractor review: B: Decomposers belongs with 'Organisms breaking down dead organic matter', whereas the tested description is 'Major reservoir of phosphorus'. C: Primary succession belongs with 'Succession beginning on a previously lifeless surface', whereas the tested description is 'Major reservoir of phosphorus'. D: Energy pyramid belongs with 'Pyramid that is always upright', whereas the tested description is 'Major reservoir of phosphorus'.

BIO-12-CH14-0005 | Hard | Decomposition steps

Correct answer: C. Enzymatic breakdown by bacteria and fungi - Catabolism; Organisms breaking down dead organic matter - Decomposers

Solution: Both parts of 'Enzymatic breakdown by bacteria and fungi - Catabolism; Organisms breaking down dead organic matter - Decomposers' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Total photosynthetic production before respiratory loss' requires Gross primary productivity, and 'Pyramid showing standing crop' requires Pyramid of biomass. B: The answers have been interchanged: 'Succession beginning on a previously lifeless surface' requires Primary succession, and 'Food chain beginning with dead organic matter' requires Detritus food chain. D: The answers have been interchanged: 'Autotrophs converting solar energy to chemical energy' requires Producers, and 'Major reservoir of phosphorus' requires Sedimentary rocks.

BIO-12-CH14-0006 | Medium | Food chains and energy

Correct answer: C. Food web

Solution: The chapter links Interconnected food chains with Food web, so Food web completes the statement.

Distractor review: A: Net primary productivity belongs with 'Gross primary productivity minus plant respiration', whereas the tested description is 'Interconnected food chains'. B: Catabolism belongs with 'Enzymatic breakdown by bacteria and fungi', whereas the tested description is 'Interconnected food chains'. D: Leaching belongs with 'Downward movement of soluble inorganic nutrients', whereas the tested description is 'Interconnected food chains'.

BIO-12-CH14-0007 | Challenging | Matter flow through an ecosystem**Correct answer: B. Producer biomass rises while higher consumers lose a major food pathway**

Solution: Reduced grazing can initially increase producer biomass, while higher consumers lose energy transferred through primary consumers.

Distractor review: A: The alternative 'Producer biomass immediately becomes zero' conflicts with the labelled relationship, scale or mechanism in the figure. Reduced grazing can initially increase producer biomass, while higher consumers lose energy transferred through primary consumers. C: The alternative 'Higher consumers gain unlimited energy' conflicts with the labelled relationship, scale or mechanism in the figure. Reduced grazing can initially increase producer biomass, while higher consumers lose energy transferred through primary consumers. D: The alternative 'Nutrient cycling stops permanently in every compartment' conflicts with the labelled relationship, scale or mechanism in the figure. Reduced grazing can initially increase producer biomass, while higher consumers lose energy transferred through primary consumers.

BIO-12-CH14-0008 | Medium | Ecosystem organization**Correct answer: D. Decomposers**

Solution: The observation 'Organisms breaking down dead organic matter' points to Decomposers; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Producers belongs with 'Autotrophs converting solar energy to chemical energy', whereas the tested description is 'Organisms breaking down dead organic matter'. B: Mineralization belongs with 'Conversion of organic nutrients to inorganic forms', whereas the tested description is 'Organisms breaking down dead organic matter'. C: Grazing food chain belongs with 'Food chain beginning with living green plants', whereas the tested description is 'Organisms breaking down dead organic matter'.

BIO-12-CH14-0009 | Medium | Decomposition steps**Correct answer: A. Fragmentation**

Solution: The chapter links Breaking detritus into smaller particles with Fragmentation, so Fragmentation completes the statement.

Distractor review: B: Climax community belongs with 'Relatively stable final community', whereas the tested description is 'Breaking detritus into smaller particles'. C: Gross primary productivity belongs with 'Total photosynthetic production before respiratory loss', whereas the tested description is 'Breaking detritus into smaller particles'. D: Ten per cent belongs with 'Approximate energy passed to the next trophic level', whereas the tested description is 'Breaking detritus into smaller particles'.

BIO-12-CH14-0010 | Medium | Ecological pyramids**Correct answer: D. Pyramid showing standing crop - Pyramid of biomass**

Solution: The accurate entry is 'Pyramid showing standing crop - Pyramid of biomass'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Total photosynthetic production before respiratory loss' should be paired with Gross primary productivity, not Sedimentary rocks. B: 'Benefit such as pollination, climate regulation or purification' should be paired with Ecosystem service, not Biogeochemical cycling. C: 'Major reservoir of phosphorus' should be paired with Sedimentary rocks, not Atmosphere.

BIO-12-CH14-0011 | Hard | Ecosystem organization**Correct answer: C. Organisms breaking down dead organic matter - Mineralization**

Solution: 'Organisms breaking down dead organic matter' should be paired with Decomposers, not Mineralization. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Major reservoir of phosphorus - Sedimentary rocks. B: This is a valid association: Interconnected food chains - Food web. D: This is a valid association: Conversion of detritus into inorganic nutrients - Mineralization.

BIO-12-CH14-0012 | Medium | Nutrient cycles**Correct answer: A. Major reservoir of atmospheric nitrogen - Atmosphere**

Solution: The accurate entry is 'Major reservoir of atmospheric nitrogen - Atmosphere'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Food chain beginning with dead organic matter' should be paired with Detritus food chain, not Producers. C: 'Pyramid that is always upright' should be paired with Energy pyramid, not Secondary succession. D: 'Living components of an ecosystem' should be paired with Biotic components, not Leaching.

BIO-12-CH14-0013 | Hard | Decomposition steps

Correct answer: D. Breaking detritus into smaller particles - Fragmentation; Approximate energy passed to the next trophic level - Ten per cent

Solution: Both parts of 'Breaking detritus into smaller particles - Fragmentation; Approximate energy passed to the next trophic level - Ten per cent' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Food chain beginning with dead organic matter' requires Detritus food chain, and 'Succession beginning on a previously lifeless surface' requires Primary succession. B: The answers have been interchanged: 'Non-living physical and chemical components' requires Abiotic components, and 'Movement of nutrients between organisms and environment' requires Biogeochemical cycling. C: The answers have been interchanged: 'Succession beginning on a previously lifeless surface' requires Primary succession, and 'Total photosynthetic production before respiratory loss' requires Gross primary productivity.

BIO-12-CH14-0014 | Medium | Food chains and energy

Correct answer: B. Ten per cent

Solution: The observation 'Approximate energy passed to the next trophic level' points to Ten per cent; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Mineralization belongs with 'Conversion of detritus into inorganic nutrients', whereas the tested description is 'Approximate energy passed to the next trophic level'. C: Pyramid of biomass belongs with 'Pyramid showing standing crop', whereas the tested description is 'Approximate energy passed to the next trophic level'. D: Net primary productivity belongs with 'Gross primary productivity minus plant respiration', whereas the tested description is 'Approximate energy passed to the next trophic level'.

BIO-12-CH14-0015 | Hard | Nutrient cycles

Correct answer: A. Major reservoir of phosphorus - Energy pyramid

Solution: 'Major reservoir of phosphorus' should be paired with Sedimentary rocks, not Energy pyramid. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Interconnected food chains - Food web. C: This is a valid association: Succession beginning on a previously lifeless surface - Primary succession. D: This is a valid association: Organisms breaking down dead organic matter - Decomposers.

BIO-12-CH14-0016 | Easy | Pyramid of energy

Correct answer: D. 1,000 kJ

Solution: The diagram applies an approximate ten-percent transfer from 10,000 kJ in producers to 1,000 kJ in primary consumers.

Distractor review: A: The alternative '10,000 kJ' conflicts with the labelled relationship, scale or mechanism in the figure. The diagram applies an approximate ten-percent transfer from 10,000 kJ in producers to 1,000 kJ in primary consumers. B: The alternative '100 kJ' conflicts with the labelled relationship, scale or mechanism in the figure. The diagram applies an approximate ten-percent transfer from 10,000 kJ in producers to 1,000 kJ in primary consumers. C: The alternative '10 kJ' conflicts with the labelled relationship, scale or mechanism in the figure. The diagram applies an approximate ten-percent transfer from 10,000 kJ in producers to 1,000 kJ in primary consumers.

BIO-12-CH14-0017 | Medium | Succession and services

Correct answer: A. Relatively stable final community - Climax community

Solution: The accurate entry is 'Relatively stable final community - Climax community'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Food chain beginning with dead organic matter' should be paired with Detritus food chain, not Net primary productivity. C: 'Gross primary productivity minus plant respiration' should be paired with Net primary productivity, not Catabolism. D: 'Conversion of organic nutrients to inorganic forms' should be paired with Mineralization, not Climax community.

BIO-12-CH14-0018 | Easy | Decomposition steps

Correct answer: D. Breaking detritus into smaller particles

Solution: Fragmentation represents 'Breaking detritus into smaller particles'. Each alternative describes a different term or result.

Distractor review: A: 'Food chain beginning with dead organic matter' identifies Detritus food chain, not Fragmentation. B: 'Total photosynthetic production before respiratory loss' identifies Gross primary productivity, not Fragmentation. C: 'Succession beginning on a previously lifeless surface' identifies Primary succession, not Fragmentation.

BIO-12-CH14-0019 | Medium | Decomposition steps**Correct answer: B. Leaching**

Solution: The observation 'Downward movement of soluble inorganic nutrients' points to Leaching; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Biogeochemical cycling belongs with 'Movement of nutrients between organisms and environment', whereas the tested description is 'Downward movement of soluble inorganic nutrients'. C: Energy pyramid belongs with 'Pyramid that is always upright', whereas the tested description is 'Downward movement of soluble inorganic nutrients'. D: Ecosystem service belongs with 'Benefit such as pollination, climate regulation or purification', whereas the tested description is 'Downward movement of soluble inorganic nutrients'.

BIO-12-CH14-0020 | Easy | Productivity and decomposition**Correct answer: D. Gross primary productivity minus plant respiration**

Solution: Net primary productivity represents 'Gross primary productivity minus plant respiration'. Each alternative describes a different term or result.

Distractor review: A: 'Microbial conversion of detritus to dark amorphous matter' identifies Humification, not Net primary productivity. B: 'Succession after disturbance where soil remains' identifies Secondary succession, not Net primary productivity. C: 'Total photosynthetic production before respiratory loss' identifies Gross primary productivity, not Net primary productivity.

BIO-12-CH14-0021 | Challenging | Ecological pyramids**Correct answer: C. I, III and IV only**

Solution: Statement I is correct because 'Pyramid showing organism counts at trophic levels' corresponds to Pyramid of numbers. Statement II is incorrect: 'Pyramid that is always upright' corresponds to Energy pyramid, not Pyramid of biomass. Statement III is correct because 'Pyramid showing standing crop' corresponds to Pyramid of biomass. Statement IV is correct because 'Conversion of detritus into inorganic nutrients' corresponds to Mineralization. Therefore, the valid combination is I, III and IV only.

Distractor review: A: This combination includes false statement(s) II. B: This combination includes false statement(s) II and omits true statement(s) I. D: This combination omits true statement(s) III, IV.

BIO-12-CH14-0022 | Easy | Food chains and energy**Correct answer: D. Ten per cent**

Solution: 'Approximate energy passed to the next trophic level' is the defining description for Ten per cent. The other choices belong to different chapter concepts.

Distractor review: A: Pyramid of biomass belongs with 'Pyramid showing standing crop', whereas the tested description is 'Approximate energy passed to the next trophic level'. B: Biotic components belongs with 'Living components of an ecosystem', whereas the tested description is 'Approximate energy passed to the next trophic level'. C: Grazing food chain belongs with 'Food chain beginning with living green plants', whereas the tested description is 'Approximate energy passed to the next trophic level'.

BIO-12-CH14-0023 | Easy | Ecological pyramids**Correct answer: B. Pyramid showing standing crop**

Solution: Pyramid of biomass represents 'Pyramid showing standing crop'. Each alternative describes a different term or result.

Distractor review: A: 'Food chain beginning with living green plants' identifies Grazing food chain, not Pyramid of biomass. C: 'Total photosynthetic production before respiratory loss' identifies Gross primary productivity, not Pyramid of biomass. D: 'Major reservoir of atmospheric nitrogen' identifies Atmosphere, not Pyramid of biomass.

BIO-12-CH14-0024 | Medium | Succession and services**Correct answer: B. Succession after disturbance where soil remains - Secondary succession**

Solution: The accurate entry is 'Succession after disturbance where soil remains - Secondary succession'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Non-living physical and chemical components' should be paired with Abiotic components, not Ten per cent. C: 'Enzymatic breakdown by bacteria and fungi' should be paired with Catabolism, not Gross primary productivity. D: 'Total photosynthetic production before respiratory loss' should be paired with Gross primary productivity, not Biotic components.

BIO-12-CH14-0025 | Easy | Ecosystem organization**Correct answer: C. Organisms breaking down dead organic matter**

Solution: Decomposers represents 'Organisms breaking down dead organic matter'. Each alternative describes a different term or result.

Distractor review: A: 'Food chain beginning with living green plants' identifies Grazing food chain, not Decomposers. B: 'Autotrophs converting solar energy to chemical energy' identifies Producers, not Decomposers. D: 'Major reservoir of phosphorus' identifies Sedimentary rocks, not Decomposers.

BIO-12-CH14-0026 | Medium | Nutrient cycles**Correct answer: A. Major reservoir of phosphorus - Sedimentary rocks**

Solution: The accurate entry is 'Major reservoir of phosphorus - Sedimentary rocks'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Organisms breaking down dead organic matter' should be paired with Decomposers, not Primary productivity. C: 'Interconnected food chains' should be paired with Food web, not Sedimentary rocks. D: 'Succession beginning on a previously lifeless surface' should be paired with Primary succession, not Biotic components.

BIO-12-CH14-0027 | Hard | Ecosystem organization**Correct answer: D. Living components of an ecosystem - Biogeochemical cycling**

Solution: 'Living components of an ecosystem' should be paired with Biotic components, not Biogeochemical cycling. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Food chain beginning with dead organic matter - Detritus food chain. B: This is a valid association: Benefit such as pollination, climate regulation or purification - Ecosystem service. C: This is a valid association: Food chain beginning with living green plants - Grazing food chain.

BIO-12-CH14-0028 | Hard | Productivity and decomposition**Correct answer: A. Rate of biomass production by producers - Primary productivity; Downward movement of soluble inorganic nutrients - Leaching**

Solution: Both parts of 'Rate of biomass production by producers - Primary productivity; Downward movement of soluble inorganic nutrients - Leaching' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Pyramid showing organism counts at trophic levels' requires Pyramid of numbers, and 'Food chain beginning with dead organic matter' requires Detritus food chain. C: The answers have been interchanged: 'Pyramid showing standing crop' requires Pyramid of biomass, and 'Interconnected food chains' requires Food web. D: The answers have been interchanged: 'Interconnected food chains' requires Food web, and 'Breaking detritus into smaller particles' requires Fragmentation.

BIO-12-CH14-0029 | Medium | Ecosystem organization**Correct answer: C. Producers**

Solution: The chapter links Autotrophs converting solar energy to chemical energy with Producers, so Producers completes the statement.

Distractor review: A: Leaching belongs with 'Downward movement of soluble inorganic nutrients', whereas the tested description is 'Autotrophs converting solar energy to chemical energy'. B: Food web belongs with 'Interconnected food chains', whereas the tested description is 'Autotrophs converting solar energy to chemical energy'. D: Decomposers belongs with 'Organisms breaking down dead organic matter', whereas the tested description is 'Autotrophs converting solar energy to chemical energy'.

BIO-12-CH14-0030 | Medium | Productivity and decomposition**Correct answer: C. Net primary productivity**

Solution: The observation 'Gross primary productivity minus plant respiration' points to Net primary productivity; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Humification belongs with 'Microbial conversion of detritus to dark amorphous matter', whereas the tested description is 'Gross primary productivity minus plant respiration'. B: Gross primary productivity belongs with 'Total photosynthetic production before respiratory loss', whereas the tested description is 'Gross primary productivity minus plant respiration'. D: Secondary succession belongs with 'Succession after disturbance where soil remains', whereas the tested description is 'Gross primary productivity minus plant respiration'.

BIO-12-CH14-0031 | Challenging | Food chains and energy**Correct answer: A. I and II only**

Solution: Statement I is correct because 'Food chain beginning with living green plants' corresponds to Grazing food chain. Statement II is correct because 'Interconnected food chains' corresponds to Food web. Statement III is incorrect: 'Approximate energy passed to the next trophic level' corresponds to Ten per cent, not Primary succession. Statement IV is incorrect: 'Succession beginning on a previously lifeless surface' corresponds to Primary succession, not Grazing food chain. Therefore, the valid combination is I and II only.

Distractor review: B: This combination includes false statement(s) III, IV and omits true statement(s) I, II. C: This combination omits true statement(s) I. D: This combination includes false statement(s) III, IV and omits true statement(s) II.

BIO-12-CH14-0032 | Challenging | Ecosystem organization**Correct answer: B. I, II and IV only**

Solution: Statement I is correct because 'Organisms breaking down dead organic matter' corresponds to Decomposers. Statement II is correct because 'Non-living physical and chemical components' corresponds to Abiotic components. Statement III is incorrect: 'Living components of an ecosystem' corresponds to Biotic components, not Primary succession. Statement IV is correct because 'Succession beginning on a previously lifeless surface' corresponds to Primary succession. Therefore, the valid combination is I, II and IV only.

Distractor review: A: This combination omits true statement(s) I, IV. C: This combination includes false statement(s) III and omits true statement(s) I. D: This combination omits true statement(s) I, II.

BIO-12-CH14-0033 | Easy | Decomposition steps**Correct answer: B. Enzymatic breakdown by bacteria and fungi**

Solution: Catabolism represents 'Enzymatic breakdown by bacteria and fungi'. Each alternative describes a different term or result.

Distractor review: A: 'Organisms breaking down dead organic matter' identifies Decomposers, not Catabolism. C: 'Downward movement of soluble inorganic nutrients' identifies Leaching, not Catabolism. D: 'Autotrophs converting solar energy to chemical energy' identifies Producers, not Catabolism.

BIO-12-CH14-0034 | Medium | Succession and services**Correct answer: A. Secondary succession**

Solution: The observation 'Succession after disturbance where soil remains' points to Secondary succession; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Producers belongs with 'Autotrophs converting solar energy to chemical energy', whereas the tested description is 'Succession after disturbance where soil remains'. C: Primary succession belongs with 'Succession beginning on a previously lifeless surface', whereas the tested description is 'Succession after disturbance where soil remains'. D: Abiotic components belongs with 'Non-living physical and chemical components', whereas the tested description is 'Succession after disturbance where soil remains'.

BIO-12-CH14-0035 | Medium | Decomposition steps**Correct answer: D. Enzymatic breakdown by bacteria and fungi - Catabolism**

Solution: The accurate entry is 'Enzymatic breakdown by bacteria and fungi - Catabolism'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Organisms breaking down dead organic matter' should be paired with Decomposers, not Detritus food chain. B: 'Succession beginning on a previously lifeless surface' should be paired with Primary succession, not Producers. C: 'Autotrophs converting solar energy to chemical energy' should be paired with Producers, not Gross primary productivity.

BIO-12-CH14-0036 | Medium | Decomposition steps**Correct answer: D. Humification**

Solution: The chapter links Microbial conversion of detritus to dark amorphous matter with Humification, so Humification completes the statement.

Distractor review: A: Fragmentation belongs with 'Breaking detritus into smaller particles', whereas the tested description is 'Microbial conversion of detritus to dark amorphous matter'. B: Producers belongs with 'Autotrophs converting solar energy to chemical energy', whereas the tested description is 'Microbial conversion of detritus to dark amorphous matter'. C: Sedimentary rocks belongs with 'Major reservoir of phosphorus', whereas the tested description is 'Microbial conversion of detritus to dark amorphous matter'.

BIO-12-CH14-0037 | Hard | Ecosystem organization

Correct answer: B. Organisms breaking down dead organic matter - Decomposers; Conversion of organic nutrients to inorganic forms - Mineralization

Solution: Both parts of 'Organisms breaking down dead organic matter - Decomposers; Conversion of organic nutrients to inorganic forms - Mineralization' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Interconnected food chains' requires Food web, and 'Succession after disturbance where soil remains' requires Secondary succession. C: The answers have been interchanged: 'Food chain beginning with living green plants' requires Grazing food chain, and 'Major reservoir of phosphorus' requires Sedimentary rocks. D: The answers have been interchanged: 'Conversion of detritus into inorganic nutrients' requires Mineralization, and 'Relatively stable final community' requires Climax community.

BIO-12-CH14-0038 | Challenging | Succession and services

Correct answer: C. IV only

Solution: Statement I is incorrect: 'Succession after disturbance where soil remains' corresponds to Secondary succession, not Ecosystem service. Statement II is incorrect: 'Benefit such as pollination, climate regulation or purification' corresponds to Ecosystem service, not Climax community. Statement III is incorrect: 'Relatively stable final community' corresponds to Climax community, not Primary productivity. Statement IV is correct because 'Rate of biomass production by producers' corresponds to Primary productivity. Therefore, the valid combination is IV only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) IV. B: This combination includes false statement(s) II, III. D: This combination includes false statement(s) II.

BIO-12-CH14-0039 | Hard | Productivity and decomposition

Correct answer: C. Gross primary productivity minus plant respiration - Net primary productivity; Microbial conversion of detritus to dark amorphous matter - Humification

Solution: Both parts of 'Gross primary productivity minus plant respiration - Net primary productivity; Microbial conversion of detritus to dark amorphous matter - Humification' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Benefit such as pollination, climate regulation or purification' requires Ecosystem service, and 'Autotrophs converting solar energy to chemical energy' requires Producers. B: The answers have been interchanged: 'Succession after disturbance where soil remains' requires Secondary succession, and 'Enzymatic breakdown by bacteria and fungi' requires Catabolism. D: The answers have been interchanged: 'Enzymatic breakdown by bacteria and fungi' requires Catabolism, and 'Pyramid showing standing crop' requires Pyramid of biomass.

BIO-12-CH14-0040 | Medium | Ecosystem organization

Correct answer: D. Autotrophs converting solar energy to chemical energy - Producers

Solution: The accurate entry is 'Autotrophs converting solar energy to chemical energy - Producers'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Microbial conversion of detritus to dark amorphous matter' should be paired with Humification, not Producers. B: 'Breaking detritus into smaller particles' should be paired with Fragmentation, not Ten per cent. C: 'Interconnected food chains' should be paired with Food web, not Leaching.

BIO-12-CH14-0041 | Hard | Food chains and energy

Correct answer: A. Interconnected food chains - Food web; Enzymatic breakdown by bacteria and fungi - Catabolism

Solution: Both parts of 'Interconnected food chains - Food web; Enzymatic breakdown by bacteria and fungi - Catabolism' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Major reservoir of phosphorus' requires Sedimentary rocks, and 'Downward movement of soluble inorganic nutrients' requires Leaching. C: The answers have been interchanged: 'Downward movement of soluble inorganic nutrients' requires Leaching, and 'Gross primary productivity minus plant respiration' requires Net primary productivity. D: The answers have been interchanged: 'Gross primary productivity minus plant respiration' requires Net primary productivity, and 'Non-living physical and chemical components' requires Abiotic components.

BIO-12-CH14-0042 | Medium | Productivity and decomposition**Correct answer: C. Gross primary productivity**

Solution: The chapter links Total photosynthetic production before respiratory loss with Gross primary productivity, so Gross primary productivity completes the statement.

Distractor review: A: First trophic level belongs with 'Trophic level occupied by green plants', whereas the tested description is 'Total photosynthetic production before respiratory loss'. B: Fragmentation belongs with 'Breaking detritus into smaller particles', whereas the tested description is 'Total photosynthetic production before respiratory loss'. D: Catabolism belongs with 'Enzymatic breakdown by bacteria and fungi', whereas the tested description is 'Total photosynthetic production before respiratory loss'.

BIO-12-CH14-0043 | Medium | Food chains and energy**Correct answer: A. Food chain beginning with living green plants - Grazing food chain**

Solution: The accurate entry is 'Food chain beginning with living green plants - Grazing food chain'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Benefit such as pollination, climate regulation or purification' should be paired with Ecosystem service, not Climax community. C: 'Relatively stable final community' should be paired with Climax community, not Primary productivity. D: 'Pyramid showing organism counts at trophic levels' should be paired with Pyramid of numbers, not Ecosystem service.

BIO-12-CH14-0044 | Medium | Ecological pyramids**Correct answer: D. Energy pyramid**

Solution: The chapter links Pyramid that is always upright with Energy pyramid, so Energy pyramid completes the statement.

Distractor review: A: First trophic level belongs with 'Trophic level occupied by green plants', whereas the tested description is 'Pyramid that is always upright'. B: Secondary succession belongs with 'Succession after disturbance where soil remains', whereas the tested description is 'Pyramid that is always upright'. C: Pyramid of biomass belongs with 'Pyramid showing standing crop', whereas the tested description is 'Pyramid that is always upright'.

BIO-12-CH14-0045 | Easy | Productivity and decomposition**Correct answer: B. Rate of biomass production by producers**

Solution: Primary productivity represents 'Rate of biomass production by producers'. Each alternative describes a different term or result.

Distractor review: A: 'Downward movement of soluble inorganic nutrients' identifies Leaching, not Primary productivity. C: 'Conversion of detritus into inorganic nutrients' identifies Mineralization, not Primary productivity. D: 'Interconnected food chains' identifies Food web, not Primary productivity.

BIO-12-CH14-0046 | Hard | Succession and services**Correct answer: D. Benefit such as pollination, climate regulation or purification - Fragmentation**

Solution: 'Benefit such as pollination, climate regulation or purification' should be paired with Ecosystem service, not Fragmentation. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Pyramid that is always upright - Energy pyramid. B: This is a valid association: Total photosynthetic production before respiratory loss - Gross primary productivity. C: This is a valid association: Approximate energy passed to the next trophic level - Ten per cent.

BIO-12-CH14-0047 | Easy | Nutrient cycles**Correct answer: B. Atmosphere**

Solution: 'Major reservoir of atmospheric nitrogen' is the defining description for Atmosphere. The other choices belong to different chapter concepts.

Distractor review: A: Secondary succession belongs with 'Succession after disturbance where soil remains', whereas the tested description is 'Major reservoir of atmospheric nitrogen'. C: Energy pyramid belongs with 'Pyramid that is always upright', whereas the tested description is 'Major reservoir of atmospheric nitrogen'. D: Detritus food chain belongs with 'Food chain beginning with dead organic matter', whereas the tested description is 'Major reservoir of atmospheric nitrogen'.

BIO-12-CH14-0048 | Medium | Succession and services**Correct answer: B. Succession beginning on a previously lifeless surface - Primary succession**

Solution: The accurate entry is 'Succession beginning on a previously lifeless surface - Primary succession'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Conversion of organic nutrients to inorganic forms' should be paired with Mineralization, not Producers. C: 'Major reservoir of atmospheric nitrogen' should be paired with Atmosphere, not Pyramid of biomass. D: 'Pyramid showing standing crop' should be paired with Pyramid of biomass, not Net primary productivity.

BIO-12-CH14-0049 | Challenging | Ecological pyramids**Correct answer: A. I, II and III only**

Solution: Statement I is correct because 'Pyramid that is always upright' corresponds to Energy pyramid. Statement II is correct because 'Pyramid showing standing crop' corresponds to Pyramid of biomass. Statement III is correct because 'Pyramid showing organism counts at trophic levels' corresponds to Pyramid of numbers. Statement IV is incorrect: 'Total photosynthetic production before respiratory loss' corresponds to Gross primary productivity, not Energy pyramid. Therefore, the valid combination is I, II and III only.

Distractor review: B: This combination omits true statement(s) I, II. C: This combination includes false statement(s) IV and omits true statement(s) I. D: This combination includes false statement(s) IV and omits true statement(s) II, III.

BIO-12-CH14-0050 | Hard | Nutrient cycles**Correct answer: C. Major reservoir of atmospheric nitrogen - Atmosphere; Pyramid that is always upright - Energy pyramid**

Solution: Both parts of 'Major reservoir of atmospheric nitrogen - Atmosphere; Pyramid that is always upright - Energy pyramid' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Food chain beginning with living green plants' requires Grazing food chain, and 'Succession after disturbance where soil remains' requires Secondary succession. B: The answers have been interchanged: 'Autotrophs converting solar energy to chemical energy' requires Producers, and 'Downward movement of soluble inorganic nutrients' requires Leaching. D: The answers have been interchanged: 'Living components of an ecosystem' requires Biotic components, and 'Autotrophs converting solar energy to chemical energy' requires Producers.

BIO-12-CH14-0051 | Medium | Nutrient cycles**Correct answer: C. Biogeochemical cycling**

Solution: The observation 'Movement of nutrients between organisms and environment' points to Biogeochemical cycling; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Catabolism belongs with 'Enzymatic breakdown by bacteria and fungi', whereas the tested description is 'Movement of nutrients between organisms and environment'. B: Mineralization belongs with 'Conversion of organic nutrients to inorganic forms', whereas the tested description is 'Movement of nutrients between organisms and environment'. D: Sedimentary rocks belongs with 'Major reservoir of phosphorus', whereas the tested description is 'Movement of nutrients between organisms and environment'.

BIO-12-CH14-0052 | Challenging | Succession and services**Correct answer: B. III only**

Solution: Statement I is incorrect: 'Relatively stable final community' corresponds to Climax community, not Primary succession. Statement II is incorrect: 'Succession beginning on a previously lifeless surface' corresponds to Primary succession, not Secondary succession. Statement III is correct because 'Succession after disturbance where soil remains' corresponds to Secondary succession. Statement IV is incorrect: 'Pyramid showing standing crop' corresponds to Pyramid of biomass, not Climax community. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) II. C: This combination includes false statement(s) I, IV and omits true statement(s) III. D: This combination includes false statement(s) IV.

BIO-12-CH14-0053 | Medium | Productivity and decomposition**Correct answer: A. Mineralization**

Solution: The chapter links Conversion of detritus into inorganic nutrients with Mineralization, so Mineralization completes the statement.

Distractor review: B: Leaching belongs with 'Downward movement of soluble inorganic nutrients', whereas the tested description is 'Conversion of detritus into inorganic nutrients'. C: Pyramid of biomass belongs with 'Pyramid showing standing crop', whereas the tested description is 'Conversion of detritus into inorganic nutrients'. D: Gross primary productivity belongs with 'Total photosynthetic production before respiratory loss', whereas the tested description is 'Conversion of detritus into inorganic nutrients'.

BIO-12-CH14-0054 | Medium | Food chains and energy**Correct answer: D. Grazing food chain**

Solution: The observation 'Food chain beginning with living green plants' points to Grazing food chain; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Climax community belongs with 'Relatively stable final community', whereas the tested description is 'Food chain beginning with living green plants'. B: Humification belongs with 'Microbial conversion of detritus to dark amorphous matter', whereas the tested description is 'Food chain beginning with living green plants'. C: Ecosystem service belongs with 'Benefit such as pollination, climate regulation or purification', whereas the tested description is 'Food chain beginning with living green plants'.

BIO-12-CH14-0055 | Easy | Ecological pyramids**Correct answer: D. Pyramid that is always upright**

Solution: Energy pyramid represents 'Pyramid that is always upright'. Each alternative describes a different term or result.

Distractor review: A: 'Total photosynthetic production before respiratory loss' identifies Gross primary productivity, not Energy pyramid. B: 'Succession after disturbance where soil remains' identifies Secondary succession, not Energy pyramid. C: 'Rate of biomass production by producers' identifies Primary productivity, not Energy pyramid.

BIO-12-CH14-0056 | Hard | Ecological pyramids**Correct answer: A. Pyramid showing standing crop - Pyramid of biomass; Total photosynthetic production before respiratory loss - Gross primary productivity**

Solution: Both parts of 'Pyramid showing standing crop - Pyramid of biomass; Total photosynthetic production before respiratory loss - Gross primary productivity' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Major reservoir of phosphorus' requires Sedimentary rocks, and 'Benefit such as pollination, climate regulation or purification' requires Ecosystem service. C: The answers have been interchanged: 'Food chain beginning with living green plants' requires Grazing food chain, and 'Major reservoir of phosphorus' requires Sedimentary rocks. D: The answers have been interchanged: 'Benefit such as pollination, climate regulation or purification' requires Ecosystem service, and 'Major reservoir of atmospheric nitrogen' requires Atmosphere.

BIO-12-CH14-0057 | Medium | Nutrient cycles**Correct answer: D. Mineralization**

Solution: The observation 'Conversion of organic nutrients to inorganic forms' points to Mineralization; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Primary succession belongs with 'Succession beginning on a previously lifeless surface', whereas the tested description is 'Conversion of organic nutrients to inorganic forms'. B: Grazing food chain belongs with 'Food chain beginning with living green plants', whereas the tested description is 'Conversion of organic nutrients to inorganic forms'. C: Biogeochemical cycling belongs with 'Movement of nutrients between organisms and environment', whereas the tested description is 'Conversion of organic nutrients to inorganic forms'.

BIO-12-CH14-0058 | Medium | Ecosystem organization**Correct answer: B. Biotic components**

Solution: The chapter links Living components of an ecosystem with Biotic components, so Biotic components completes the statement.

Distractor review: A: Grazing food chain belongs with 'Food chain beginning with living green plants', whereas the tested description is 'Living components of an ecosystem'. C: Detritus food chain belongs with 'Food chain beginning with dead organic matter', whereas the tested description is 'Living components of an ecosystem'. D: Abiotic components belongs with 'Non-living physical and chemical components', whereas the tested description is 'Living components of an ecosystem'.

BIO-12-CH14-0059 | Easy | Productivity and decomposition**Correct answer: C. Total photosynthetic production before respiratory loss**

Solution: Gross primary productivity represents 'Total photosynthetic production before respiratory loss'. Each alternative describes a different term or result.

Distractor review: A: 'Breaking detritus into smaller particles' identifies Fragmentation, not Gross primary productivity. B: 'Movement of nutrients between organisms and environment' identifies Biogeochemical cycling, not Gross primary productivity. D: 'Rate of biomass production by producers' identifies Primary productivity, not Gross primary productivity.

BIO-12-CH14-0060 | Medium | Food chains and energy**Correct answer: D. Grazing food chain**

Solution: The chapter links Food chain beginning with living green plants with Grazing food chain, so Grazing food chain completes the statement.

Distractor review: A: Climax community belongs with 'Relatively stable final community', whereas the tested description is 'Food chain beginning with living green plants'. B: Ten per cent belongs with 'Approximate energy passed to the next trophic level', whereas the tested description is 'Food chain beginning with living green plants'. C: Pyramid of numbers belongs with 'Pyramid showing organism counts at trophic levels', whereas the tested description is 'Food chain beginning with living green plants'.

BIO-12-CH14-0061 | Hard | Succession and services

Correct answer: C. Succession beginning on a previously lifeless surface - Primary succession; Pyramid showing organism counts at trophic levels - Pyramid of numbers

Solution: Both parts of 'Succession beginning on a previously lifeless surface - Primary succession; Pyramid showing organism counts at trophic levels - Pyramid of numbers' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Conversion of organic nutrients to inorganic forms' requires Mineralization, and 'Gross primary productivity minus plant respiration' requires Net primary productivity. B: The answers have been interchanged: 'Non-living physical and chemical components' requires Abiotic components, and 'Pyramid showing standing crop' requires Pyramid of biomass. D: The answers have been interchanged: 'Pyramid showing standing crop' requires Pyramid of biomass, and 'Conversion of organic nutrients to inorganic forms' requires Mineralization.

BIO-12-CH14-0062 | Hard | Food chains and energy

Correct answer: B. Food chain beginning with living green plants - Pyramid of numbers

Solution: 'Food chain beginning with living green plants' should be paired with Grazing food chain, not Pyramid of numbers. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Microbial conversion of detritus to dark amorphous matter - Humification. C: This is a valid association: Breaking detritus into smaller particles - Fragmentation. D: This is a valid association: Benefit such as pollination, climate regulation or purification - Ecosystem service.

BIO-12-CH14-0063 | Medium | Succession and services

Correct answer: C. Climax community

Solution: The observation 'Relatively stable final community' points to Climax community; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Detritus food chain belongs with 'Food chain beginning with dead organic matter', whereas the tested description is 'Relatively stable final community'. B: Fragmentation belongs with 'Breaking detritus into smaller particles', whereas the tested description is 'Relatively stable final community'. D: Primary succession belongs with 'Succession beginning on a previously lifeless surface', whereas the tested description is 'Relatively stable final community'.

BIO-12-CH14-0064 | Hard | Ecological pyramids

Correct answer: D. Pyramid that is always upright - Energy pyramid; Breaking detritus into smaller particles - Fragmentation

Solution: Both parts of 'Pyramid that is always upright - Energy pyramid; Breaking detritus into smaller particles - Fragmentation' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Rate of biomass production by producers' requires Primary productivity, and 'Movement of nutrients between organisms and environment' requires Biogeochemical cycling. B: The answers have been interchanged: 'Succession after disturbance where soil remains' requires Secondary succession, and 'Enzymatic breakdown by bacteria and fungi' requires Catabolism. C: The answers have been interchanged: 'Total photosynthetic production before respiratory loss' requires Gross primary productivity, and 'Succession after disturbance where soil remains' requires Secondary succession.

BIO-12-CH14-0065 | Hard | Nutrient cycles

Correct answer: A. Major reservoir of atmospheric nitrogen - Energy pyramid

Solution: 'Major reservoir of atmospheric nitrogen' should be paired with Atmosphere, not Energy pyramid. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Food chain beginning with living green plants - Grazing food chain. C: This is a valid association: Food chain beginning with dead organic matter - Detritus food chain. D: This is a valid association: Autotrophs converting solar energy to chemical energy - Producers.

BIO-12-CH14-0066 | Hard | Productivity and decomposition

Correct answer: D. Total photosynthetic production before respiratory loss - Gross primary productivity; Movement of nutrients between organisms and environment - Biogeochemical cycling

Solution: Both parts of 'Total photosynthetic production before respiratory loss - Gross primary productivity; Movement of nutrients between organisms and environment - Biogeochemical cycling' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Breaking detritus into smaller particles' requires Fragmentation, and 'Enzymatic breakdown by bacteria and fungi' requires Catabolism. B: The answers have been interchanged: 'Enzymatic breakdown by bacteria and fungi' requires Catabolism, and 'Microbial conversion of detritus to dark amorphous matter' requires Humification. C: The answers have been interchanged: 'Succession after disturbance where soil remains' requires Secondary succession, and 'Benefit such as pollination, climate regulation or purification' requires Ecosystem service.

BIO-12-CH14-0067 | Easy | Ecological pyramids**Correct answer: D. Trophic level occupied by green plants**

Solution: First trophic level represents 'Trophic level occupied by green plants'. Each alternative describes a different term or result.

Distractor review: A: 'Breaking detritus into smaller particles' identifies Fragmentation, not First trophic level. B: 'Pyramid showing standing crop' identifies Pyramid of biomass, not First trophic level. C: 'Pyramid showing organism counts at trophic levels' identifies Pyramid of numbers, not First trophic level.

BIO-12-CH14-0068 | Challenging | Decomposition steps**Correct answer: A. III and IV only**

Solution: Statement I is incorrect: 'Breaking detritus into smaller particles' corresponds to Fragmentation, not Humification. Statement II is incorrect: 'Microbial conversion of detritus to dark amorphous matter' corresponds to Humification, not Catabolism. Statement III is correct because 'Enzymatic breakdown by bacteria and fungi' corresponds to Catabolism. Statement IV is correct because 'Conversion of organic nutrients to inorganic forms' corresponds to Mineralization. Therefore, the valid combination is III and IV only.

Distractor review: B: This combination includes false statement(s) II and omits true statement(s) IV. C: This combination includes false statement(s) I, II and omits true statement(s) III, IV. D: This combination includes false statement(s) I.

BIO-12-CH14-0069 | Easy | Ecosystem organization**Correct answer: B. Biotic components**

Solution: 'Living components of an ecosystem' is the defining description for Biotic components. The other choices belong to different chapter concepts.

Distractor review: A: Grazing food chain belongs with 'Food chain beginning with living green plants', whereas the tested description is 'Living components of an ecosystem'. C: Pyramid of biomass belongs with 'Pyramid showing standing crop', whereas the tested description is 'Living components of an ecosystem'. D: Detritus food chain belongs with 'Food chain beginning with dead organic matter', whereas the tested description is 'Living components of an ecosystem'.

BIO-12-CH14-0070 | Medium | Productivity and decomposition**Correct answer: B. Gross primary productivity minus plant respiration - Net primary productivity**

Solution: The accurate entry is 'Gross primary productivity minus plant respiration - Net primary productivity'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Enzymatic breakdown by bacteria and fungi' should be paired with Catabolism, not Ecosystem service. C: 'Microbial conversion of detritus to dark amorphous matter' should be paired with Humification, not Secondary succession. D: 'Pyramid showing standing crop' should be paired with Pyramid of biomass, not Producers.

BIO-12-CH14-0071 | Medium | Ecological pyramids**Correct answer: A. Pyramid that is always upright - Energy pyramid**

Solution: The accurate entry is 'Pyramid that is always upright - Energy pyramid'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Breaking detritus into smaller particles' should be paired with Fragmentation, not Secondary succession. C: 'Total photosynthetic production before respiratory loss' should be paired with Gross primary productivity, not Catabolism. D: 'Movement of nutrients between organisms and environment' should be paired with Biogeochemical cycling, not Energy pyramid.

BIO-12-CH14-0072 | Hard | Decomposition steps**Correct answer: A. Breaking detritus into smaller particles - Ten per cent**

Solution: 'Breaking detritus into smaller particles' should be paired with Fragmentation, not Ten per cent. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Food chain beginning with dead organic matter - Detritus food chain. C: This is a valid association: Non-living physical and chemical components - Abiotic components. D: This is a valid association: Relatively stable final community - Climax community.

BIO-12-CH14-0073 | Challenging | Nutrient cycles**Correct answer: B. II only**

Solution: Statement I is incorrect: 'Conversion of organic nutrients to inorganic forms' corresponds to Mineralization, not Sedimentary rocks. Statement II is correct because 'Major reservoir of phosphorus' corresponds to Sedimentary rocks. Statement III is incorrect: 'Movement of nutrients between organisms and environment' corresponds to Biogeochemical cycling, not Catabolism. Statement IV is incorrect: 'Enzymatic breakdown by bacteria and fungi' corresponds to Catabolism, not Mineralization. Therefore, the valid combination is II only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) II. C: This combination includes false statement(s) I. D: This combination includes false statement(s) III.

BIO-12-CH14-0074 | Easy | Succession and services**Correct answer: C. Succession beginning on a previously lifeless surface**

Solution: Primary succession represents 'Succession beginning on a previously lifeless surface'. Each alternative describes a different term or result.

Distractor review: A: 'Major reservoir of atmospheric nitrogen' identifies Atmosphere, not Primary succession. B: 'Relatively stable final community' identifies Climax community, not Primary succession. D: 'Conversion of organic nutrients to inorganic forms' identifies Mineralization, not Primary succession.

BIO-12-CH14-0075 | Easy | Ecosystem organization**Correct answer: B. Decomposers**

Solution: 'Organisms breaking down dead organic matter' is the defining description for Decomposers. The other choices belong to different chapter concepts.

Distractor review: A: Grazing food chain belongs with 'Food chain beginning with living green plants', whereas the tested description is 'Organisms breaking down dead organic matter'. C: Sedimentary rocks belongs with 'Major reservoir of phosphorus', whereas the tested description is 'Organisms breaking down dead organic matter'. D: Mineralization belongs with 'Conversion of organic nutrients to inorganic forms', whereas the tested description is 'Organisms breaking down dead organic matter'.

BIO-12-CH14-0076 | Easy | Food chains and energy**Correct answer: B. Grazing food chain**

Solution: 'Food chain beginning with living green plants' is the defining description for Grazing food chain. The other choices belong to different chapter concepts.

Distractor review: A: Pyramid of numbers belongs with 'Pyramid showing organism counts at trophic levels', whereas the tested description is 'Food chain beginning with living green plants'. C: Food web belongs with 'Interconnected food chains', whereas the tested description is 'Food chain beginning with living green plants'. D: Ten per cent belongs with 'Approximate energy passed to the next trophic level', whereas the tested description is 'Food chain beginning with living green plants'.

BIO-12-CH14-0077 | Hard | Succession and services**Correct answer: A. Relatively stable final community - Mineralization**

Solution: 'Relatively stable final community' should be paired with Climax community, not Mineralization. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Conversion of organic nutrients to inorganic forms - Mineralization. C: This is a valid association: Breaking detritus into smaller particles - Fragmentation. D: This is a valid association: Food chain beginning with dead organic matter - Detritus food chain.

BIO-12-CH14-0078 | Easy | Succession and services**Correct answer: A. Ecosystem service**

Solution: 'Benefit such as pollination, climate regulation or purification' is the defining description for Ecosystem service. The other choices belong to different chapter concepts.

Distractor review: B: Primary succession belongs with 'Succession beginning on a previously lifeless surface', whereas the tested description is 'Benefit such as pollination, climate regulation or purification'. C: Ten per cent belongs with 'Approximate energy passed to the next trophic level', whereas the tested description is 'Benefit such as pollination, climate regulation or purification'. D: Fragmentation belongs with 'Breaking detritus into smaller particles', whereas the tested description is 'Benefit such as pollination, climate regulation or purification'.

BIO-12-CH14-0079 | Hard | Ecological pyramids

Correct answer: C. Trophic level occupied by green plants - First trophic level; Breaking detritus into smaller particles - Fragmentation

Solution: Both parts of 'Trophic level occupied by green plants - First trophic level; Breaking detritus into smaller particles - Fragmentation' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Enzymatic breakdown by bacteria and fungi' requires Catabolism, and 'Non-living physical and chemical components' requires Abiotic components. B: The answers have been interchanged: 'Approximate energy passed to the next trophic level' requires Ten per cent, and 'Gross primary productivity minus plant respiration' requires Net primary productivity. D: The answers have been interchanged: 'Non-living physical and chemical components' requires Abiotic components, and 'Approximate energy passed to the next trophic level' requires Ten per cent.

BIO-12-CH14-0080 | Medium | Ecological pyramids

Correct answer: C. Pyramid of biomass

Solution: The chapter links Pyramid showing standing crop with Pyramid of biomass, so Pyramid of biomass completes the statement.

Distractor review: A: Ecosystem service belongs with 'Benefit such as pollination, climate regulation or purification', whereas the tested description is 'Pyramid showing standing crop'. B: Gross primary productivity belongs with 'Total photosynthetic production before respiratory loss', whereas the tested description is 'Pyramid showing standing crop'. D: Sedimentary rocks belongs with 'Major reservoir of phosphorus', whereas the tested description is 'Pyramid showing standing crop'.

BIO-12-CH14-0081 | Easy | Food chains and energy

Correct answer: B. Interconnected food chains

Solution: Food web represents 'Interconnected food chains'. Each alternative describes a different term or result.

Distractor review: A: 'Gross primary productivity minus plant respiration' identifies Net primary productivity, not Food web. C: 'Food chain beginning with living green plants' identifies Grazing food chain, not Food web. D: 'Major reservoir of phosphorus' identifies Sedimentary rocks, not Food web.

BIO-12-CH14-0082 | Easy | Succession and services

Correct answer: D. Climax community

Solution: 'Relatively stable final community' is the defining description for Climax community. The other choices belong to different chapter concepts.

Distractor review: A: Ecosystem service belongs with 'Benefit such as pollination, climate regulation or purification', whereas the tested description is 'Relatively stable final community'. B: Net primary productivity belongs with 'Gross primary productivity minus plant respiration', whereas the tested description is 'Relatively stable final community'. C: Mineralization belongs with 'Conversion of detritus into inorganic nutrients', whereas the tested description is 'Relatively stable final community'.

BIO-12-CH14-0083 | Medium | Productivity and decomposition

Correct answer: B. Primary productivity

Solution: The observation 'Rate of biomass production by producers' points to Primary productivity; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Food web belongs with 'Interconnected food chains', whereas the tested description is 'Rate of biomass production by producers'. C: Fragmentation belongs with 'Breaking detritus into smaller particles', whereas the tested description is 'Rate of biomass production by producers'. D: Mineralization belongs with 'Conversion of detritus into inorganic nutrients', whereas the tested description is 'Rate of biomass production by producers'.

BIO-12-CH14-0084 | Easy | Nutrient cycles

Correct answer: C. Major reservoir of atmospheric nitrogen

Solution: Atmosphere represents 'Major reservoir of atmospheric nitrogen'. Each alternative describes a different term or result.

Distractor review: A: 'Food chain beginning with dead organic matter' identifies Detritus food chain, not Atmosphere. B: 'Pyramid that is always upright' identifies Energy pyramid, not Atmosphere. D: 'Food chain beginning with living green plants' identifies Grazing food chain, not Atmosphere.

BIO-12-CH14-0085 | Easy | Decomposition steps**Correct answer: C. Fragmentation**

Solution: 'Breaking detritus into smaller particles' is the defining description for Fragmentation. The other choices belong to different chapter concepts.

Distractor review: A: Humification belongs with 'Microbial conversion of detritus to dark amorphous matter', whereas the tested description is 'Breaking detritus into smaller particles'. B: Detritus food chain belongs with 'Food chain beginning with dead organic matter', whereas the tested description is 'Breaking detritus into smaller particles'. D: Gross primary productivity belongs with 'Total photosynthetic production before respiratory loss', whereas the tested description is 'Breaking detritus into smaller particles'.

BIO-12-CH14-0086 | Medium | Ecosystem organization**Correct answer: A. Living components of an ecosystem - Biotic components**

Solution: The accurate entry is 'Living components of an ecosystem - Biotic components'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Food chain beginning with dead organic matter' should be paired with Detritus food chain, not Grazing food chain. C: 'Food chain beginning with living green plants' should be paired with Grazing food chain, not Atmosphere. D: 'Pyramid showing standing crop' should be paired with Pyramid of biomass, not Ecosystem service.

BIO-12-CH14-0087 | Medium | Decomposition steps**Correct answer: D. Leaching**

Solution: The chapter links Downward movement of soluble inorganic nutrients with Leaching, so Leaching completes the statement.

Distractor review: A: Fragmentation belongs with 'Breaking detritus into smaller particles', whereas the tested description is 'Downward movement of soluble inorganic nutrients'. B: Ecosystem service belongs with 'Benefit such as pollination, climate regulation or purification', whereas the tested description is 'Downward movement of soluble inorganic nutrients'. C: Catabolism belongs with 'Enzymatic breakdown by bacteria and fungi', whereas the tested description is 'Downward movement of soluble inorganic nutrients'.

BIO-12-CH14-0088 | Easy | Nutrient cycles**Correct answer: C. Sedimentary rocks**

Solution: 'Major reservoir of phosphorus' is the defining description for Sedimentary rocks. The other choices belong to different chapter concepts.

Distractor review: A: Mineralization belongs with 'Conversion of organic nutrients to inorganic forms', whereas the tested description is 'Major reservoir of phosphorus'. B: Decomposers belongs with 'Organisms breaking down dead organic matter', whereas the tested description is 'Major reservoir of phosphorus'. D: Energy pyramid belongs with 'Pyramid that is always upright', whereas the tested description is 'Major reservoir of phosphorus'.

Standard 12 | Chapter 15

Biodiversity, Conservation and Environmental Issues

BIO-12-CH15-0001 | Hard | Species-area relationship

Correct answer: C. Slope

Solution: On the log-log plot, Z is the slope and log C is the intercept.

Distractor review: A: The alternative 'Y-intercept log C' conflicts with the labelled relationship, scale or mechanism in the figure. On the log-log plot, Z is the slope and log C is the intercept. B: The alternative 'Total habitat area only' conflicts with the labelled relationship, scale or mechanism in the figure. On the log-log plot, Z is the slope and log C is the intercept. D: The alternative 'Number of trophic levels' conflicts with the labelled relationship, scale or mechanism in the figure. On the log-log plot, Z is the slope and log C is the intercept.

BIO-12-CH15-0002 | Easy | Species-area relationship

Correct answer: A. Species richness increases with area

Solution: The species-area relationship predicts greater richness in larger sampled areas.

Distractor review: B: The alternative 'Species richness always falls with area' conflicts with the labelled relationship, scale or mechanism in the figure. The species-area relationship predicts greater richness in larger sampled areas. C: The alternative 'Area and richness are unrelated' conflicts with the labelled relationship, scale or mechanism in the figure. The species-area relationship predicts greater richness in larger sampled areas. D: The alternative 'Every area contains exactly one species' conflicts with the labelled relationship, scale or mechanism in the figure. The species-area relationship predicts greater richness in larger sampled areas.

BIO-12-CH15-0003 | Medium | Catalytic ozone destruction by chlorine

Correct answer: C. Chlorofluorocarbons

Solution: UV radiation can release chlorine radicals from stratospheric chlorofluorocarbons.

Distractor review: A: The alternative 'Glucose' conflicts with the labelled relationship, scale or mechanism in the figure. UV radiation can release chlorine radicals from stratospheric chlorofluorocarbons. B: The alternative 'Cellulose' conflicts with the labelled relationship, scale or mechanism in the figure. UV radiation can release chlorine radicals from stratospheric chlorofluorocarbons. D: The alternative 'Amino acids' conflicts with the labelled relationship, scale or mechanism in the figure. UV radiation can release chlorine radicals from stratospheric chlorofluorocarbons.

BIO-12-CH15-0004 | Easy | Levels of biodiversity

Correct answer: B. Species richness

Solution: 'Number of species present in a community' is the defining description for Species richness. The other choices belong to different chapter concepts.

Distractor review: A: Species diversity belongs with 'Variety of species in a region', whereas the tested description is 'Number of species present in a community'. C: Endemic species belongs with 'Species restricted to a particular geographic area', whereas the tested description is 'Number of species present in a community'. D: It decreases belongs with 'General change in species richness from equator to poles', whereas the tested description is 'Number of species present in a community'.

BIO-12-CH15-0005 | Easy | Forests and law

Correct answer: C. Indian law providing a framework for conservation and benefit sharing

Solution: Biological Diversity Act 2002 represents 'Indian law providing a framework for conservation and benefit sharing'. Each alternative describes a different term or result.

Distractor review: A: 'Pollutant load measured through oxygen consumed by microbes' identifies Biochemical oxygen demand, not Biological Diversity Act 2002. B: 'Destruction and fragmentation of natural habitat' identifies Major cause of biodiversity loss, not Biological Diversity Act 2002. D: 'Rise in average global temperature due to enhanced greenhouse effect' identifies Global warming, not Biological Diversity Act 2002.

BIO-12-CH15-0006 | Easy | Biodiversity patterns

Correct answer: C. Relationship in which larger areas contain more species

Solution: Species-area relationship represents 'Relationship in which larger areas contain more species'. Each alternative describes a different term or result.

Distractor review: A: 'Variation among genes within a species' identifies Genetic diversity, not Species-area relationship. B: 'Ex situ storage of viable seeds' identifies Seed bank, not Species-area relationship. D: 'General change in species richness from equator to poles' identifies It decreases, not Species-area relationship.

BIO-12-CH15-0007 | Medium | Levels of biodiversity**Correct answer: A. Variety of species in a region - Species diversity**

Solution: The accurate entry is 'Variety of species in a region - Species diversity'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Nutrient enrichment causing excessive aquatic plant growth' should be paired with Eutrophication, not Overexploitation. C: 'Harvesting organisms faster than they recover' should be paired with Overexploitation, not Montreal Protocol. D: 'Chemicals releasing chlorine that destroys stratospheric ozone' should be paired with Chlorofluorocarbons, not It decreases.

BIO-12-CH15-0008 | Easy | Threats and extinction**Correct answer: D. Co-extinction**

Solution: 'Secondary extinction following loss of a dependent species' is the defining description for Co-extinction. The other choices belong to different chapter concepts.

Distractor review: A: Mission Harit Maharashtra belongs with 'Maharashtra initiative promoting large-scale tree planting', whereas the tested description is 'Secondary extinction following loss of a dependent species'. B: Invasive alien species belongs with 'Non-native organism spreading and causing harm', whereas the tested description is 'Secondary extinction following loss of a dependent species'. C: Species diversity belongs with 'Variety of species in a region', whereas the tested description is 'Secondary extinction following loss of a dependent species'.

BIO-12-CH15-0009 | Medium | Forests and law**Correct answer: D. Mission Harit Maharashtra**

Solution: The chapter links Maharashtra initiative promoting large-scale tree planting with Mission Harit Maharashtra, so Mission Harit Maharashtra completes the statement.

Distractor review: A: Biomagnification belongs with 'Increase in concentration of persistent toxins at higher trophic levels', whereas the tested description is 'Maharashtra initiative promoting large-scale tree planting'. B: Montreal Protocol belongs with 'Agreement aimed at controlling ozone-depleting substances', whereas the tested description is 'Maharashtra initiative promoting large-scale tree planting'. C: It decreases belongs with 'General change in species richness from equator to poles', whereas the tested description is 'Maharashtra initiative promoting large-scale tree planting'.

BIO-12-CH15-0010 | Hard | Forests and law**Correct answer: B. Large-scale removal of forest cover - In situ conservation**

Solution: 'Large-scale removal of forest cover' should be paired with Deforestation, not In situ conservation. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Chemicals releasing chlorine that destroys stratospheric ozone - Chlorofluorocarbons. C: This is a valid association: Variety of species in a region - Species diversity. D: This is a valid association: Variety of habitats and ecological processes - Ecosystem diversity.

BIO-12-CH15-0011 | Medium | Levels of biodiversity**Correct answer: D. Variation among genes within a species - Genetic diversity**

Solution: The accurate entry is 'Variation among genes within a species - Genetic diversity'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Non-native organism spreading and causing harm' should be paired with Invasive alien species, not Carbon dioxide. B: 'Gas chiefly responsible for anthropogenic greenhouse warming' should be paired with Carbon dioxide, not Mission Harit Maharashtra. C: 'Nutrient enrichment causing excessive aquatic plant growth' should be paired with Eutrophication, not Species-area relationship.

BIO-12-CH15-0012 | Medium | Forests and law**Correct answer: B. Reforestation**

Solution: The observation 'Restoring a forest that once existed but was removed' points to Reforestation; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Mission Harit Maharashtra belongs with 'Maharashtra initiative promoting large-scale tree planting', whereas the tested description is 'Restoring a forest that once existed but was removed'. C: Montreal Protocol belongs with 'Agreement aimed at controlling ozone-depleting substances', whereas the tested description is 'Restoring a forest that once existed but was removed'. D: Biodiversity hotspot belongs with 'Region with many endemic species under exceptional threat', whereas the tested description is 'Restoring a forest that once existed but was removed'.

BIO-12-CH15-0013 | Medium | Threats and extinction**Correct answer: A. Overexploitation**

Solution: The chapter links Harvesting organisms faster than they recover with Overexploitation, so Overexploitation completes the statement.

Distractor review: B: Co-extinction belongs with 'Secondary extinction following loss of a dependent species', whereas the tested description is 'Harvesting organisms faster than they recover'. C: Montreal Protocol belongs with 'Agreement aimed at controlling ozone-depleting substances', whereas the tested description is 'Harvesting organisms faster than they recover'. D: Eutrophication belongs with 'Nutrient enrichment causing excessive aquatic plant growth', whereas the tested description is 'Harvesting organisms faster than they recover'.

BIO-12-CH15-0014 | Easy | Threats and extinction**Correct answer: D. Major cause of biodiversity loss**

Solution: 'Destruction and fragmentation of natural habitat' is the defining description for Major cause of biodiversity loss. The other choices belong to different chapter concepts.

Distractor review: A: Invasive alien species belongs with 'Non-native organism spreading and causing harm', whereas the tested description is 'Destruction and fragmentation of natural habitat'. B: Ecosystem diversity belongs with 'Variety of habitats and ecological processes', whereas the tested description is 'Destruction and fragmentation of natural habitat'. C: Reforestation belongs with 'Restoring a forest that once existed but was removed', whereas the tested description is 'Destruction and fragmentation of natural habitat'.

BIO-12-CH15-0015 | Medium | Biodiversity patterns**Correct answer: C. Species-area relationship**

Solution: The observation 'Relationship in which larger areas contain more species' points to Species-area relationship; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Genetic diversity belongs with 'Variation among genes within a species', whereas the tested description is 'Relationship in which larger areas contain more species'. B: Biosphere reserve belongs with 'In situ protected area permitting regulated human activity', whereas the tested description is 'Relationship in which larger areas contain more species'. D: Co-extinction belongs with 'Secondary extinction following loss of a dependent species', whereas the tested description is 'Relationship in which larger areas contain more species'.

BIO-12-CH15-0016 | Hard | Biodiversity patterns**Correct answer: B. Relationship in which larger areas contain more species - Species-area relationship; Increase in concentration of persistent toxins at higher trophic levels - Biomagnification**

Solution: Both parts of 'Relationship in which larger areas contain more species - Species-area relationship; Increase in concentration of persistent toxins at higher trophic levels - Biomagnification' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Ex situ storage of viable seeds' requires Seed bank, and 'Variation among genes within a species' requires Genetic diversity. C: The answers have been interchanged: 'In situ protected area permitting regulated human activity' requires Biosphere reserve, and 'Secondary extinction following loss of a dependent species' requires Co-extinction. D: The answers have been interchanged: 'Variation among genes within a species' requires Genetic diversity, and 'Variety of habitats and ecological processes' requires Ecosystem diversity.

BIO-12-CH15-0017 | Medium | Threats and extinction**Correct answer: D. Co-extinction**

Solution: The observation 'Secondary extinction following loss of a dependent species' points to Co-extinction; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Invasive alien species belongs with 'Non-native organism spreading and causing harm', whereas the tested description is 'Secondary extinction following loss of a dependent species'. B: Montreal Protocol belongs with 'Agreement aimed at controlling ozone-depleting substances', whereas the tested description is 'Secondary extinction following loss of a dependent species'. C: Particulate matter belongs with 'Solid particles and droplets suspended in air', whereas the tested description is 'Secondary extinction following loss of a dependent species'.

BIO-12-CH15-0018 | Hard | Levels of biodiversity**Correct answer: A. Number of species present in a community - Co-extinction**

Solution: 'Number of species present in a community' should be paired with Species richness, not Co-extinction. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Species restricted to a particular geographic area - Endemic species. C: This is a valid association: Large-scale removal of forest cover - Deforestation. D: This is a valid association: Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra.

BIO-12-CH15-0019 | Easy | Pollution**Correct answer: C. Biomagnification**

Solution: 'Increase in concentration of persistent toxins at higher trophic levels' is the defining description for Biomagnification. The other choices belong to different chapter concepts.

Distractor review: A: Reforestation belongs with 'Restoring a forest that once existed but was removed', whereas the tested description is 'Increase in concentration of persistent toxins at higher trophic levels'. B: Species richness belongs with 'Number of species present in a community', whereas the tested description is 'Increase in concentration of persistent toxins at higher trophic levels'. D: Particulate matter belongs with 'Solid particles and droplets suspended in air', whereas the tested description is 'Increase in concentration of persistent toxins at higher trophic levels'.

BIO-12-CH15-0020 | Hard | Biodiversity patterns**Correct answer: B. General change in species richness from equator to poles - It decreases; Variety of species in a region - Species diversity**

Solution: Both parts of 'General change in species richness from equator to poles - It decreases; Variety of species in a region - Species diversity' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Gas chiefly responsible for anthropogenic greenhouse warming' requires Carbon dioxide, and 'In situ protected area permitting regulated human activity' requires Biosphere reserve. C: The answers have been interchanged: 'Rise in average global temperature due to enhanced greenhouse effect' requires Global warming, and 'Protection of species in their natural habitats' requires In situ conservation. D: The answers have been interchanged: 'Protection of species in their natural habitats' requires In situ conservation, and 'Indian law providing a framework for conservation and benefit sharing' requires Biological Diversity Act 2002.

BIO-12-CH15-0021 | Challenging | Catalytic ozone destruction by chlorine**Correct answer: B. Ozone depletion chiefly increases surface UV-B; it is not itself the direct mechanism of greenhouse warming**

Solution: Loss of stratospheric ozone increases UV-B penetration, whereas greenhouse warming arises mainly from infrared absorption by greenhouse gases.

Distractor review: A: The alternative 'Ozone depletion and greenhouse warming are exactly the same process' conflicts with the labelled relationship, scale or mechanism in the figure. Loss of stratospheric ozone increases UV-B penetration, whereas greenhouse warming arises mainly from infrared absorption by greenhouse gases. C: The alternative 'Ozone depletion blocks all ultraviolet radiation' conflicts with the labelled relationship, scale or mechanism in the figure. Loss of stratospheric ozone increases UV-B penetration, whereas greenhouse warming arises mainly from infrared absorption by greenhouse gases. D: The alternative 'Greenhouse gases destroy ozone only by the chlorine-radical cycle shown' conflicts with the labelled relationship, scale or mechanism in the figure. Loss of stratospheric ozone increases UV-B penetration, whereas greenhouse warming arises mainly from infrared absorption by greenhouse gases.

BIO-12-CH15-0022 | Hard | Conservation**Correct answer: D. Protection of species in their natural habitats - In situ conservation; Nutrient enrichment causing excessive aquatic plant growth - Eutrophication**

Solution: Both parts of 'Protection of species in their natural habitats - In situ conservation; Nutrient enrichment causing excessive aquatic plant growth - Eutrophication' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Relationship in which larger areas contain more species' requires Species-area relationship, and 'Variety of species in a region' requires Species diversity. B: The answers have been interchanged: 'Maharashtra initiative promoting large-scale tree planting' requires Mission Harit Maharashtra, and 'Chemicals releasing chlorine that destroys stratospheric ozone' requires Chlorofluorocarbons. C: The answers have been interchanged: 'Variation among genes within a species' requires Genetic diversity, and 'Maharashtra initiative promoting large-scale tree planting' requires Mission Harit Maharashtra.

BIO-12-CH15-0023 | Medium | Biodiversity patterns**Correct answer: C. General change in species richness from equator to poles - It decreases**

Solution: The accurate entry is 'General change in species richness from equator to poles - It decreases'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Rise in average global temperature due to enhanced greenhouse effect' should be paired with Global warming, not Biological Diversity Act 2002. B: 'Gas chiefly responsible for anthropogenic greenhouse warming' should be paired with Carbon dioxide, not Seed bank. D: 'Ex situ storage of viable seeds' should be paired with Seed bank, not In situ conservation.

BIO-12-CH15-0024 | Hard | Levels of biodiversity**Correct answer: B. Variation among genes within a species - Invasive alien species**

Solution: 'Variation among genes within a species' should be paired with Genetic diversity, not Invasive alien species. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra. C: This is a valid association: Relationship in which larger areas contain more species - Species-area relationship. D: This is a valid association: Pollutant load measured through oxygen consumed by microbes - Biochemical oxygen demand.

BIO-12-CH15-0025 | Medium | Climate and ozone**Correct answer: D. Chemicals releasing chlorine that destroys stratospheric ozone - Chlorofluorocarbons**

Solution: The accurate entry is 'Chemicals releasing chlorine that destroys stratospheric ozone - Chlorofluorocarbons'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Ex situ storage of viable seeds' should be paired with Seed bank, not Overexploitation. B: 'Restoring a forest that once existed but was removed' should be paired with Reforestation, not Species-area relationship. C: 'Protection outside natural habitats' should be paired with Ex situ conservation, not Chlorofluorocarbons.

BIO-12-CH15-0026 | Hard | Biodiversity patterns**Correct answer: A. Region with many endemic species under exceptional threat - Biodiversity hotspot; Secondary extinction following loss of a dependent species - Co-extinction**

Solution: Both parts of 'Region with many endemic species under exceptional threat - Biodiversity hotspot; Secondary extinction following loss of a dependent species - Co-extinction' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Agreement aimed at controlling ozone-depleting substances' requires Montreal Protocol, and 'Variety of species in a region' requires Species diversity. C: The answers have been interchanged: 'In situ protected area permitting regulated human activity' requires Biosphere reserve, and 'Large-scale removal of forest cover' requires Deforestation. D: The answers have been interchanged: 'Large-scale removal of forest cover' requires Deforestation, and 'Agreement aimed at controlling ozone-depleting substances' requires Montreal Protocol.

BIO-12-CH15-0027 | Challenging | Forests and law**Correct answer: B. IV only**

Solution: Statement I is incorrect: 'Large-scale removal of forest cover' corresponds to Deforestation, not Biological Diversity Act 2002. Statement II is incorrect: 'Indian law providing a framework for conservation and benefit sharing' corresponds to Biological Diversity Act 2002, not Reforestation. Statement III is incorrect: 'Restoring a forest that once existed but was removed' corresponds to Reforestation, not Global warming. Statement IV is correct because 'Rise in average global temperature due to enhanced greenhouse effect' corresponds to Global warming. Therefore, the valid combination is IV only.

Distractor review: A: This combination includes false statement(s) I, II, III and omits true statement(s) IV. C: This combination includes false statement(s) I, III. D: This combination includes false statement(s) I, III and omits true statement(s) IV.

BIO-12-CH15-0028 | Easy | Conservation**Correct answer: A. Biosphere reserve**

Solution: 'In situ protected area permitting regulated human activity' is the defining description for Biosphere reserve. The other choices belong to different chapter concepts.

Distractor review: B: Biodiversity hotspot belongs with 'Region with many endemic species under exceptional threat', whereas the tested description is 'In situ protected area permitting regulated human activity'. C: Mission Harit Maharashtra belongs with 'Maharashtra initiative promoting large-scale tree planting', whereas the tested description is 'In situ protected area permitting regulated human activity'. D: Global warming belongs with 'Rise in average global temperature due to enhanced greenhouse effect', whereas the tested description is 'In situ protected area permitting regulated human activity'.

BIO-12-CH15-0029 | Medium | Levels of biodiversity**Correct answer: A. Genetic diversity**

Solution: The observation 'Variation among genes within a species' points to Genetic diversity; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Carbon dioxide belongs with 'Gas chiefly responsible for anthropogenic greenhouse warming', whereas the tested description is 'Variation among genes within a species'. C: Species richness belongs with 'Number of species present in a community', whereas the tested description is 'Variation among genes within a species'. D: Mission Harit Maharashtra belongs with 'Maharashtra initiative promoting large-scale tree planting', whereas the tested description is 'Variation among genes within a species'.

BIO-12-CH15-0030 | Medium | Pollution**Correct answer: D. Biomagnification**

Solution: The observation 'Increase in concentration of persistent toxins at higher trophic levels' points to Biomagnification; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Particulate matter belongs with 'Solid particles and droplets suspended in air', whereas the tested description is 'Increase in concentration of persistent toxins at higher trophic levels'. B: Ecosystem diversity belongs with 'Variety of habitats and ecological processes', whereas the tested description is 'Increase in concentration of persistent toxins at higher trophic levels'. C: Species richness belongs with 'Number of species present in a community', whereas the tested description is 'Increase in concentration of persistent toxins at higher trophic levels'.

BIO-12-CH15-0031 | Medium | Biodiversity patterns**Correct answer: D. Relationship in which larger areas contain more species - Species-area relationship**

Solution: The accurate entry is 'Relationship in which larger areas contain more species - Species-area relationship'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Increase in concentration of persistent toxins at higher trophic levels' should be paired with Biomagnification, not Co-extinction. B: 'In situ protected area permitting regulated human activity' should be paired with Biosphere reserve, not Seed bank. C: 'Secondary extinction following loss of a dependent species' should be paired with Co-extinction, not Genetic diversity.

BIO-12-CH15-0032 | Medium | Conservation**Correct answer: C. Ex situ conservation**

Solution: The observation 'Protection outside natural habitats' points to Ex situ conservation; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Biosphere reserve belongs with 'In situ protected area permitting regulated human activity', whereas the tested description is 'Protection outside natural habitats'. B: Chlorofluorocarbons belongs with 'Chemicals releasing chlorine that destroys stratospheric ozone', whereas the tested description is 'Protection outside natural habitats'. D: Endemic species belongs with 'Species restricted to a particular geographic area', whereas the tested description is 'Protection outside natural habitats'.

BIO-12-CH15-0033 | Medium | Pollution**Correct answer: A. Eutrophication**

Solution: The chapter links Nutrient enrichment causing excessive aquatic plant growth with Eutrophication, so Eutrophication completes the statement.

Distractor review: B: Co-extinction belongs with 'Secondary extinction following loss of a dependent species', whereas the tested description is 'Nutrient enrichment causing excessive aquatic plant growth'. C: Species-area relationship belongs with 'Relationship in which larger areas contain more species', whereas the tested description is 'Nutrient enrichment causing excessive aquatic plant growth'. D: Carbon dioxide belongs with 'Gas chiefly responsible for anthropogenic greenhouse warming', whereas the tested description is 'Nutrient enrichment causing excessive aquatic plant growth'.

BIO-12-CH15-0034 | Easy | Forests and law**Correct answer: A. Mission Harit Maharashtra**

Solution: 'Maharashtra initiative promoting large-scale tree planting' is the defining description for Mission Harit Maharashtra. The other choices belong to different chapter concepts.

Distractor review: B: In situ conservation belongs with 'Protection of species in their natural habitats', whereas the tested description is 'Maharashtra initiative promoting large-scale tree planting'. C: Biomagnification belongs with 'Increase in concentration of persistent toxins at higher trophic levels', whereas the tested description is 'Maharashtra initiative promoting large-scale tree planting'. D: Deforestation belongs with 'Large-scale removal of forest cover', whereas the tested description is 'Maharashtra initiative promoting large-scale tree planting'.

BIO-12-CH15-0035 | Easy | Pollution**Correct answer: B. Nutrient enrichment causing excessive aquatic plant growth**

Solution: Eutrophication represents 'Nutrient enrichment causing excessive aquatic plant growth'. Each alternative describes a different term or result.

Distractor review: A: 'Pollutant load measured through oxygen consumed by microbes' identifies Biochemical oxygen demand, not Eutrophication. C: 'Relationship in which larger areas contain more species' identifies Species-area relationship, not Eutrophication. D: 'Secondary extinction following loss of a dependent species' identifies Co-extinction, not Eutrophication.

BIO-12-CH15-0036 | Hard | Pollution

Correct answer: D. Increase in concentration of persistent toxins at higher trophic levels - Biomagnification; Variety of habitats and ecological processes - Ecosystem diversity

Solution: Both parts of 'Increase in concentration of persistent toxins at higher trophic levels - Biomagnification; Variety of habitats and ecological processes - Ecosystem diversity' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Number of species present in a community' requires Species richness, and 'Restoring a forest that once existed but was removed' requires Reforestation. B: The answers have been interchanged: 'Region with many endemic species under exceptional threat' requires Biodiversity hotspot, and 'Ex situ storage of viable seeds' requires Seed bank. C: The answers have been interchanged: 'Species restricted to a particular geographic area' requires Endemic species, and 'Number of species present in a community' requires Species richness.

BIO-12-CH15-0037 | Easy | Conservation

Correct answer: D. In situ conservation

Solution: 'Protection of species in their natural habitats' is the defining description for In situ conservation. The other choices belong to different chapter concepts.

Distractor review: A: Chlorofluorocarbons belongs with 'Chemicals releasing chlorine that destroys stratospheric ozone', whereas the tested description is 'Protection of species in their natural habitats'. B: Seed bank belongs with 'Ex situ storage of viable seeds', whereas the tested description is 'Protection of species in their natural habitats'. C: Eutrophication belongs with 'Nutrient enrichment causing excessive aquatic plant growth', whereas the tested description is 'Protection of species in their natural habitats'.

BIO-12-CH15-0038 | Medium | Biodiversity patterns

Correct answer: A. It decreases

Solution: The observation 'General change in species richness from equator to poles' points to It decreases; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Global warming belongs with 'Rise in average global temperature due to enhanced greenhouse effect', whereas the tested description is 'General change in species richness from equator to poles'. C: Species diversity belongs with 'Variety of species in a region', whereas the tested description is 'General change in species richness from equator to poles'. D: Biosphere reserve belongs with 'In situ protected area permitting regulated human activity', whereas the tested description is 'General change in species richness from equator to poles'.

BIO-12-CH15-0039 | Easy | Levels of biodiversity

Correct answer: D. Variation among genes within a species

Solution: Genetic diversity represents 'Variation among genes within a species'. Each alternative describes a different term or result.

Distractor review: A: 'Relationship in which larger areas contain more species' identifies Species-area relationship, not Genetic diversity. B: 'Non-native organism spreading and causing harm' identifies Invasive alien species, not Genetic diversity. C: 'Maharashtra initiative promoting large-scale tree planting' identifies Mission Harit Maharashtra, not Genetic diversity.

BIO-12-CH15-0040 | Medium | Pollution

Correct answer: A. Biochemical oxygen demand

Solution: The chapter links Pollutant load measured through oxygen consumed by microbes with Biochemical oxygen demand, so Biochemical oxygen demand completes the statement.

Distractor review: B: Species richness belongs with 'Number of species present in a community', whereas the tested description is 'Pollutant load measured through oxygen consumed by microbes'. C: Eutrophication belongs with 'Nutrient enrichment causing excessive aquatic plant growth', whereas the tested description is 'Pollutant load measured through oxygen consumed by microbes'. D: Particulate matter belongs with 'Solid particles and droplets suspended in air', whereas the tested description is 'Pollutant load measured through oxygen consumed by microbes'.

BIO-12-CH15-0041 | Hard | Conservation

Correct answer: C. In situ protected area permitting regulated human activity - Biodiversity hotspot

Solution: 'In situ protected area permitting regulated human activity' should be paired with Biosphere reserve, not Biodiversity hotspot. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Rise in average global temperature due to enhanced greenhouse effect - Global warming. B: This is a valid association: Chemicals releasing chlorine that destroys stratospheric ozone - Chlorofluorocarbons. D: This is a valid association: Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra.

BIO-12-CH15-0042 | Hard | Threats and extinction

Correct answer: D. Harvesting organisms faster than they recover - Overexploitation; Indian law providing a framework for conservation and benefit sharing - Biological Diversity Act 2002

Solution: Both parts of 'Harvesting organisms faster than they recover - Overexploitation; Indian law providing a framework for conservation and benefit sharing - Biological Diversity Act 2002' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Maharashtra initiative promoting large-scale tree planting' requires Mission Harit Maharashtra, and 'Gas chiefly responsible for anthropogenic greenhouse warming' requires Carbon dioxide. B: The answers have been interchanged: 'Protection outside natural habitats' requires Ex situ conservation, and 'Maharashtra initiative promoting large-scale tree planting' requires Mission Harit Maharashtra. C: The answers have been interchanged: 'Nutrient enrichment causing excessive aquatic plant growth' requires Eutrophication, and 'Protection outside natural habitats' requires Ex situ conservation.

BIO-12-CH15-0043 | Hard | Conservation

Correct answer: C. Ex situ storage of viable seeds - Species diversity

Solution: 'Ex situ storage of viable seeds' should be paired with Seed bank, not Species diversity. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide. B: This is a valid association: Secondary extinction following loss of a dependent species - Co-extinction. D: This is a valid association: Region with many endemic species under exceptional threat - Biodiversity hotspot.

BIO-12-CH15-0044 | Medium | Conservation

Correct answer: B. Ex situ conservation

Solution: The chapter links Protection outside natural habitats with Ex situ conservation, so Ex situ conservation completes the statement.

Distractor review: A: Seed bank belongs with 'Ex situ storage of viable seeds', whereas the tested description is 'Protection outside natural habitats'. C: Biosphere reserve belongs with 'In situ protected area permitting regulated human activity', whereas the tested description is 'Protection outside natural habitats'. D: Montreal Protocol belongs with 'Agreement aimed at controlling ozone-depleting substances', whereas the tested description is 'Protection outside natural habitats'.

BIO-12-CH15-0045 | Medium | Conservation

Correct answer: C. Protection outside natural habitats - Ex situ conservation

Solution: The accurate entry is 'Protection outside natural habitats - Ex situ conservation'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Variety of species in a region' should be paired with Species diversity, not Co-extinction. B: 'Chemicals releasing chlorine that destroys stratospheric ozone' should be paired with Chlorofluorocarbons, not Biological Diversity Act 2002. D: 'Species restricted to a particular geographic area' should be paired with Endemic species, not Chlorofluorocarbons.

BIO-12-CH15-0046 | Easy | Biodiversity patterns

Correct answer: D. General change in species richness from equator to poles

Solution: It decreases represents 'General change in species richness from equator to poles'. Each alternative describes a different term or result.

Distractor review: A: 'Region with many endemic species under exceptional threat' identifies Biodiversity hotspot, not It decreases. B: 'Variety of species in a region' identifies Species diversity, not It decreases. C: 'Rise in average global temperature due to enhanced greenhouse effect' identifies Global warming, not It decreases.

BIO-12-CH15-0047 | Hard | Climate and ozone

Correct answer: D. Agreement aimed at controlling ozone-depleting substances - Invasive alien species

Solution: 'Agreement aimed at controlling ozone-depleting substances' should be paired with Montreal Protocol, not Invasive alien species. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Variation among genes within a species - Genetic diversity. B: This is a valid association: Solid particles and droplets suspended in air - Particulate matter. C: This is a valid association: Harvesting organisms faster than they recover - Overexploitation.

BIO-12-CH15-0048 | Medium | Forests and law**Correct answer: C. Deforestation**

Solution: The chapter links Large-scale removal of forest cover with Deforestation, so Deforestation completes the statement.

Distractor review: A: Ecosystem diversity belongs with 'Variety of habitats and ecological processes', whereas the tested description is 'Large-scale removal of forest cover'. B: Major cause of biodiversity loss belongs with 'Destruction and fragmentation of natural habitat', whereas the tested description is 'Large-scale removal of forest cover'. D: In situ conservation belongs with 'Protection of species in their natural habitats', whereas the tested description is 'Large-scale removal of forest cover'.

BIO-12-CH15-0049 | Medium | Conservation**Correct answer: D. Seed bank**

Solution: The observation 'Ex situ storage of viable seeds' points to Seed bank; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Species diversity belongs with 'Variety of species in a region', whereas the tested description is 'Ex situ storage of viable seeds'. B: Invasive alien species belongs with 'Non-native organism spreading and causing harm', whereas the tested description is 'Ex situ storage of viable seeds'. C: Ex situ conservation belongs with 'Protection outside natural habitats', whereas the tested description is 'Ex situ storage of viable seeds'.

BIO-12-CH15-0050 | Easy | Threats and extinction**Correct answer: C. Harvesting organisms faster than they recover**

Solution: Overexploitation represents 'Harvesting organisms faster than they recover'. Each alternative describes a different term or result.

Distractor review: A: 'Maharashtra initiative promoting large-scale tree planting' identifies Mission Harit Maharashtra, not Overexploitation. B: 'Indian law providing a framework for conservation and benefit sharing' identifies Biological Diversity Act 2002, not Overexploitation. D: 'Nutrient enrichment causing excessive aquatic plant growth' identifies Eutrophication, not Overexploitation.

BIO-12-CH15-0051 | Challenging | Levels of biodiversity**Correct answer: B. I, II and III only**

Solution: Statement I is correct because 'Number of species present in a community' corresponds to Species richness. Statement II is correct because 'Variety of habitats and ecological processes' corresponds to Ecosystem diversity. Statement III is correct because 'Variation among genes within a species' corresponds to Genetic diversity. Statement IV is incorrect: 'Maharashtra initiative promoting large-scale tree planting' corresponds to Mission Harit Maharashtra, not Species richness. Therefore, the valid combination is I, II and III only.

Distractor review: A: This combination includes false statement(s) IV. C: This combination includes false statement(s) IV and omits true statement(s) I, III. D: This combination omits true statement(s) II.

BIO-12-CH15-0052 | Challenging | Climate and ozone**Correct answer: B. III only**

Solution: Statement I is incorrect: 'Chemicals releasing chlorine that destroys stratospheric ozone' corresponds to Chlorofluorocarbons, not Montreal Protocol. Statement II is incorrect: 'Agreement aimed at controlling ozone-depleting substances' corresponds to Montreal Protocol, not Carbon dioxide. Statement III is correct because 'Gas chiefly responsible for anthropogenic greenhouse warming' corresponds to Carbon dioxide. Statement IV is incorrect: 'Number of species present in a community' corresponds to Species richness, not Chlorofluorocarbons. Therefore, the valid combination is III only.

Distractor review: A: This combination includes false statement(s) I and omits true statement(s) III. C: This combination includes false statement(s) I, II, IV and omits true statement(s) III. D: This combination includes false statement(s) I, IV and omits true statement(s) III.

BIO-12-CH15-0053 | Medium | Levels of biodiversity**Correct answer: A. Species richness**

Solution: The observation 'Number of species present in a community' points to Species richness; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: It decreases belongs with 'General change in species richness from equator to poles', whereas the tested description is 'Number of species present in a community'. C: Ecosystem diversity belongs with 'Variety of habitats and ecological processes', whereas the tested description is 'Number of species present in a community'. D: Deforestation belongs with 'Large-scale removal of forest cover', whereas the tested description is 'Number of species present in a community'.

BIO-12-CH15-0054 | Medium | Forests and law**Correct answer: B. Biological Diversity Act 2002**

Solution: The observation 'Indian law providing a framework for conservation and benefit sharing' points to Biological Diversity Act 2002; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Biochemical oxygen demand belongs with 'Pollutant load measured through oxygen consumed by microbes', whereas the tested description is 'Indian law providing a framework for conservation and benefit sharing'. C: Mission Harit Maharashtra belongs with 'Maharashtra initiative promoting large-scale tree planting', whereas the tested description is 'Indian law providing a framework for conservation and benefit sharing'. D: Global warming belongs with 'Rise in average global temperature due to enhanced greenhouse effect', whereas the tested description is 'Indian law providing a framework for conservation and benefit sharing'.

BIO-12-CH15-0055 | Medium | Pollution**Correct answer: B. Biomagnification**

Solution: The chapter links Increase in concentration of persistent toxins at higher trophic levels with Biomagnification, so Biomagnification completes the statement.

Distractor review: A: Ecosystem diversity belongs with 'Variety of habitats and ecological processes', whereas the tested description is 'Increase in concentration of persistent toxins at higher trophic levels'. C: Eutrophication belongs with 'Nutrient enrichment causing excessive aquatic plant growth', whereas the tested description is 'Increase in concentration of persistent toxins at higher trophic levels'. D: Particulate matter belongs with 'Solid particles and droplets suspended in air', whereas the tested description is 'Increase in concentration of persistent toxins at higher trophic levels'.

BIO-12-CH15-0056 | Challenging | Forests and law**Correct answer: C. III and IV only**

Solution: Statement I is incorrect: 'Indian law providing a framework for conservation and benefit sharing' corresponds to Biological Diversity Act 2002, not Mission Harit Maharashtra. Statement II is incorrect: 'Maharashtra initiative promoting large-scale tree planting' corresponds to Mission Harit Maharashtra, not Deforestation. Statement III is correct because 'Large-scale removal of forest cover' corresponds to Deforestation. Statement IV is correct because 'Solid particles and droplets suspended in air' corresponds to Particulate matter. Therefore, the valid combination is III and IV only.

Distractor review: A: This combination includes false statement(s) II and omits true statement(s) III. B: This combination includes false statement(s) II and omits true statement(s) IV. D: This combination includes false statement(s) II.

BIO-12-CH15-0057 | Easy | Forests and law**Correct answer: D. Deforestation**

Solution: 'Large-scale removal of forest cover' is the defining description for Deforestation. The other choices belong to different chapter concepts.

Distractor review: A: Carbon dioxide belongs with 'Gas chiefly responsible for anthropogenic greenhouse warming', whereas the tested description is 'Large-scale removal of forest cover'. B: In situ conservation belongs with 'Protection of species in their natural habitats', whereas the tested description is 'Large-scale removal of forest cover'. C: Biological Diversity Act 2002 belongs with 'Indian law providing a framework for conservation and benefit sharing', whereas the tested description is 'Large-scale removal of forest cover'.

BIO-12-CH15-0058 | Medium | Levels of biodiversity**Correct answer: A. Genetic diversity**

Solution: The chapter links Variation among genes within a species with Genetic diversity, so Genetic diversity completes the statement.

Distractor review: B: Species diversity belongs with 'Variety of species in a region', whereas the tested description is 'Variation among genes within a species'. C: Species-area relationship belongs with 'Relationship in which larger areas contain more species', whereas the tested description is 'Variation among genes within a species'. D: Carbon dioxide belongs with 'Gas chiefly responsible for anthropogenic greenhouse warming', whereas the tested description is 'Variation among genes within a species'.

BIO-12-CH15-0059 | Medium | Pollution**Correct answer: D. Eutrophication**

Solution: The observation 'Nutrient enrichment causing excessive aquatic plant growth' points to Eutrophication; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Co-extinction belongs with 'Secondary extinction following loss of a dependent species', whereas the tested description is 'Nutrient enrichment causing excessive aquatic plant growth'. B: Species-area relationship belongs with 'Relationship in which larger areas contain more species', whereas the tested description is 'Nutrient enrichment causing excessive aquatic plant growth'. C: Particulate matter belongs with 'Solid particles and droplets suspended in air', whereas the tested description is 'Nutrient enrichment causing excessive aquatic plant growth'.

BIO-12-CH15-0060 | Hard | Levels of biodiversity

Correct answer: C. Variation among genes within a species - Genetic diversity; Non-native organism spreading and causing harm - Invasive alien species

Solution: Both parts of 'Variation among genes within a species - Genetic diversity; Non-native organism spreading and causing harm - Invasive alien species' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Gas chiefly responsible for anthropogenic greenhouse warming' requires Carbon dioxide, and 'Relationship in which larger areas contain more species' requires Species-area relationship. B: The answers have been interchanged: 'Relationship in which larger areas contain more species' requires Species-area relationship, and 'Maharashtra initiative promoting large-scale tree planting' requires Mission Harit Maharashtra. D: The answers have been interchanged: 'Nutrient enrichment causing excessive aquatic plant growth' requires Eutrophication, and 'Gas chiefly responsible for anthropogenic greenhouse warming' requires Carbon dioxide.

BIO-12-CH15-0061 | Easy | Forests and law

Correct answer: A. Maharashtra initiative promoting large-scale tree planting

Solution: Mission Harit Maharashtra represents 'Maharashtra initiative promoting large-scale tree planting'. Each alternative describes a different term or result.

Distractor review: B: 'Large-scale removal of forest cover' identifies Deforestation, not Mission Harit Maharashtra. C: 'Protection of species in their natural habitats' identifies In situ conservation, not Mission Harit Maharashtra. D: 'Increase in concentration of persistent toxins at higher trophic levels' identifies Biomagnification, not Mission Harit Maharashtra.

BIO-12-CH15-0062 | Medium | Threats and extinction

Correct answer: D. Destruction and fragmentation of natural habitat - Major cause of biodiversity loss

Solution: The accurate entry is 'Destruction and fragmentation of natural habitat - Major cause of biodiversity loss'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Nutrient enrichment causing excessive aquatic plant growth' should be paired with Eutrophication, not Ecosystem diversity. B: 'Agreement aimed at controlling ozone-depleting substances' should be paired with Montreal Protocol, not Seed bank. C: 'Protection outside natural habitats' should be paired with Ex situ conservation, not Major cause of biodiversity loss.

BIO-12-CH15-0063 | Challenging | Threats and extinction

Correct answer: A. II only

Solution: Statement I is incorrect: 'Harvesting organisms faster than they recover' corresponds to Overexploitation, not Major cause of biodiversity loss. Statement II is correct because 'Destruction and fragmentation of natural habitat' corresponds to Major cause of biodiversity loss. Statement III is incorrect: 'Non-native organism spreading and causing harm' corresponds to Invasive alien species, not Biosphere reserve. Statement IV is incorrect: 'In situ protected area permitting regulated human activity' corresponds to Biosphere reserve, not Overexploitation. Therefore, the valid combination is II only.

Distractor review: B: This combination includes false statement(s) I, III and omits true statement(s) II. C: This combination includes false statement(s) III, IV. D: This combination includes false statement(s) I, III.

BIO-12-CH15-0064 | Medium | Climate and ozone

Correct answer: A. Global warming

Solution: The observation 'Rise in average global temperature due to enhanced greenhouse effect' points to Global warming; that association, rather than a neighbouring term, completes the analysis.

Distractor review: B: Biochemical oxygen demand belongs with 'Pollutant load measured through oxygen consumed by microbes', whereas the tested description is 'Rise in average global temperature due to enhanced greenhouse effect'. C: Deforestation belongs with 'Large-scale removal of forest cover', whereas the tested description is 'Rise in average global temperature due to enhanced greenhouse effect'. D: Genetic diversity belongs with 'Variation among genes within a species', whereas the tested description is 'Rise in average global temperature due to enhanced greenhouse effect'.

BIO-12-CH15-0065 | Medium | Biodiversity patterns

Correct answer: C. Biodiversity hotspot

Solution: The chapter links Region with many endemic species under exceptional threat with Biodiversity hotspot, so Biodiversity hotspot completes the statement.

Distractor review: A: Deforestation belongs with 'Large-scale removal of forest cover', whereas the tested description is 'Region with many endemic species under exceptional threat'. B: Ex situ conservation belongs with 'Protection outside natural habitats', whereas the tested description is 'Region with many endemic species under exceptional threat'. D: Montreal Protocol belongs with 'Agreement aimed at controlling ozone-depleting substances', whereas the tested description is 'Region with many endemic species under exceptional threat'.

BIO-12-CH15-0066 | Easy | Pollution**Correct answer: A. Increase in concentration of persistent toxins at higher trophic levels**

Solution: Biomagnification represents 'Increase in concentration of persistent toxins at higher trophic levels'. Each alternative describes a different term or result.

Distractor review: B: 'Nutrient enrichment causing excessive aquatic plant growth' identifies Eutrophication, not Biomagnification. C: 'Species restricted to a particular geographic area' identifies Endemic species, not Biomagnification. D: 'Restoring a forest that once existed but was removed' identifies Reforestation, not Biomagnification.

BIO-12-CH15-0067 | Easy | Threats and extinction**Correct answer: D. Non-native organism spreading and causing harm**

Solution: Invasive alien species represents 'Non-native organism spreading and causing harm'. Each alternative describes a different term or result.

Distractor review: A: 'Protection outside natural habitats' identifies Ex situ conservation, not Invasive alien species. B: 'Restoring a forest that once existed but was removed' identifies Reforestation, not Invasive alien species. C: 'Nutrient enrichment causing excessive aquatic plant growth' identifies Eutrophication, not Invasive alien species.

BIO-12-CH15-0068 | Hard | Climate and ozone**Correct answer: B. Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide; Harvesting organisms faster than they recover - Overexploitation**

Solution: Both parts of 'Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide; Harvesting organisms faster than they recover - Overexploitation' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Region with many endemic species under exceptional threat' requires Biodiversity hotspot, and 'Destruction and fragmentation of natural habitat' requires Major cause of biodiversity loss. C: The answers have been interchanged: 'Secondary extinction following loss of a dependent species' requires Co-extinction, and 'Region with many endemic species under exceptional threat' requires Biodiversity hotspot. D: The answers have been interchanged: 'Protection of species in their natural habitats' requires In situ conservation, and 'Variety of habitats and ecological processes' requires Ecosystem diversity.

BIO-12-CH15-0069 | Hard | Conservation**Correct answer: A. Protection of species in their natural habitats - Eutrophication**

Solution: 'Protection of species in their natural habitats' should be paired with In situ conservation, not Eutrophication. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Variation among genes within a species - Genetic diversity. C: This is a valid association: Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra. D: This is a valid association: Relationship in which larger areas contain more species - Species-area relationship.

BIO-12-CH15-0070 | Hard | Pollution**Correct answer: C. Solid particles and droplets suspended in air - Chlorofluorocarbons**

Solution: 'Solid particles and droplets suspended in air' should be paired with Particulate matter, not Chlorofluorocarbons. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Indian law providing a framework for conservation and benefit sharing - Biological Diversity Act 2002. B: This is a valid association: Number of species present in a community - Species richness. D: This is a valid association: Variation among genes within a species - Genetic diversity.

BIO-12-CH15-0071 | Challenging | Biodiversity patterns**Correct answer: D. II and IV only**

Solution: Statement I is incorrect: 'Region with many endemic species under exceptional threat' corresponds to Biodiversity hotspot, not It decreases. Statement II is correct because 'General change in species richness from equator to poles' corresponds to It decreases. Statement III is incorrect: 'Relationship in which larger areas contain more species' corresponds to Species-area relationship, not Deforestation. Statement IV is correct because 'Large-scale removal of forest cover' corresponds to Deforestation. Therefore, the valid combination is II and IV only.

Distractor review: A: This combination includes false statement(s) III and omits true statement(s) IV. B: This combination includes false statement(s) I, III and omits true statement(s) II. C: This combination includes false statement(s) I.

BIO-12-CH15-0072 | Medium | Threats and extinction**Correct answer: C. Invasive alien species**

Solution: The observation 'Non-native organism spreading and causing harm' points to Invasive alien species; that association, rather than a neighbouring term, completes the analysis.

Distractor review: A: Eutrophication belongs with 'Nutrient enrichment causing excessive aquatic plant growth', whereas the tested description is 'Non-native organism spreading and causing harm'. B: Ex situ conservation belongs with 'Protection outside natural habitats', whereas the tested description is 'Non-native organism spreading and causing harm'. D: Reforestation belongs with 'Restoring a forest that once existed but was removed', whereas the tested description is 'Non-native organism spreading and causing harm'.

BIO-12-CH15-0073 | Easy | Climate and ozone**Correct answer: B. Montreal Protocol**

Solution: 'Agreement aimed at controlling ozone-depleting substances' is the defining description for Montreal Protocol. The other choices belong to different chapter concepts.

Distractor review: A: Carbon dioxide belongs with 'Gas chiefly responsible for anthropogenic greenhouse warming', whereas the tested description is 'Agreement aimed at controlling ozone-depleting substances'. C: Overexploitation belongs with 'Harvesting organisms faster than they recover', whereas the tested description is 'Agreement aimed at controlling ozone-depleting substances'. D: Ex situ conservation belongs with 'Protection outside natural habitats', whereas the tested description is 'Agreement aimed at controlling ozone-depleting substances'.

BIO-12-CH15-0074 | Hard | Threats and extinction**Correct answer: B. Non-native organism spreading and causing harm - Invasive alien species; Protection outside natural habitats - Ex situ conservation**

Solution: Both parts of 'Non-native organism spreading and causing harm - Invasive alien species; Protection outside natural habitats - Ex situ conservation' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Chemicals releasing chlorine that destroys stratospheric ozone' requires Chlorofluorocarbons, and 'Restoring a forest that once existed but was removed' requires Reforestation. C: The answers have been interchanged: 'Nutrient enrichment causing excessive aquatic plant growth' requires Eutrophication, and 'Chemicals releasing chlorine that destroys stratospheric ozone' requires Chlorofluorocarbons. D: The answers have been interchanged: 'Pollutant load measured through oxygen consumed by microbes' requires Biochemical oxygen demand, and 'Nutrient enrichment causing excessive aquatic plant growth' requires Eutrophication.

BIO-12-CH15-0075 | Medium | Forests and law**Correct answer: A. Reforestation**

Solution: The chapter links Restoring a forest that once existed but was removed with Reforestation, so Reforestation completes the statement.

Distractor review: B: Biodiversity hotspot belongs with 'Region with many endemic species under exceptional threat', whereas the tested description is 'Restoring a forest that once existed but was removed'. C: Montreal Protocol belongs with 'Agreement aimed at controlling ozone-depleting substances', whereas the tested description is 'Restoring a forest that once existed but was removed'. D: Biosphere reserve belongs with 'In situ protected area permitting regulated human activity', whereas the tested description is 'Restoring a forest that once existed but was removed'.

BIO-12-CH15-0076 | Medium | Climate and ozone**Correct answer: B. Global warming**

Solution: The chapter links Rise in average global temperature due to enhanced greenhouse effect with Global warming, so Global warming completes the statement.

Distractor review: A: Genetic diversity belongs with 'Variation among genes within a species', whereas the tested description is 'Rise in average global temperature due to enhanced greenhouse effect'. C: Chlorofluorocarbons belongs with 'Chemicals releasing chlorine that destroys stratospheric ozone', whereas the tested description is 'Rise in average global temperature due to enhanced greenhouse effect'. D: Particulate matter belongs with 'Solid particles and droplets suspended in air', whereas the tested description is 'Rise in average global temperature due to enhanced greenhouse effect'.

BIO-12-CH15-0077 | Hard | Forests and law

Correct answer: B. Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra; Variation among genes within a species - Genetic diversity

Solution: Both parts of 'Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra; Variation among genes within a species - Genetic diversity' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Agreement aimed at controlling ozone-depleting substances' requires Montreal Protocol, and 'Protection of species in their natural habitats' requires In situ conservation. C: The answers have been interchanged: 'Solid particles and droplets suspended in air' requires Particulate matter, and 'Non-native organism spreading and causing harm' requires Invasive alien species. D: The answers have been interchanged: 'General change in species richness from equator to poles' requires It decreases, and 'Agreement aimed at controlling ozone-depleting substances' requires Montreal Protocol.

BIO-12-CH15-0078 | Challenging | Conservation

Correct answer: C. I only

Solution: Statement I is correct because 'In situ protected area permitting regulated human activity' corresponds to Biosphere reserve. Statement II is incorrect: 'Protection of species in their natural habitats' corresponds to In situ conservation, not Seed bank. Statement III is incorrect: 'Ex situ storage of viable seeds' corresponds to Seed bank, not It decreases. Statement IV is incorrect: 'General change in species richness from equator to poles' corresponds to It decreases, not Biosphere reserve. Therefore, the valid combination is I only.

Distractor review: A: This combination includes false statement(s) IV and omits true statement(s) I. B: This combination includes false statement(s) IV. D: This combination includes false statement(s) II and omits true statement(s) I.

BIO-12-CH15-0079 | Easy | Biodiversity patterns

Correct answer: C. Region with many endemic species under exceptional threat

Solution: Biodiversity hotspot represents 'Region with many endemic species under exceptional threat'. Each alternative describes a different term or result.

Distractor review: A: 'Agreement aimed at controlling ozone-depleting substances' identifies Montreal Protocol, not Biodiversity hotspot. B: 'Protection outside natural habitats' identifies Ex situ conservation, not Biodiversity hotspot. D: 'Secondary extinction following loss of a dependent species' identifies Co-extinction, not Biodiversity hotspot.

BIO-12-CH15-0080 | Hard | Threats and extinction

Correct answer: B. Destruction and fragmentation of natural habitat - Eutrophication

Solution: 'Destruction and fragmentation of natural habitat' should be paired with Major cause of biodiversity loss, not Eutrophication. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Variety of habitats and ecological processes - Ecosystem diversity. C: This is a valid association: Restoring a forest that once existed but was removed - Reforestation. D: This is a valid association: Protection outside natural habitats - Ex situ conservation.

BIO-12-CH15-0081 | Medium | Climate and ozone

Correct answer: C. Montreal Protocol

Solution: The chapter links Agreement aimed at controlling ozone-depleting substances with Montreal Protocol, so Montreal Protocol completes the statement.

Distractor review: A: Biosphere reserve belongs with 'In situ protected area permitting regulated human activity', whereas the tested description is 'Agreement aimed at controlling ozone-depleting substances'. B: Carbon dioxide belongs with 'Gas chiefly responsible for anthropogenic greenhouse warming', whereas the tested description is 'Agreement aimed at controlling ozone-depleting substances'. D: Ex situ conservation belongs with 'Protection outside natural habitats', whereas the tested description is 'Agreement aimed at controlling ozone-depleting substances'.

BIO-12-CH15-0082 | Easy | Climate and ozone

Correct answer: A. Rise in average global temperature due to enhanced greenhouse effect

Solution: Global warming represents 'Rise in average global temperature due to enhanced greenhouse effect'. Each alternative describes a different term or result.

Distractor review: B: 'Large-scale removal of forest cover' identifies Deforestation, not Global warming. C: 'Ex situ storage of viable seeds' identifies Seed bank, not Global warming. D: 'Chemicals releasing chlorine that destroys stratospheric ozone' identifies Chlorofluorocarbons, not Global warming.

BIO-12-CH15-0083 | Medium | Threats and extinction**Correct answer: C. Secondary extinction following loss of a dependent species - Co-extinction**

Solution: The accurate entry is 'Secondary extinction following loss of a dependent species - Co-extinction'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Maharashtra initiative promoting large-scale tree planting' should be paired with Mission Harit Maharashtra, not Eutrophication. B: 'Variety of species in a region' should be paired with Species diversity, not Montreal Protocol. D: 'Pollutant load measured through oxygen consumed by microbes' should be paired with Biochemical oxygen demand, not Particulate matter.

BIO-12-CH15-0084 | Challenging | Climate and ozone**Correct answer: C. II and III only**

Solution: Statement I is incorrect: 'Rise in average global temperature due to enhanced greenhouse effect' corresponds to Global warming, not Carbon dioxide. Statement II is correct because 'Gas chiefly responsible for anthropogenic greenhouse warming' corresponds to Carbon dioxide. Statement III is correct because 'Chemicals releasing chlorine that destroys stratospheric ozone' corresponds to Chlorofluorocarbons. Statement IV is incorrect: 'Variation among genes within a species' corresponds to Genetic diversity, not Global warming. Therefore, the valid combination is II and III only.

Distractor review: A: This combination includes false statement(s) I. B: This combination includes false statement(s) IV and omits true statement(s) III. D: This combination includes false statement(s) I, IV and omits true statement(s) II, III.

BIO-12-CH15-0085 | Easy | Levels of biodiversity**Correct answer: B. Genetic diversity**

Solution: 'Variation among genes within a species' is the defining description for Genetic diversity. The other choices belong to different chapter concepts.

Distractor review: A: Invasive alien species belongs with 'Non-native organism spreading and causing harm', whereas the tested description is 'Variation among genes within a species'. C: Carbon dioxide belongs with 'Gas chiefly responsible for anthropogenic greenhouse warming', whereas the tested description is 'Variation among genes within a species'. D: Mission Harit Maharashtra belongs with 'Maharashtra initiative promoting large-scale tree planting', whereas the tested description is 'Variation among genes within a species'.

BIO-12-CH15-0086 | Medium | Forests and law**Correct answer: C. Large-scale removal of forest cover - Deforestation**

Solution: The accurate entry is 'Large-scale removal of forest cover - Deforestation'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Chemicals releasing chlorine that destroys stratospheric ozone' should be paired with Chlorofluorocarbons, not Endemic species. B: 'Destruction and fragmentation of natural habitat' should be paired with Major cause of biodiversity loss, not Species diversity. D: 'Variety of species in a region' should be paired with Species diversity, not Deforestation.

BIO-12-CH15-0087 | Hard | Climate and ozone**Correct answer: B. Rise in average global temperature due to enhanced greenhouse effect - Global warming; Pollutant load measured through oxygen consumed by microbes - Biochemical oxygen demand**

Solution: Both parts of 'Rise in average global temperature due to enhanced greenhouse effect - Global warming; Pollutant load measured through oxygen consumed by microbes - Biochemical oxygen demand' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Large-scale removal of forest cover' requires Deforestation, and 'Variety of species in a region' requires Species diversity. C: The answers have been interchanged: 'Solid particles and droplets suspended in air' requires Particulate matter, and 'Variation among genes within a species' requires Genetic diversity. D: The answers have been interchanged: 'Variety of species in a region' requires Species diversity, and 'In situ protected area permitting regulated human activity' requires Biosphere reserve.

BIO-12-CH15-0088 | Easy | Conservation**Correct answer: B. Protection of species in their natural habitats**

Solution: In situ conservation represents 'Protection of species in their natural habitats'. Each alternative describes a different term or result.

Distractor review: A: 'Nutrient enrichment causing excessive aquatic plant growth' identifies Eutrophication, not In situ conservation. C: 'Ex situ storage of viable seeds' identifies Seed bank, not In situ conservation. D: 'In situ protected area permitting regulated human activity' identifies Biosphere reserve, not In situ conservation.

BIO-12-CH15-0089 | Medium | Threats and extinction**Correct answer: A. Harvesting organisms faster than they recover - Overexploitation**

Solution: The accurate entry is 'Harvesting organisms faster than they recover - Overexploitation'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Agreement aimed at controlling ozone-depleting substances' should be paired with Montreal Protocol, not Ex situ conservation. C: 'Maharashtra initiative promoting large-scale tree planting' should be paired with Mission Harit Maharashtra, not Reforestation. D: 'Indian law providing a framework for conservation and benefit sharing' should be paired with Biological Diversity Act 2002, not Eutrophication.

BIO-12-CH15-0090 | Easy | Pollution**Correct answer: D. Particulate matter**

Solution: 'Solid particles and droplets suspended in air' is the defining description for Particulate matter. The other choices belong to different chapter concepts.

Distractor review: A: Eutrophication belongs with 'Nutrient enrichment causing excessive aquatic plant growth', whereas the tested description is 'Solid particles and droplets suspended in air'. B: Genetic diversity belongs with 'Variation among genes within a species', whereas the tested description is 'Solid particles and droplets suspended in air'. C: Biological Diversity Act 2002 belongs with 'Indian law providing a framework for conservation and benefit sharing', whereas the tested description is 'Solid particles and droplets suspended in air'.

BIO-12-CH15-0091 | Challenging | Pollution**Correct answer: C. I, II, III and IV**

Solution: Statement I is correct because 'Nutrient enrichment causing excessive aquatic plant growth' corresponds to Eutrophication. Statement II is correct because 'Pollutant load measured through oxygen consumed by microbes' corresponds to Biochemical oxygen demand. Statement III is correct because 'Increase in concentration of persistent toxins at higher trophic levels' corresponds to Biomagnification. Statement IV is correct because 'Protection outside natural habitats' corresponds to Ex situ conservation. Therefore, the valid combination is I, II, III and IV.

Distractor review: A: This combination omits true statement(s) I, III, IV. B: This combination omits true statement(s) I, II, III. D: This combination omits true statement(s) II, III.

BIO-12-CH15-0092 | Hard | Pollution**Correct answer: A. Nutrient enrichment causing excessive aquatic plant growth - Eutrophication; Variety of habitats and ecological processes - Ecosystem diversity**

Solution: Both parts of 'Nutrient enrichment causing excessive aquatic plant growth - Eutrophication; Variety of habitats and ecological processes - Ecosystem diversity' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: B: The answers have been interchanged: 'Relationship in which larger areas contain more species' requires Species-area relationship, and 'Gas chiefly responsible for anthropogenic greenhouse warming' requires Carbon dioxide. C: The answers have been interchanged: 'Large-scale removal of forest cover' requires Deforestation, and 'Harvesting organisms faster than they recover' requires Overexploitation. D: The answers have been interchanged: 'Secondary extinction following loss of a dependent species' requires Co-extinction, and 'Large-scale removal of forest cover' requires Deforestation.

BIO-12-CH15-0093 | Medium | Levels of biodiversity**Correct answer: D. Number of species present in a community - Species richness**

Solution: The accurate entry is 'Number of species present in a community - Species richness'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Large-scale removal of forest cover' should be paired with Deforestation, not Species-area relationship. B: 'General change in species richness from equator to poles' should be paired with It decreases, not Global warming. C: 'Secondary extinction following loss of a dependent species' should be paired with Co-extinction, not Endemic species.

BIO-12-CH15-0094 | Hard | Forests and law**Correct answer: A. Maharashtra initiative promoting large-scale tree planting - Genetic diversity**

Solution: 'Maharashtra initiative promoting large-scale tree planting' should be paired with Mission Harit Maharashtra, not Genetic diversity. The option showing that mismatch is the entry that requires correction.

Distractor review: B: This is a valid association: Protection of species in their natural habitats - In situ conservation. C: This is a valid association: General change in species richness from equator to poles - It decreases. D: This is a valid association: Solid particles and droplets suspended in air - Particulate matter.

BIO-12-CH15-0095 | Hard | Forests and law

Correct answer: D. Indian law providing a framework for conservation and benefit sharing - Biological Diversity Act 2002; Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide

Solution: Both parts of 'Indian law providing a framework for conservation and benefit sharing - Biological Diversity Act 2002; Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide' preserve their respective chapter relationships; each alternative contains swapped answers.

Distractor review: A: The answers have been interchanged: 'Pollutant load measured through oxygen consumed by microbes' requires Biochemical oxygen demand, and 'Agreement aimed at controlling ozone-depleting substances' requires Montreal Protocol. B: The answers have been interchanged: 'In situ protected area permitting regulated human activity' requires Biosphere reserve, and 'General change in species richness from equator to poles' requires It decreases. C: The answers have been interchanged: 'Agreement aimed at controlling ozone-depleting substances' requires Montreal Protocol, and 'Rise in average global temperature due to enhanced greenhouse effect' requires Global warming.

BIO-12-CH15-0096 | Hard | Threats and extinction

Correct answer: B. Secondary extinction following loss of a dependent species - Species diversity

Solution: 'Secondary extinction following loss of a dependent species' should be paired with Co-extinction, not Species diversity. The option showing that mismatch is the entry that requires correction.

Distractor review: A: This is a valid association: Solid particles and droplets suspended in air - Particulate matter. C: This is a valid association: Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra. D: This is a valid association: Species restricted to a particular geographic area - Endemic species.

BIO-12-CH15-0097 | Medium | Conservation

Correct answer: B. Ex situ storage of viable seeds - Seed bank

Solution: The accurate entry is 'Ex situ storage of viable seeds - Seed bank'. The other entries combine a description with the answer from a different concept.

Distractor review: A: 'Agreement aimed at controlling ozone-depleting substances' should be paired with Montreal Protocol, not Invasive alien species. C: 'Non-native organism spreading and causing harm' should be paired with Invasive alien species, not Overexploitation. D: 'Secondary extinction following loss of a dependent species' should be paired with Co-extinction, not Biodiversity hotspot.

BIO-12-CH15-0098 | Medium | Conservation

Correct answer: A. In situ protected area permitting regulated human activity - Biosphere reserve

Solution: The accurate entry is 'In situ protected area permitting regulated human activity - Biosphere reserve'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Nutrient enrichment causing excessive aquatic plant growth' should be paired with Eutrophication, not Carbon dioxide. C: 'Chemicals releasing chlorine that destroys stratospheric ozone' should be paired with Chlorofluorocarbons, not Biosphere reserve. D: 'Rise in average global temperature due to enhanced greenhouse effect' should be paired with Global warming, not Montreal Protocol.

BIO-12-CH15-0099 | Easy | Conservation

Correct answer: D. Seed bank

Solution: 'Ex situ storage of viable seeds' is the defining description for Seed bank. The other choices belong to different chapter concepts.

Distractor review: A: Montreal Protocol belongs with 'Agreement aimed at controlling ozone-depleting substances', whereas the tested description is 'Ex situ storage of viable seeds'. B: Invasive alien species belongs with 'Non-native organism spreading and causing harm', whereas the tested description is 'Ex situ storage of viable seeds'. C: Ex situ conservation belongs with 'Protection outside natural habitats', whereas the tested description is 'Ex situ storage of viable seeds'.

BIO-12-CH15-0100 | Medium | Climate and ozone

Correct answer: A. Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide

Solution: The accurate entry is 'Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide'. The other entries combine a description with the answer from a different concept.

Distractor review: B: 'Variety of habitats and ecological processes' should be paired with Ecosystem diversity, not Biodiversity hotspot. C: 'Region with many endemic species under exceptional threat' should be paired with Biodiversity hotspot, not Eutrophication. D: 'Protection of species in their natural habitats' should be paired with In situ conservation, not Co-extinction.